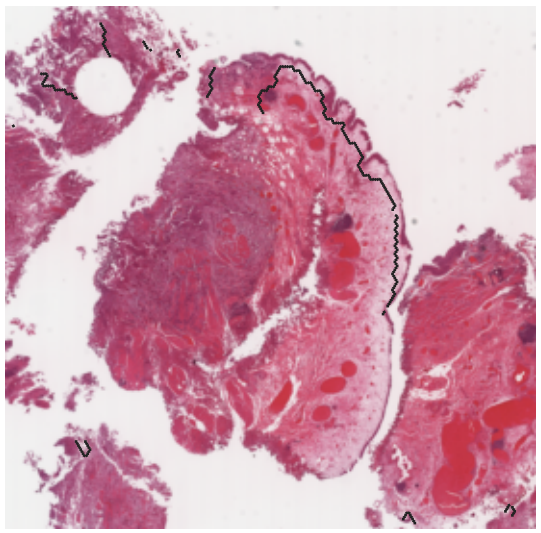
GSE319536_BC01 | Tissue context

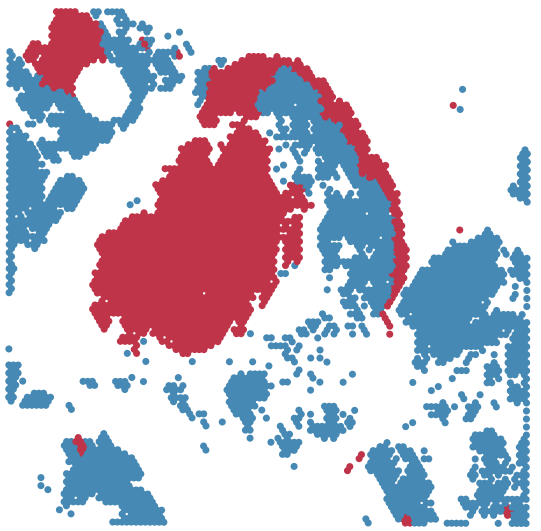
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



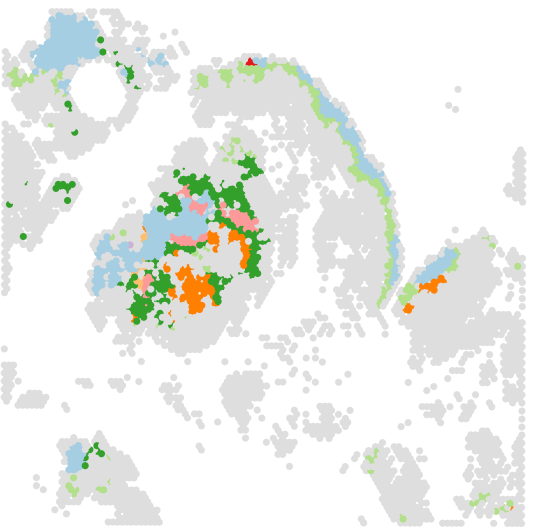
Black line: observed candidate outer edge

Final candidate location



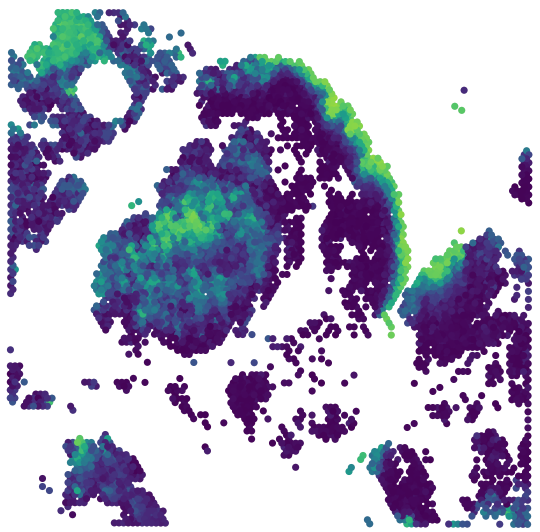
inside outside

Accepted R12 (original 0–8)



0 3 5 7 8
2 4 6

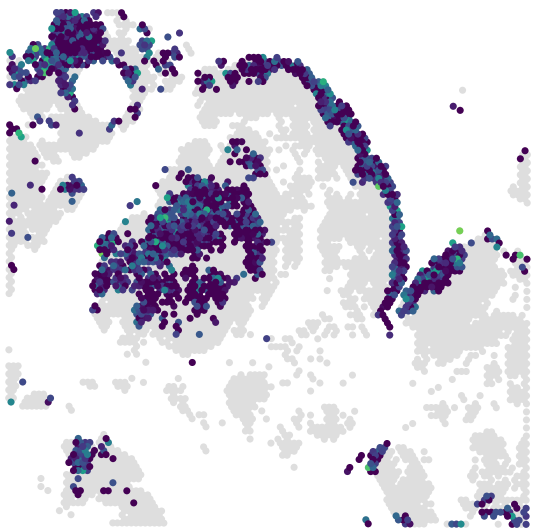
Epithelial support



0 1

Fraction of all 30

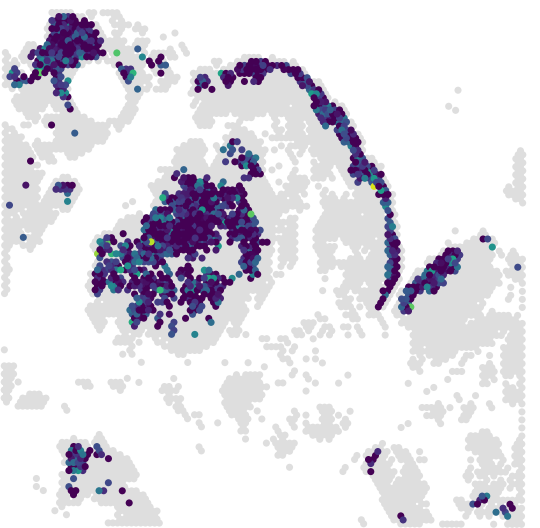
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



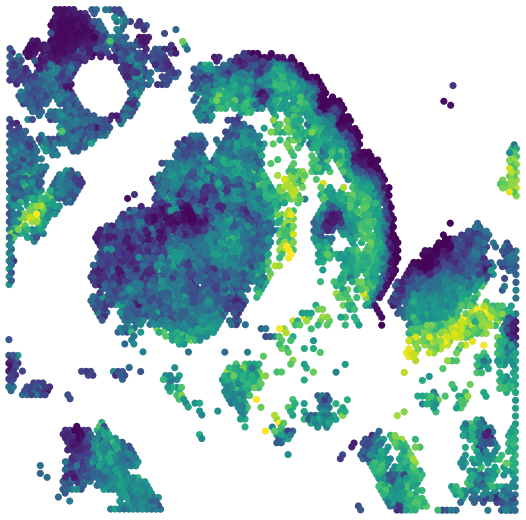
0 0.14

Matched-model AUCell

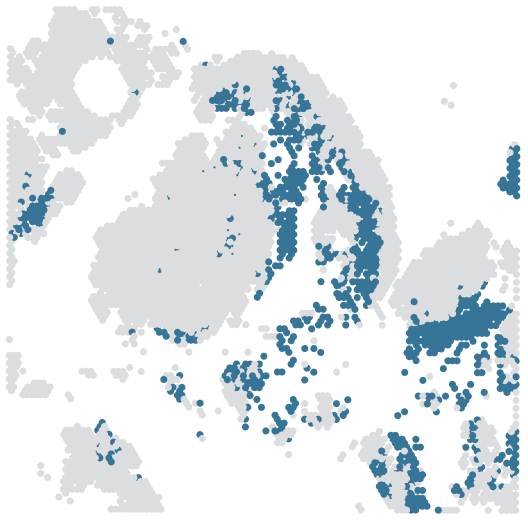
GSE319536_BC01 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

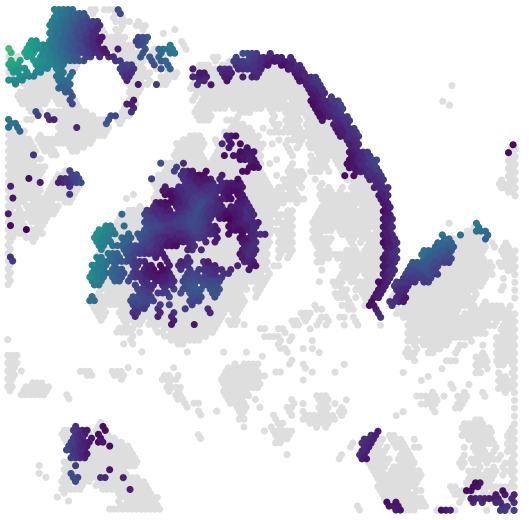
CAF composition

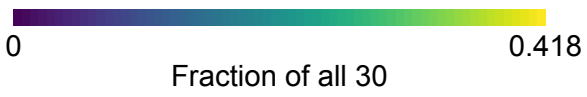


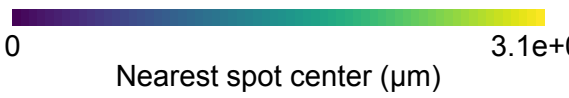
CAF P80 targets



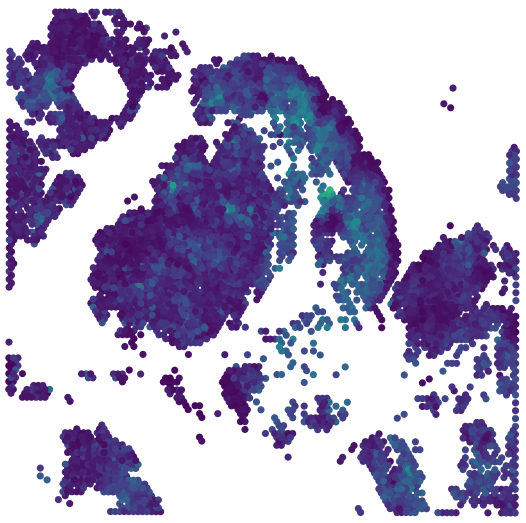
CAF distance





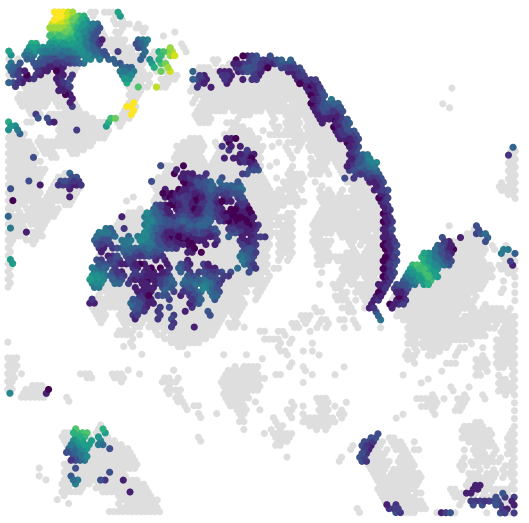


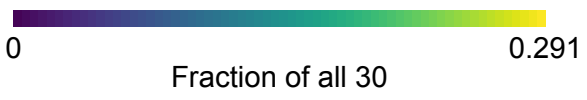
TAM composition

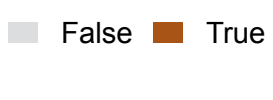


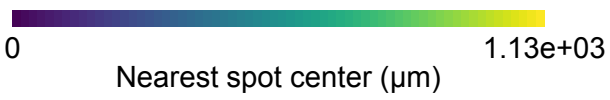
TAM P80 targets

TAM distance

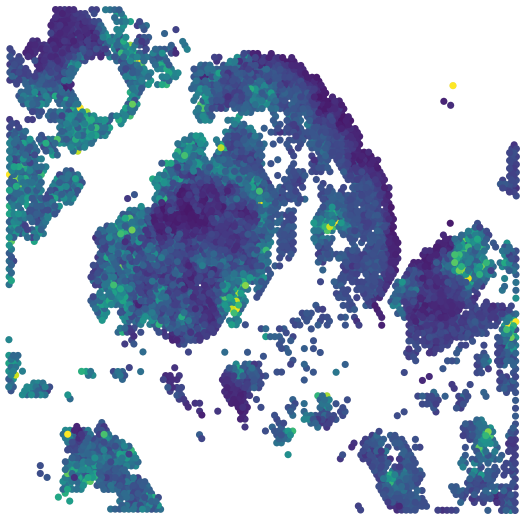




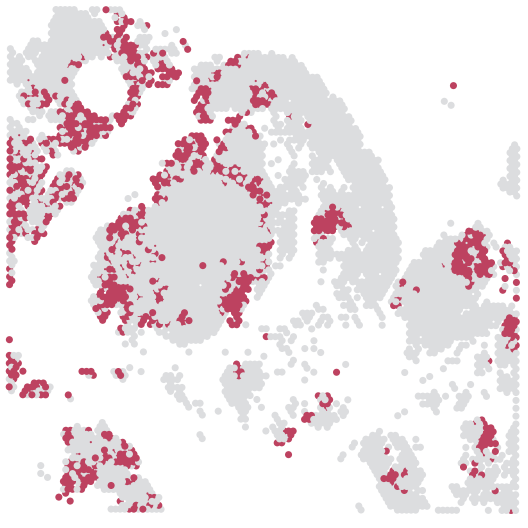




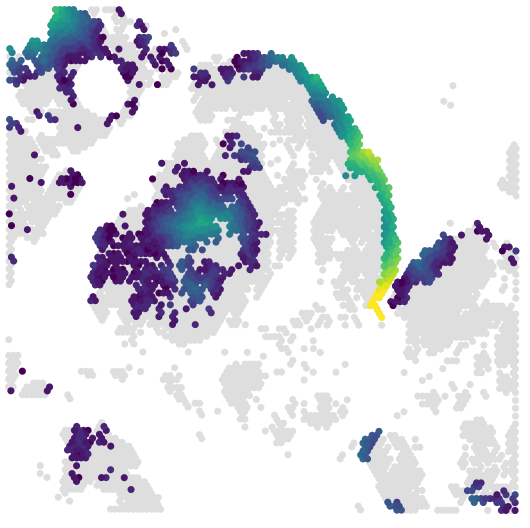
Treg composition

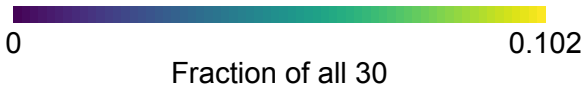


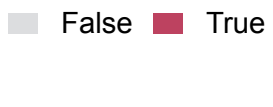
Treg P80 targets

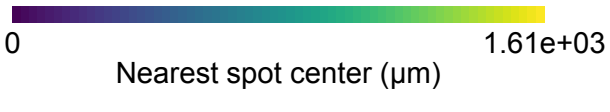


Treg distance



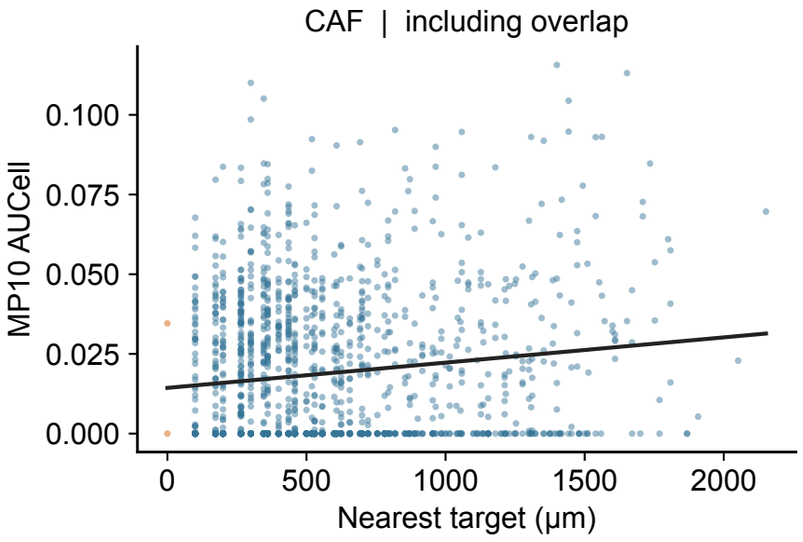




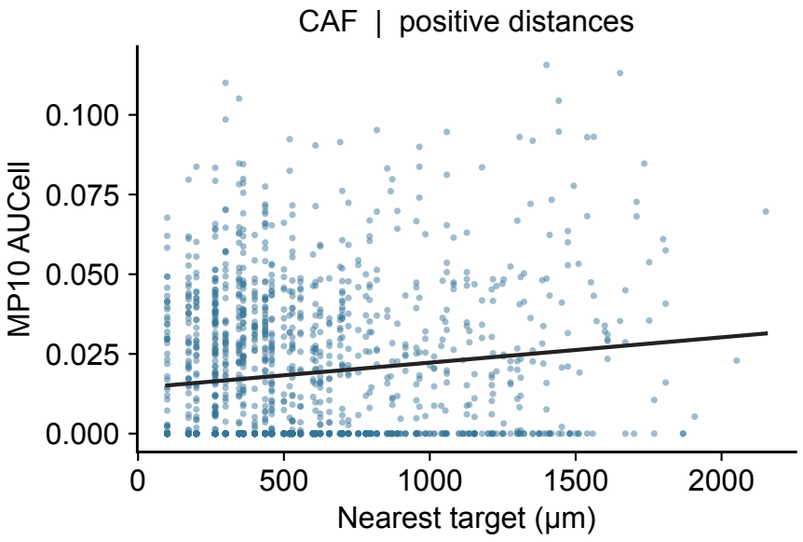


GSE319536_BC01 | MP10 versus nearest target

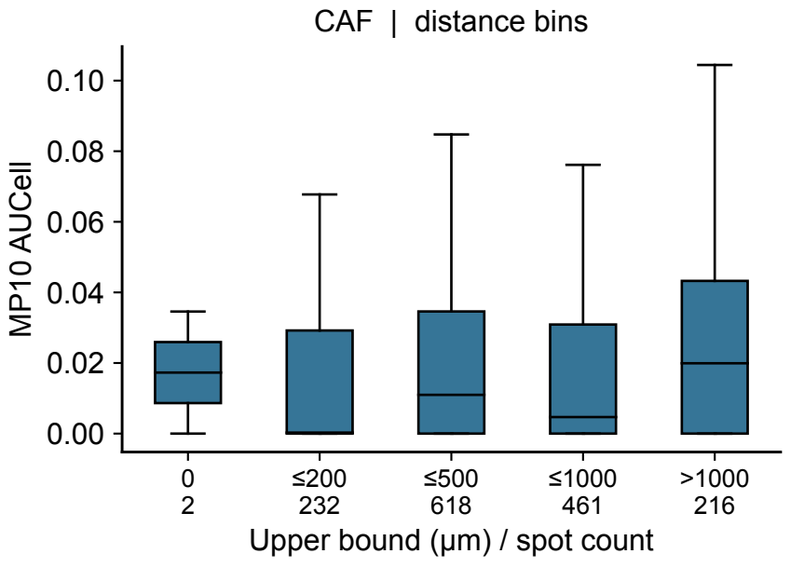
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



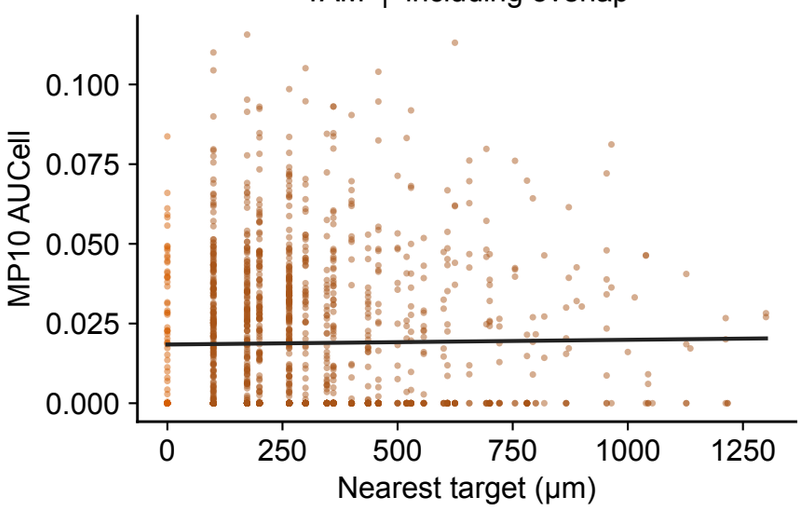
r +0.13 ρ +0.07
n 1529/1567 queries • 1056 targets



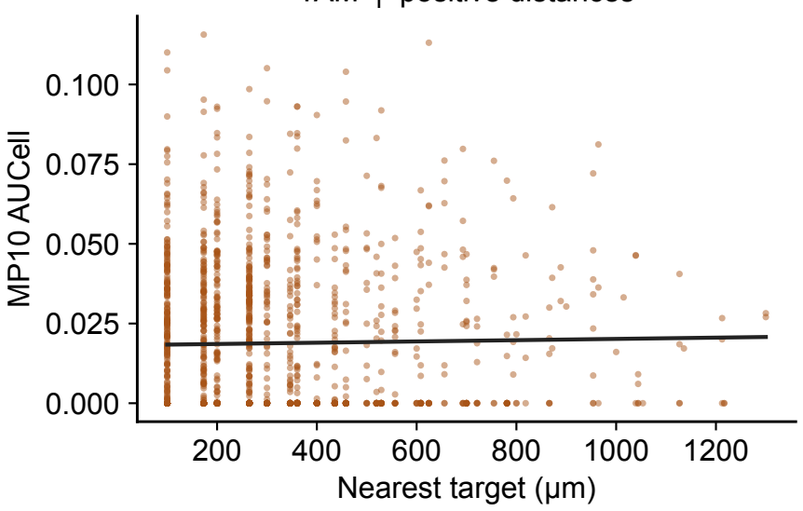
r +0.13 ρ +0.07
n 1527/1567 queries • 1056 targets



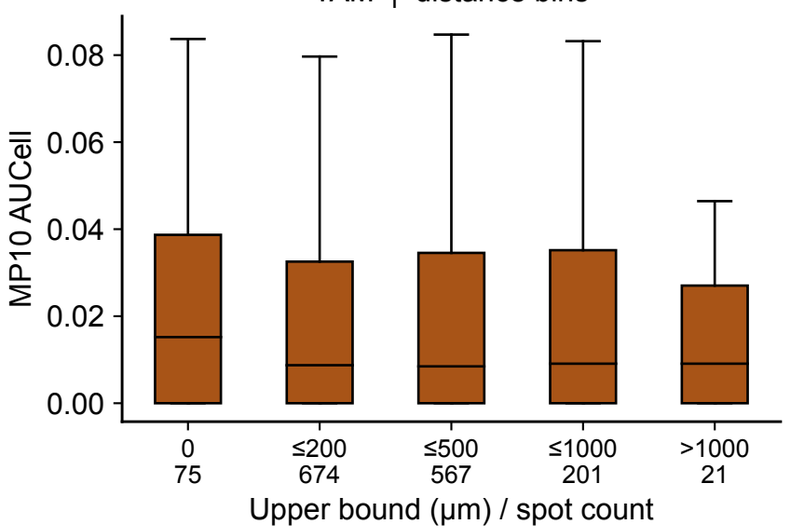
Orange = zero distance
Line = within-section linear fit



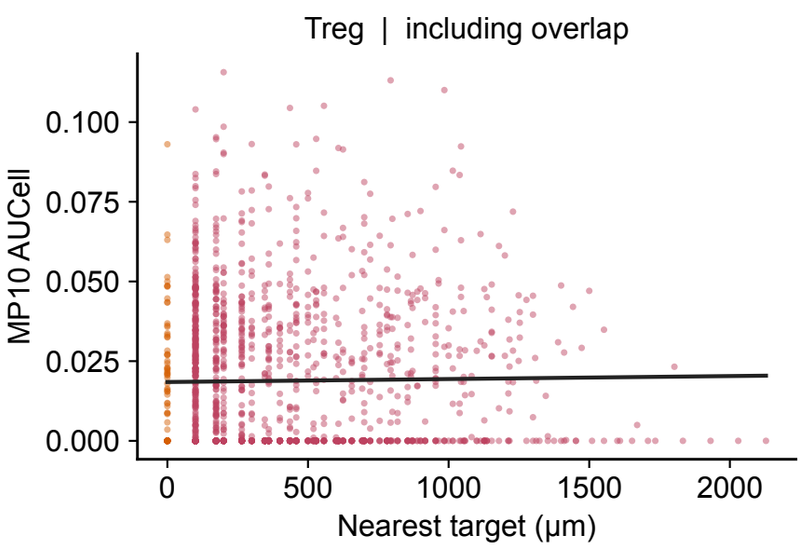
r +0.01 ρ +0.01
n 1538/1567 queries • 1056 targets



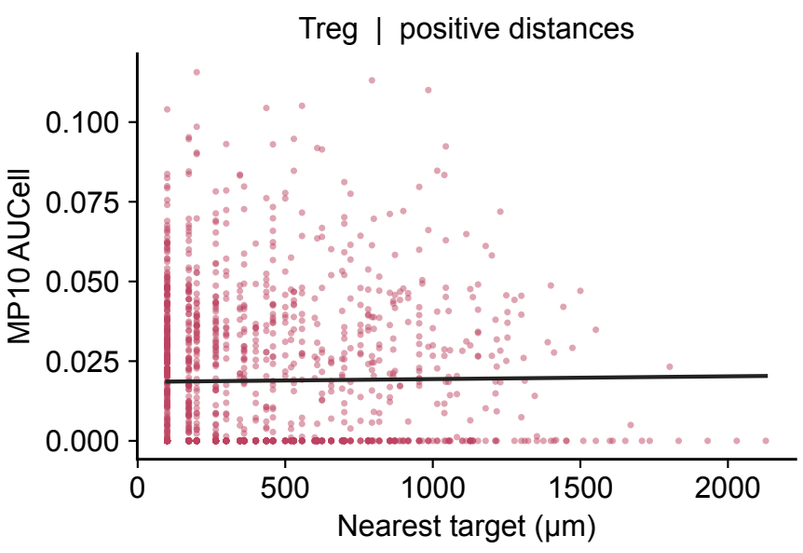
r +0.02 ρ +0.01
n 1463/1567 queries • 1056 targets



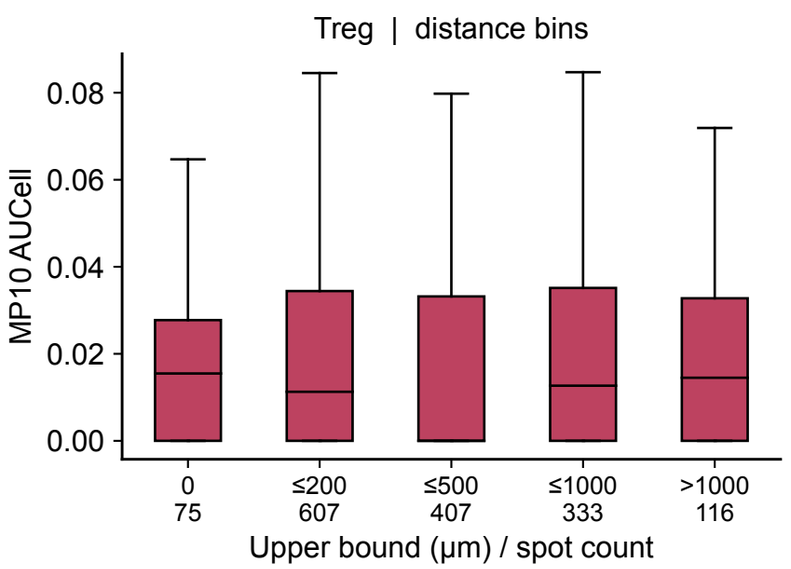
Orange = zero distance
Line = within-section linear fit



r +0.01 ρ -0.00
n 1538/1567 queries • 1056 targets



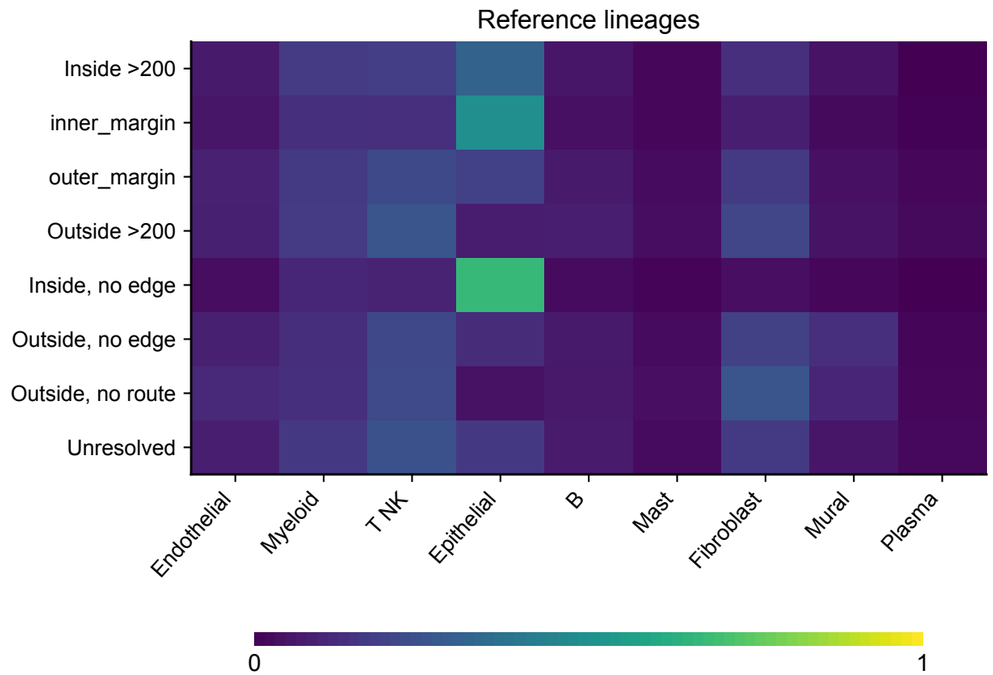
r +0.01 ρ -0.00
n 1463/1567 queries • 1056 targets

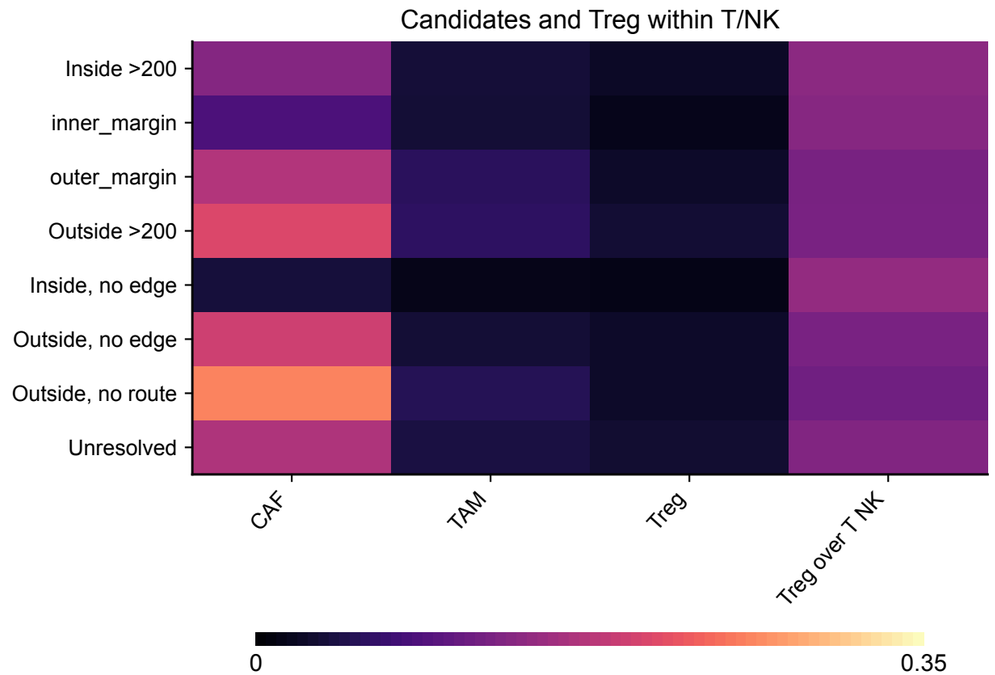


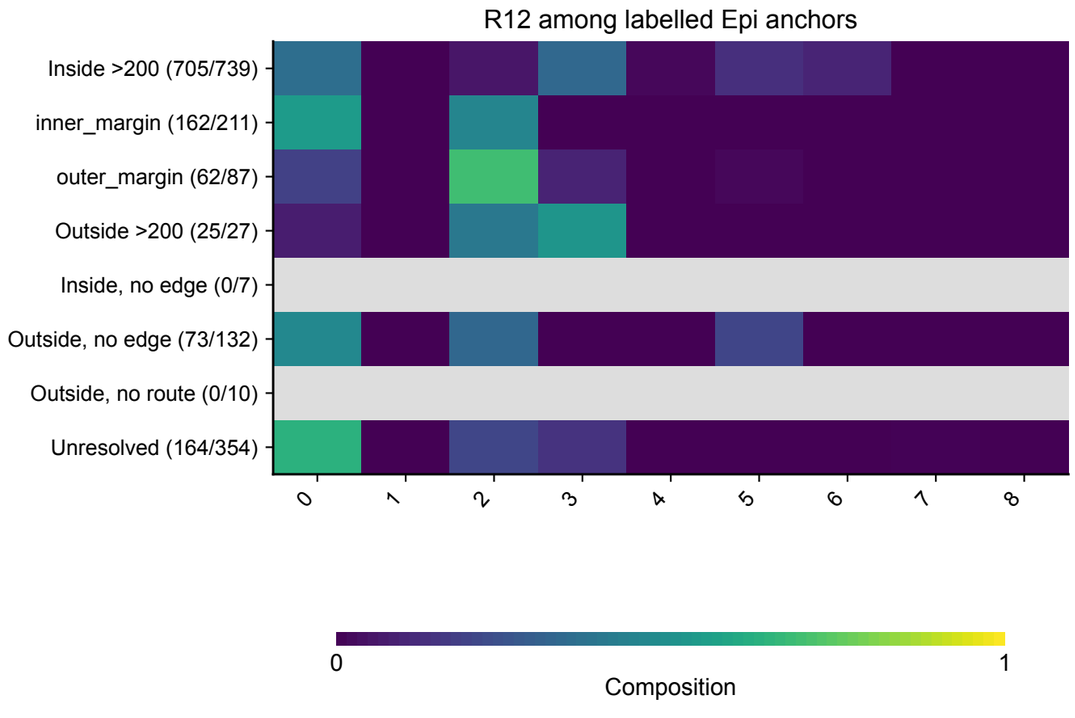
Orange = zero distance
Line = within-section linear fit

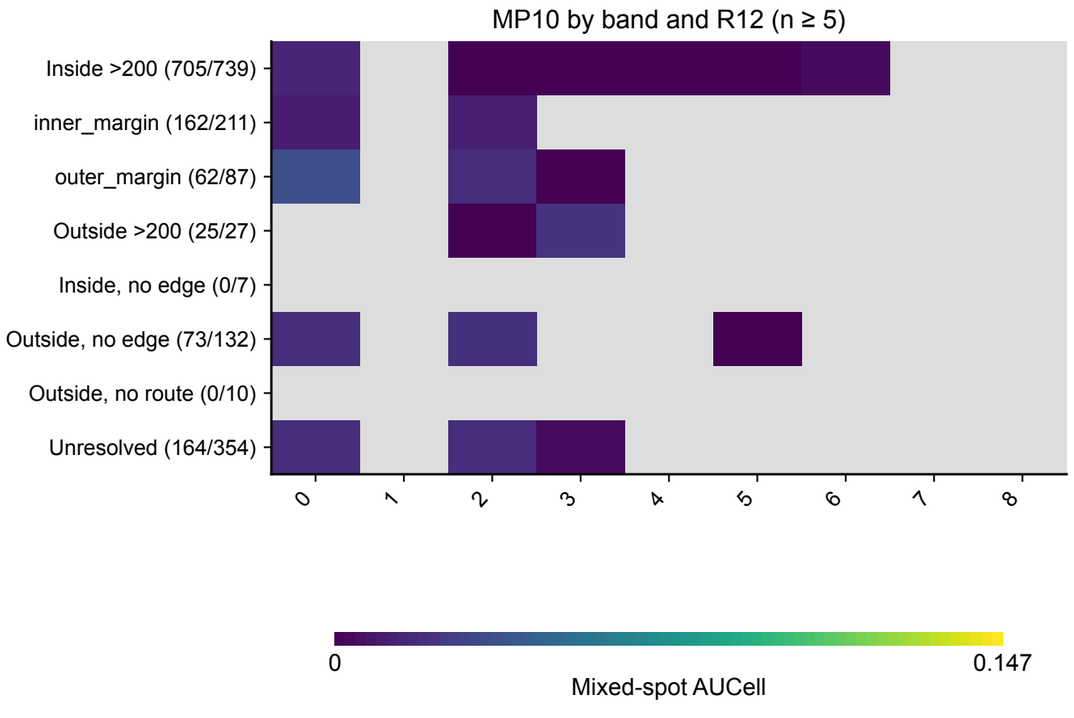
GSE319536_BC01 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





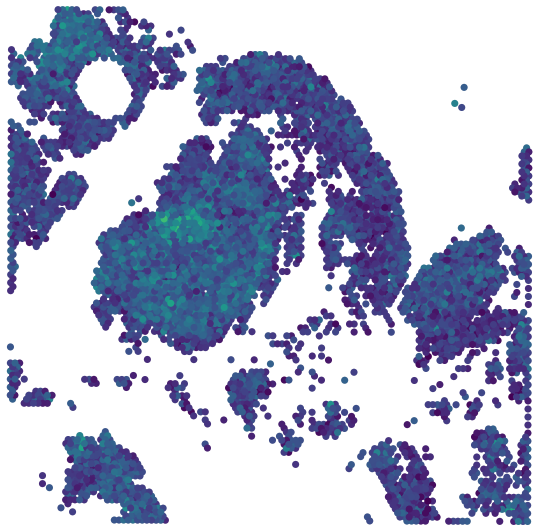




GSE319536_BC01 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

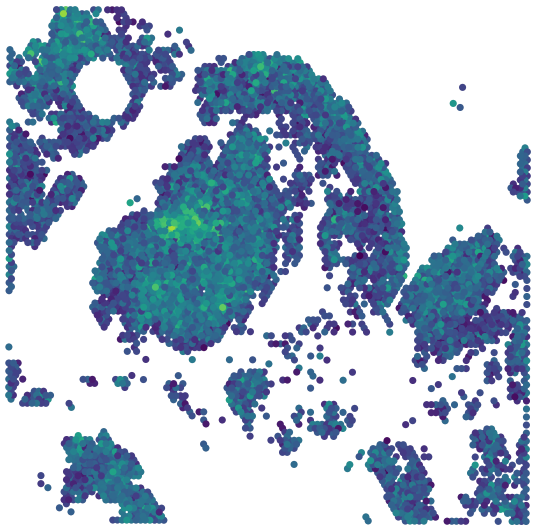
E2F Targets



0 0.152

UCell score

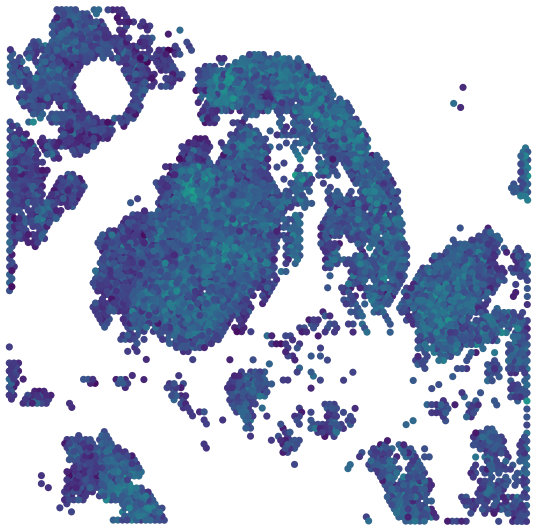
G2M Checkpoint



0 0.134

UCell score

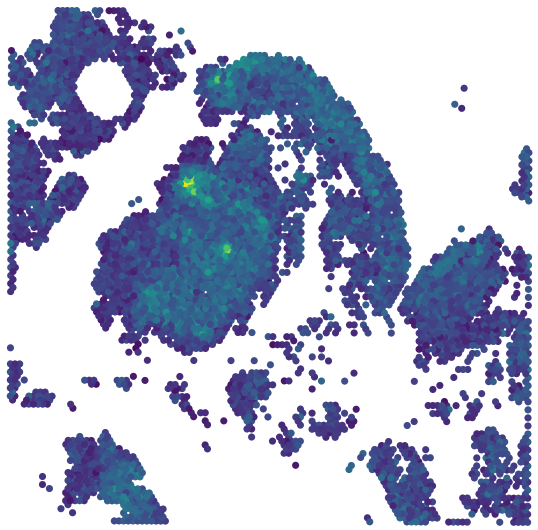
Hypoxia



0 0.211

UCell score

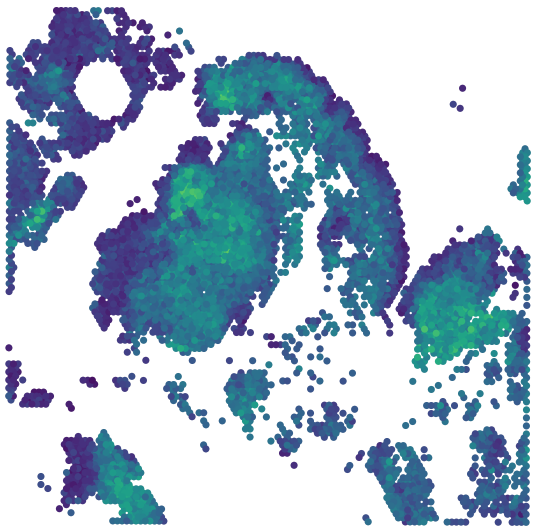
Tnfa Signaling Via Nfkb



0 0.275

UCell score

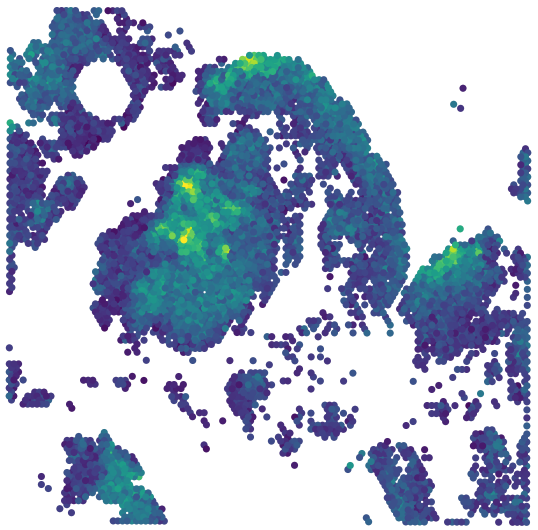
Emt



0 0.386

UCell score

Interferon Gamma Response



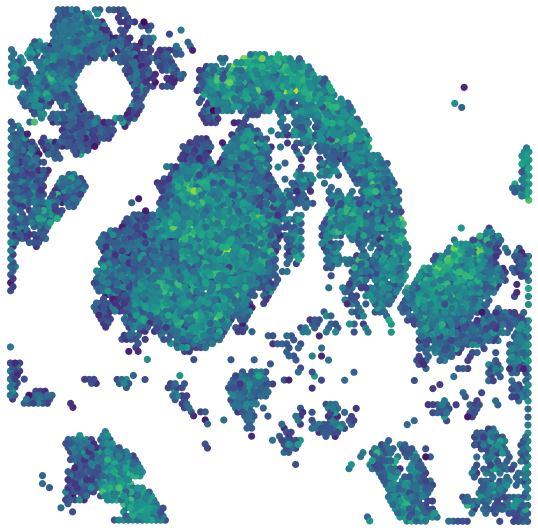
0 0.275

UCell score

GSE319536_BC01 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

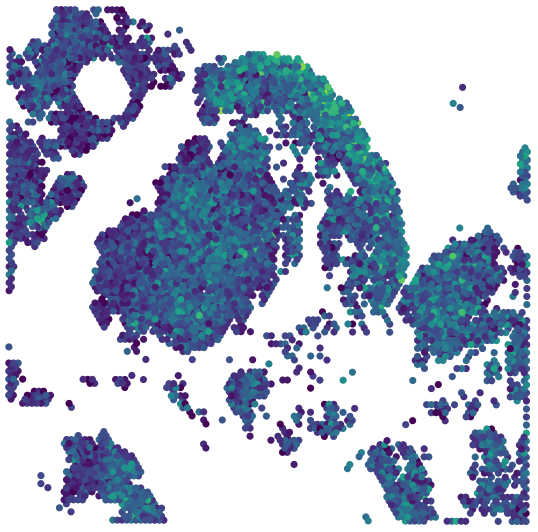
Il2 Stat5 Signaling



0 0.144

UCell score

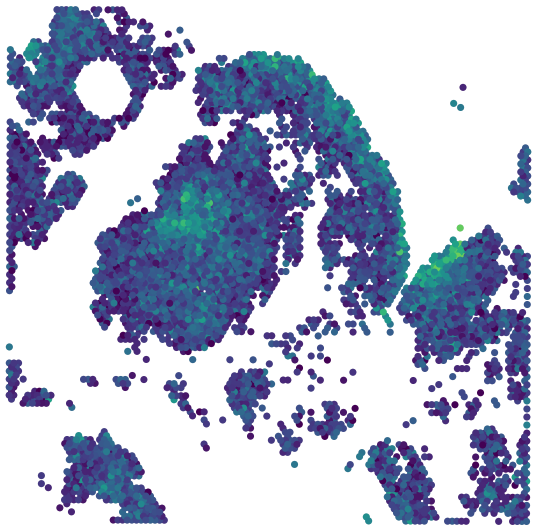
Tgf Beta Signaling



0 0.232

UCell score

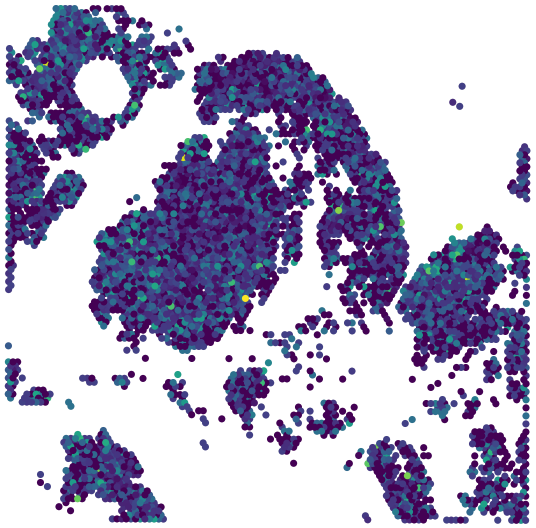
Cholesterol Homeostasis



0 0.21

UCell score

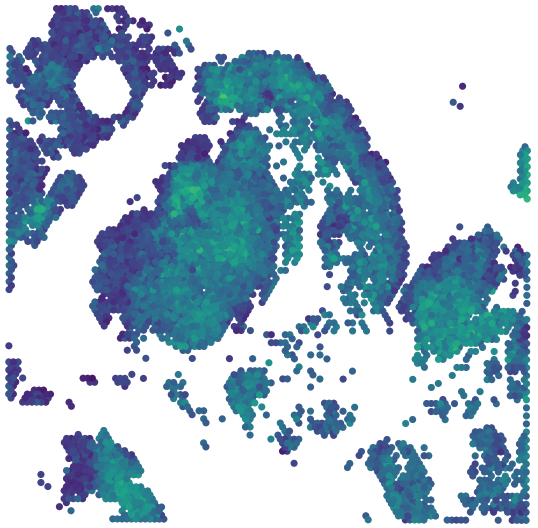
Cholesterol Biosynthesis



0 0.166

UCell score

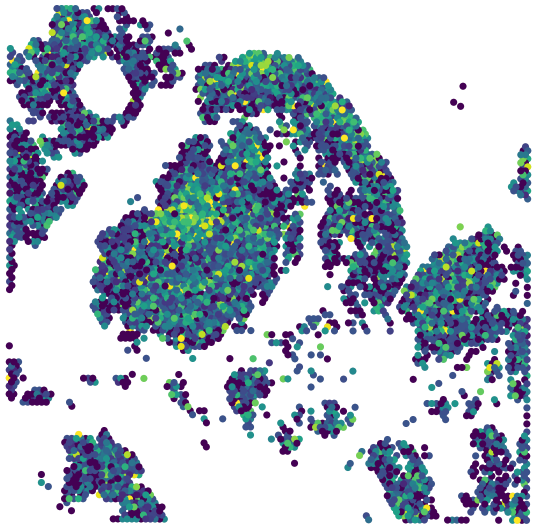
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



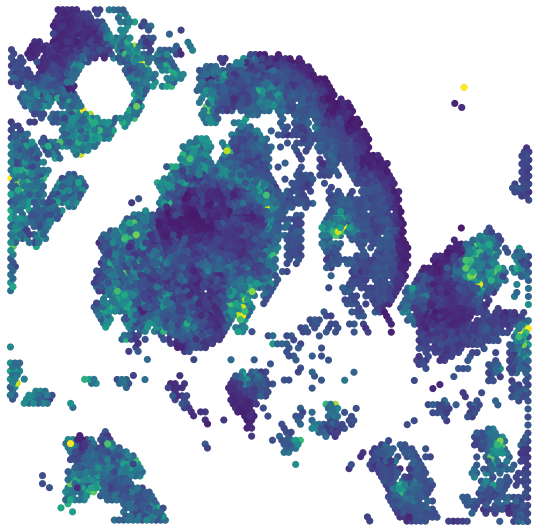
0 0.151

UCell score

GSE319536_BC01 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

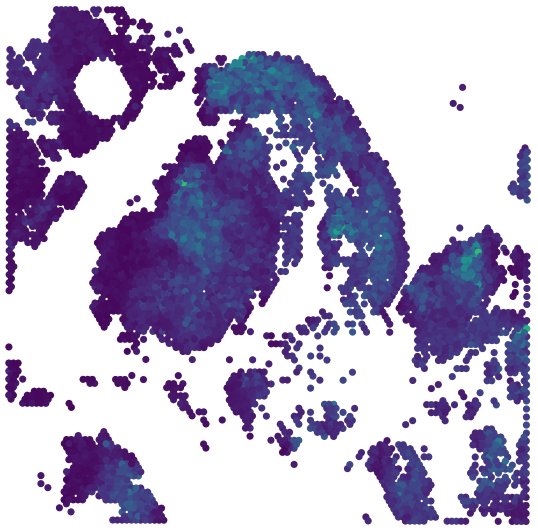
Treg composition



0 0.102

Fraction of all 30

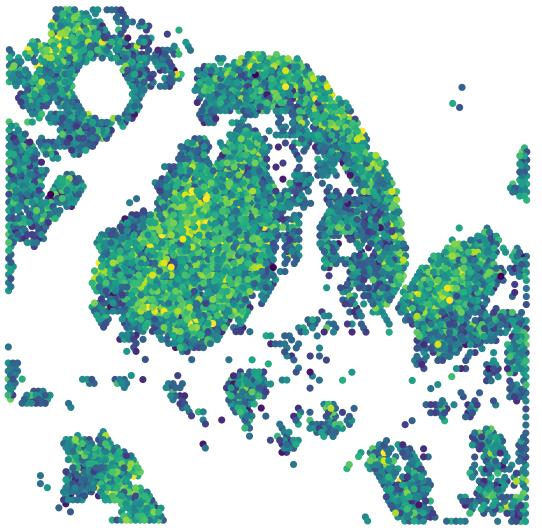
Treg estimated abundance



0 2.85

Posterior mean estimate

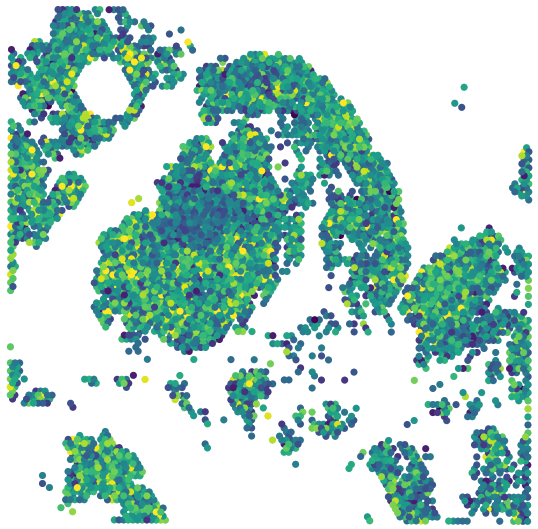
Treg Resting Vs Activated Dn



0 0.0778

UCell score

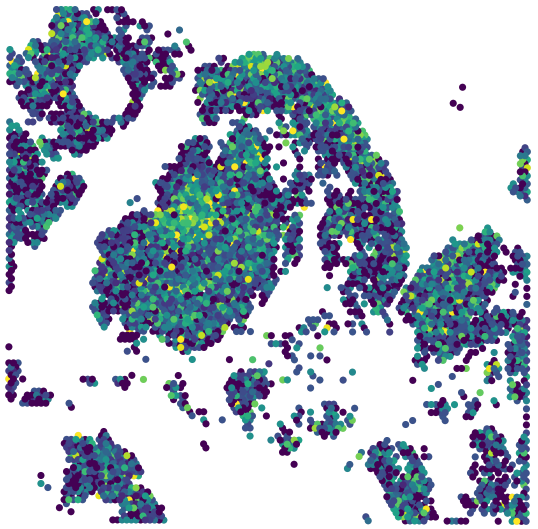
Treg Resting Vs Activated Up



0 0.0609

UCell score

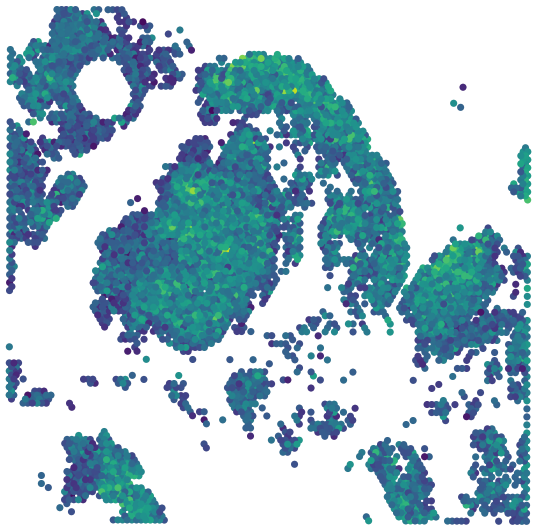
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

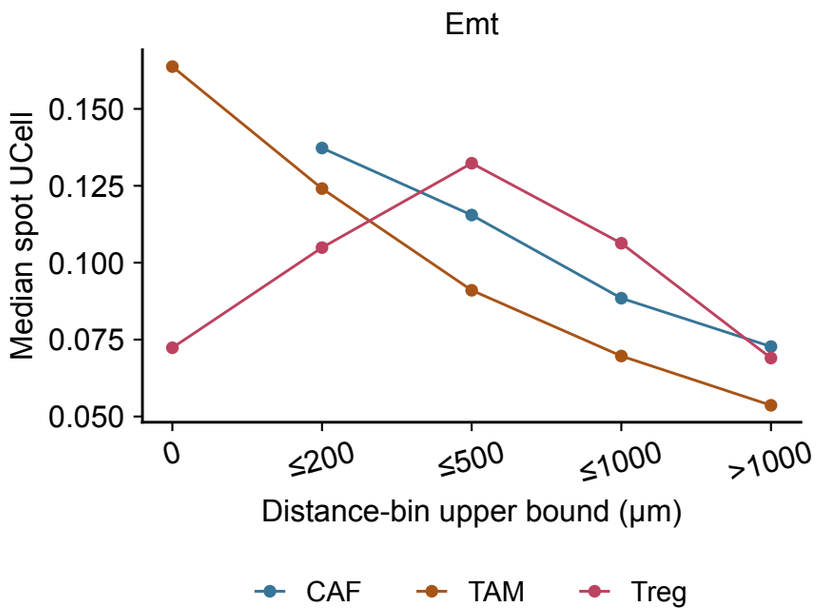


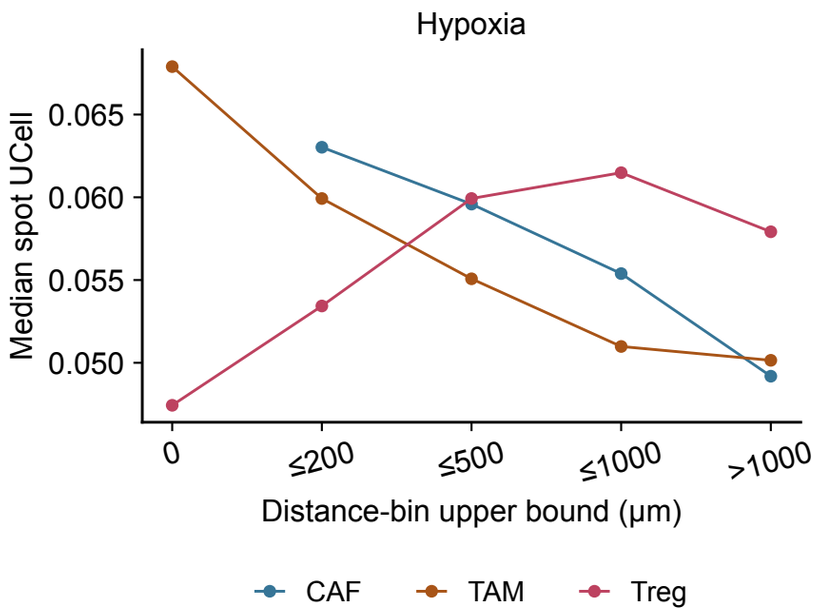
0 0.144

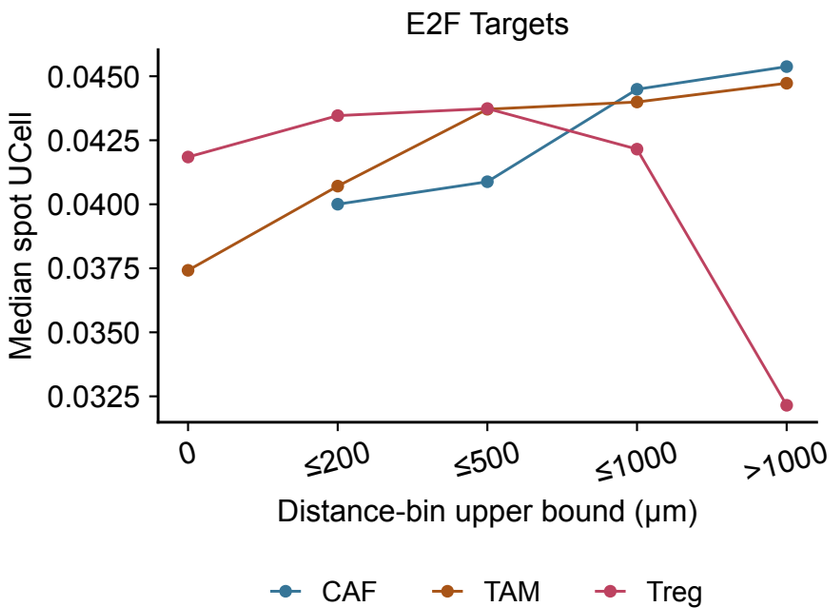
UCell score

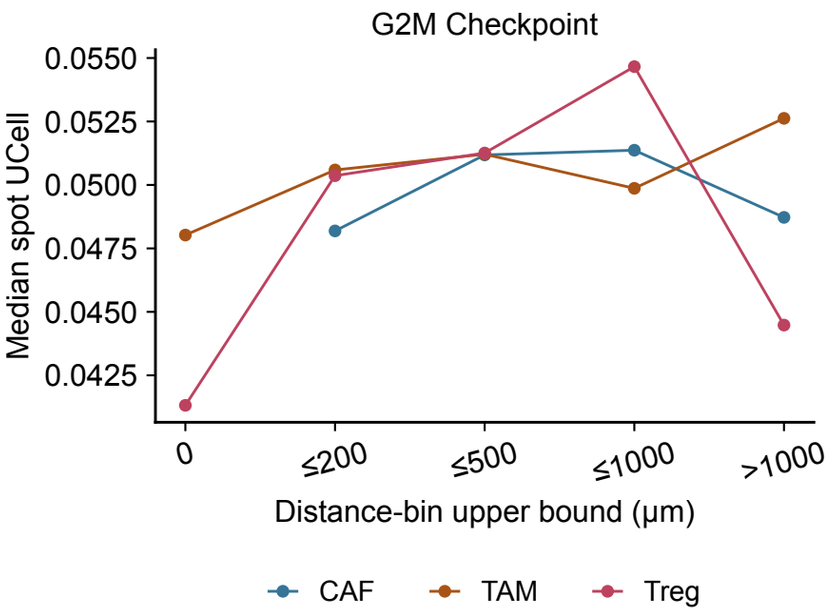
GSE319536_BC01 | Programs by target distance

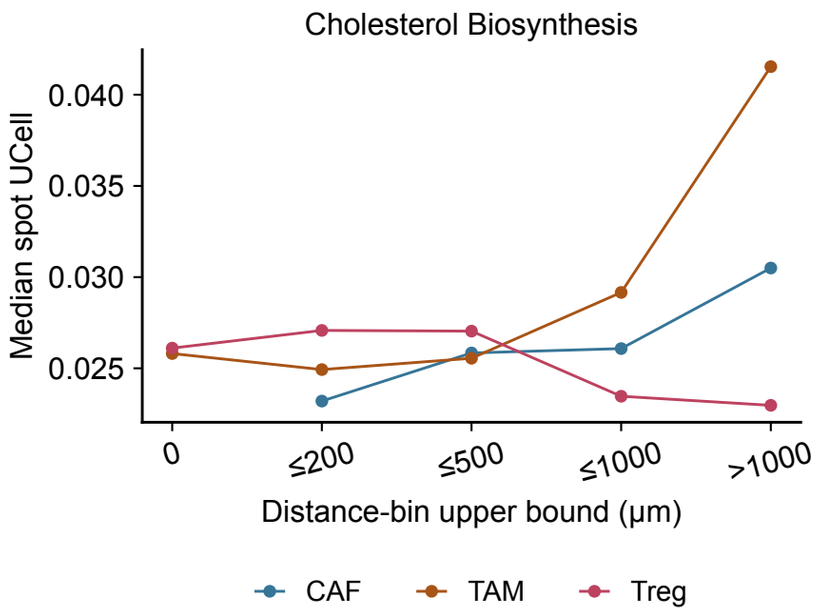
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

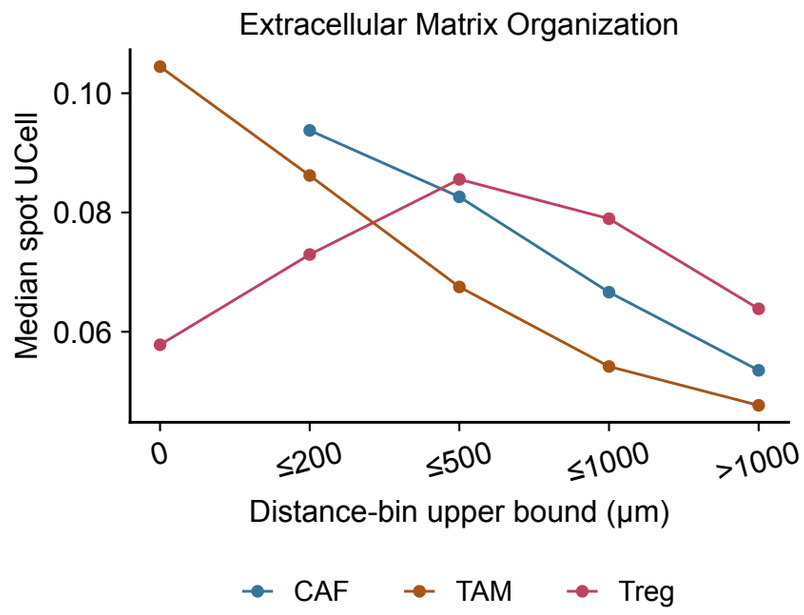






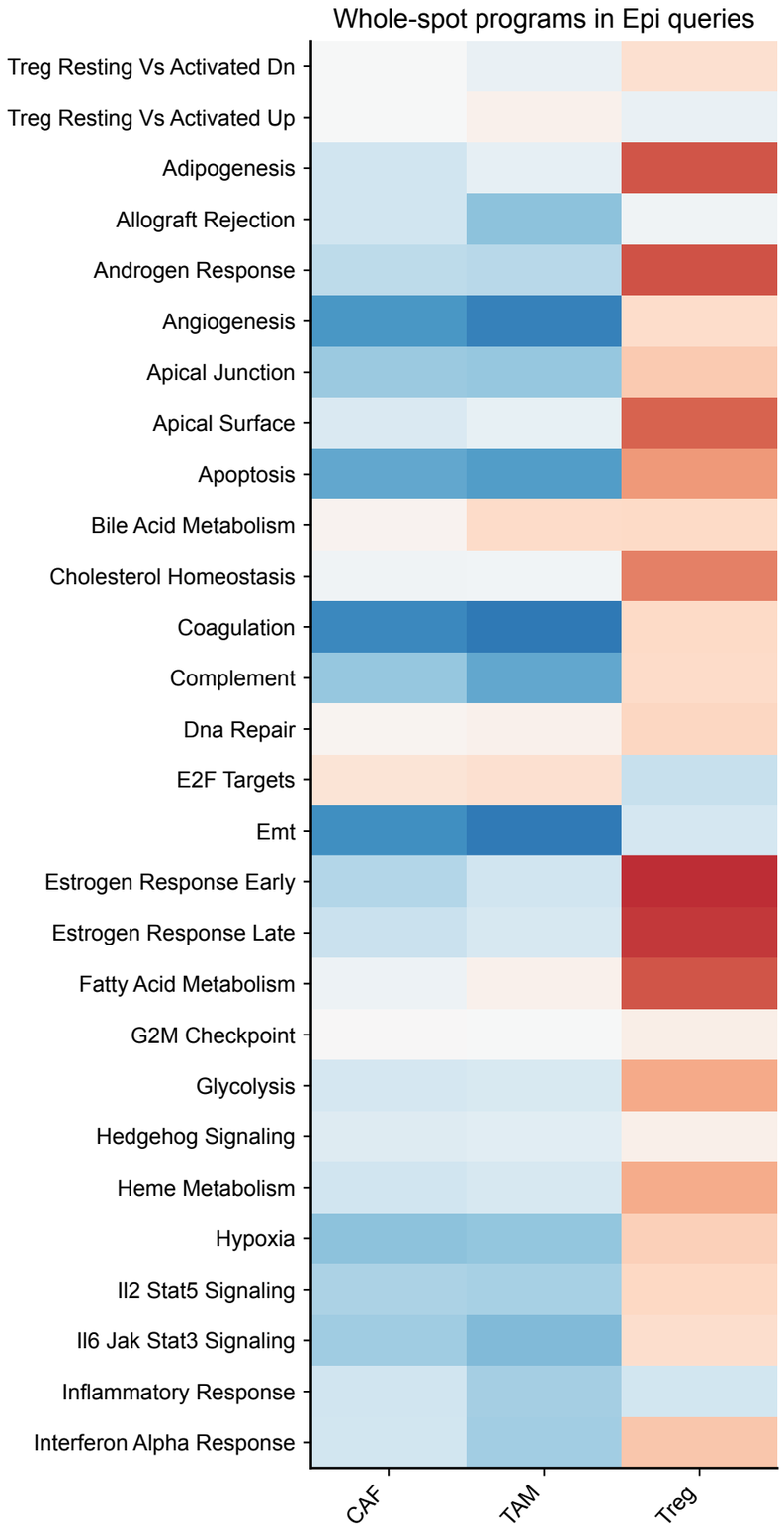


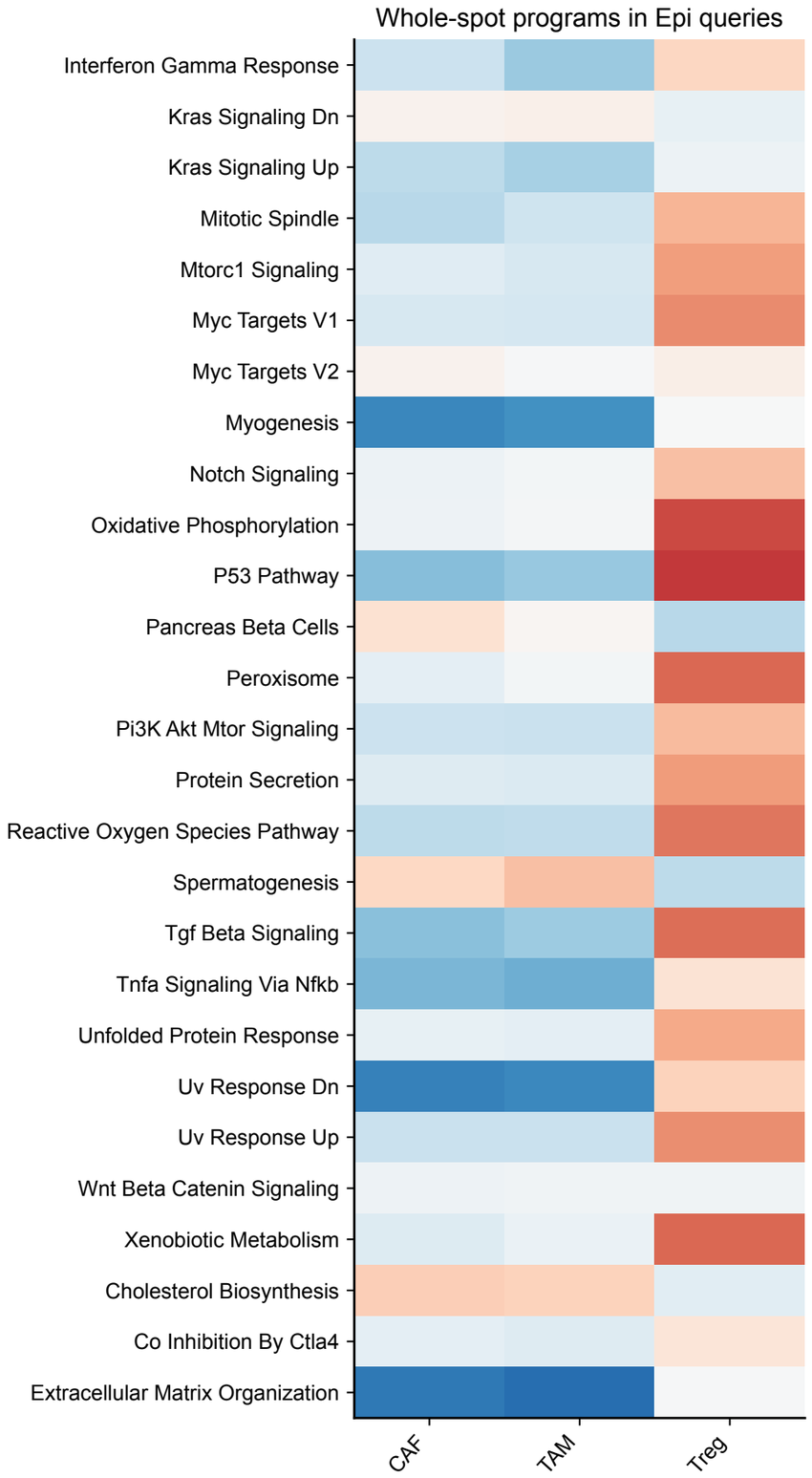




GSE319536_BC01 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

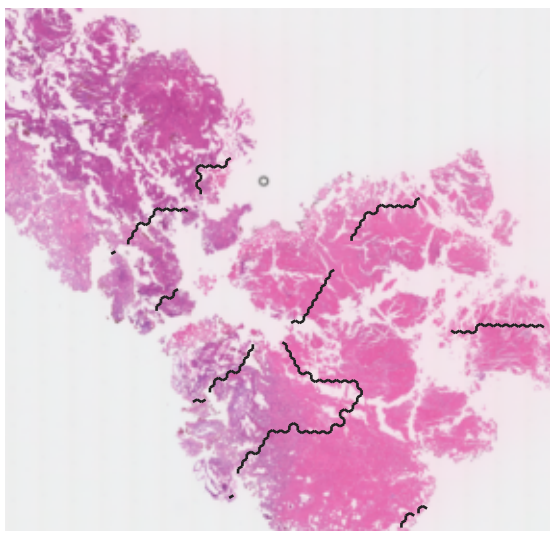




GSE319536_BC02 | Tissue context

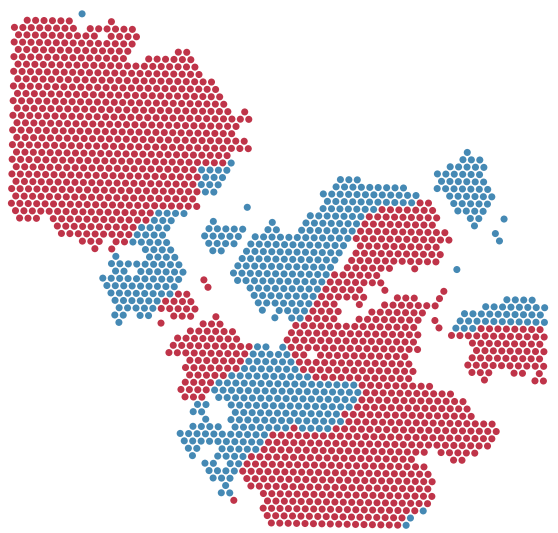
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



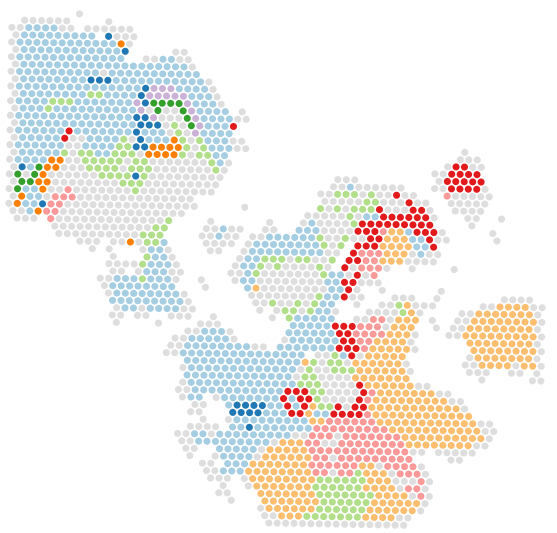
Black line: observed candidate outer edge

Final candidate location



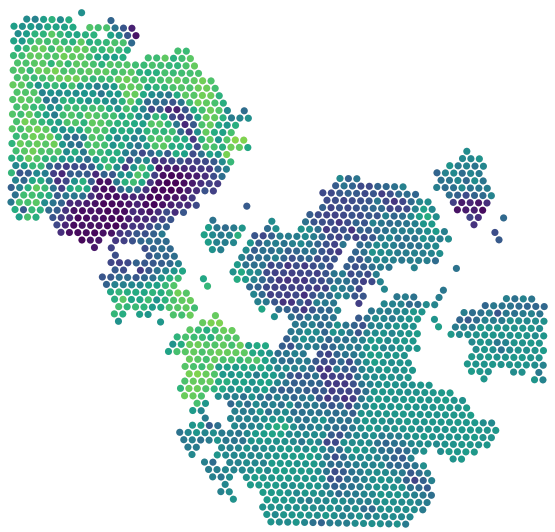
inside outside

Accepted R12 (original 0–8)



0 2 4 6 8
1 3 5 7

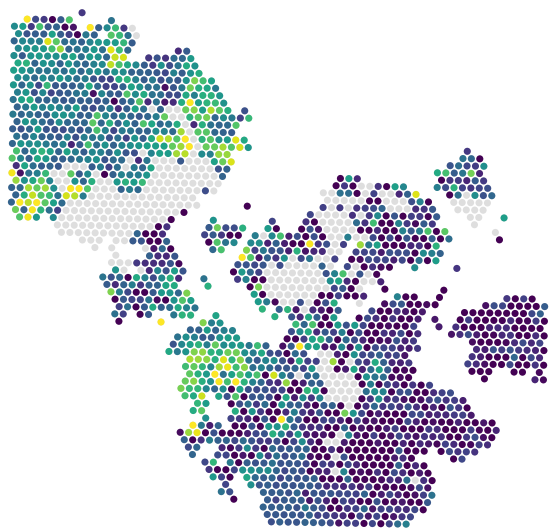
Epithelial support



0 1

Fraction of all 30

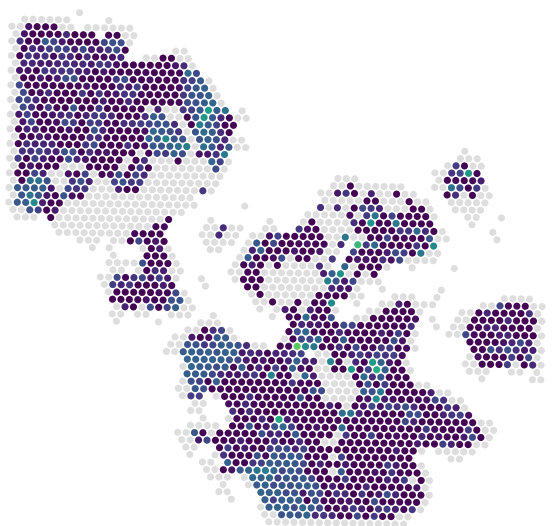
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



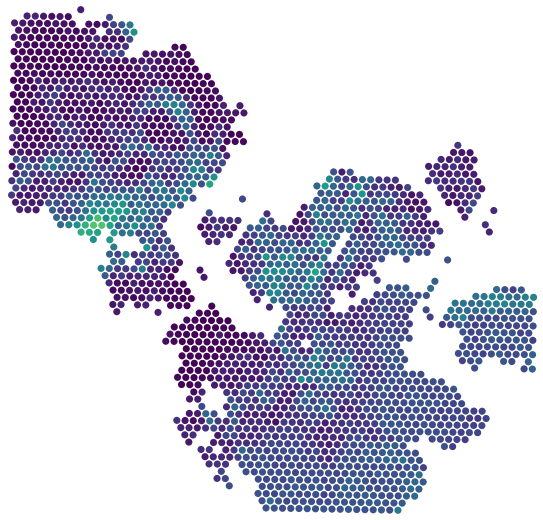
0 0.14

Matched-model AUCell

GSE319536_BC02 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

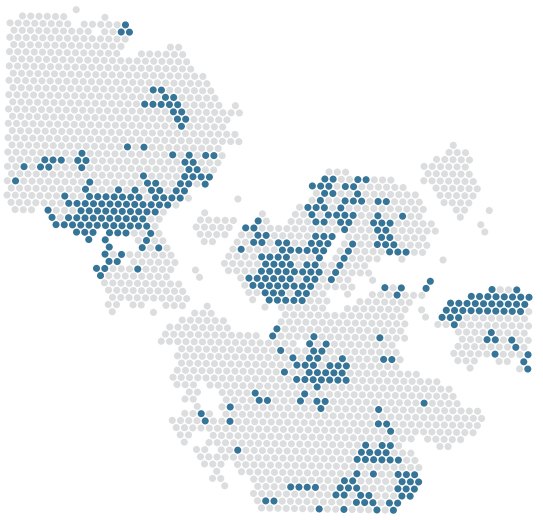
CAF composition



0 0.418

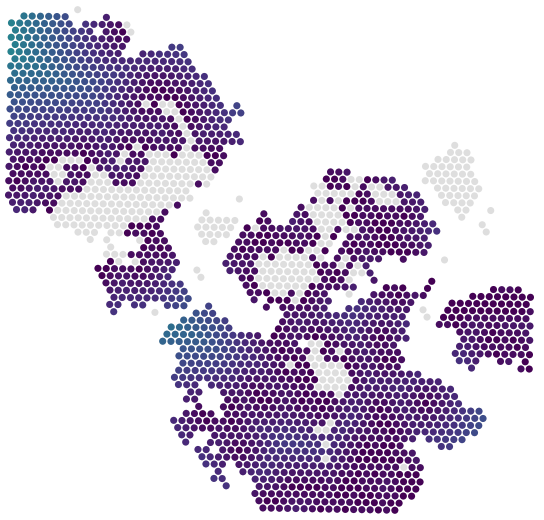
Fraction of all 30

CAF P80 targets



False True

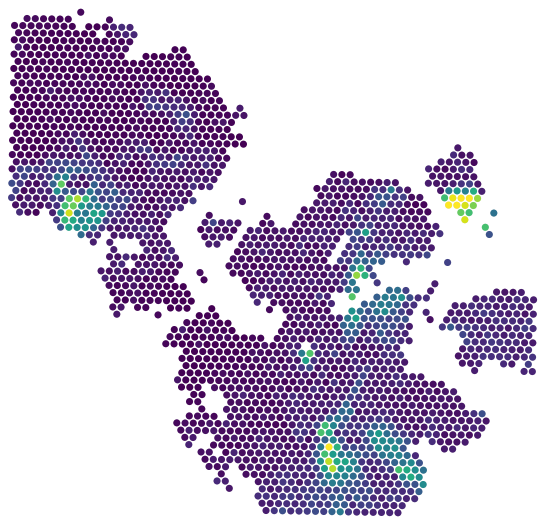
CAF distance



0 3.1e+03

Nearest spot center (μm)

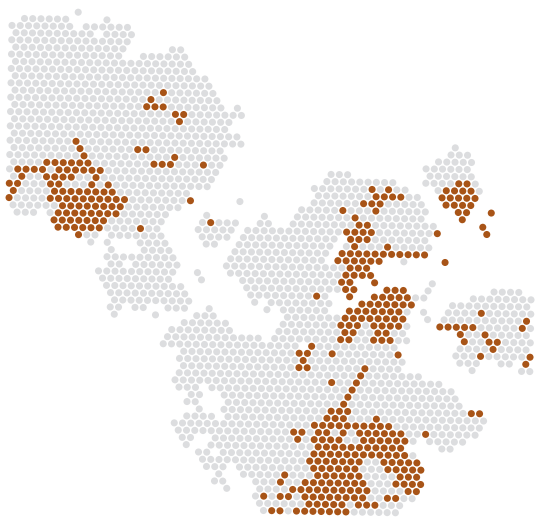
TAM composition



0 0.291

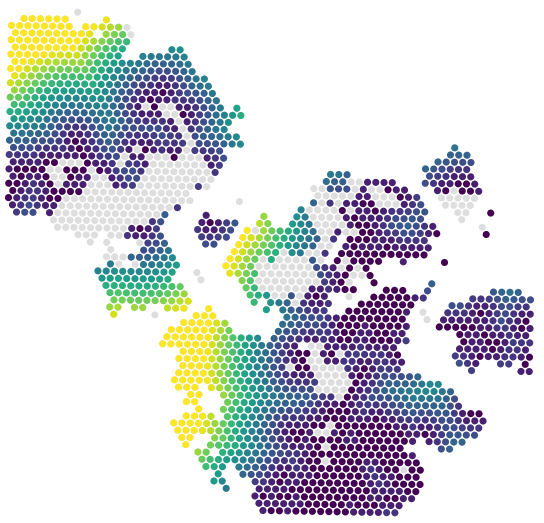
Fraction of all 30

TAM P80 targets



False True

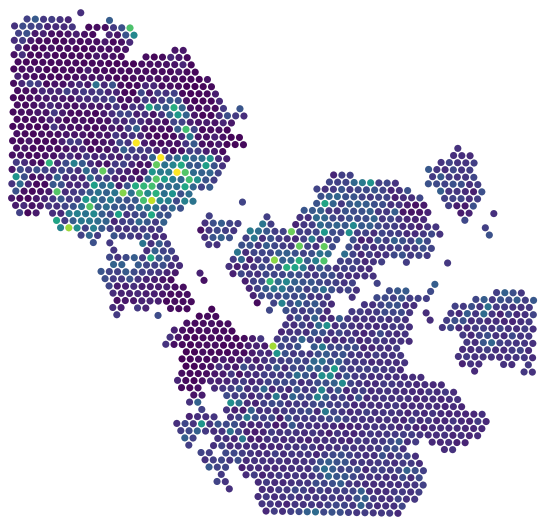
TAM distance



0 1.13e+03

Nearest spot center (μm)

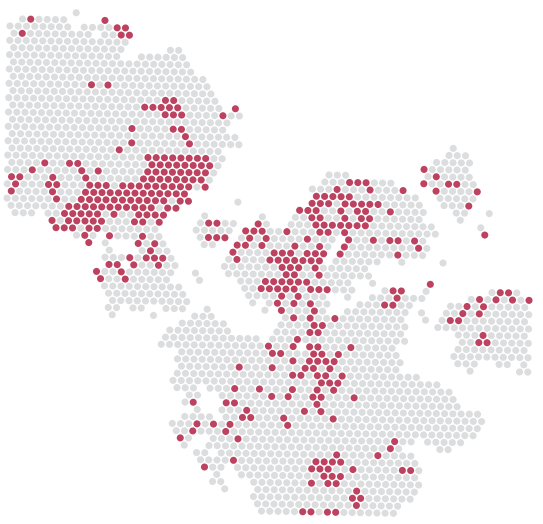
Treg composition



0 0.102

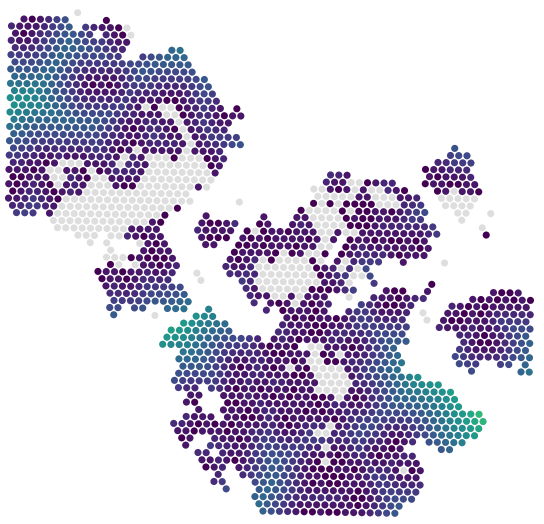
Fraction of all 30

Treg P80 targets



False True

Treg distance

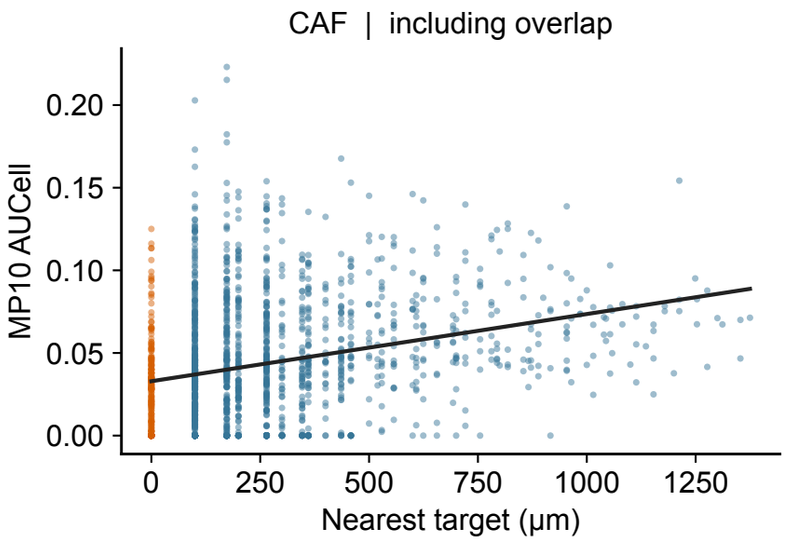


0 1.61e+03

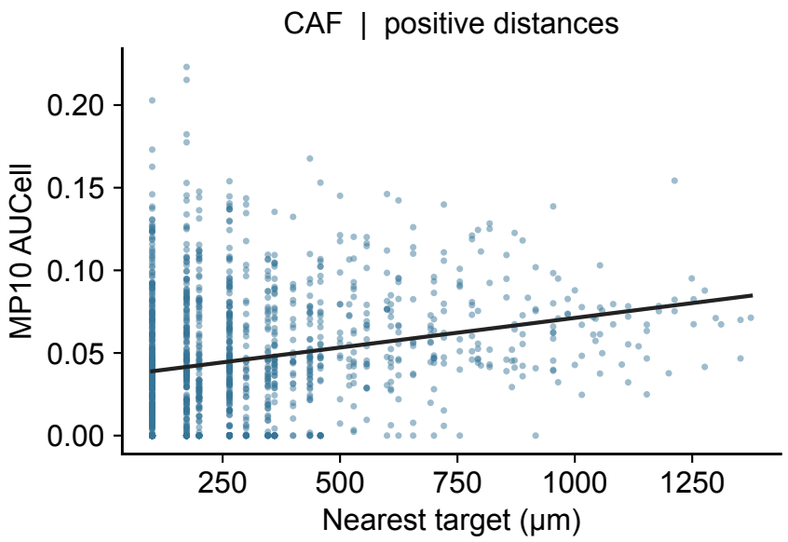
Nearest spot center (μm)

GSE319536_BC02 | MP10 versus nearest target

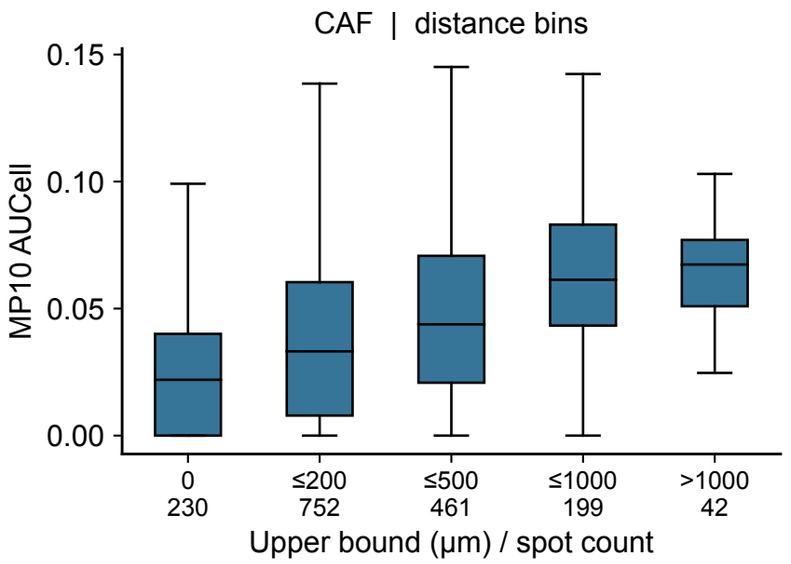
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



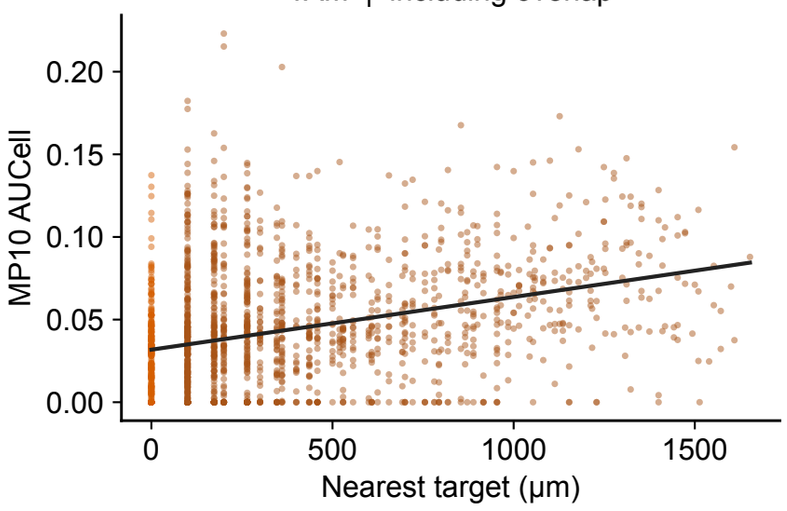
r +0.28 ρ +0.33
n 1684/1744 queries • 407 targets



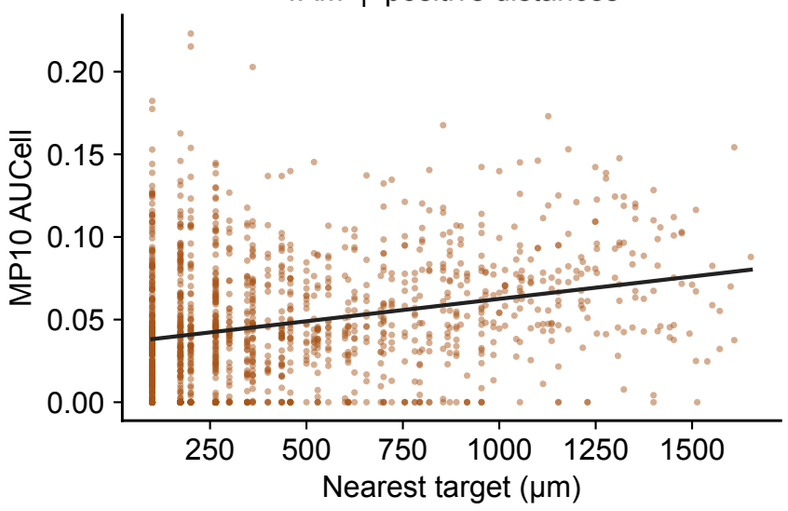
r +0.24 ρ +0.29
n 1454/1744 queries • 407 targets



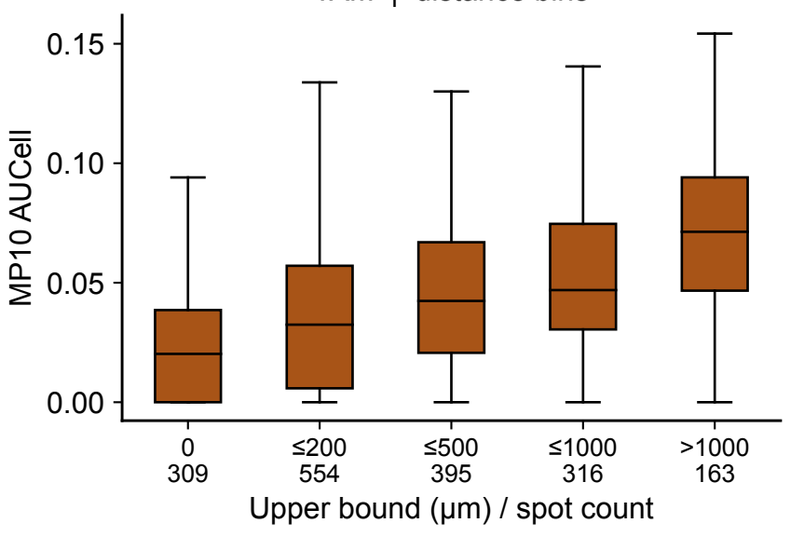
Orange = zero distance
Line = within-section linear fit



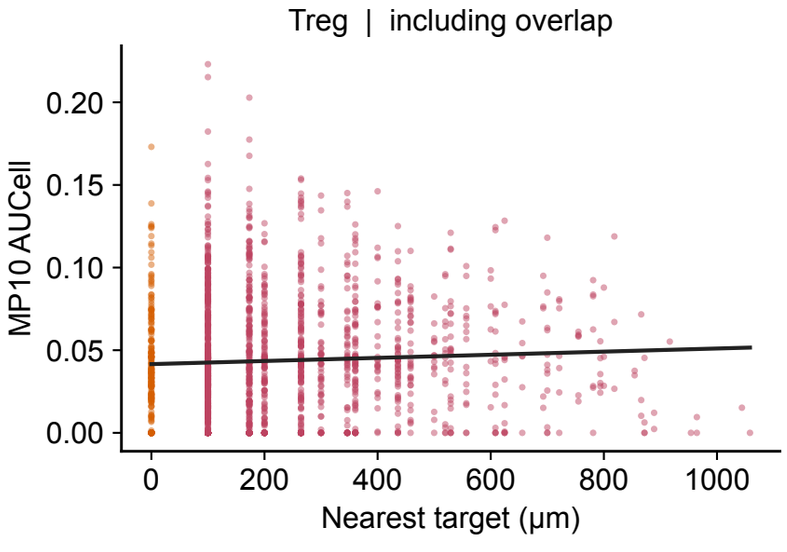
r +0.33 ρ +0.37
n 1737/1744 queries • 407 targets



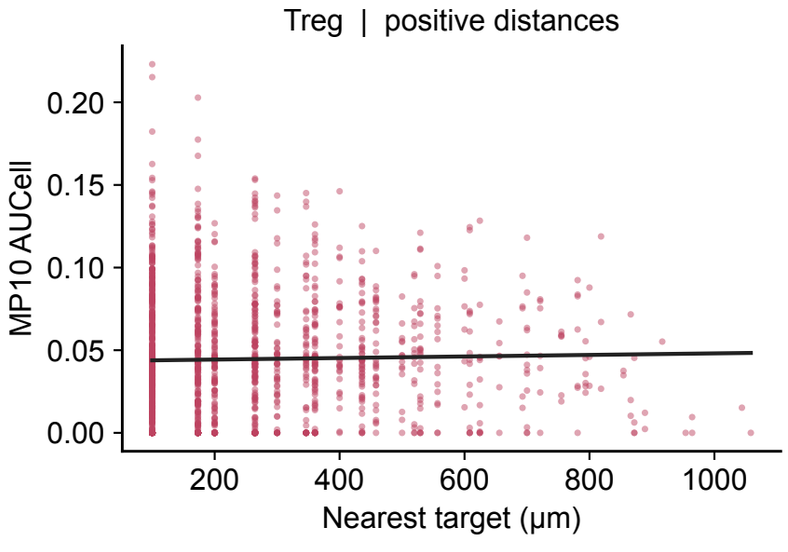
r +0.27 ρ +0.29
n 1428/1744 queries • 407 targets



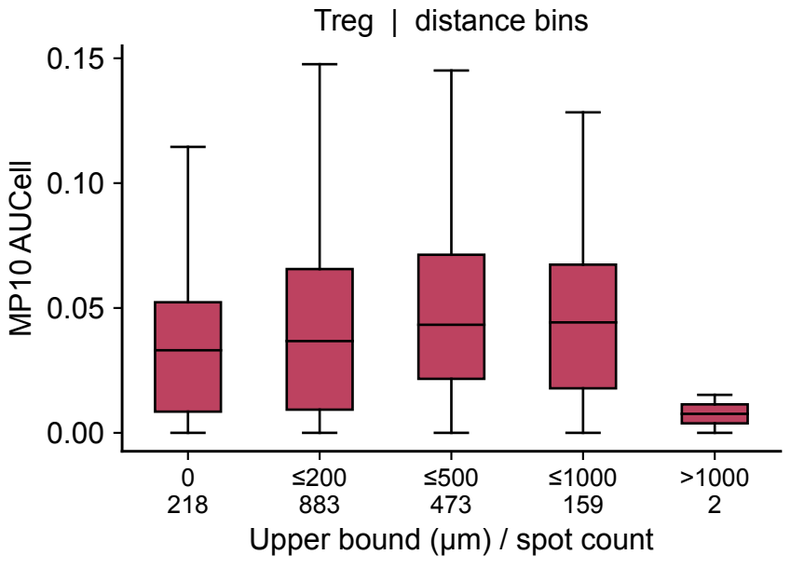
Orange = zero distance
Line = within-section linear fit



r +0.05 ρ +0.10
n 1735/1744 queries • 407 targets



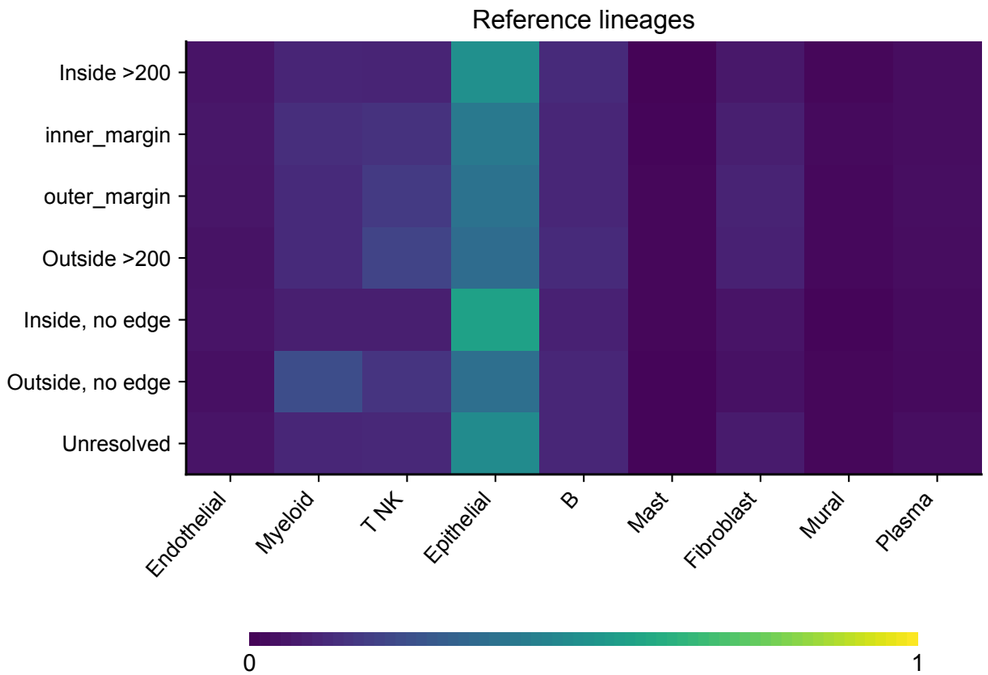
r +0.02 ρ +0.08
n 1517/1744 queries • 407 targets

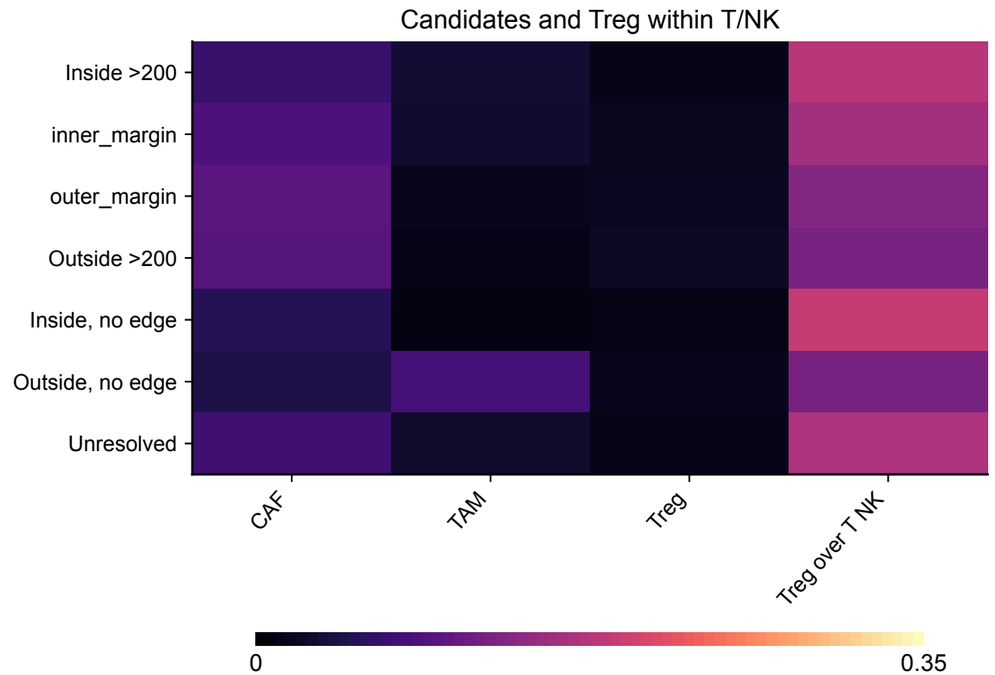


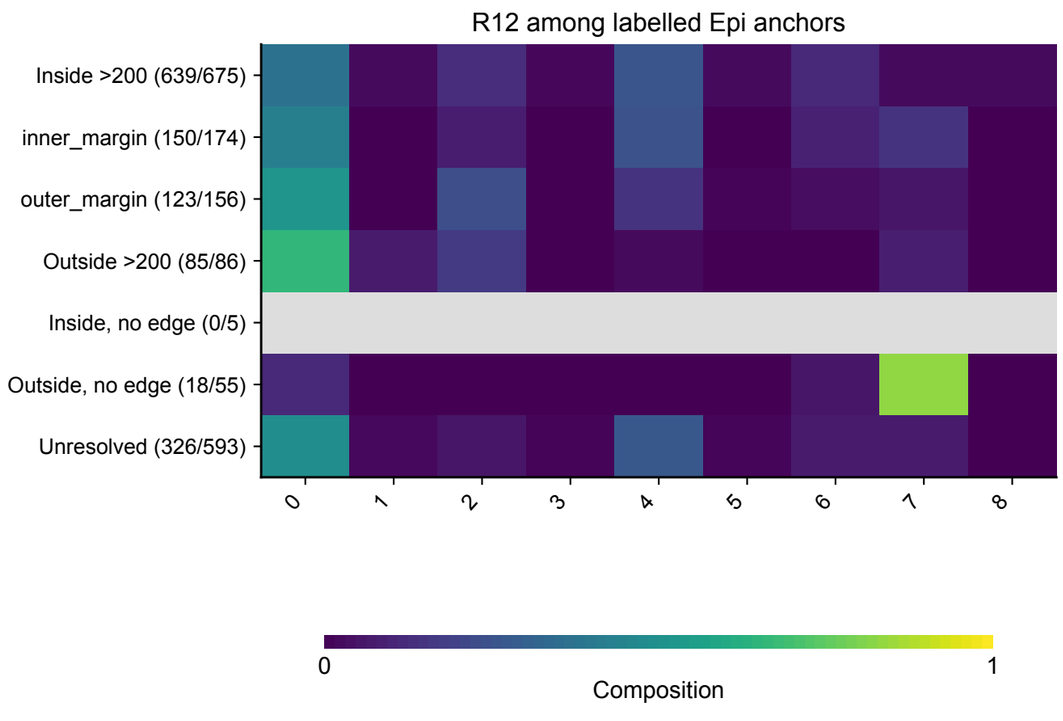
Orange = zero distance
Line = within-section linear fit

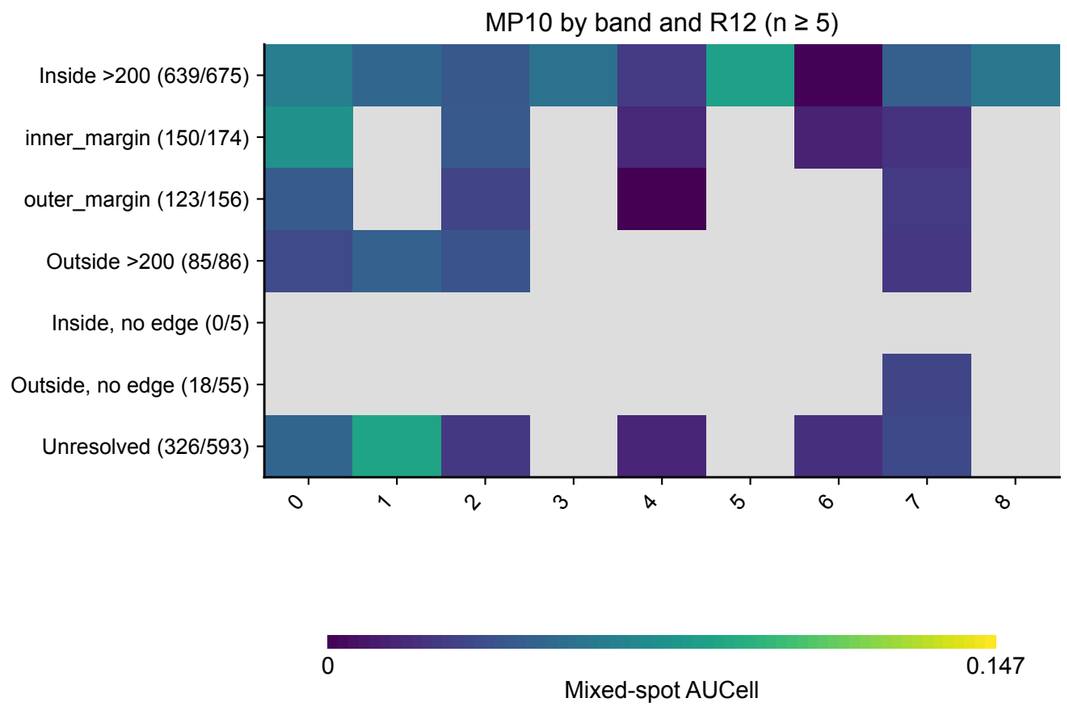
GSE319536_BC02 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





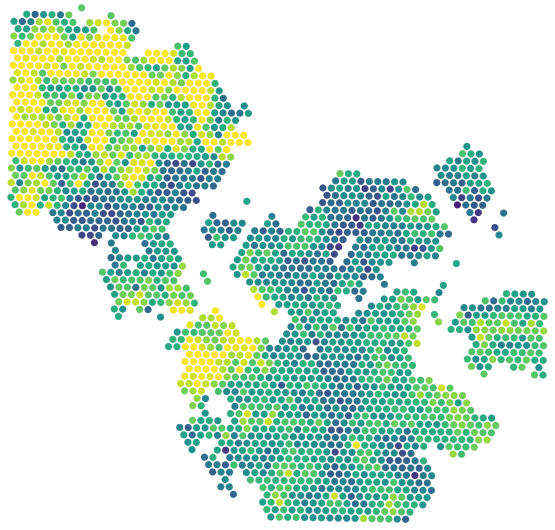




GSE319536_BC02 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

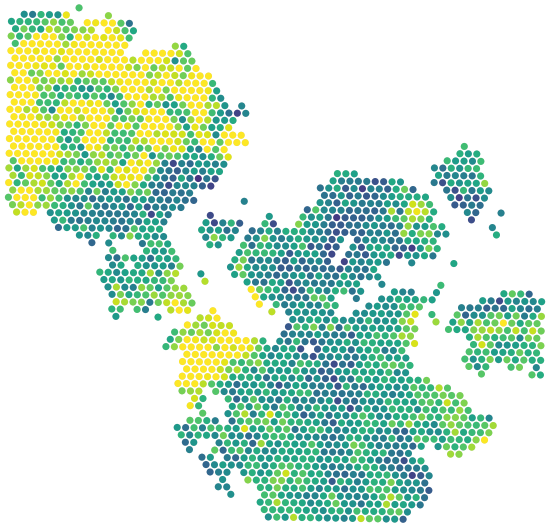
E2F Targets



0 0.152

UCell score

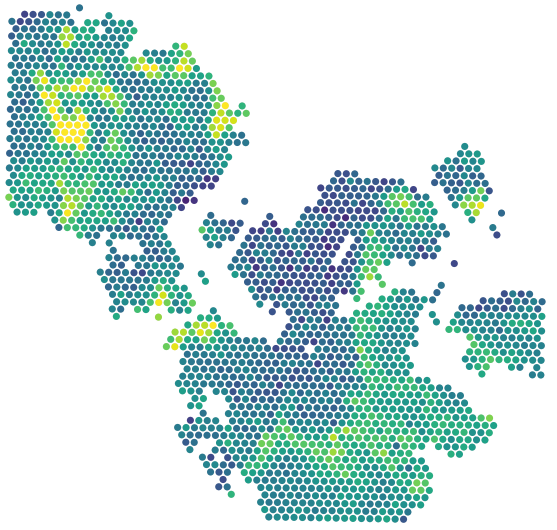
G2M Checkpoint



0 0.134

UCell score

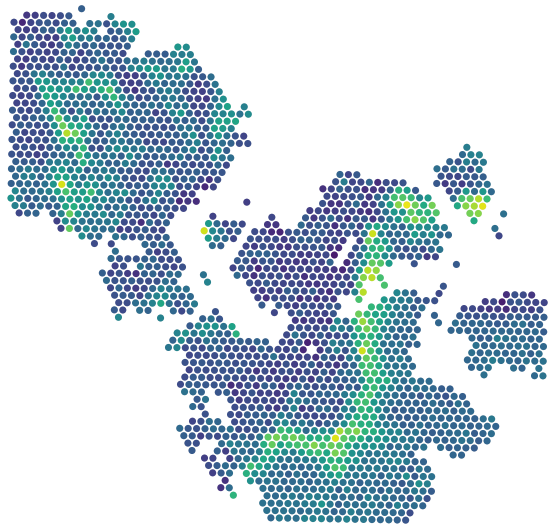
Hypoxia



0 0.211

UCell score

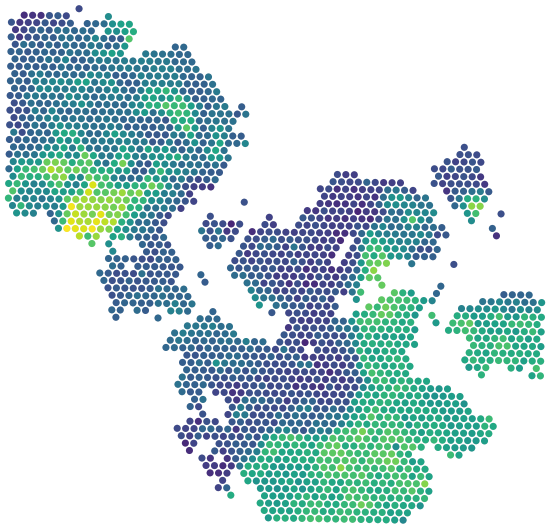
Tnfa Signaling Via Nfkb



0 0.275

UCell score

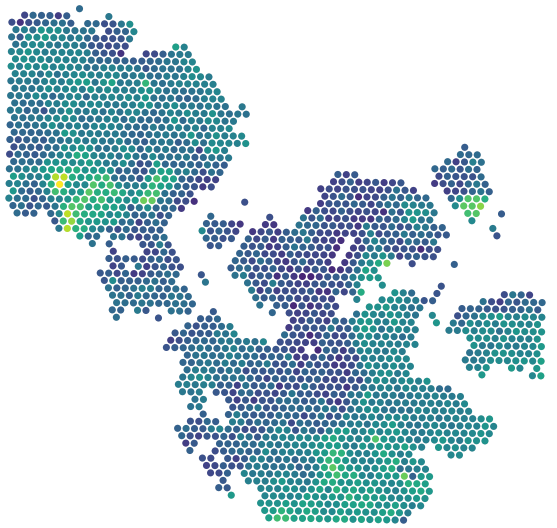
Emt



0 0.386

UCell score

Interferon Gamma Response



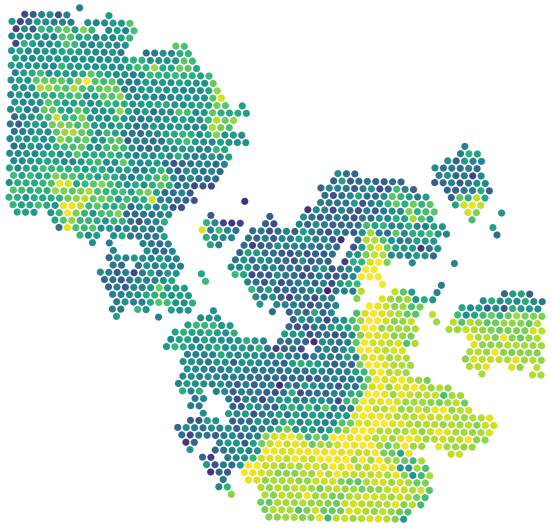
0 0.275

UCell score

GSE319536_BC02 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

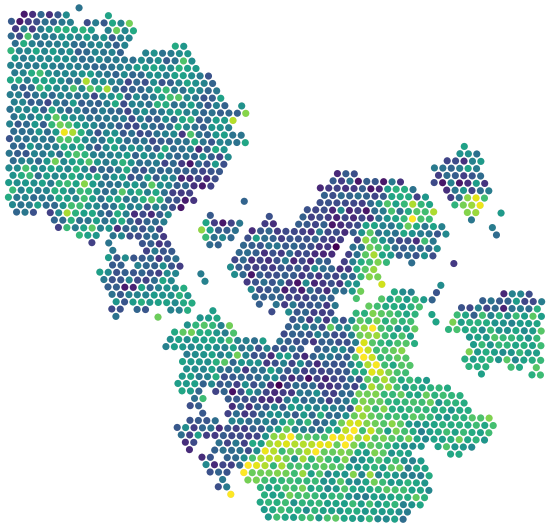
Il2 Stat5 Signaling



0 0.144

UCell score

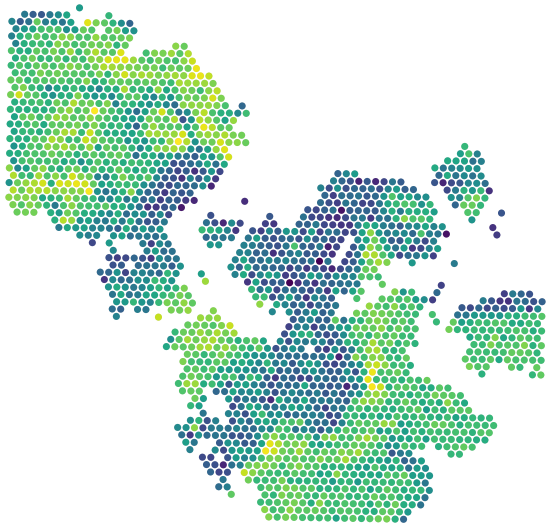
Tgf Beta Signaling



0 0.232

UCell score

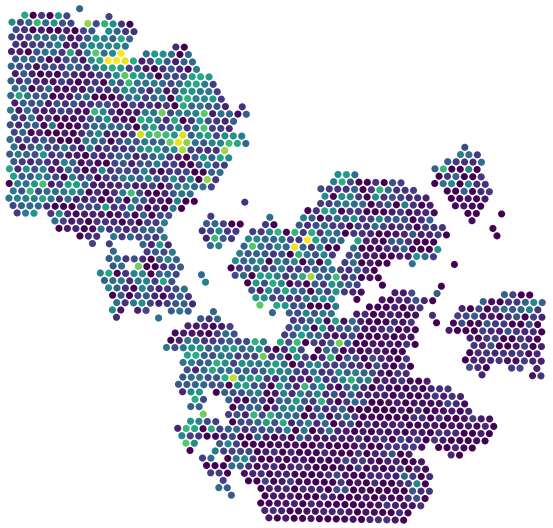
Cholesterol Homeostasis



0 0.21

UCell score

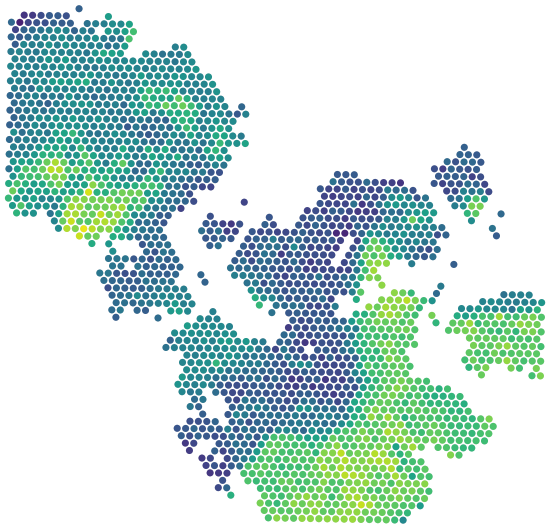
Cholesterol Biosynthesis



0 0.166

UCell score

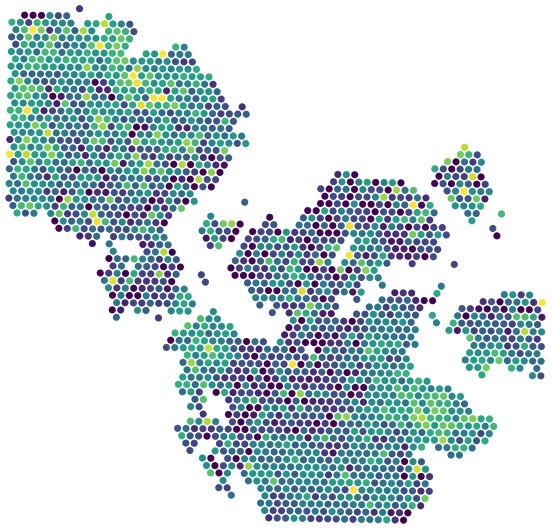
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



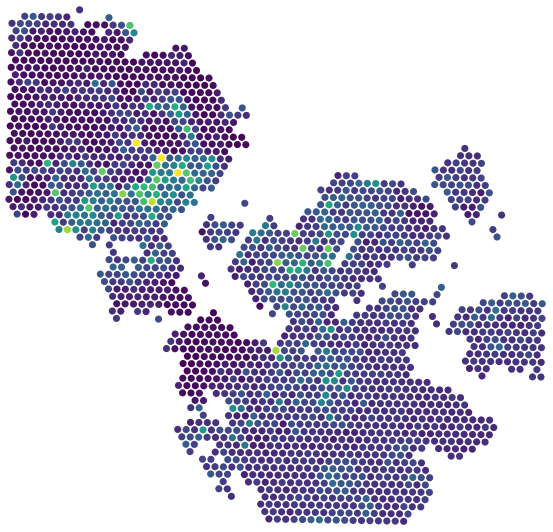
0 0.151

UCell score

GSE319536_BC02 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

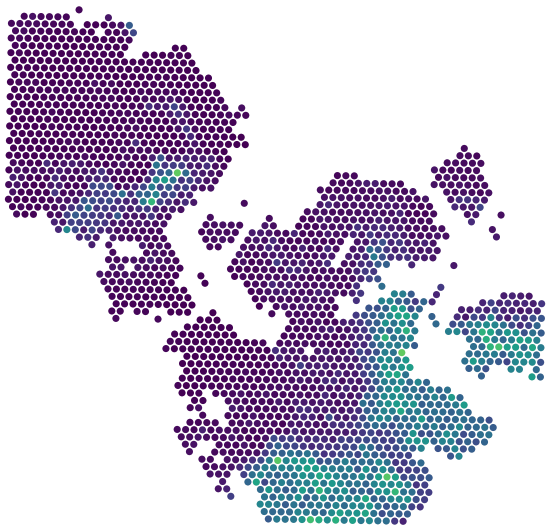
Treg composition



0 0.102

Fraction of all 30

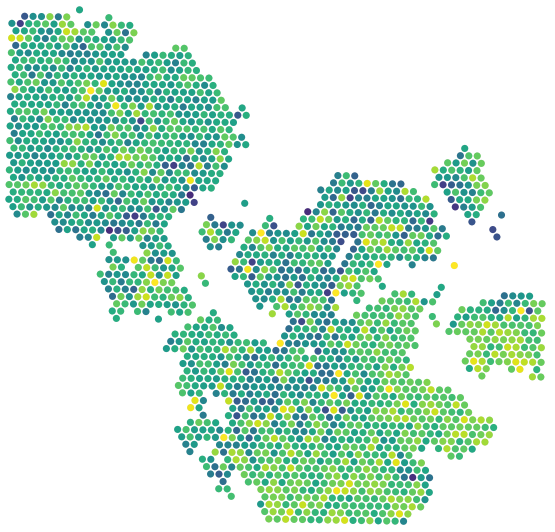
Treg estimated abundance



0 2.85

Posterior mean estimate

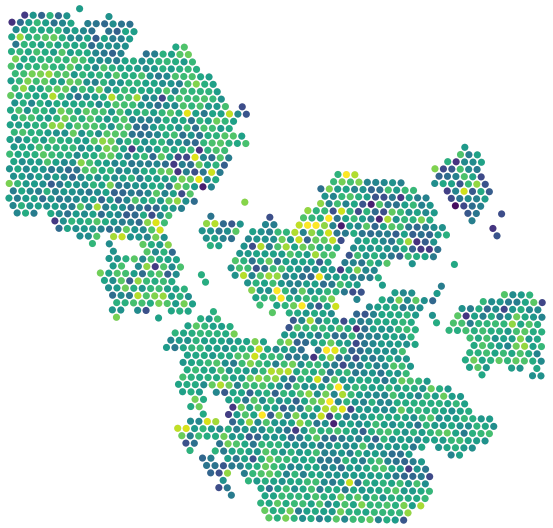
Treg Resting Vs Activated Dn



0 0.0778

UCell score

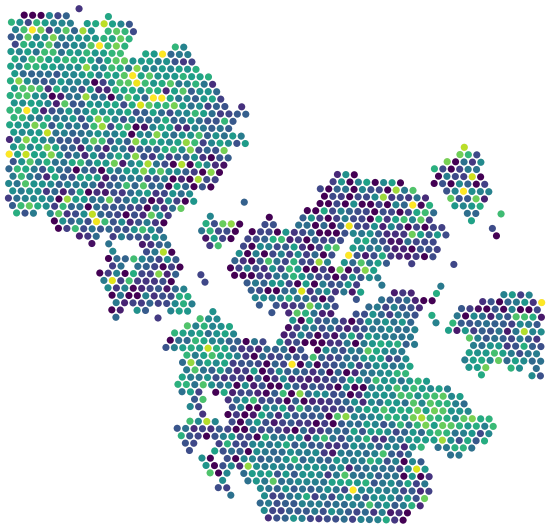
Treg Resting Vs Activated Up



0 0.0609

UCell score

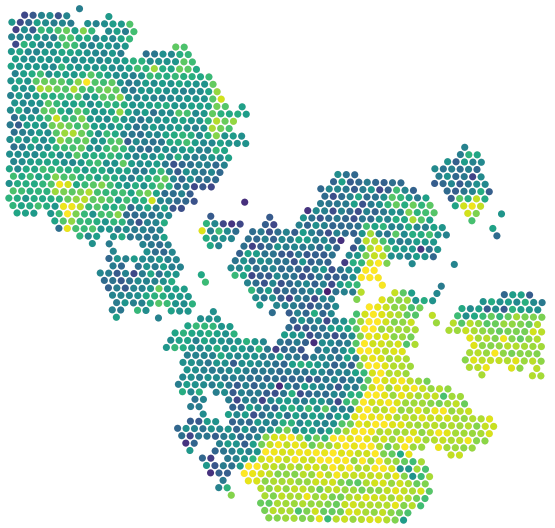
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

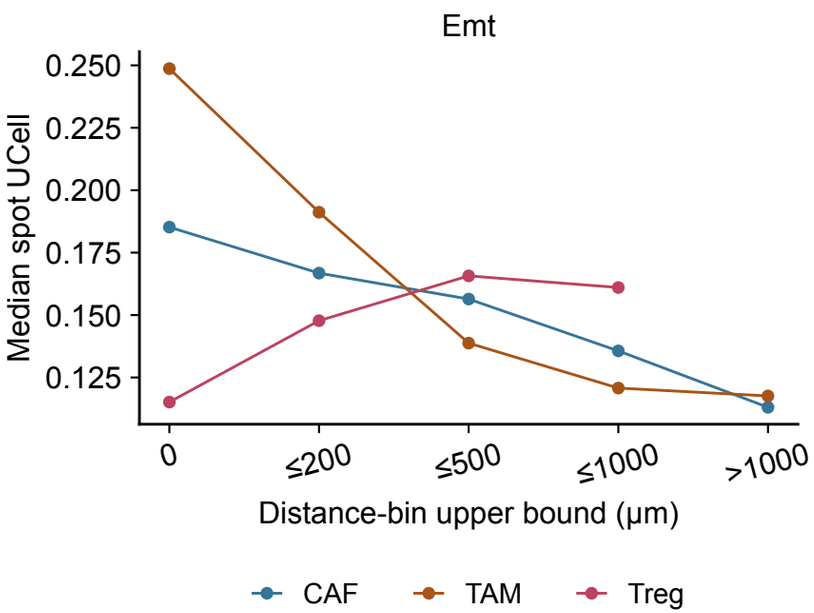


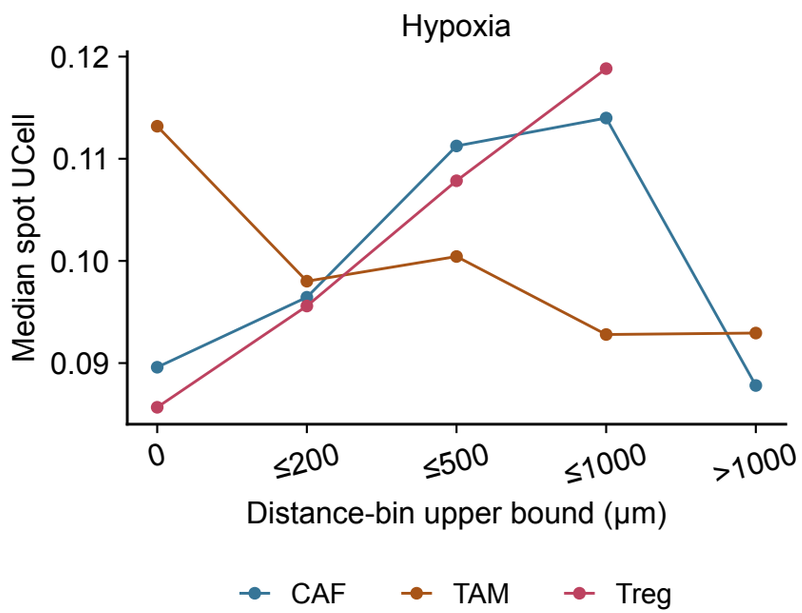
0 0.144

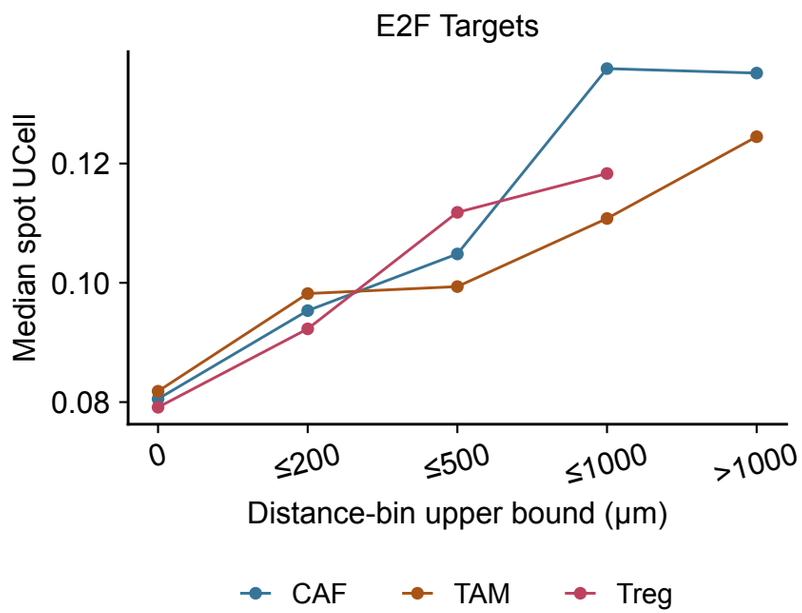
UCell score

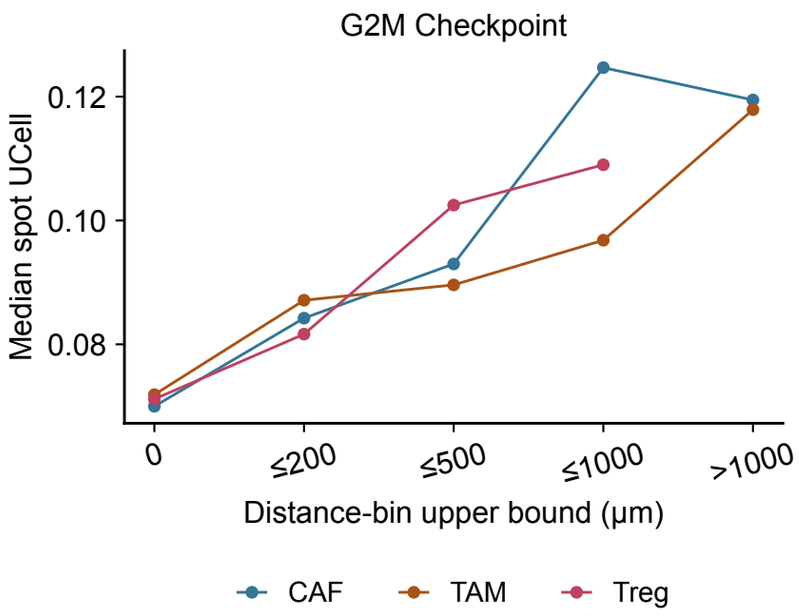
GSE319536_BC02 | Programs by target distance

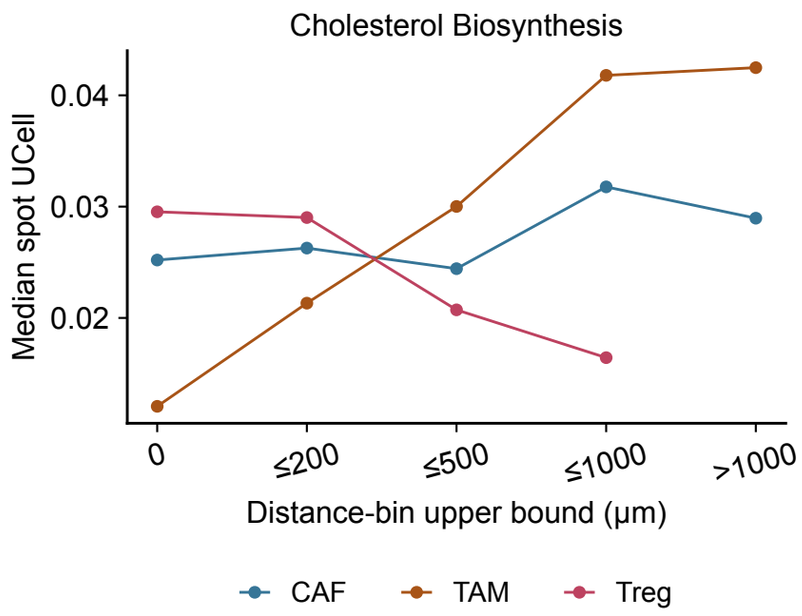
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

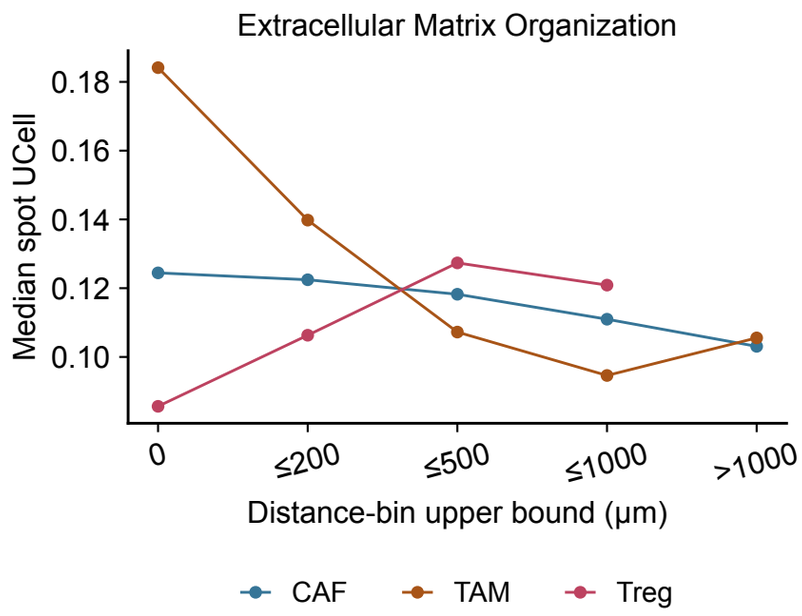






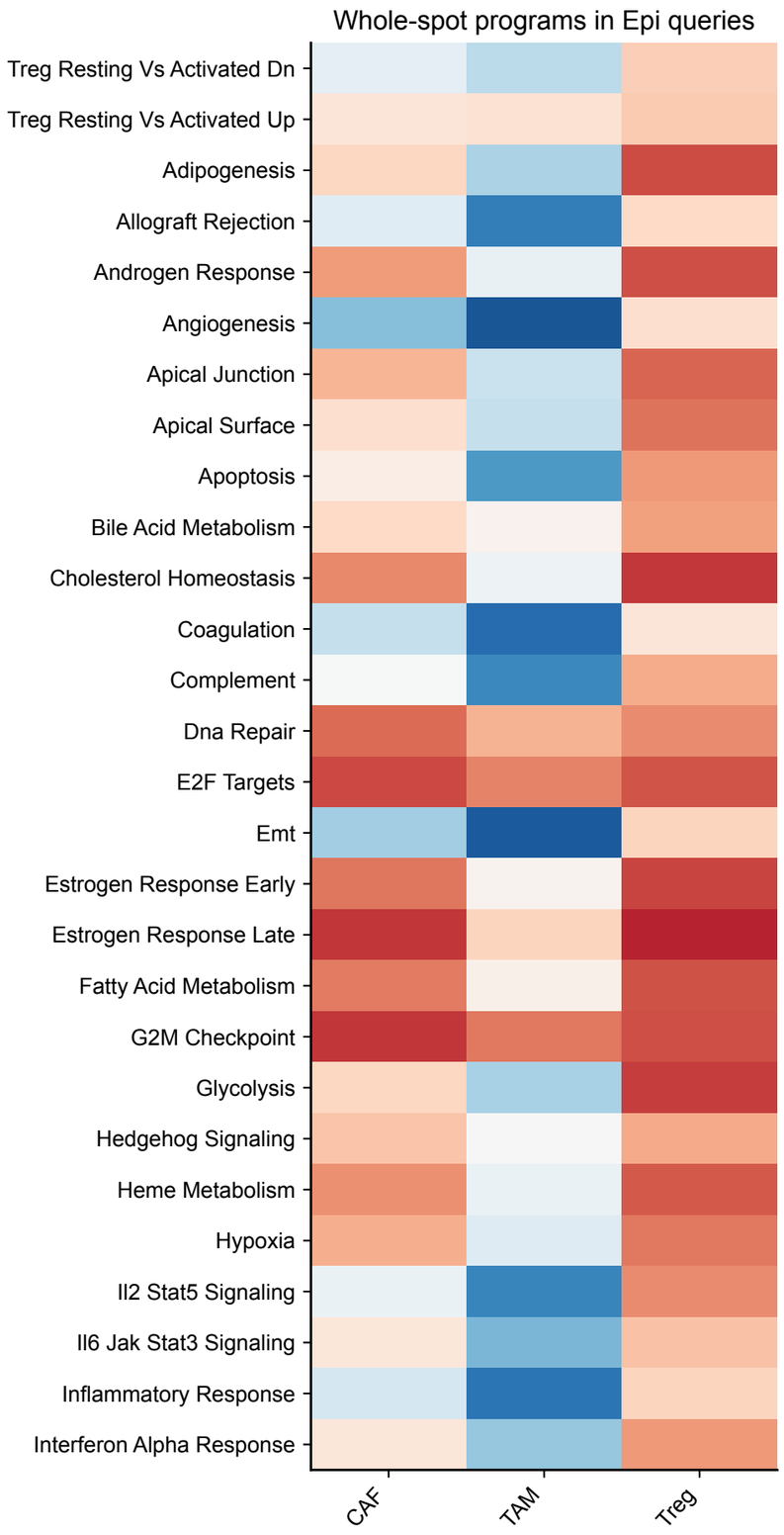


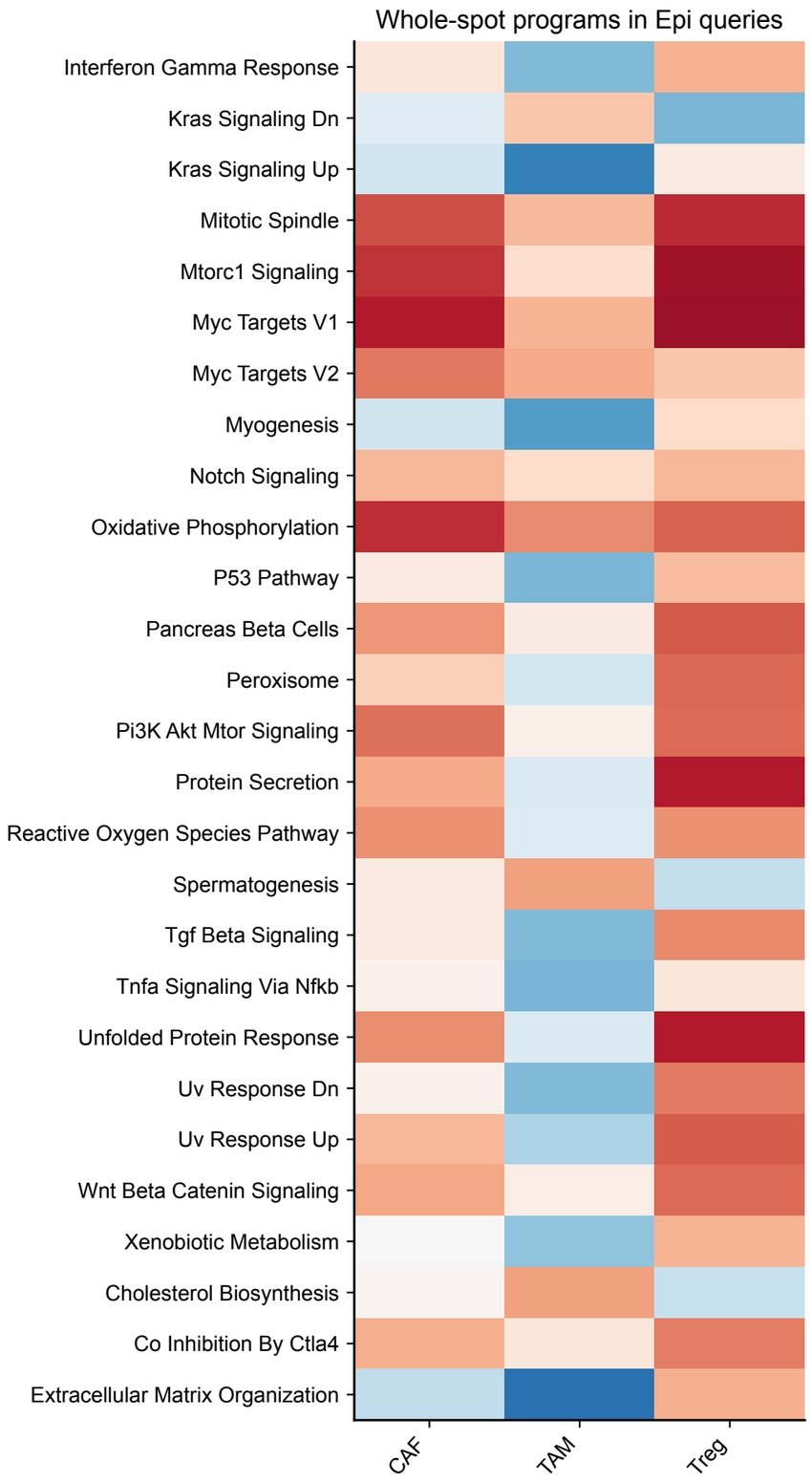




GSE319536_BC02 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

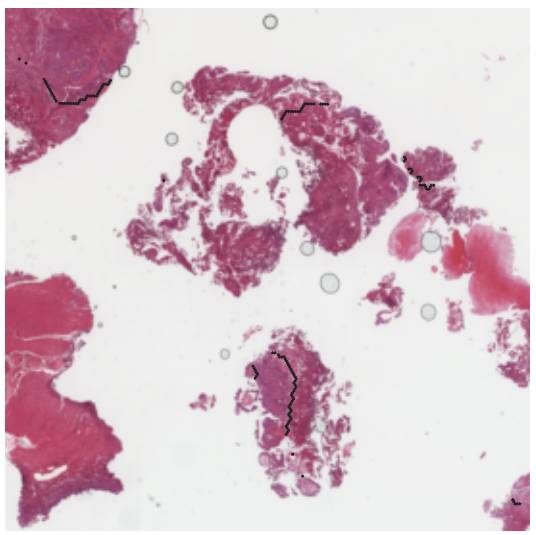




GSE319536_BC04 | Tissue context

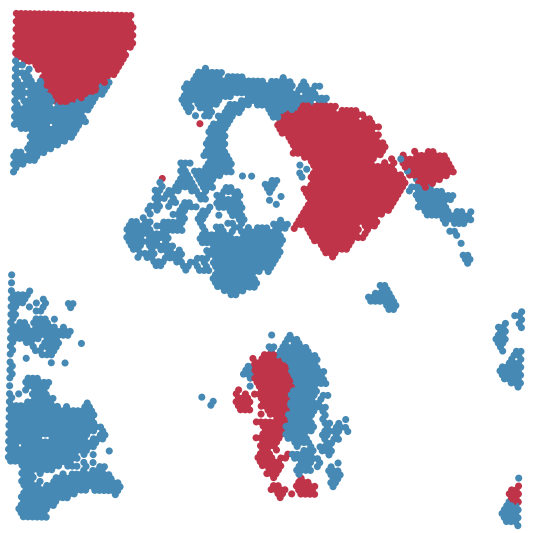
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



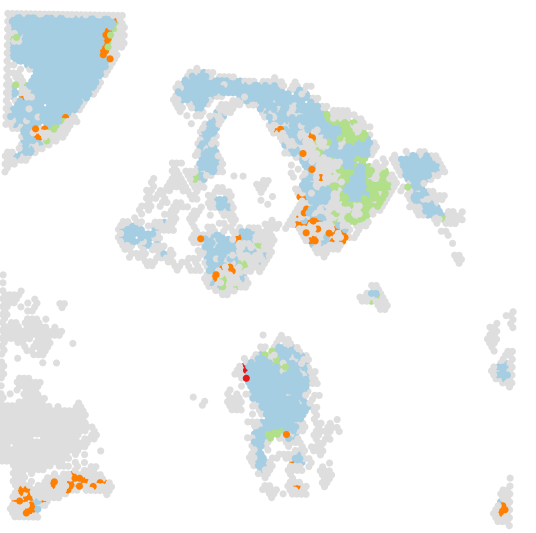
Black line: observed candidate outer edge

Final candidate location



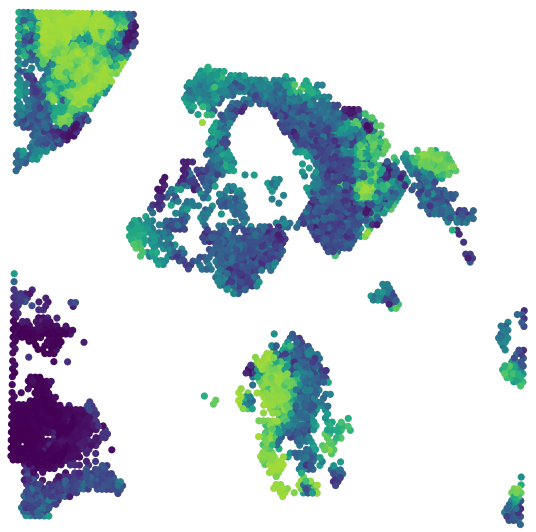
inside outside

Accepted R12 (original 0–8)



0 2 5 7

Epithelial support



0 1

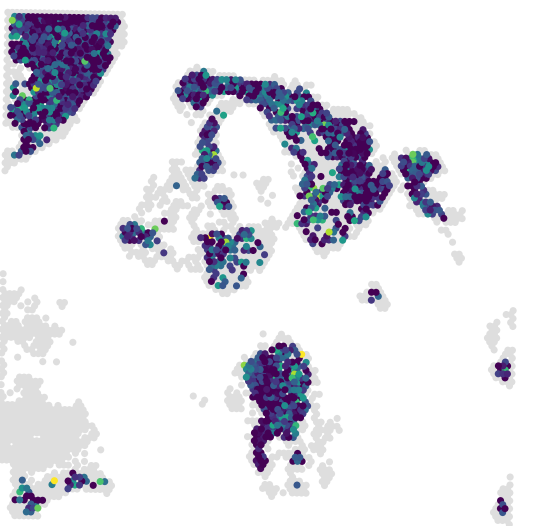
Fraction of all 30

MP10 in Epi queries

0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)

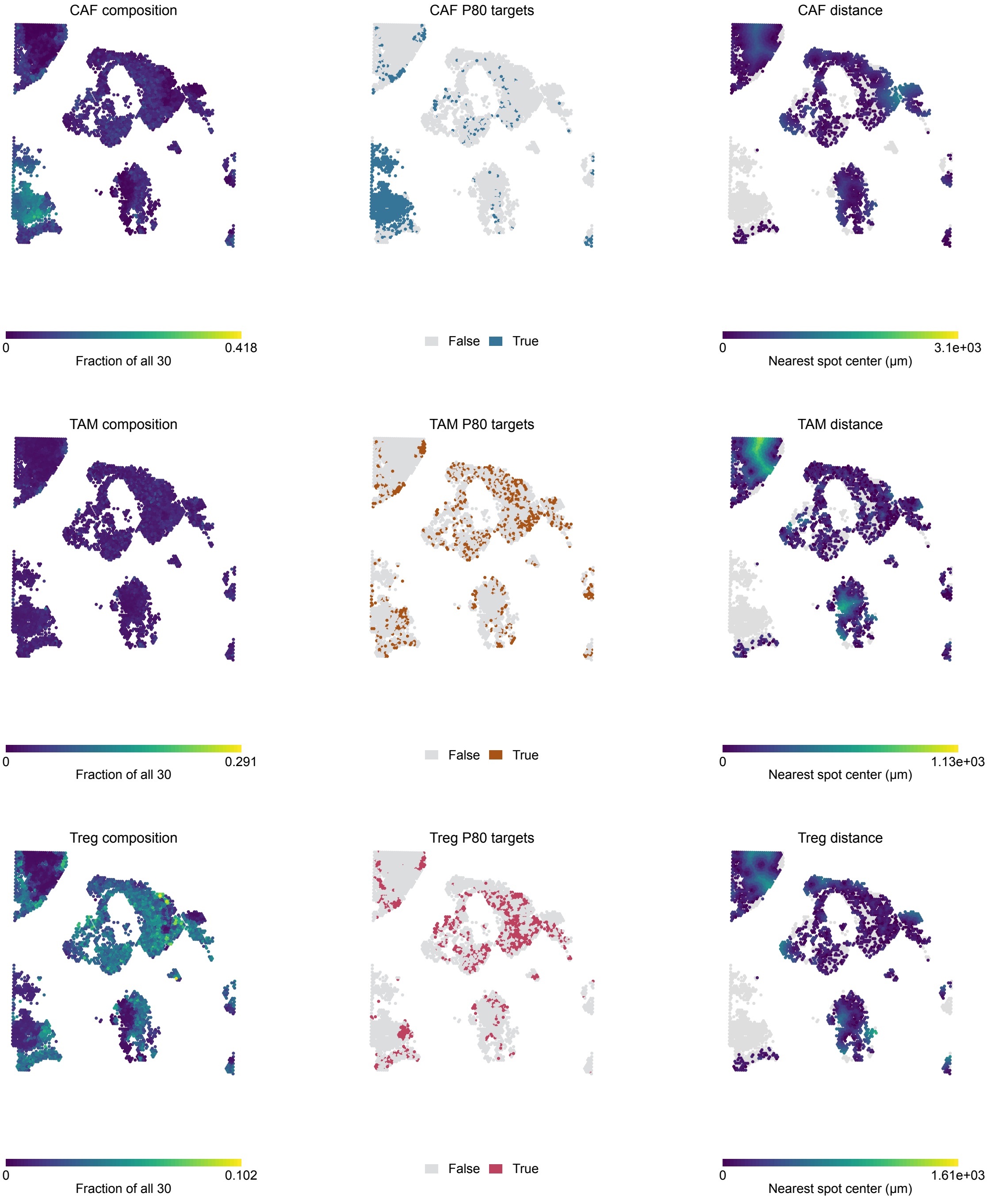


0 0.14

Matched-model AUCell

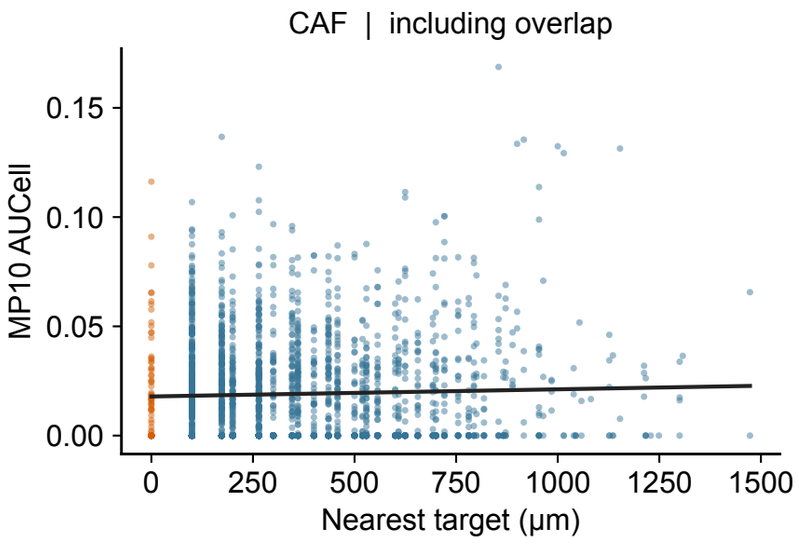
GSE319536_BC04 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

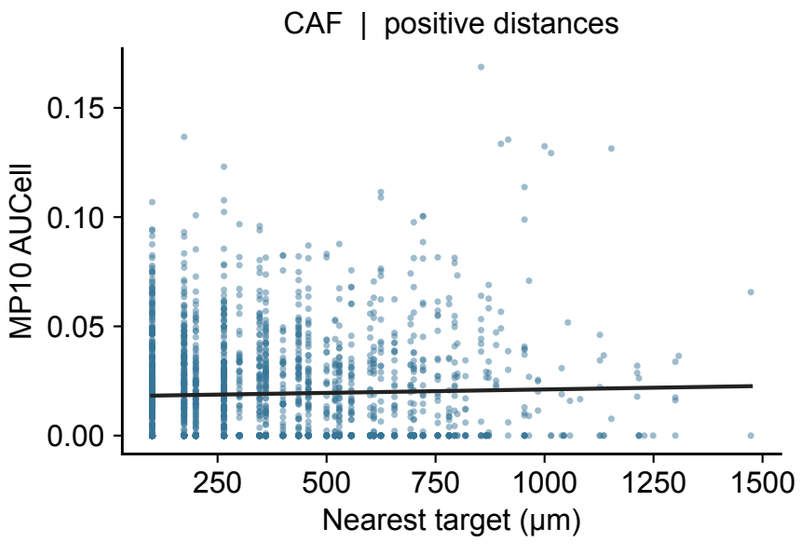


GSE319536_BC04 | MP10 versus nearest target

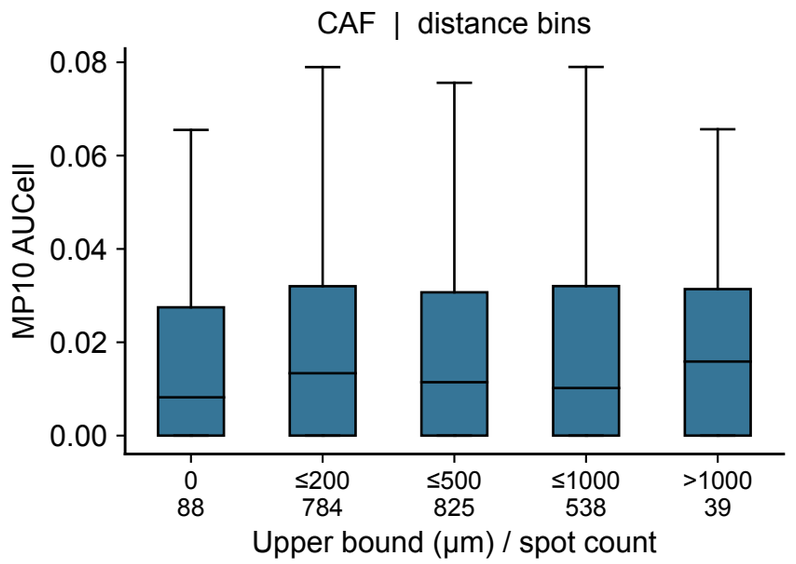
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



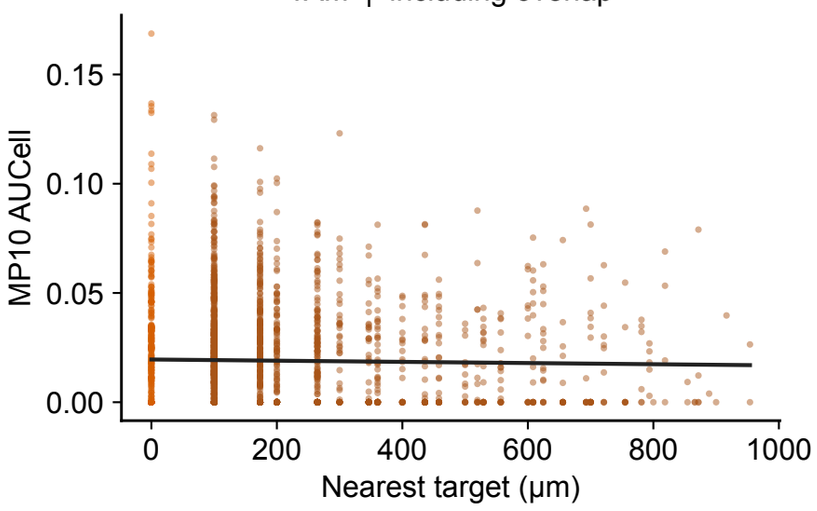
r +0.04 ρ -0.01
n 2274/2365 queries • 697 targets



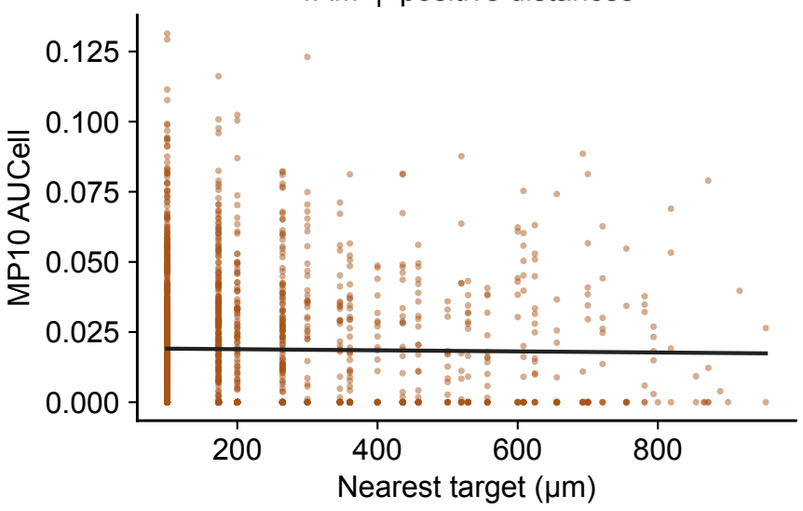
r +0.03 ρ -0.01
n 2186/2365 queries • 697 targets



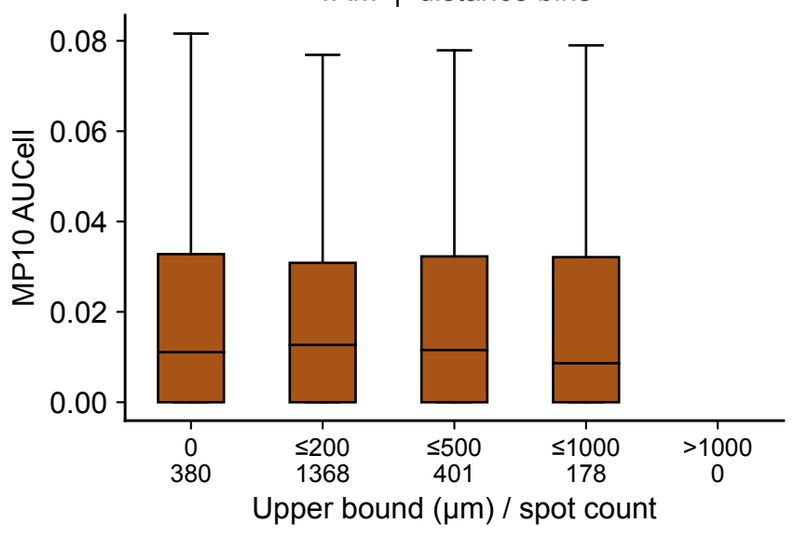
Orange = zero distance
Line = within-section linear fit



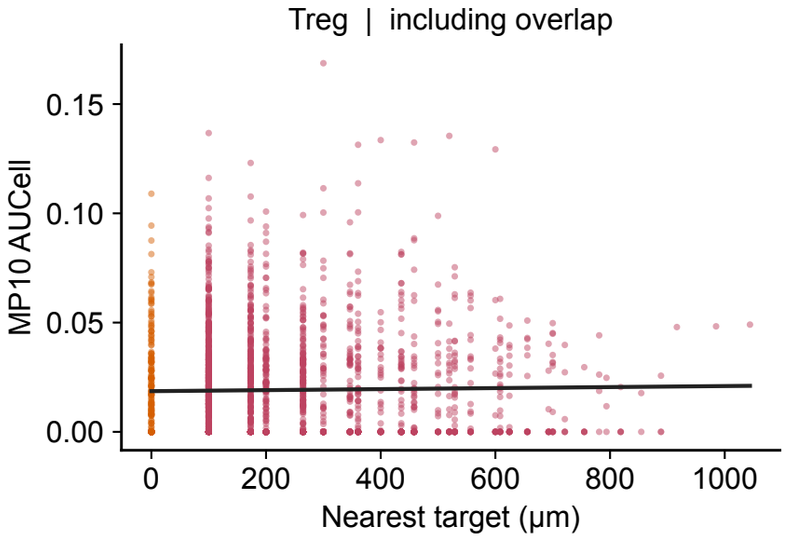
r -0.02 ρ +0.01
n 2327/2365 queries • 697 targets



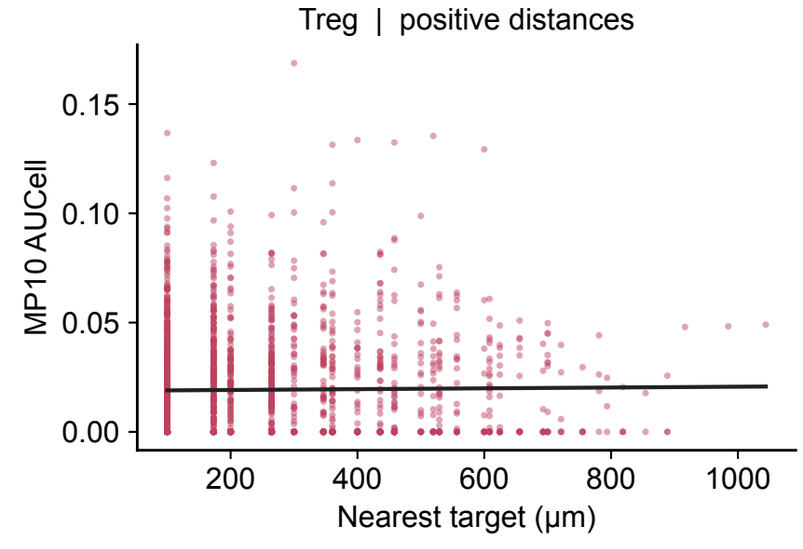
r -0.01 ρ +0.01
n 1947/2365 queries • 697 targets



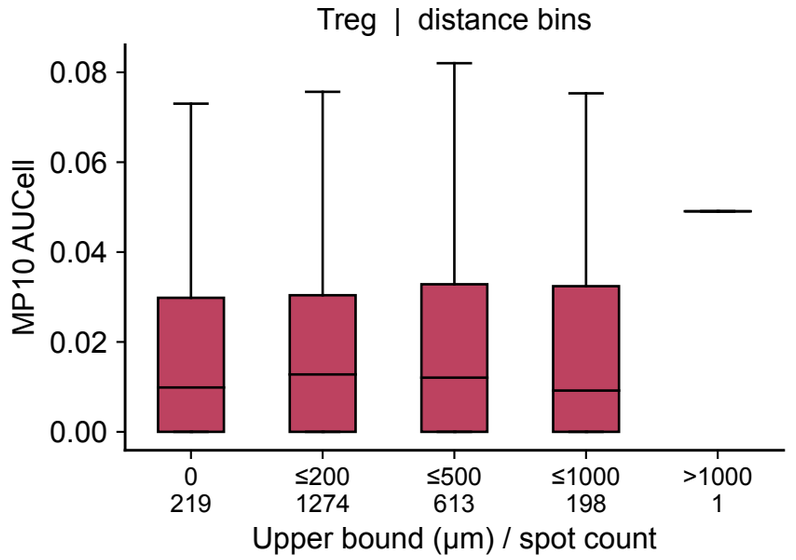
Orange = zero distance
Line = within-section linear fit



r +0.02 ρ +0.02
n 2305/2365 queries • 697 targets



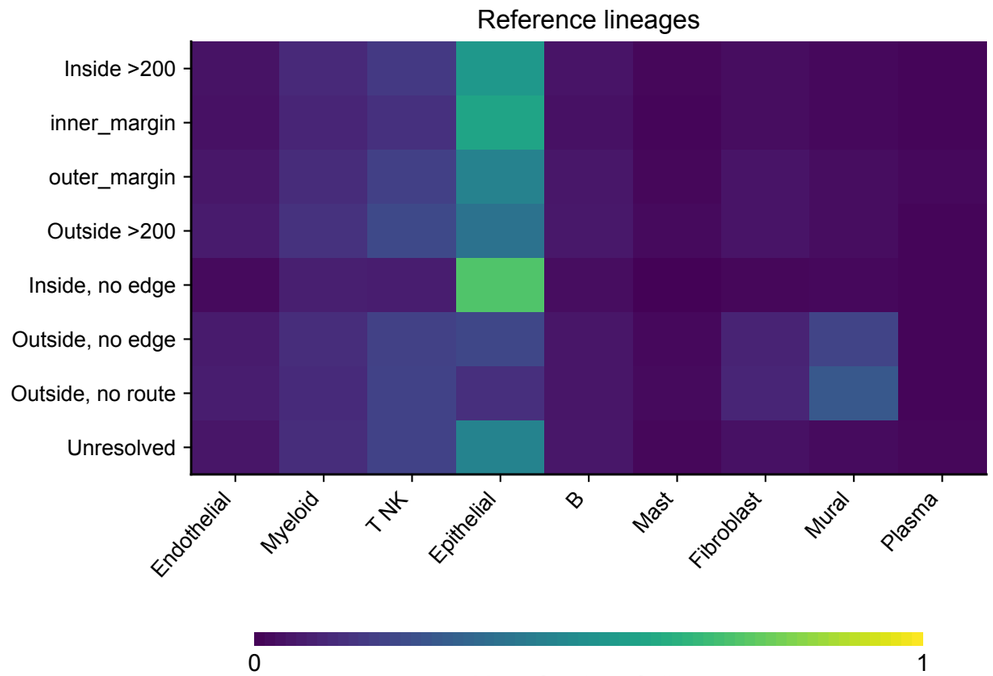
r +0.01 ρ +0.01
n 2086/2365 queries • 697 targets

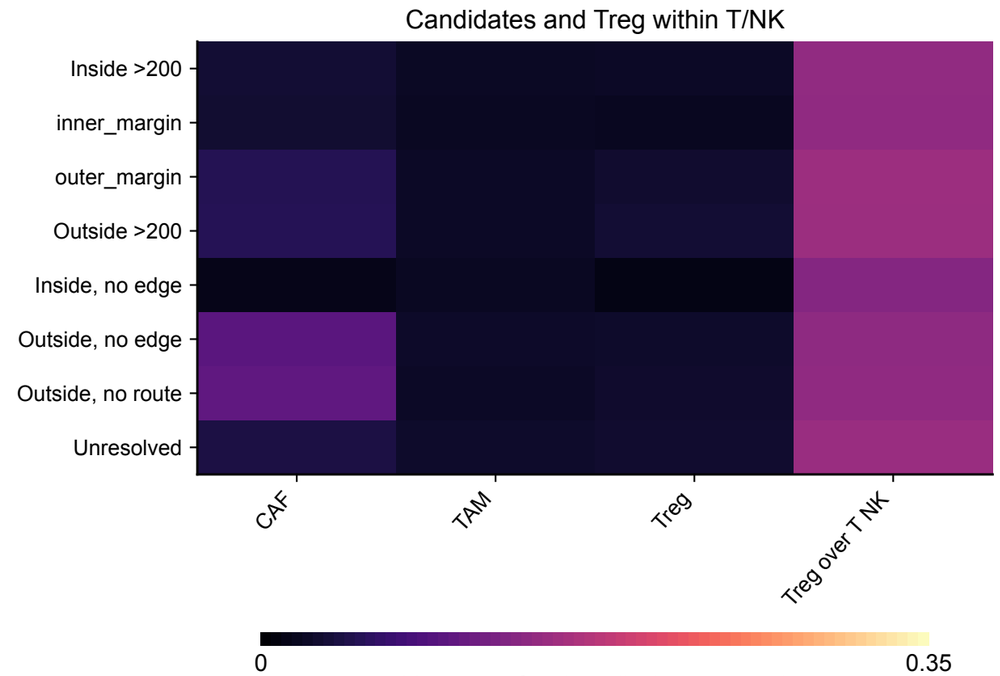


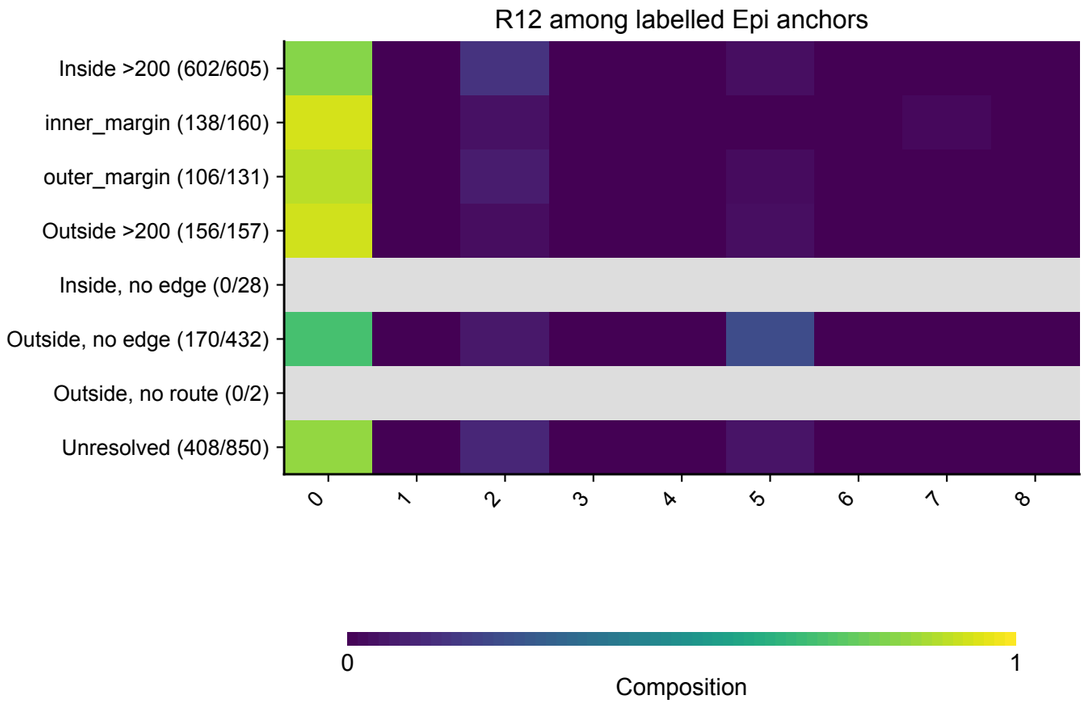
Orange = zero distance
Line = within-section linear fit

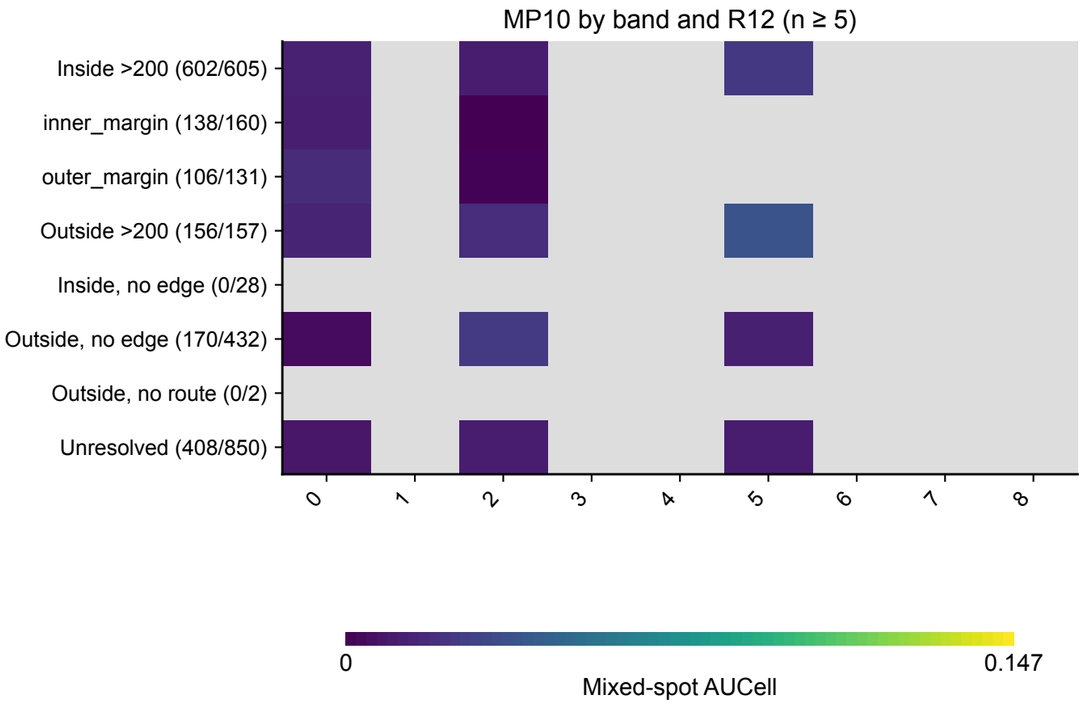
GSE319536_BC04 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable



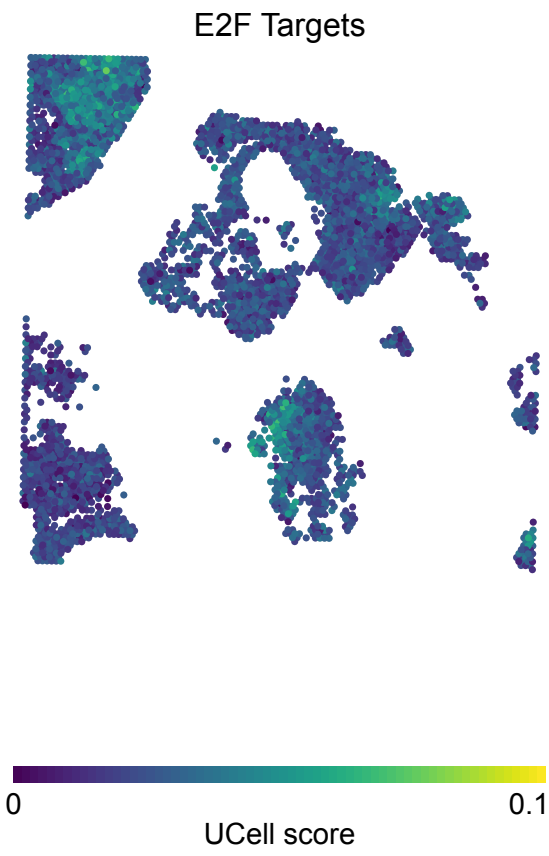


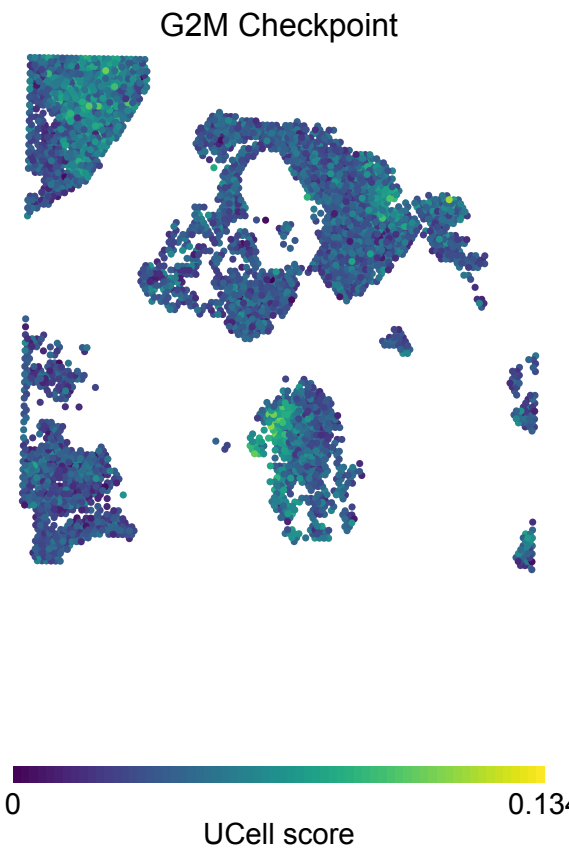




GSE319536_BC04 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

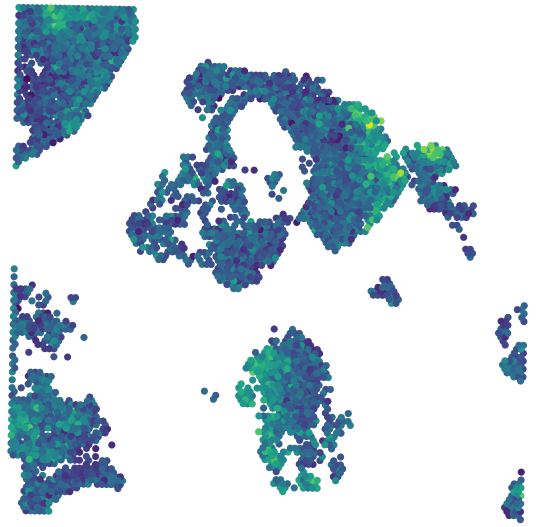




GSE319536_BC04 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

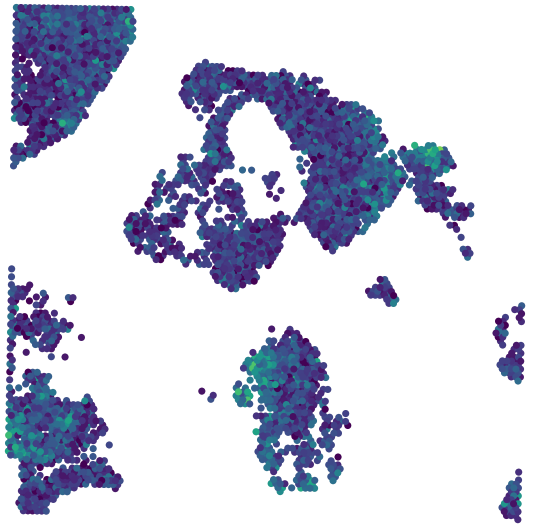
Il2 Stat5 Signaling



0 0.144

UCell score

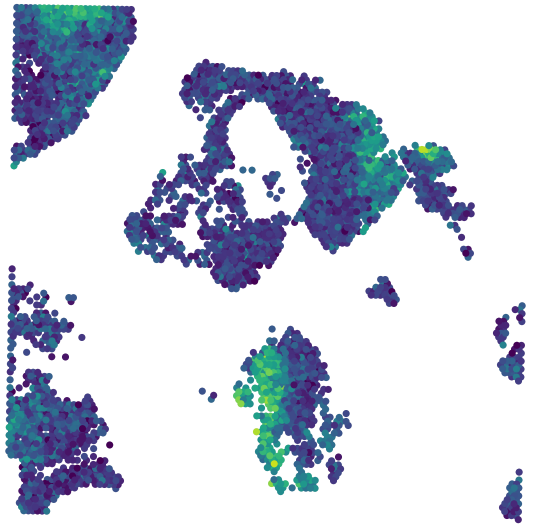
Tgf Beta Signaling



0 0.232

UCell score

Cholesterol Homeostasis



0 0.21

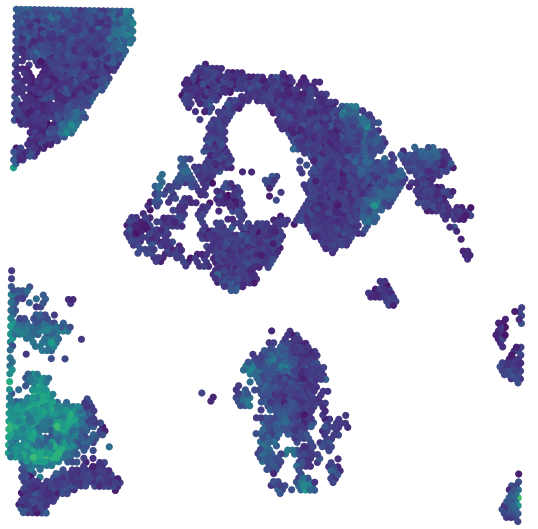
UCell score

Cholesterol Biosynthesis

0 0.166

UCell score

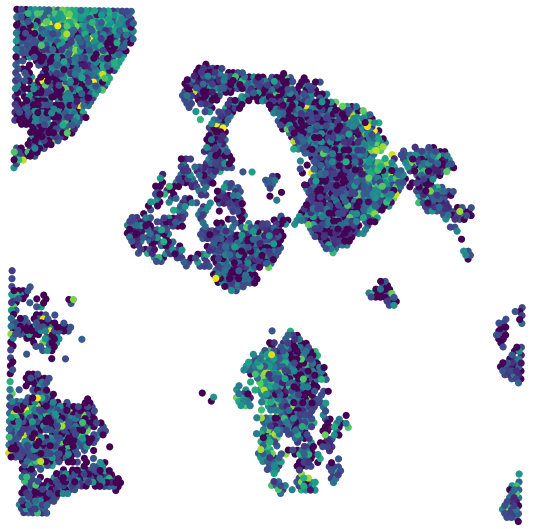
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



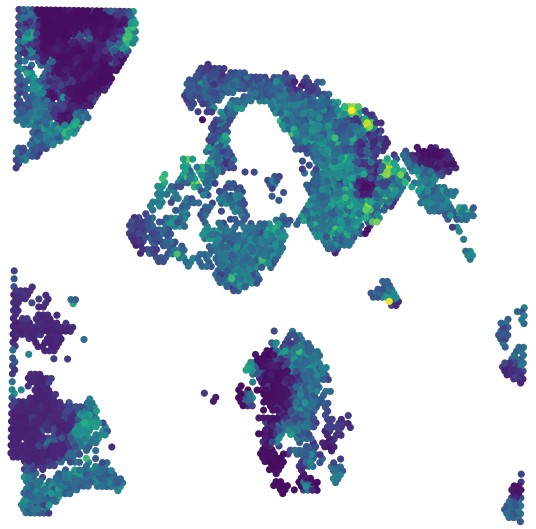
0 0.151

UCell score

GSE319536_BC04 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

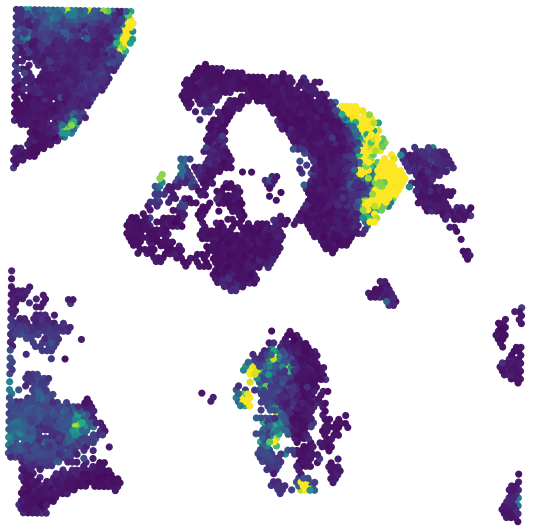
Treg composition



0 0.102

Fraction of all 30

Treg estimated abundance



0 2.85

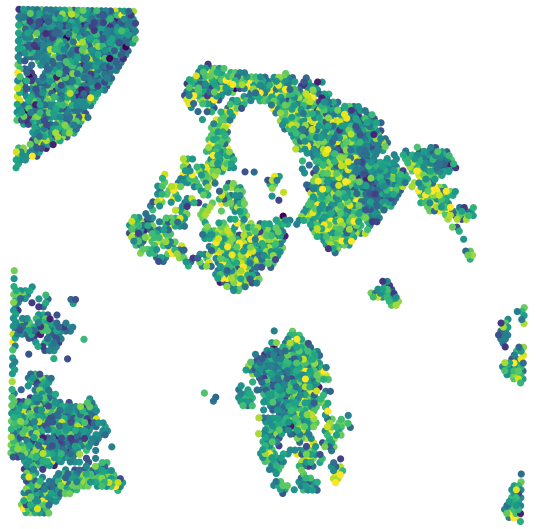
Posterior mean estimate

Treg Resting Vs Activated Dn

0 0.0778

UCell score

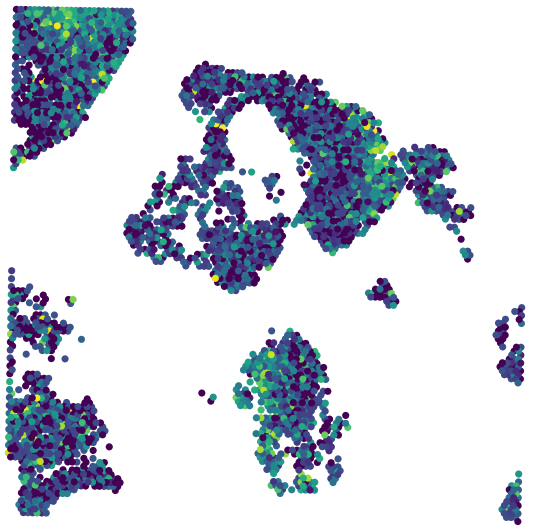
Treg Resting Vs Activated Up



0 0.0609

UCell score

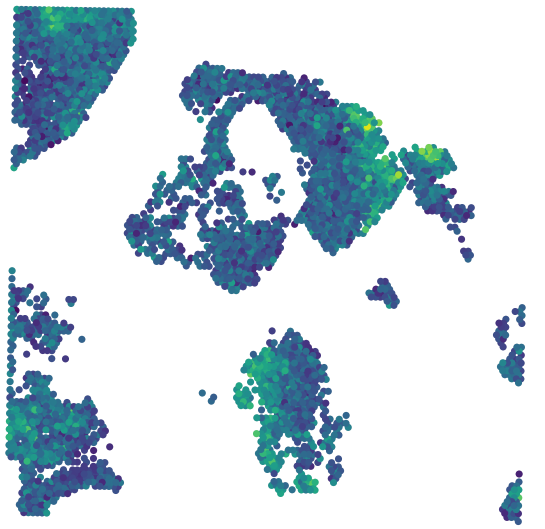
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

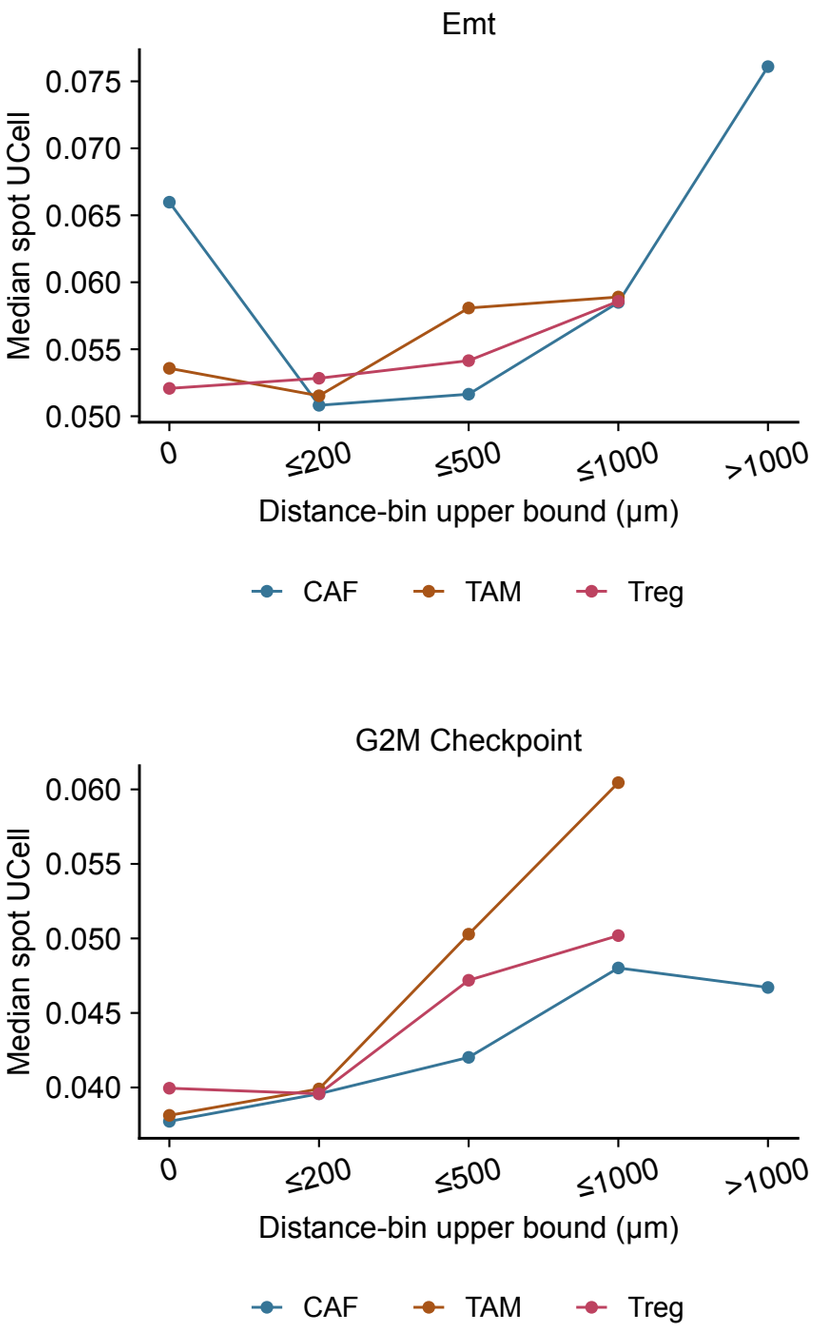


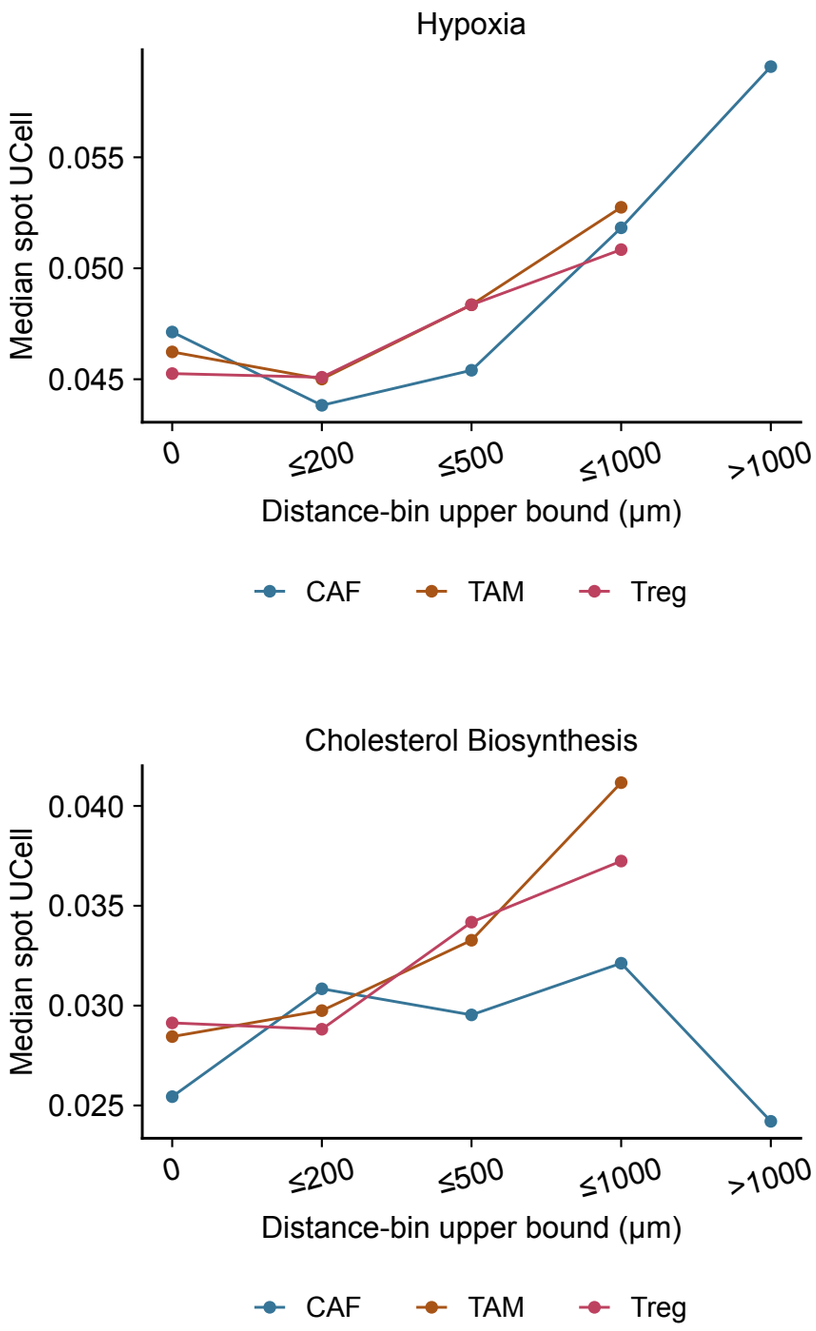
0 0.144

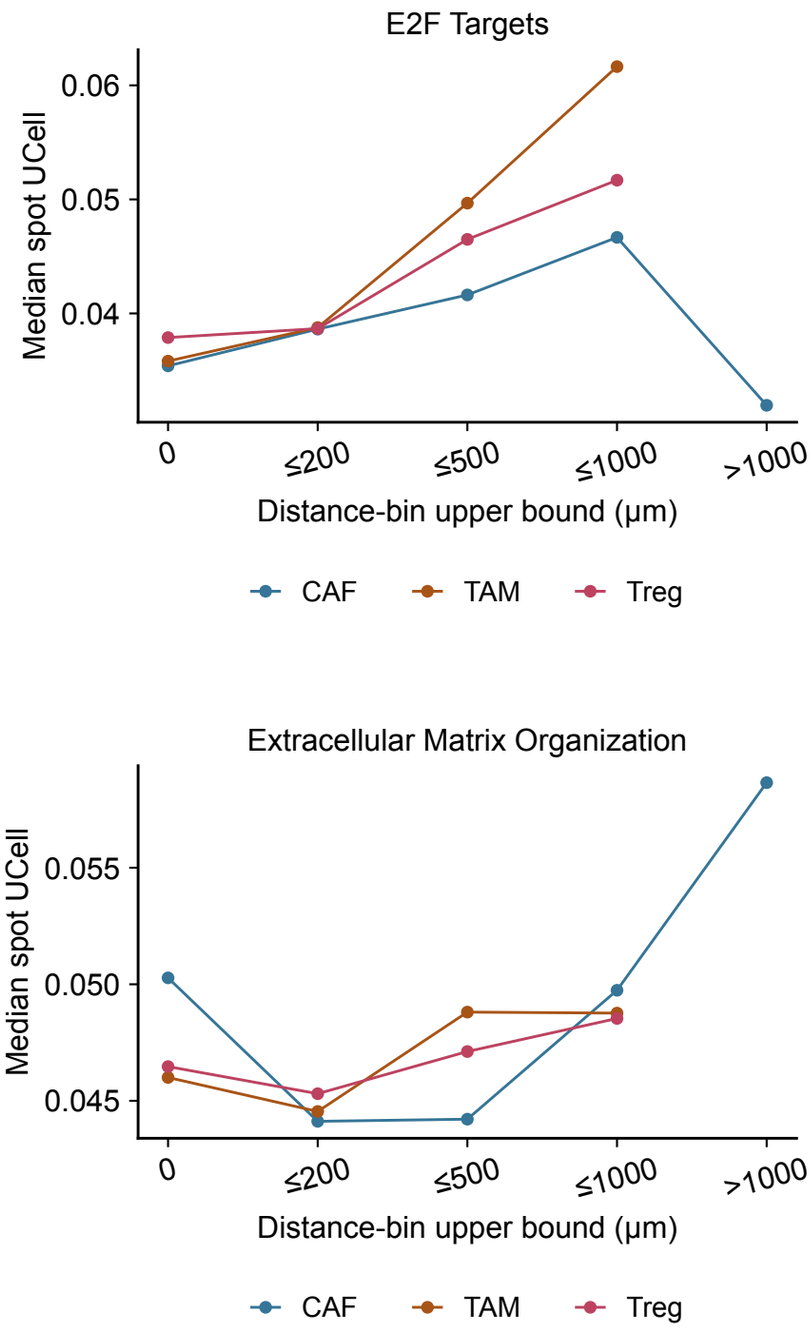
UCell score

GSE319536_BC04 | Programs by target distance

Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

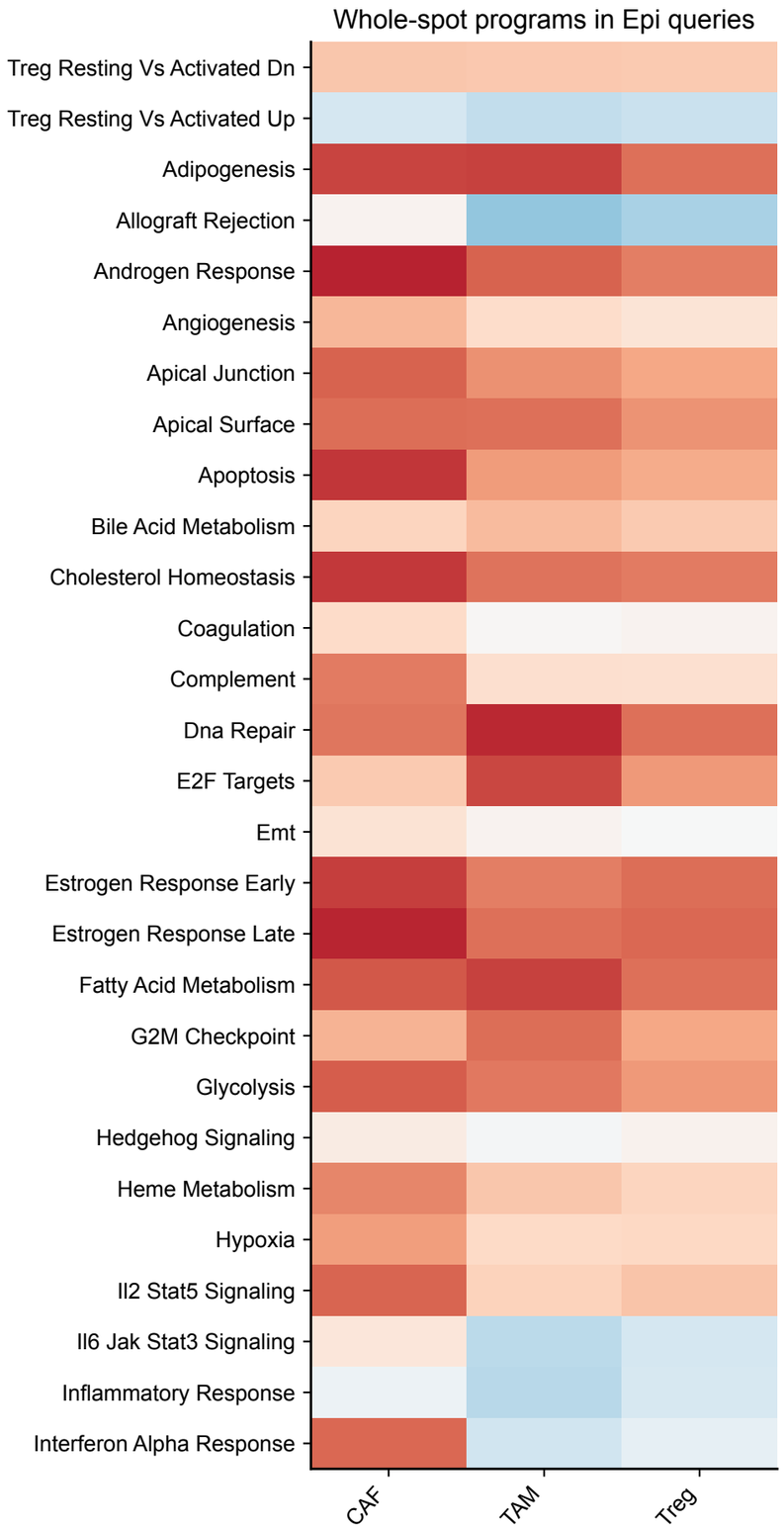


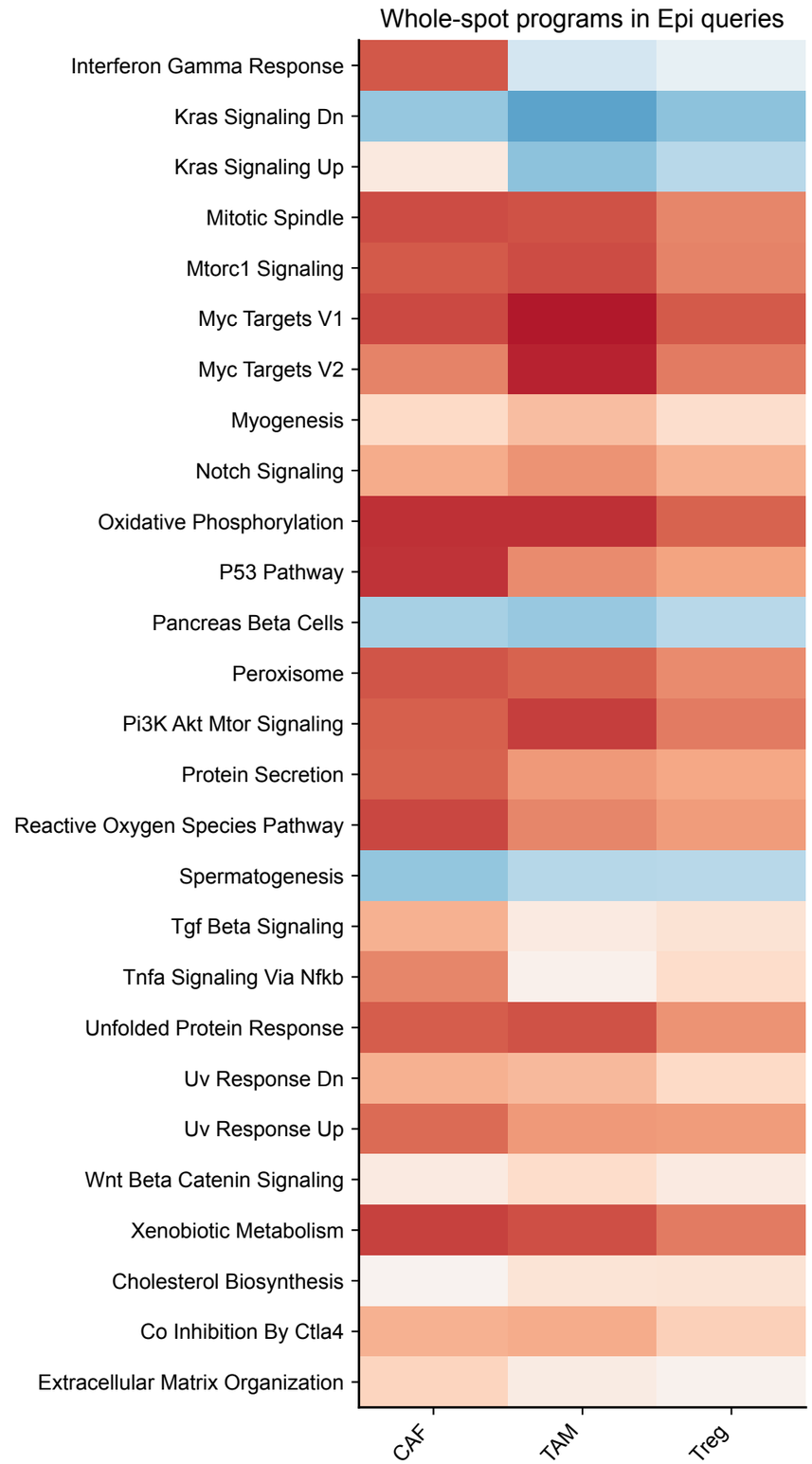




GSE319536_BC04 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

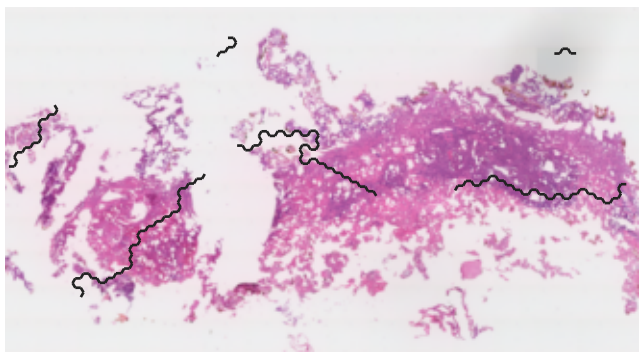




GSE319536_BC05 | Tissue context

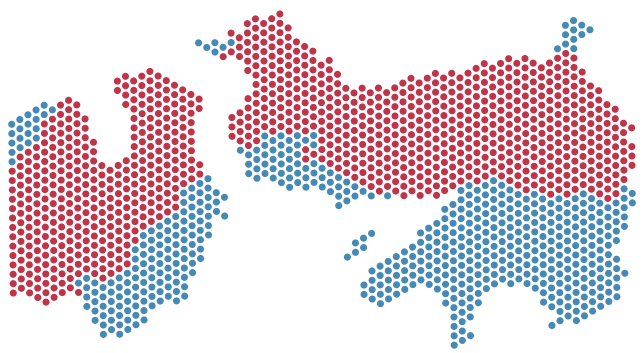
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



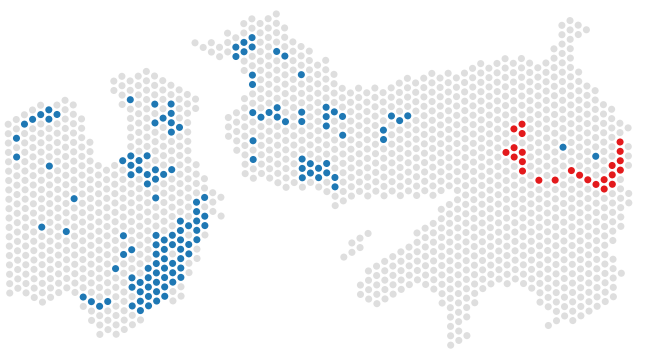
Black line: observed candidate outer edge

Final candidate location



inside outside

Accepted R12 (original 0–8)



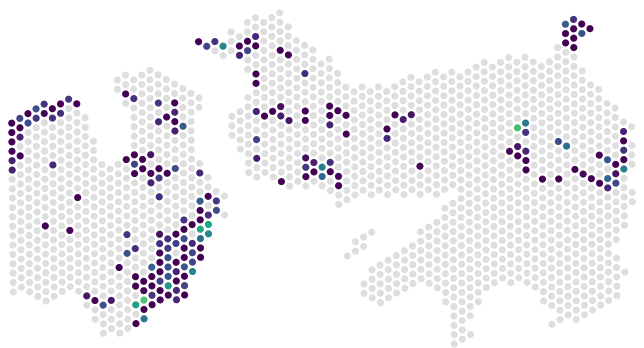
1 7

Epithelial support

0 1

Fraction of all 30

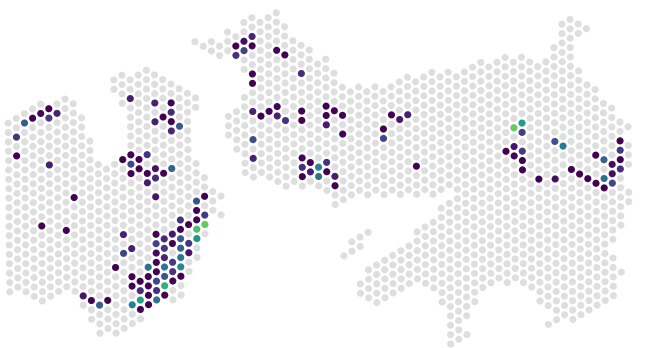
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



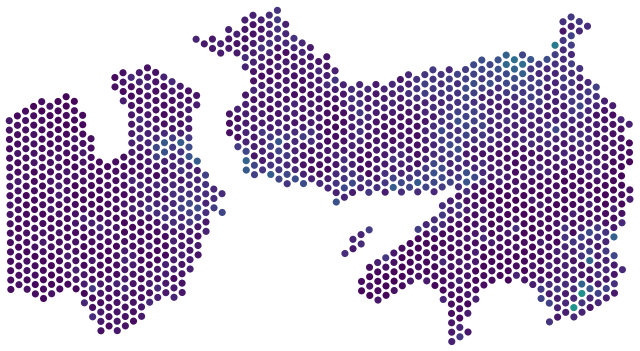
0 0.14

Matched-model AUCell

GSE319536_BC05 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

CAF composition



0 0.418
Fraction of all 30

CAF P80 targets

False True

CAF distance

0 3.1e+03
Nearest spot center (μm)

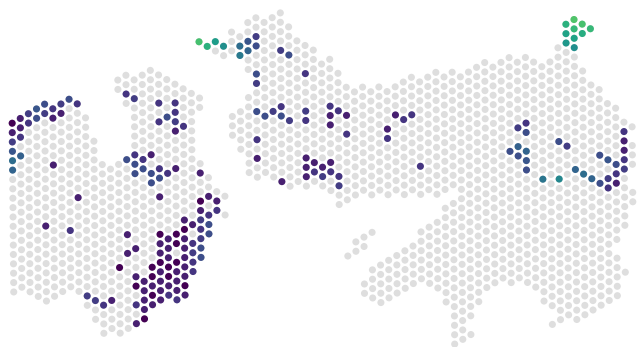
TAM composition

0 0.291
Fraction of all 30

TAM P80 targets

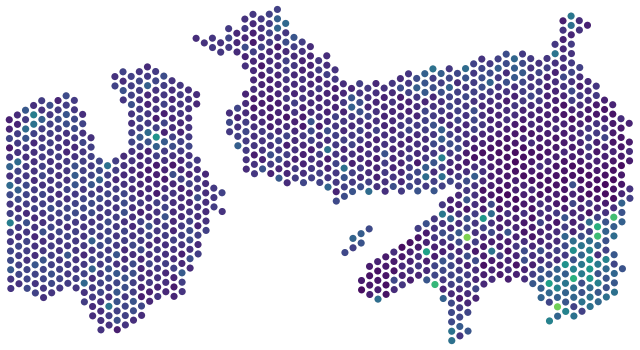
False True

TAM distance



0 1.13e+03
Nearest spot center (μm)

Treg composition



0 0.102
Fraction of all 30

Treg P80 targets

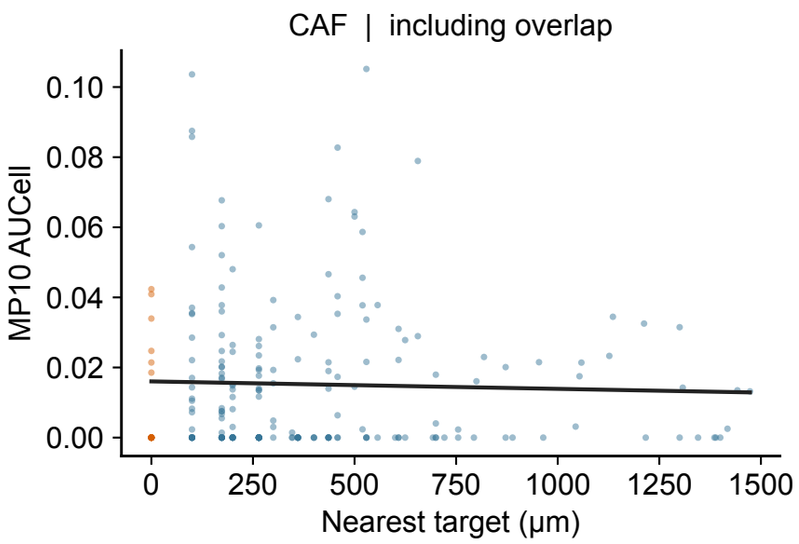
False True

Treg distance

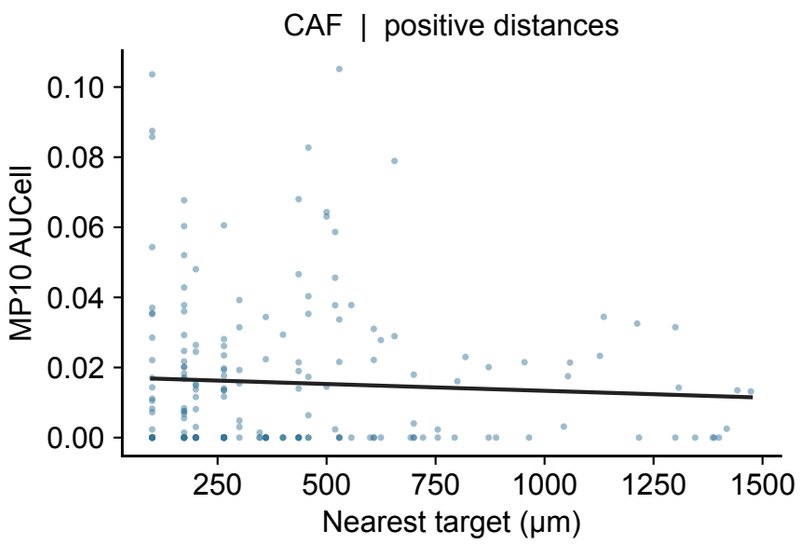
0 1.61e+03
Nearest spot center (μm)

GSE319536_BC05 | MP10 versus nearest target

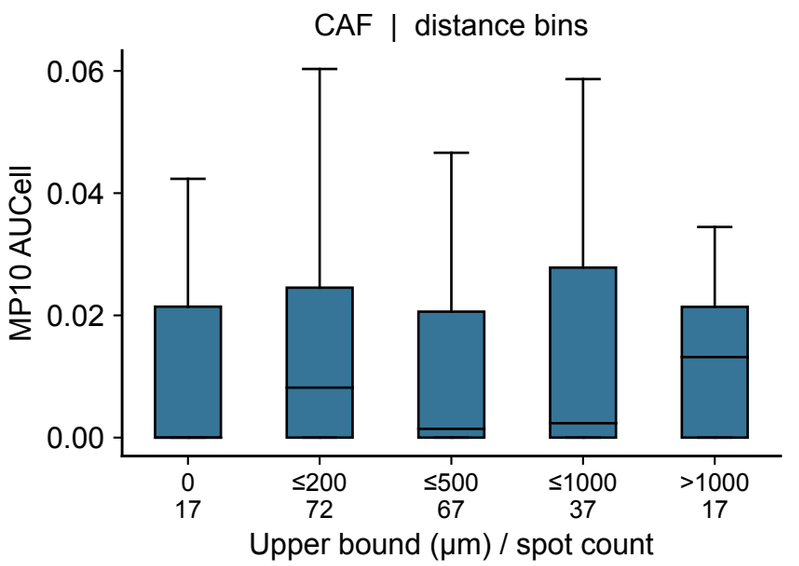
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



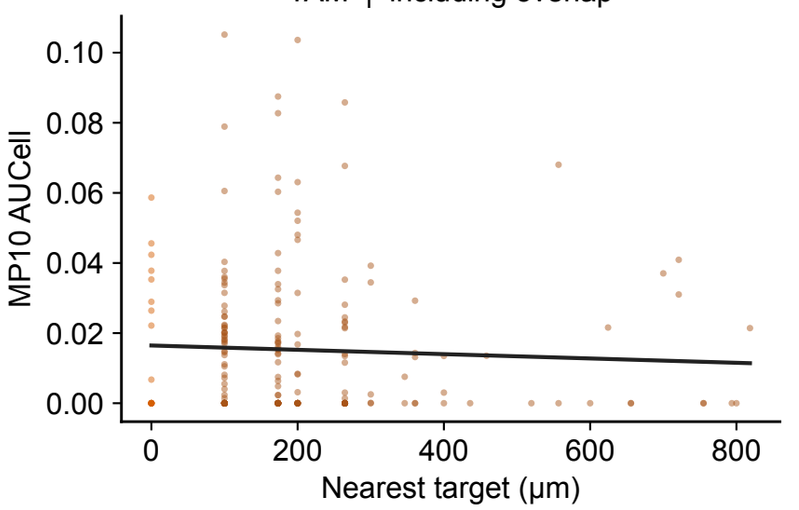
r -0.03 ρ +0.01
n 210/210 queries • 310 targets



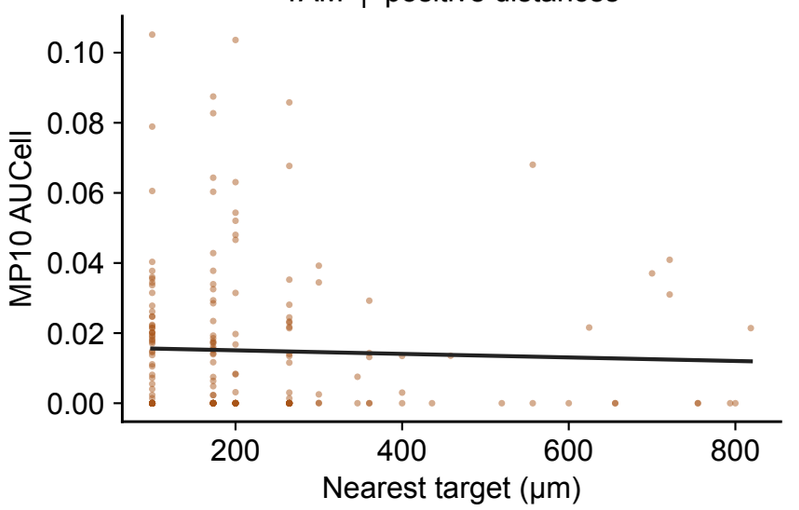
r -0.06 ρ -0.04
n 193/210 queries • 310 targets



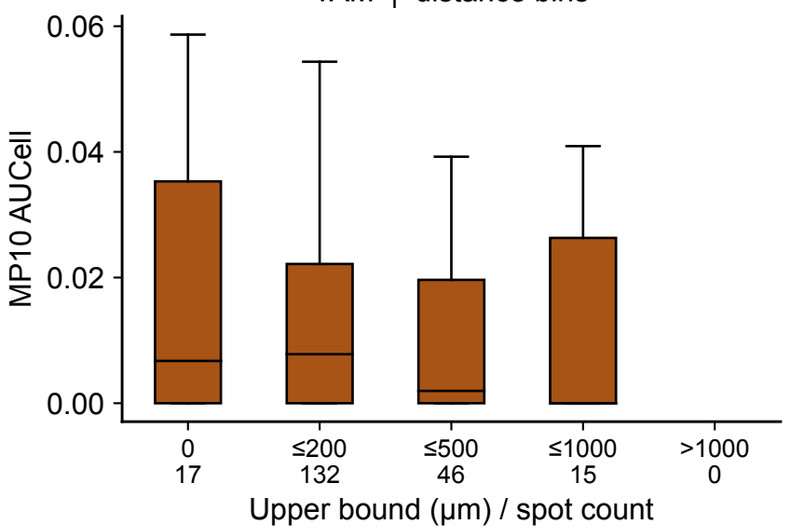
Orange = zero distance
Line = within-section linear fit



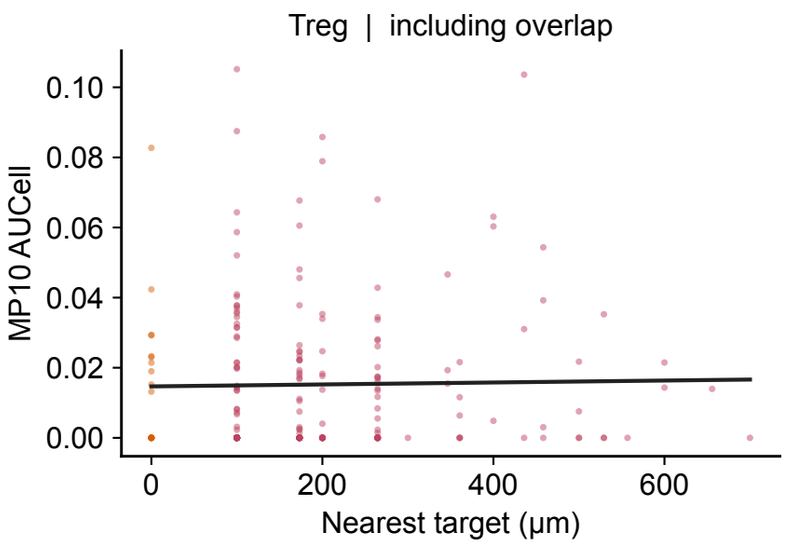
r -0.05 ρ -0.07
n 210/210 queries • 310 targets



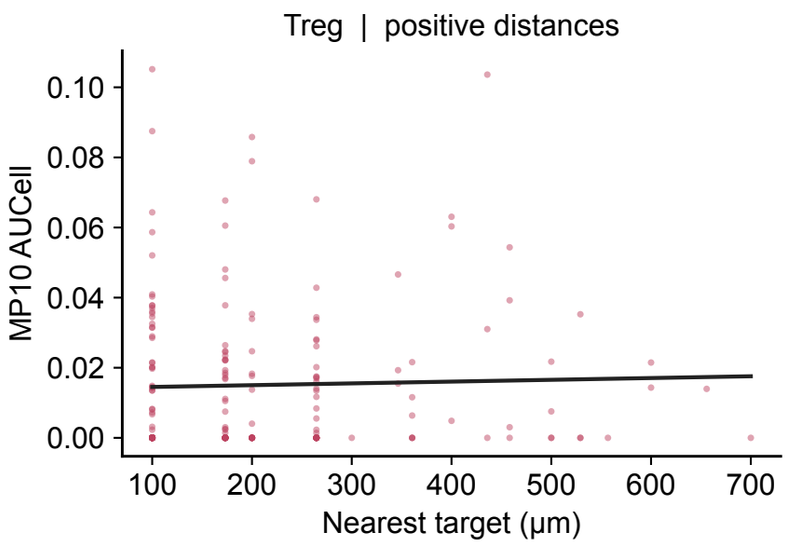
r -0.04 ρ -0.07
n 193/210 queries • 310 targets



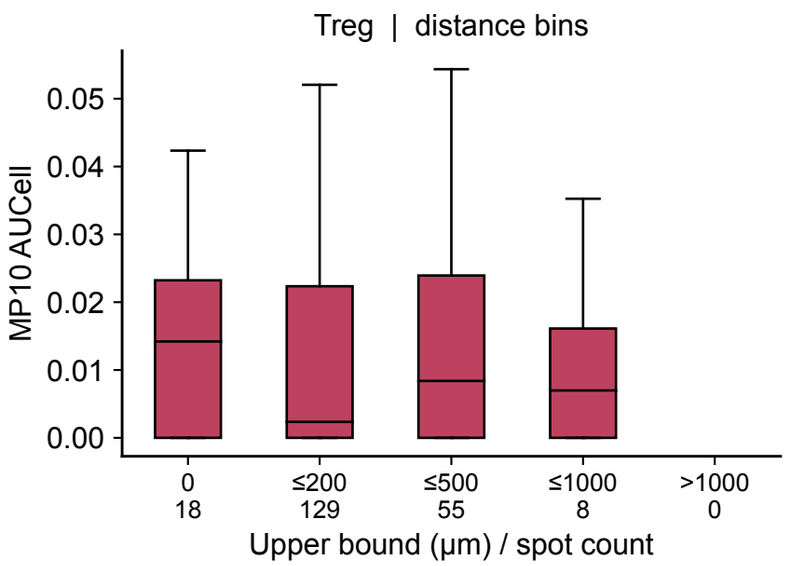
Orange = zero distance
Line = within-section linear fit



r +0.02 ρ +0.01
n 210/210 queries • 310 targets



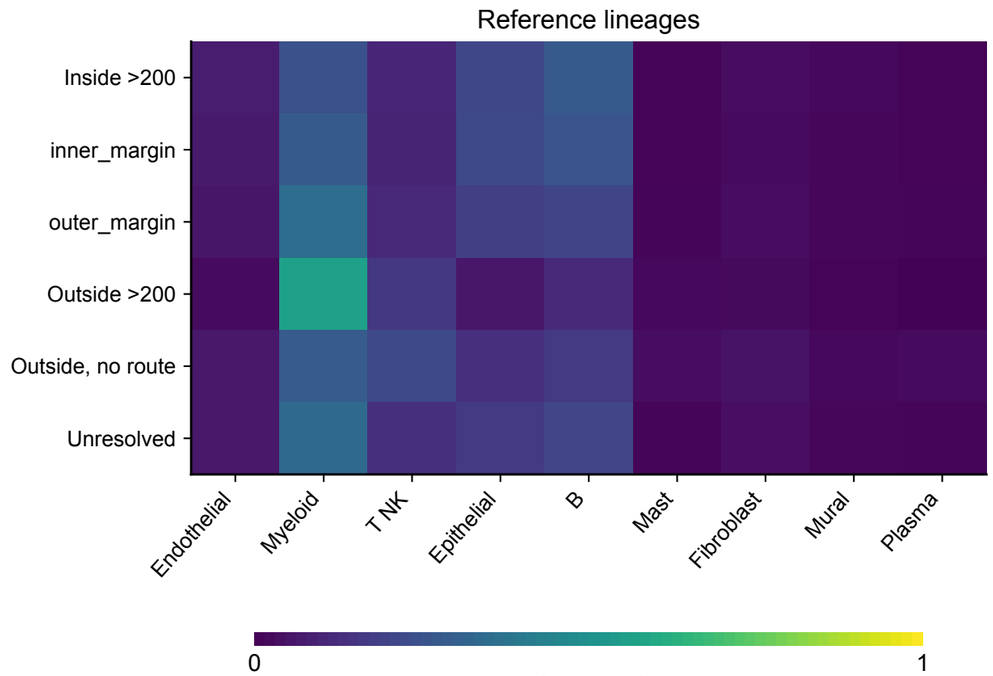
r +0.03 ρ +0.03
n 192/210 queries • 310 targets

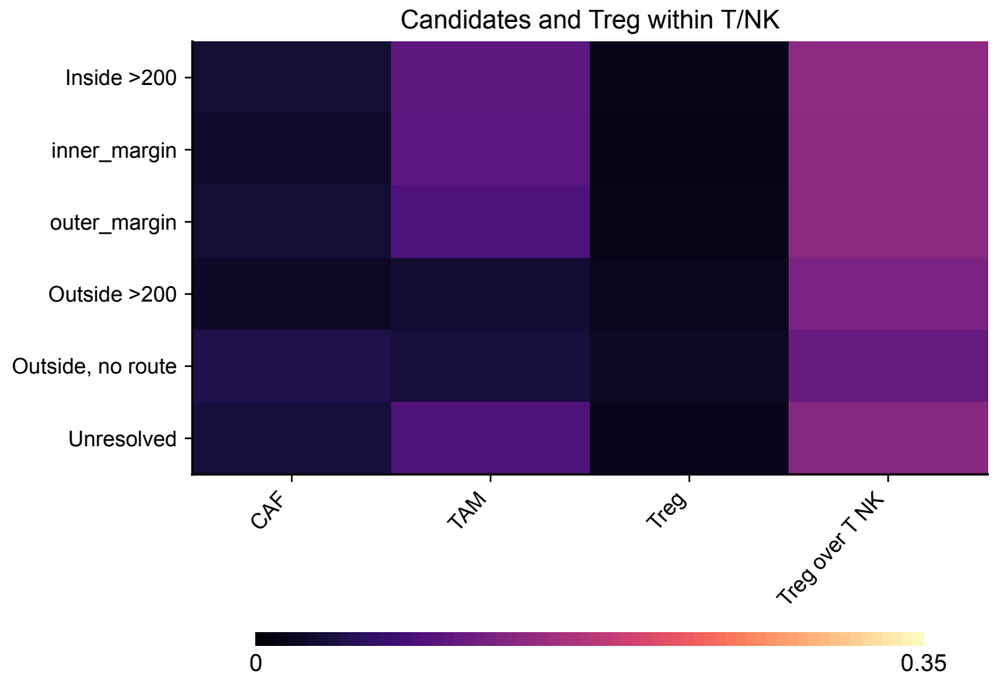


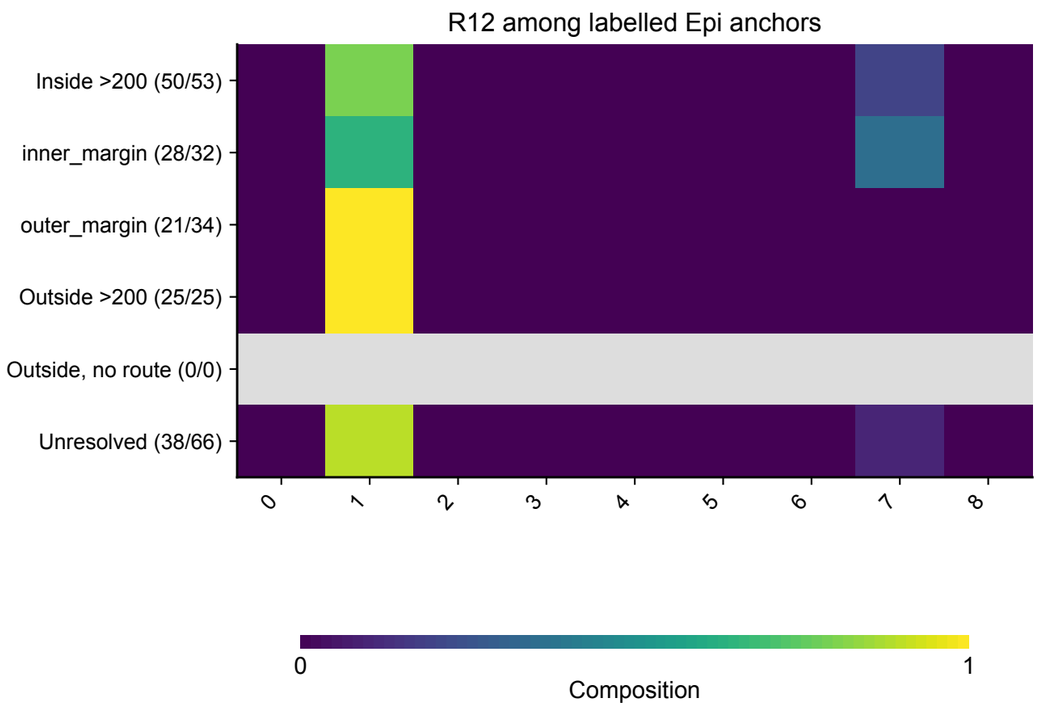
Orange = zero distance
Line = within-section linear fit

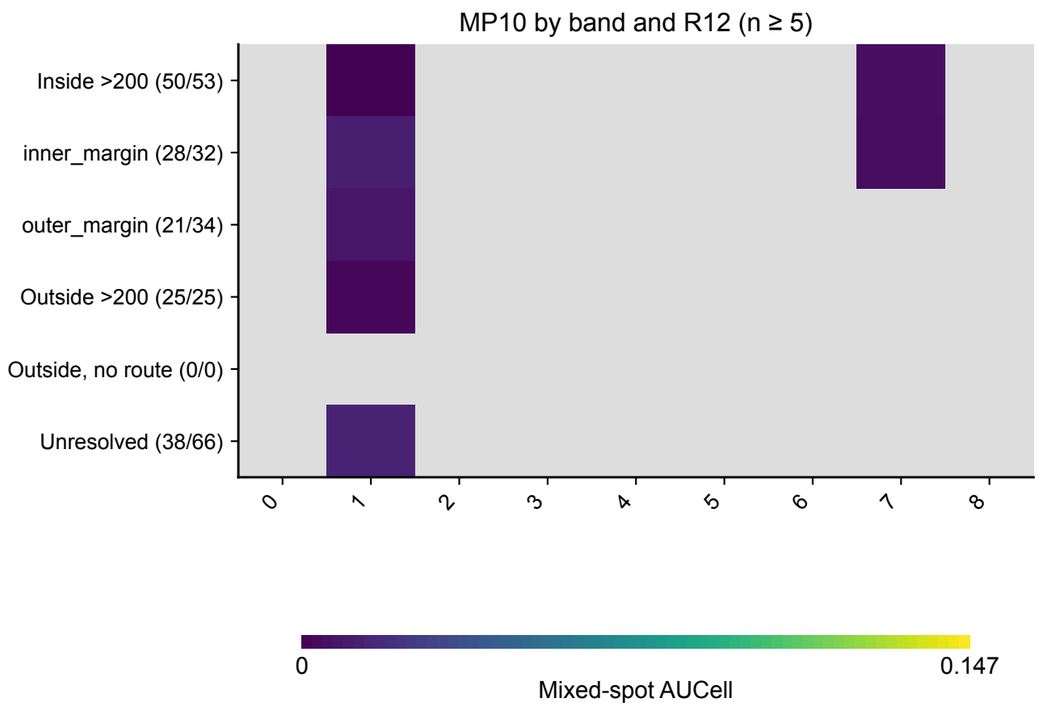
GSE319536_BC05 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable



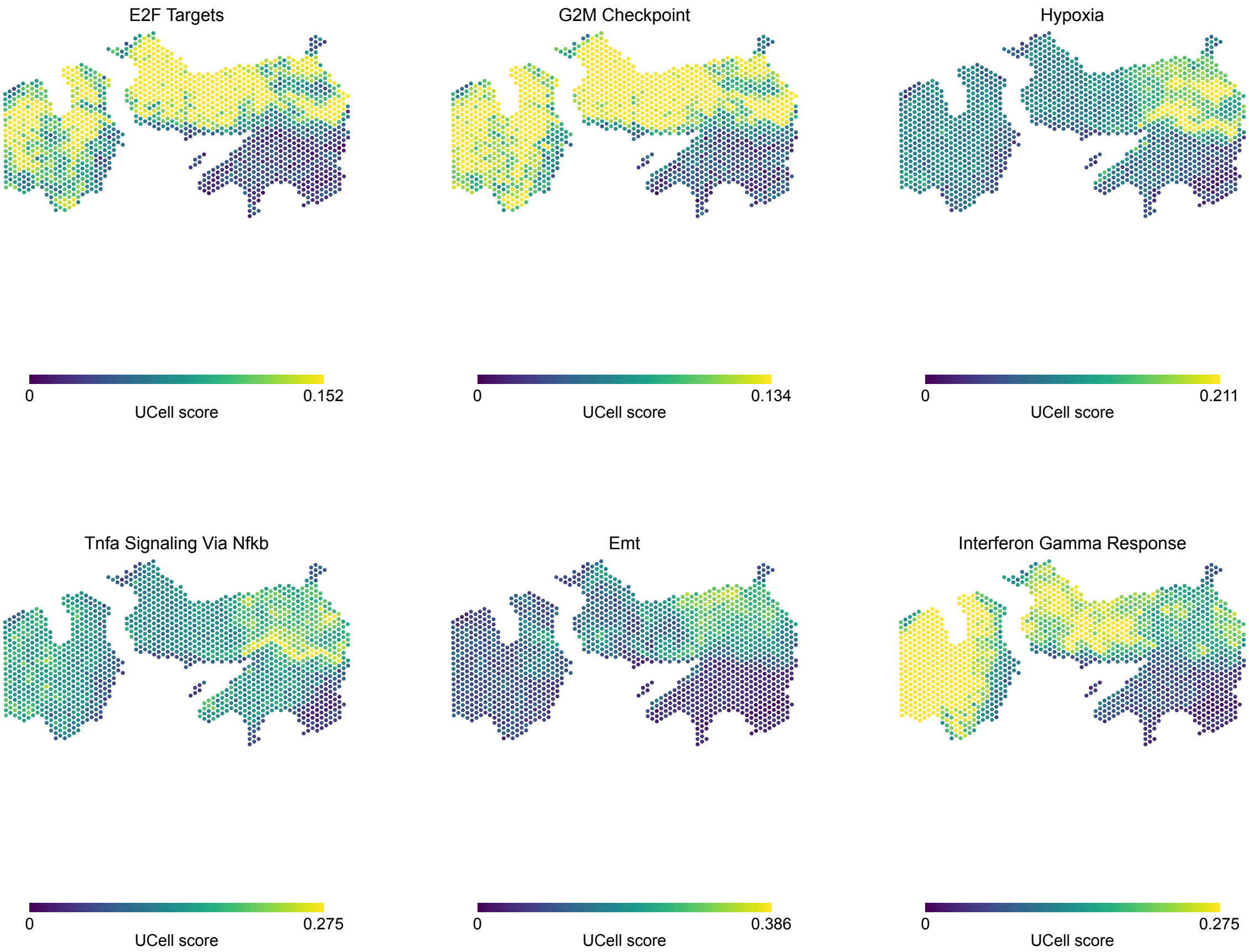






GSE319536_BC05 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections



GSE319536_BC05 | Immune, lipid and matrix programs

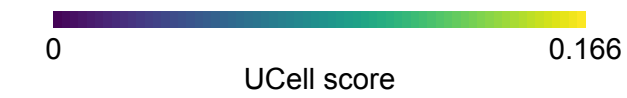
Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Il2 Stat5 Signaling

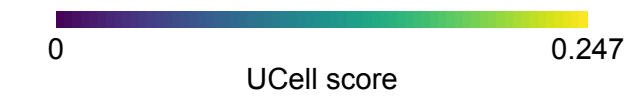
Tgf Beta Signaling

Cholesterol Homeostasis

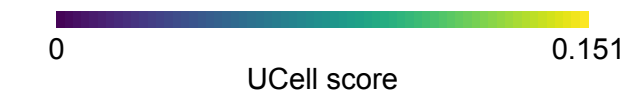
Cholesterol Biosynthesis



Extracellular Matrix Organization



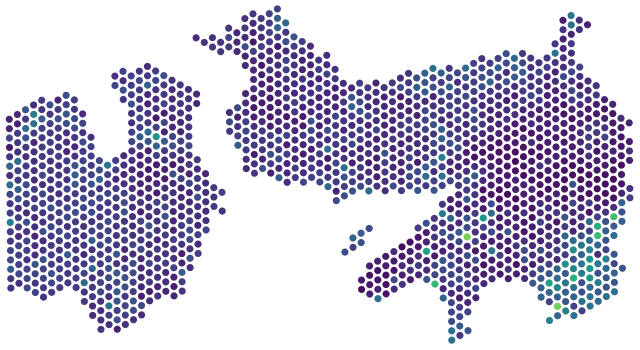
Co Inhibition By Ctla4



GSE319536_BC05 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Treg composition



0 0.102

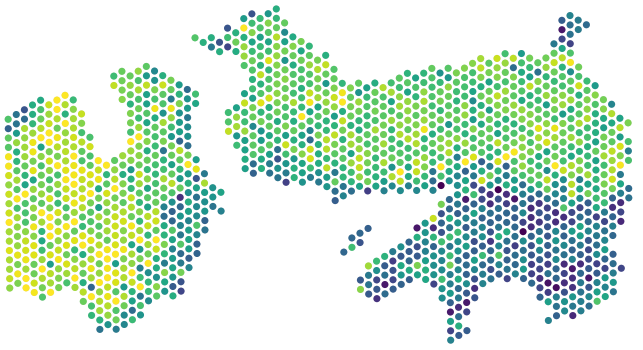
Fraction of all 30

Treg estimated abundance

0 2.85

Posterior mean estimate

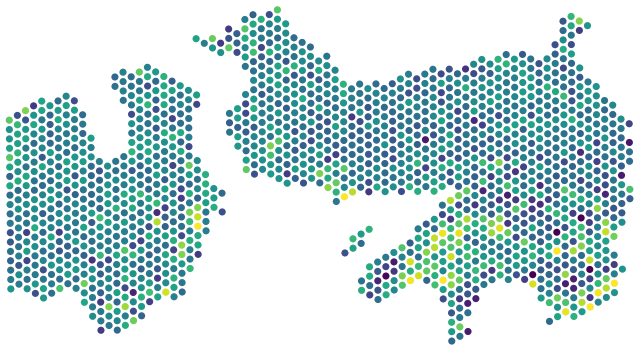
Treg Resting Vs Activated Dn



0 0.0778

UCell score

Treg Resting Vs Activated Up



0 0.0609

UCell score

Co Inhibition By Ctla4

0 0.151

UCell score

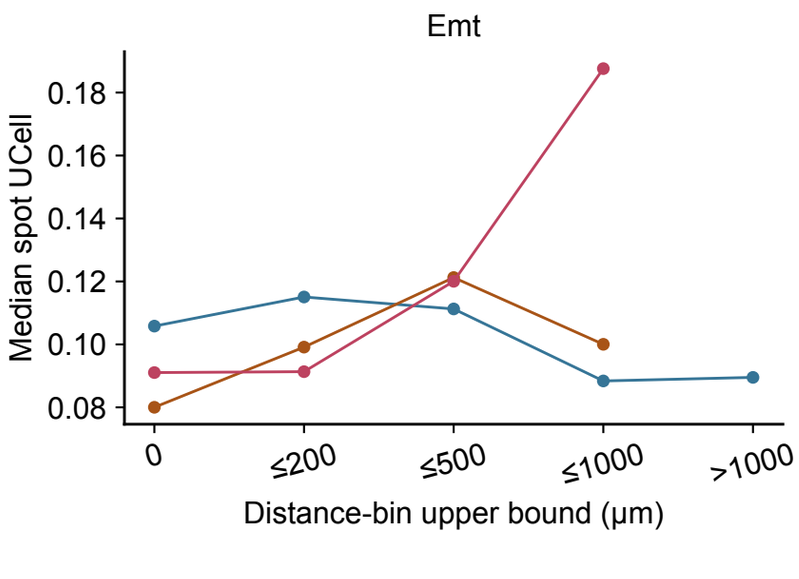
Il2 Stat5 Signaling

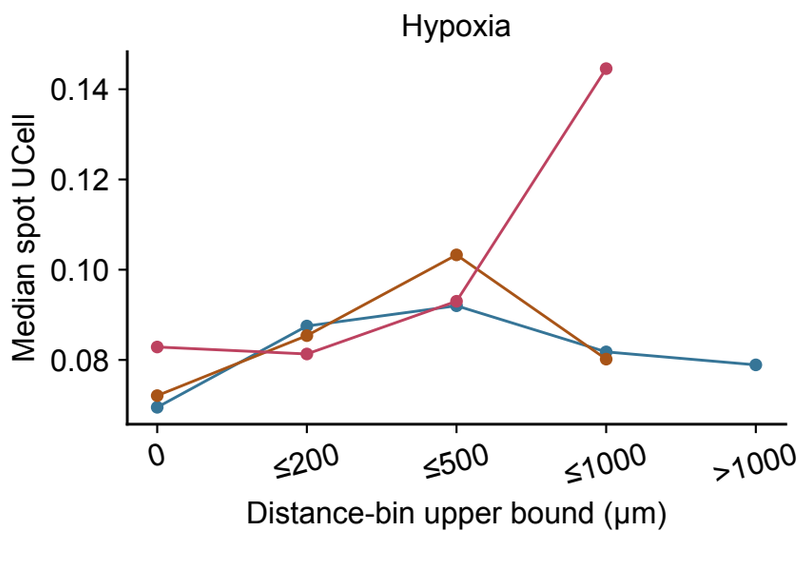
0 0.144

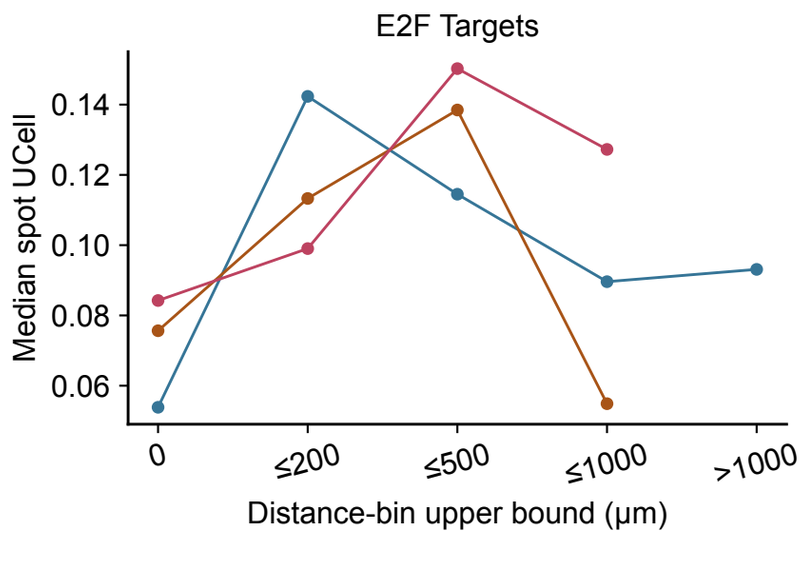
UCell score

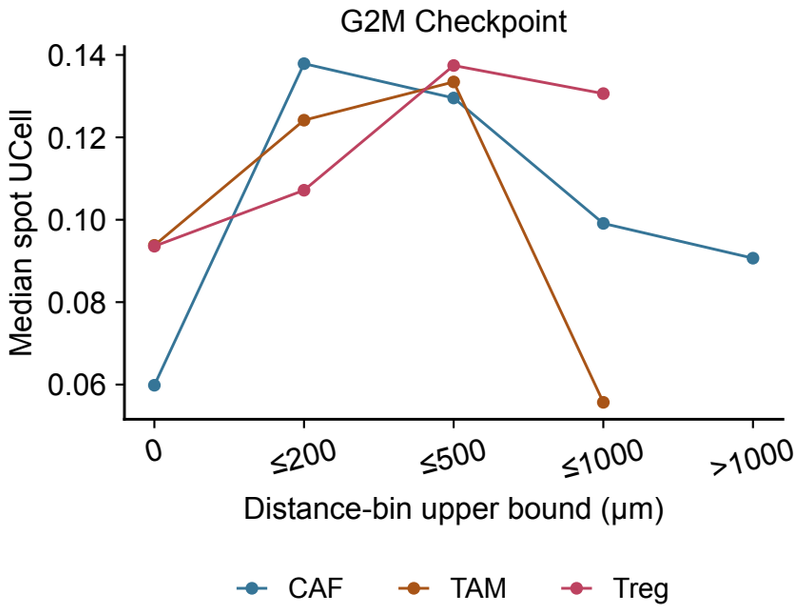
GSE319536_BC05 | Programs by target distance

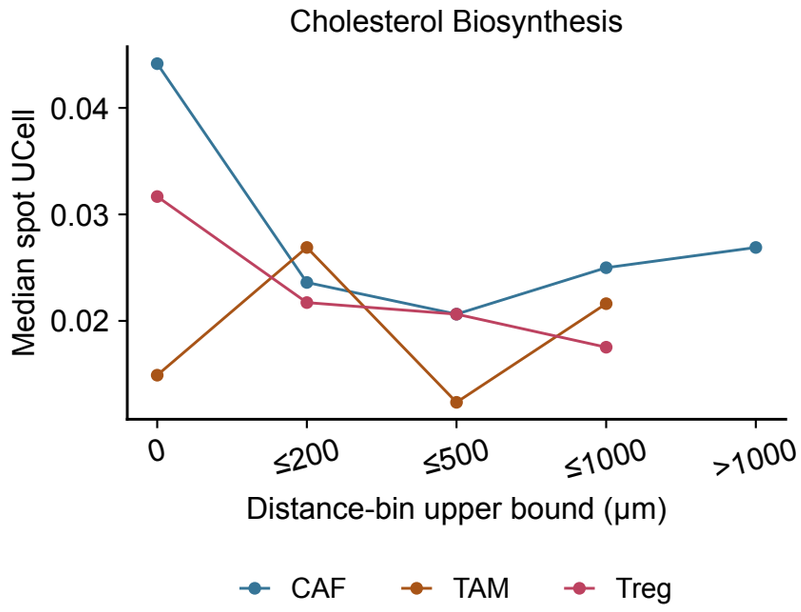
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

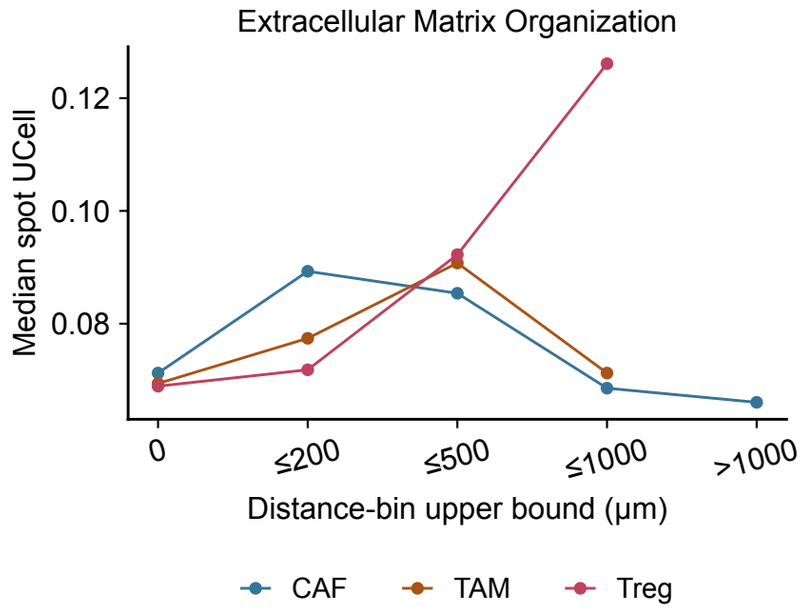






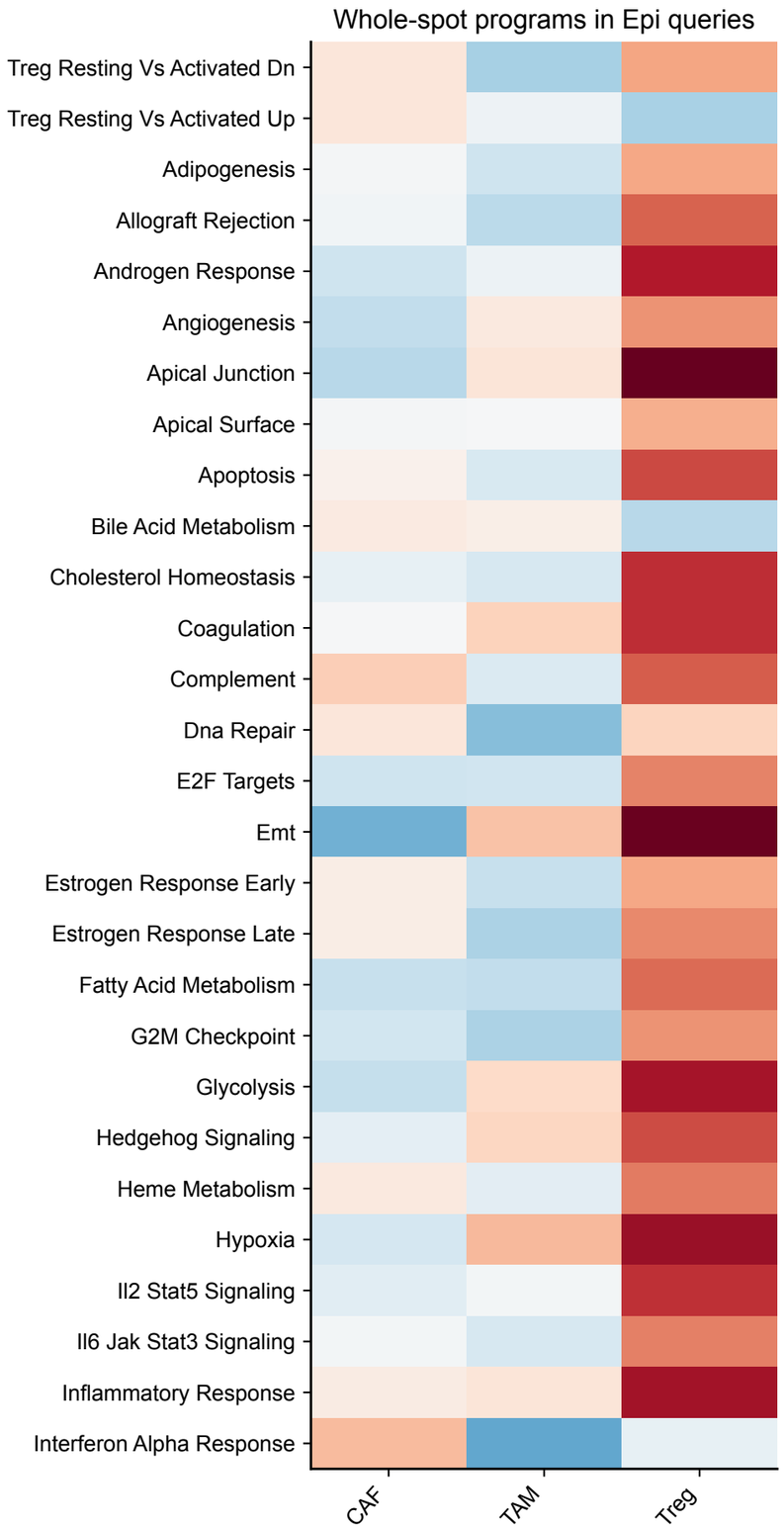


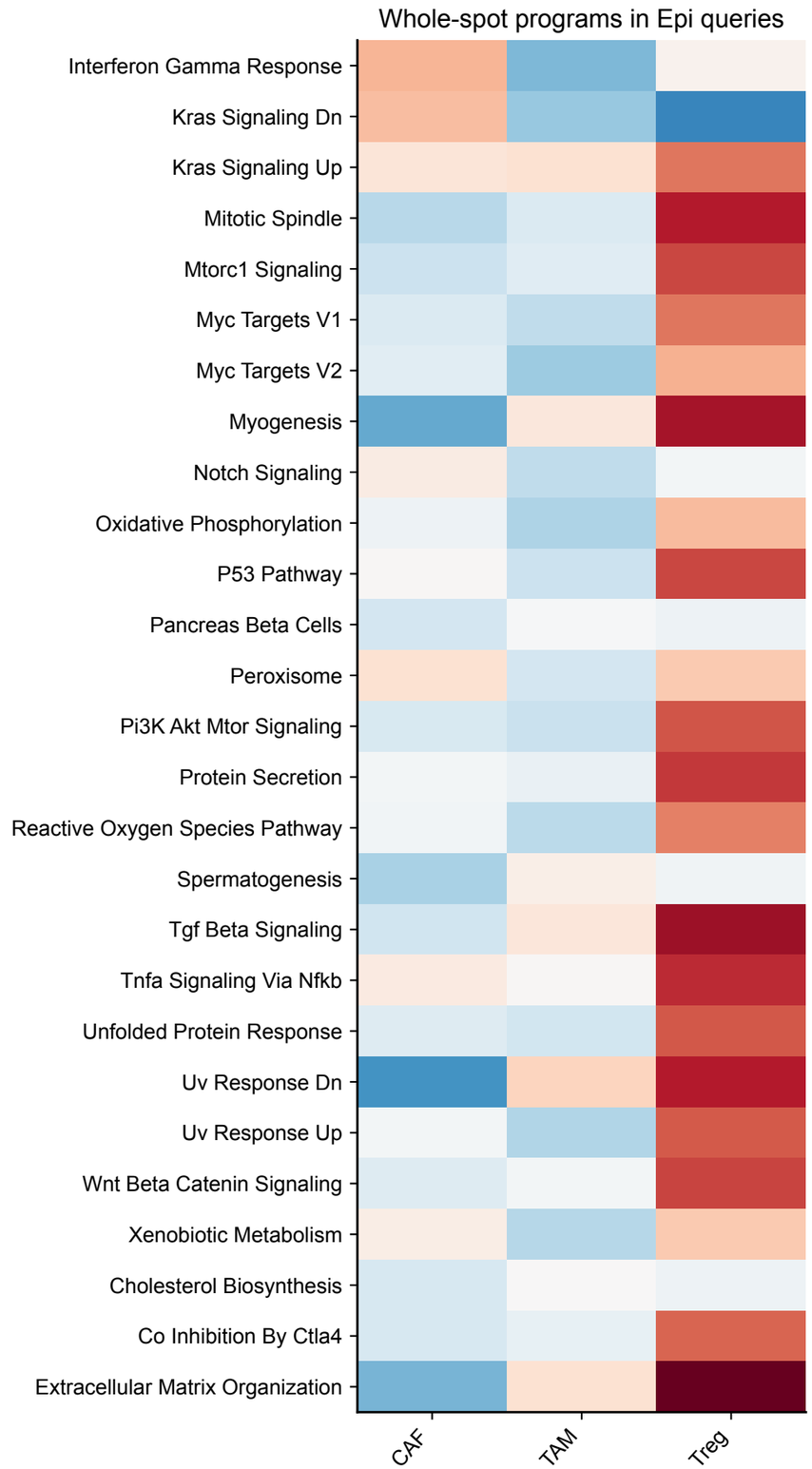




GSE319536_BC05 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

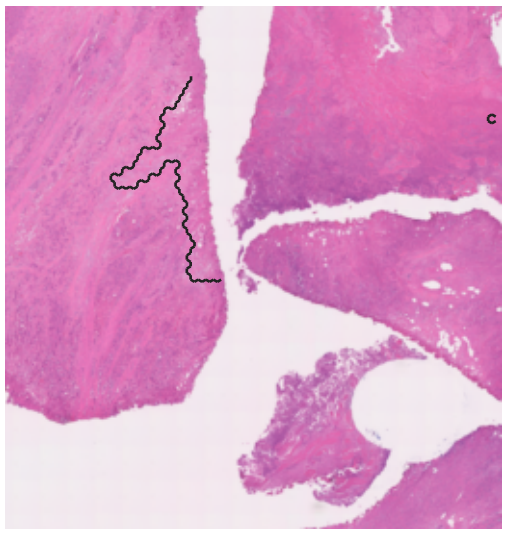




GSE319536_BC08 | Tissue context

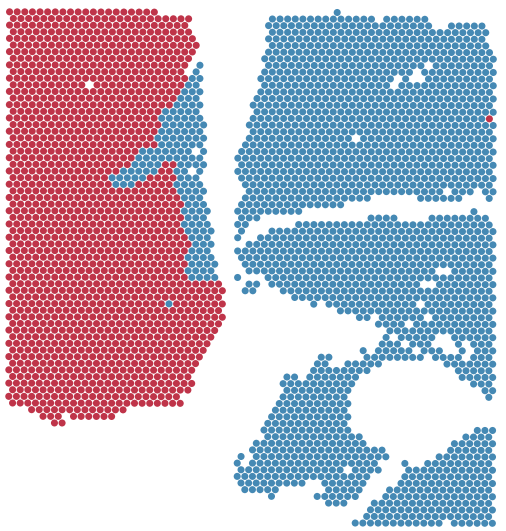
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



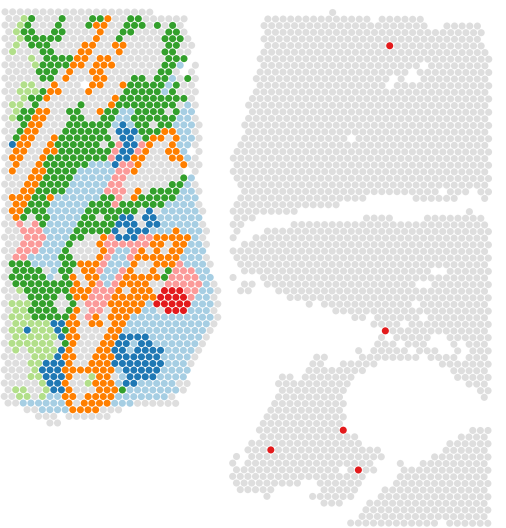
Black line: observed candidate outer edge

Final candidate location



inside outside

Accepted R12 (original 0–8)



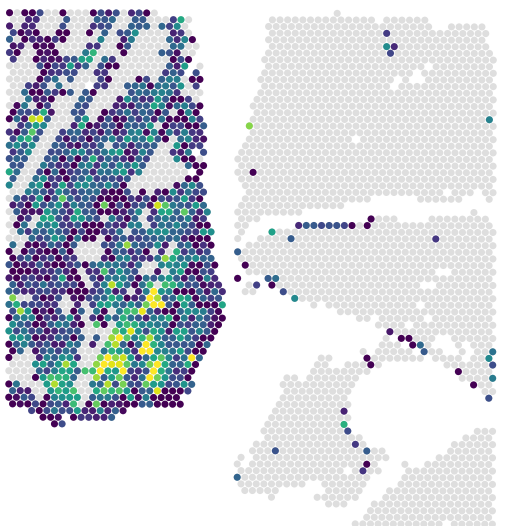
0 1 2 5 6 7
1 3

Epithelial support

0 1

Fraction of all 30

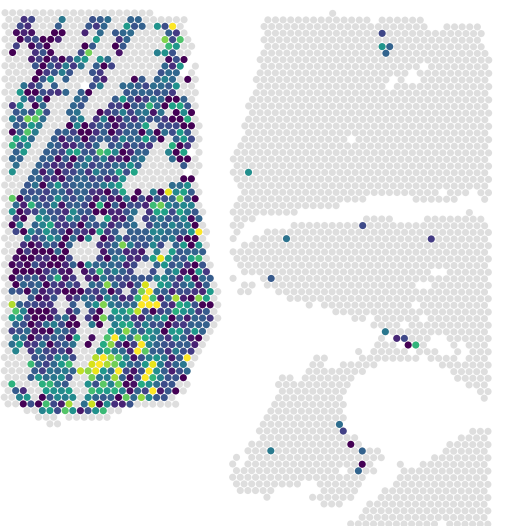
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



0 0.14

Matched-model AUCell

GSE319536_BC08 | Candidate neighborhoods

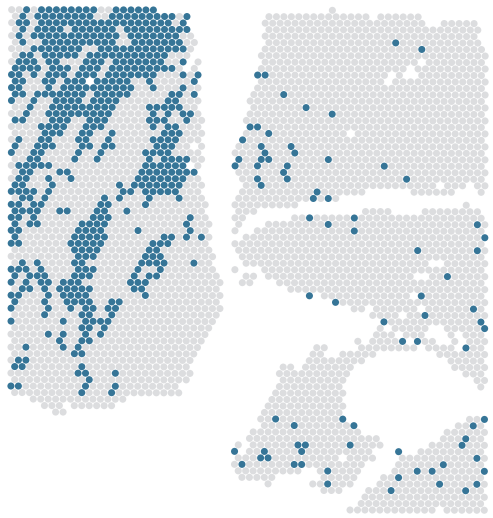
Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

CAF composition

0 0.418

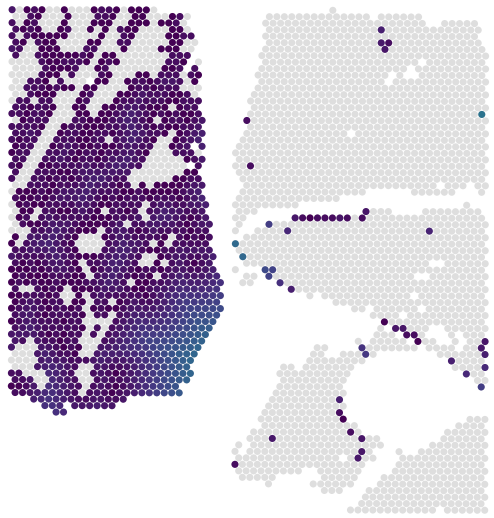
Fraction of all 30

CAF P80 targets



False True

CAF distance



0 3.1e+03

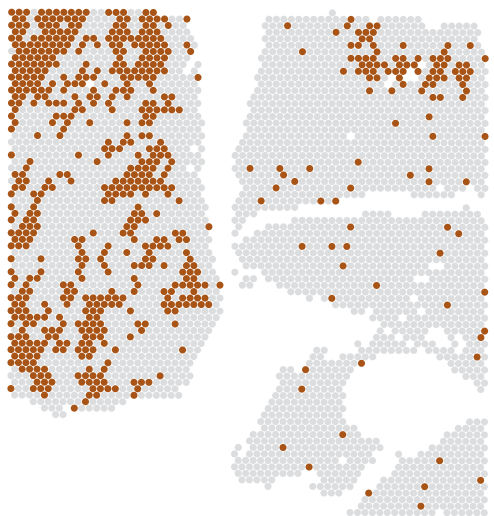
Nearest spot center (μm)

TAM composition

0 0.291

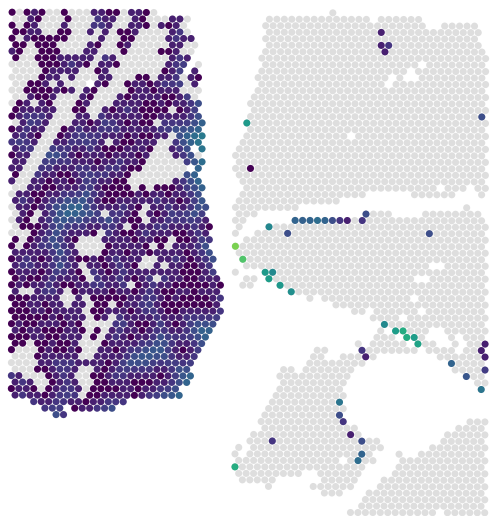
Fraction of all 30

TAM P80 targets



False True

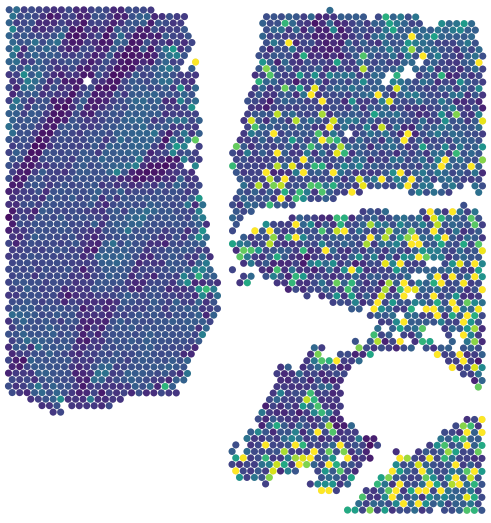
TAM distance



0 1.13e+03

Nearest spot center (μm)

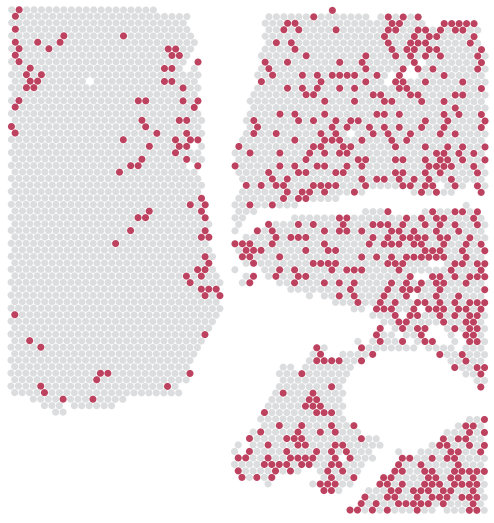
Treg composition



0 0.102

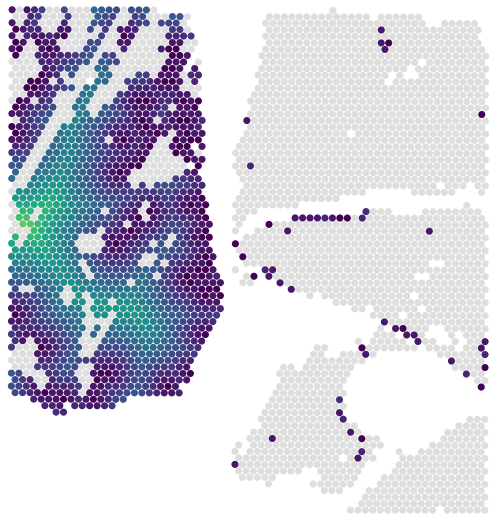
Fraction of all 30

Treg P80 targets



False True

Treg distance

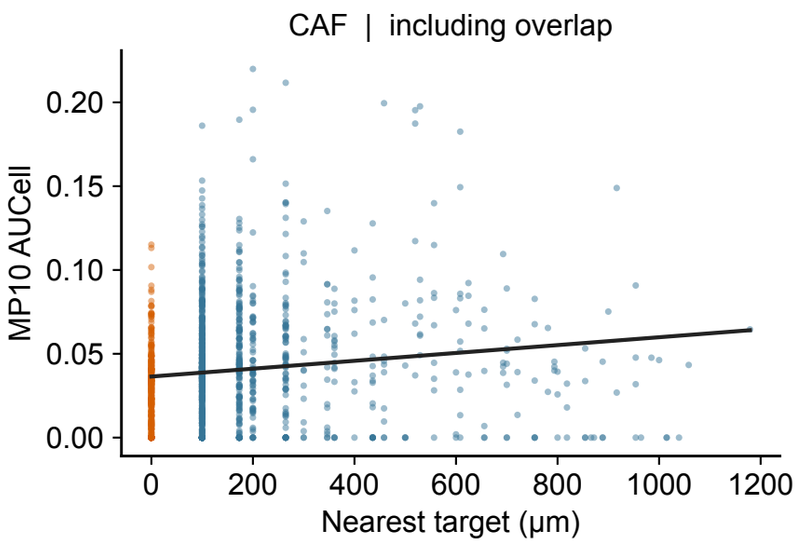


0 1.61e+03

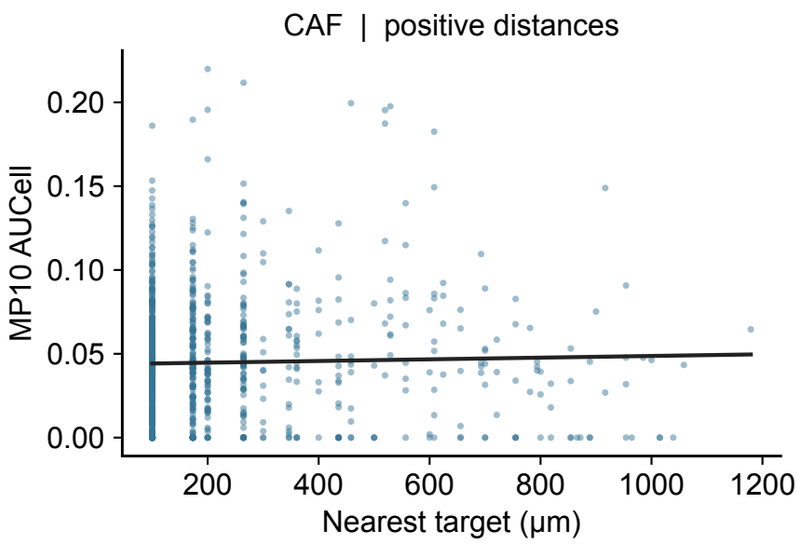
Nearest spot center (μm)

GSE319536_BC08 | MP10 versus nearest target

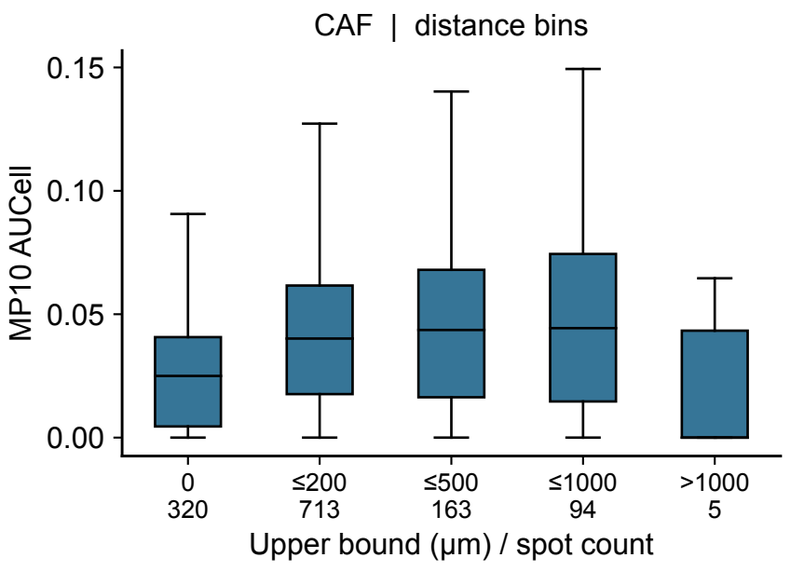
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



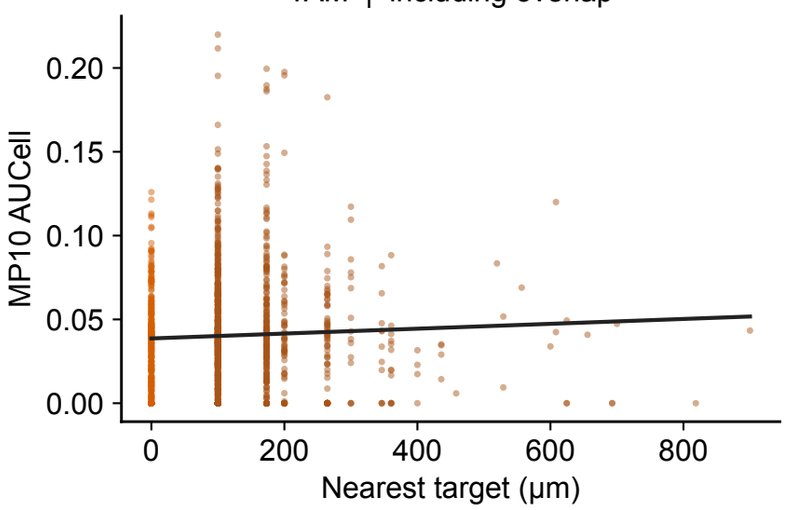
r +0.13 ρ +0.20
n 1295/1297 queries • 689 targets



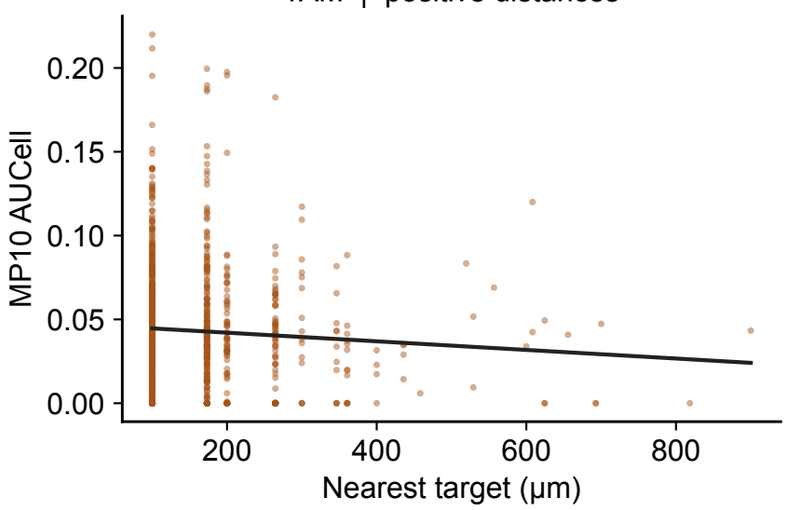
r +0.03 ρ +0.05
n 975/1297 queries • 689 targets



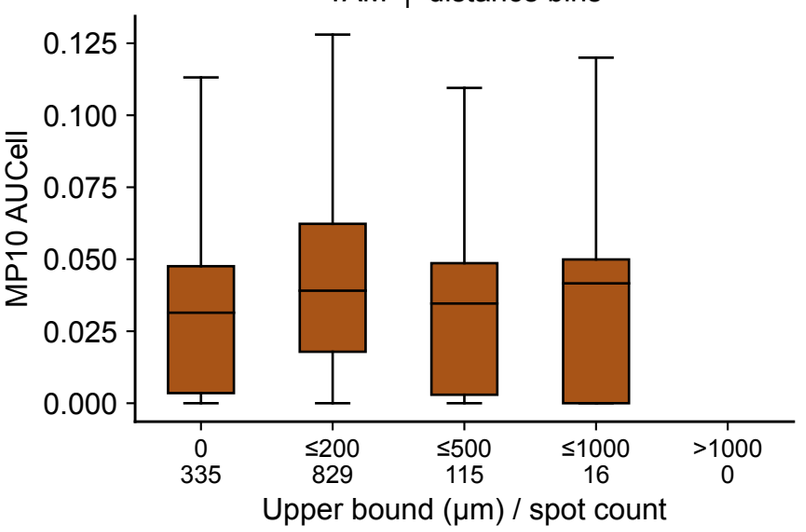
Orange = zero distance
Line = within-section linear fit



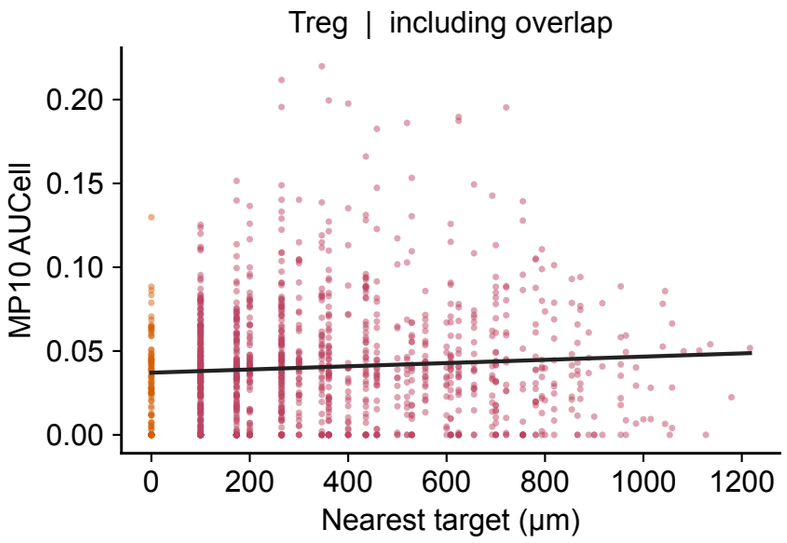
r +0.04 ρ +0.09
n 1295/1297 queries • 689 targets



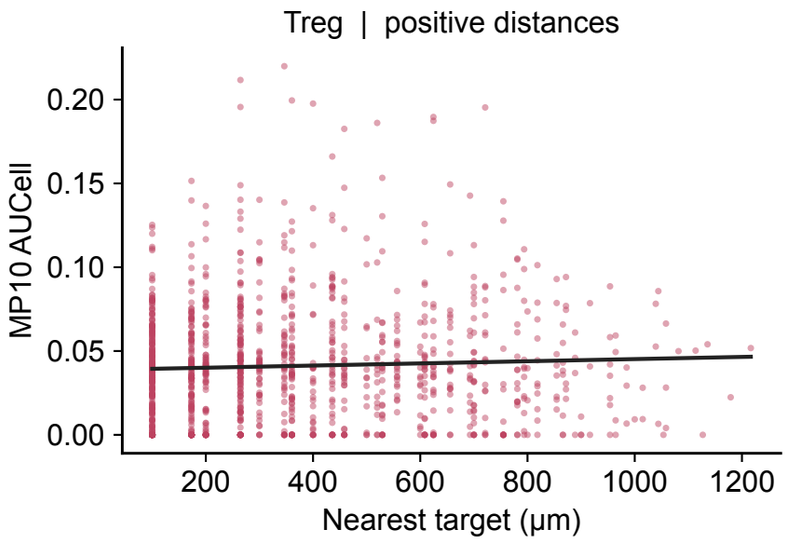
r -0.07 ρ -0.02
n 960/1297 queries • 689 targets



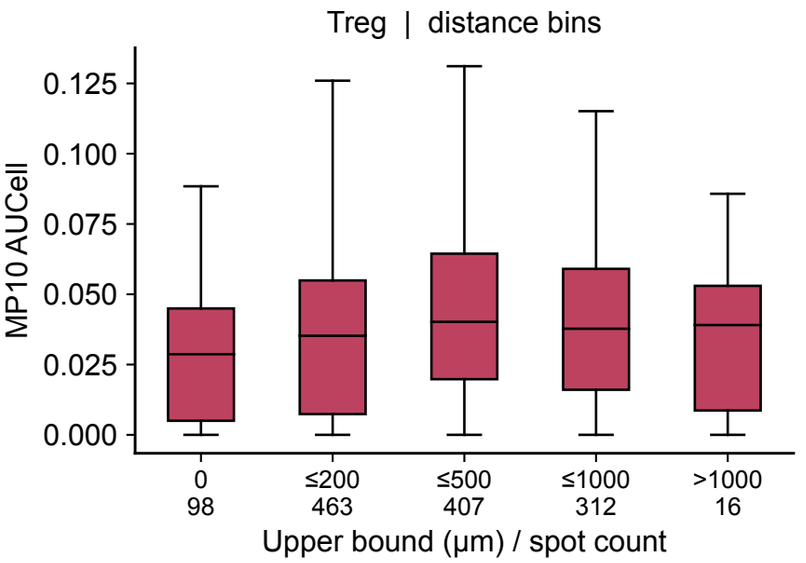
Orange = zero distance
Line = within-section linear fit



r +0.07 ρ +0.09
n 1296/1297 queries • 689 targets



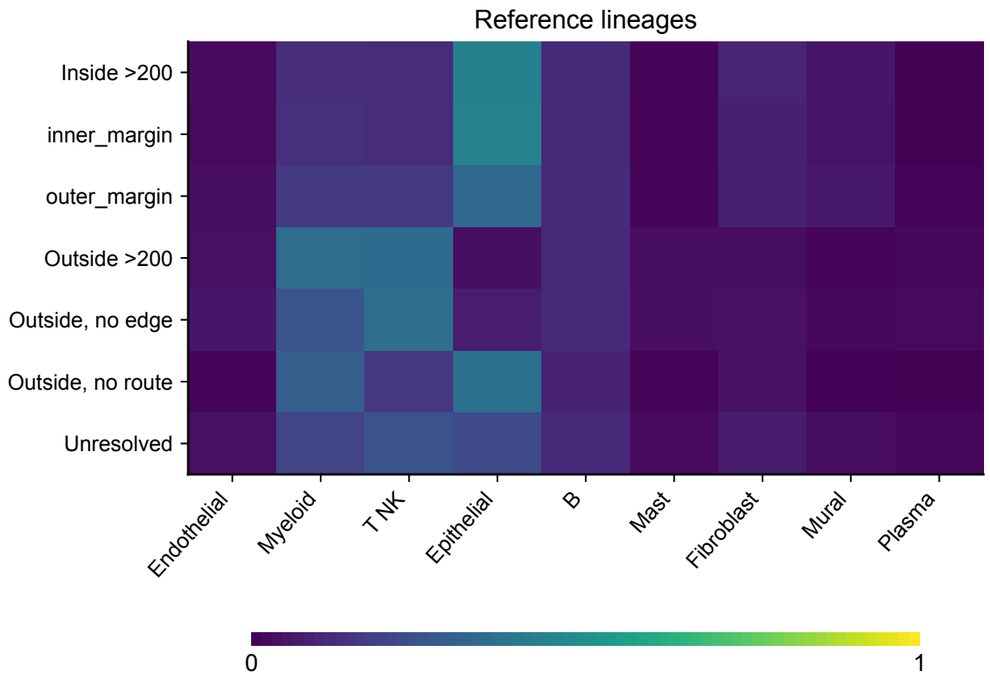
r +0.05 ρ +0.06
n 1198/1297 queries • 689 targets

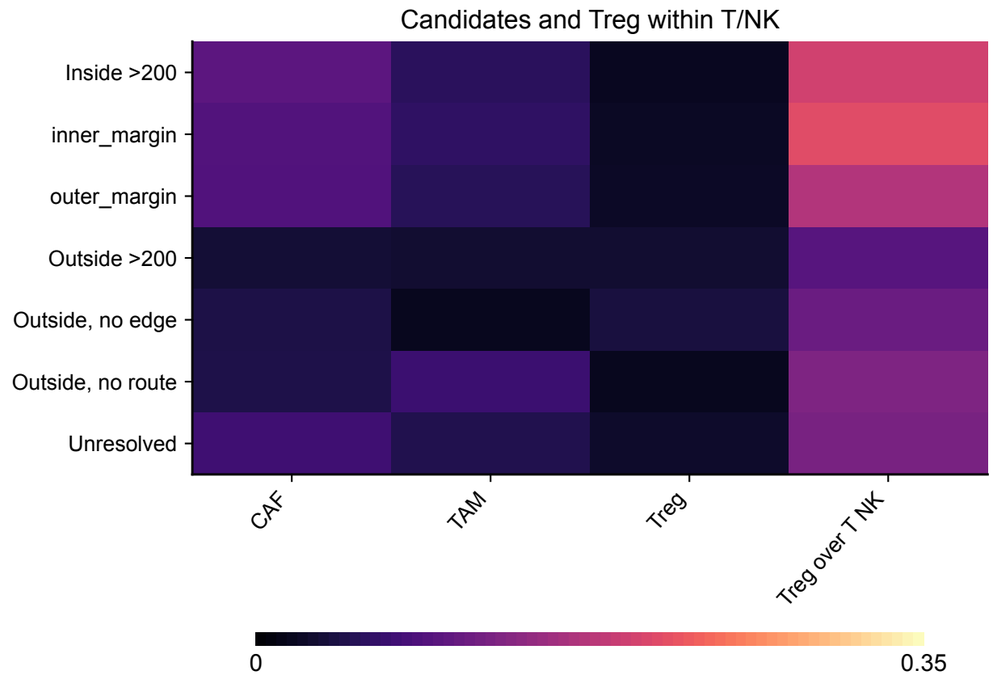


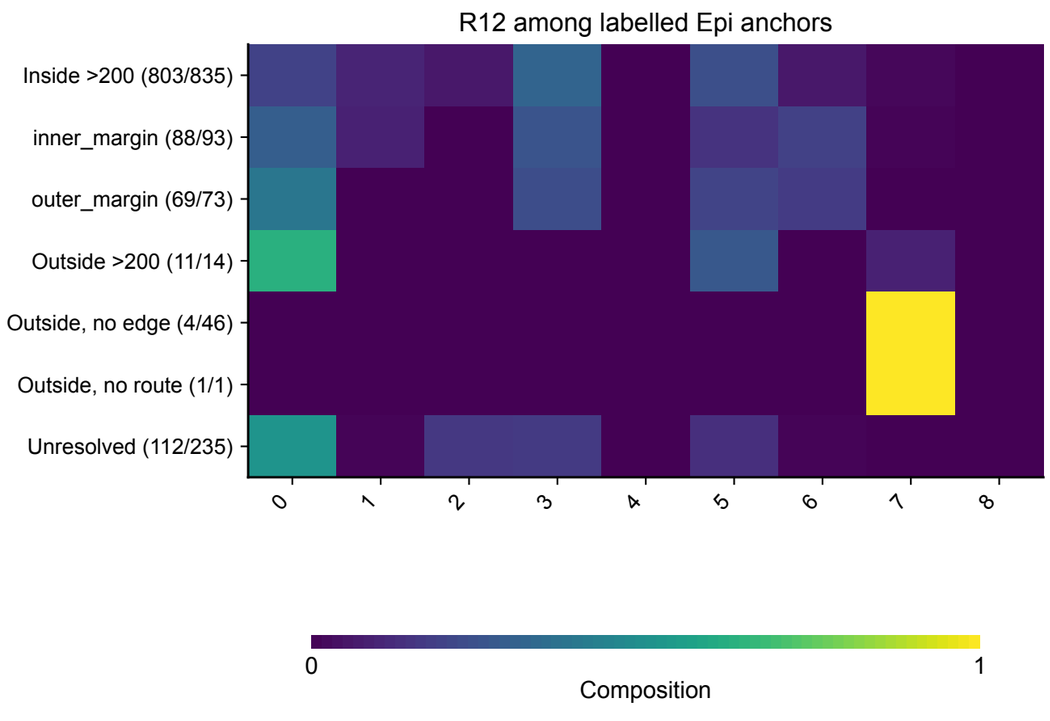
Orange = zero distance
Line = within-section linear fit

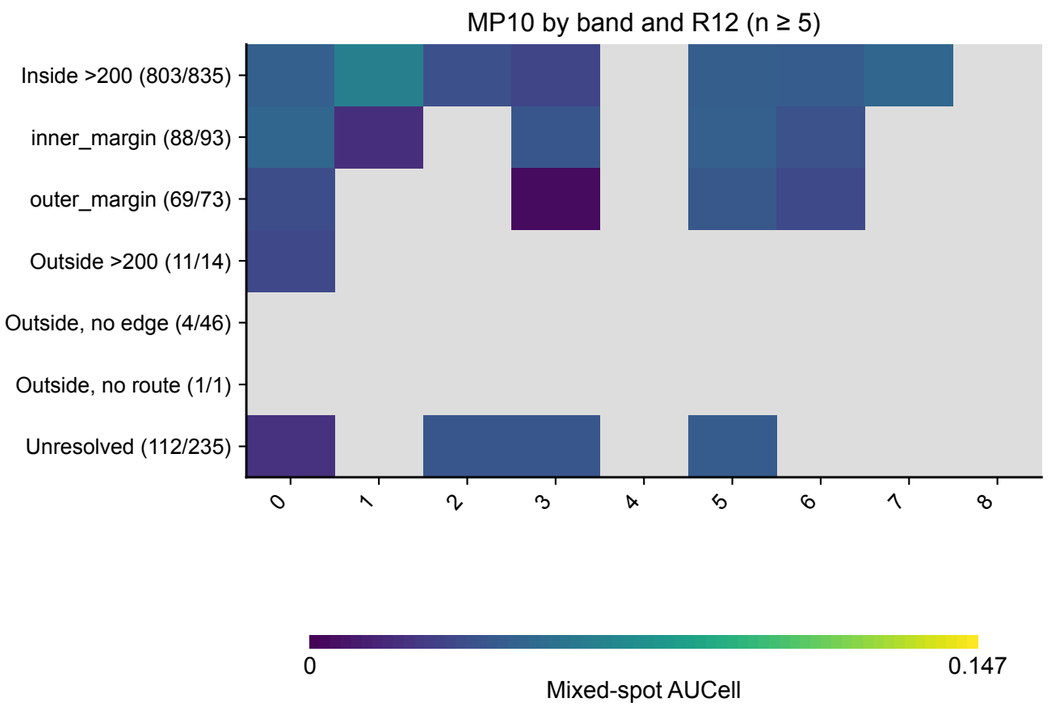
GSE319536_BC08 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





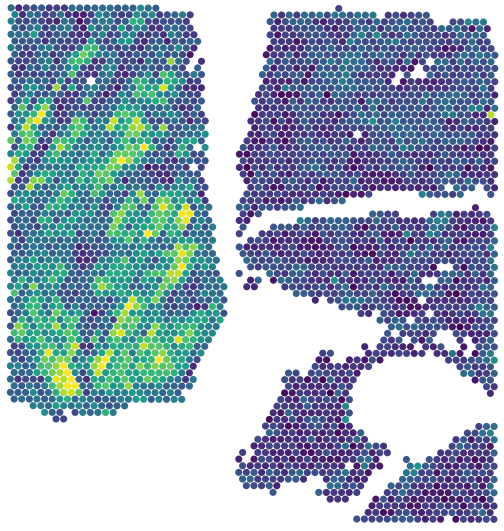




GSE319536_BC08 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

E2F Targets



0 0.152

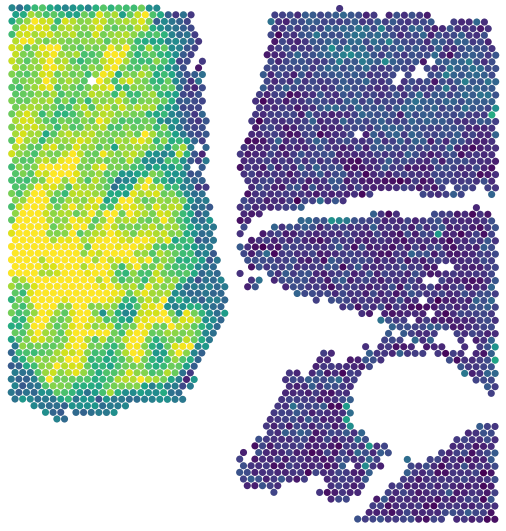
UCell score

G2M Checkpoint

0 0.134

UCell score

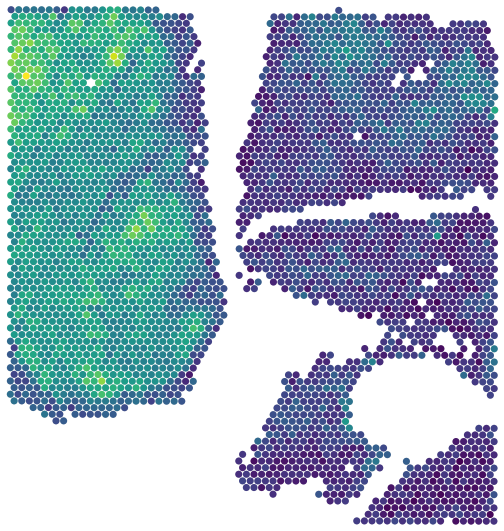
Hypoxia



0 0.211

UCell score

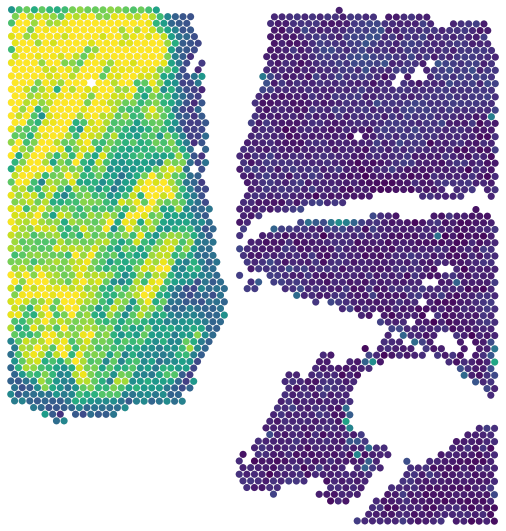
Tnfa Signaling Via Nfkb



0 0.275

UCell score

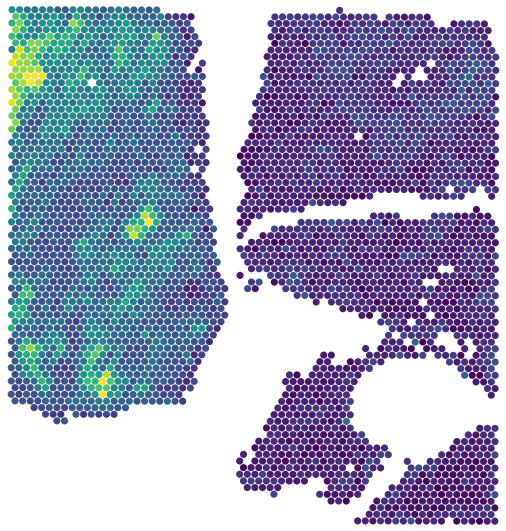
Emt



0 0.386

UCell score

Interferon Gamma Response



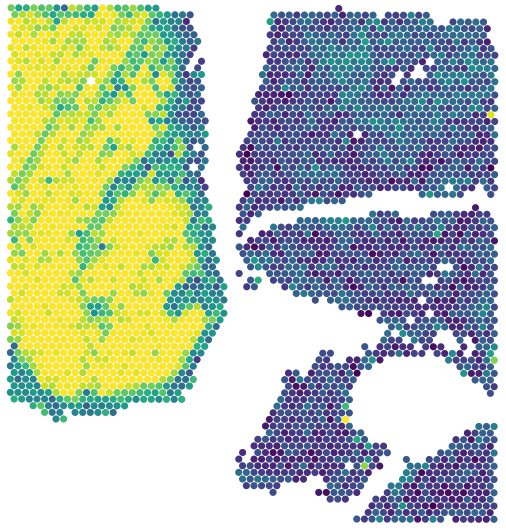
0 0.275

UCell score

GSE319536_BC08 | Immune, lipid and matrix programs

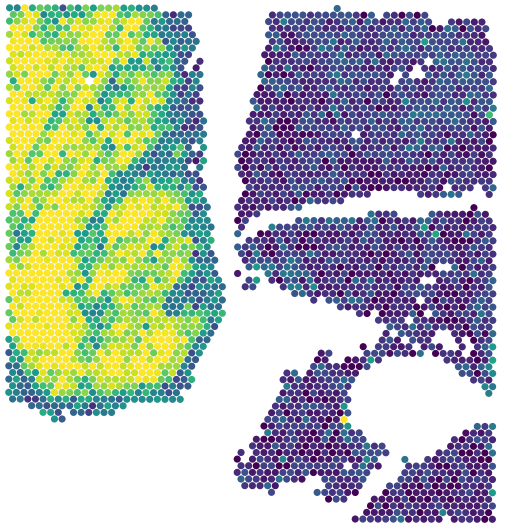
Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Il2 Stat5 Signaling



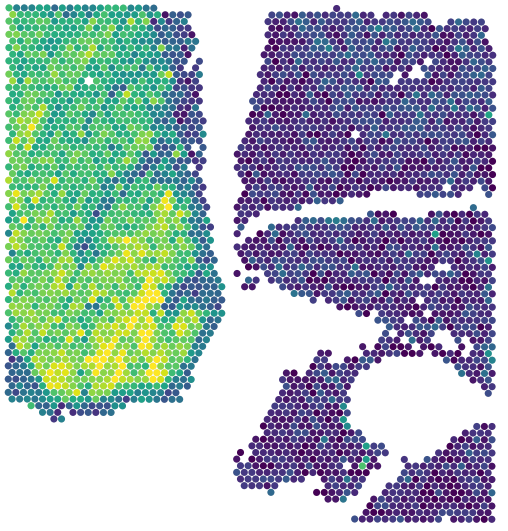
0 0.144 UCell score

Tgf Beta Signaling



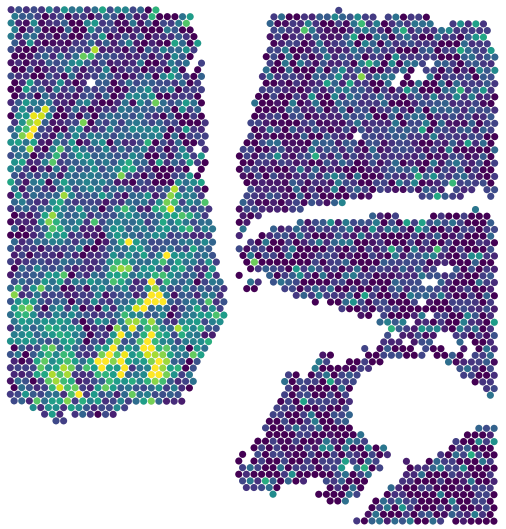
0 0.232 UCell score

Cholesterol Homeostasis



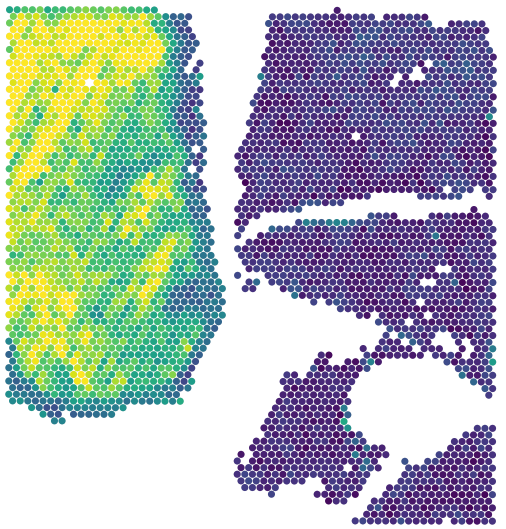
0 0.21 UCell score

Cholesterol Biosynthesis



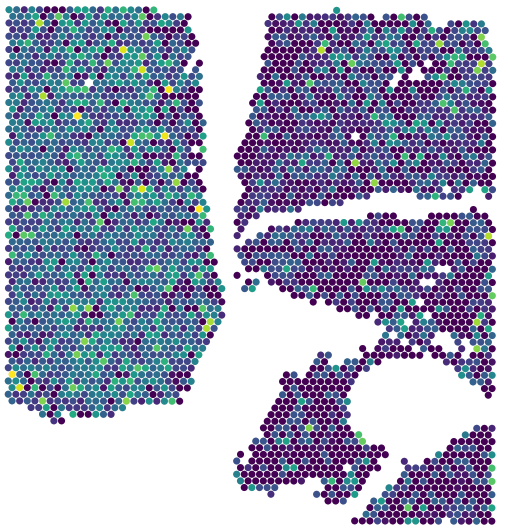
0 0.166 UCell score

Extracellular Matrix Organization



0 0.247 UCell score

Co Inhibition By Ctla4

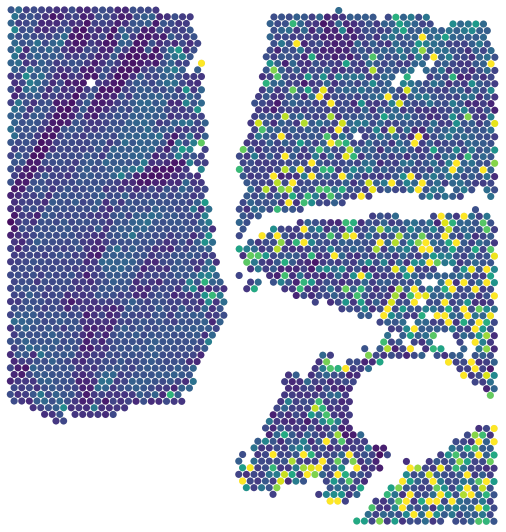


0 0.151 UCell score

GSE319536_BC08 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

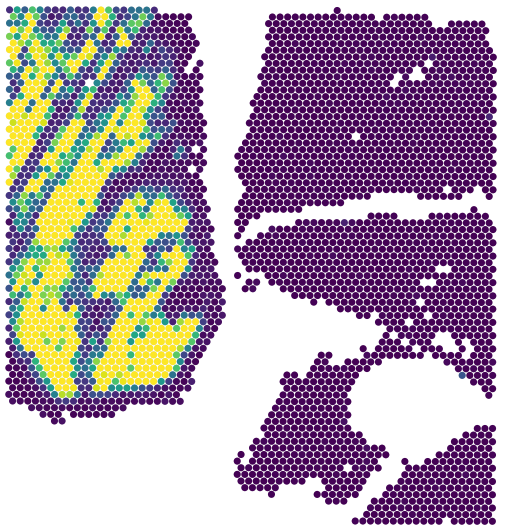
Treg composition



0 0.102

Fraction of all 30

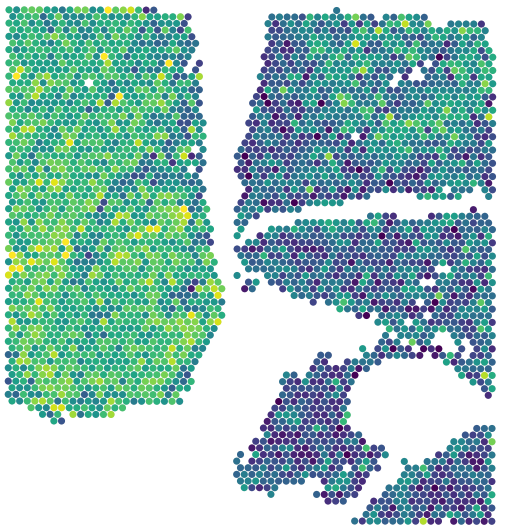
Treg estimated abundance



0 2.85

Posterior mean estimate

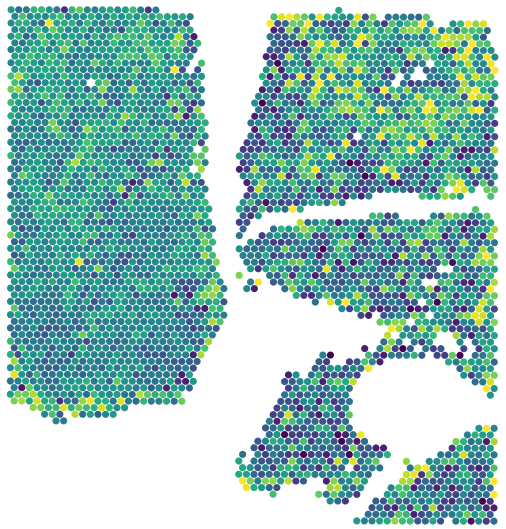
Treg Resting Vs Activated Dn



0 0.0778

UCell score

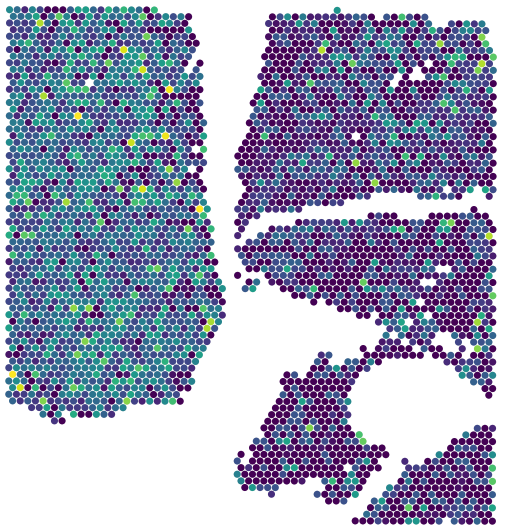
Treg Resting Vs Activated Up



0 0.0609

UCell score

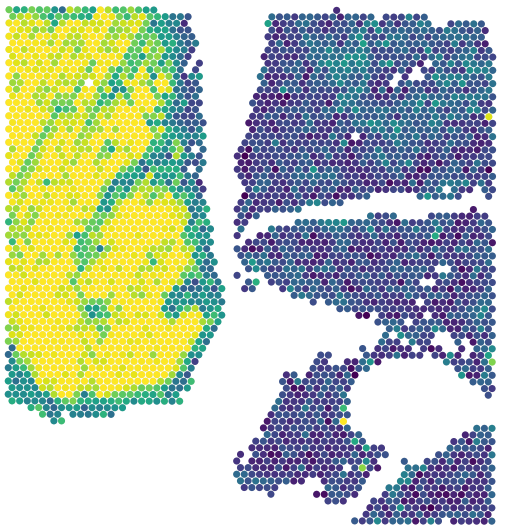
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

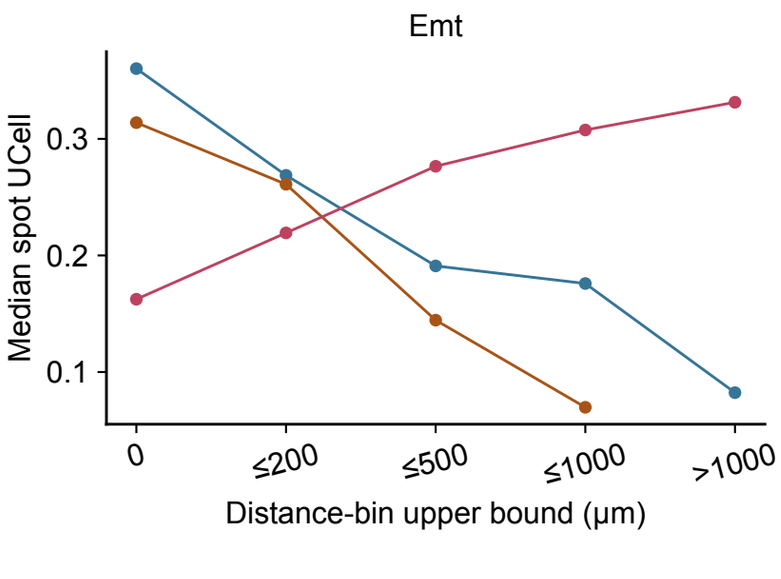


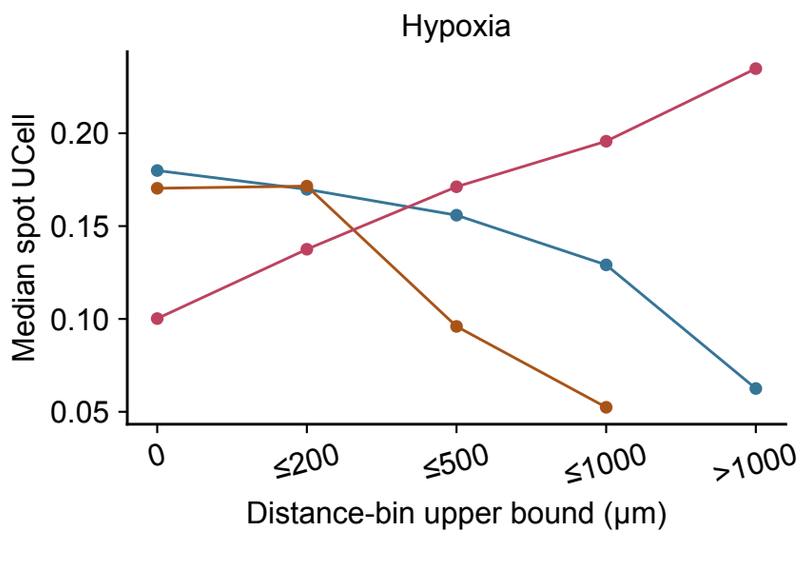
0 0.144

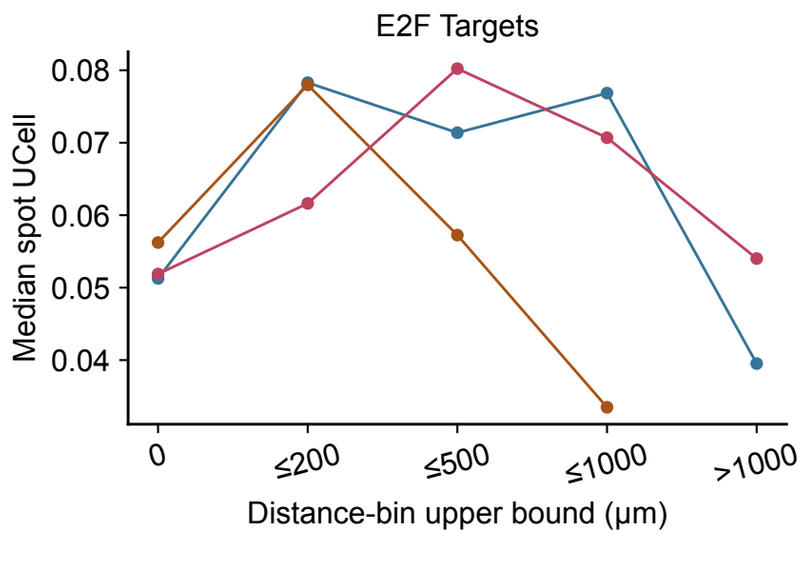
UCell score

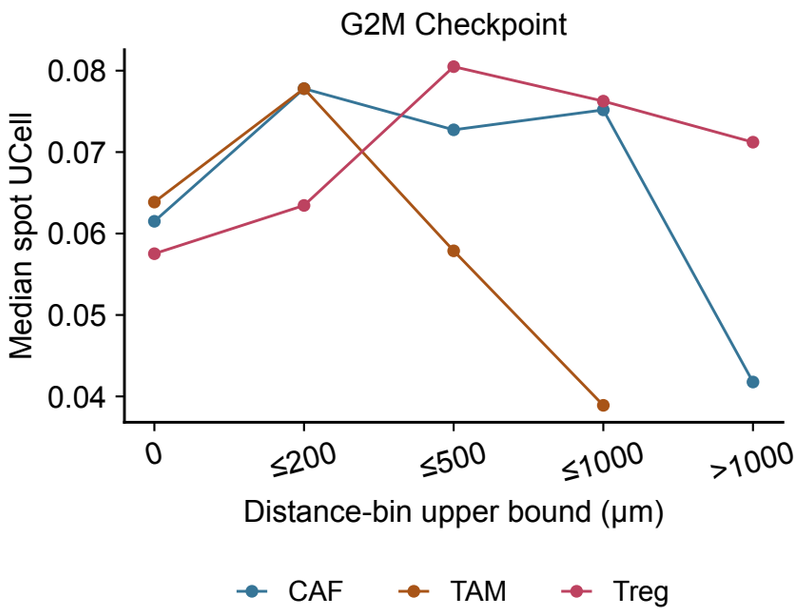
GSE319536_BC08 | Programs by target distance

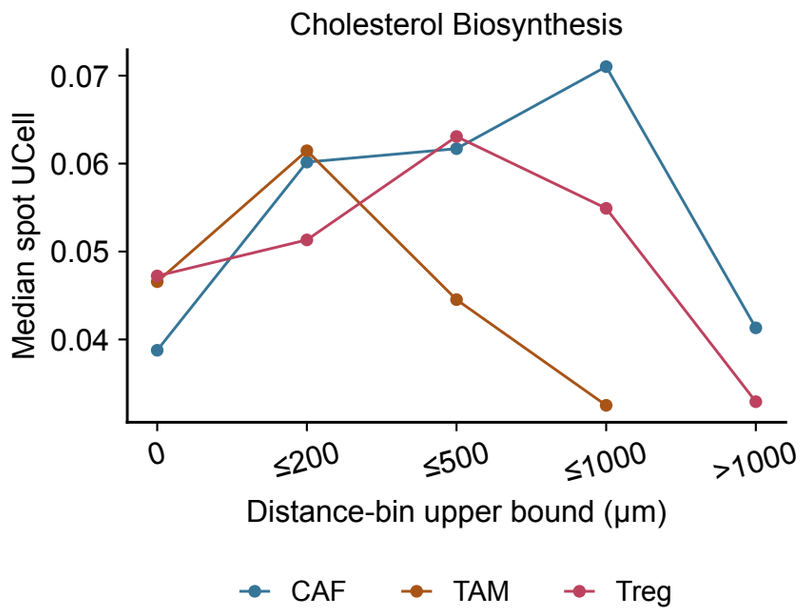
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

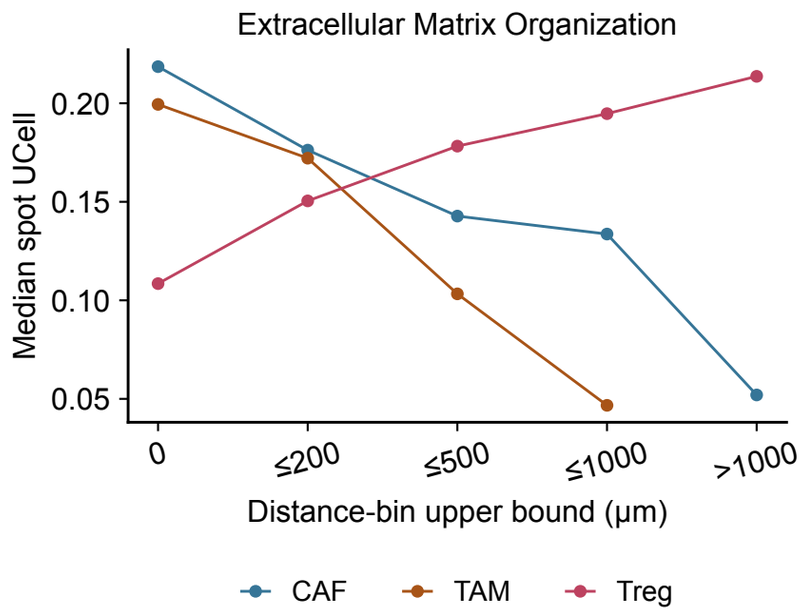






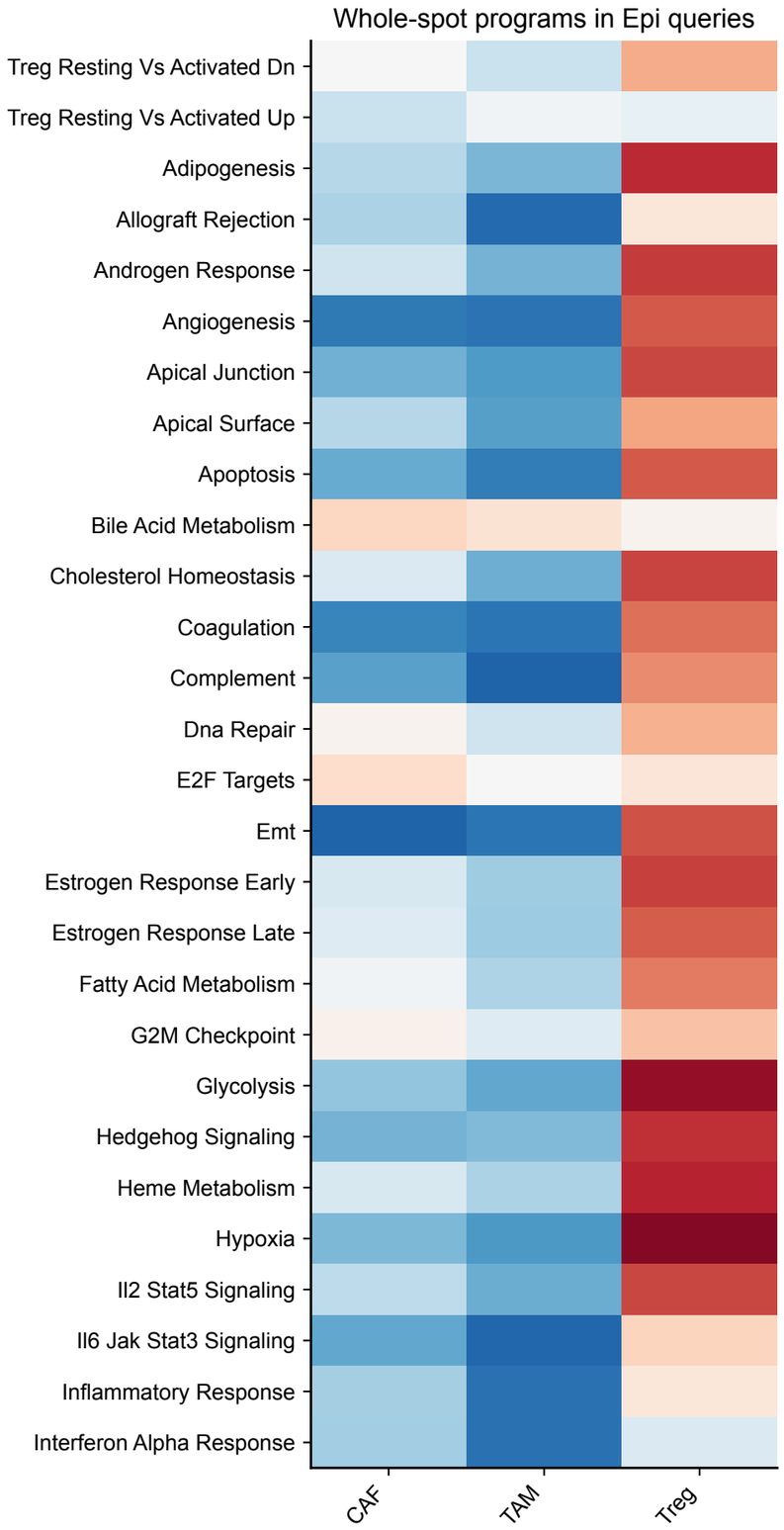


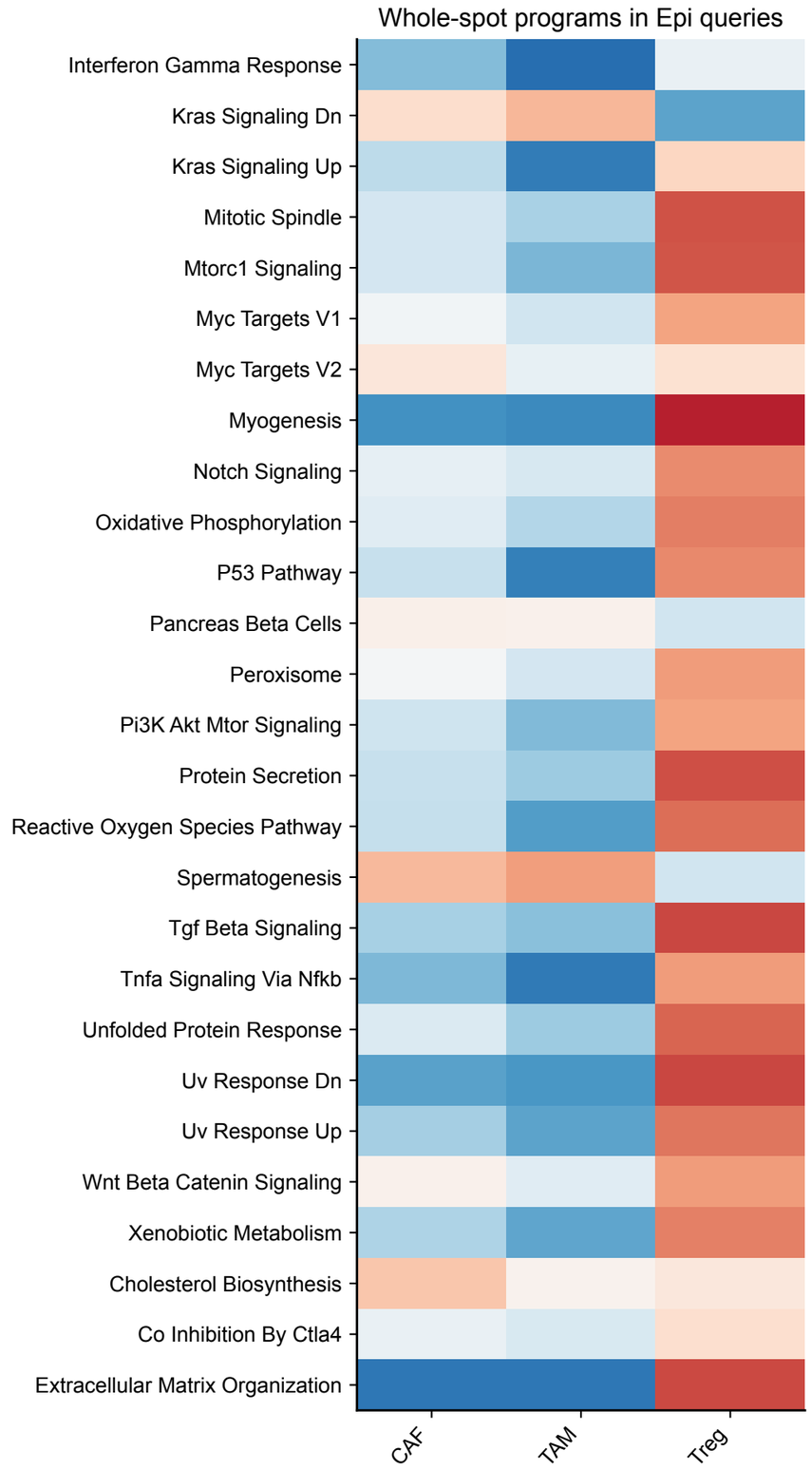




GSE319536_BC08 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

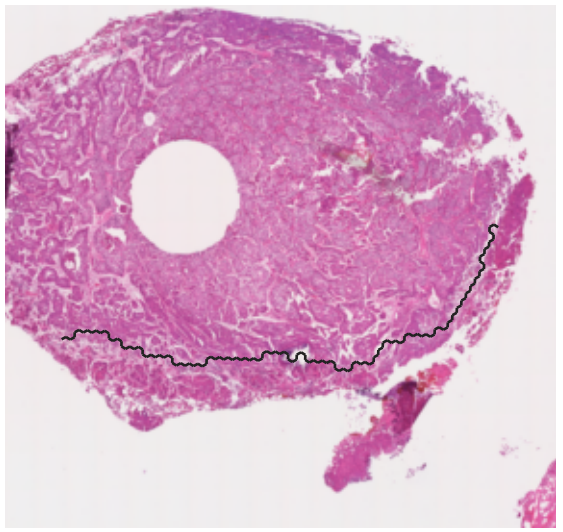




GSE319536_BC10 | Tissue context

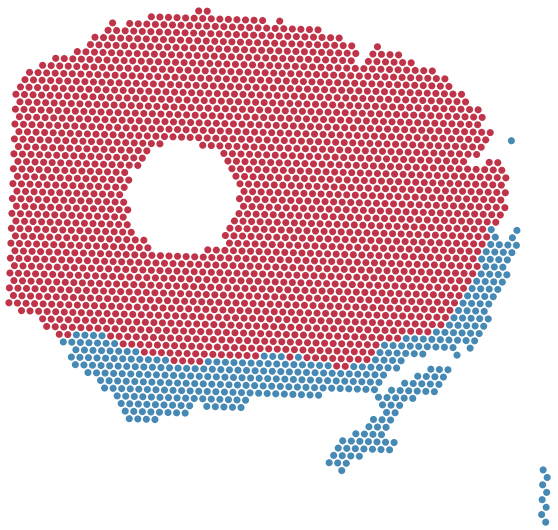
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



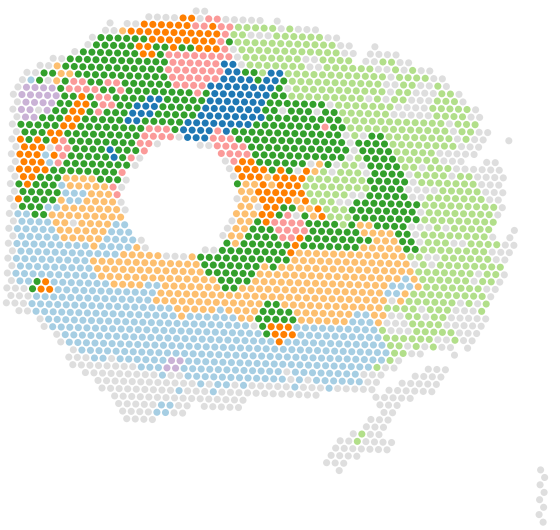
Black line: observed candidate outer edge

Final candidate location



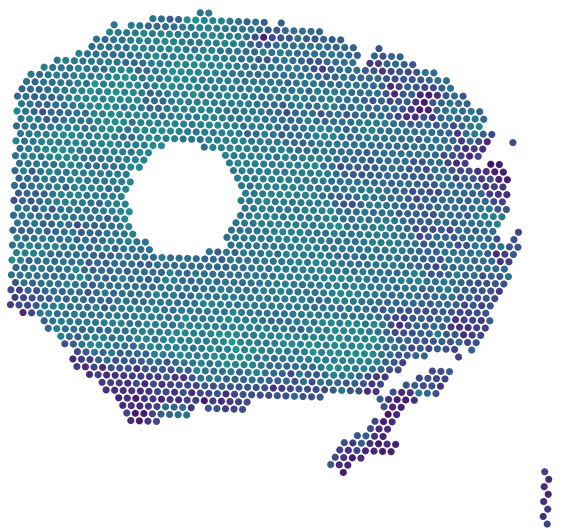
inside outside

Accepted R12 (original 0–8)



0 1 2 3 4 5 6 8

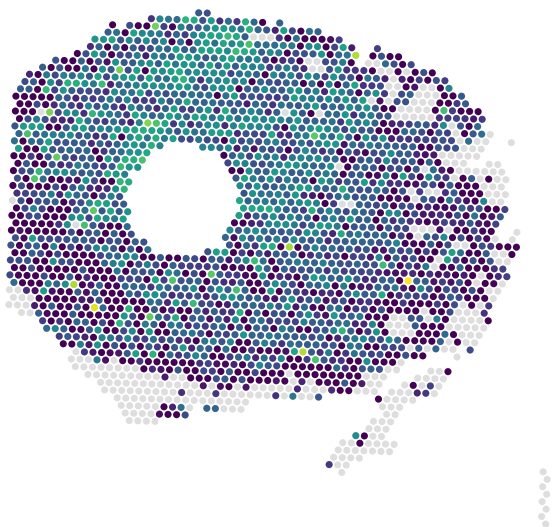
Epithelial support



0 1

Fraction of all 30

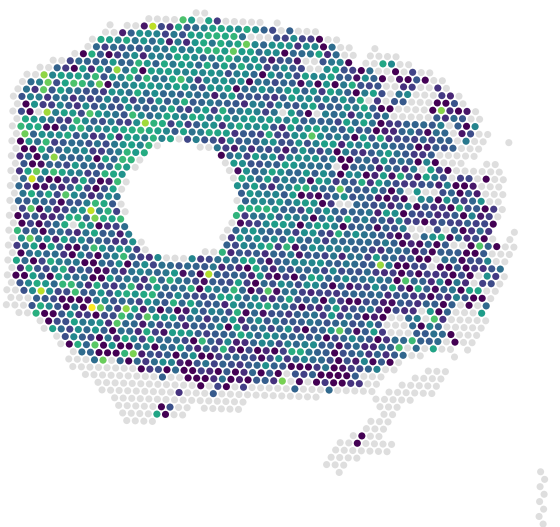
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



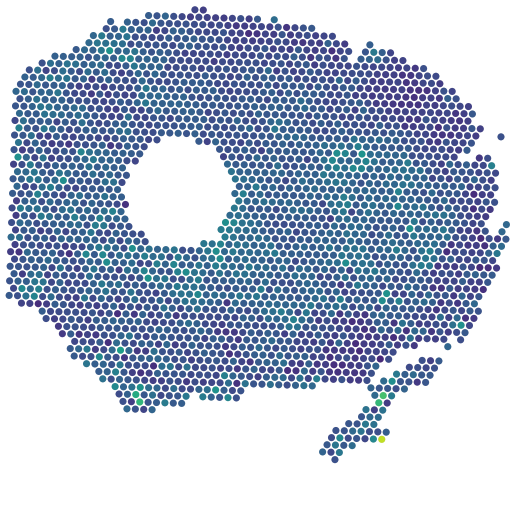
0 0.14

Matched-model AUCell

GSE319536_BC10 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

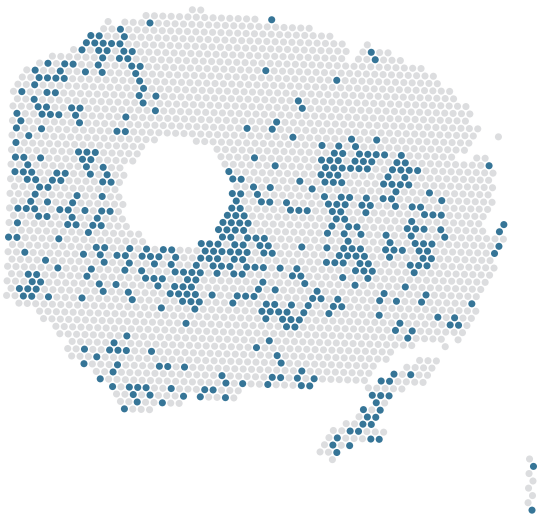
CAF composition



0 0.418

Fraction of all 30

CAF P80 targets



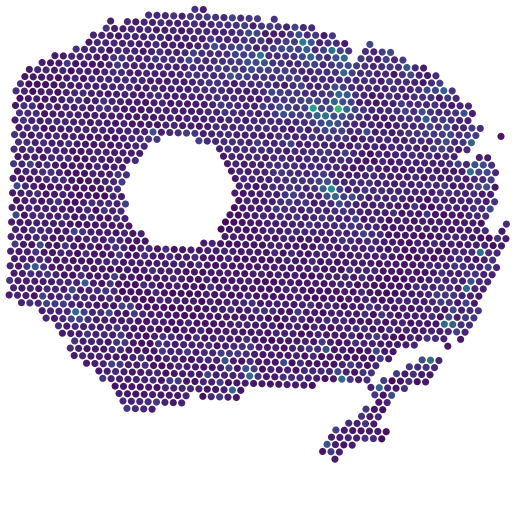
False True

CAF distance

0 3.1e+03

Nearest spot center (µm)

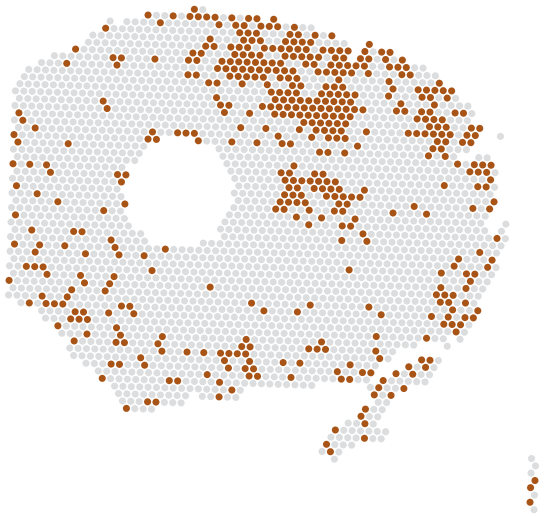
TAM composition



0 0.291

Fraction of all 30

TAM P80 targets



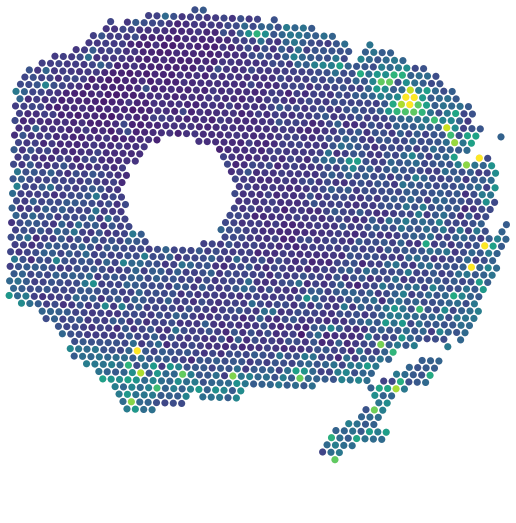
False True

TAM distance

0 1.13e+03

Nearest spot center (µm)

Treg composition



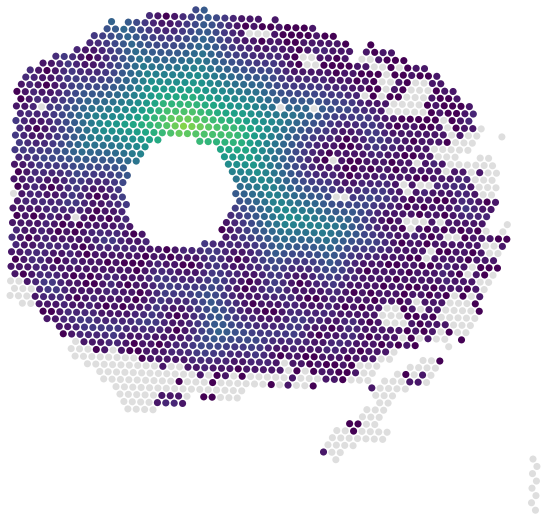
0 0.102

Fraction of all 30

Treg P80 targets

False True

Treg distance

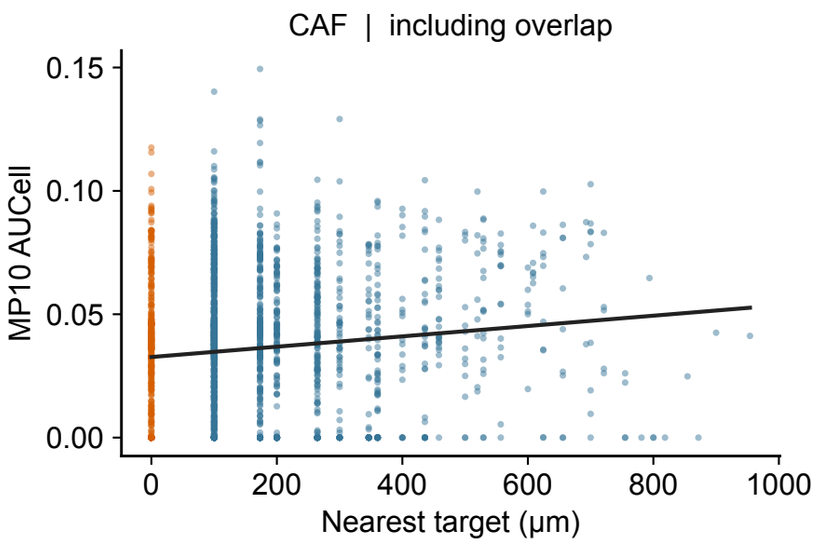


0 1.61e+03

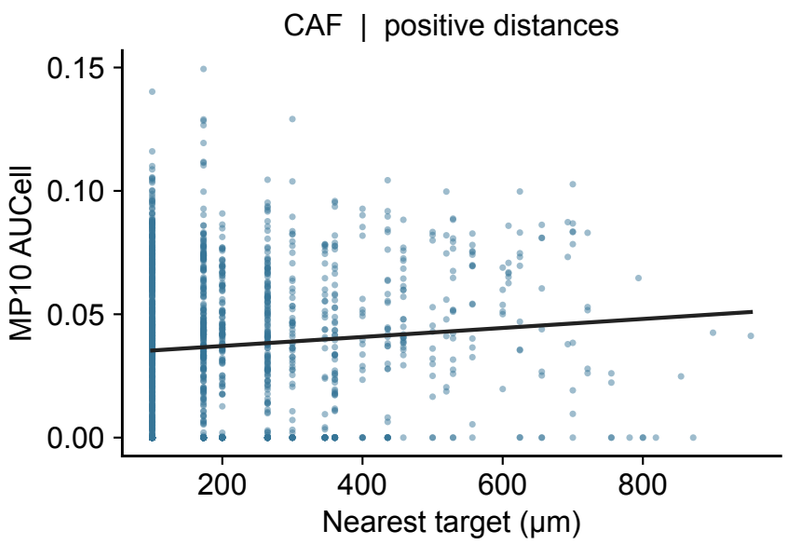
Nearest spot center (µm)

GSE319536_BC10 | MP10 versus nearest target

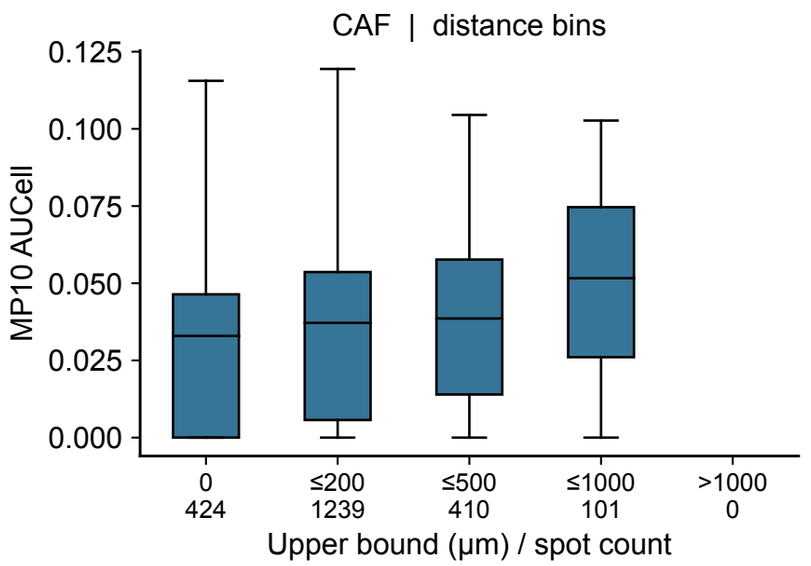
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



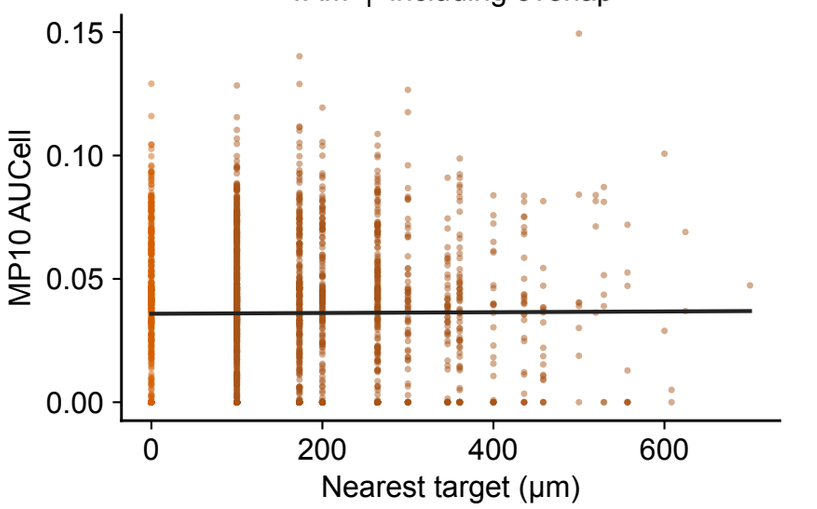
r +0.11 ρ +0.10
n 2174/2174 queries • 500 targets



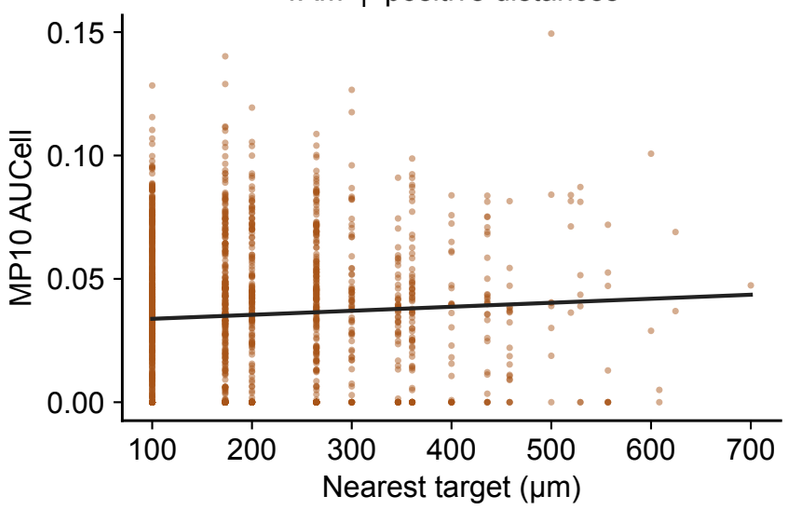
r +0.09 ρ +0.07
n 1750/2174 queries • 500 targets



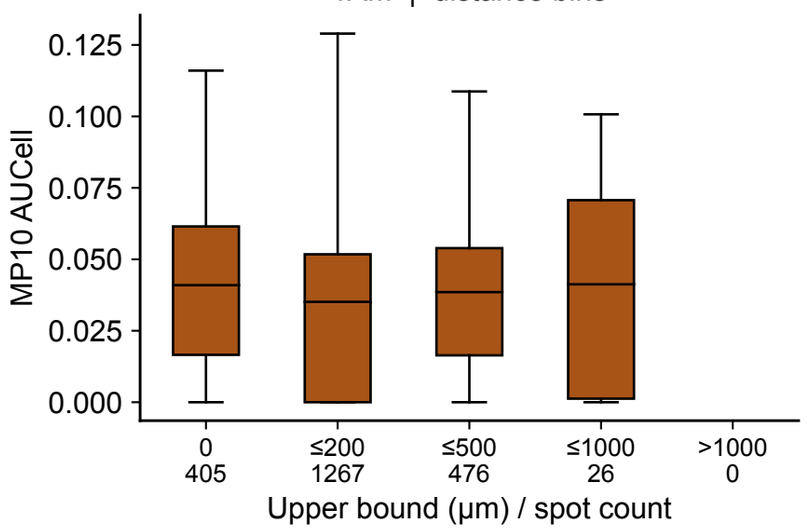
Orange = zero distance
Line = within-section linear fit



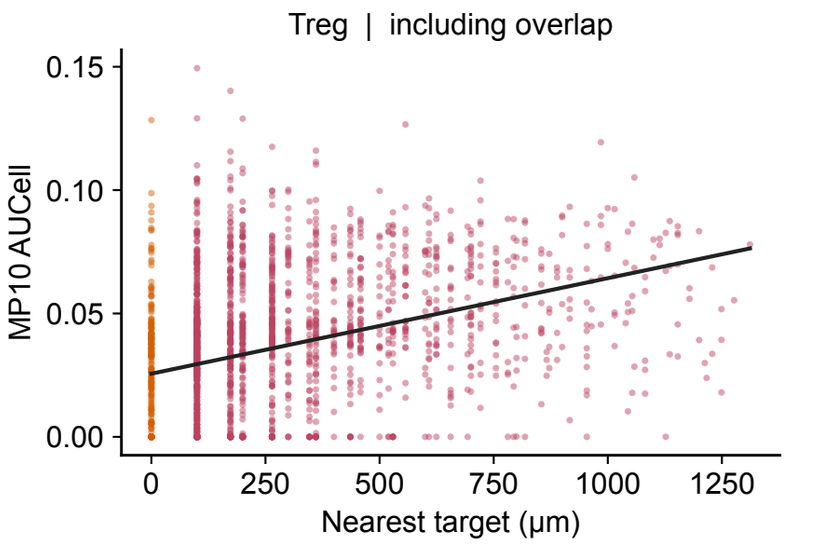
r +0.01 ρ +0.01
n 2174/2174 queries • 500 targets



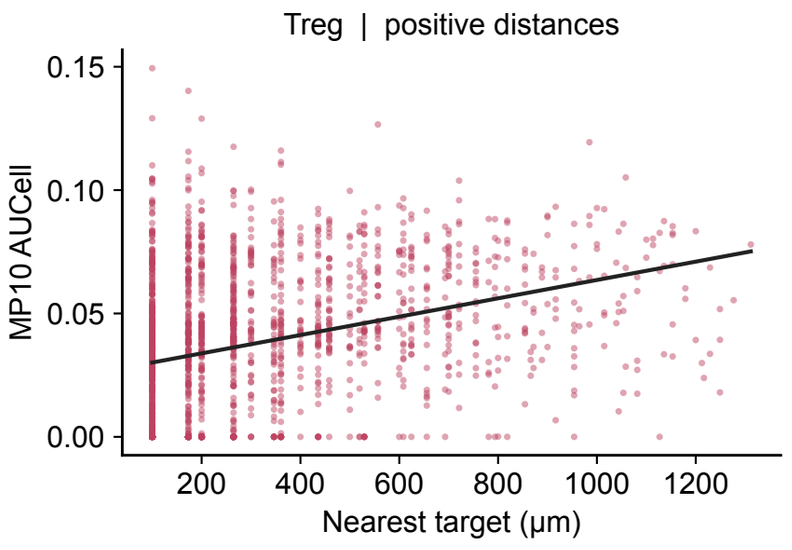
r +0.06 ρ +0.08
n 1769/2174 queries • 500 targets



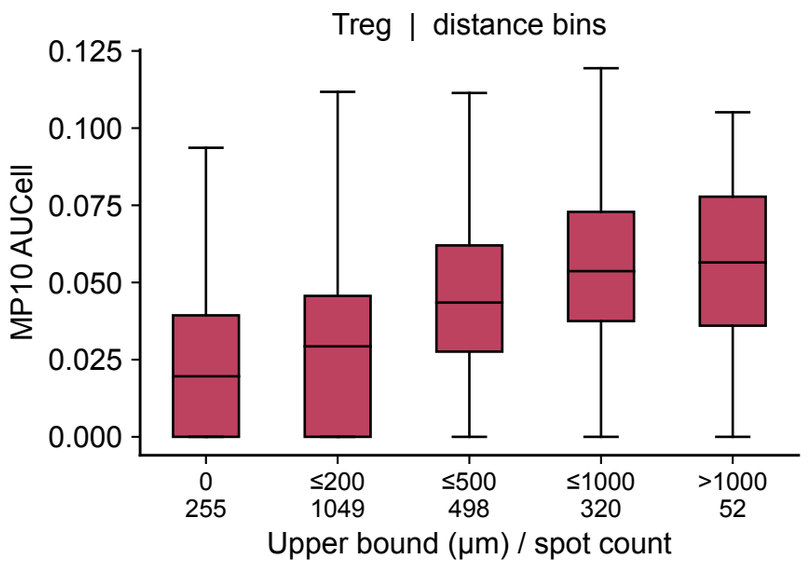
Orange = zero distance
Line = within-section linear fit



r +0.35 ρ +0.39
n 2174/2174 queries • 500 targets



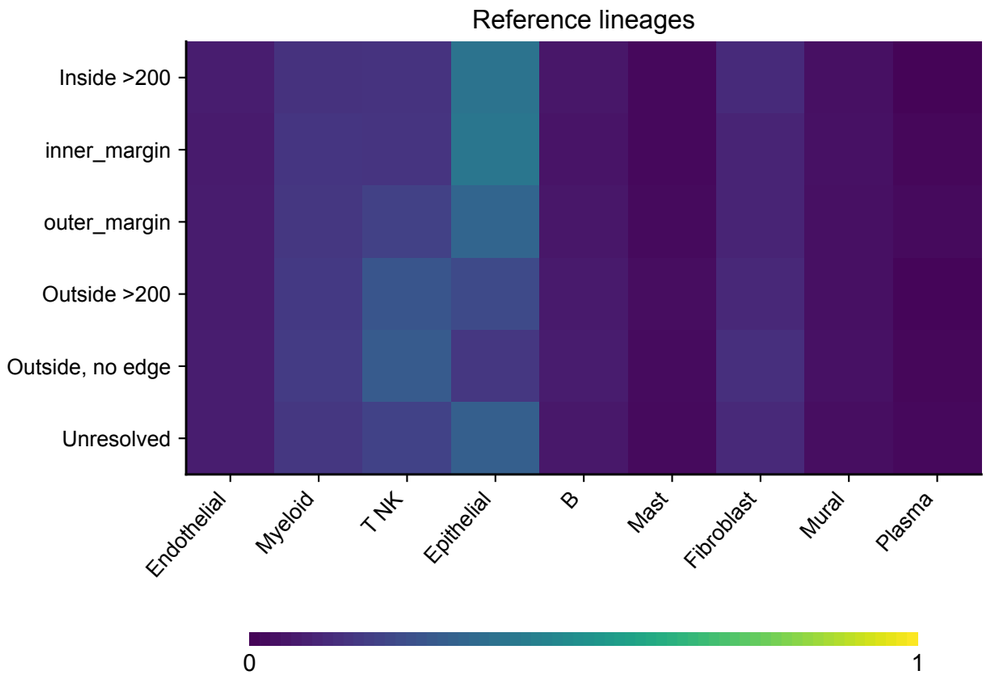
r +0.33 ρ +0.38
n 1919/2174 queries • 500 targets

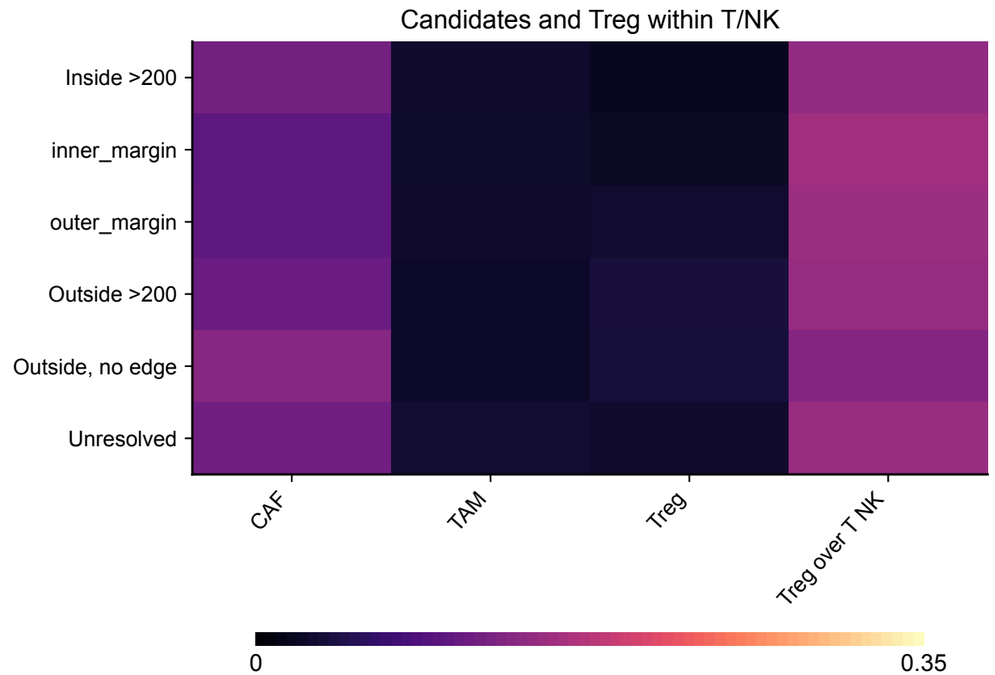


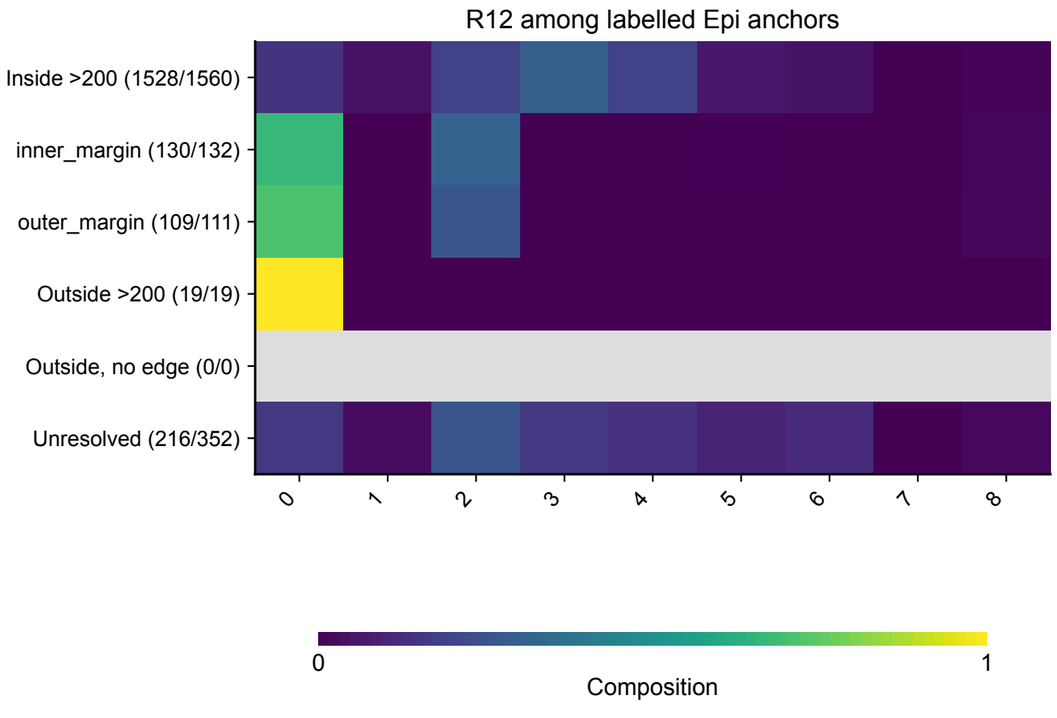
Orange = zero distance
Line = within-section linear fit

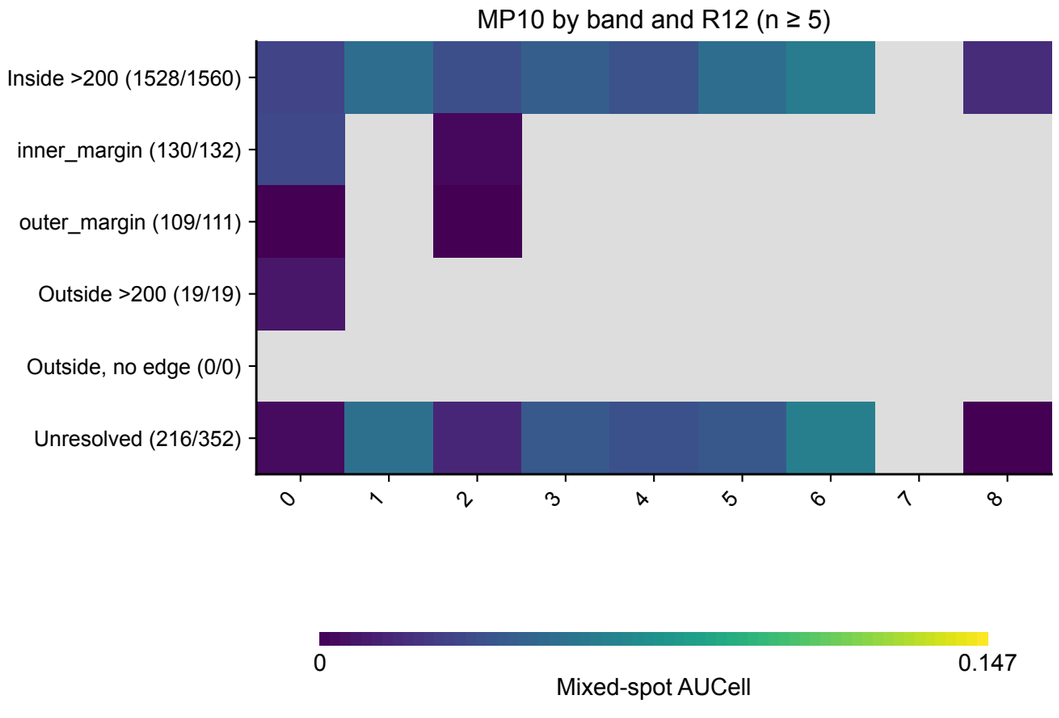
GSE319536_BC10 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





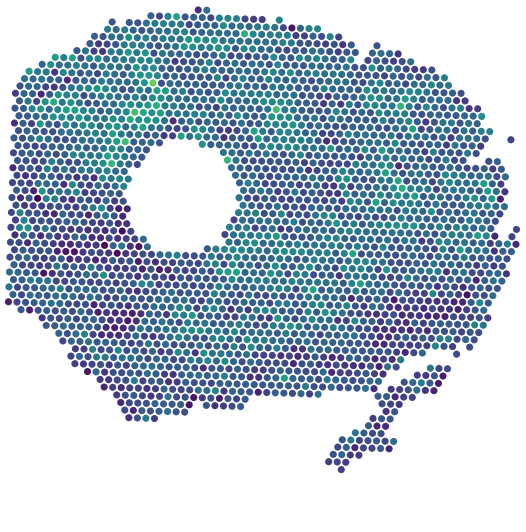




GSE319536_BC10 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

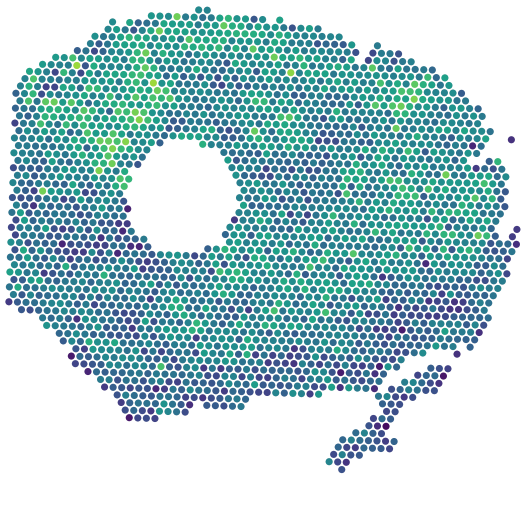
E2F Targets



0 0.152

UCell score

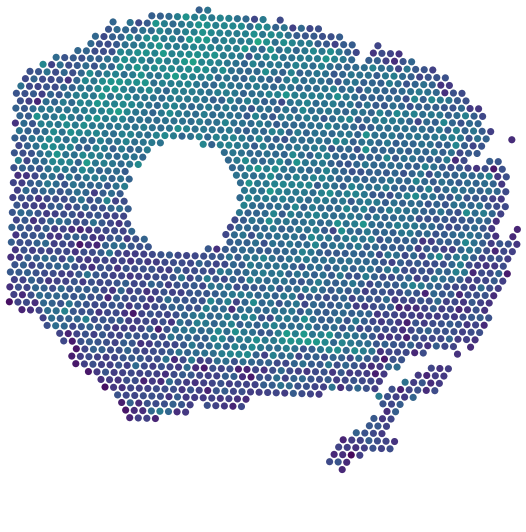
G2M Checkpoint



0 0.134

UCell score

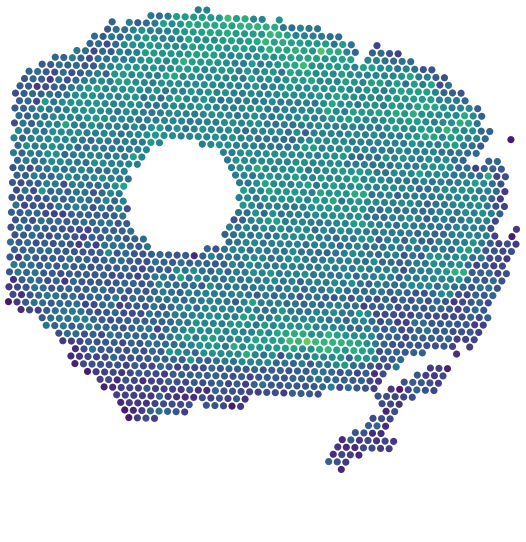
Hypoxia



0 0.211

UCell score

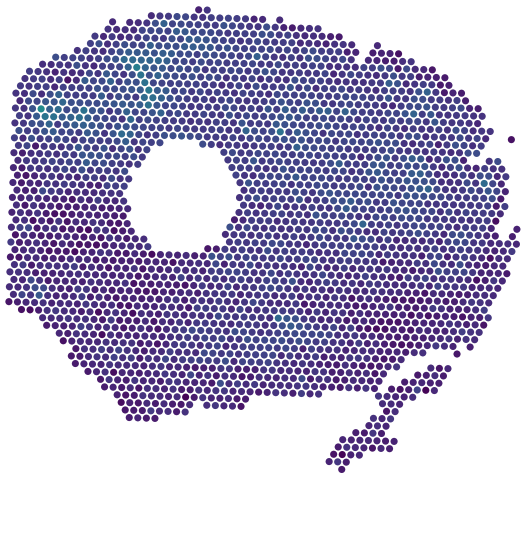
Tnfa Signaling Via Nfkb



0 0.275

UCell score

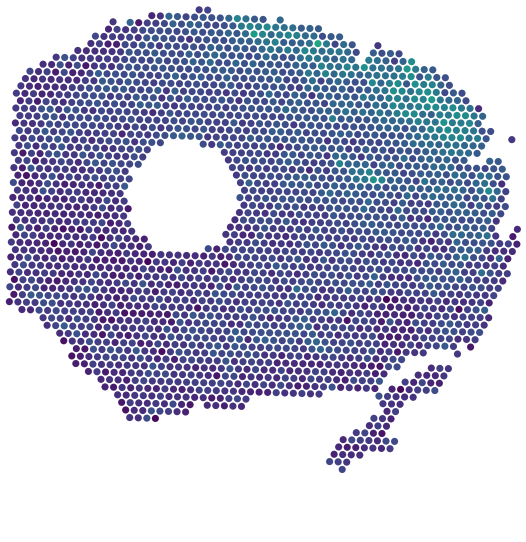
Emt



0 0.386

UCell score

Interferon Gamma Response



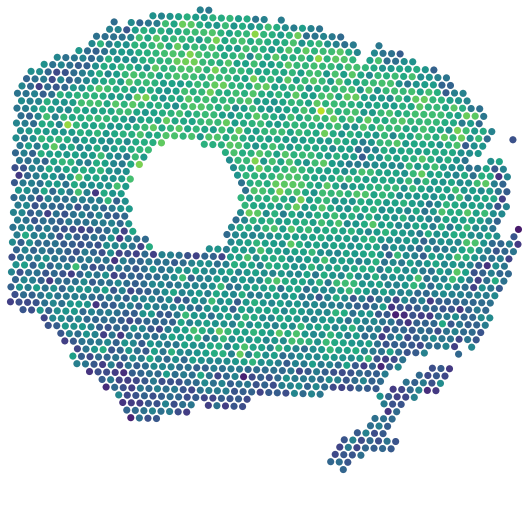
0 0.275

UCell score

GSE319536_BC10 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

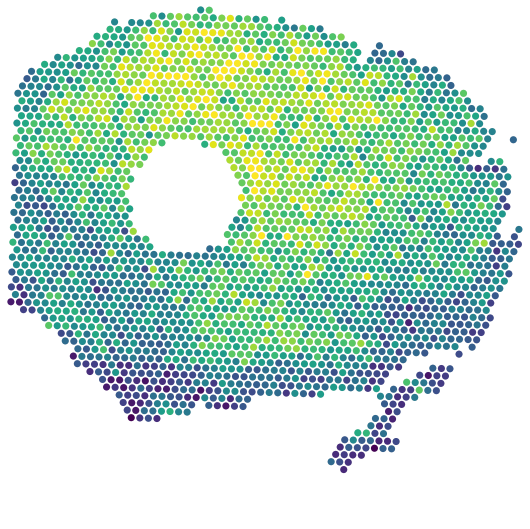
Il2 Stat5 Signaling



0 0.144

UCell score

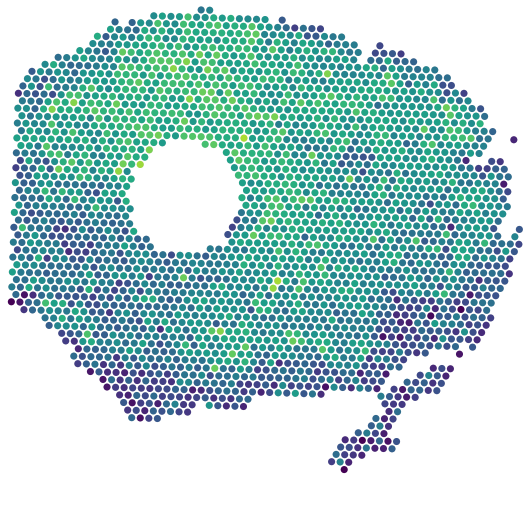
Tgf Beta Signaling



0 0.232

UCell score

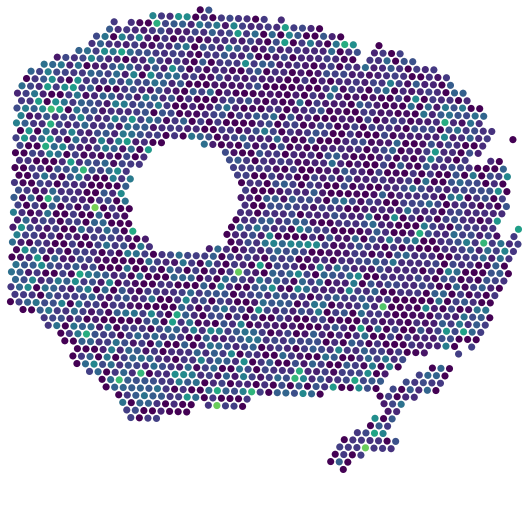
Cholesterol Homeostasis



0 0.21

UCell score

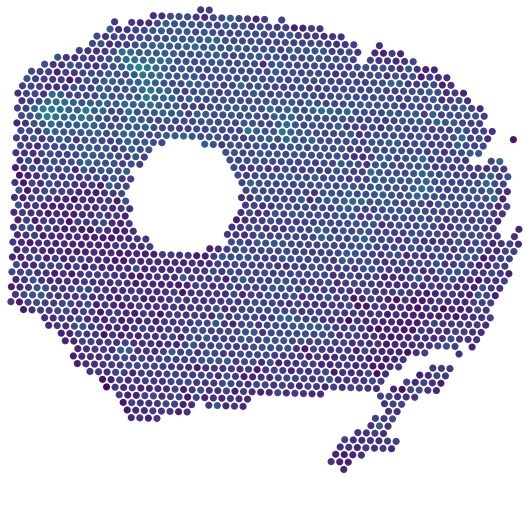
Cholesterol Biosynthesis



0 0.166

UCell score

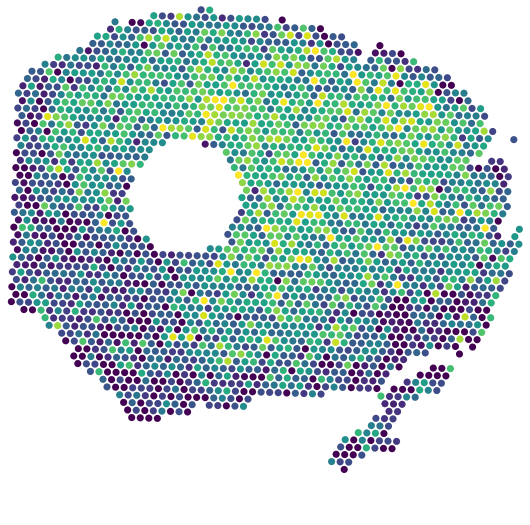
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



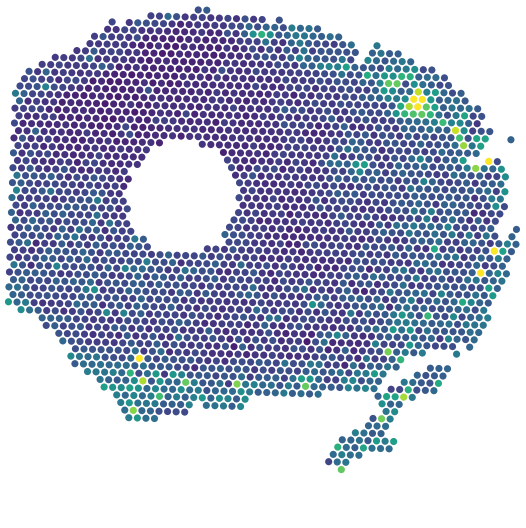
0 0.151

UCell score

GSE319536_BC10 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

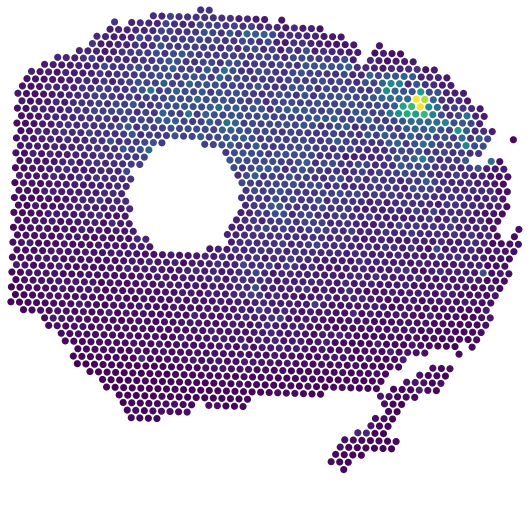
Treg composition



0 0.102

Fraction of all 30

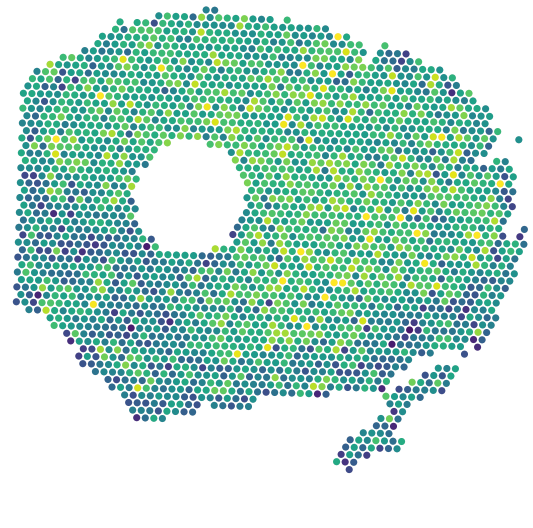
Treg estimated abundance



0 2.85

Posterior mean estimate

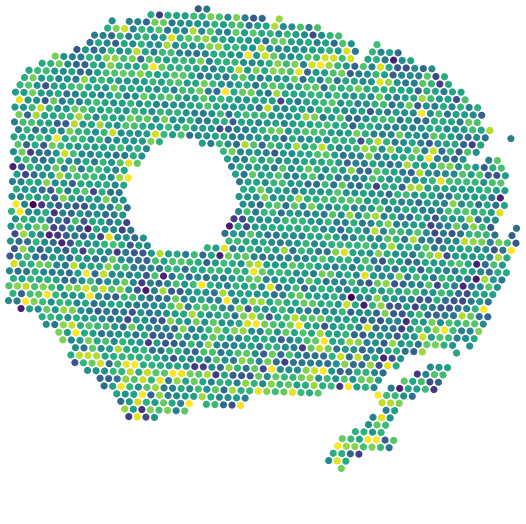
Treg Resting Vs Activated Dn



0 0.0778

UCell score

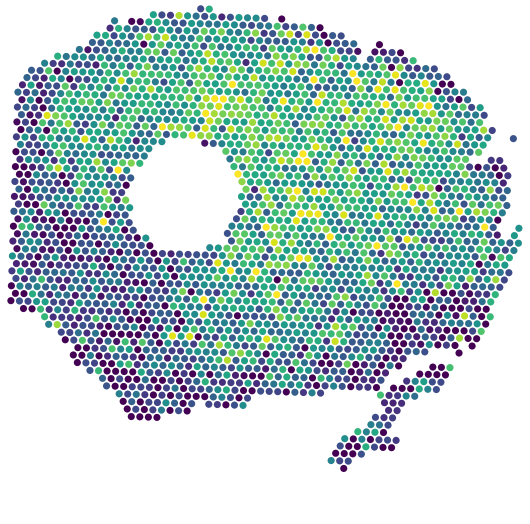
Treg Resting Vs Activated Up



0 0.0609

UCell score

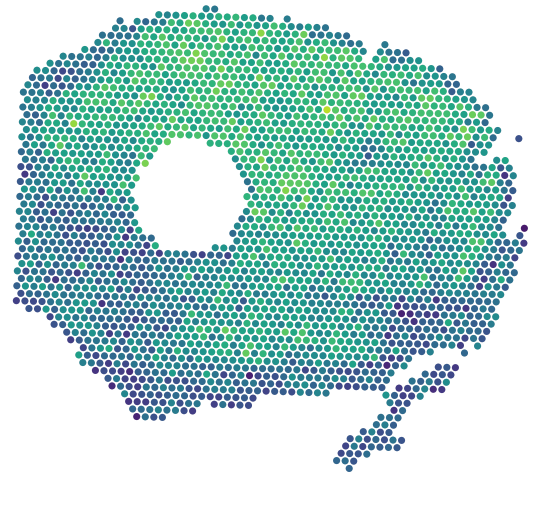
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

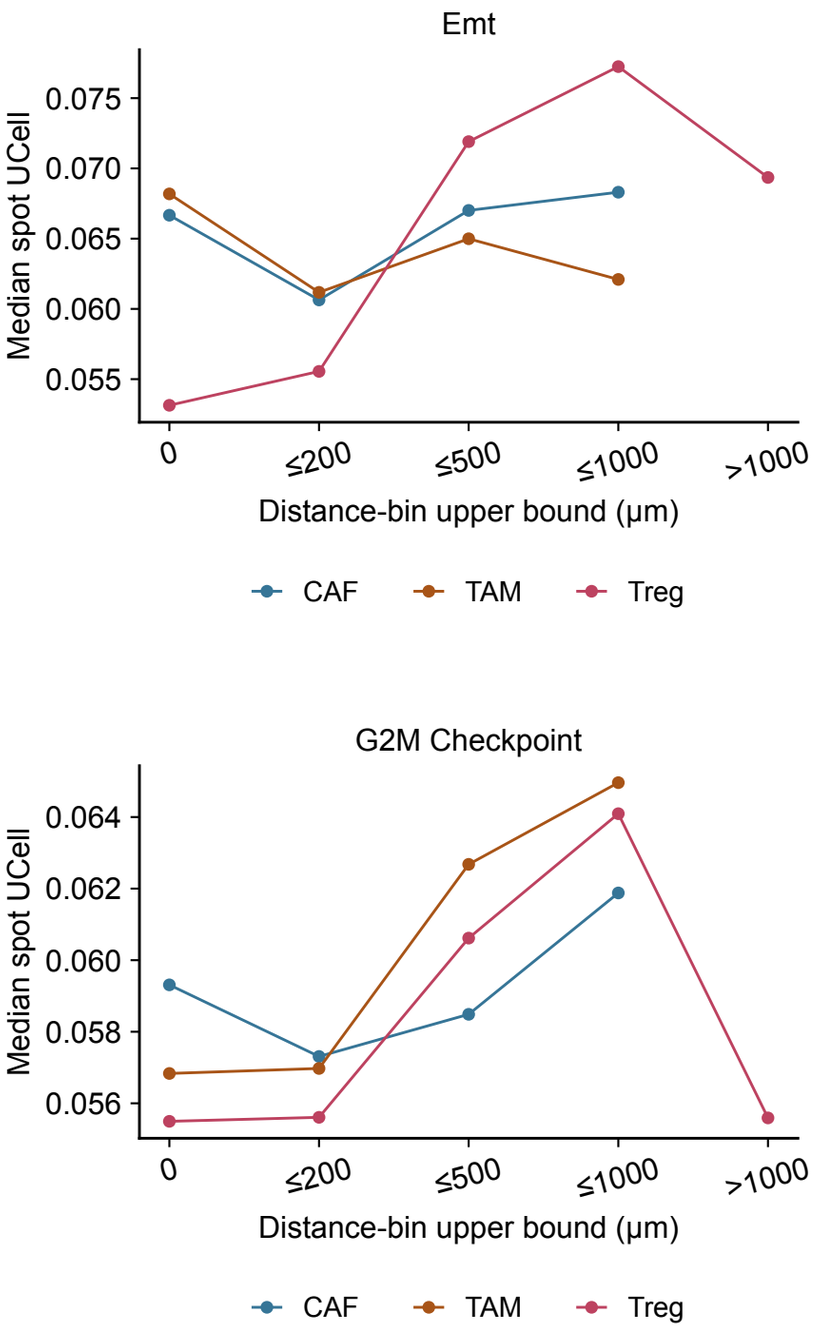


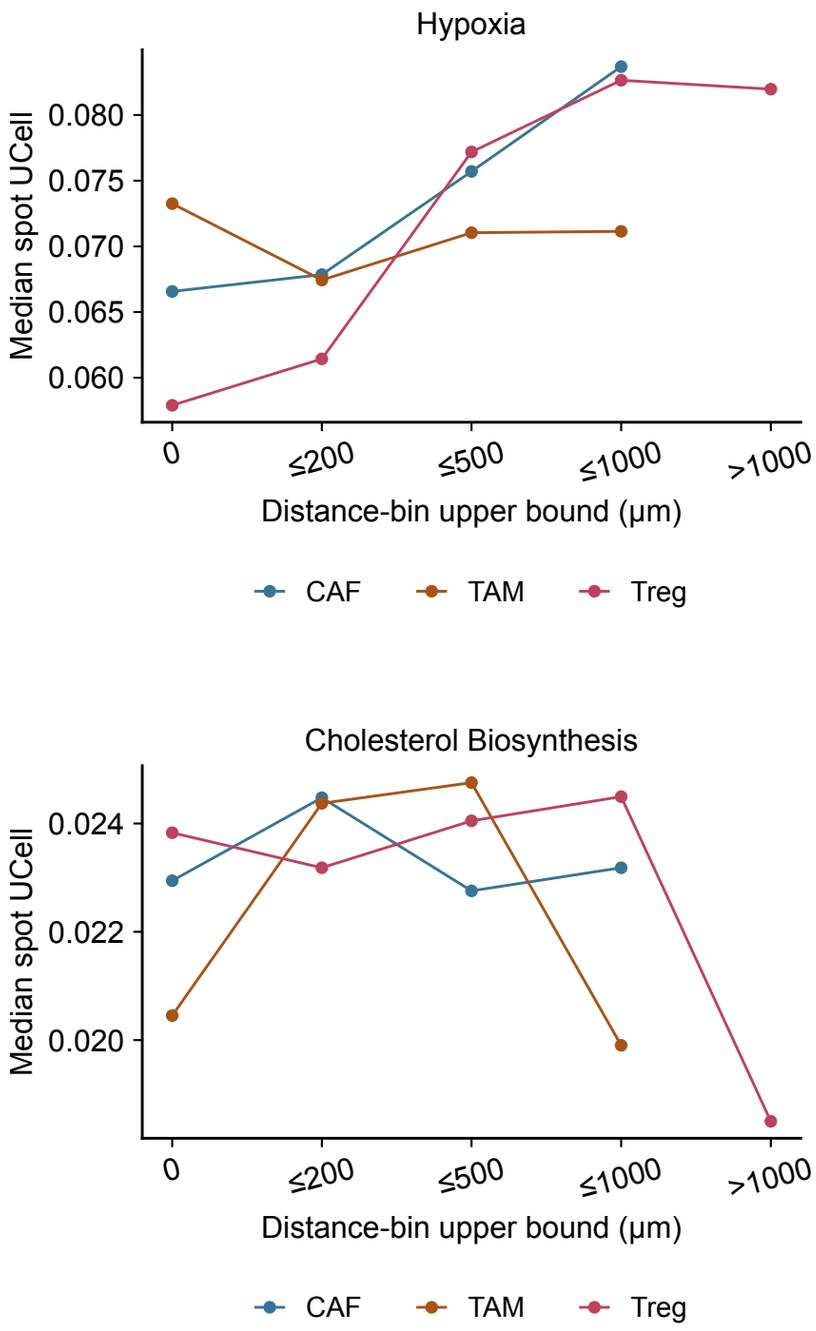
0 0.144

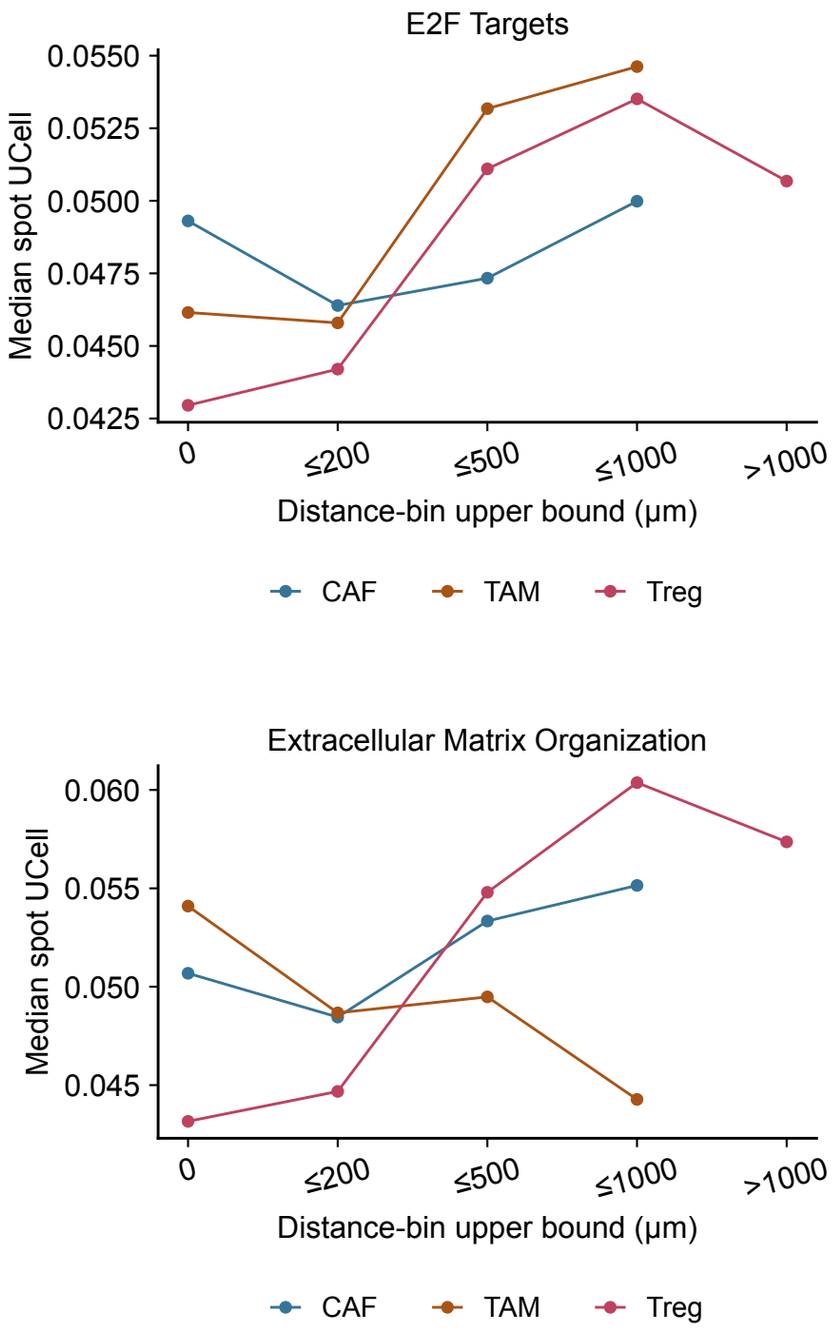
UCell score

GSE319536_BC10 | Programs by target distance

Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

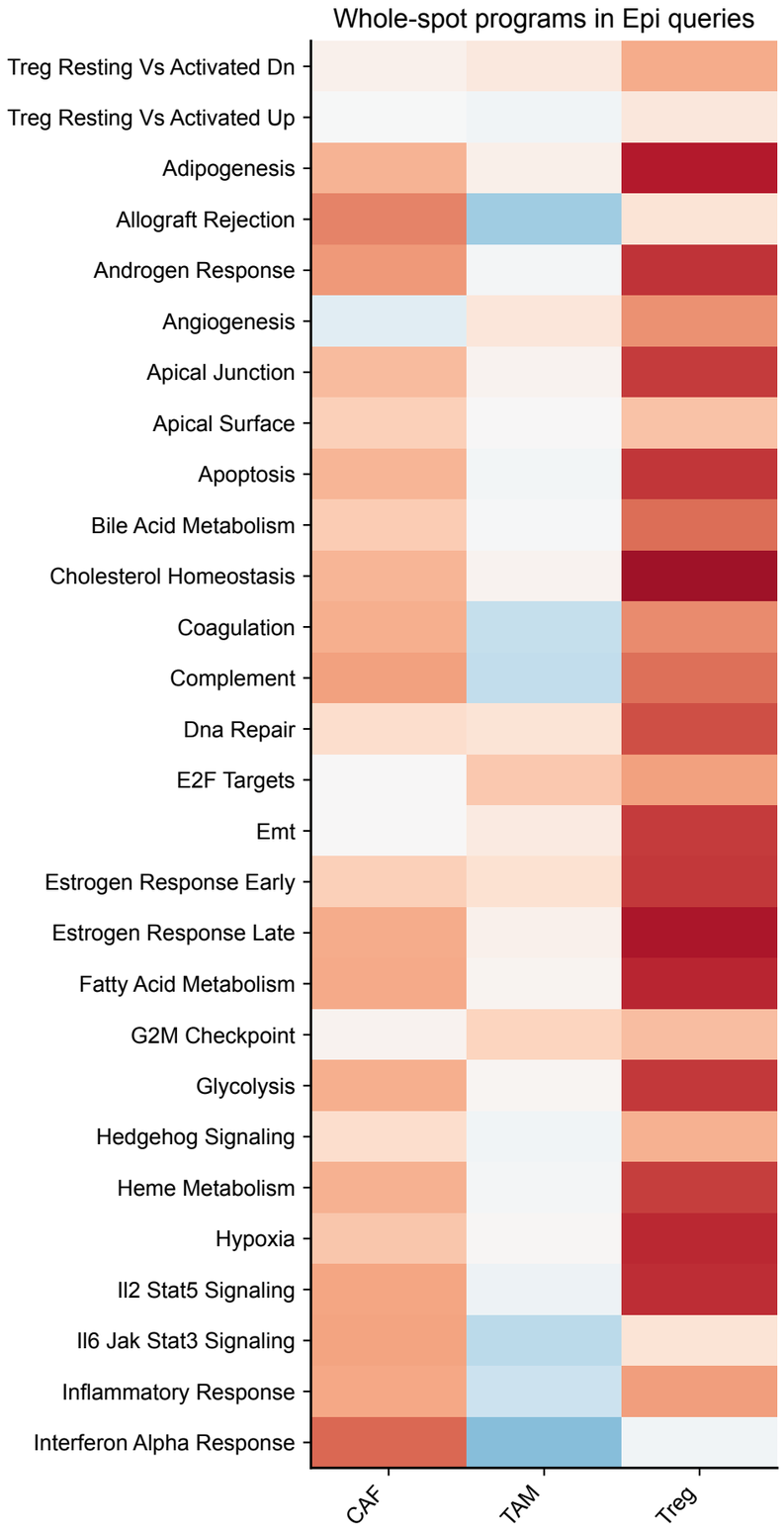


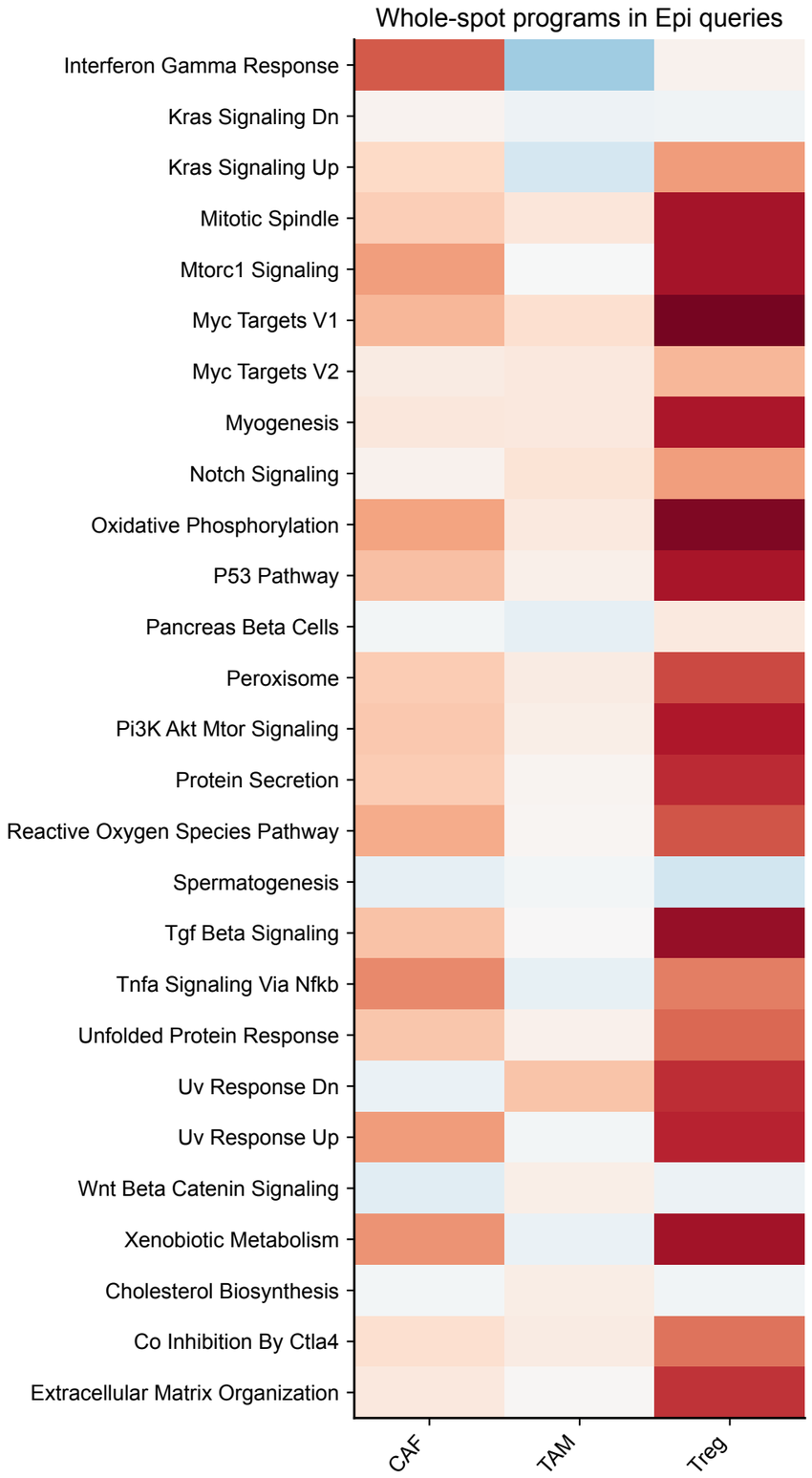




GSE319536_BC10 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

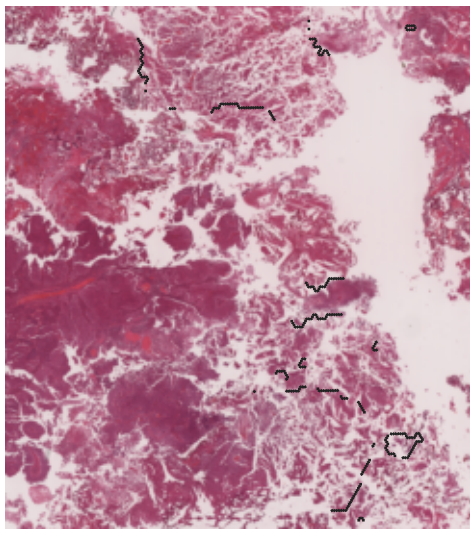




GSE319536_BC11 | Tissue context

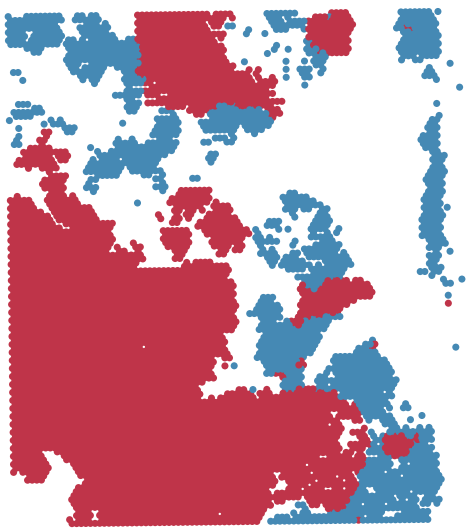
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



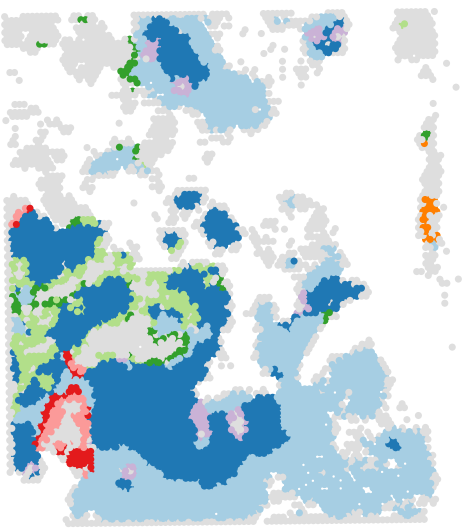
Black line: observed candidate outer edge

Final candidate location



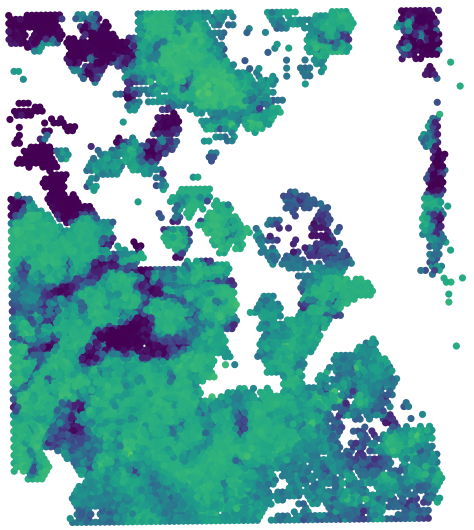
inside outside

Accepted R12 (original 0–8)



0 2 5 7 8
1 3 6

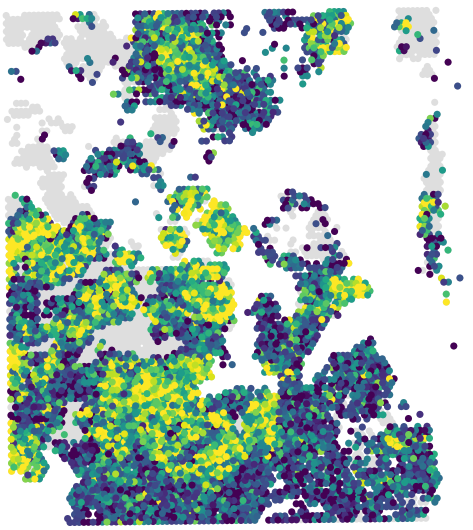
Epithelial support



0 1

Fraction of all 30

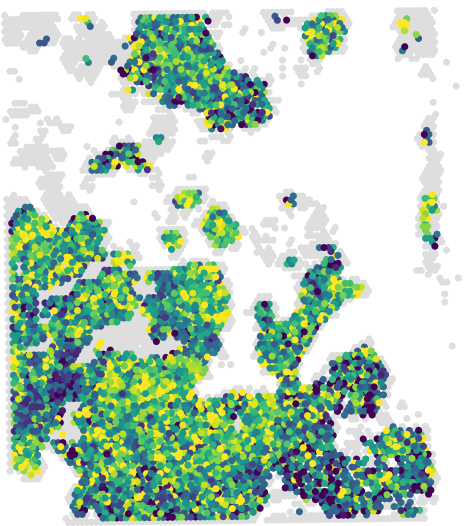
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



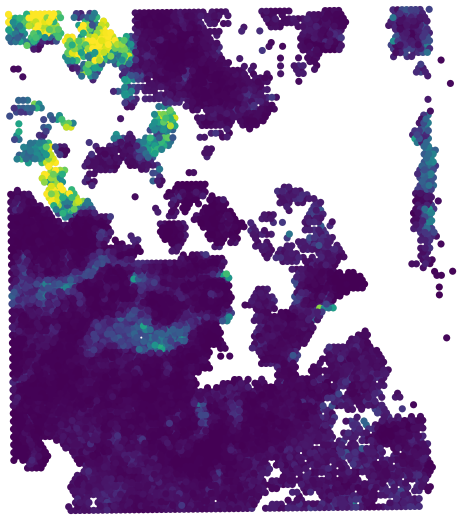
0 0.14

Matched-model AUCell

GSE319536_BC11 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

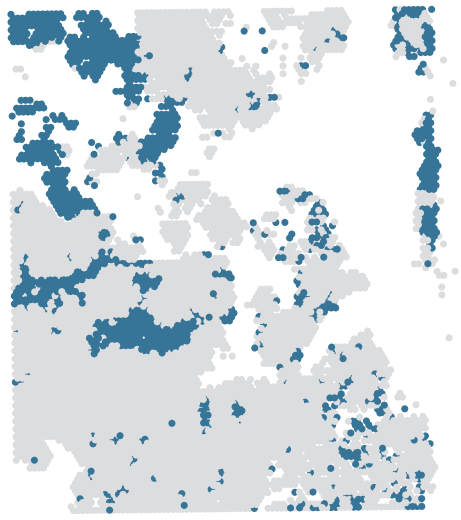
CAF composition



0 0.418

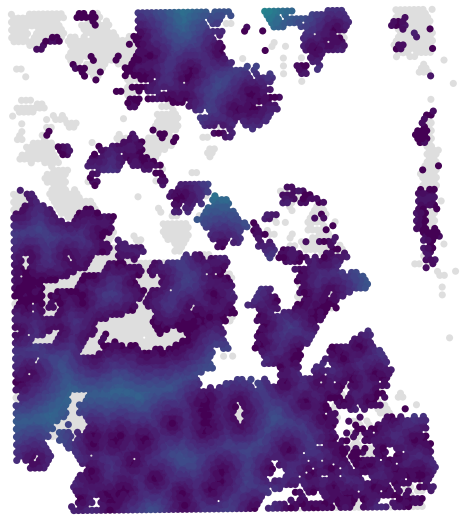
Fraction of all 30

CAF P80 targets



False True

CAF distance



0 3.1e+03

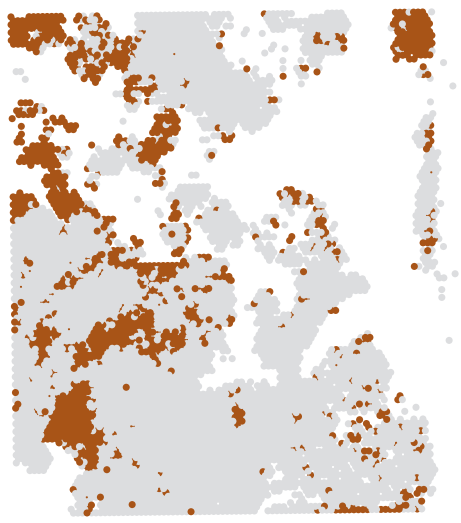
Nearest spot center (μm)

TAM composition

0 0.291

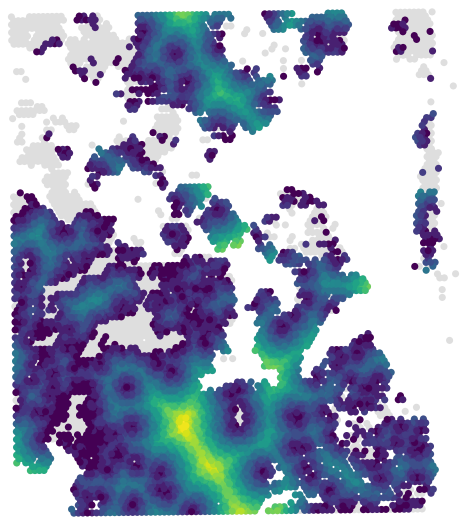
Fraction of all 30

TAM P80 targets



False True

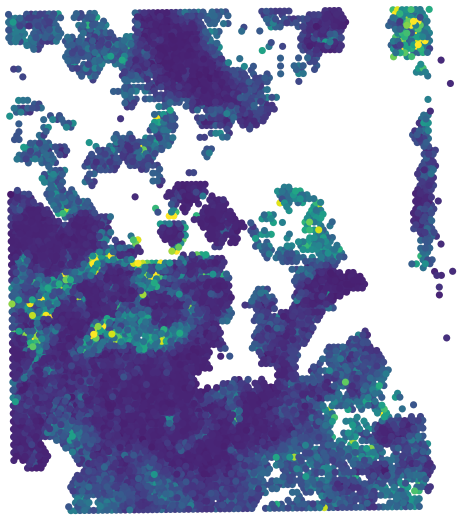
TAM distance



0 1.13e+03

Nearest spot center (μm)

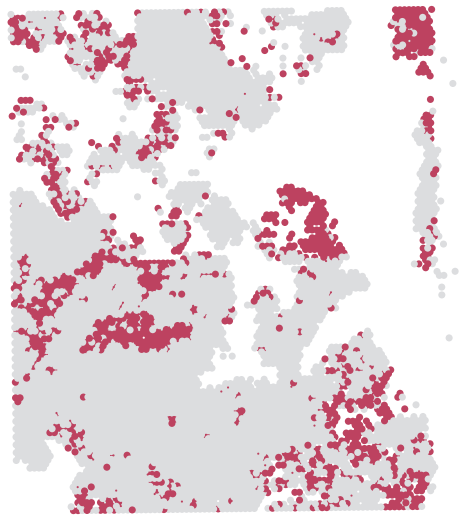
Treg composition



0 0.102

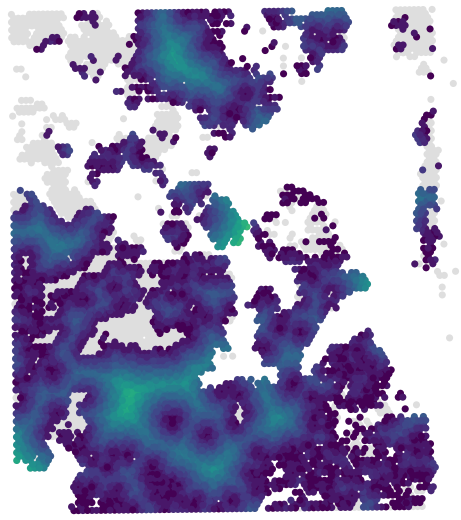
Fraction of all 30

Treg P80 targets



False True

Treg distance

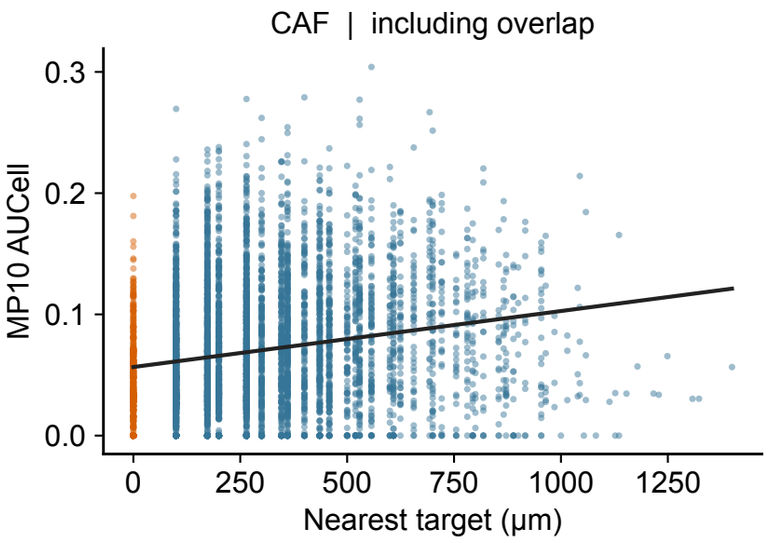


0 1.61e+03

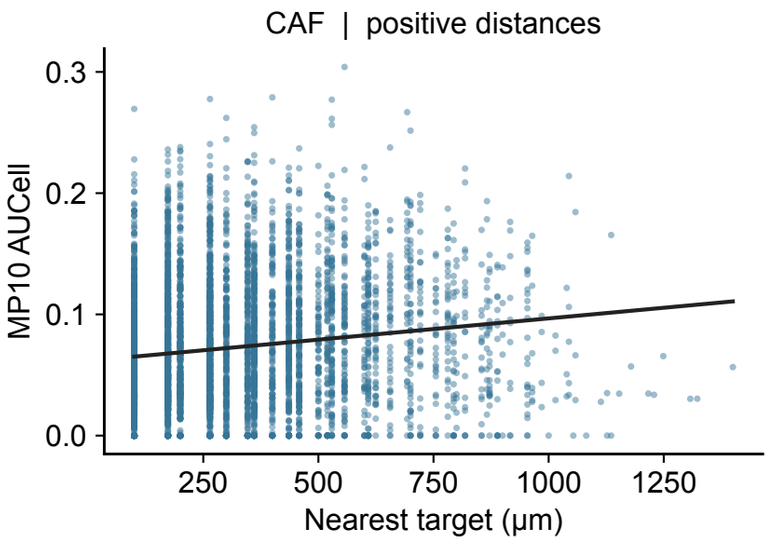
Nearest spot center (μm)

GSE319536_BC11 | MP10 versus nearest target

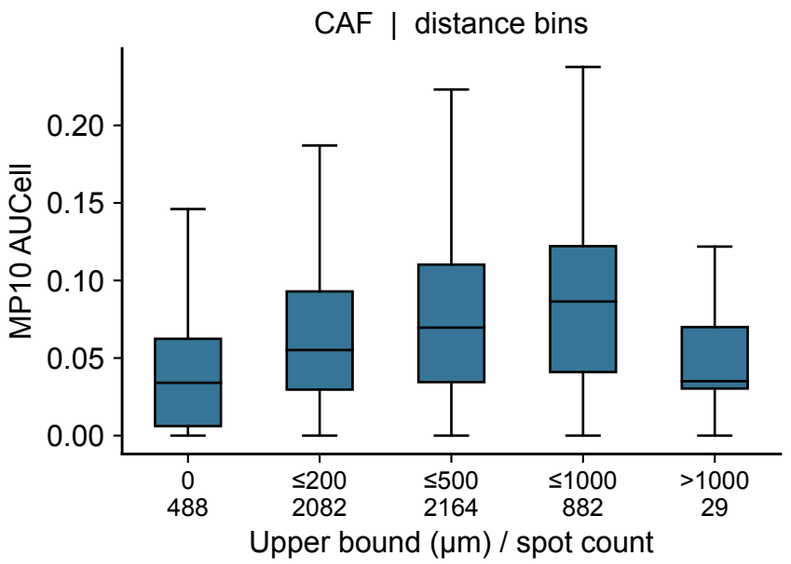
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



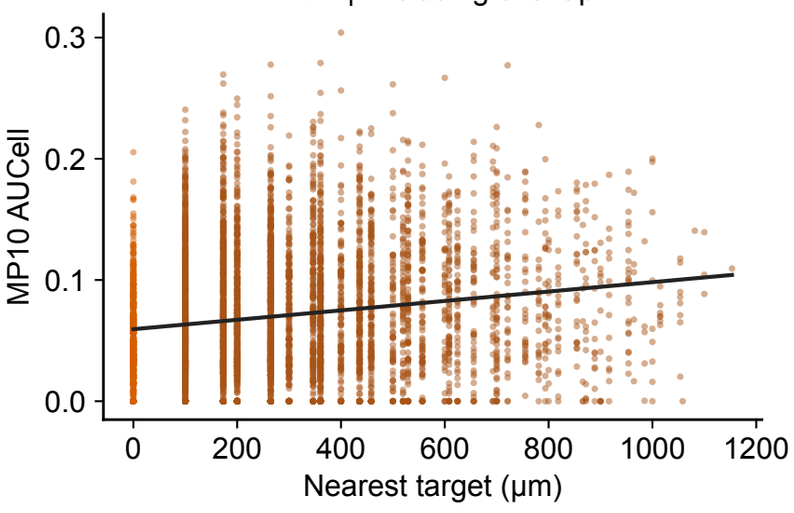
r +0.19 ρ +0.22
n 5645/5719 queries • 1372 targets



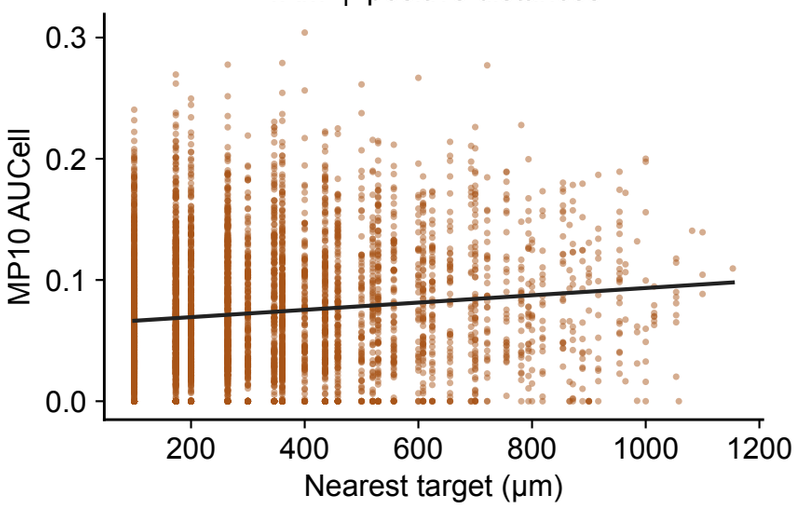
r +0.14 ρ +0.16
n 5157/5719 queries • 1372 targets



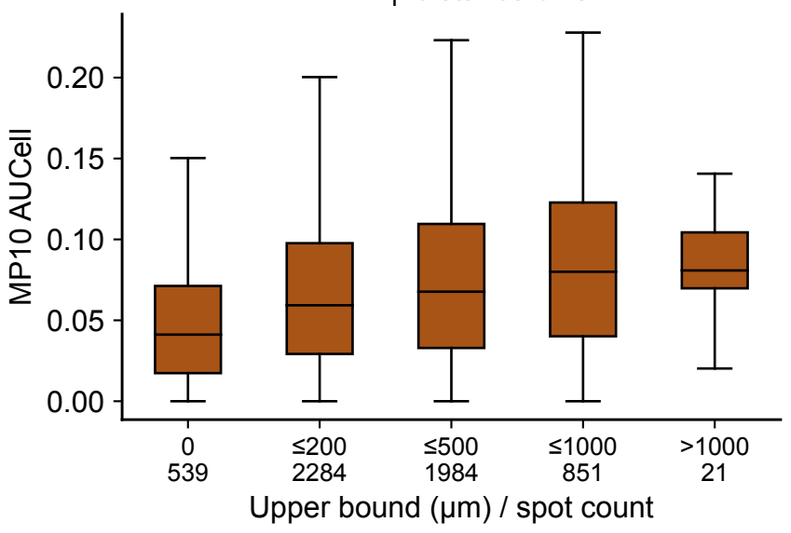
Orange = zero distance
Line = within-section linear fit



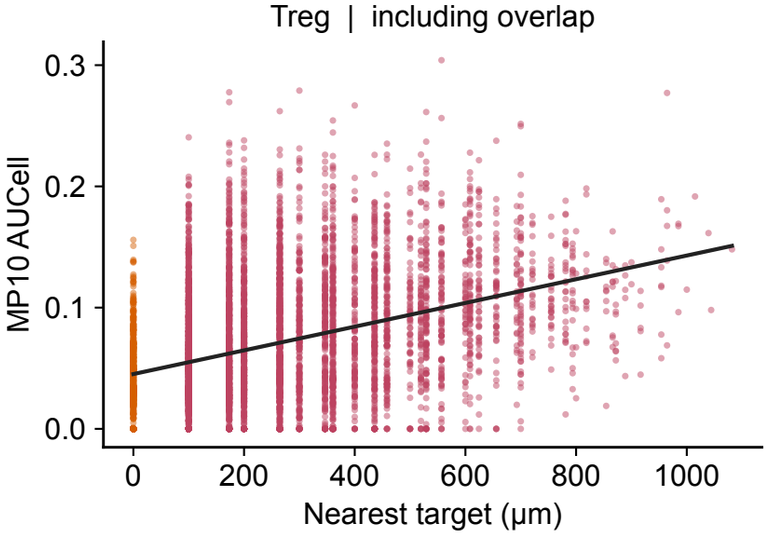
r +0.16 ρ +0.17
n 5679/5719 queries • 1372 targets



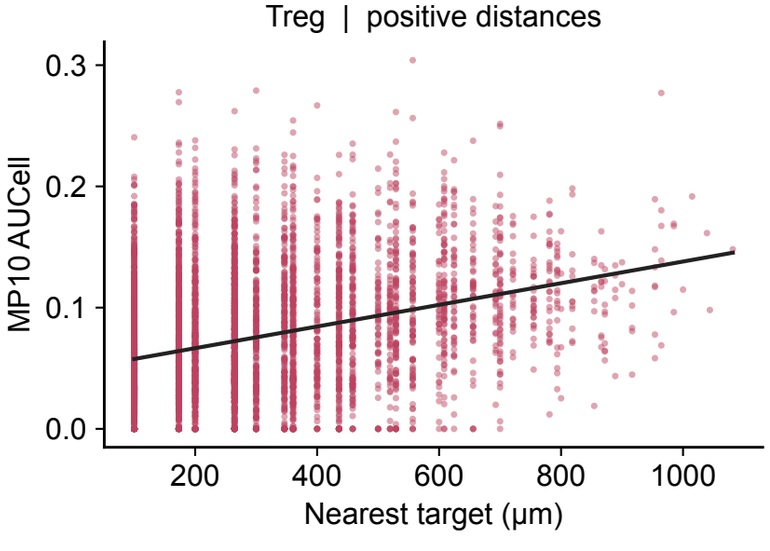
r +0.12 ρ +0.12
n 5140/5719 queries • 1372 targets



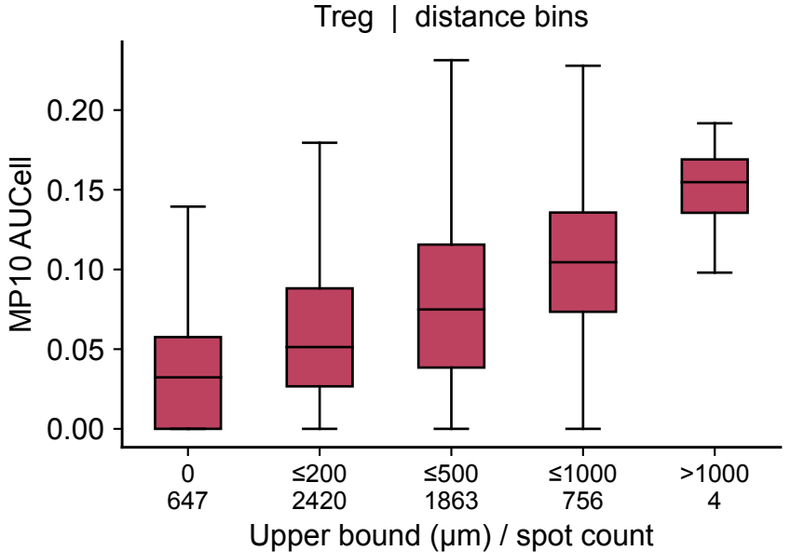
Orange = zero distance
Line = within-section linear fit



r +0.38 ρ +0.39
n 5690/5719 queries • 1372 targets



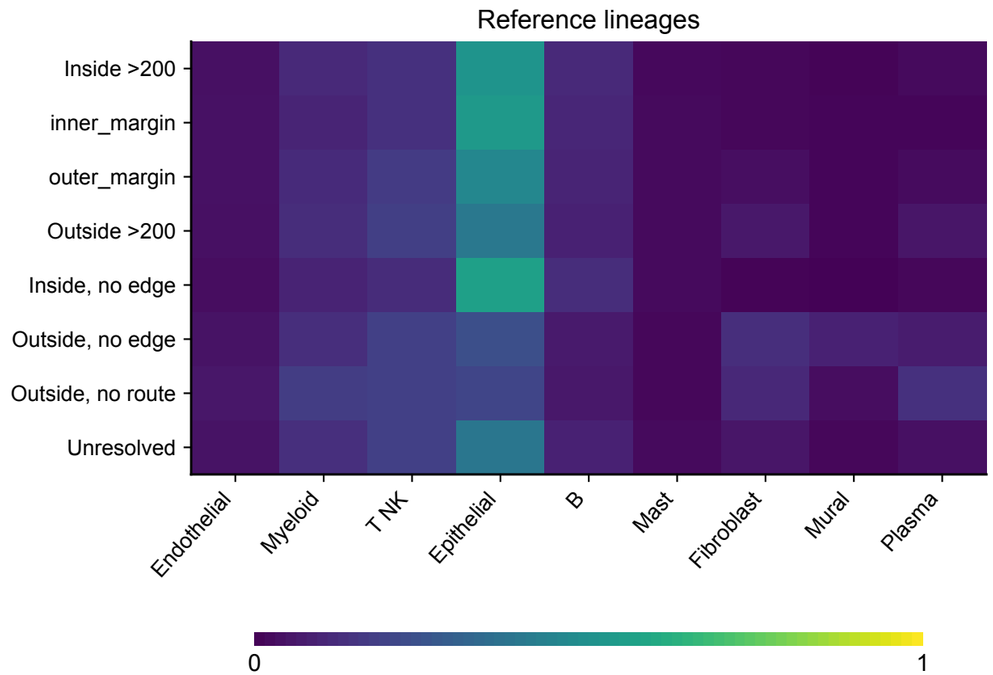
r +0.32 ρ +0.33
n 5043/5719 queries • 1372 targets

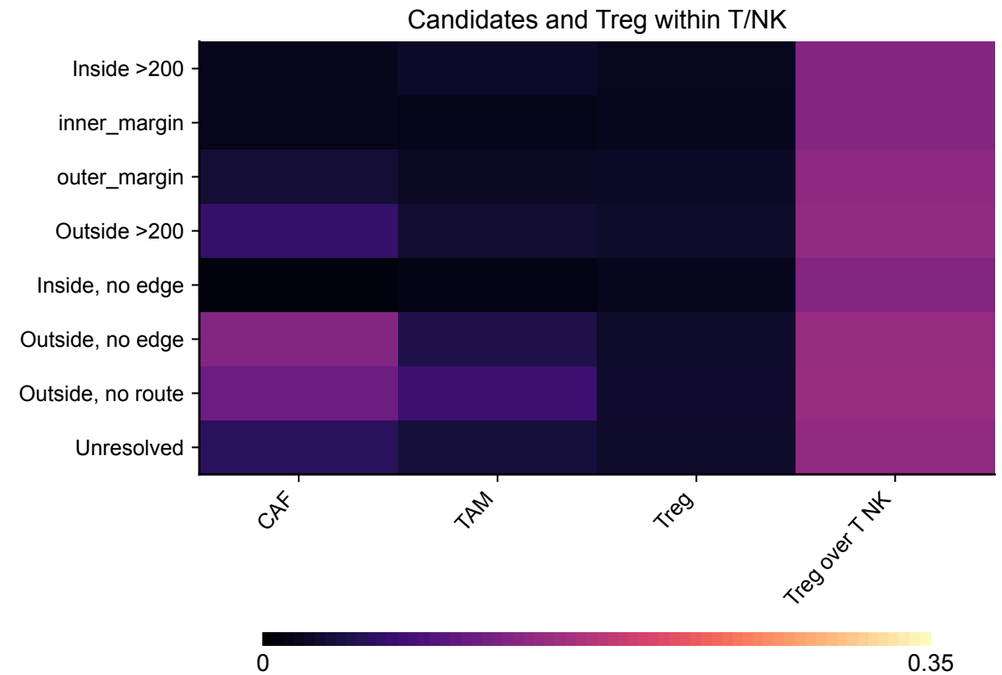


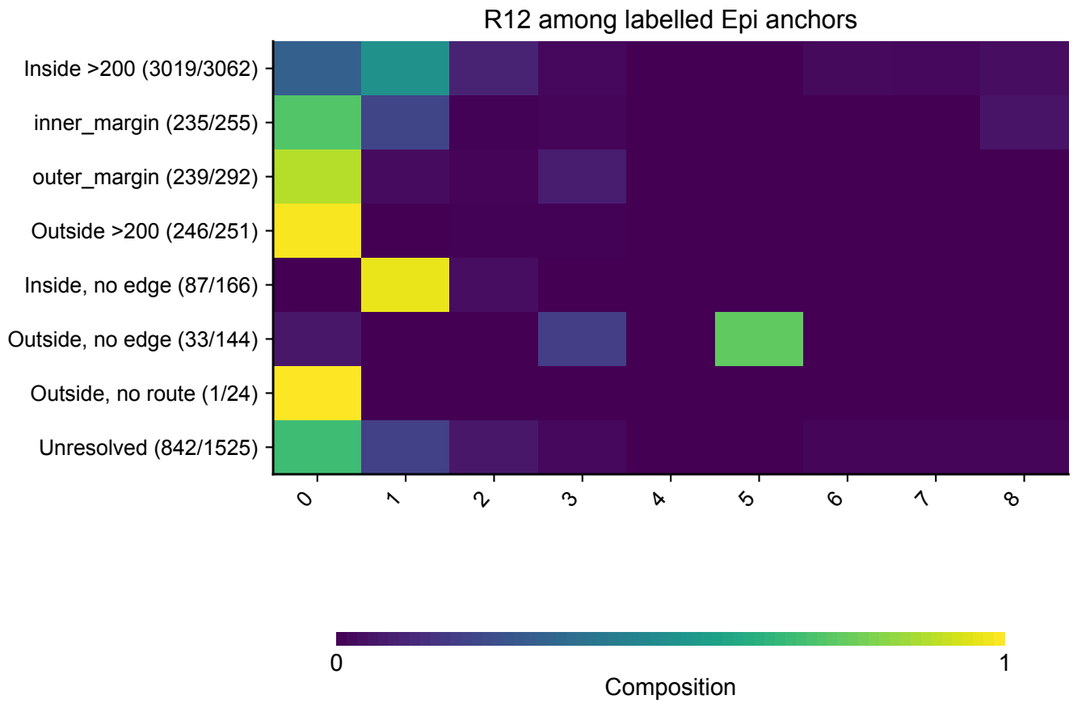
Orange = zero distance
Line = within-section linear fit

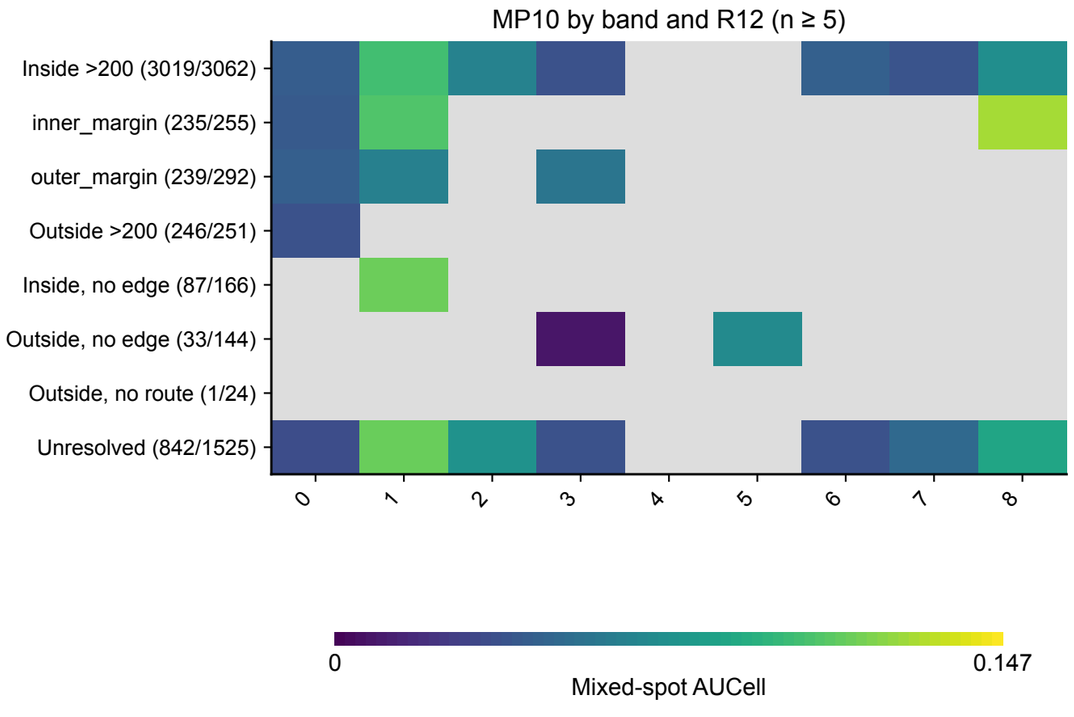
GSE319536_BC11 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





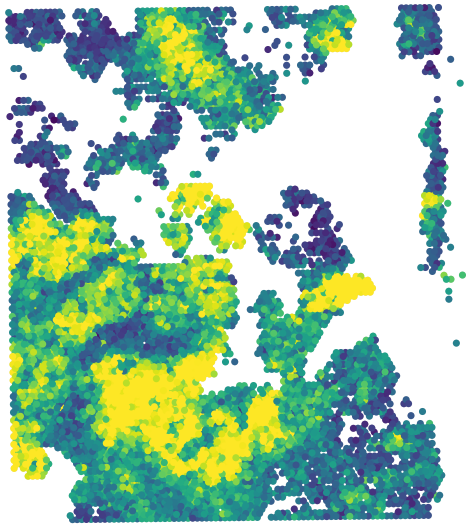




GSE319536_BC11 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

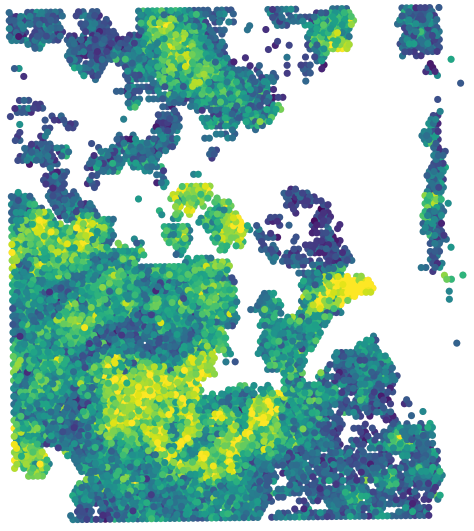
E2F Targets



0 0.152

UCell score

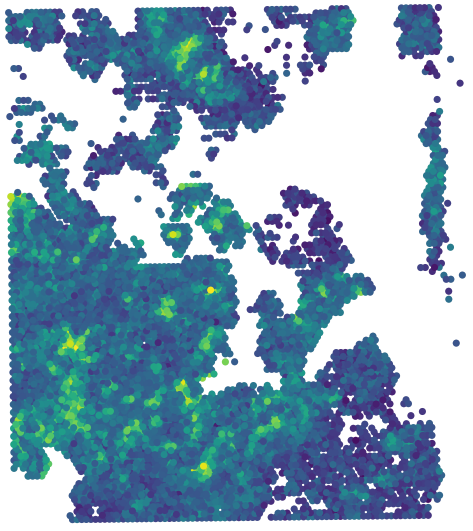
G2M Checkpoint



0 0.134

UCell score

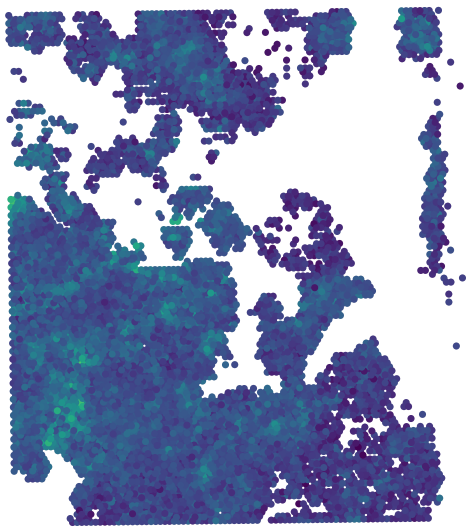
Hypoxia



0 0.211

UCell score

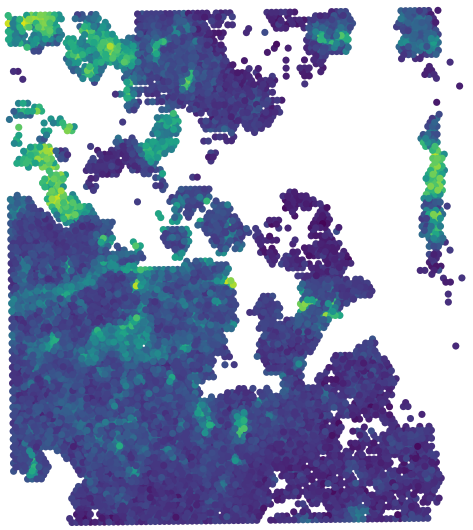
Tnfa Signaling Via Nfkb



0 0.275

UCell score

Emt



0 0.386

UCell score

Interferon Gamma Response

0 0.275

UCell score

GSE319536_BC11 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Il2 Stat5 Signaling

0 0.144

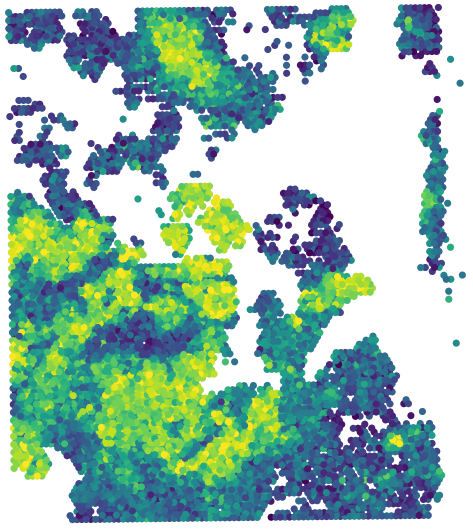
UCell score

Tgf Beta Signaling

0 0.232

UCell score

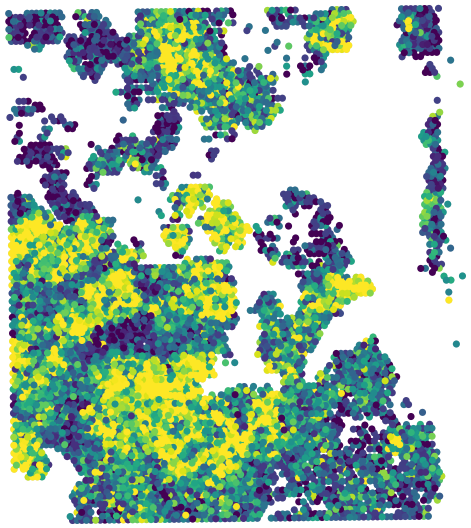
Cholesterol Homeostasis



0 0.21

UCell score

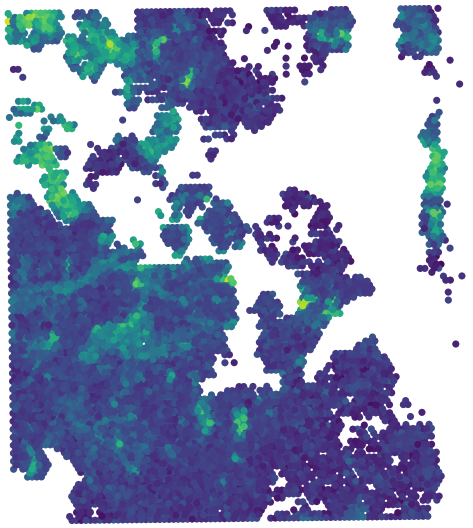
Cholesterol Biosynthesis



0 0.166

UCell score

Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4

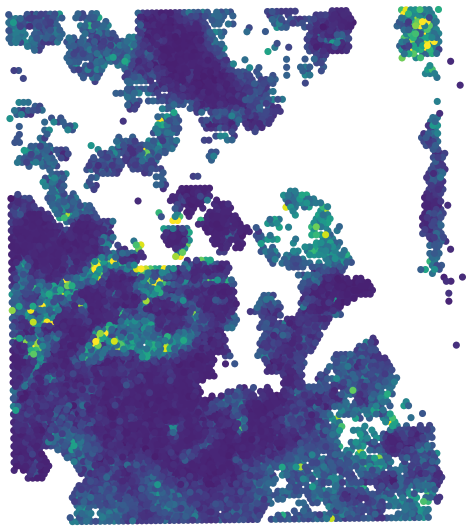
0 0.151

UCell score

GSE319536_BC11 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

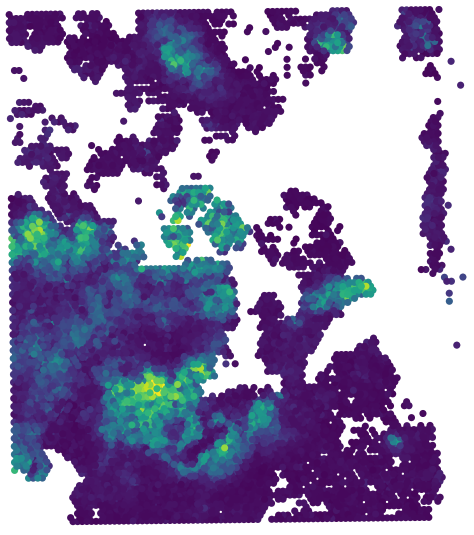
Treg composition



0 0.102

Fraction of all 30

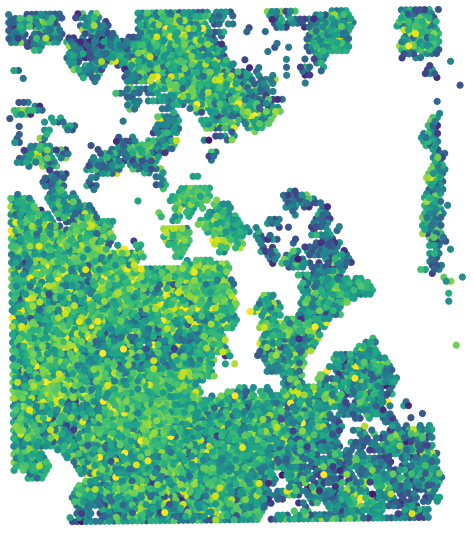
Treg estimated abundance



0 2.85

Posterior mean estimate

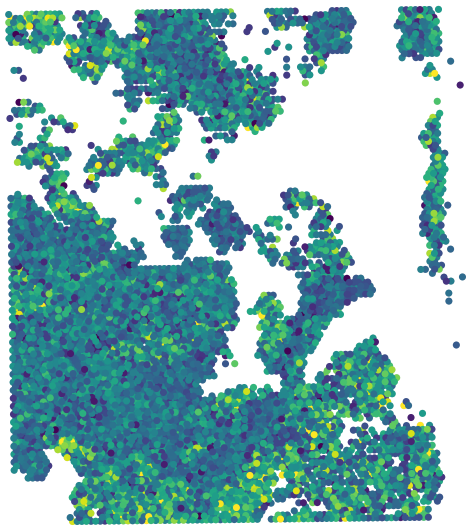
Treg Resting Vs Activated Dn



0 0.0778

UCell score

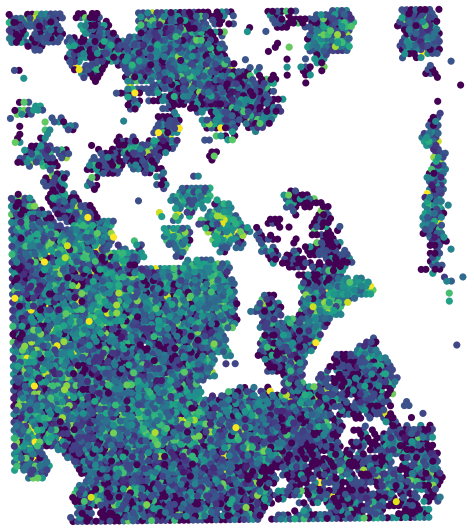
Treg Resting Vs Activated Up



0 0.0609

UCell score

Co Inhibition By Ctla4



0 0.151

UCell score

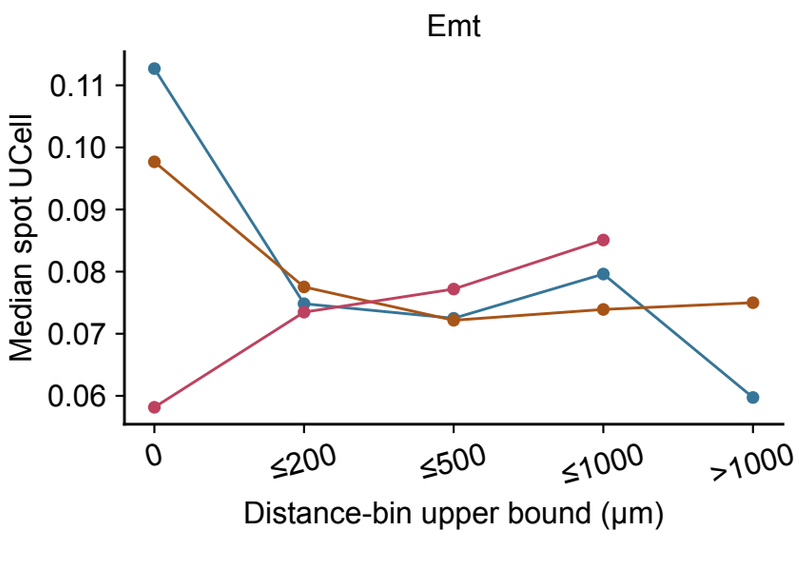
Il2 Stat5 Signaling

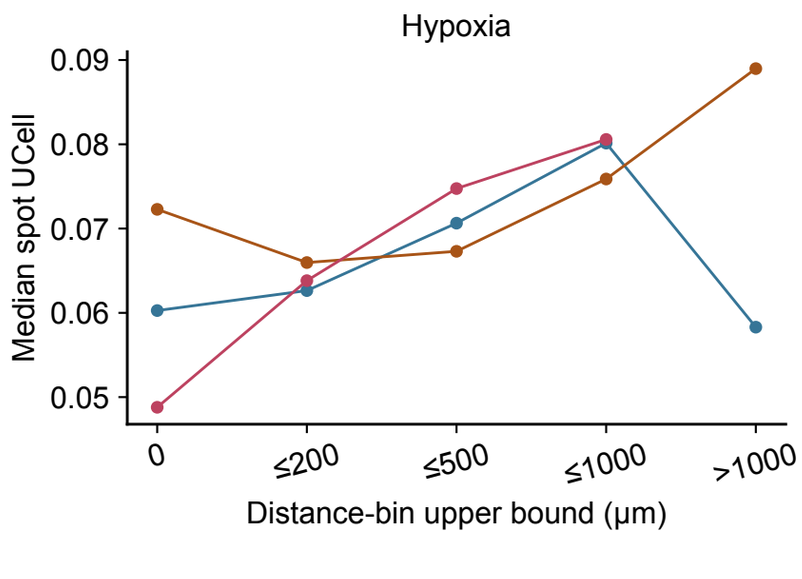
0 0.144

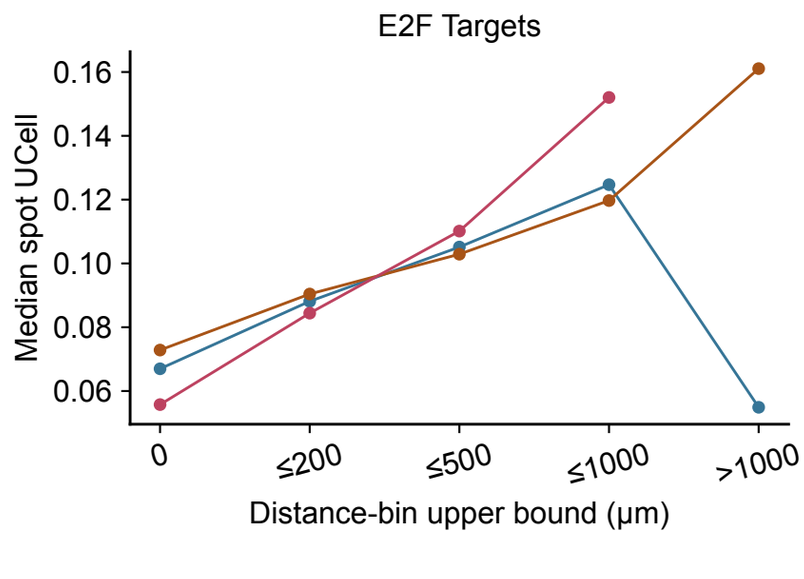
UCell score

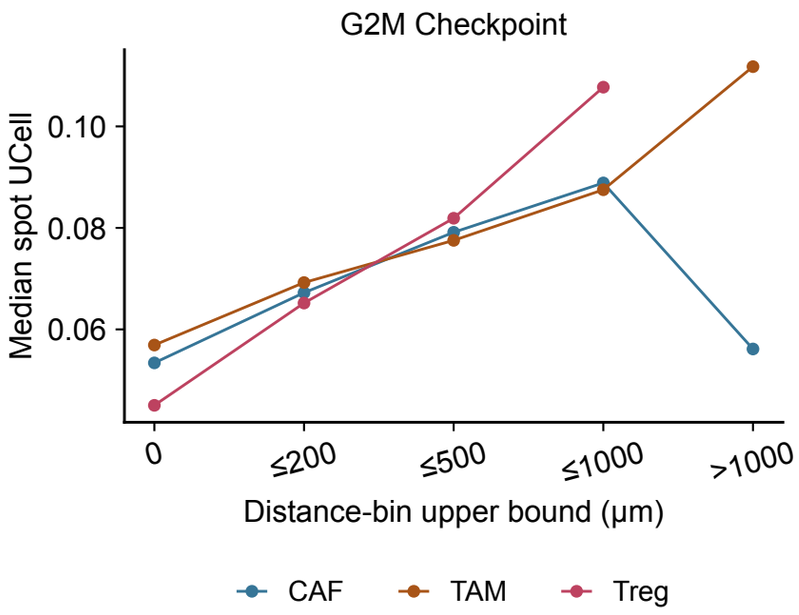
GSE319536_BC11 | Programs by target distance

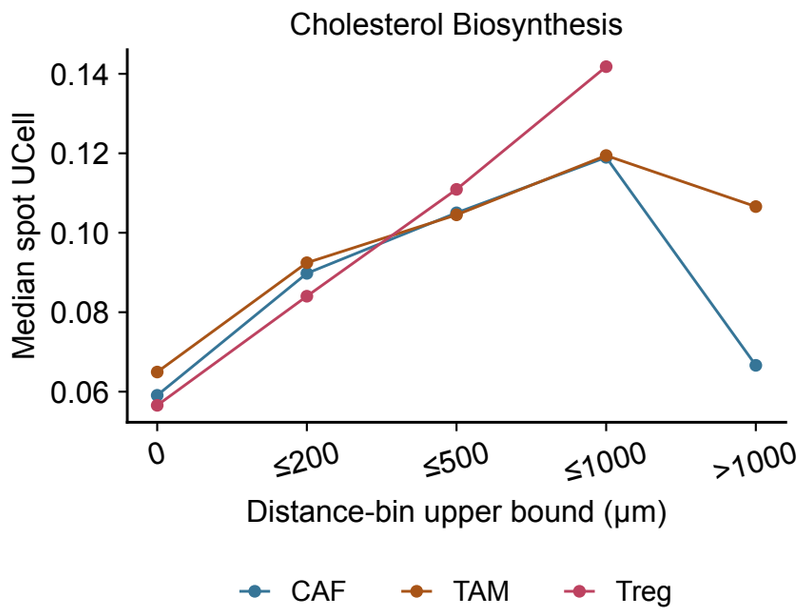
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

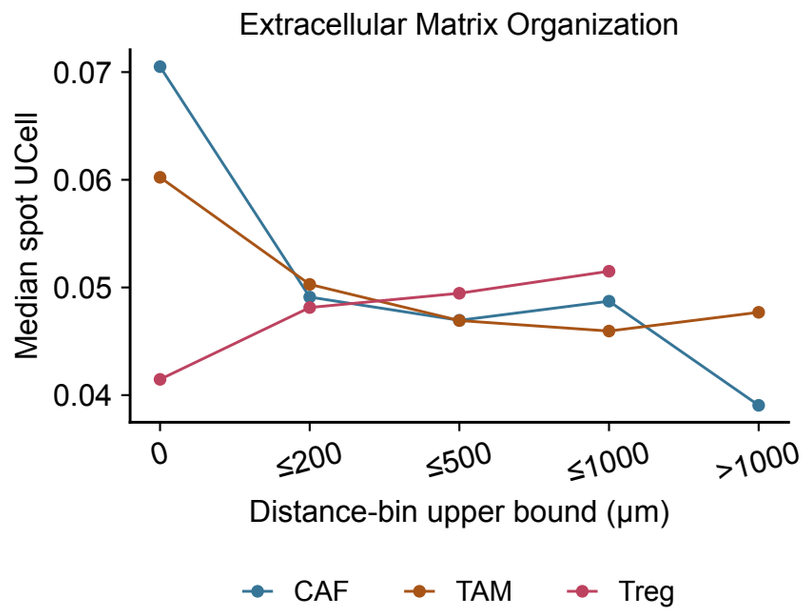






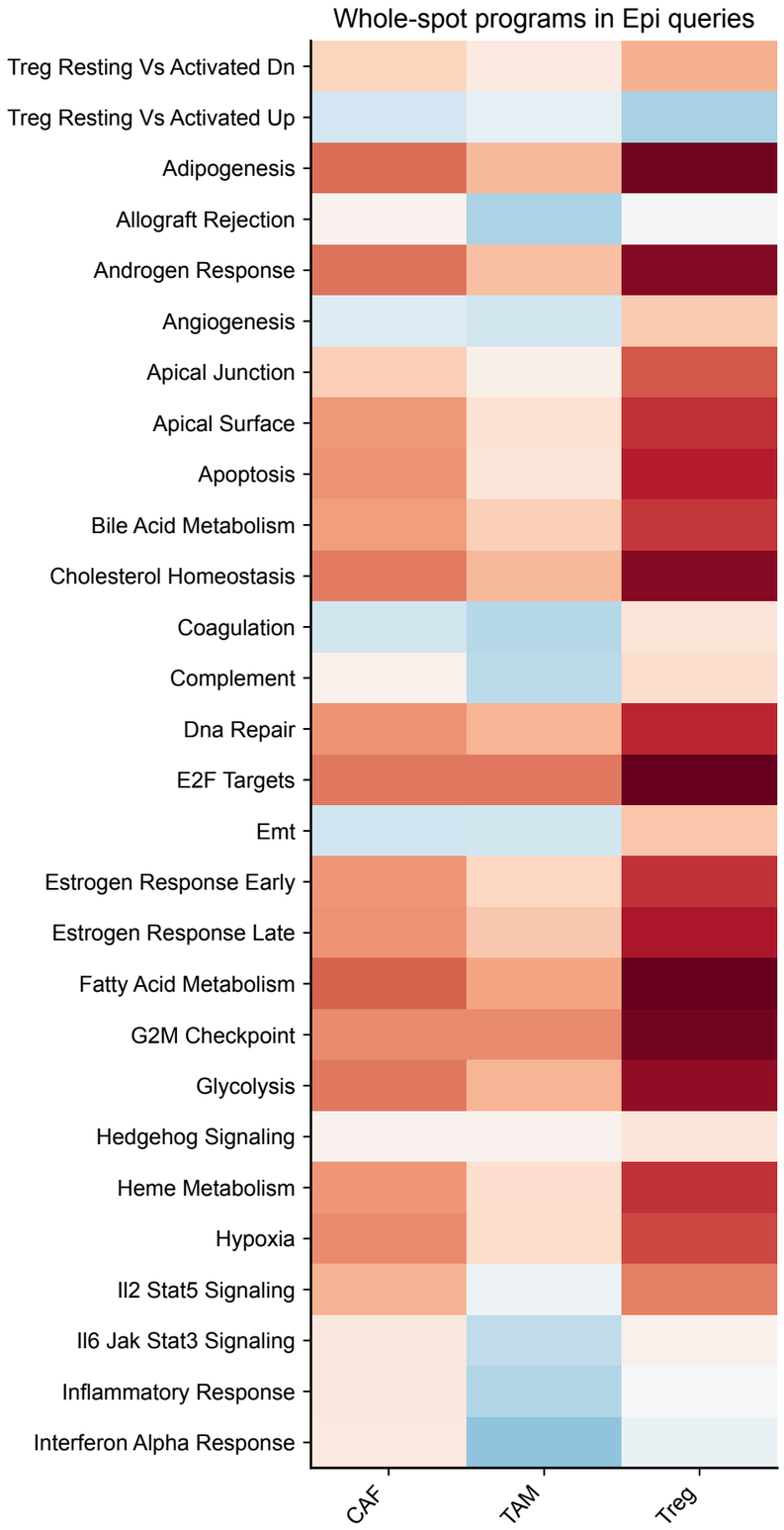


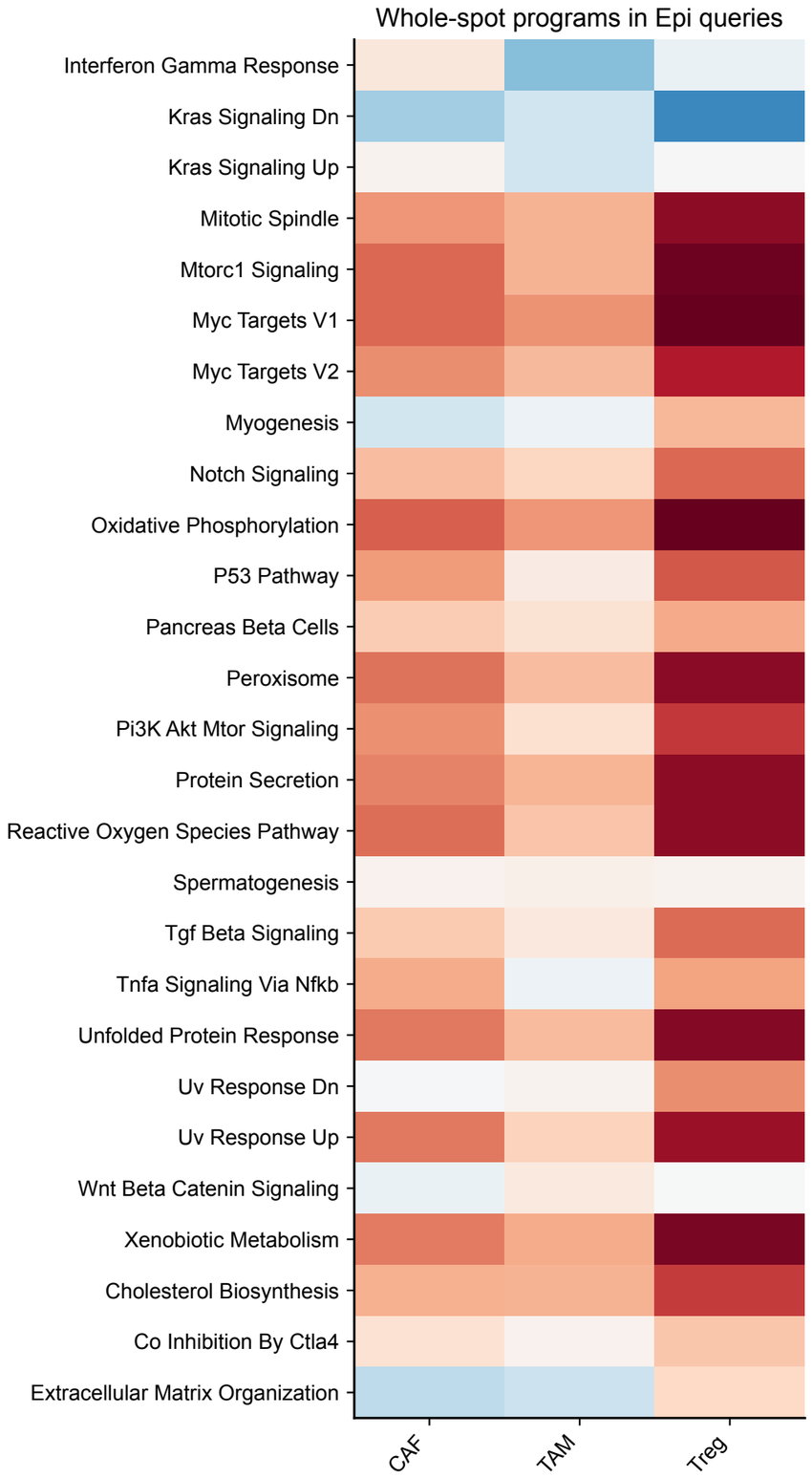




GSE319536_BC11 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

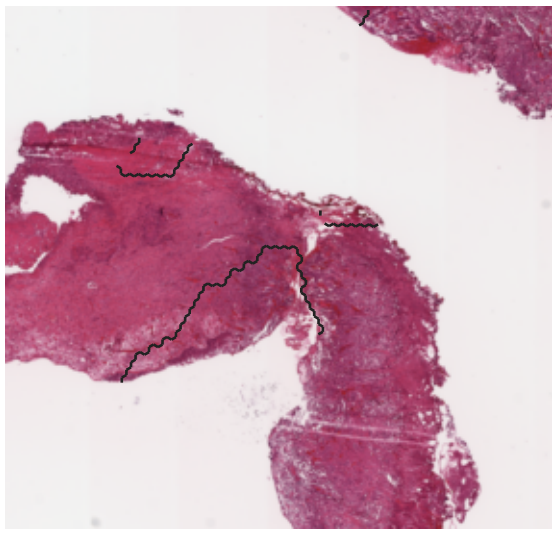




GSE319536_BC12 | Tissue context

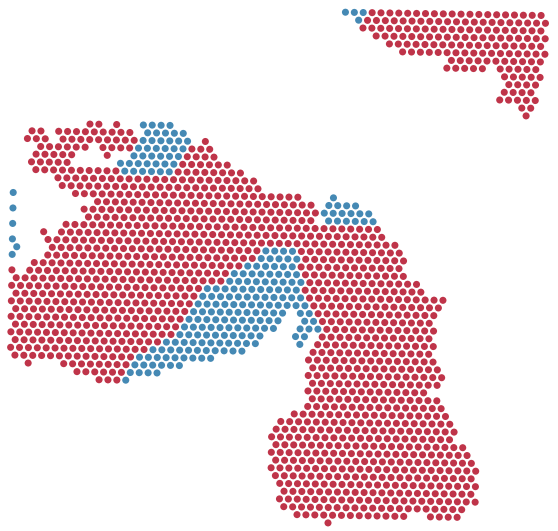
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



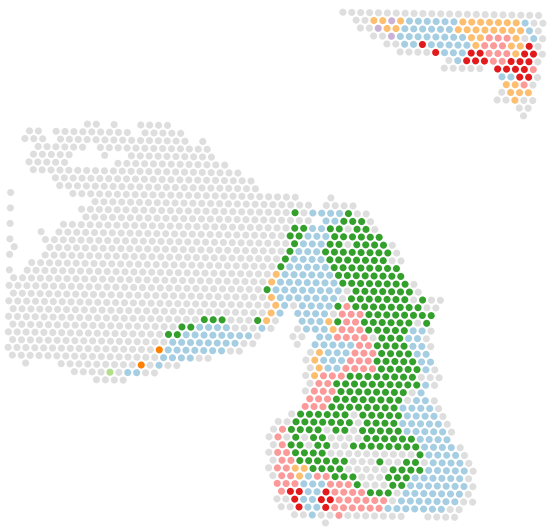
Black line: observed candidate outer edge

Final candidate location



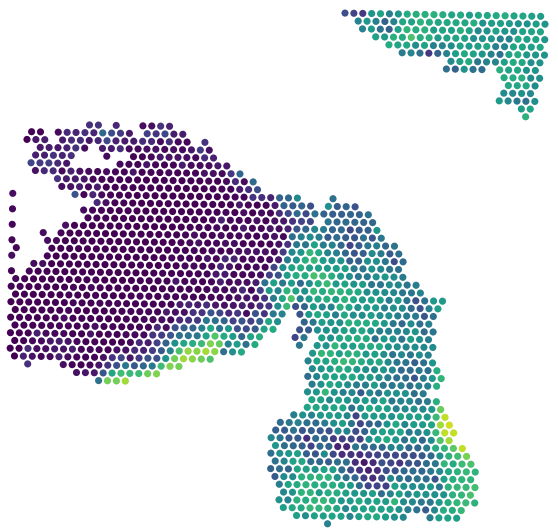
inside outside

Accepted R12 (original 0–8)



0 3 5 7 8
2 4 6

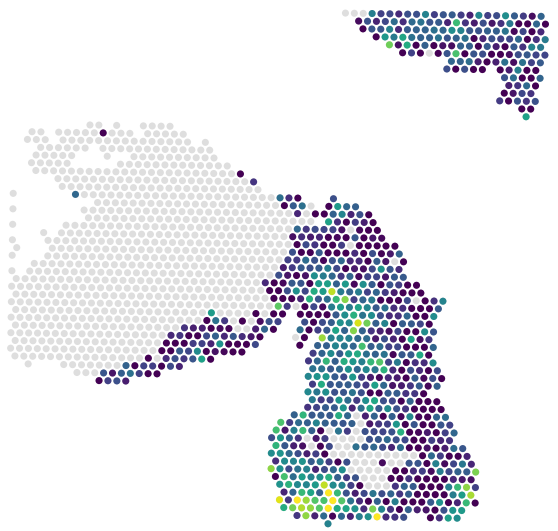
Epithelial support



0 1

Fraction of all 30

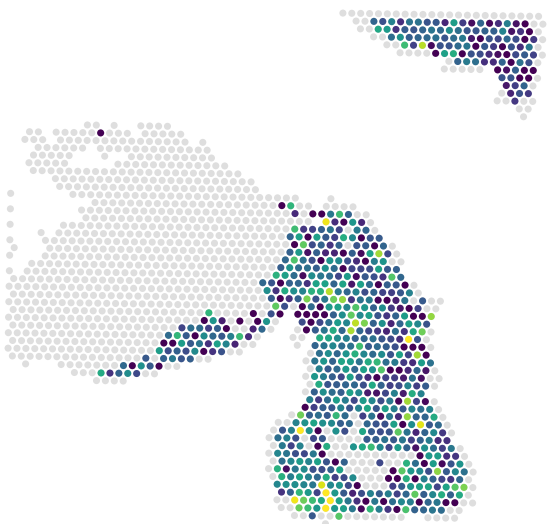
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



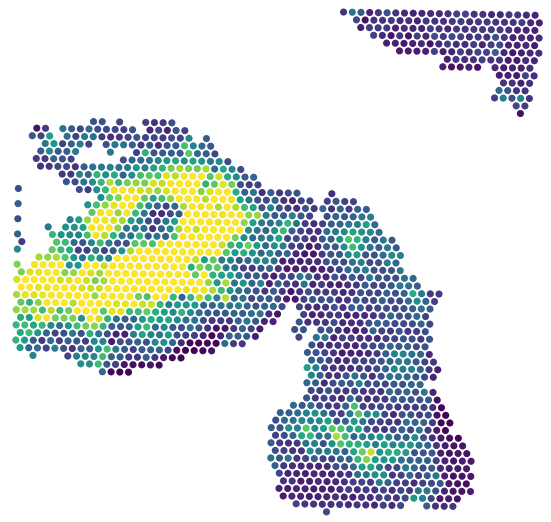
0 0.14

Matched-model AUCell

GSE319536_BC12 | Candidate neighborhoods

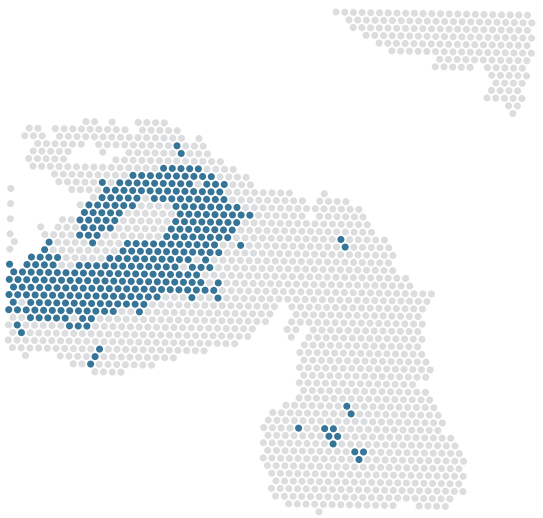
Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

CAF composition



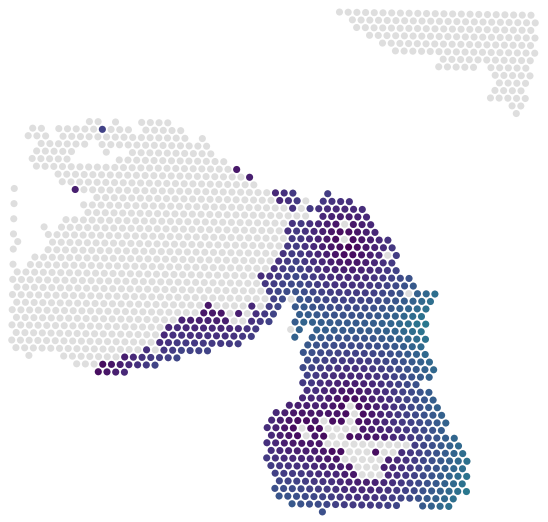
0 0.418
Fraction of all 30

CAF P80 targets



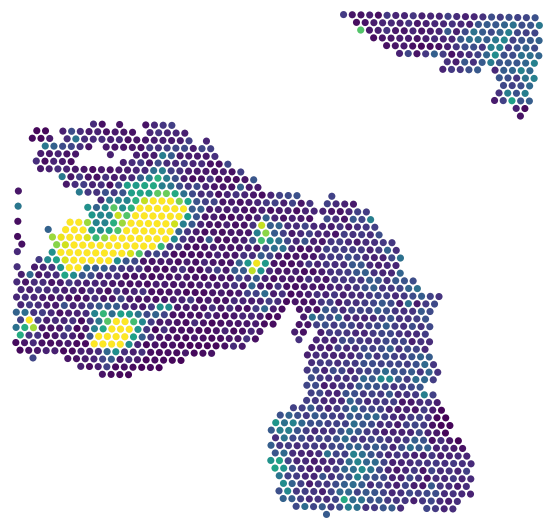
False True

CAF distance



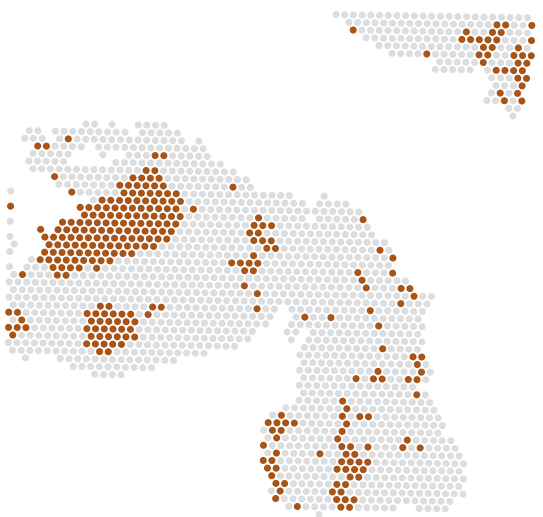
0 3.1e+03
Nearest spot center (µm)

TAM composition



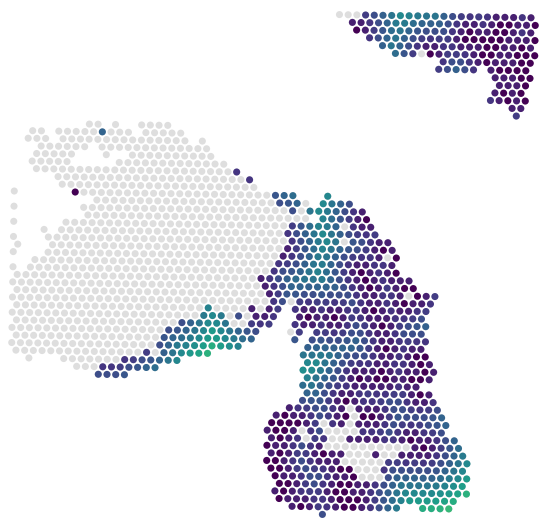
0 0.291
Fraction of all 30

TAM P80 targets



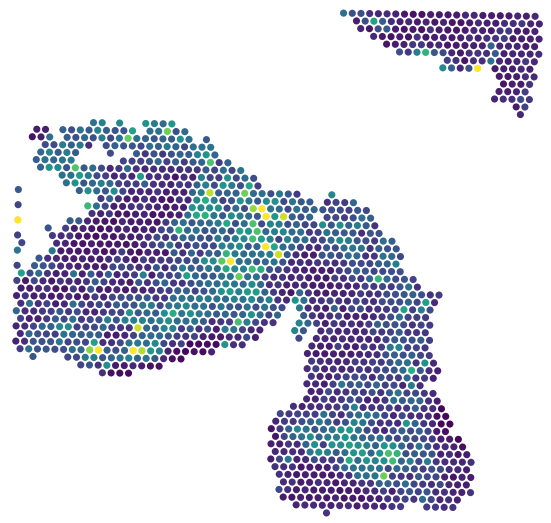
False True

TAM distance



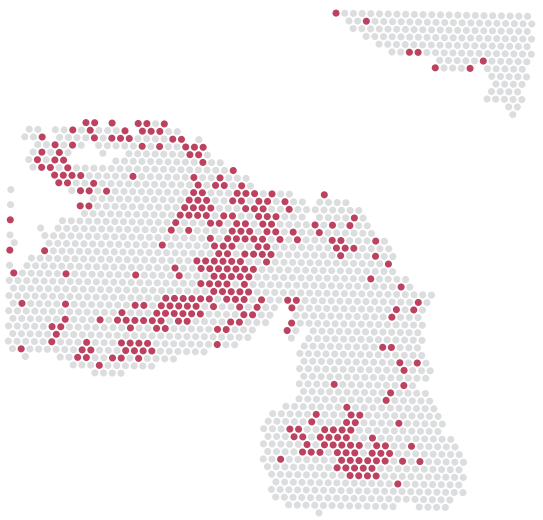
0 1.13e+03
Nearest spot center (µm)

Treg composition



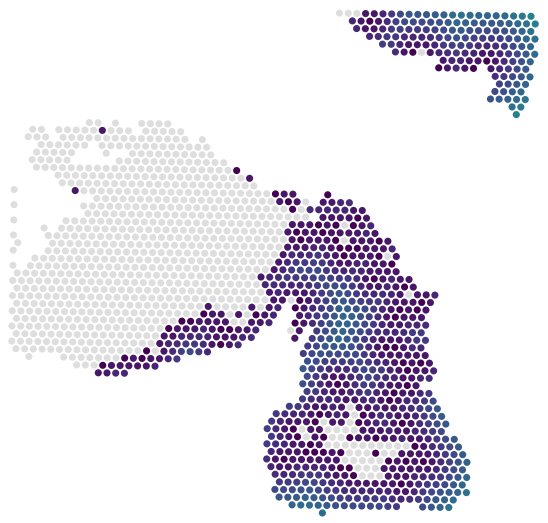
0 0.102
Fraction of all 30

Treg P80 targets



False True

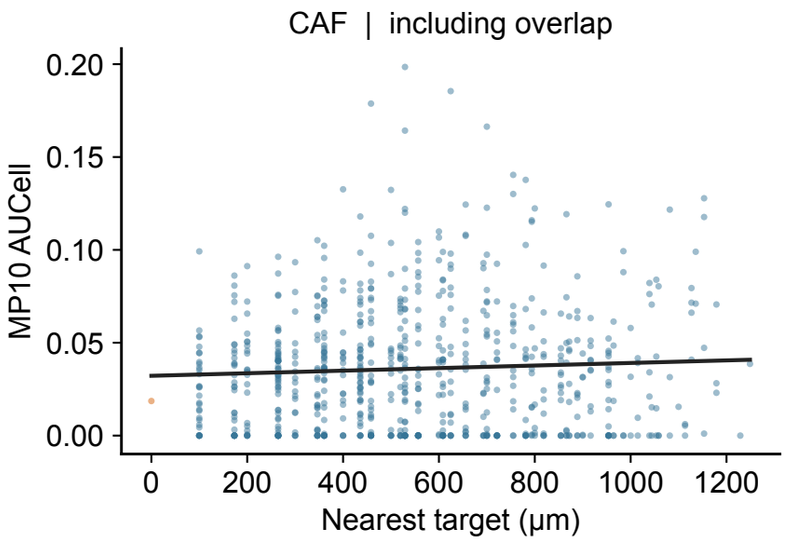
Treg distance



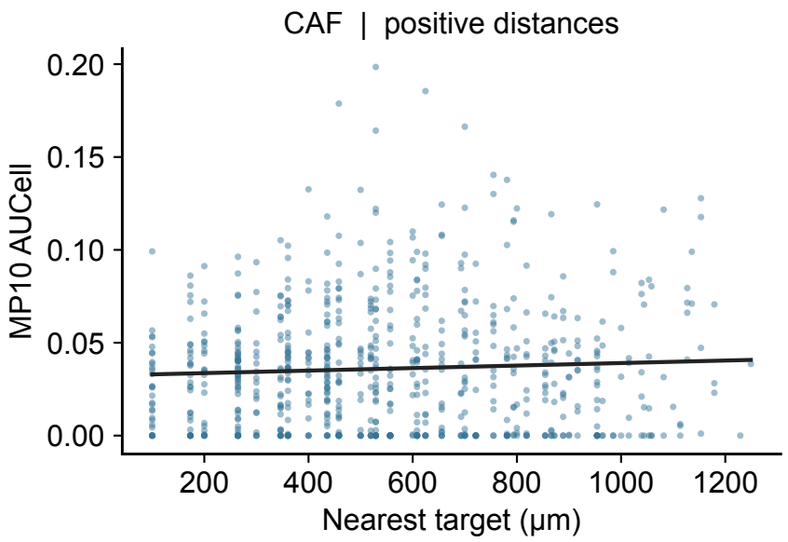
0 1.61e+03
Nearest spot center (µm)

GSE319536_BC12 | MP10 versus nearest target

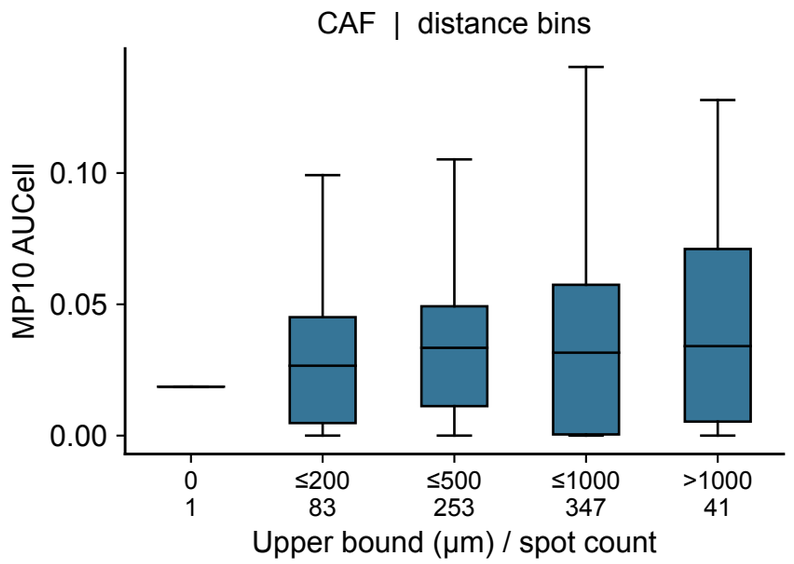
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



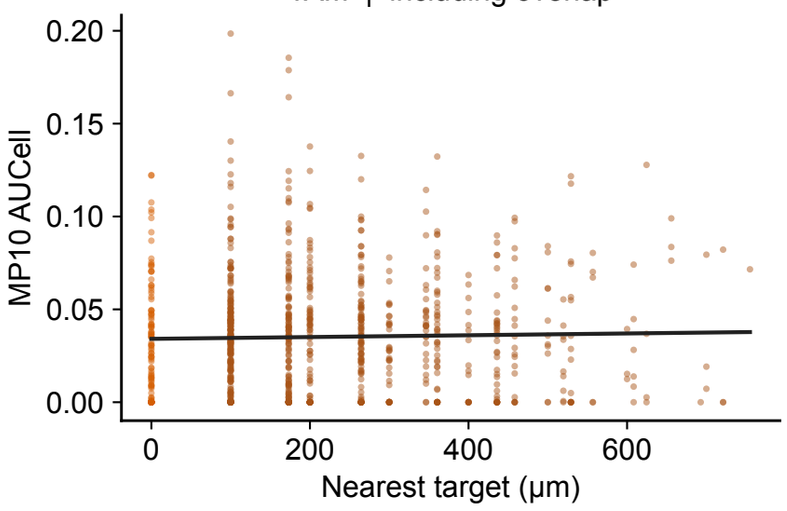
r +0.05 ρ +0.02
n 725/884 queries • 331 targets



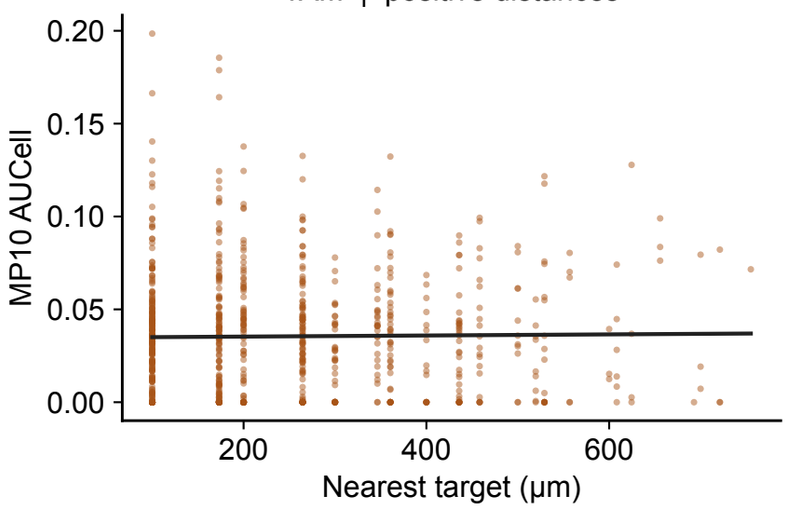
r +0.05 ρ +0.02
n 724/884 queries • 331 targets



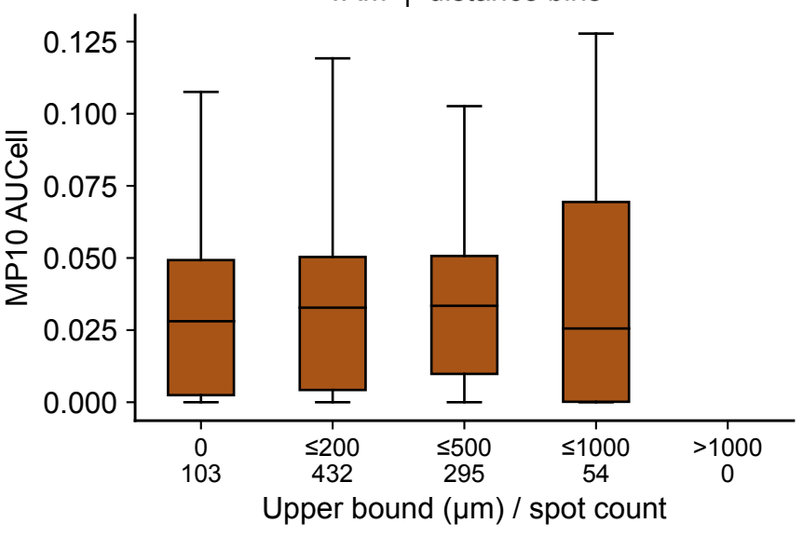
Orange = zero distance
Line = within-section linear fit



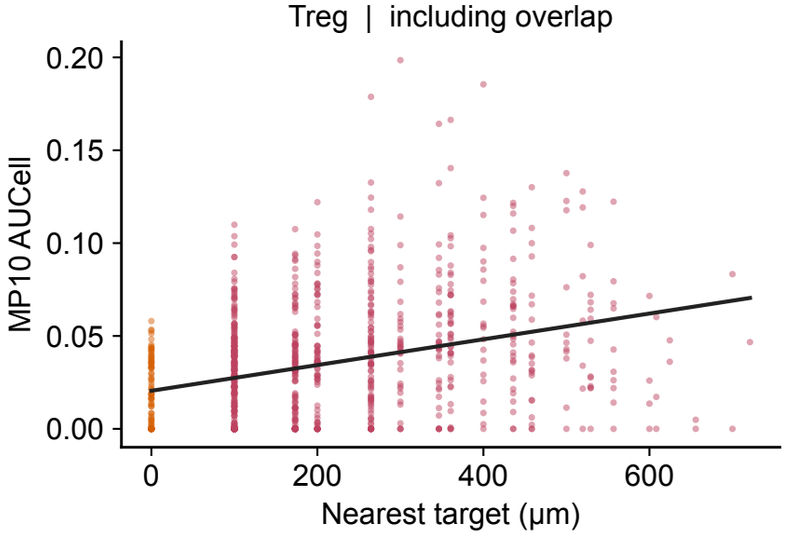
r +0.02 ρ +0.04
n 884/884 queries • 331 targets



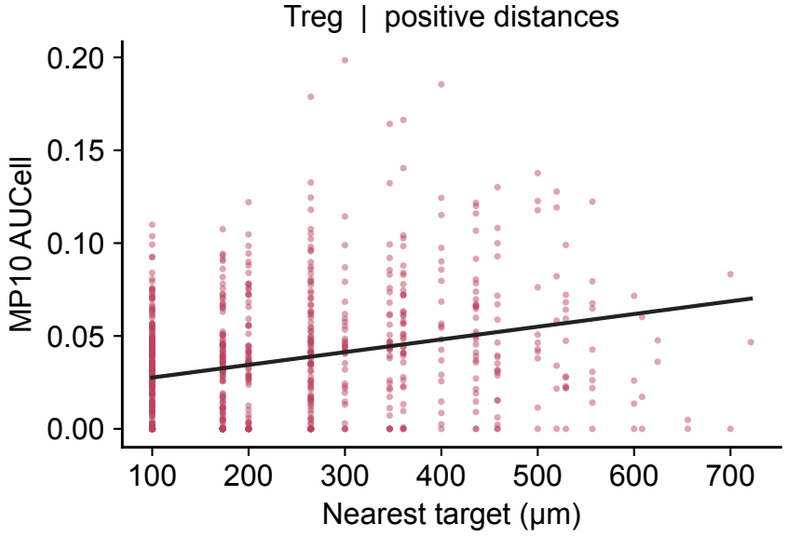
r +0.01 ρ +0.03
n 781/884 queries • 331 targets



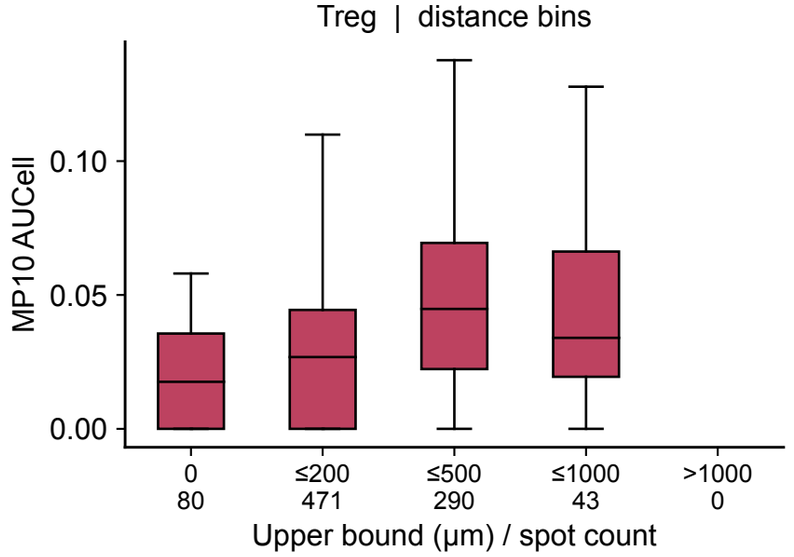
Orange = zero distance
Line = within-section linear fit



r +0.32 ρ +0.33
n 884/884 queries • 331 targets



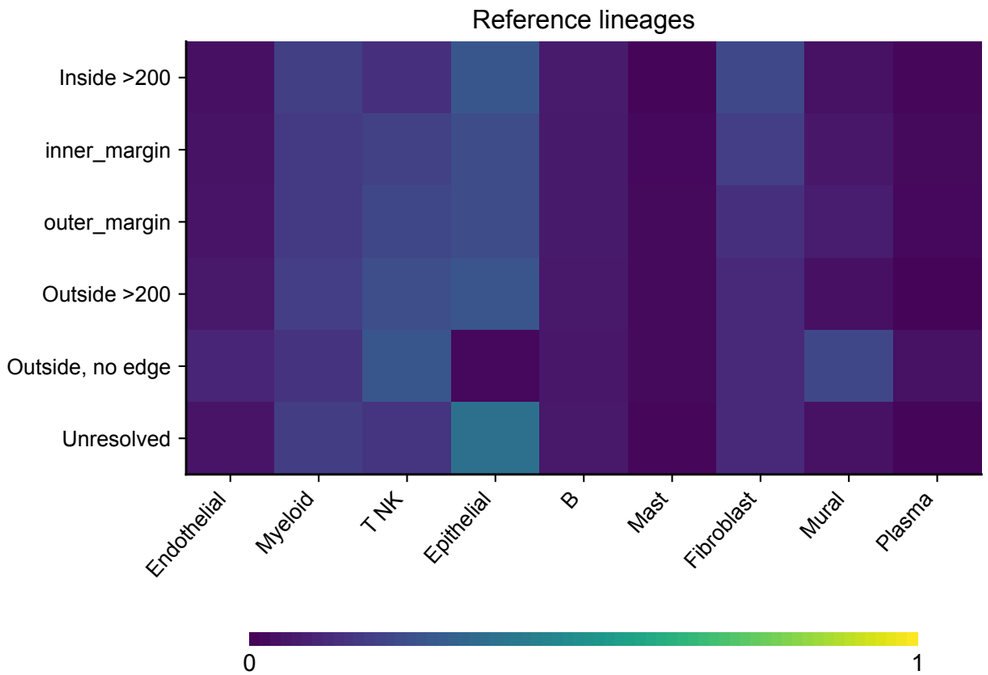
r +0.29 ρ +0.31
n 804/884 queries • 331 targets

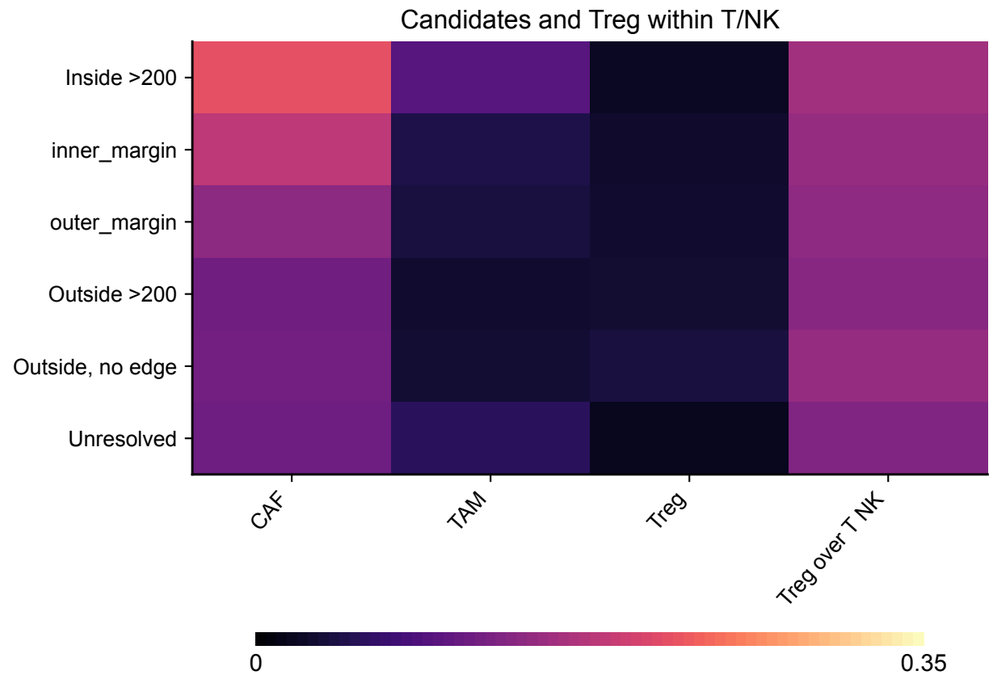


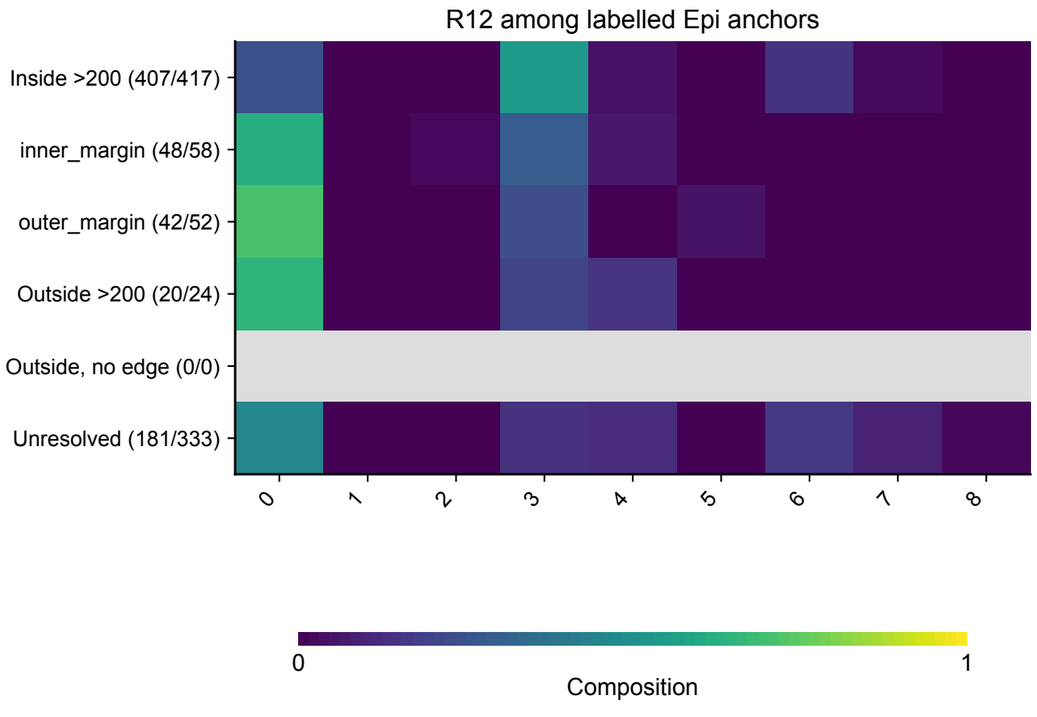
Orange = zero distance
Line = within-section linear fit

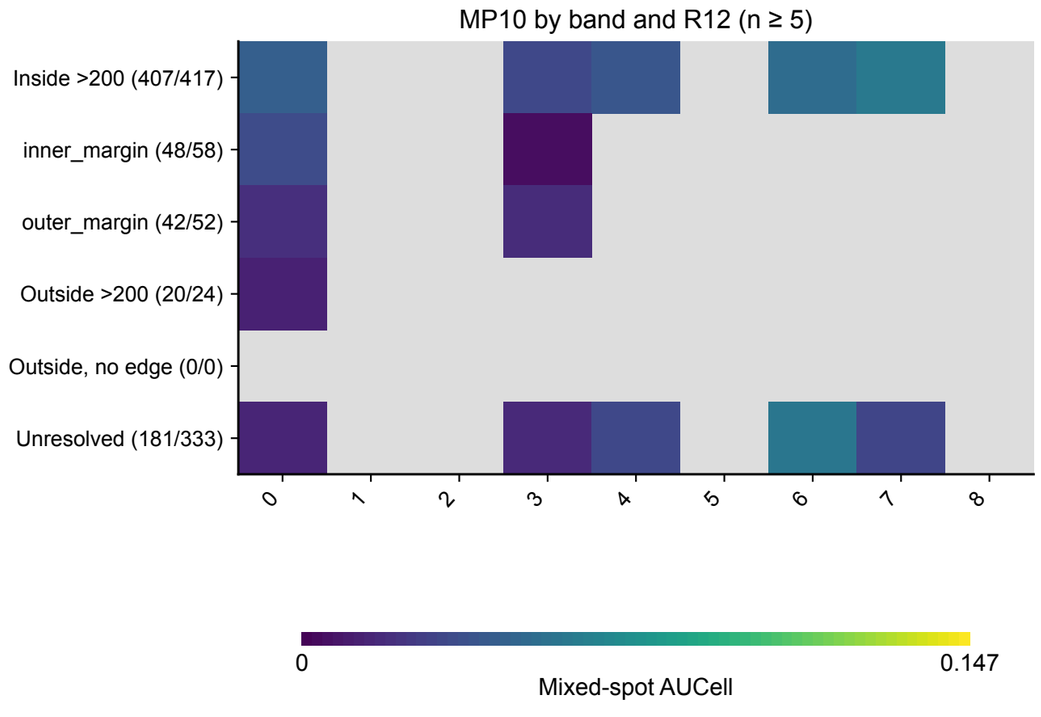
GSE319536_BC12 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable



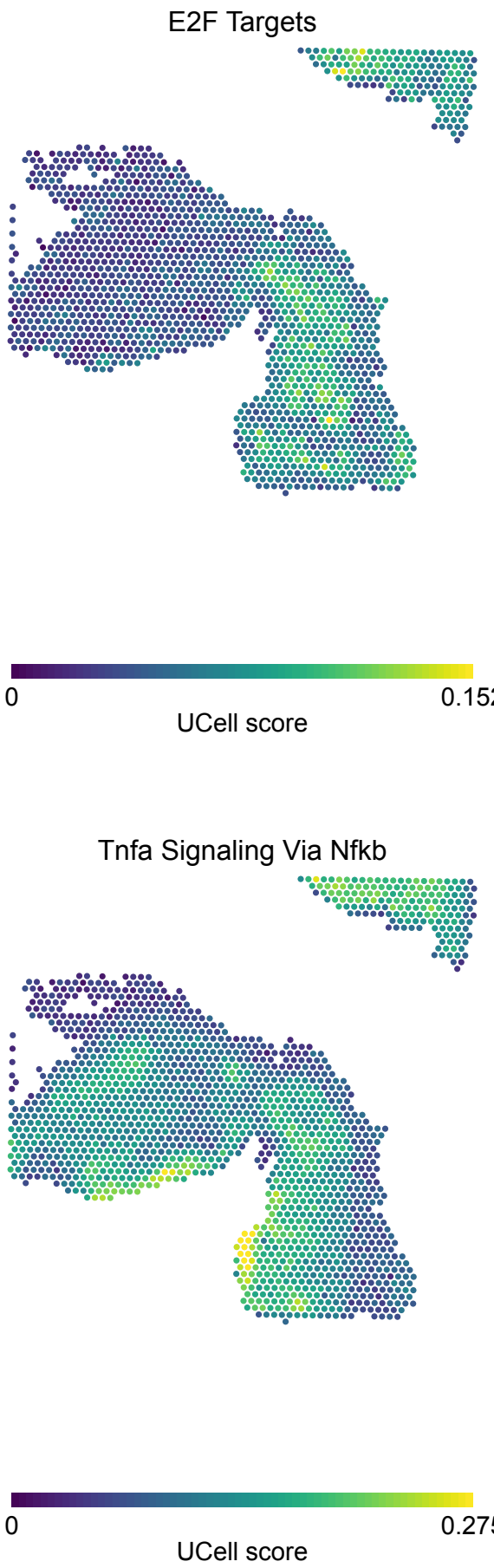


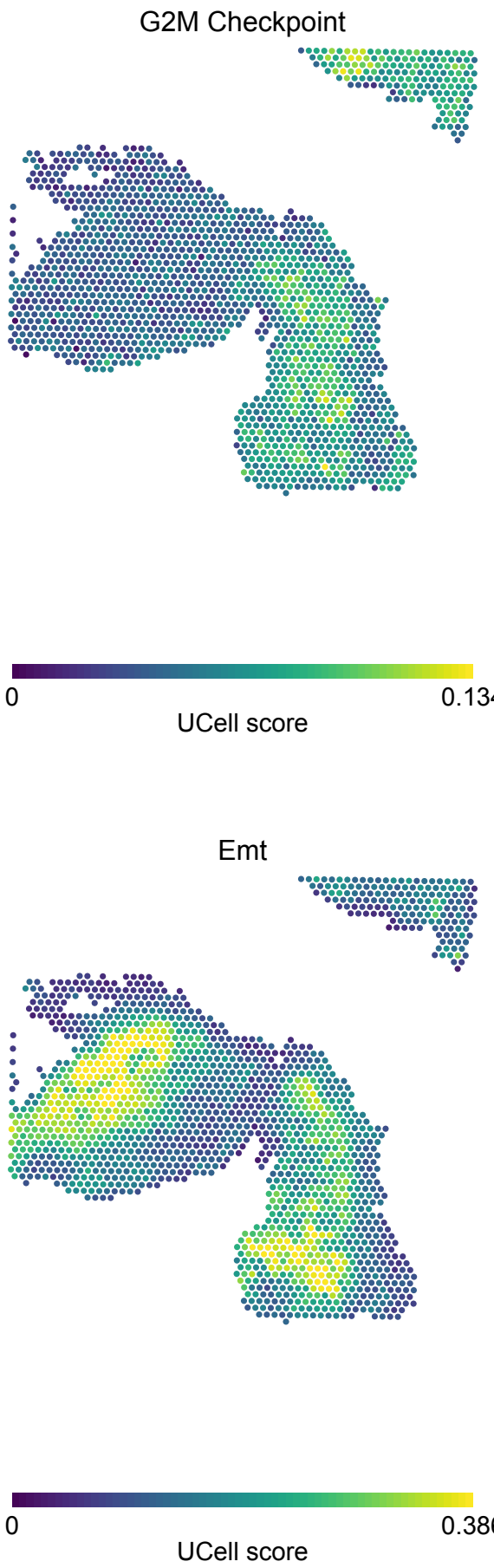


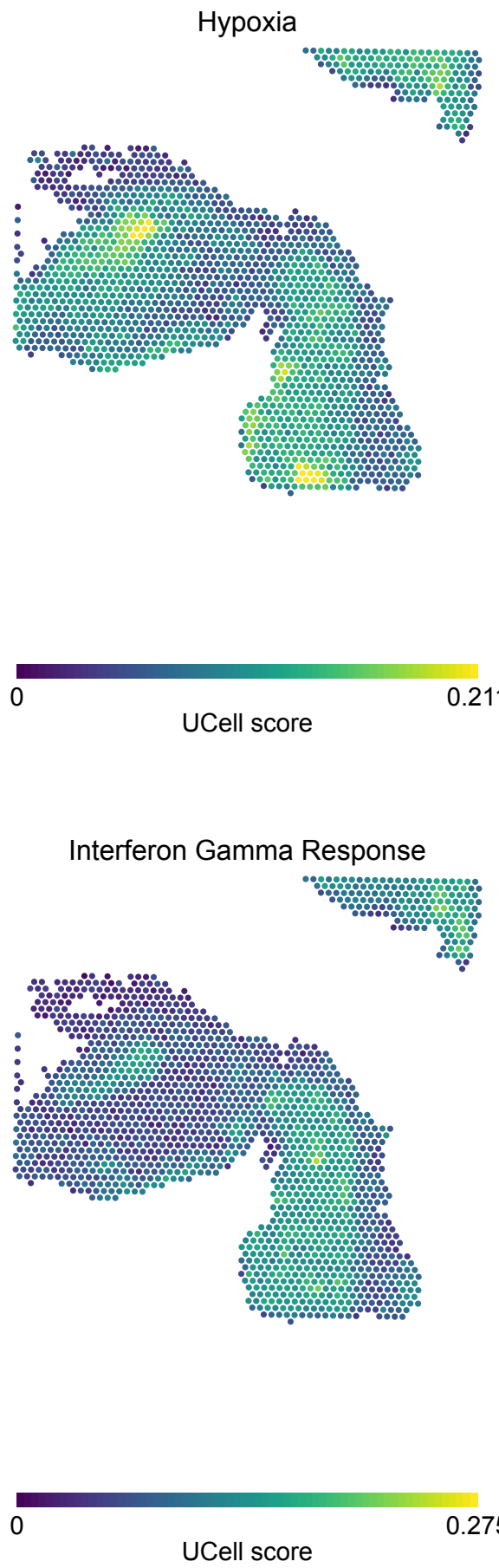


GSE319536_BC12 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections



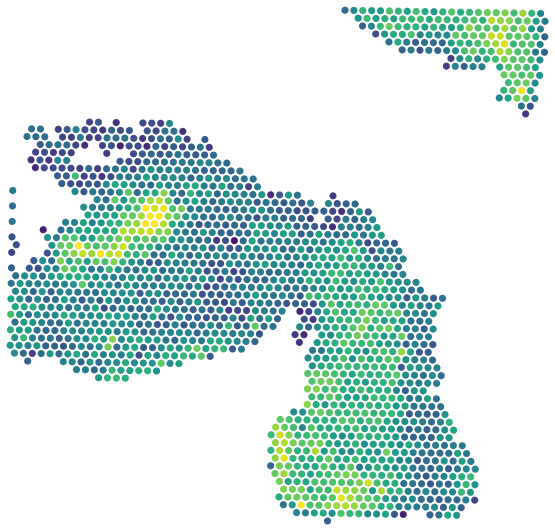




GSE319536_BC12 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

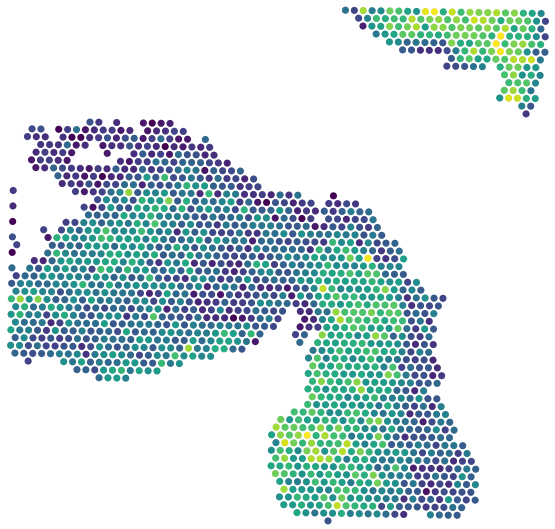
Il2 Stat5 Signaling



0 0.144

UCell score

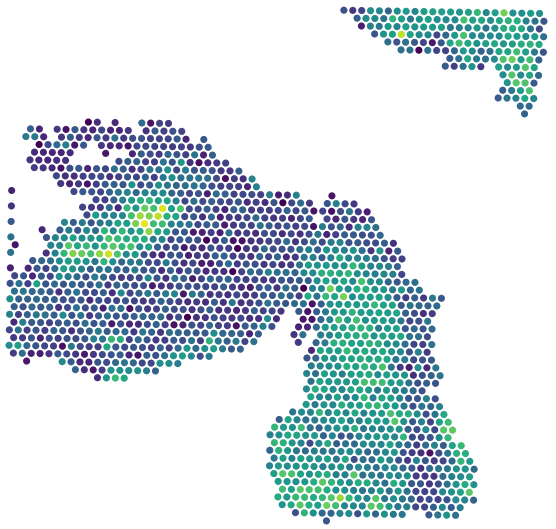
Tgf Beta Signaling



0 0.232

UCell score

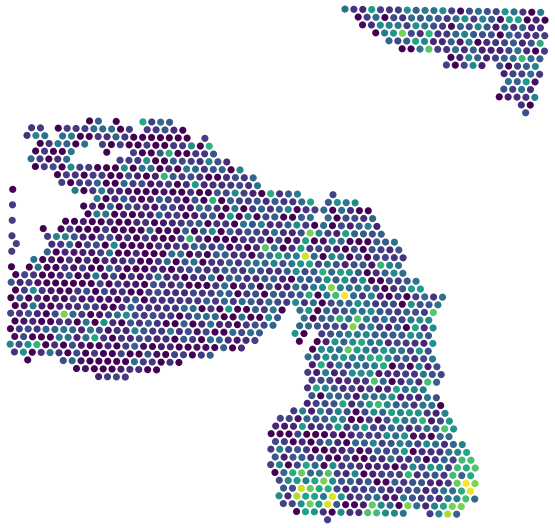
Cholesterol Homeostasis



0 0.21

UCell score

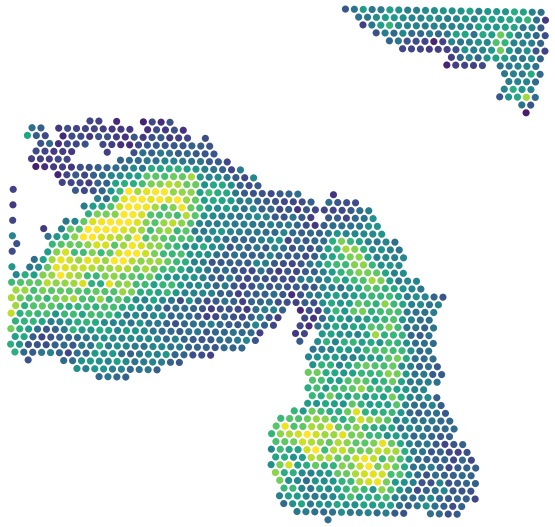
Cholesterol Biosynthesis



0 0.166

UCell score

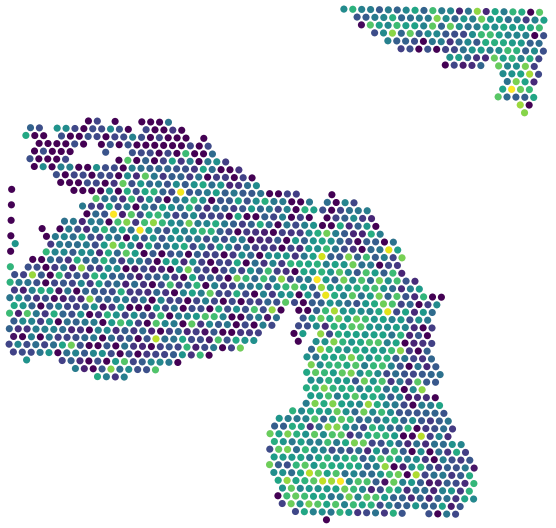
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



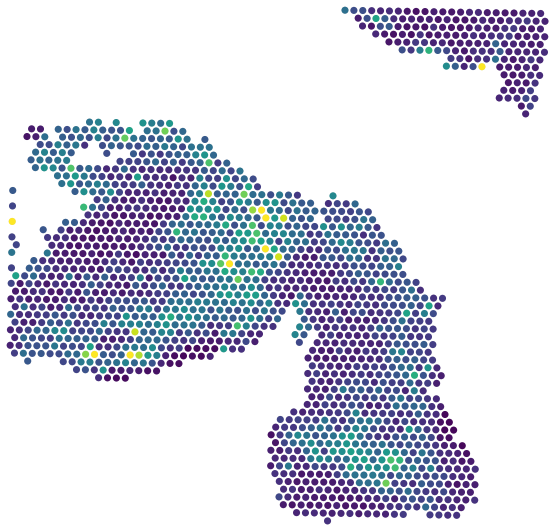
0 0.151

UCell score

GSE319536_BC12 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

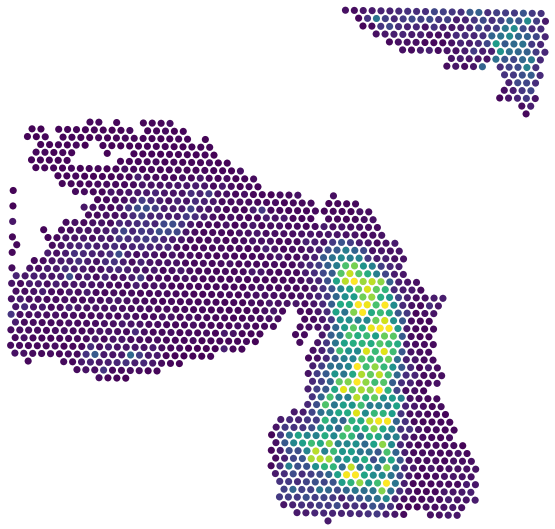
Treg composition



0 0.102

Fraction of all 30

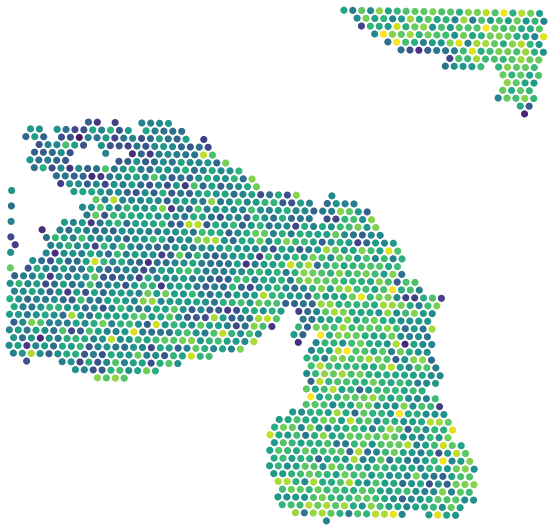
Treg estimated abundance



0 2.85

Posterior mean estimate

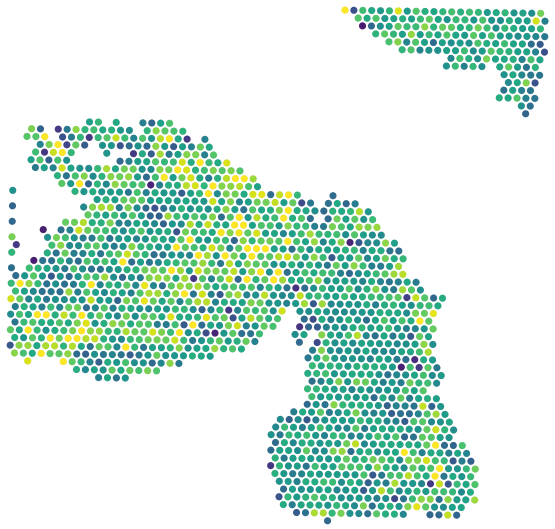
Treg Resting Vs Activated Dn



0 0.0778

UCell score

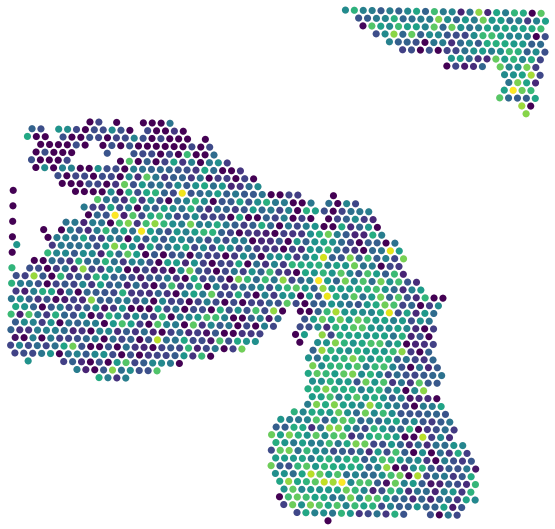
Treg Resting Vs Activated Up



0 0.0609

UCell score

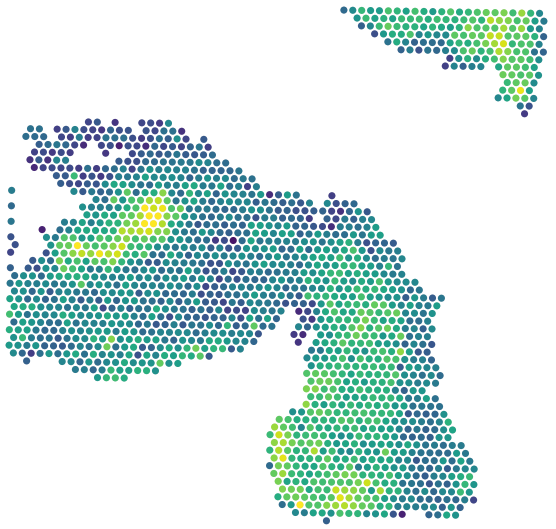
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

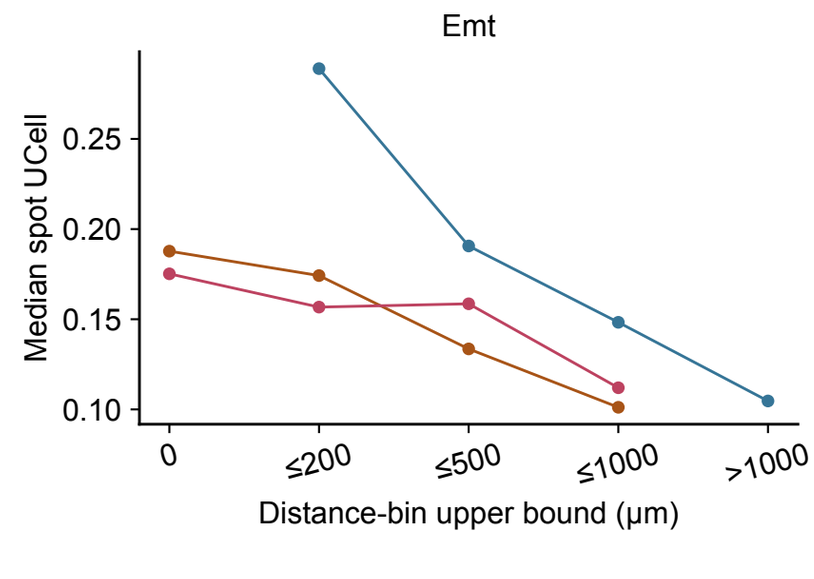


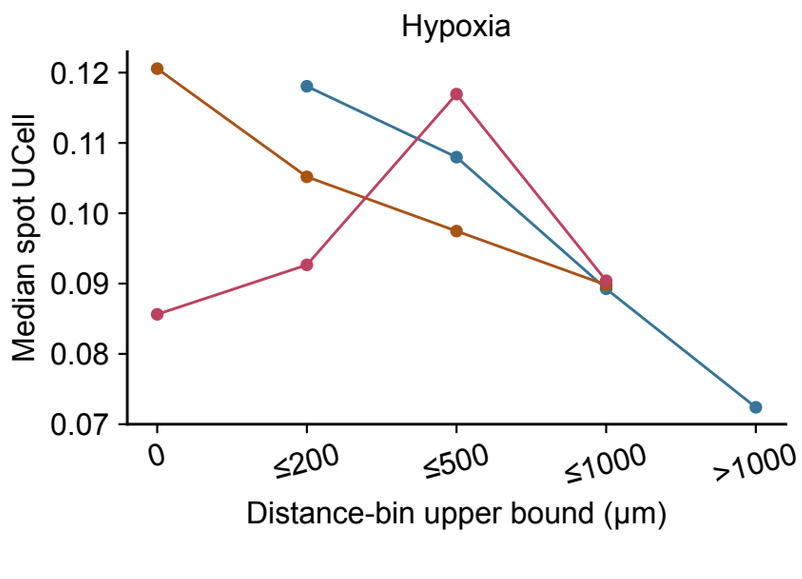
0 0.144

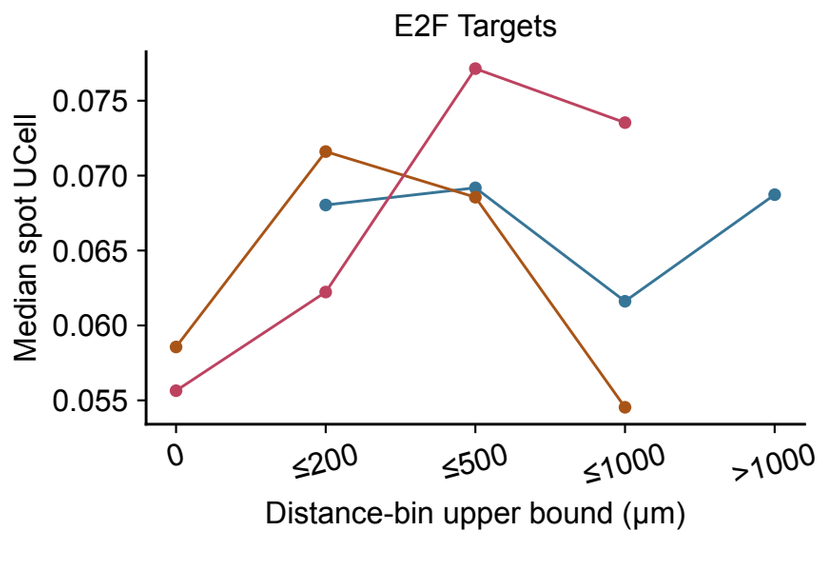
UCell score

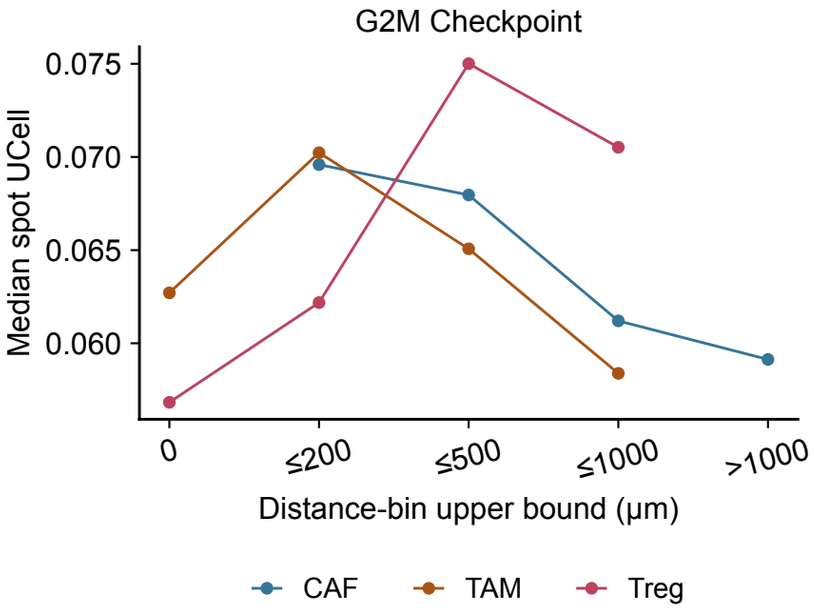
GSE319536_BC12 | Programs by target distance

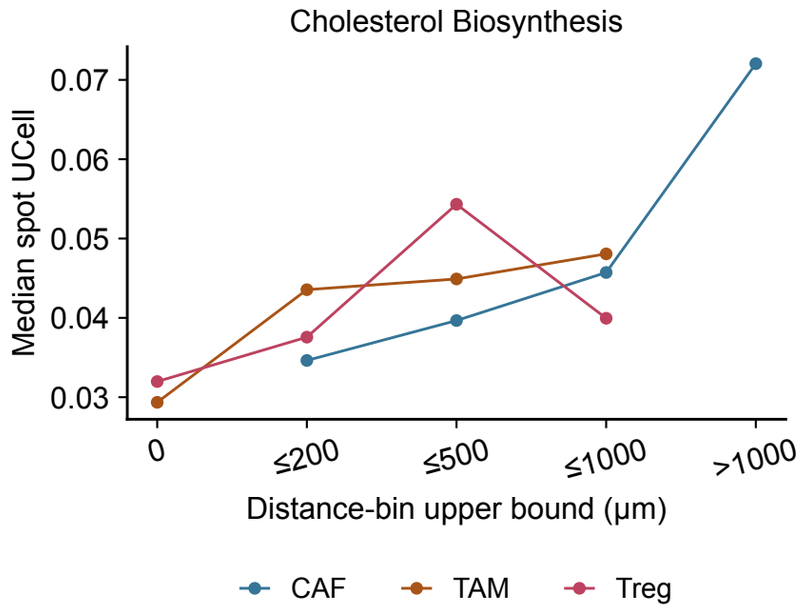
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

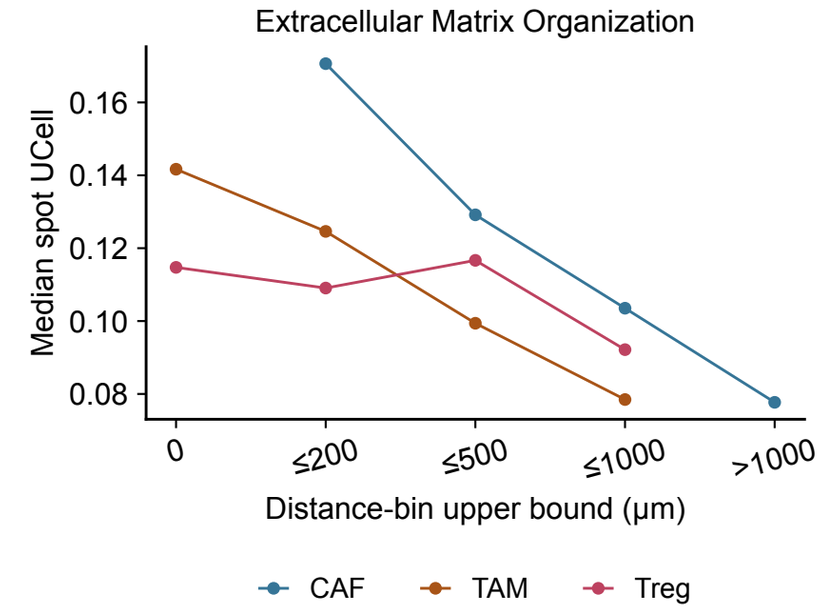






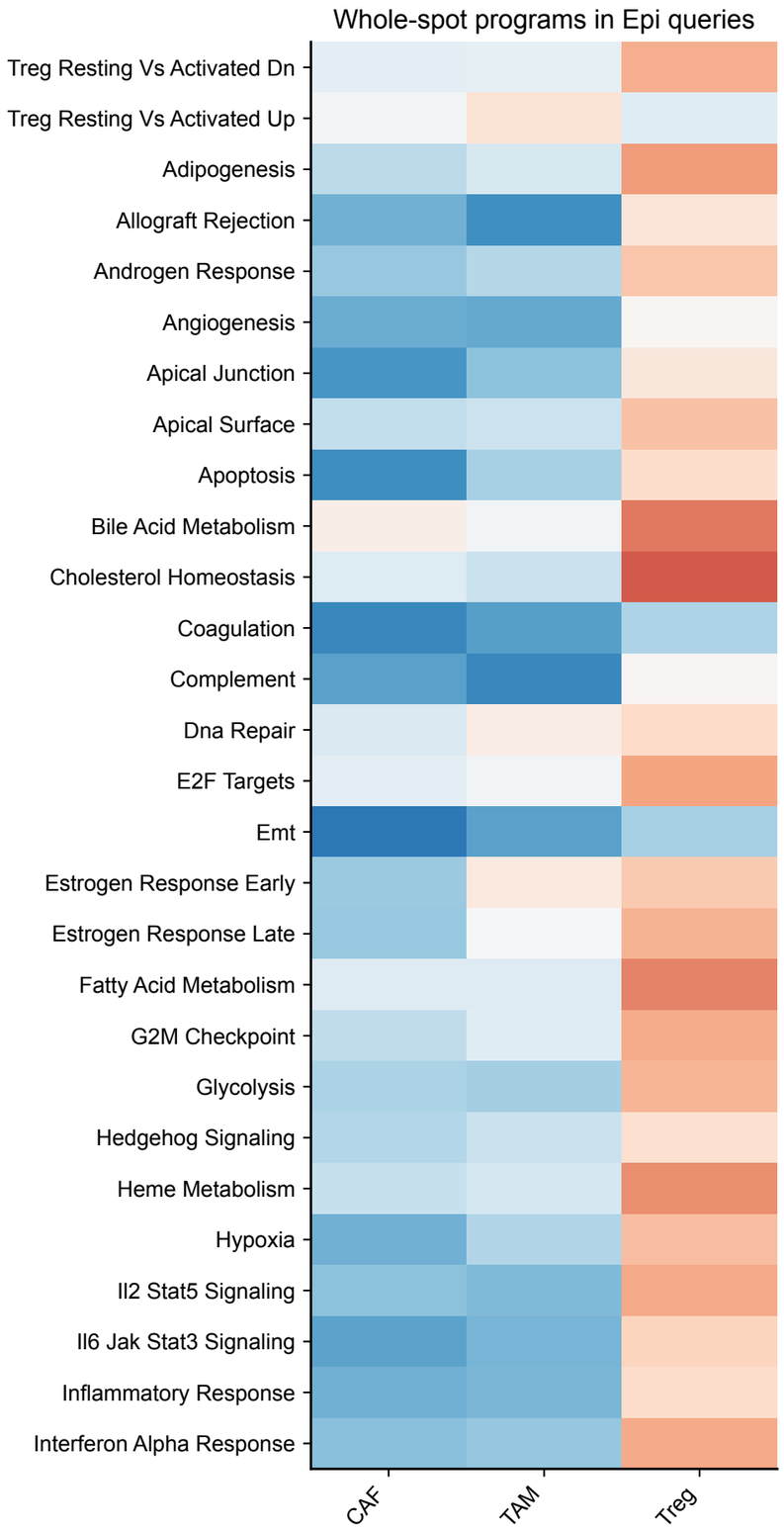


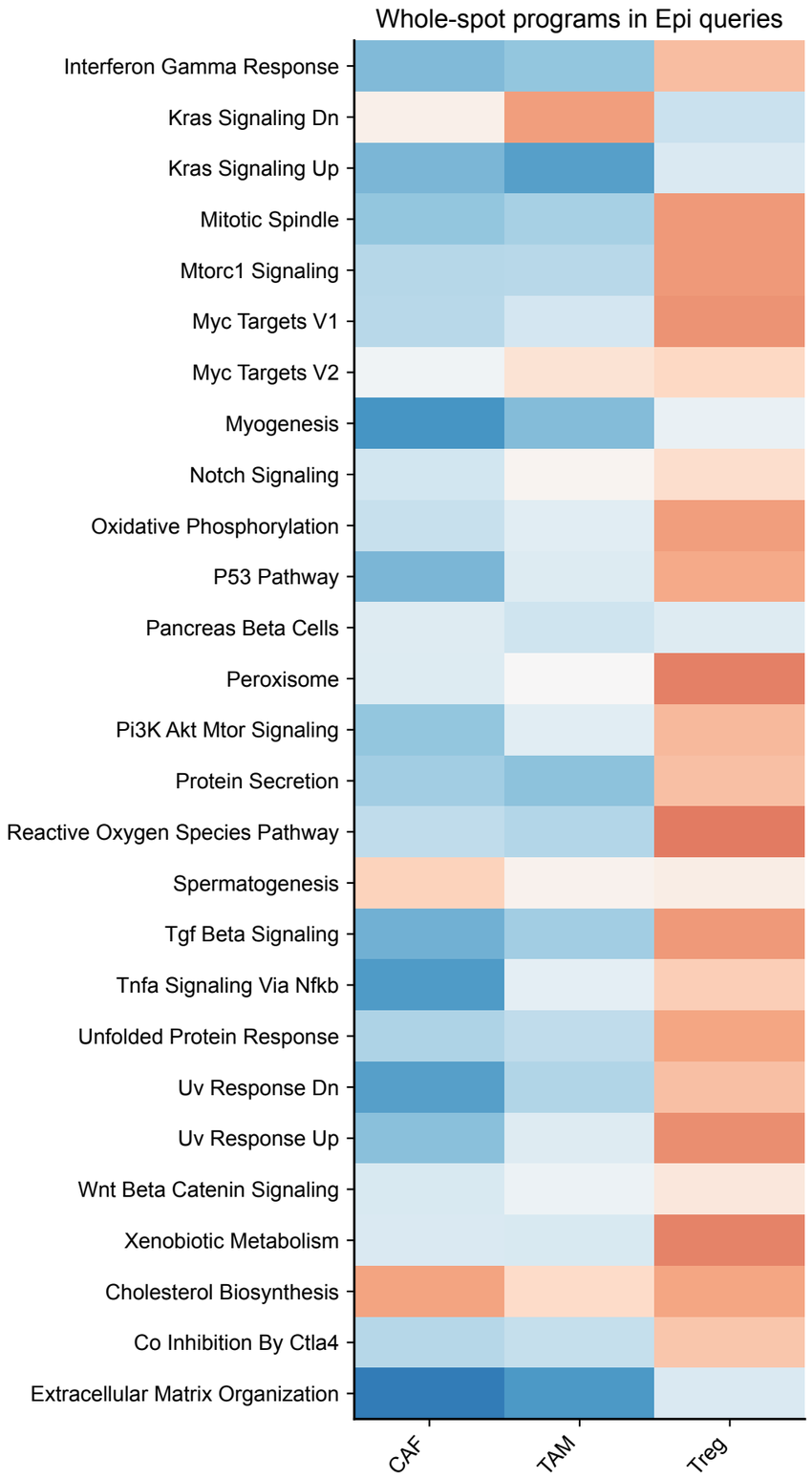




GSE319536_BC12 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

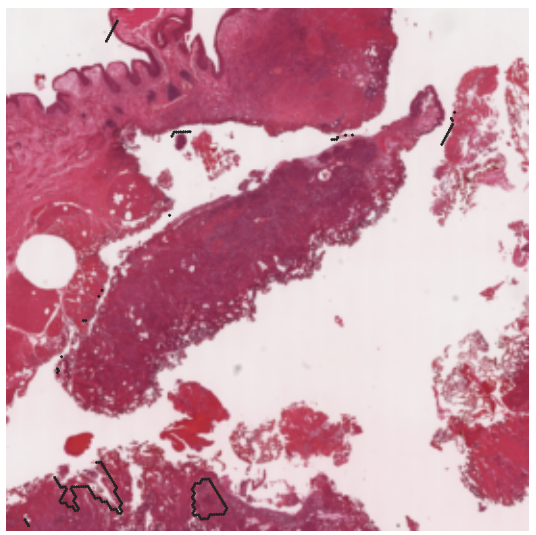




GSE319536_BC13 | Tissue context

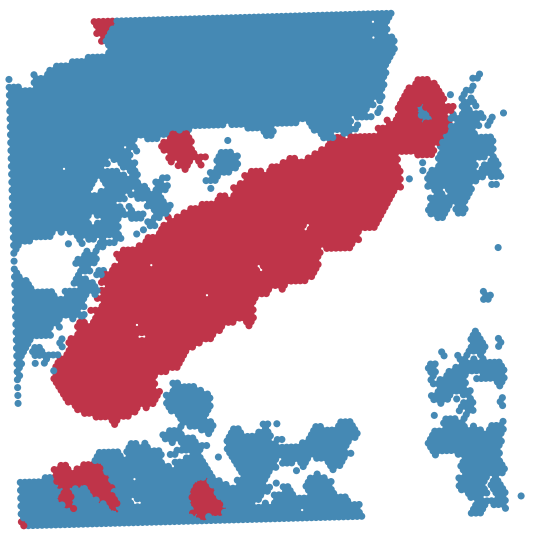
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



Black line: observed candidate outer edge

Final candidate location

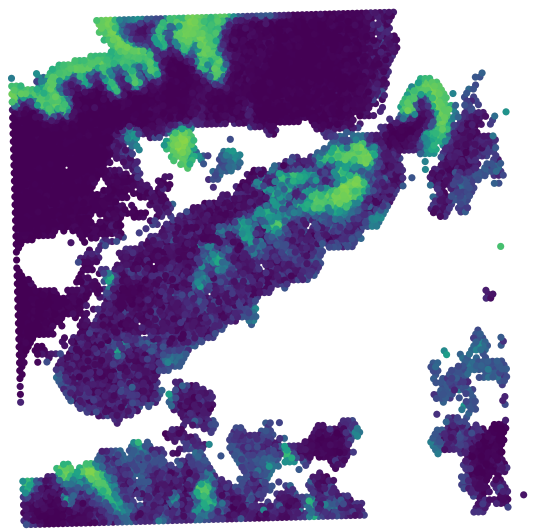


inside outside

Accepted R12 (original 0–8)

0 3 5 6 7
2 4

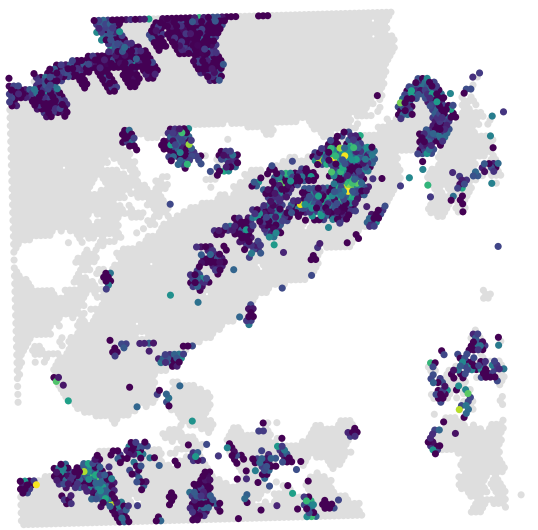
Epithelial support



0 1

Fraction of all 30

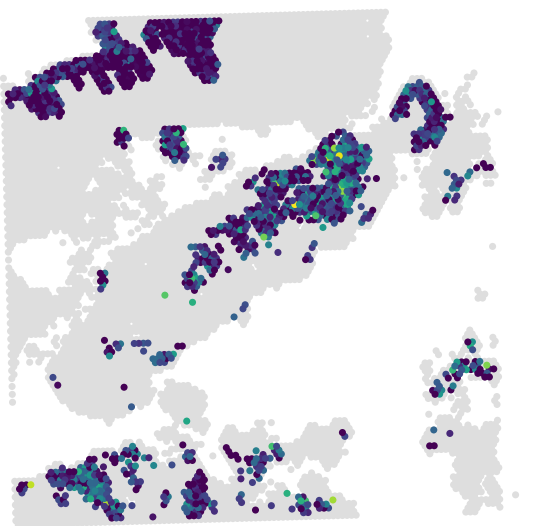
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



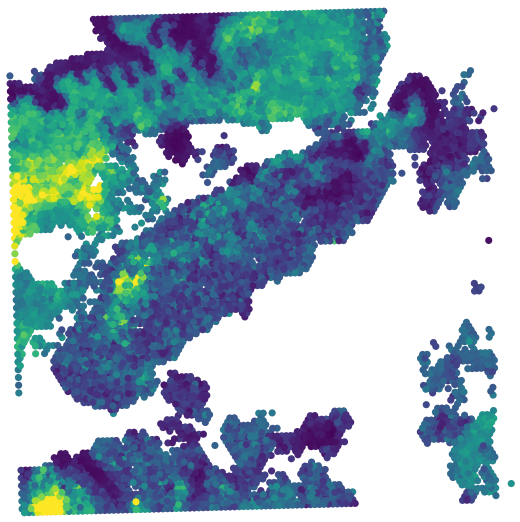
0 0.14

Matched-model AUCell

GSE319536_BC13 | Candidate neighborhoods

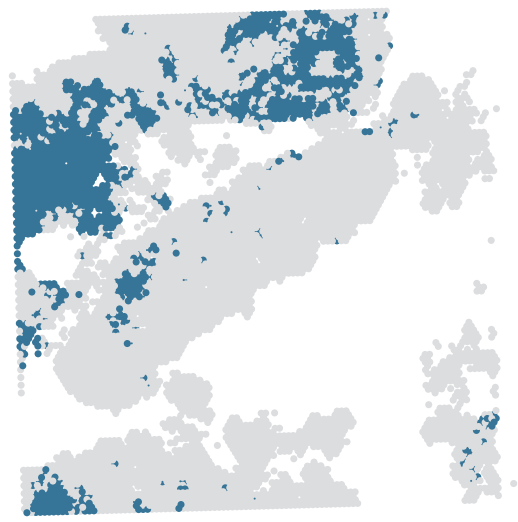
Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

CAF composition



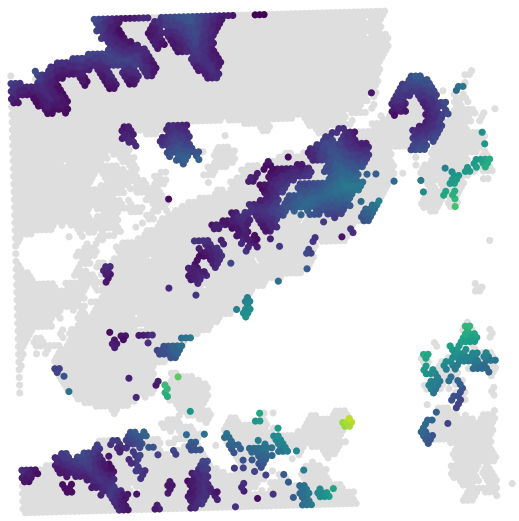
0 0.418
Fraction of all 30

CAF P80 targets



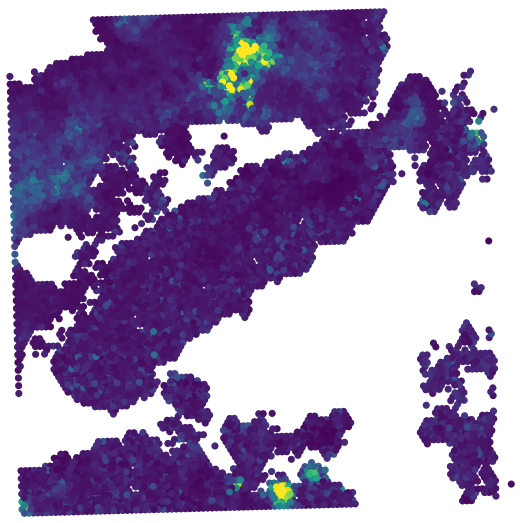
False True

CAF distance



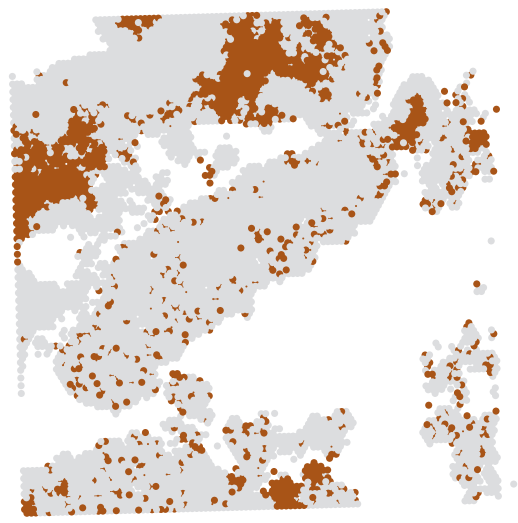
0 3.1e+03
Nearest spot center (µm)

TAM composition



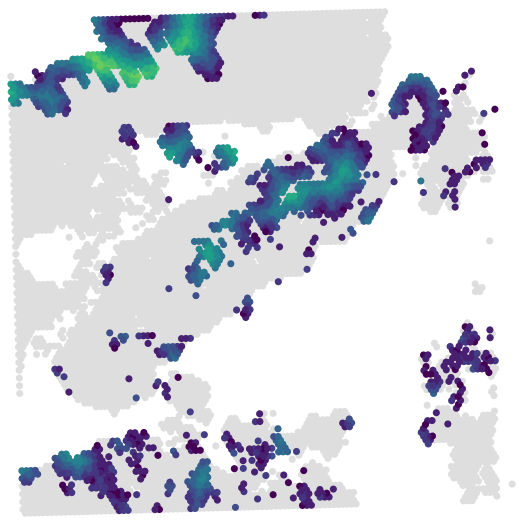
0 0.291
Fraction of all 30

TAM P80 targets



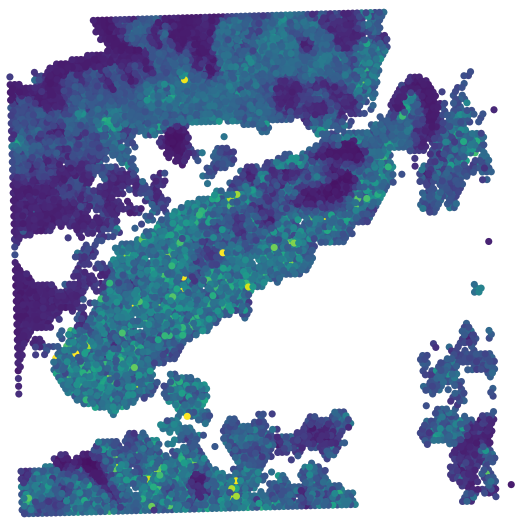
False True

TAM distance



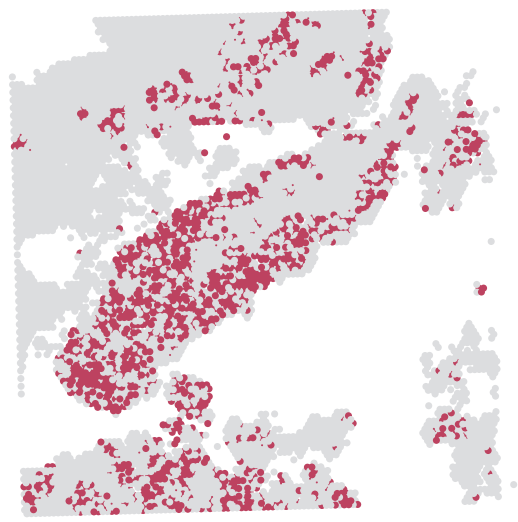
0 1.13e+03
Nearest spot center (µm)

Treg composition



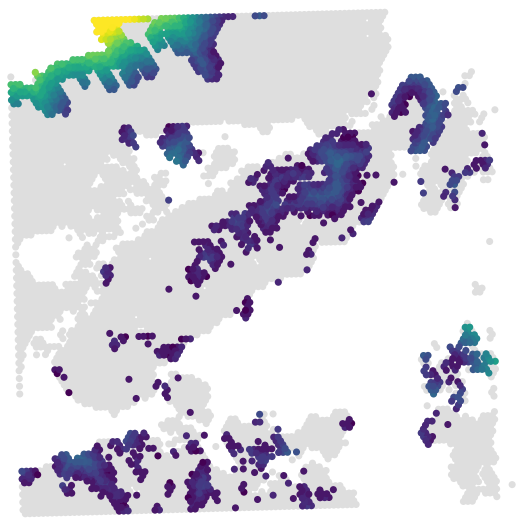
0 0.102
Fraction of all 30

Treg P80 targets



False True

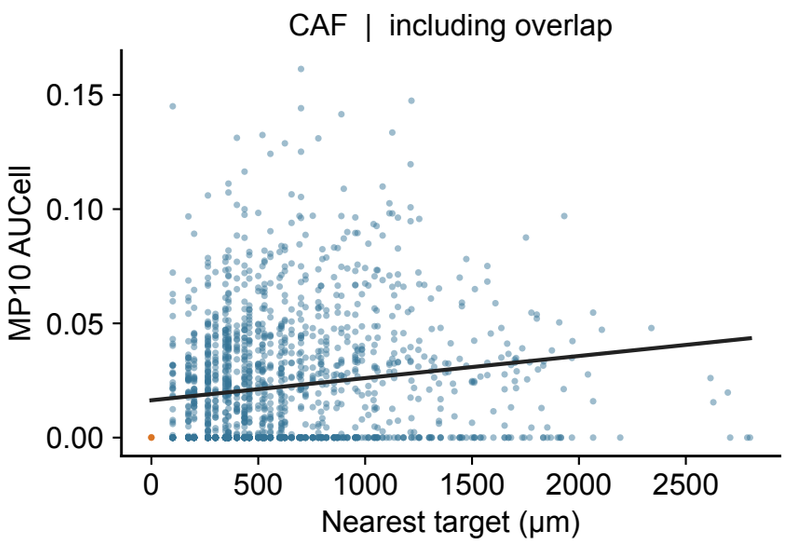
Treg distance



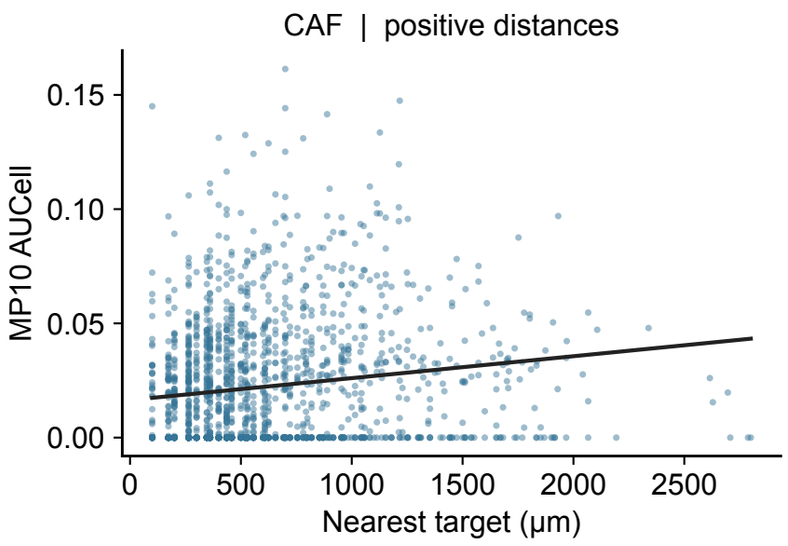
0 1.61e+03
Nearest spot center (µm)

GSE319536_BC13 | MP10 versus nearest target

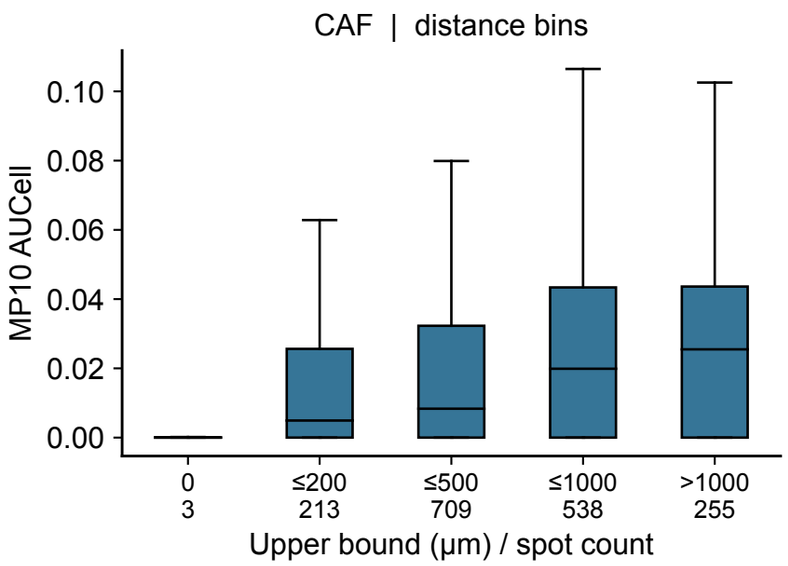
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



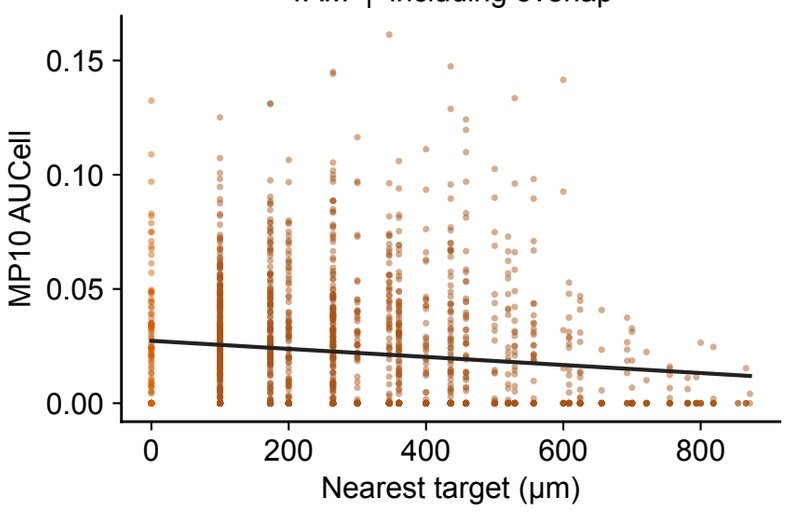
r +0.15 ρ +0.19
n 1718/1759 queries • 1552 targets



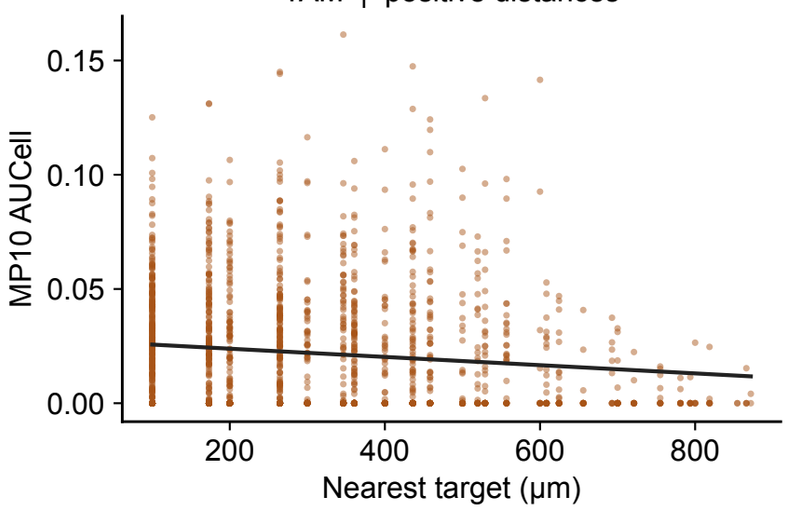
r +0.15 ρ +0.19
n 1715/1759 queries • 1552 targets



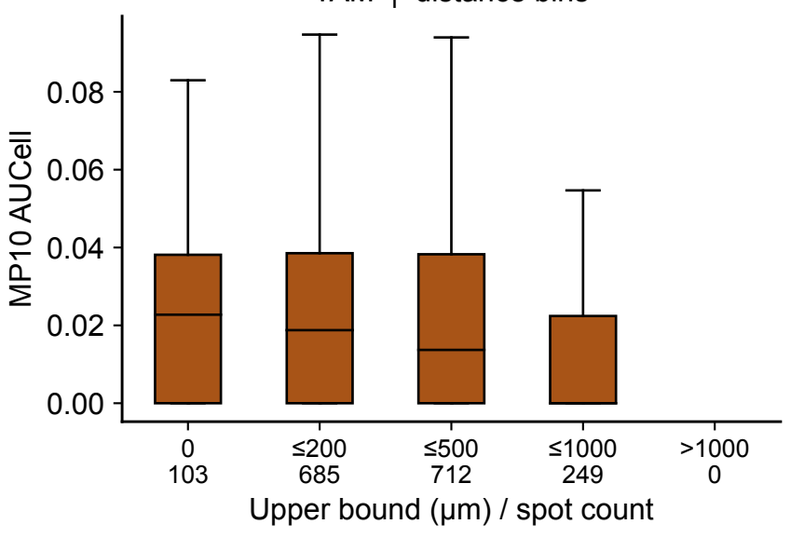
Orange = zero distance
Line = within-section linear fit



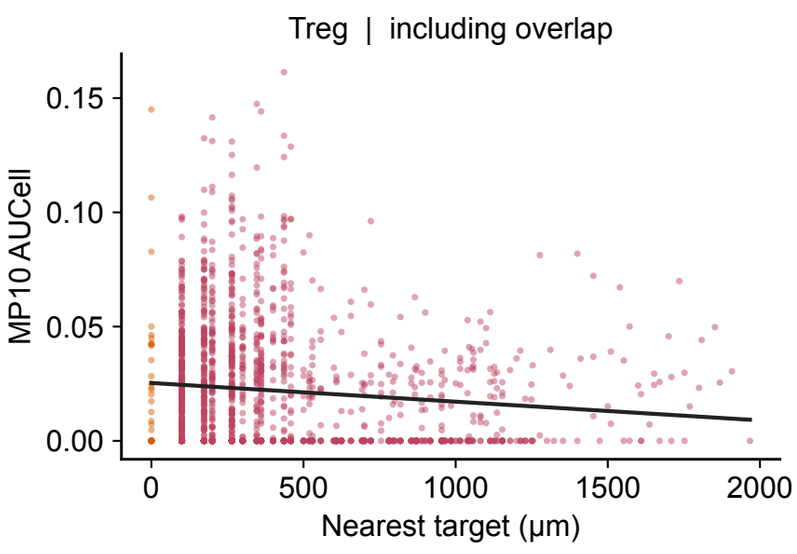
r -0.13 ρ -0.14
n 1749/1759 queries • 1552 targets



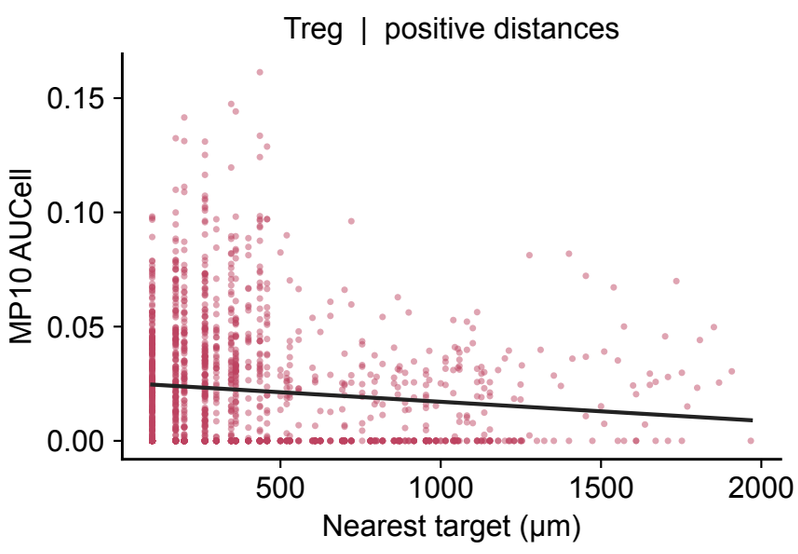
r -0.12 ρ -0.13
n 1646/1759 queries • 1552 targets



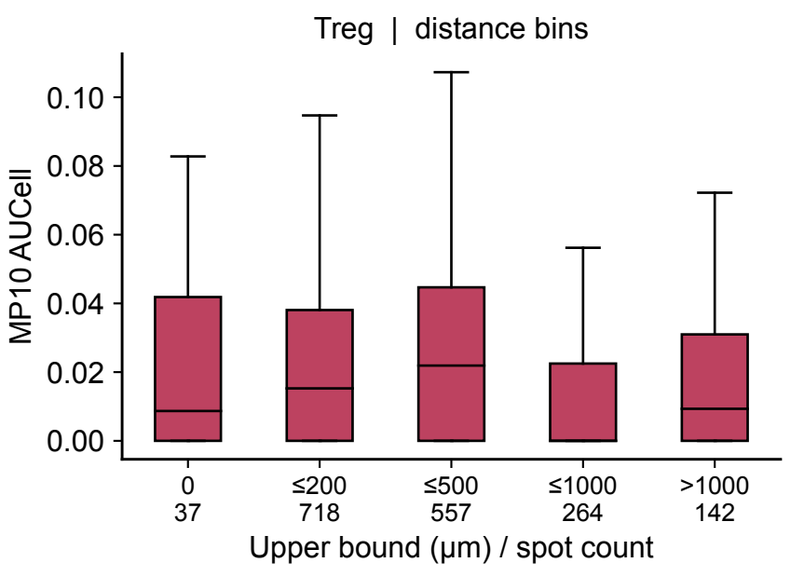
Orange = zero distance
Line = within-section linear fit



r -0.11 ρ -0.09
n 1718/1759 queries • 1552 targets



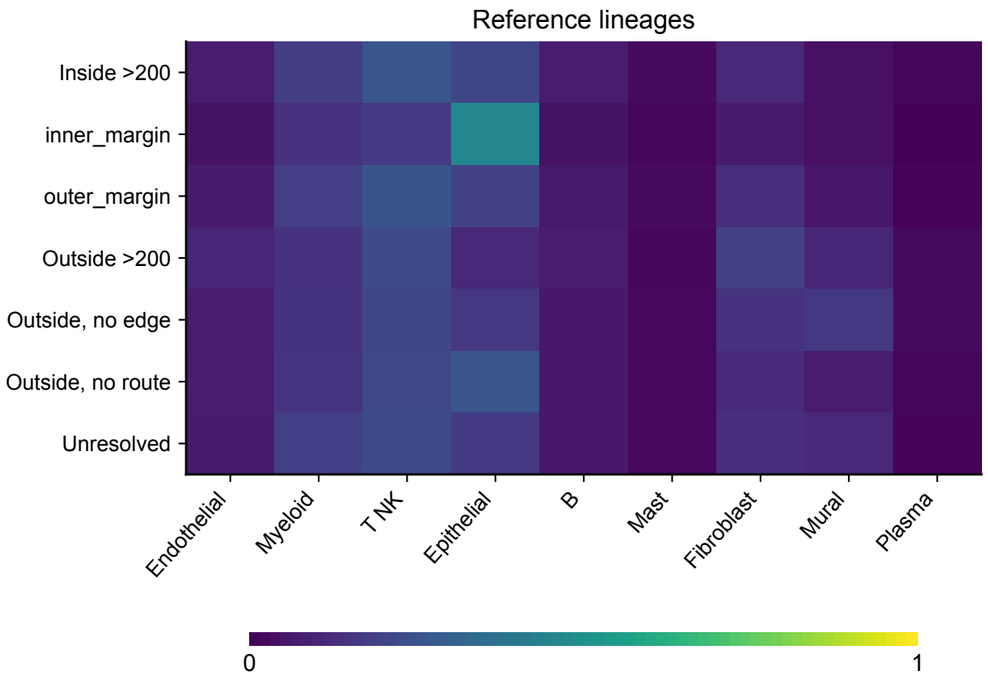
r -0.11 ρ -0.09
n 1681/1759 queries • 1552 targets

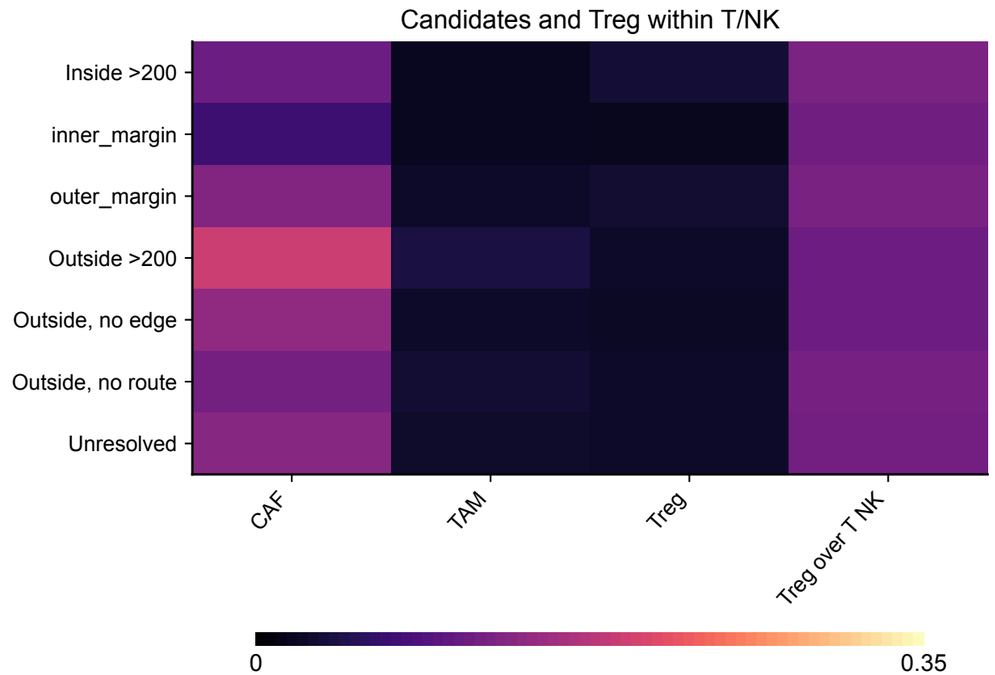


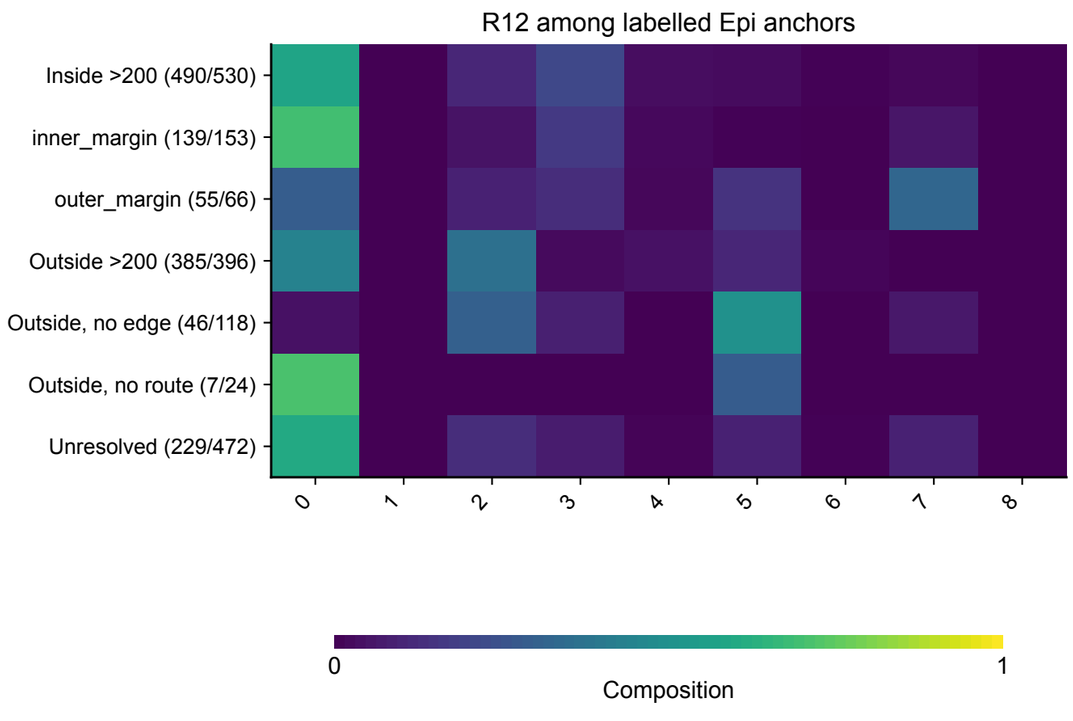
Orange = zero distance
Line = within-section linear fit

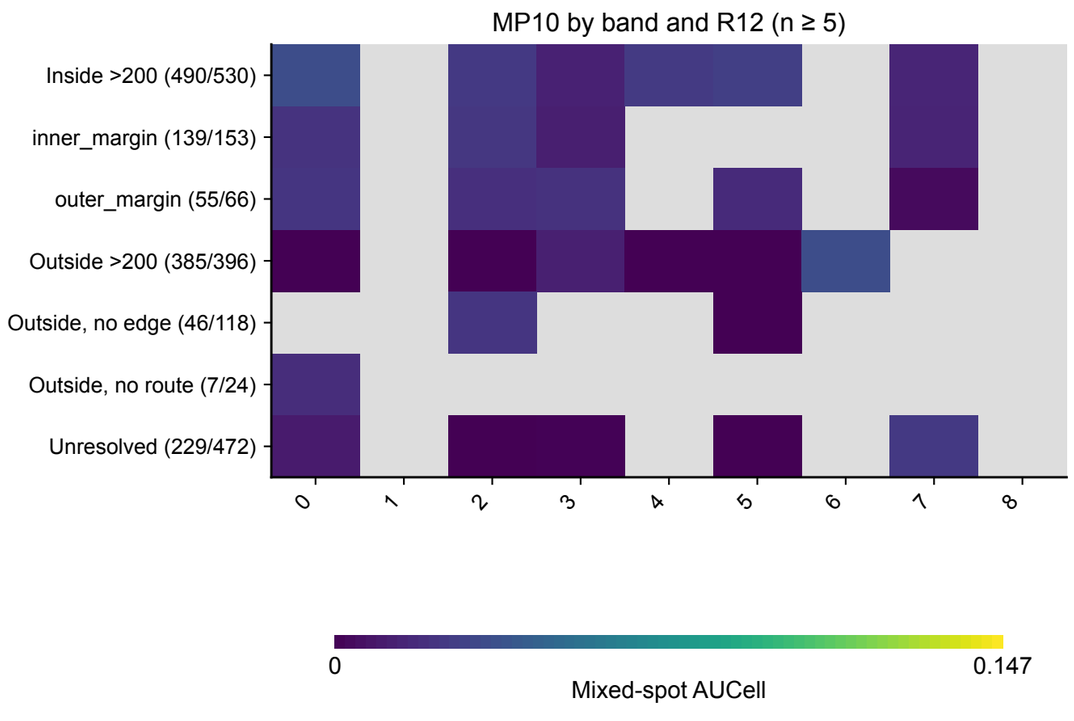
GSE319536_BC13 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





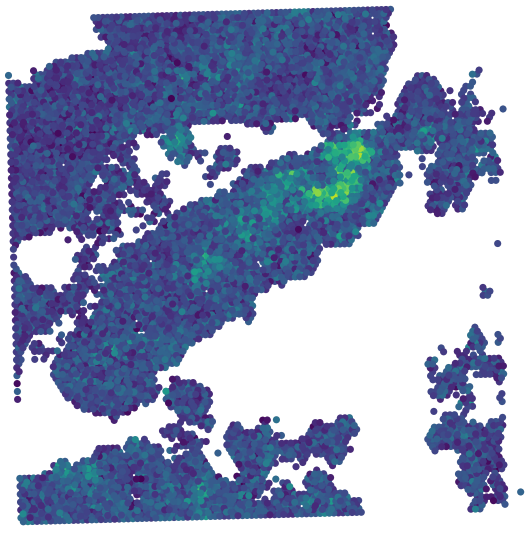




GSE319536_BC13 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

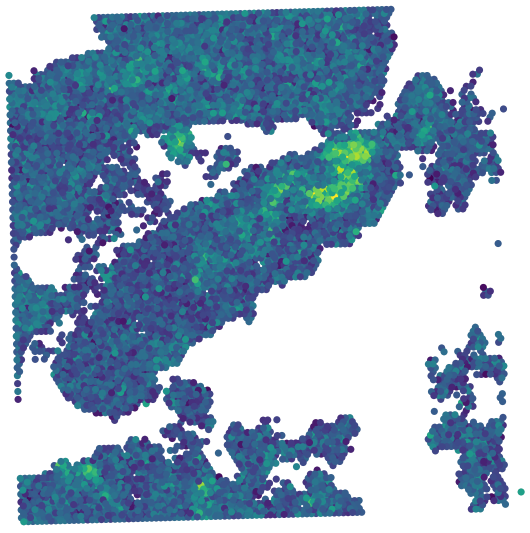
E2F Targets



0 0.152

UCell score

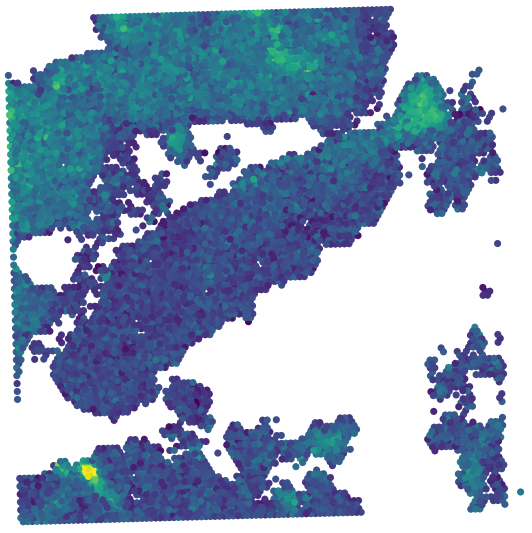
G2M Checkpoint



0 0.134

UCell score

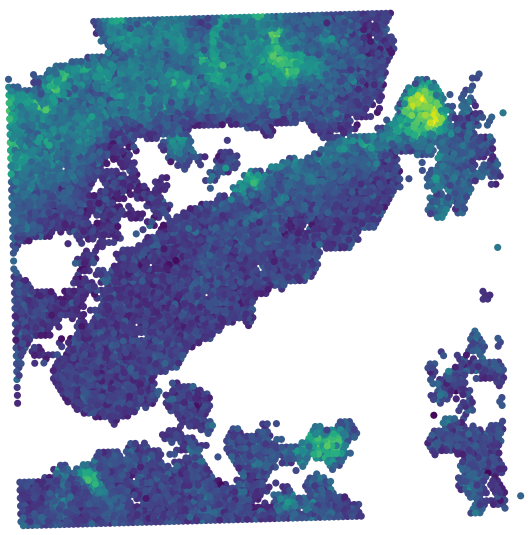
Hypoxia



0 0.211

UCell score

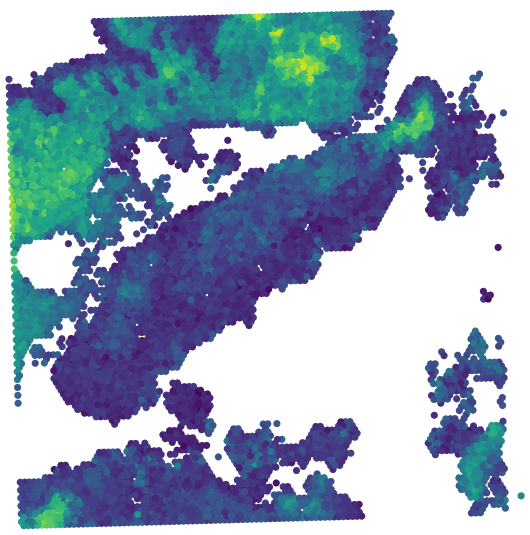
Tnfa Signaling Via Nfkb



0 0.275

UCell score

Emt



0 0.386

UCell score

Interferon Gamma Response

0 0.275

UCell score

GSE319536_BC13 | Immune, lipid and matrix programs

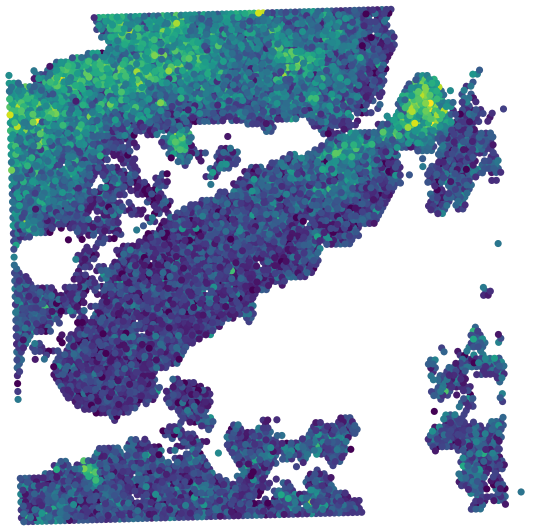
Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Il2 Stat5 Signaling

0 0.144

UCell score

Tgf Beta Signaling



0 0.232

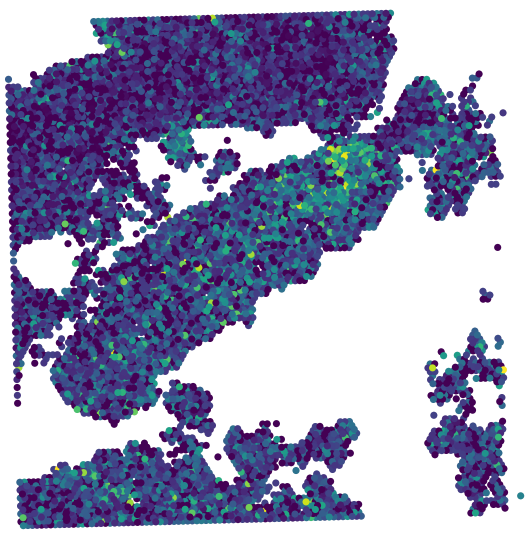
UCell score

Cholesterol Homeostasis

0 0.21

UCell score

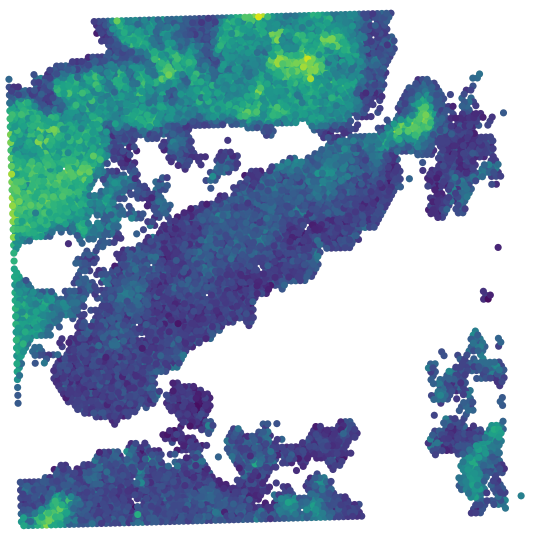
Cholesterol Biosynthesis



0 0.166

UCell score

Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4

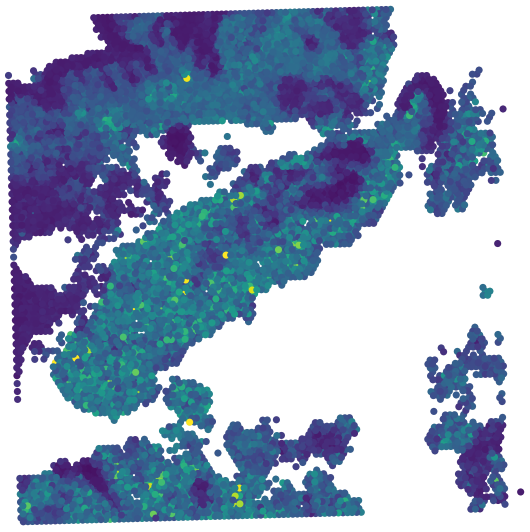
0 0.151

UCell score

GSE319536_BC13 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

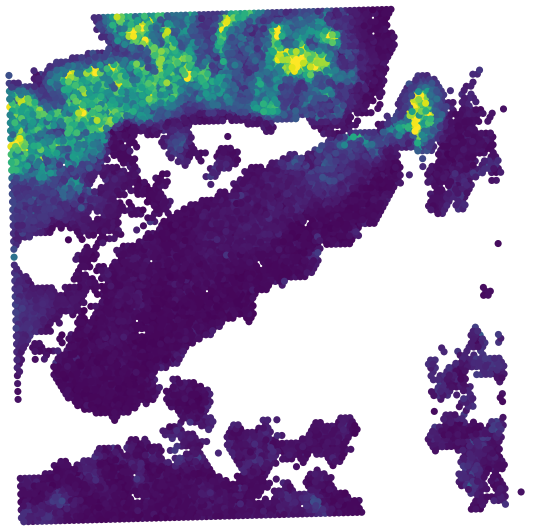
Treg composition



0 0.102

Fraction of all 30

Treg estimated abundance



0 2.85

Posterior mean estimate

Treg Resting Vs Activated Dn

0 0.0778

UCell score

Treg Resting Vs Activated Up

0 0.0609

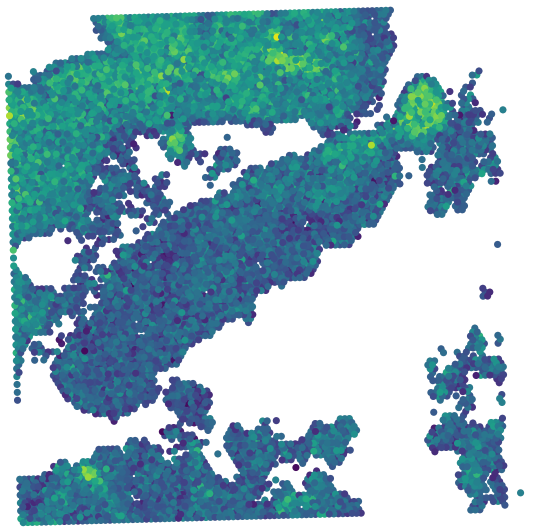
UCell score

Co Inhibition By Ctla4

0 0.151

UCell score

Il2 Stat5 Signaling

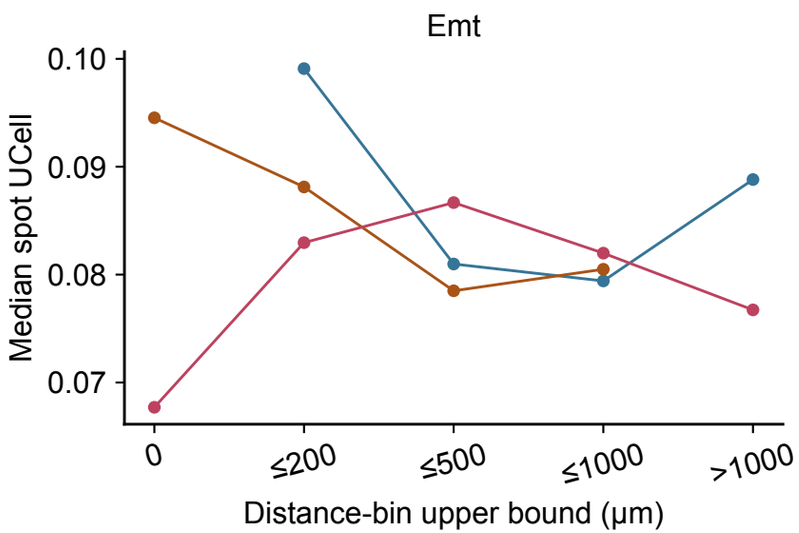


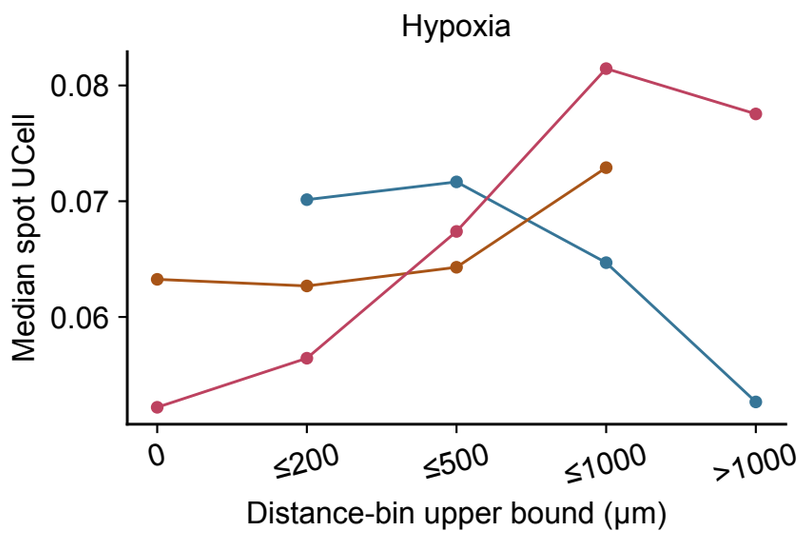
0 0.144

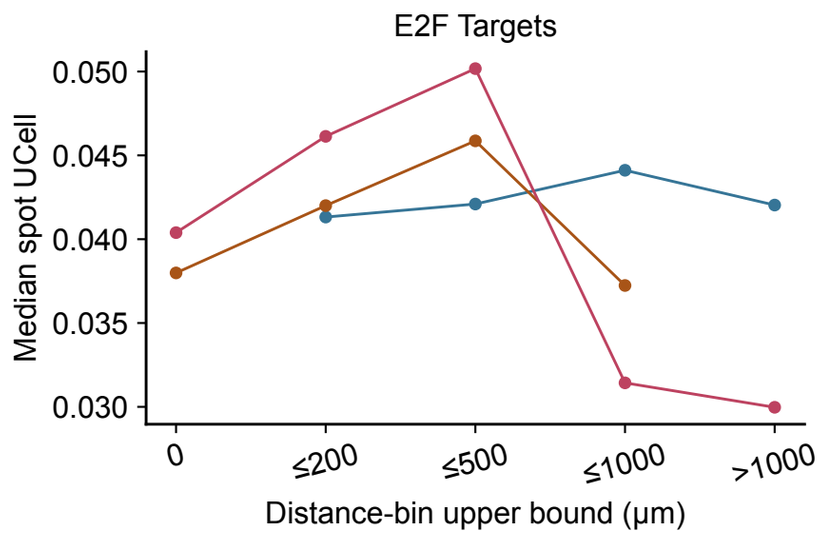
UCell score

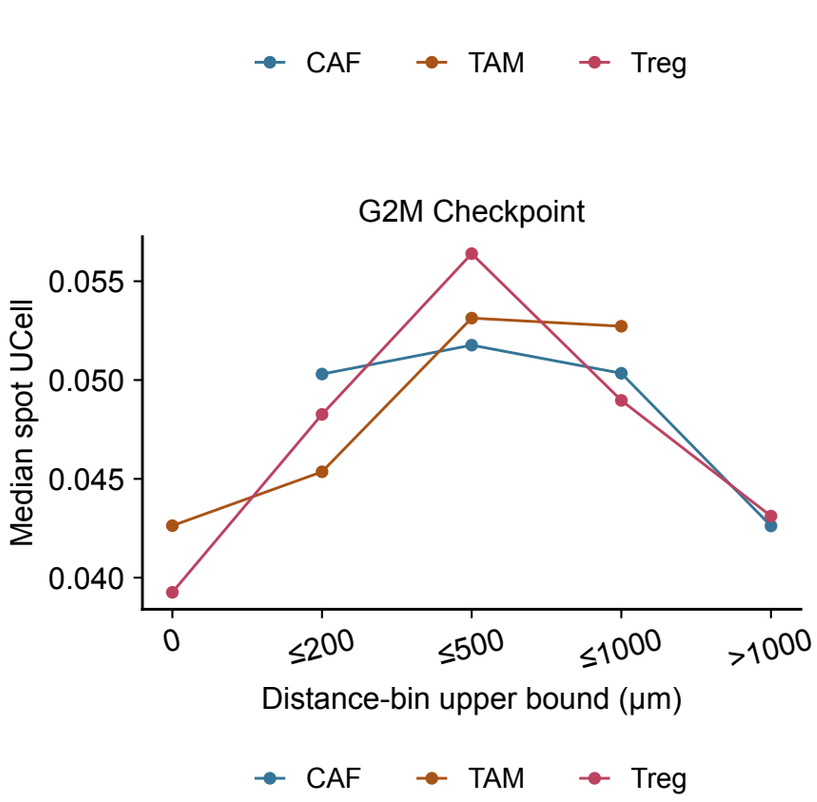
GSE319536_BC13 | Programs by target distance

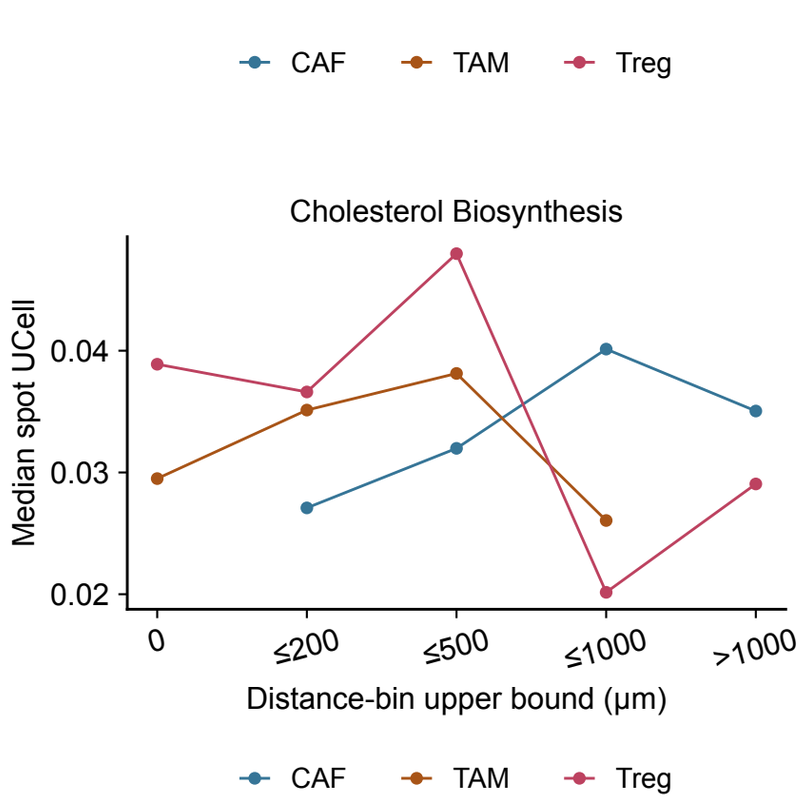
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

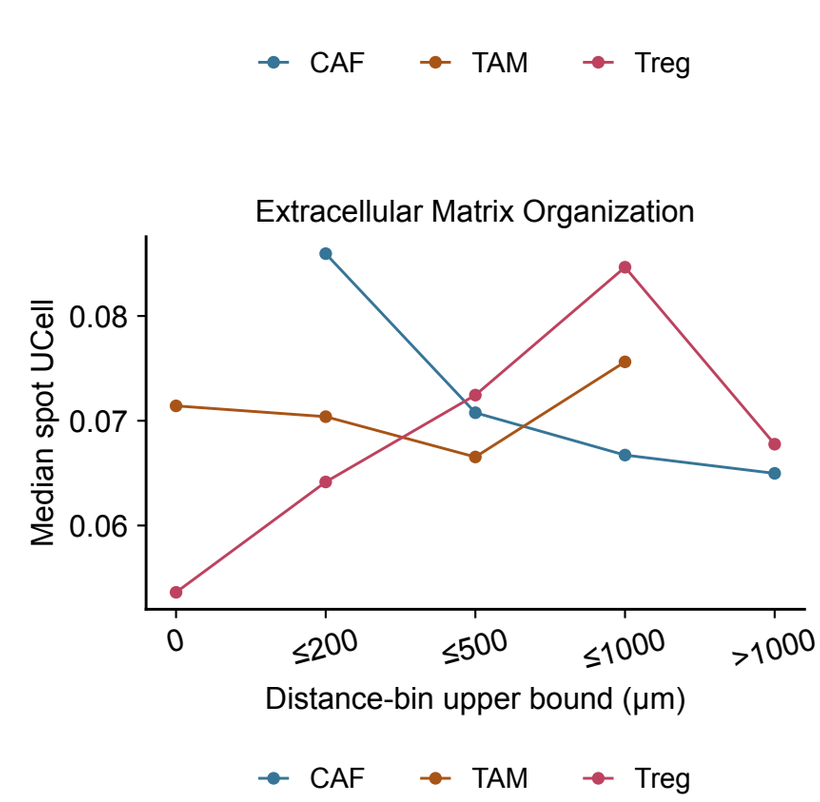






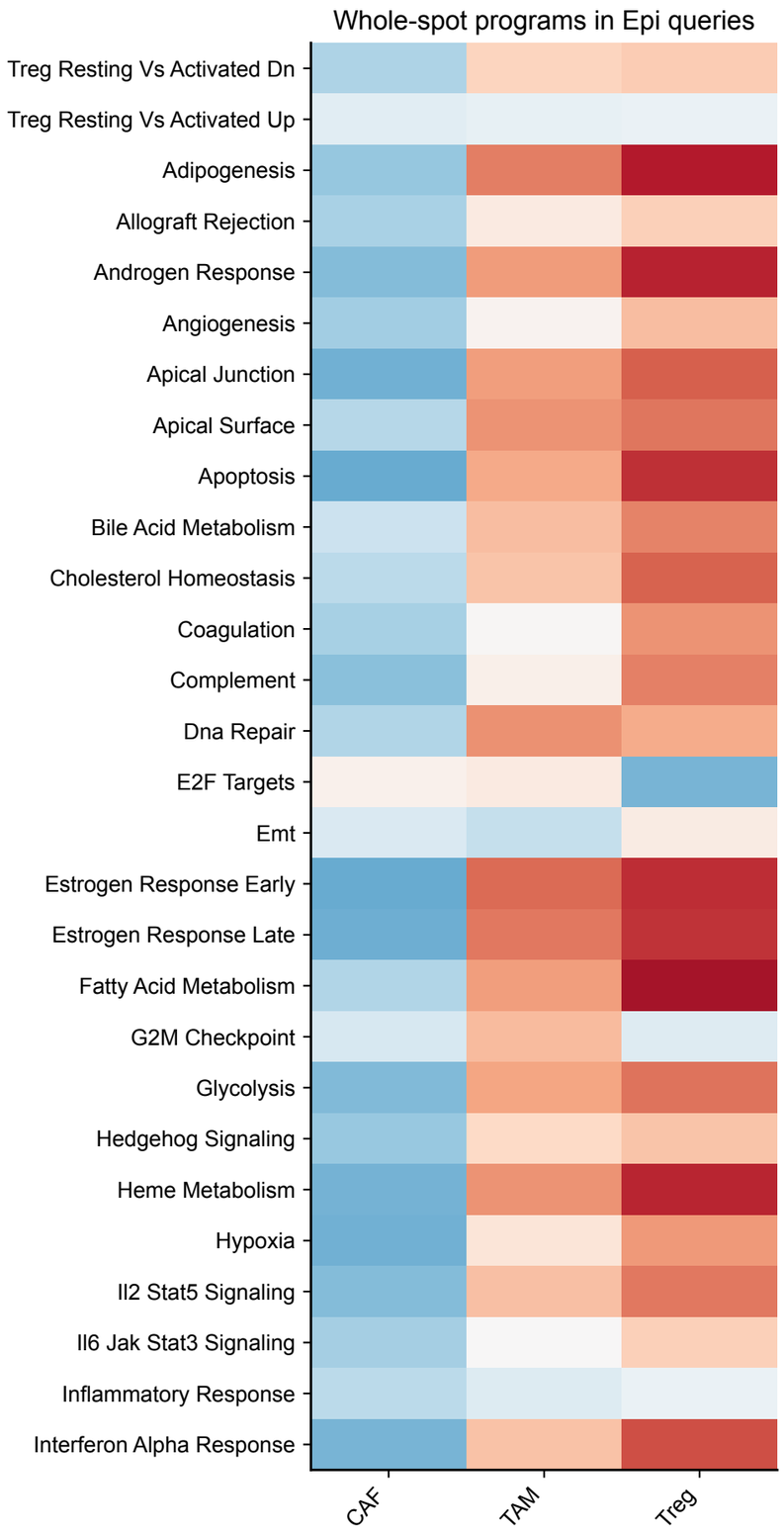


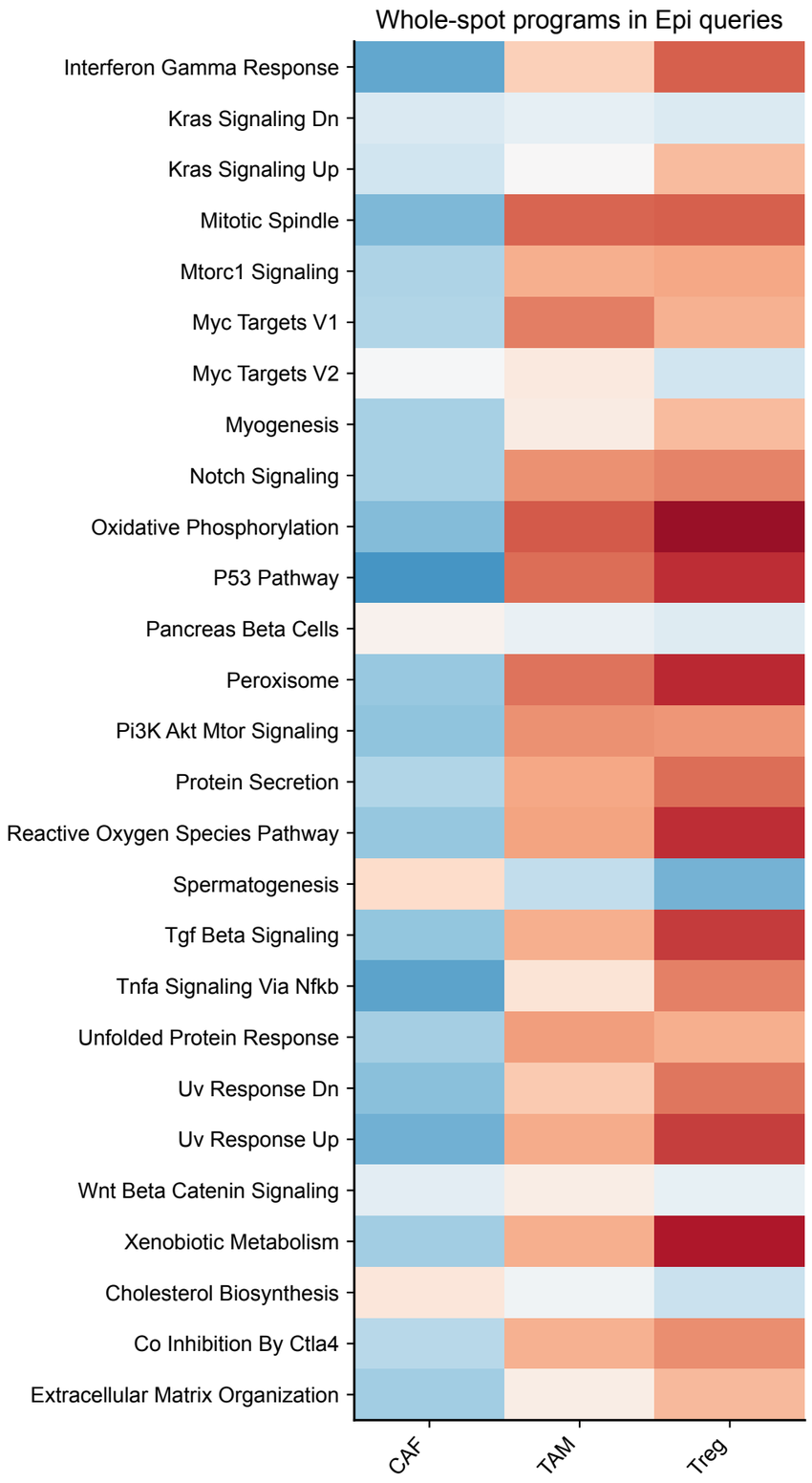




GSE319536_BC13 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

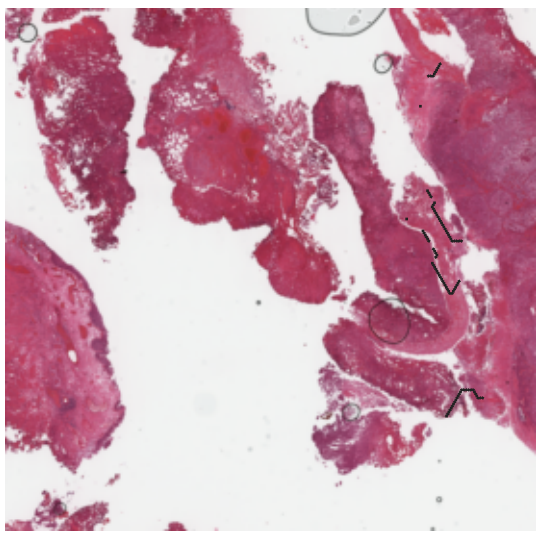




GSE319536_BC14 | Tissue context

Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge

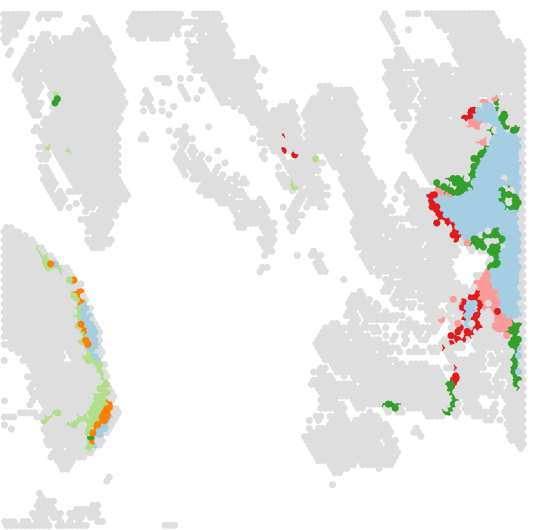


Black line: observed candidate outer edge

Final candidate location

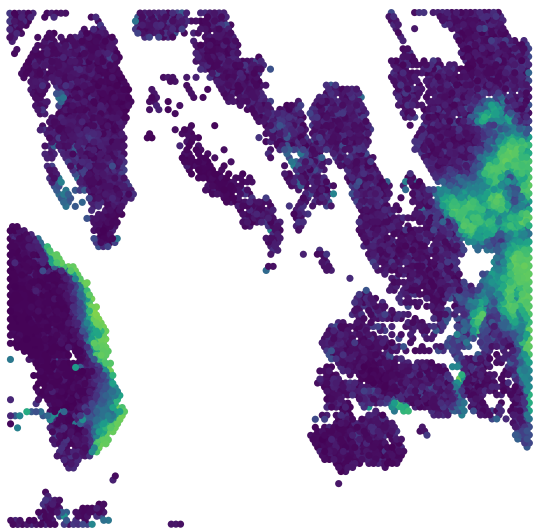
inside outside

Accepted R12 (original 0–8)



0 3 5 6 7 2

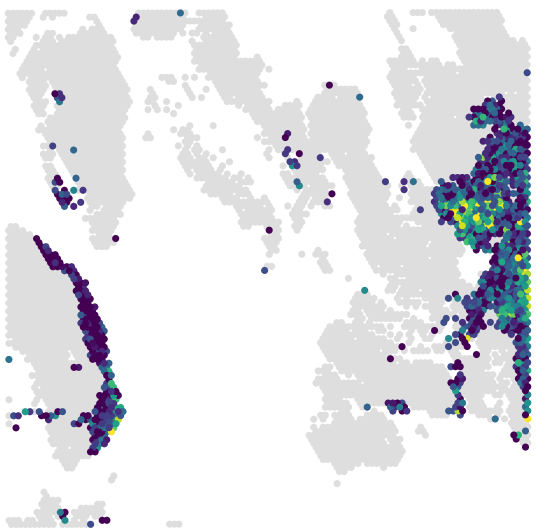
Epithelial support



0 1

Fraction of all 30

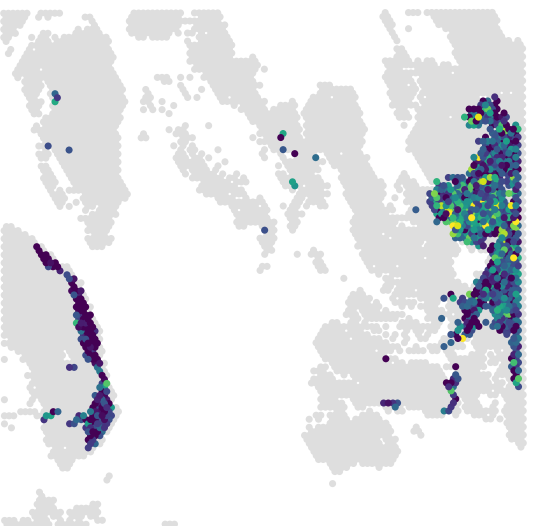
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)

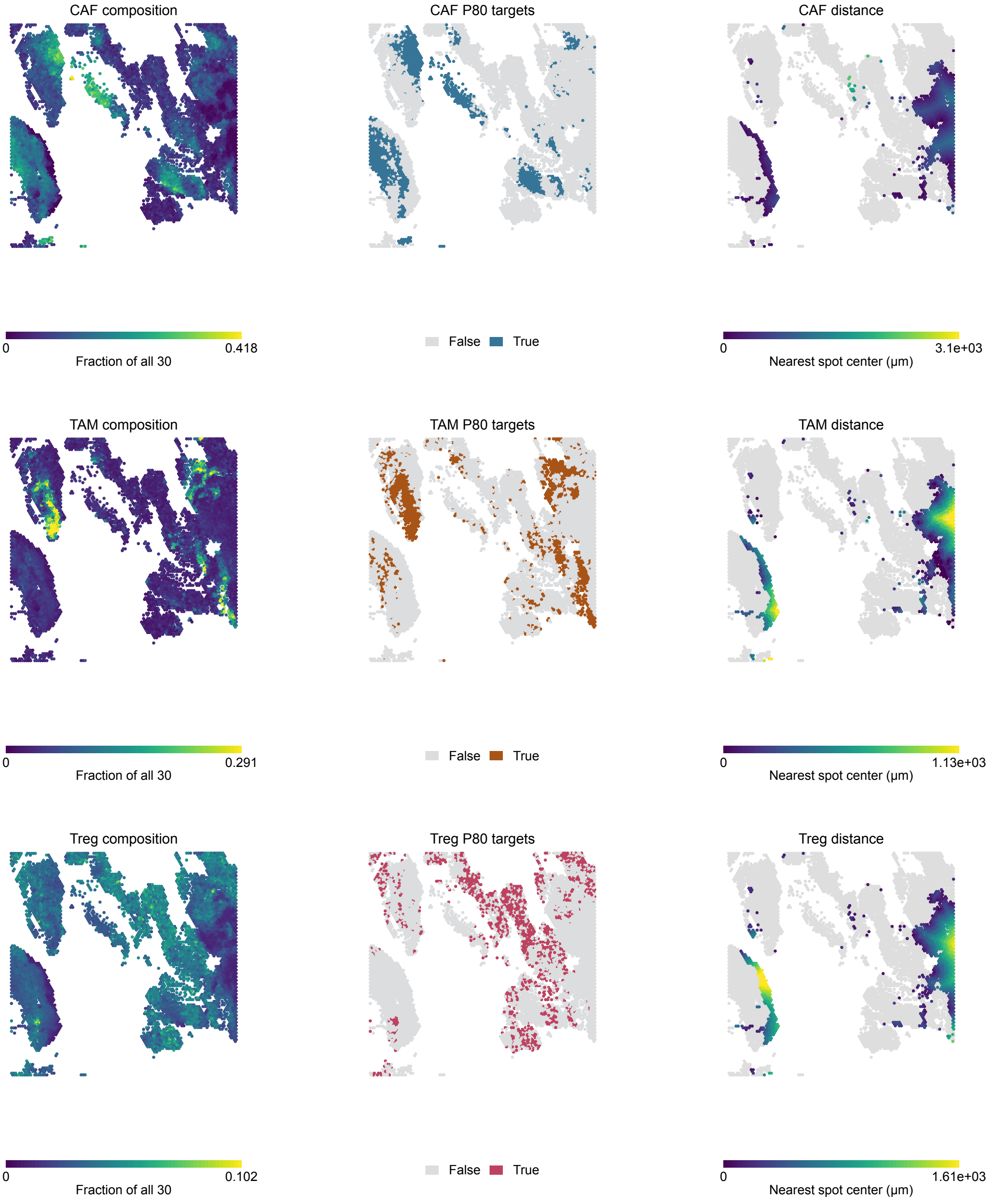


0 0.14

Matched-model AUCell

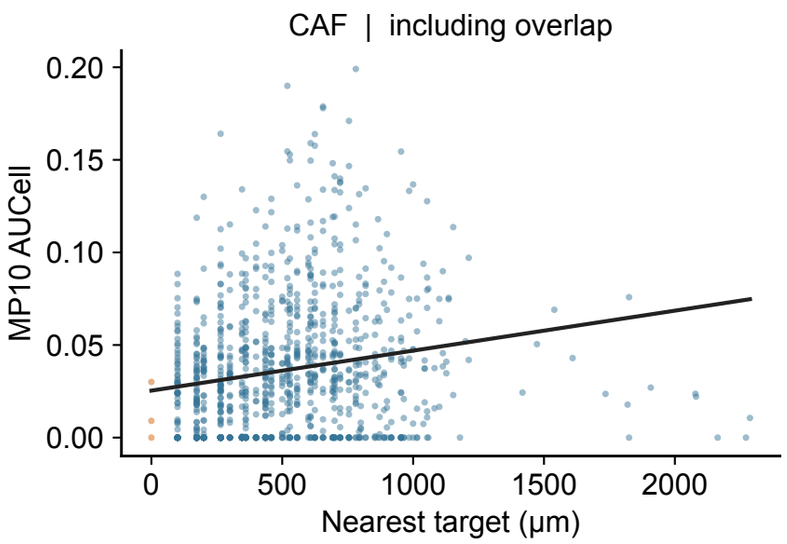
GSE319536_BC14 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

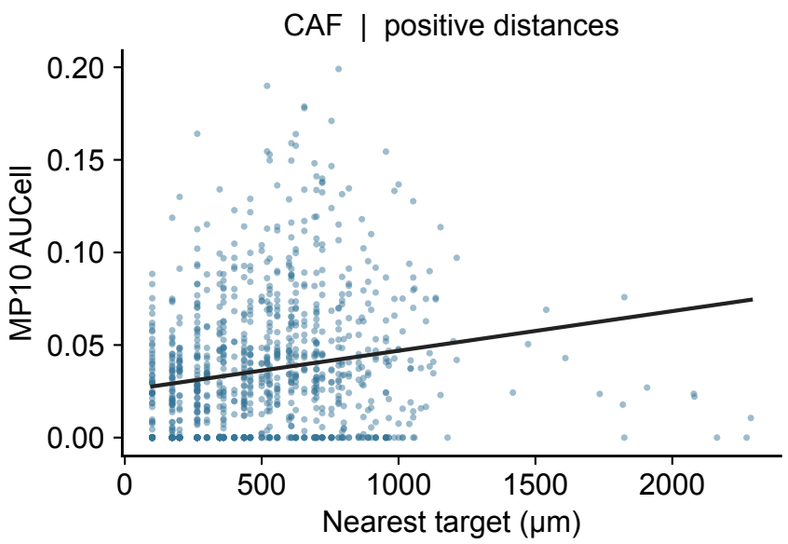


GSE319536_BC14 | MP10 versus nearest target

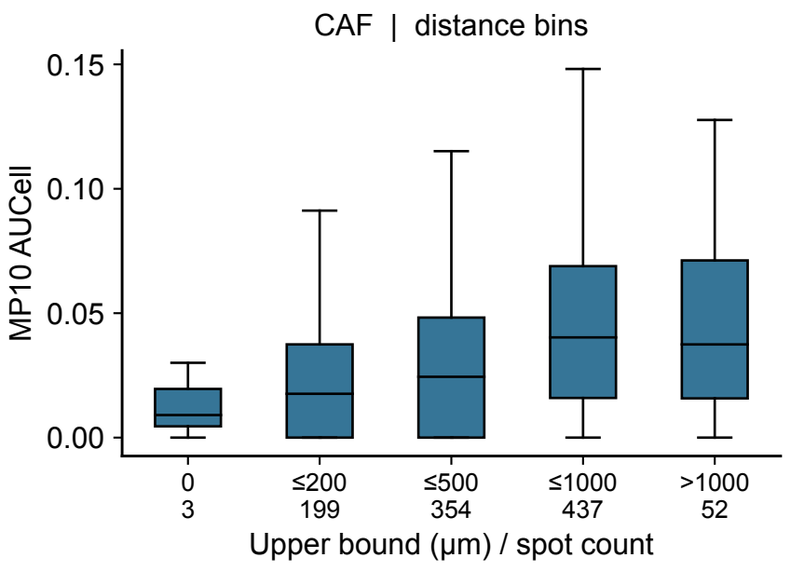
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



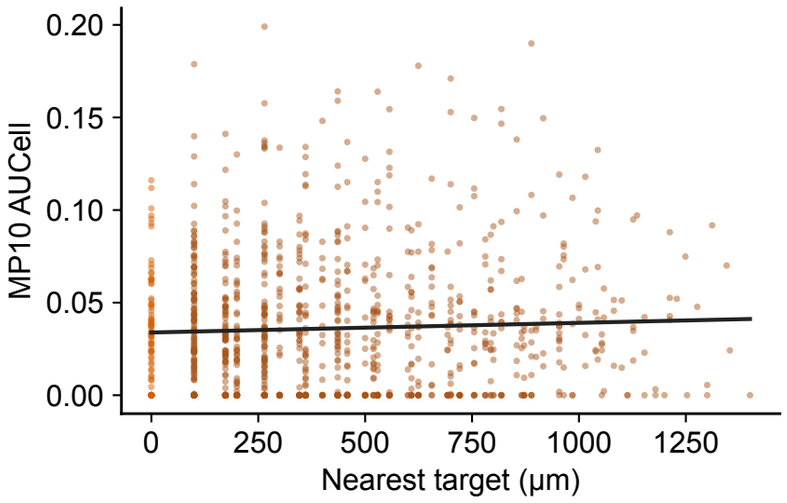
r +0.18 ρ +0.24
n 1045/1066 queries • 1173 targets



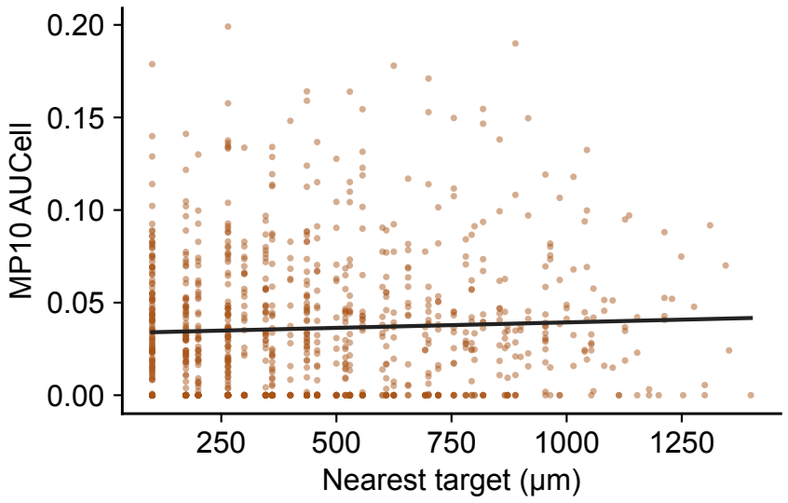
r +0.18 ρ +0.24
n 1042/1066 queries • 1173 targets



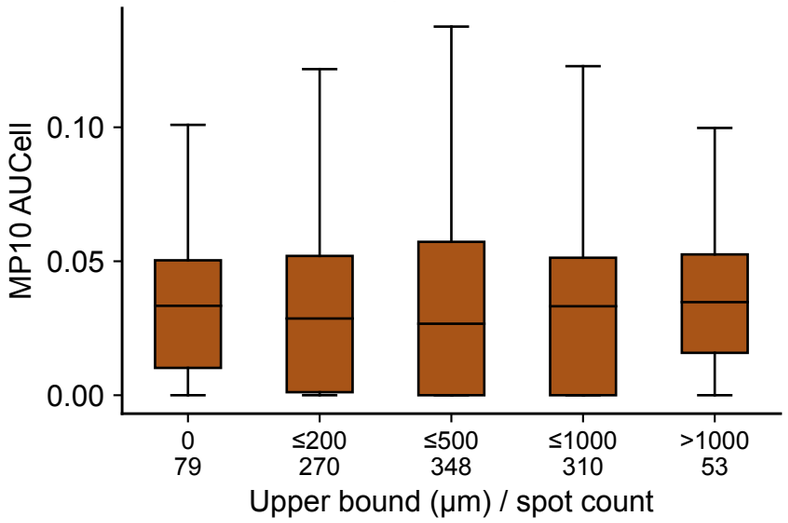
Orange = zero distance
Line = within-section linear fit



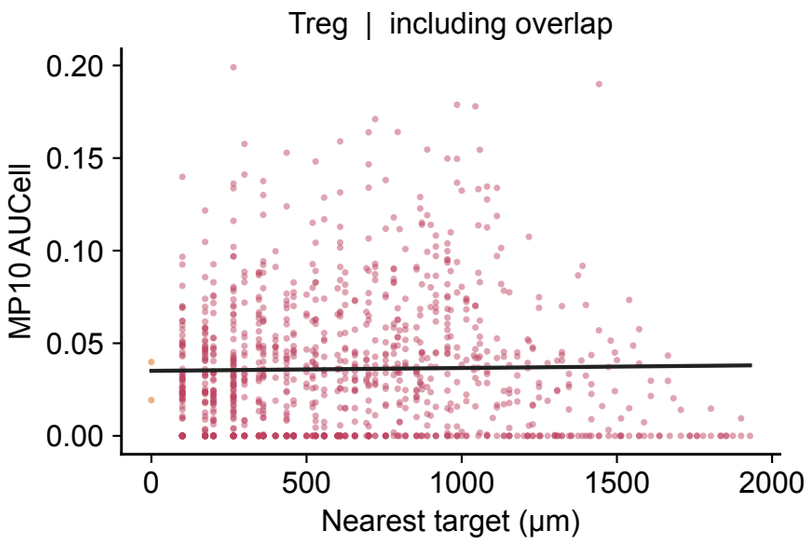
r +0.04 ρ +0.02
n 1060/1066 queries • 1173 targets



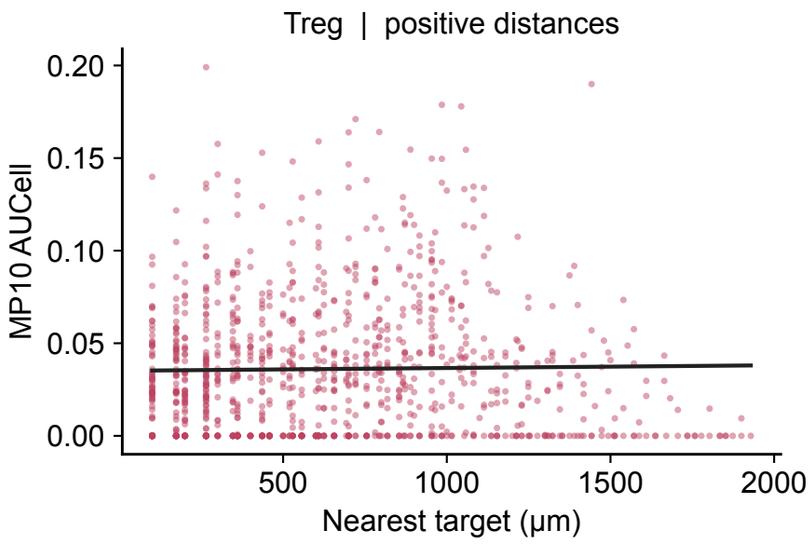
r +0.05 ρ +0.03
n 981/1066 queries • 1173 targets



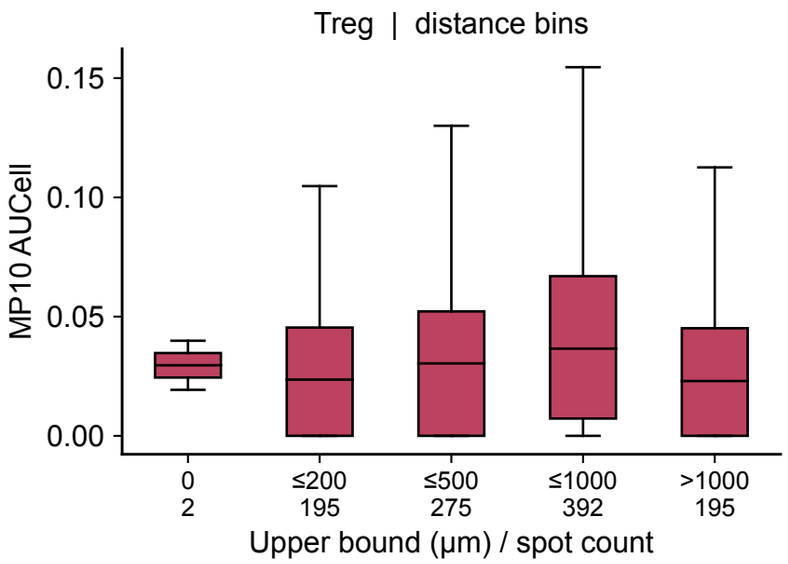
Orange = zero distance
Line = within-section linear fit



r +0.02 ρ +0.05
n 1059/1066 queries • 1173 targets



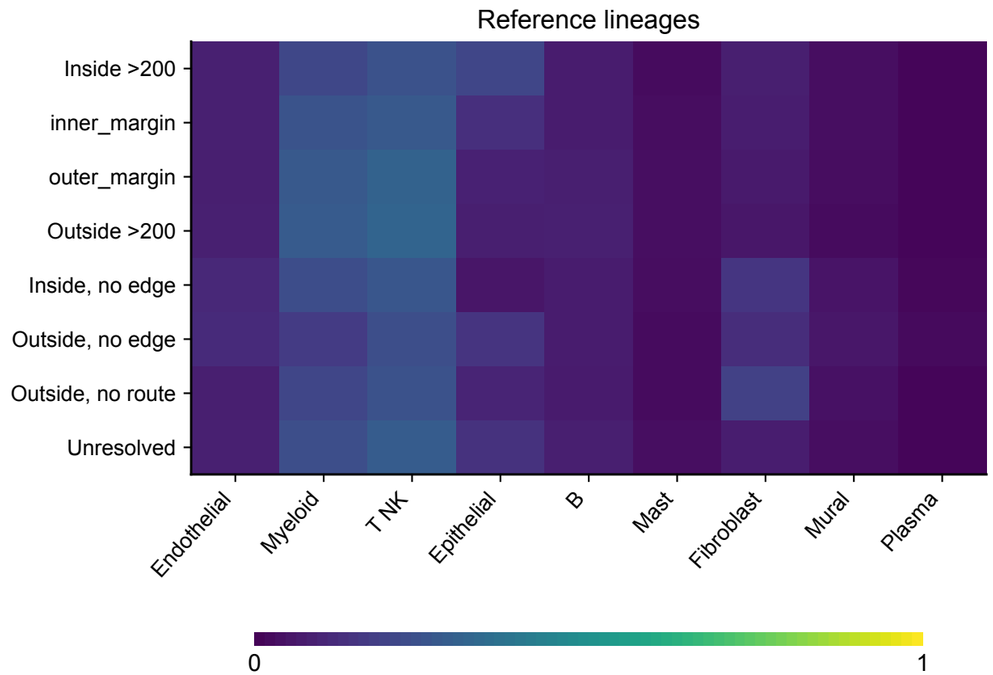
r +0.02 ρ +0.05
n 1057/1066 queries • 1173 targets

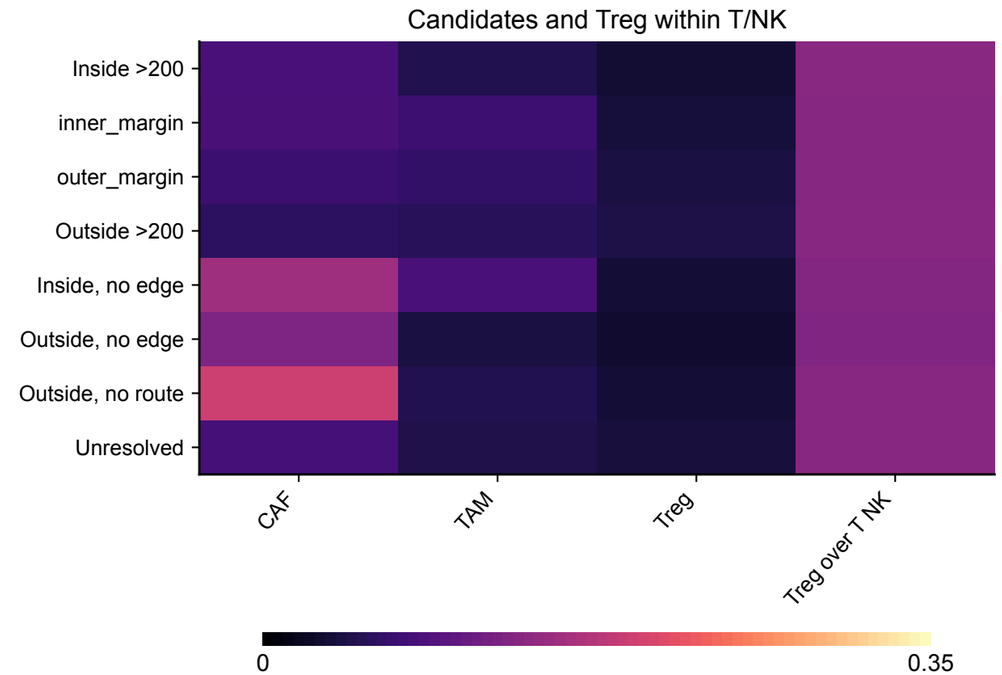


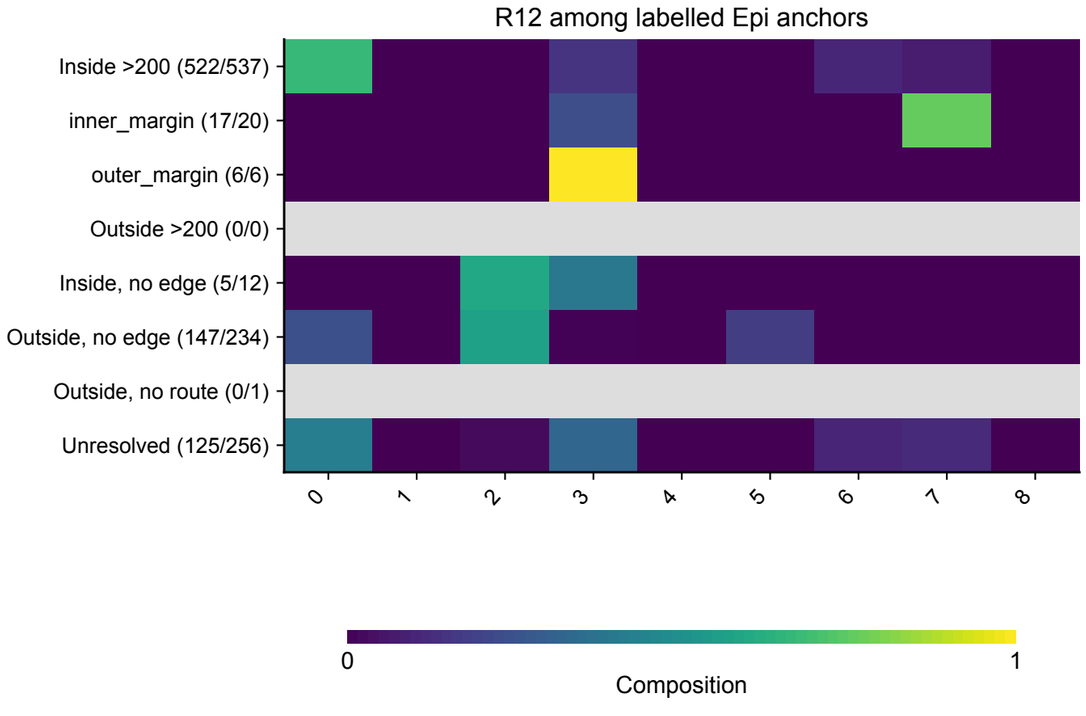
Orange = zero distance
Line = within-section linear fit

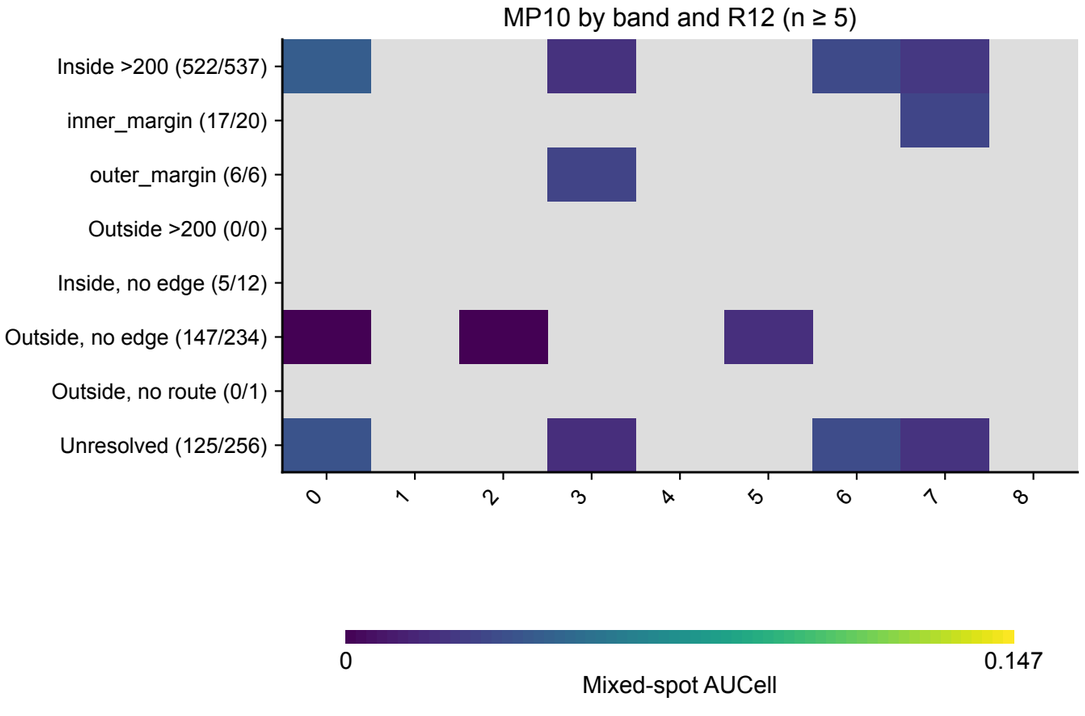
GSE319536_BC14 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





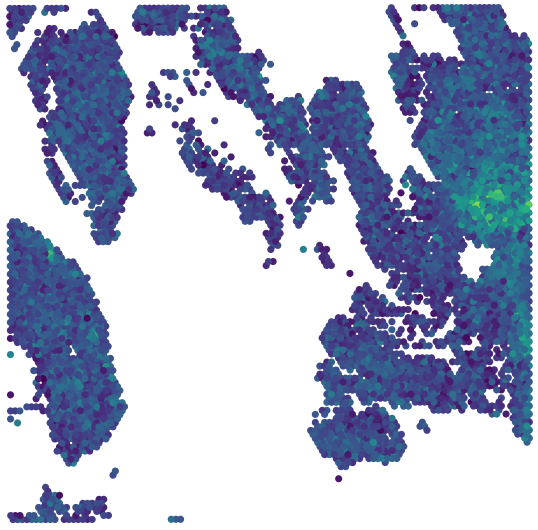




GSE319536_BC14 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

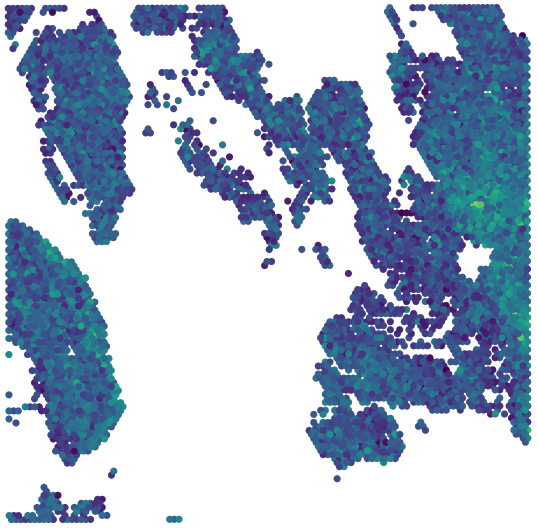
E2F Targets



0 0.152

UCell score

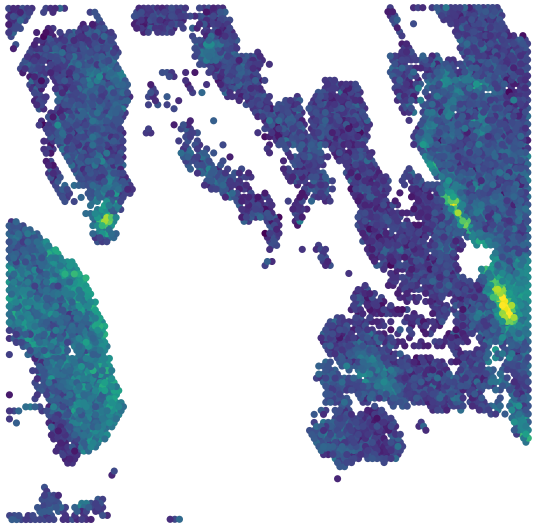
G2M Checkpoint



0 0.134

UCell score

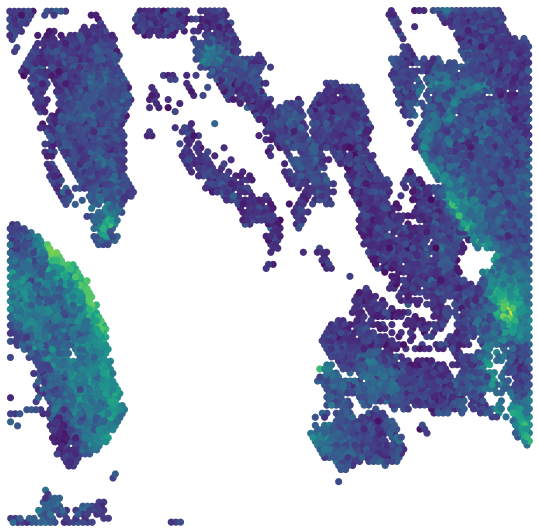
Hypoxia



0 0.211

UCell score

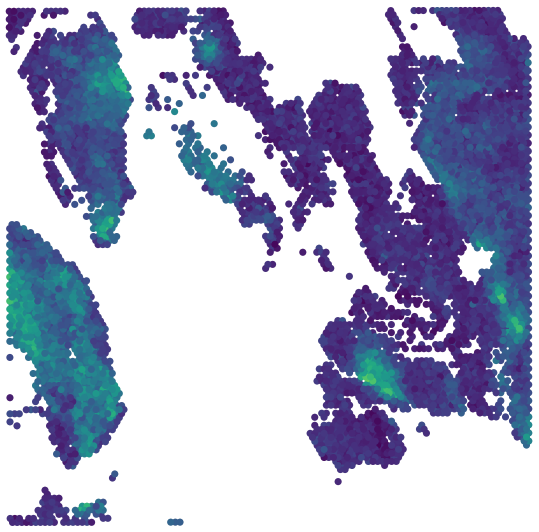
Tnfa Signaling Via Nfkb



0 0.275

UCell score

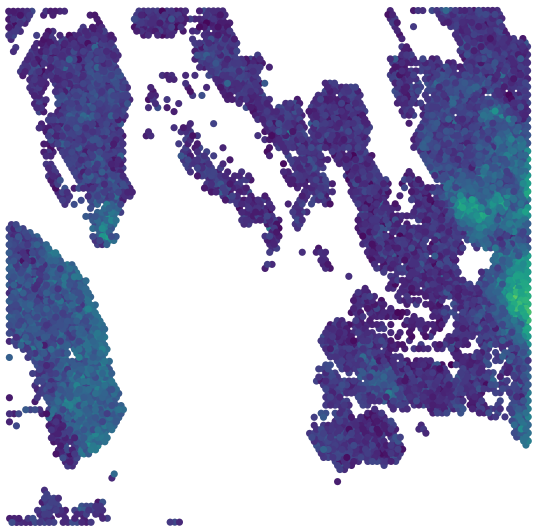
Emt



0 0.386

UCell score

Interferon Gamma Response



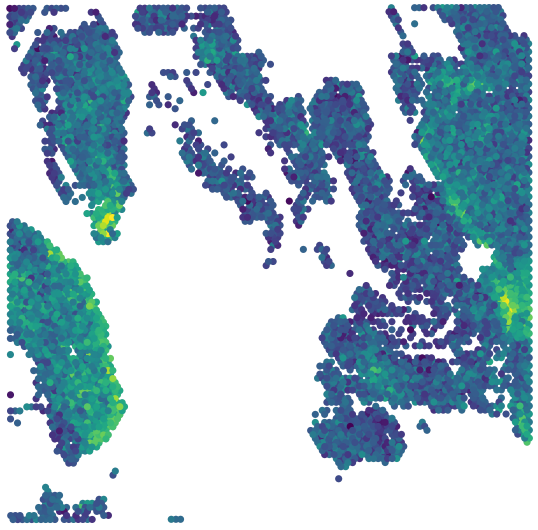
0 0.275

UCell score

GSE319536_BC14 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

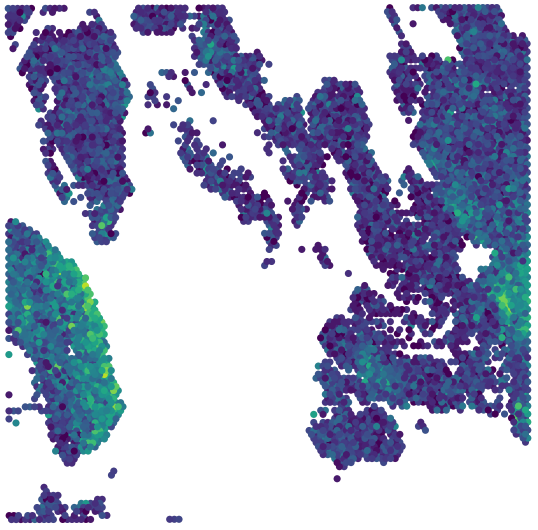
Il2 Stat5 Signaling



0 0.144

UCell score

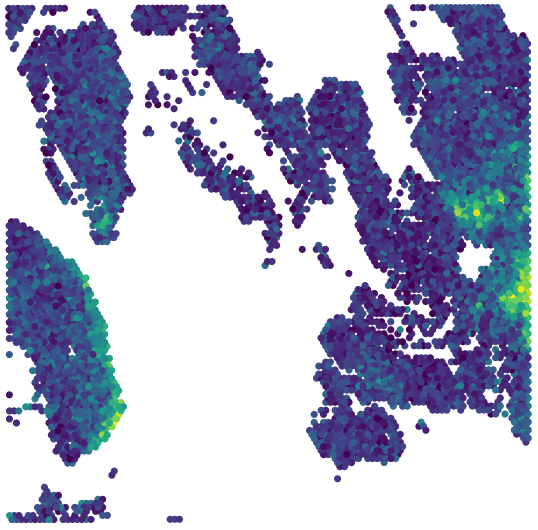
Tgf Beta Signaling



0 0.232

UCell score

Cholesterol Homeostasis



0 0.21

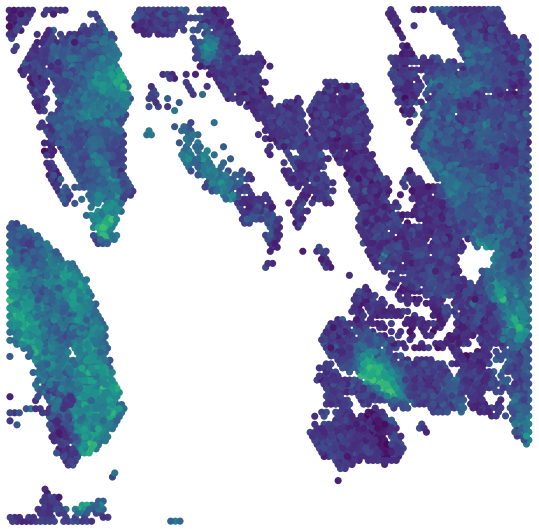
UCell score

Cholesterol Biosynthesis

0 0.166

UCell score

Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4

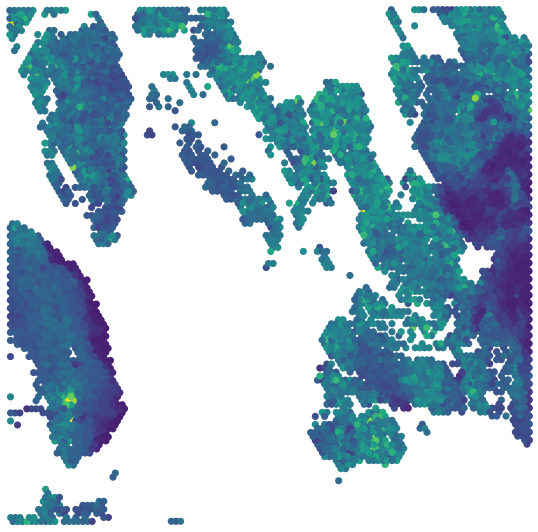
0 0.151

UCell score

GSE319536_BC14 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

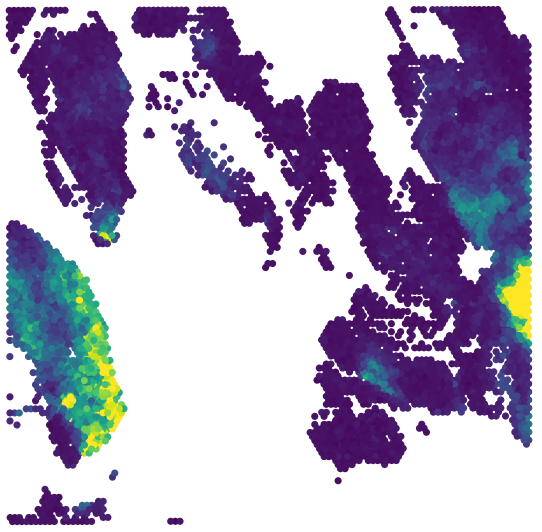
Treg composition



0 0.102

Fraction of all 30

Treg estimated abundance



0 2.85

Posterior mean estimate

Treg Resting Vs Activated Dn

0 0.0778

UCell score

Treg Resting Vs Activated Up

0 0.0609

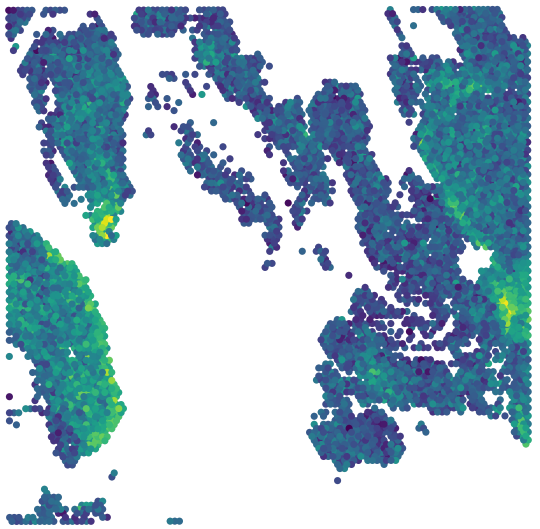
UCell score

Co Inhibition By Ctla4

0 0.151

UCell score

Il2 Stat5 Signaling

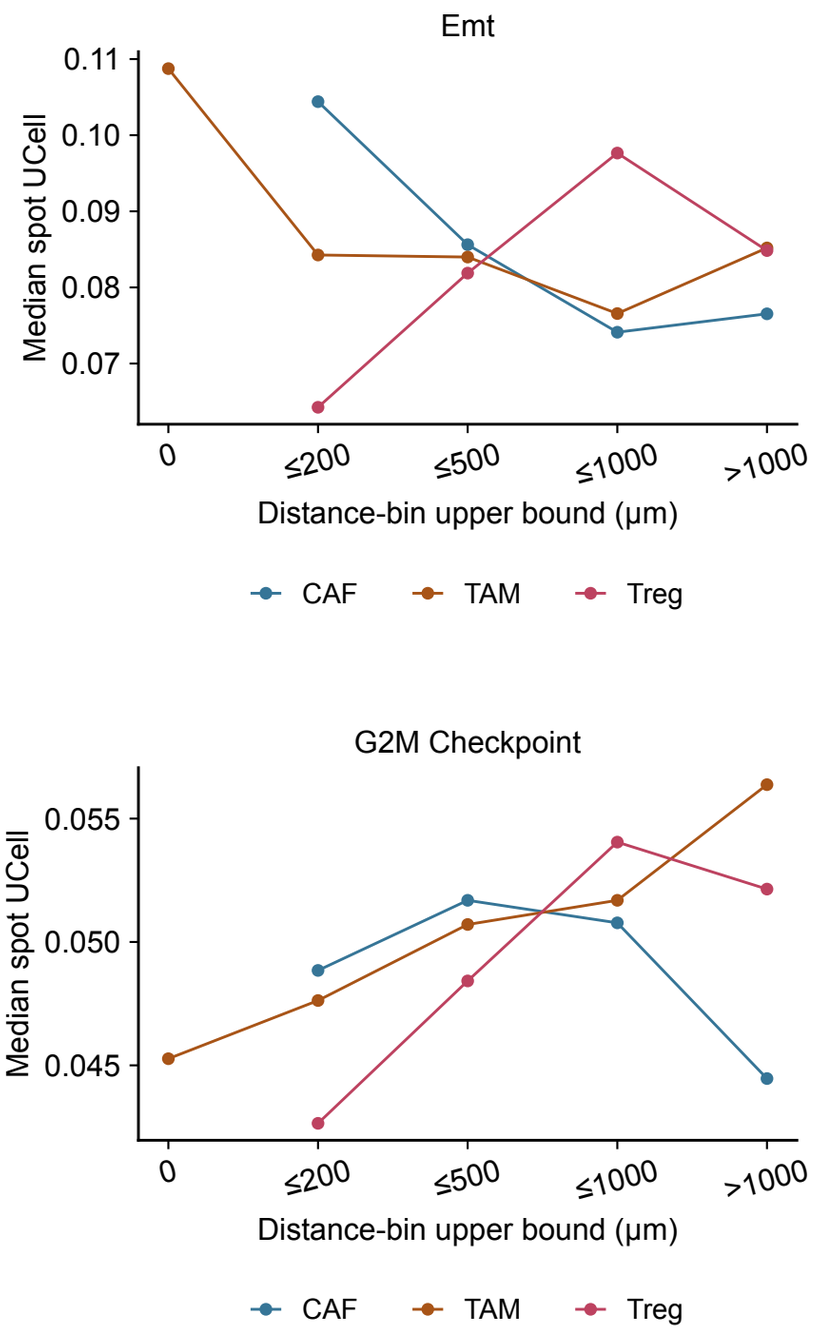


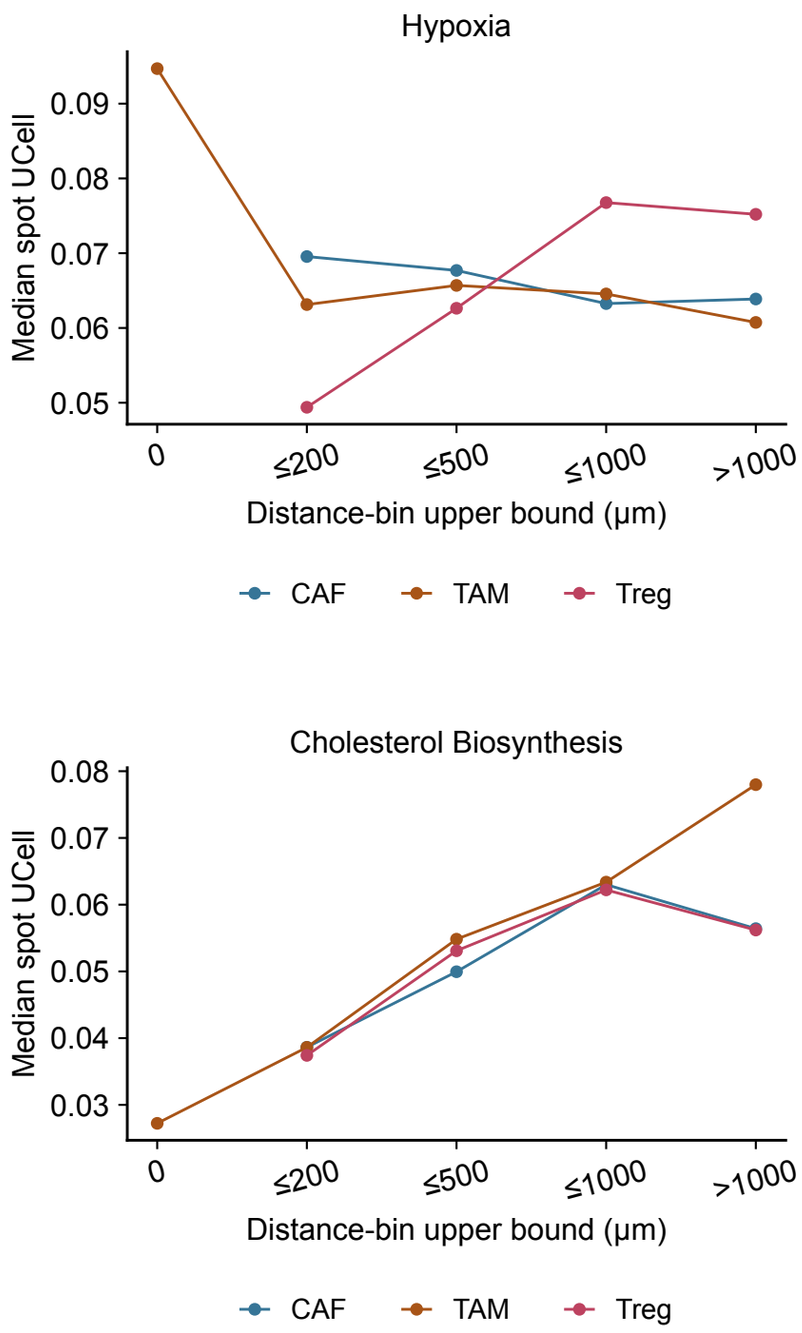
0 0.144

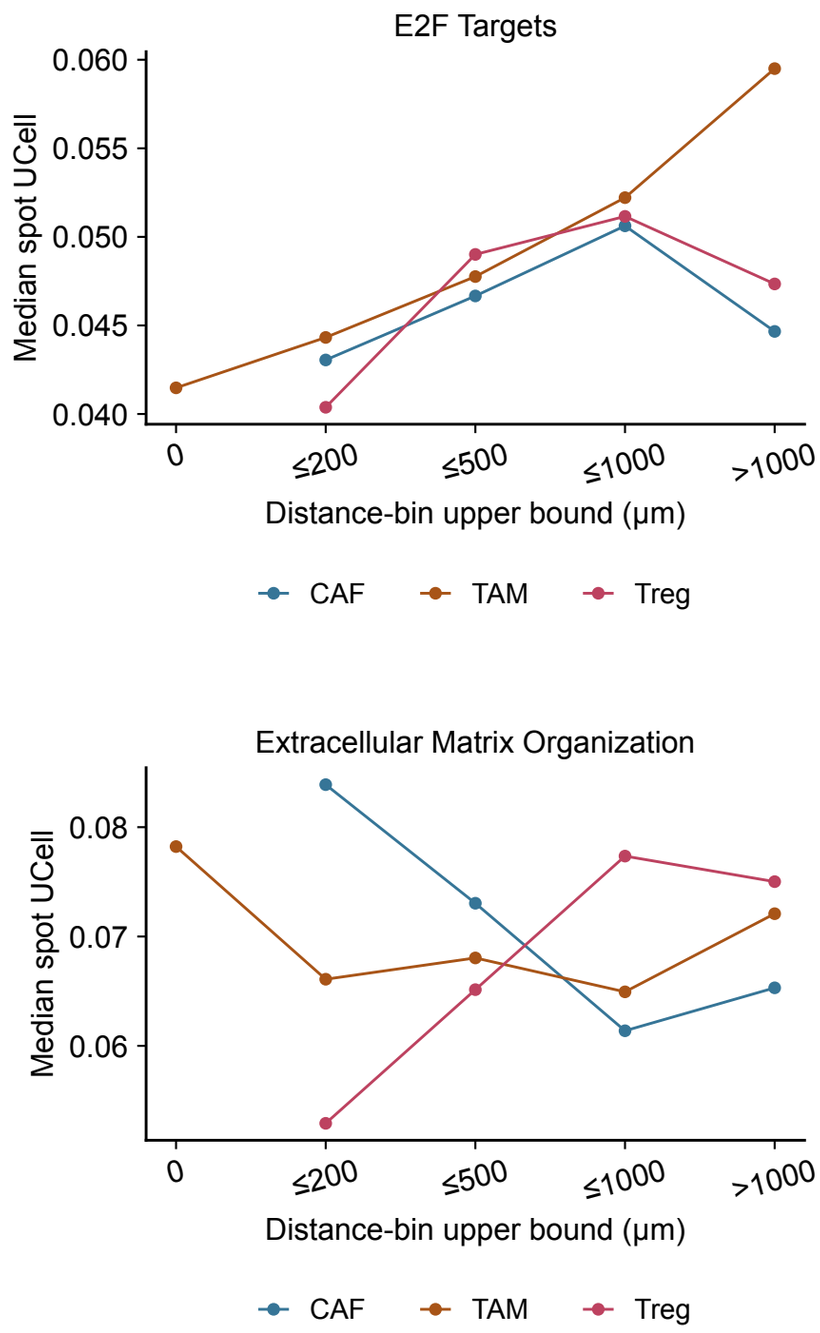
UCell score

GSE319536_BC14 | Programs by target distance

Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

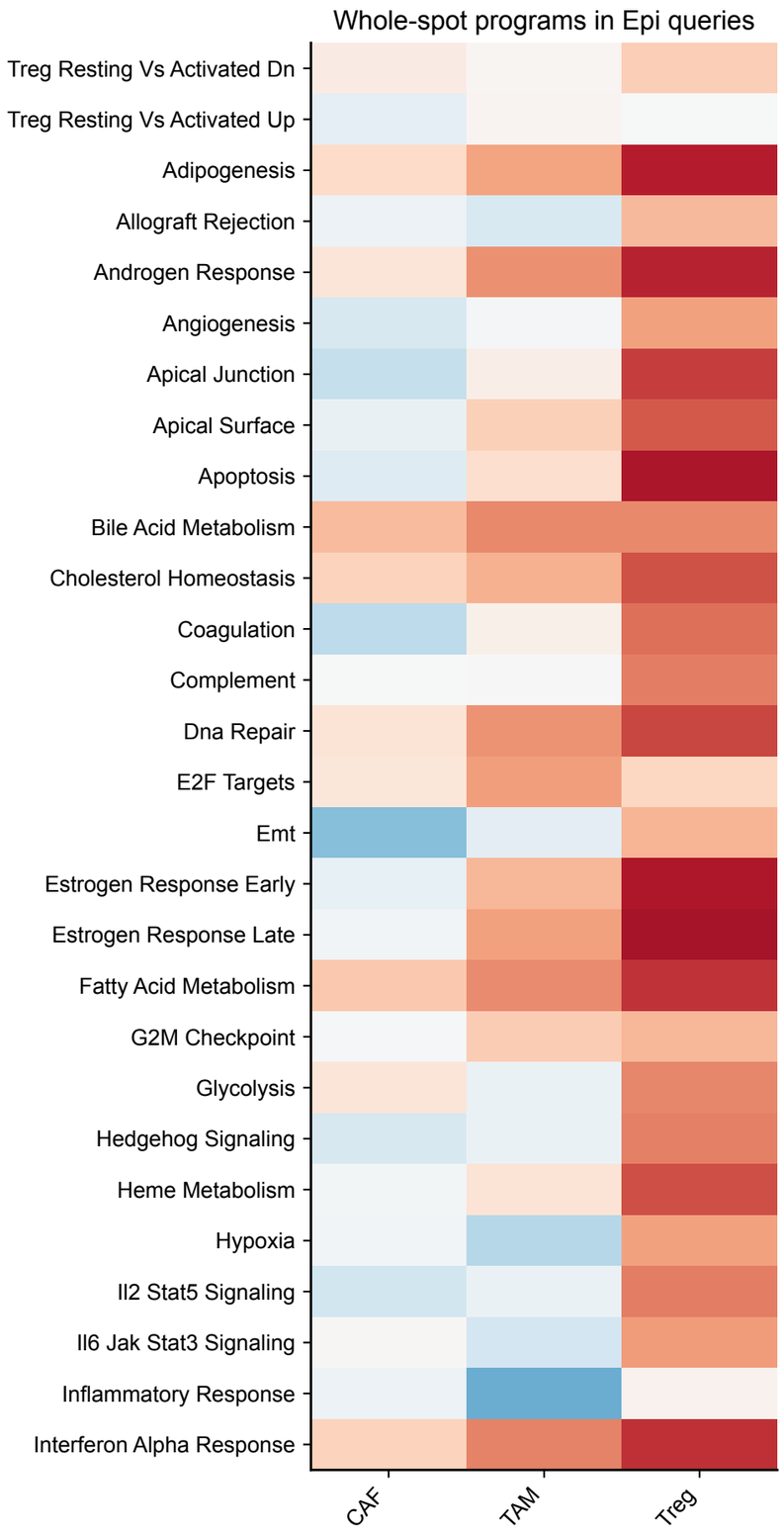


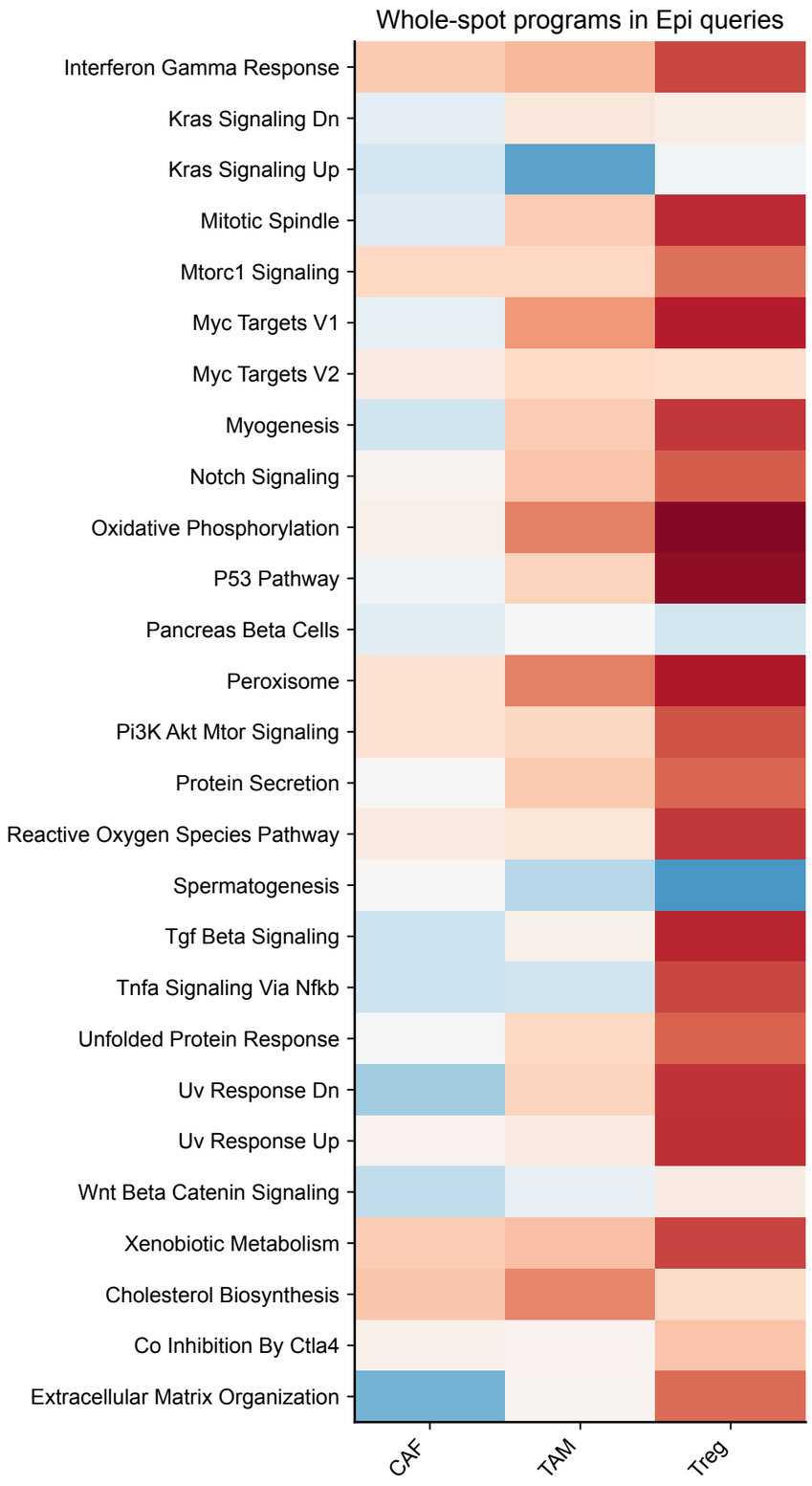




GSE319536_BC14 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

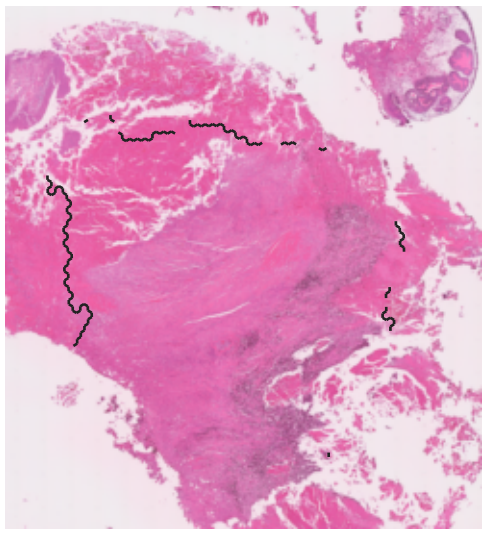




GSE319536_BC15 | Tissue context

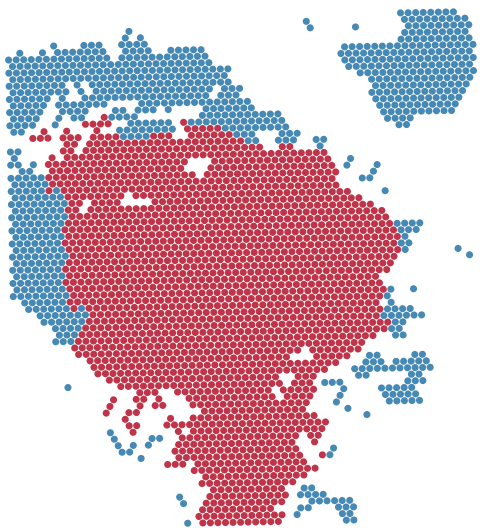
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



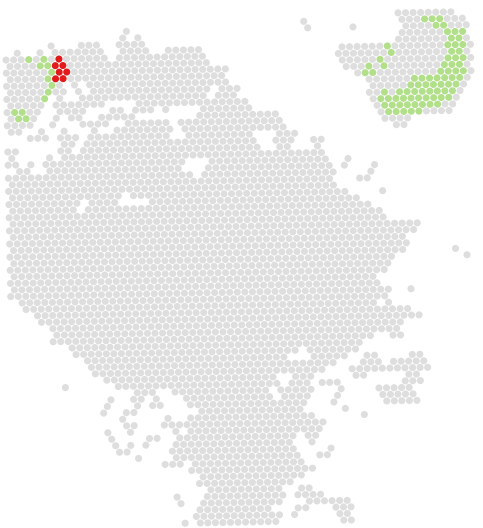
Black line: observed candidate outer edge

Final candidate location



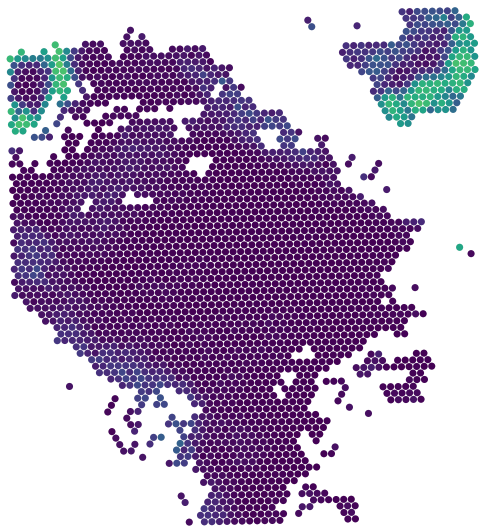
inside outside

Accepted R12 (original 0–8)



2 7

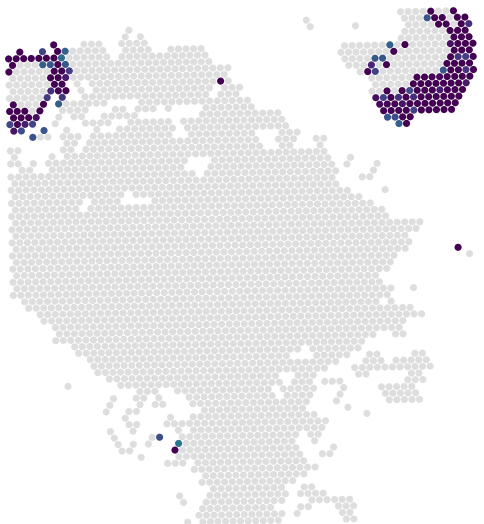
Epithelial support



0 1

Fraction of all 30

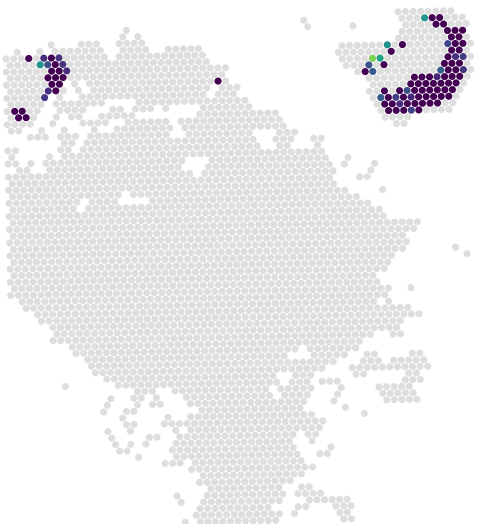
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



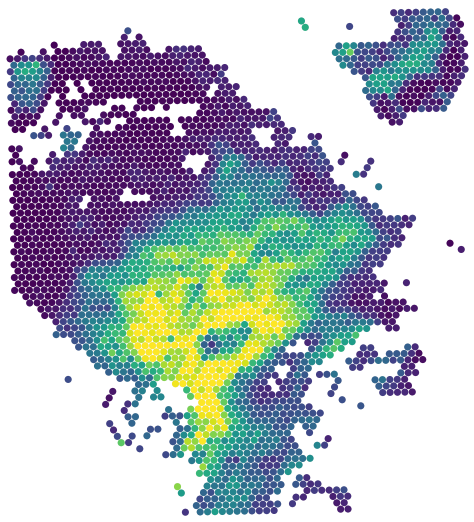
0 0.14

Matched-model AUCell

GSE319536_BC15 | Candidate neighborhoods

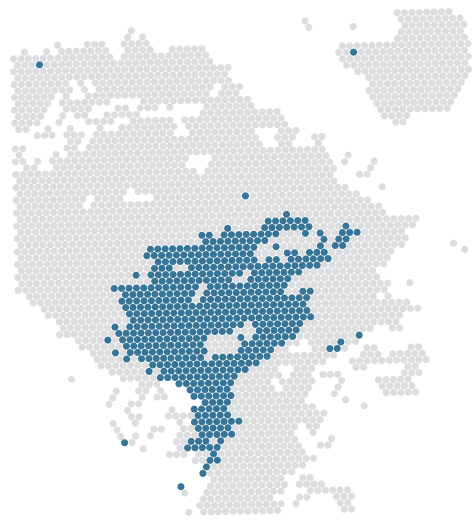
Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

CAF composition



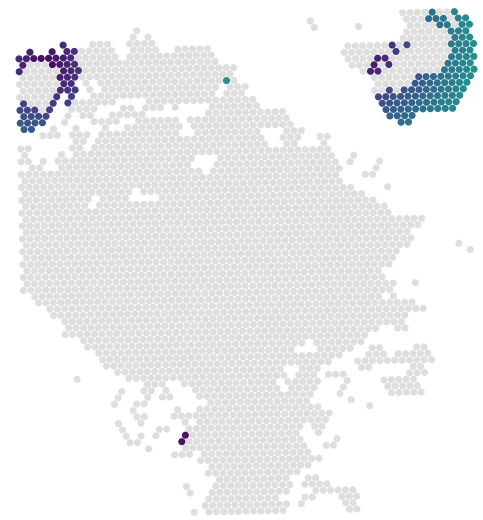
0 0.418
Fraction of all 30

CAF P80 targets



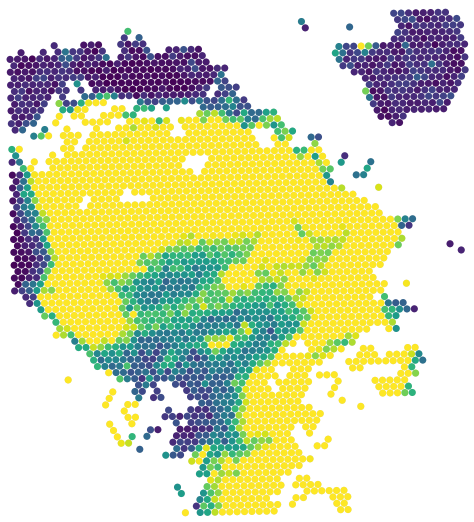
False True

CAF distance



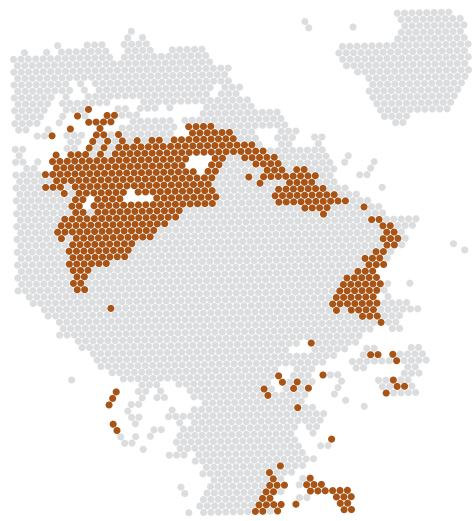
0 3.1e+03
Nearest spot center (μm)

TAM composition



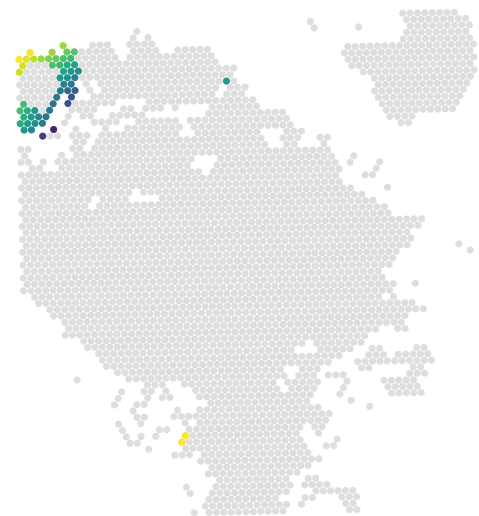
0 0.291
Fraction of all 30

TAM P80 targets



False True

TAM distance

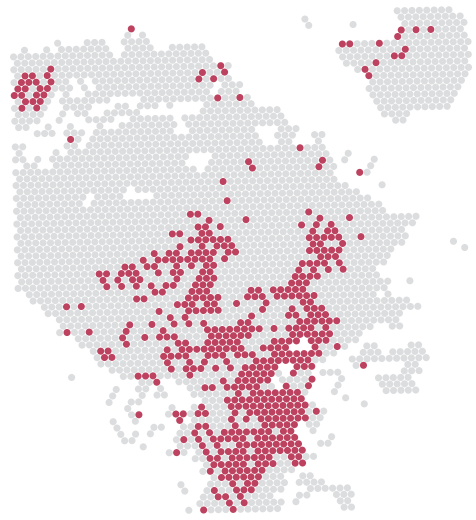


0 1.13e+03
Nearest spot center (μm)

Treg composition

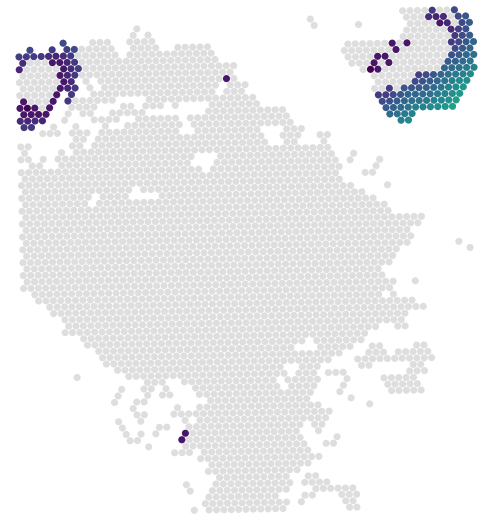
0 0.102
Fraction of all 30

Treg P80 targets



False True

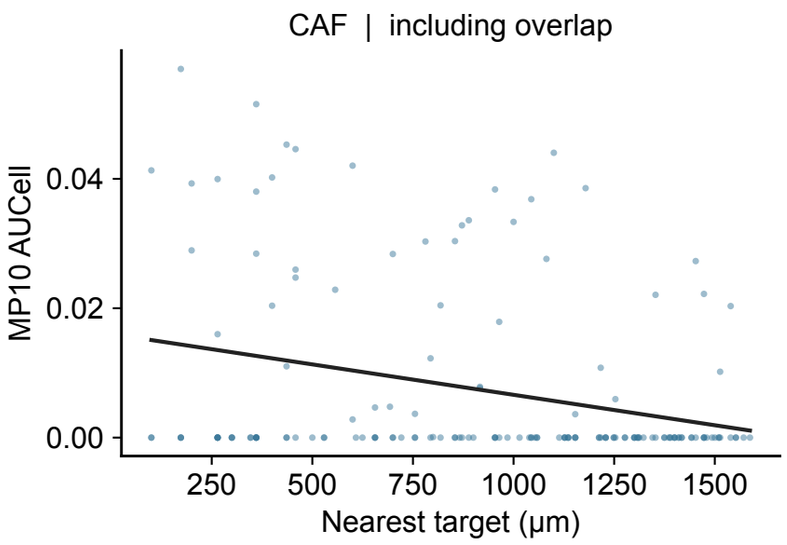
Treg distance



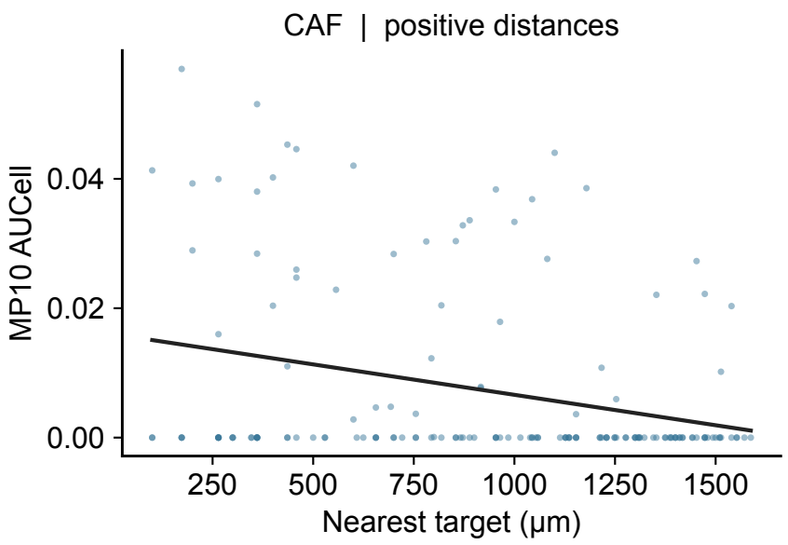
0 1.61e+03
Nearest spot center (μm)

GSE319536_BC15 | MP10 versus nearest target

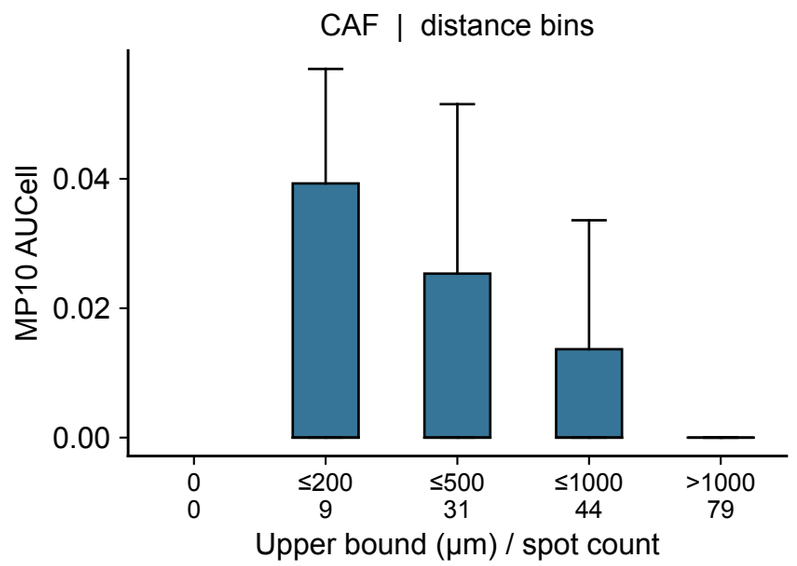
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



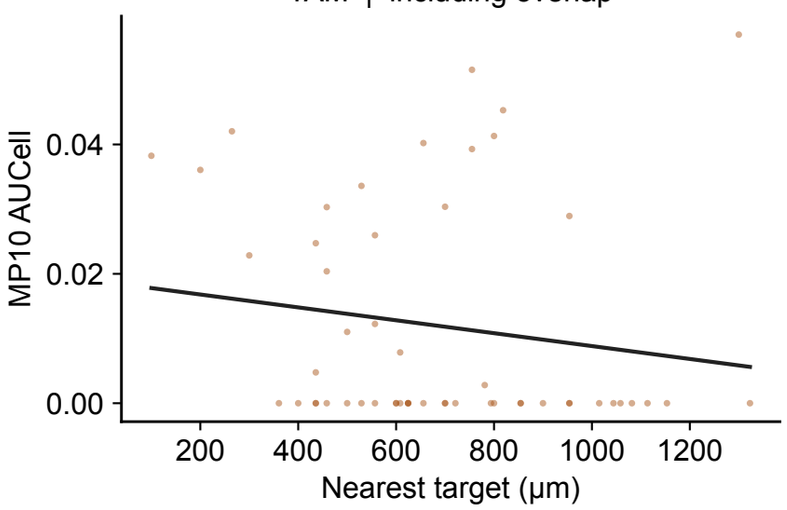
r -0.29 ρ -0.27
n 163/167 queries • 540 targets



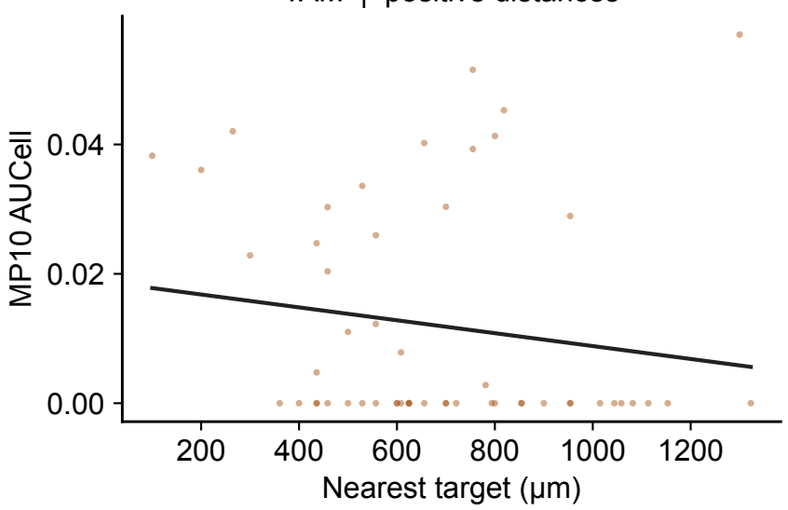
r -0.29 ρ -0.27
n 163/167 queries • 540 targets



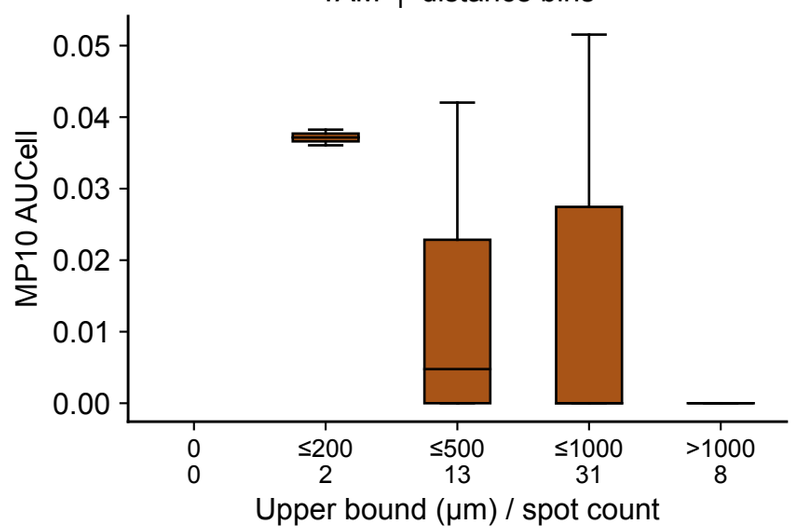
Orange = zero distance
Line = within-section linear fit



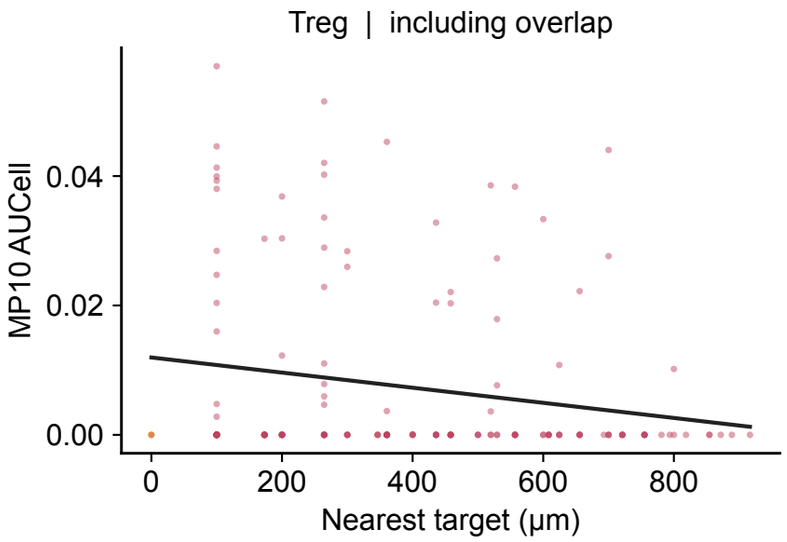
r -0.15 ρ -0.22
n 54/167 queries • 540 targets



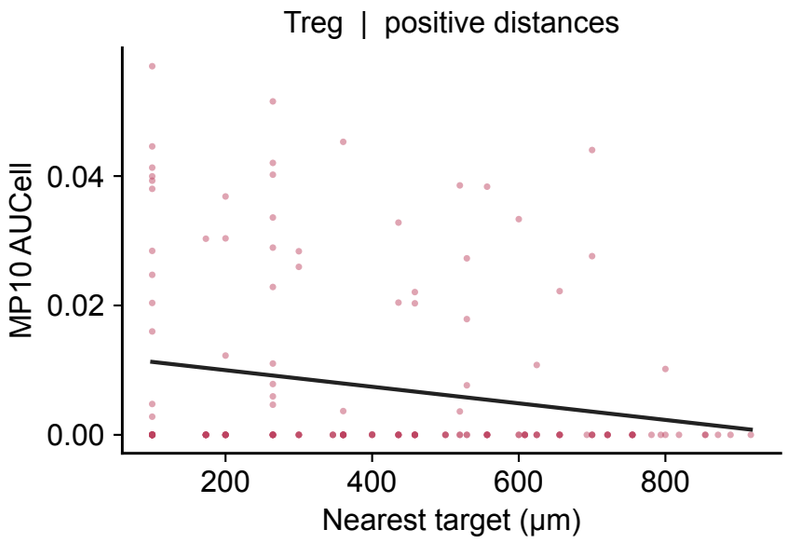
r -0.15 ρ -0.22
n 54/167 queries • 540 targets



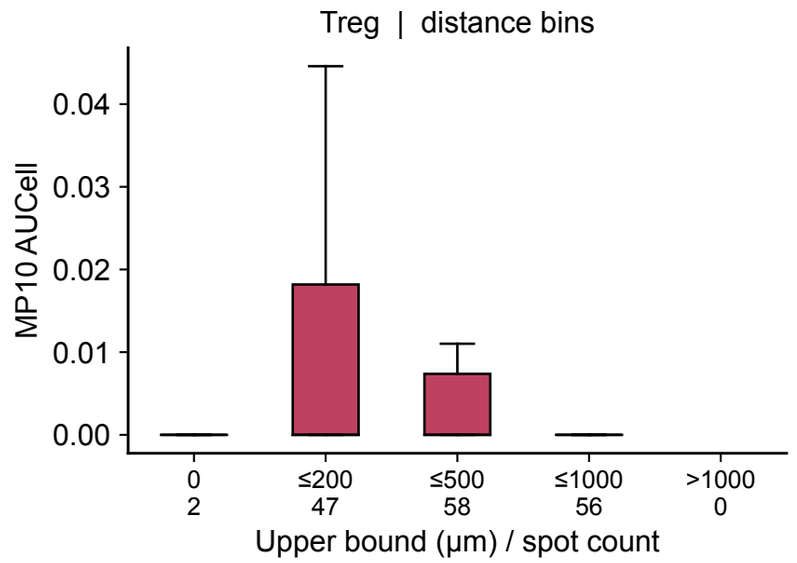
Orange = zero distance
Line = within-section linear fit



r -0.20 ρ -0.18
n 163/167 queries • 540 targets



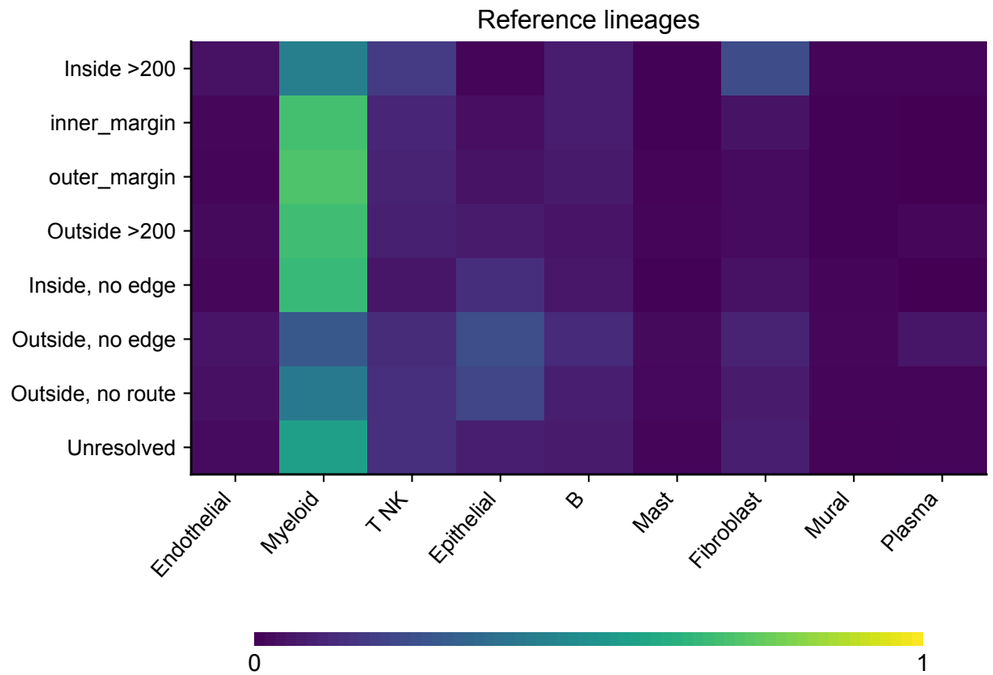
r -0.21 ρ -0.20
n 161/167 queries • 540 targets

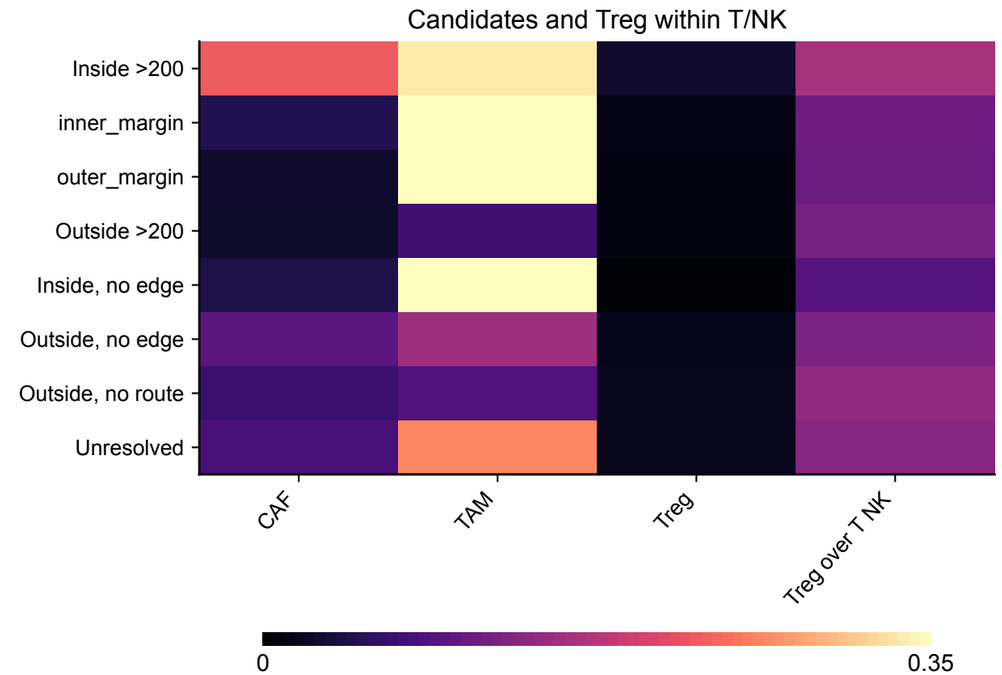


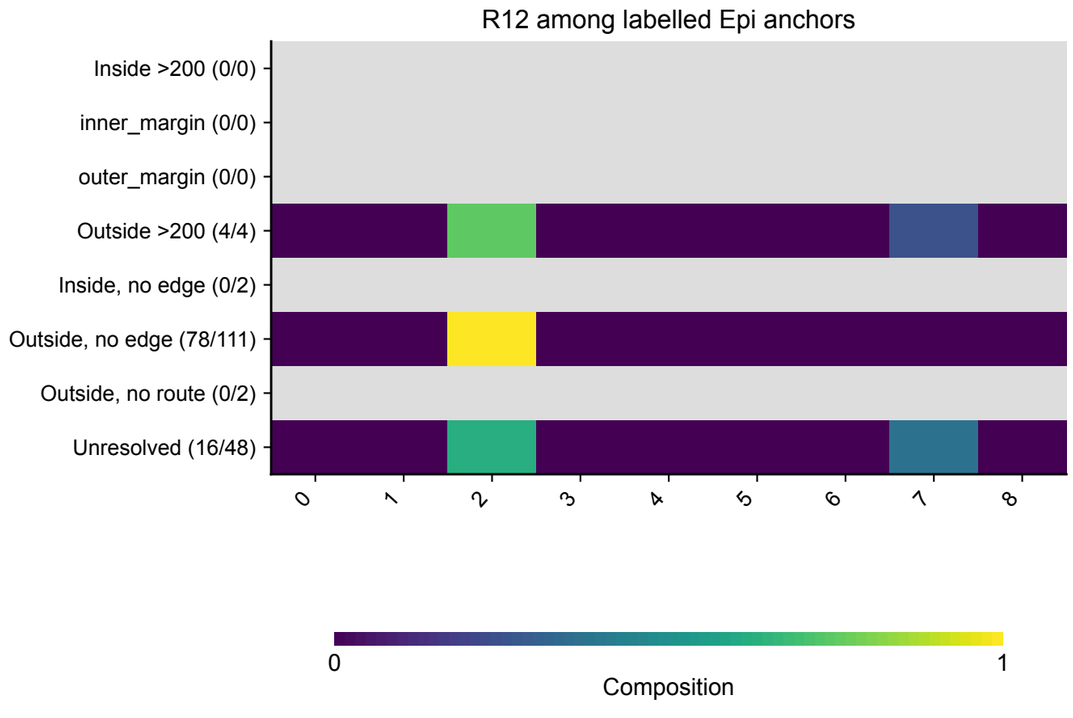
Orange = zero distance
Line = within-section linear fit

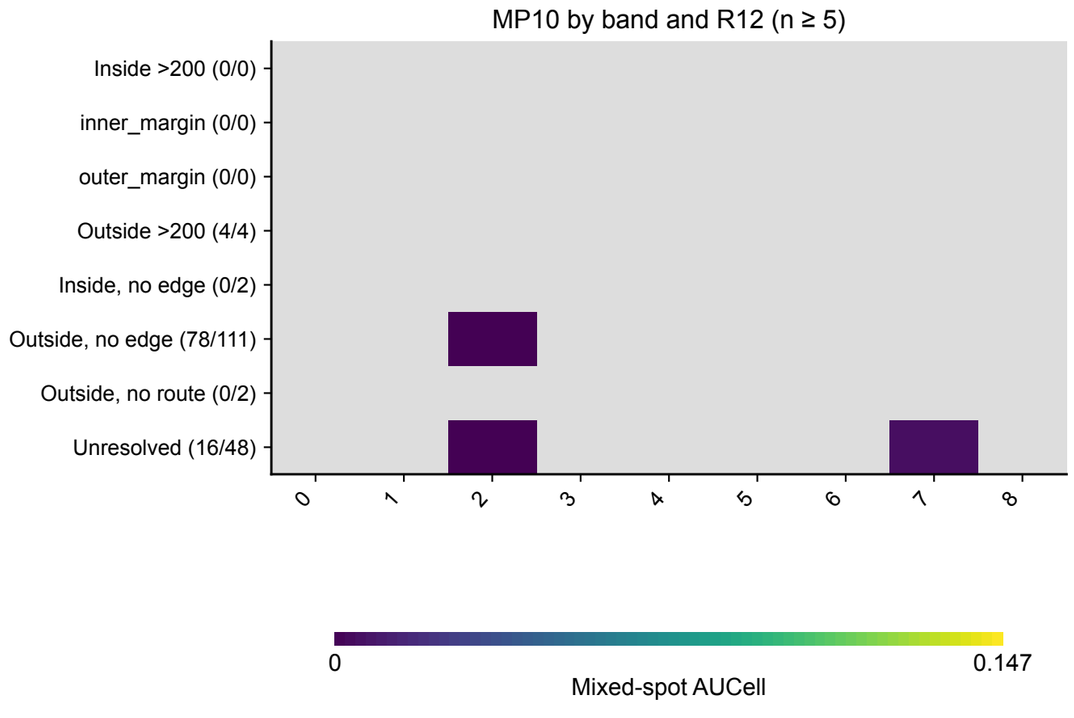
GSE319536_BC15 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





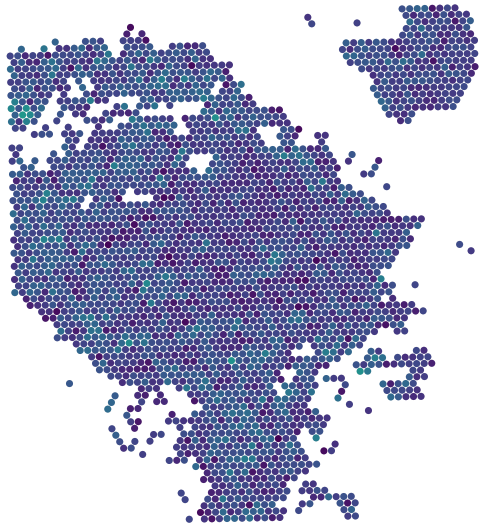




GSE319536_BC15 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

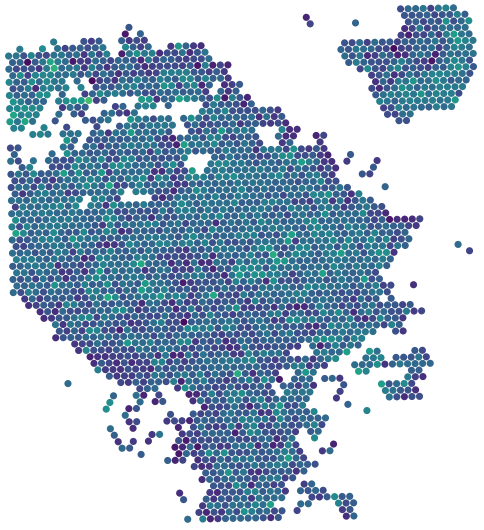
E2F Targets



0 0.152

UCell score

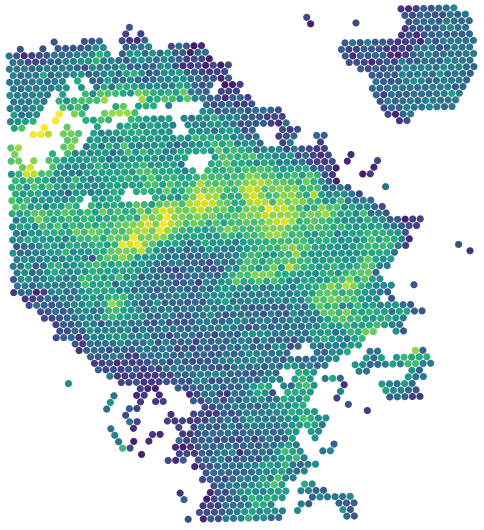
G2M Checkpoint



0 0.134

UCell score

Hypoxia



0 0.211

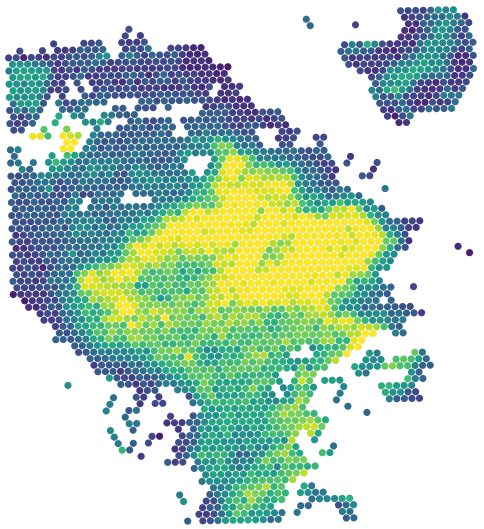
UCell score

Tnfa Signaling Via Nfkb

0 0.275

UCell score

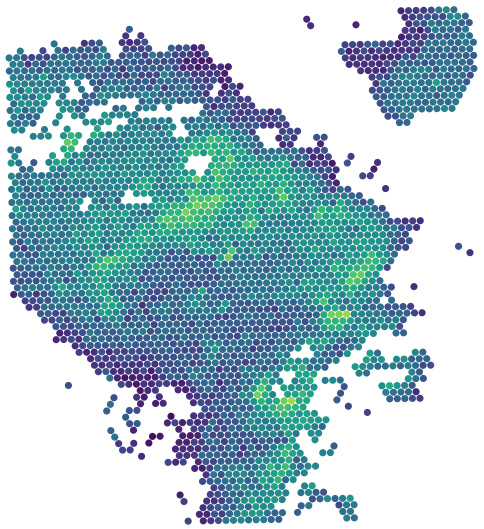
Emt



0 0.386

UCell score

Interferon Gamma Response



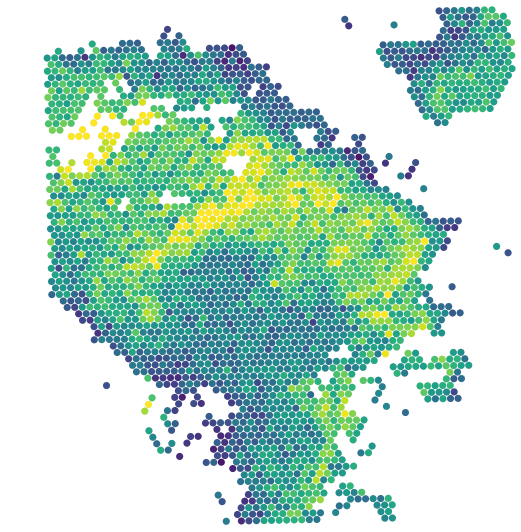
0 0.275

UCell score

GSE319536_BC15 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

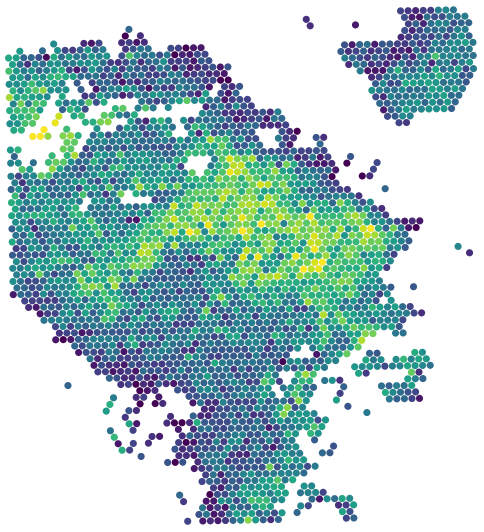
Il2 Stat5 Signaling



0 0.144

UCell score

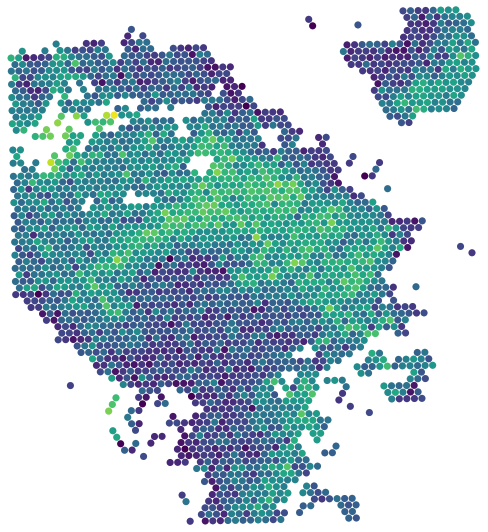
Tgf Beta Signaling



0 0.232

UCell score

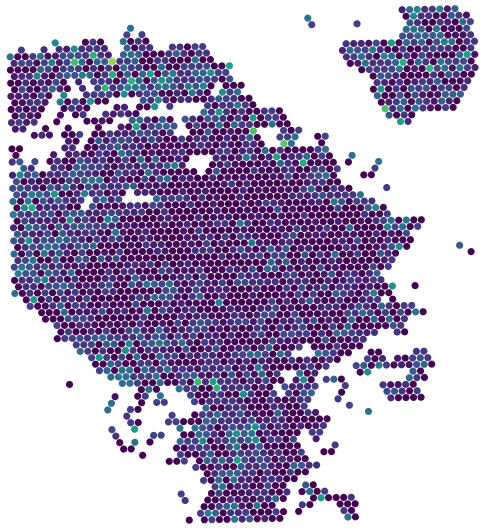
Cholesterol Homeostasis



0 0.21

UCell score

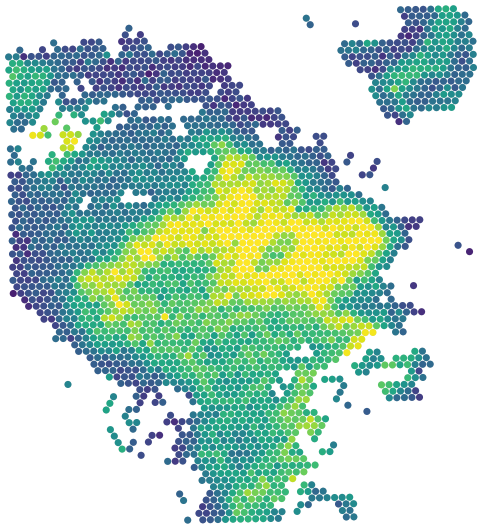
Cholesterol Biosynthesis



0 0.166

UCell score

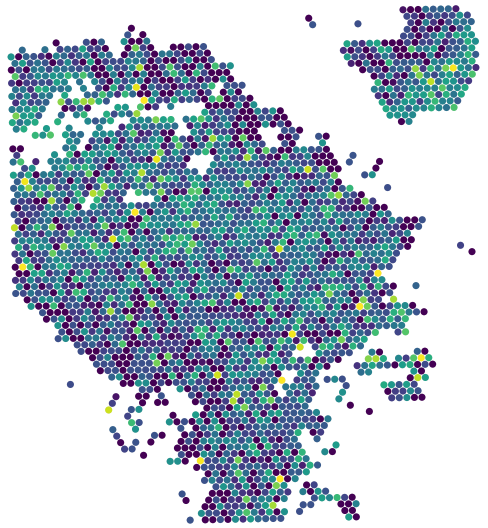
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



0 0.151

UCell score

GSE319536_BC15 | Treg context across the full tissue

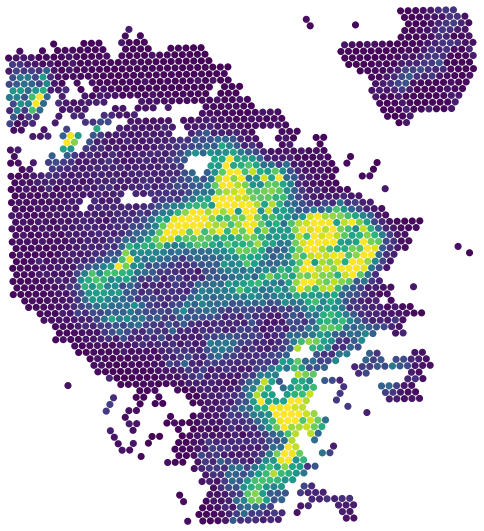
Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Treg composition

0 0.102

Fraction of all 30

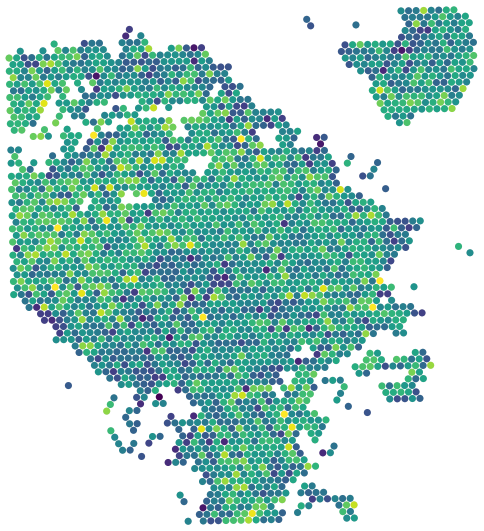
Treg estimated abundance



0 2.85

Posterior mean estimate

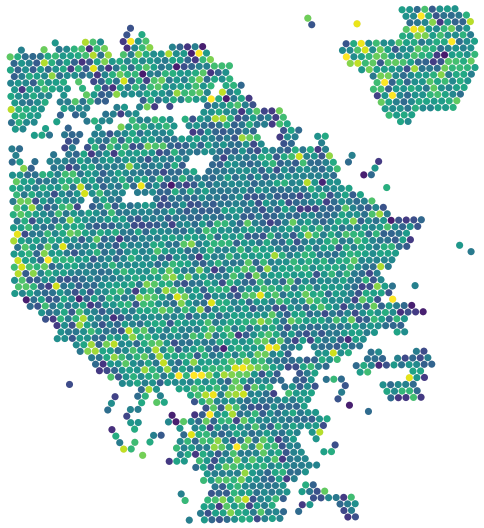
Treg Resting Vs Activated Dn



0 0.0778

UCell score

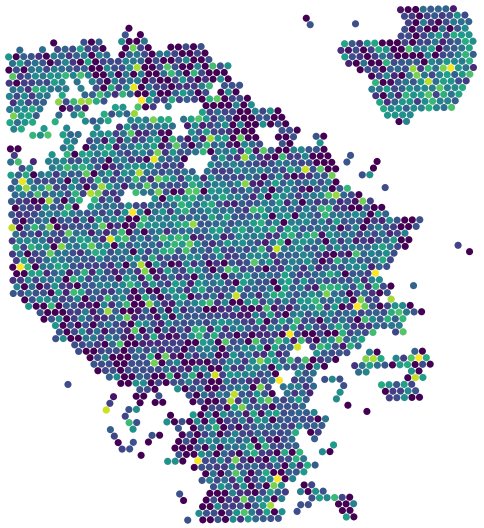
Treg Resting Vs Activated Up



0 0.0609

UCell score

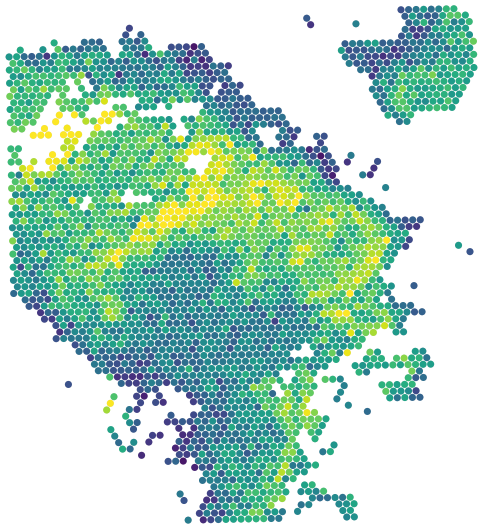
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

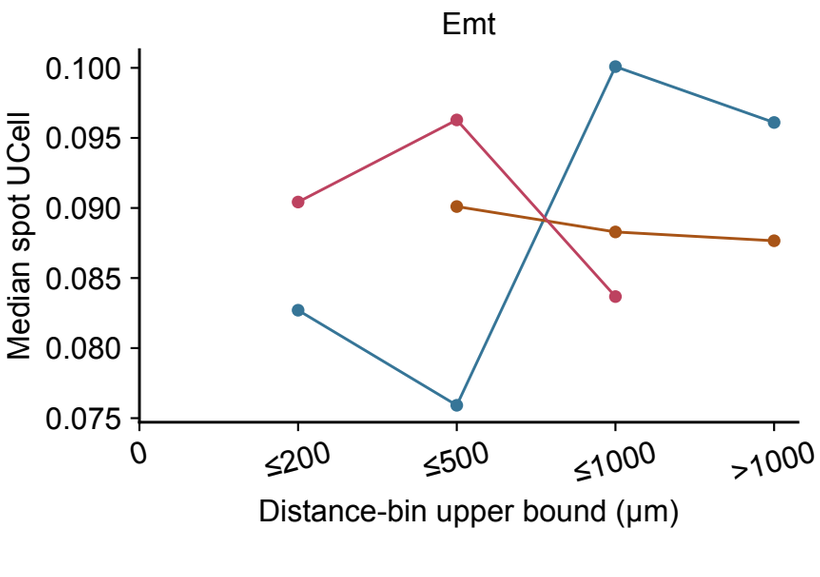


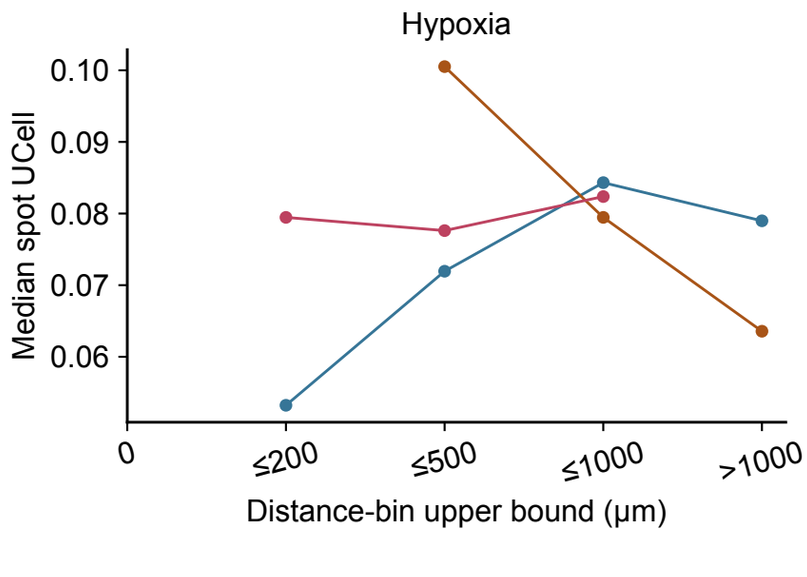
0 0.144

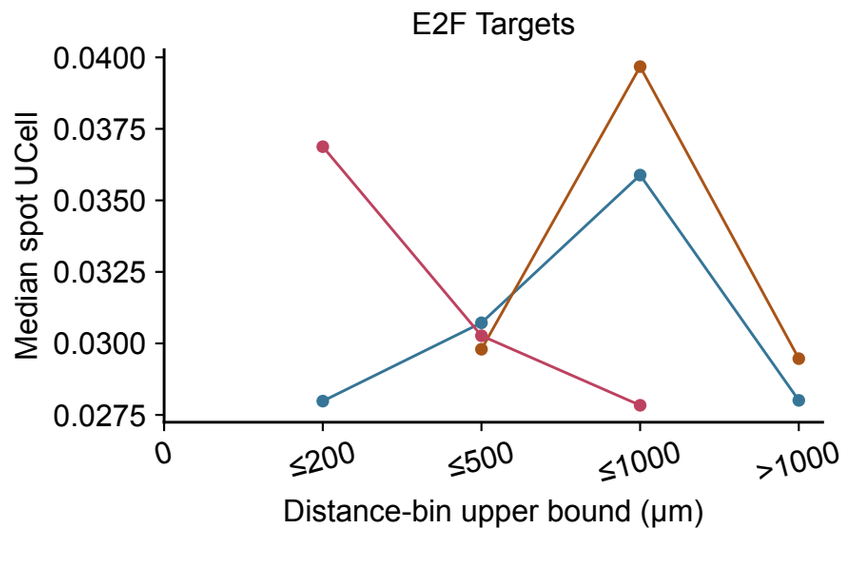
UCell score

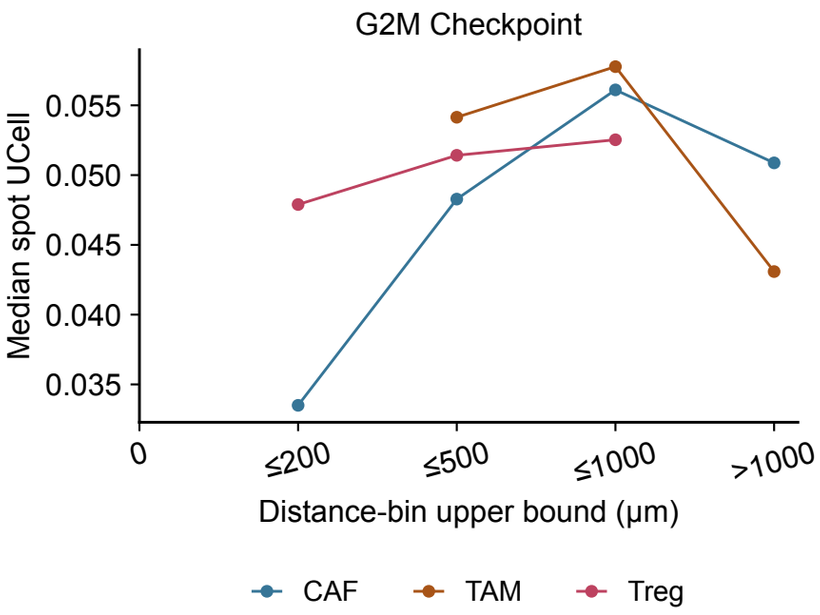
GSE319536_BC15 | Programs by target distance

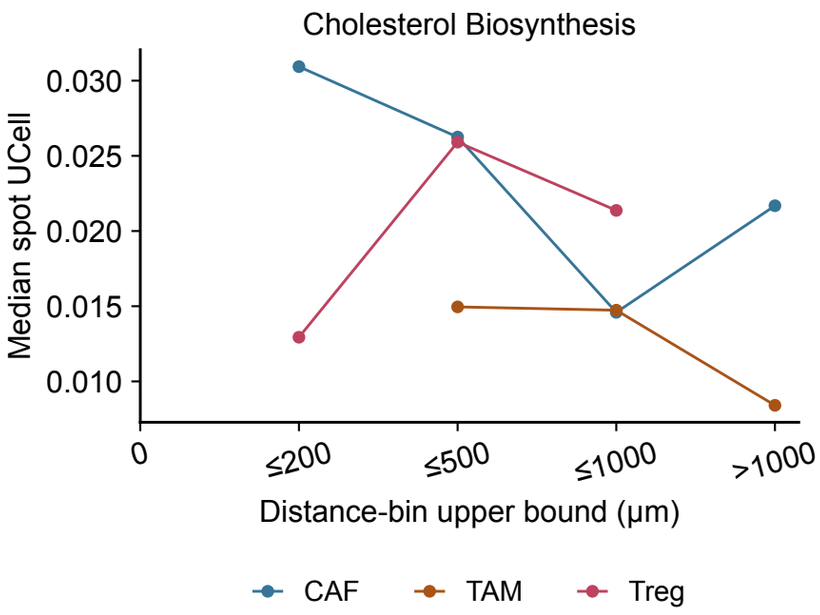
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

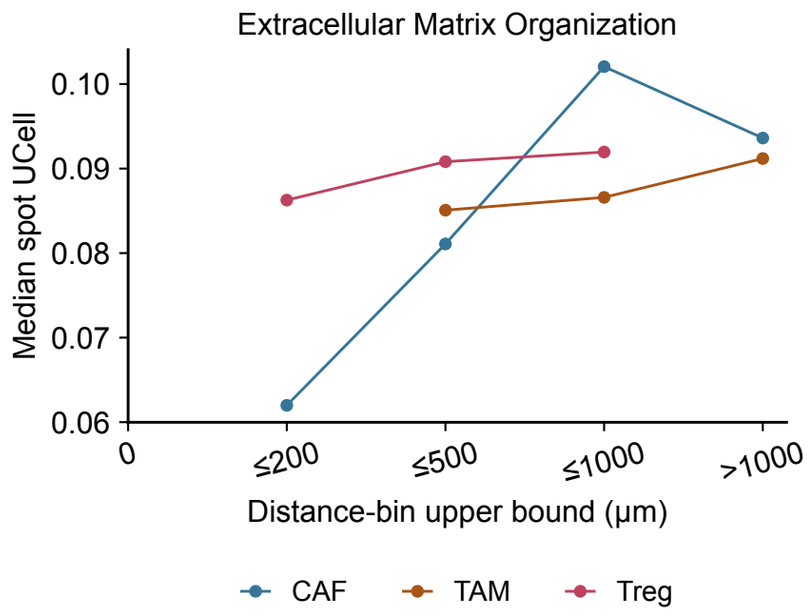






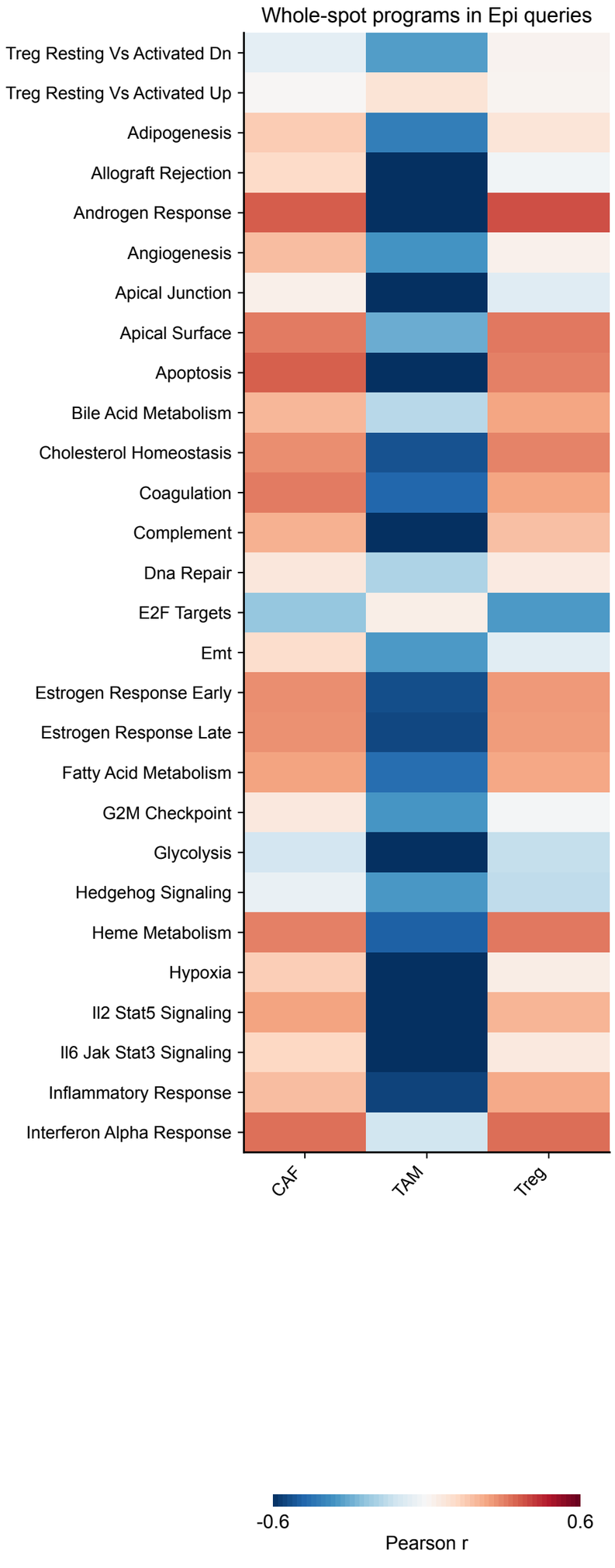






GSE319536_BC15 | Program–distance associations

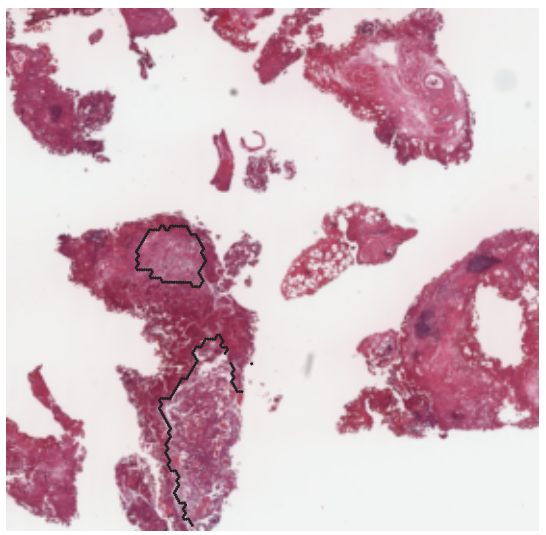
All 55 UCell programs • one section • including zero distances • display cap ±0.6



GSE319536_BC16 | Tissue context

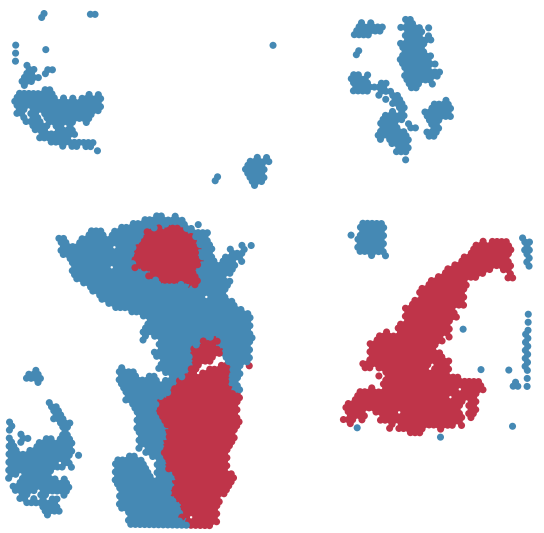
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



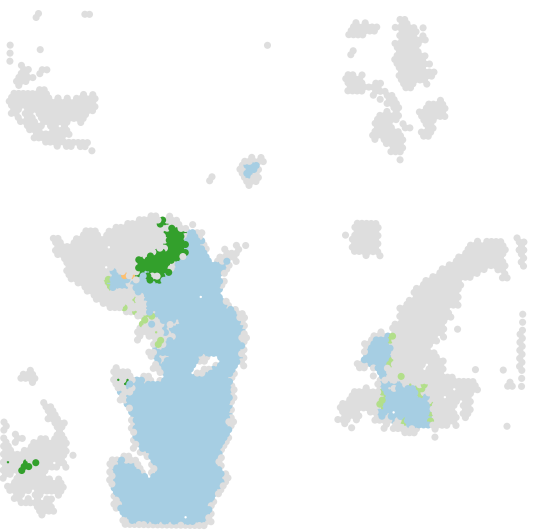
Black line: observed candidate outer edge

Final candidate location



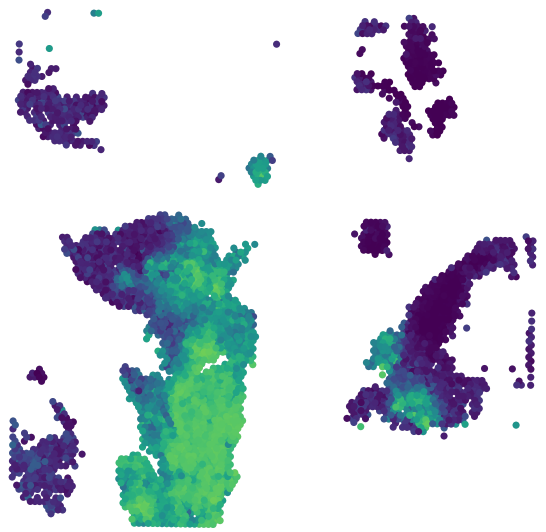
inside outside

Accepted R12 (original 0–8)



0 2 3 4

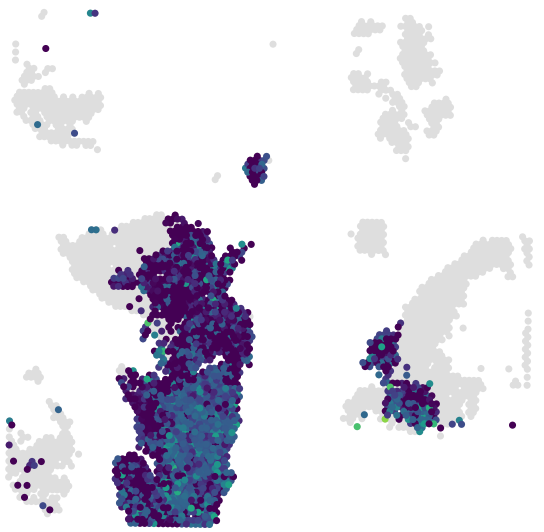
Epithelial support



0 1

Fraction of all 30

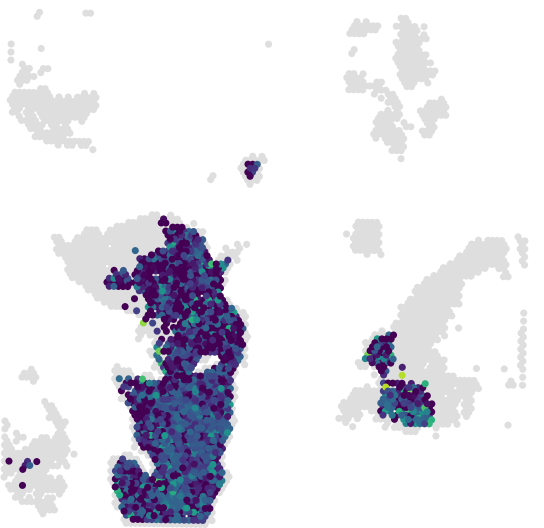
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)

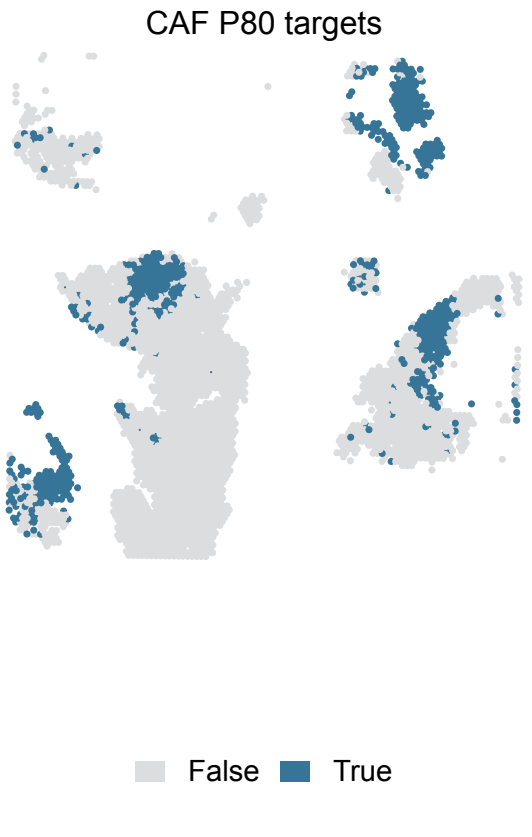


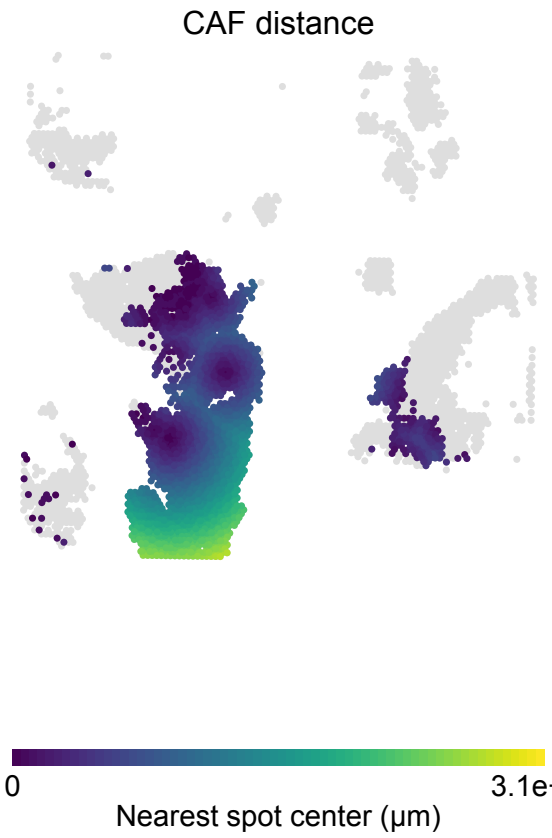
0 0.14

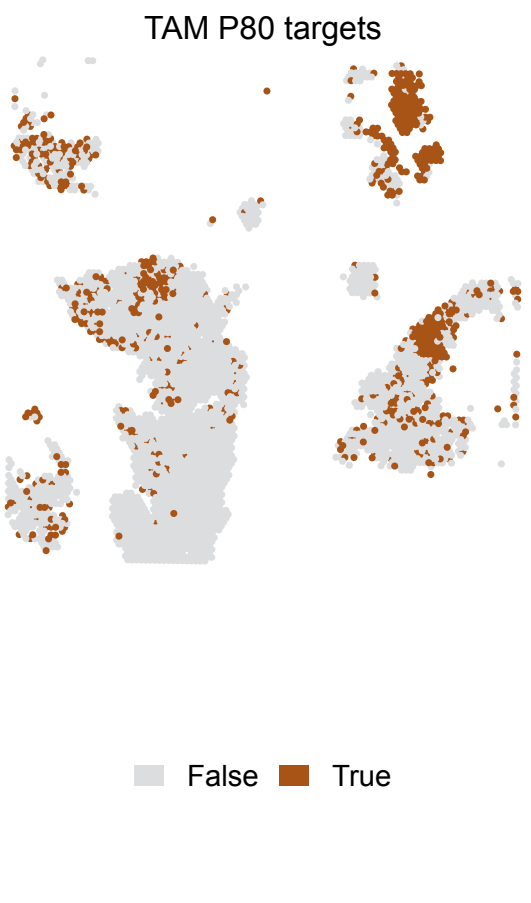
Matched-model AUCell

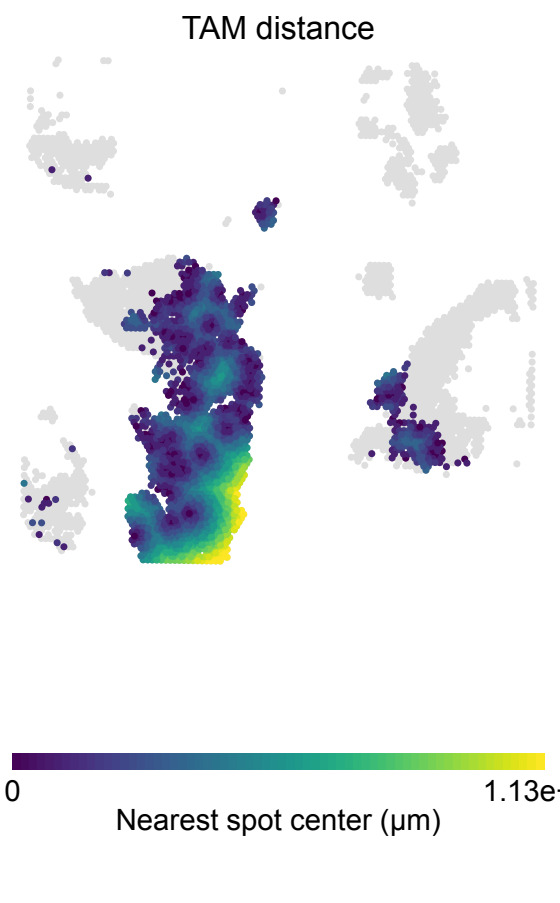
GSE319536_BC16 | Candidate neighborhoods

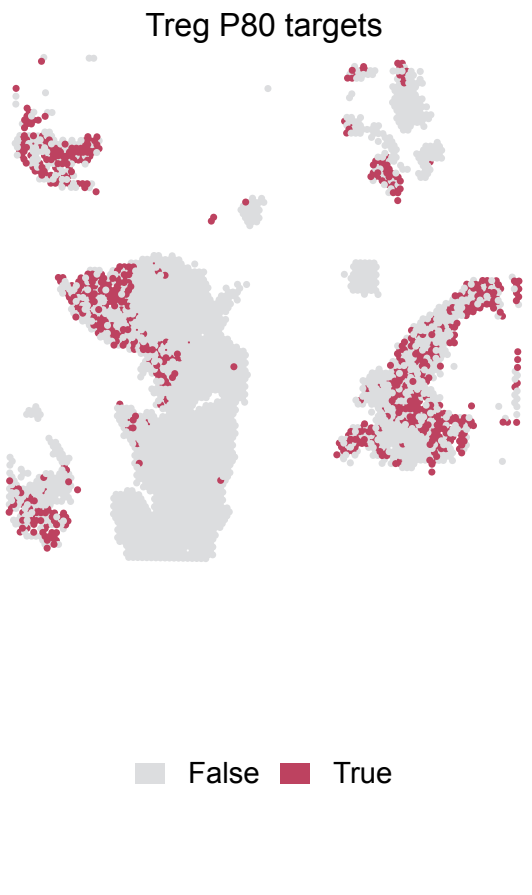
Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

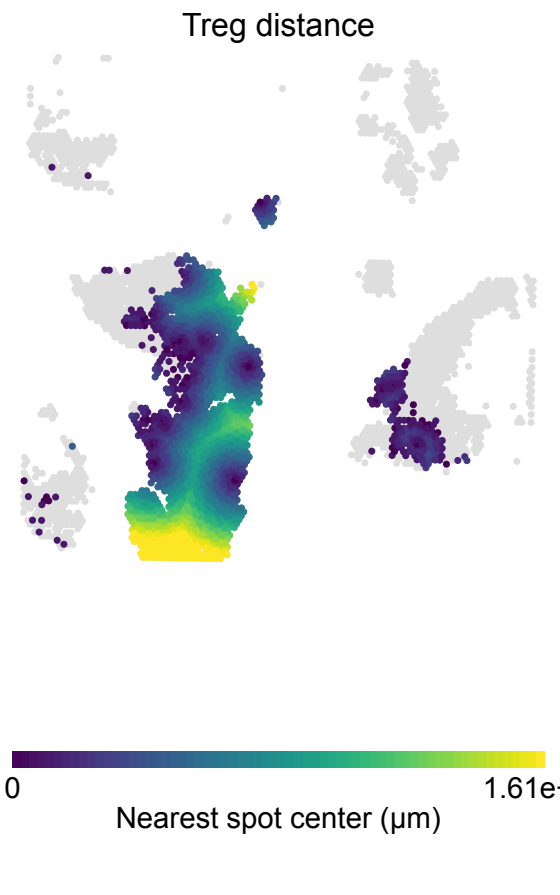






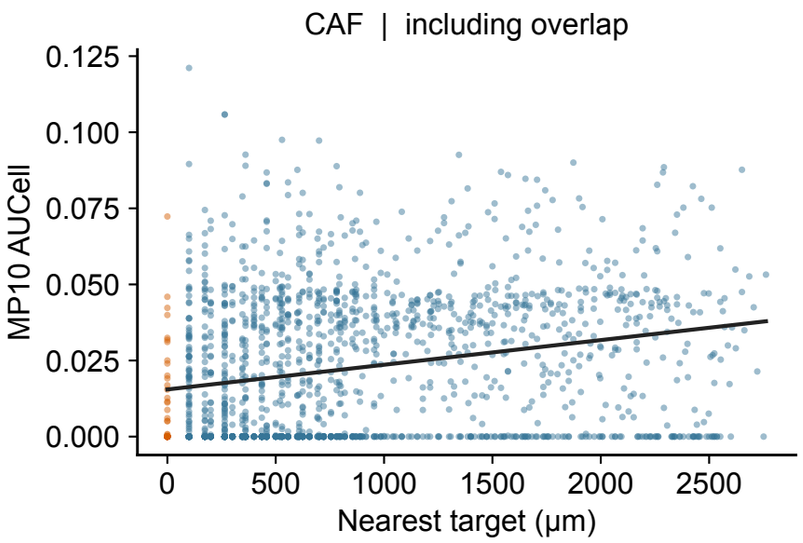




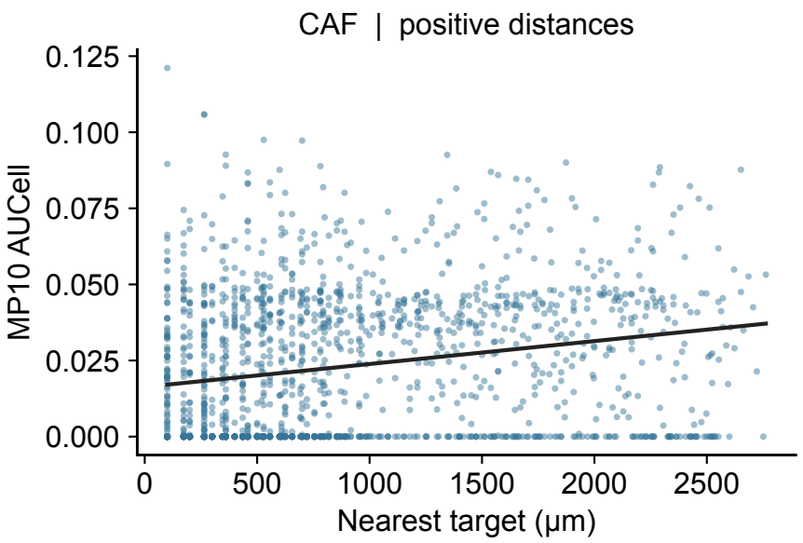


GSE319536_BC16 | MP10 versus nearest target

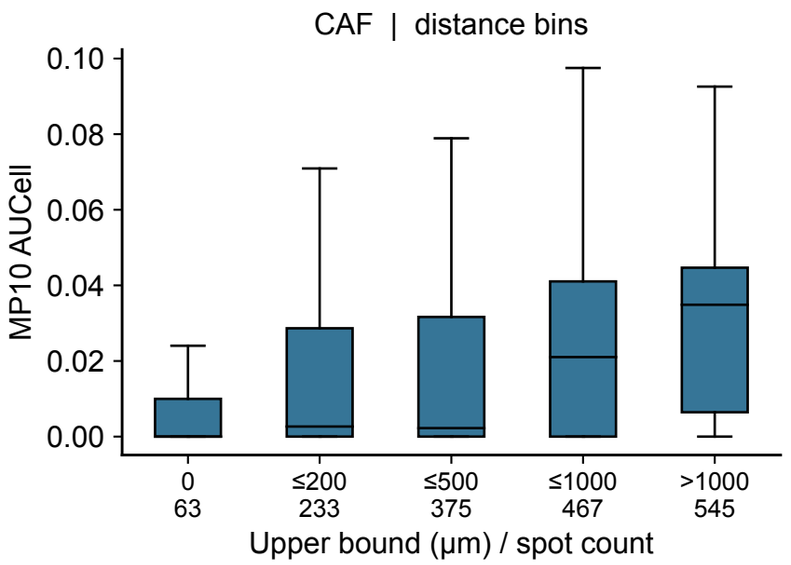
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



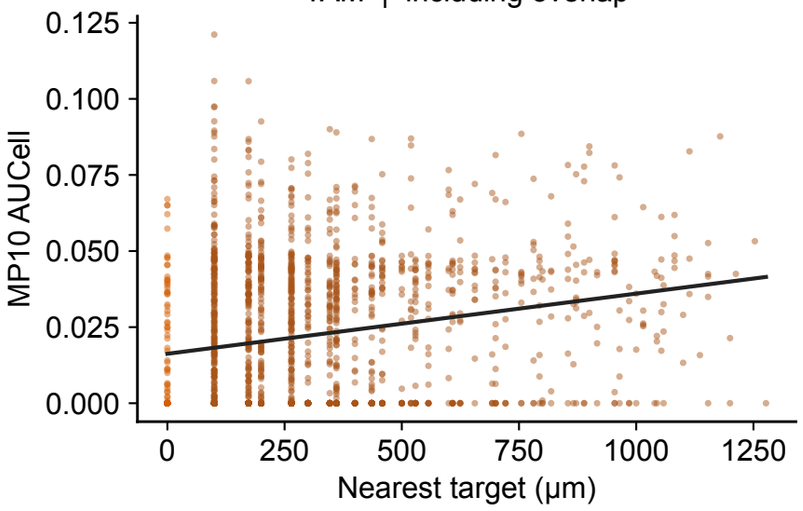
r +0.24 ρ +0.28
n 1683/1718 queries • 666 targets



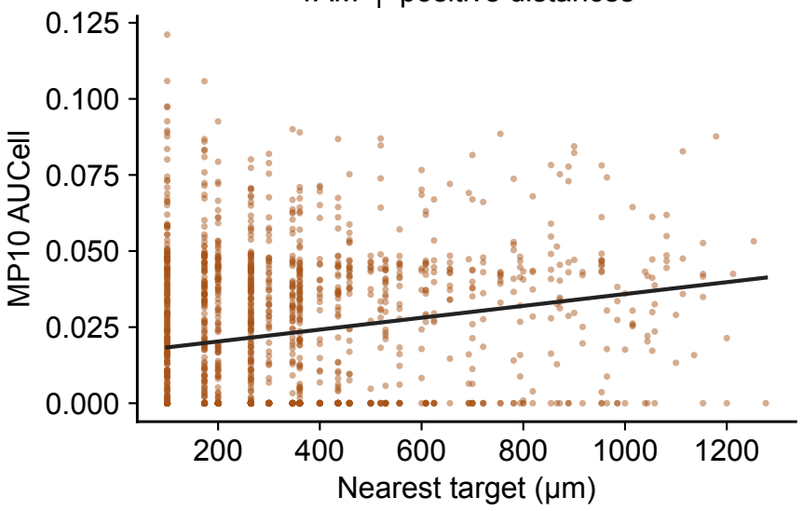
r +0.22 ρ +0.25
n 1620/1718 queries • 666 targets



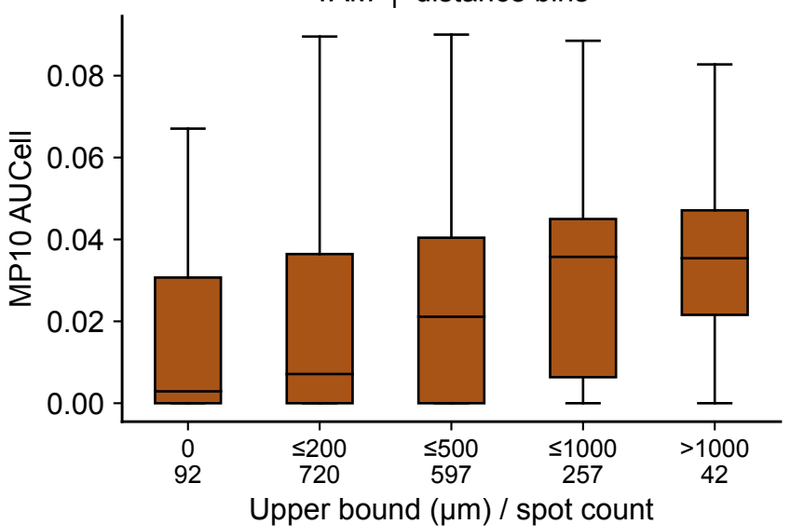
Orange = zero distance
Line = within-section linear fit



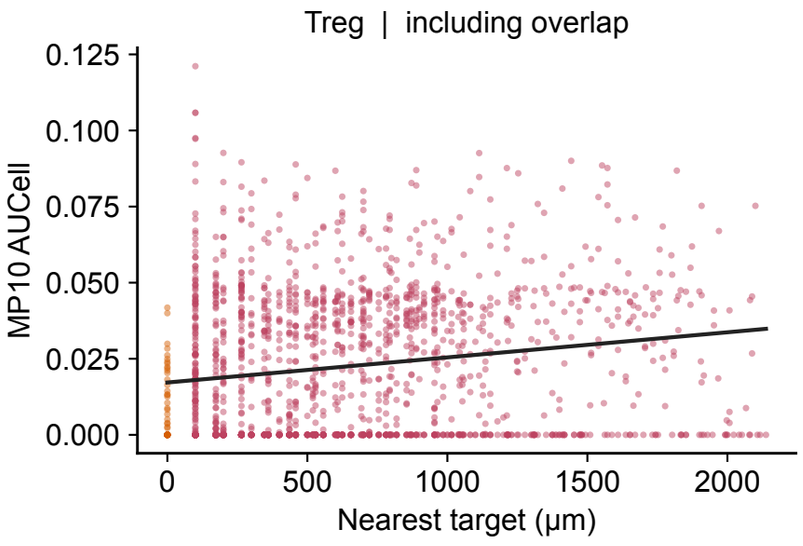
r +0.21 ρ +0.21
n 1708/1718 queries • 666 targets



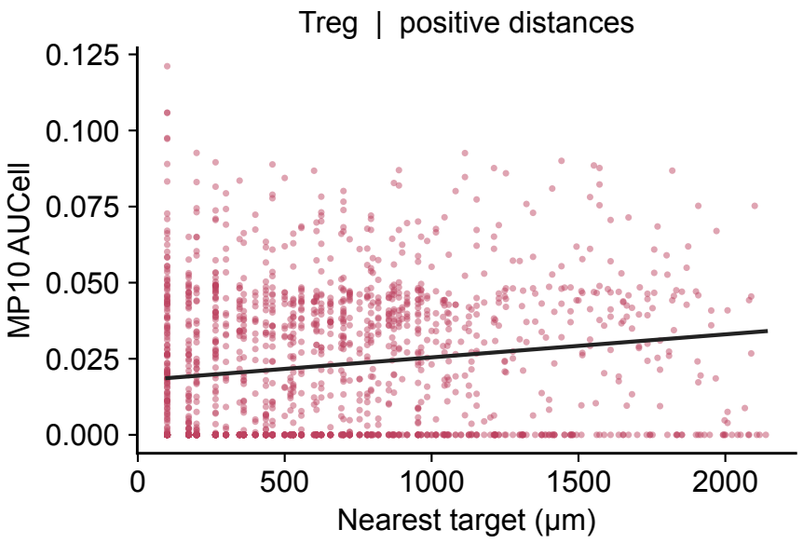
r +0.20 ρ +0.20
n 1616/1718 queries • 666 targets



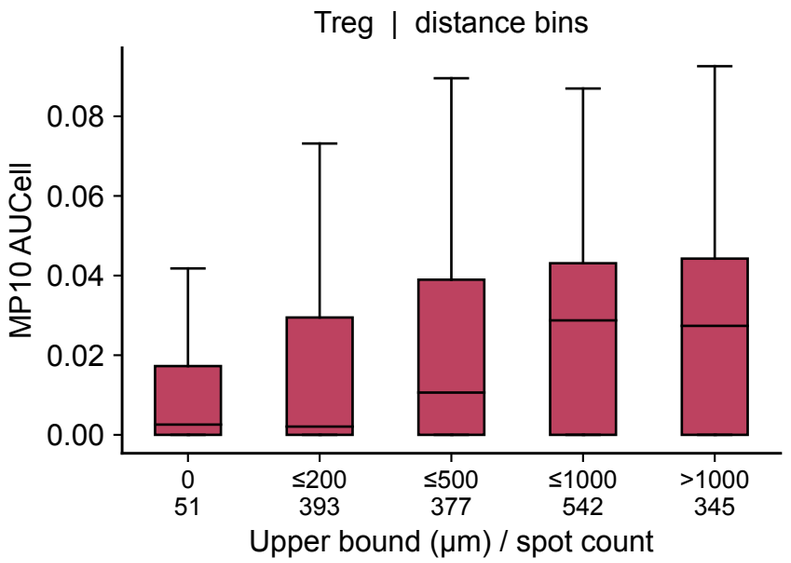
Orange = zero distance
Line = within-section linear fit



r +0.17 ρ +0.20
n 1708/1718 queries • 666 targets



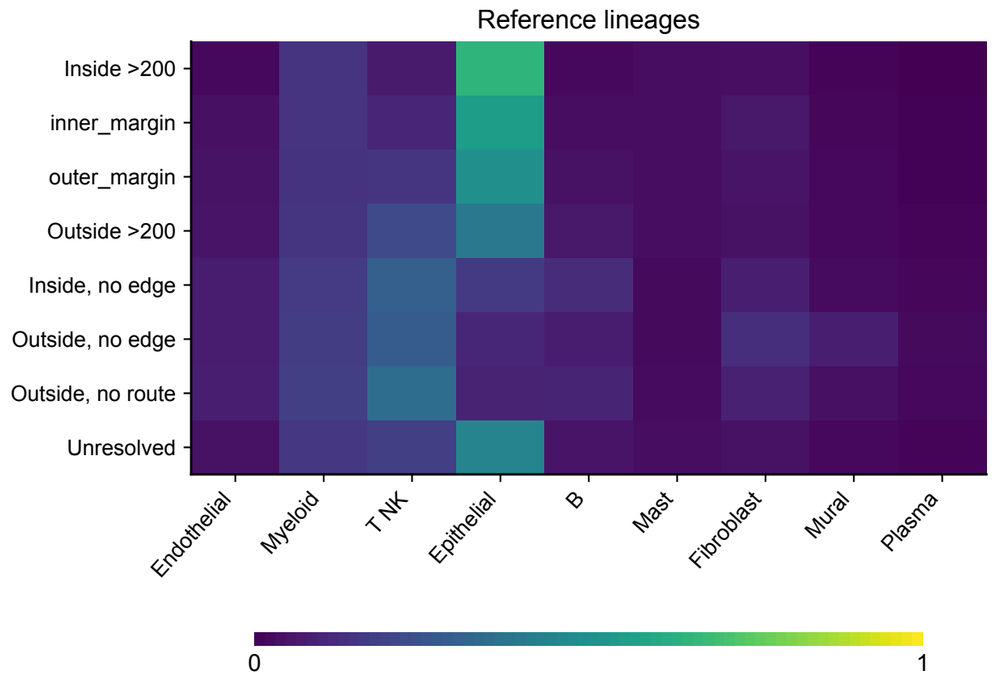
r +0.16 ρ +0.18
n 1657/1718 queries • 666 targets

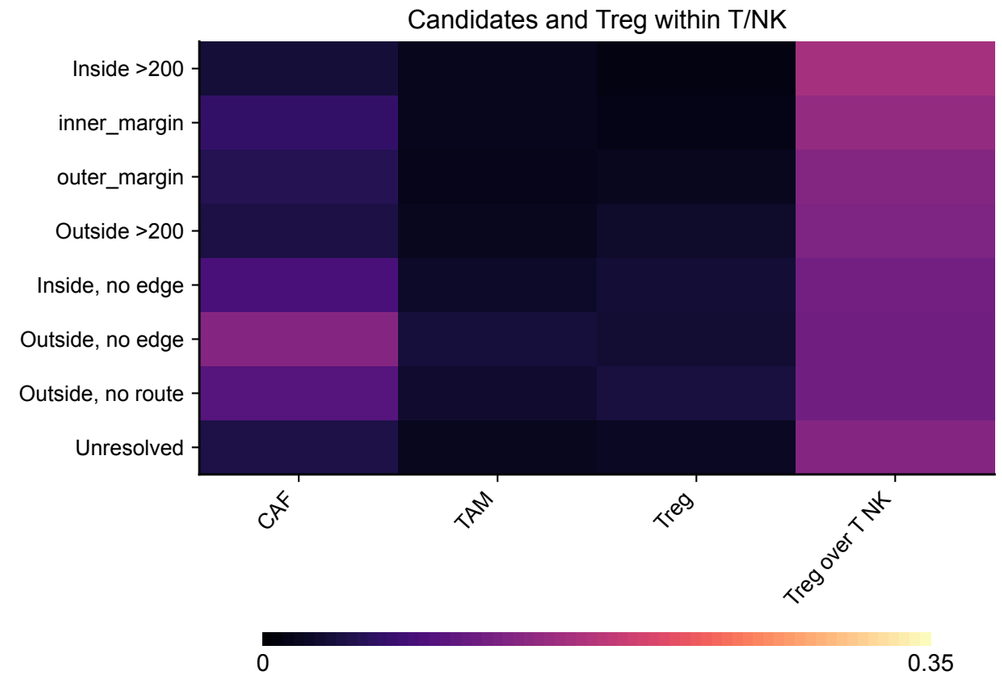


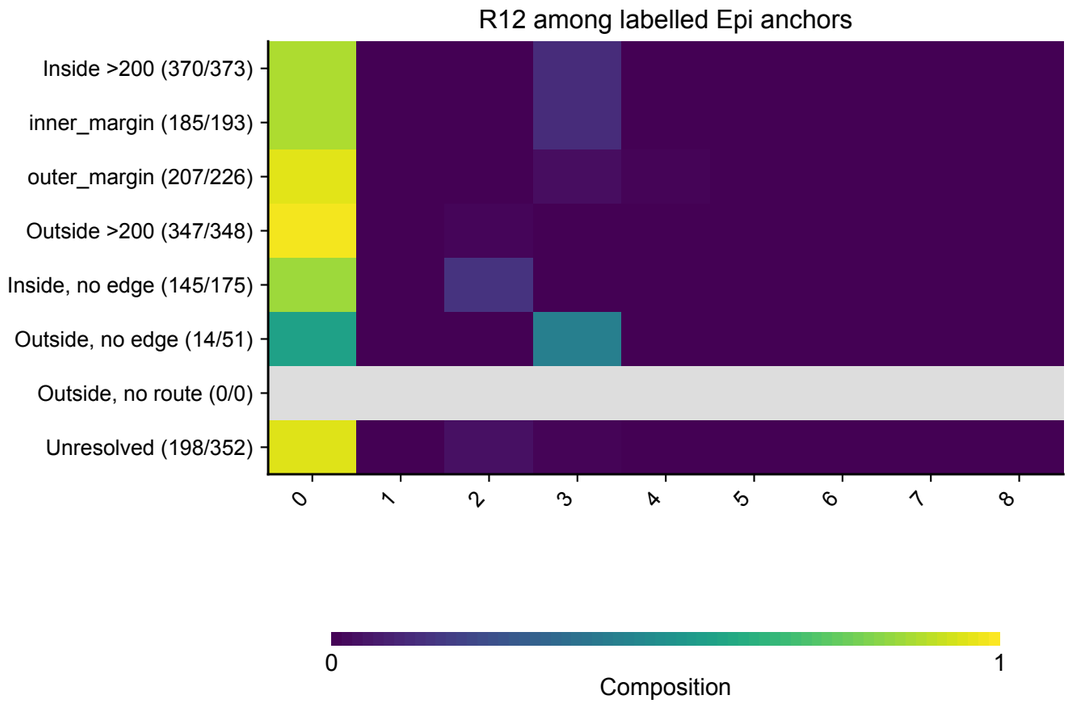
Orange = zero distance
Line = within-section linear fit

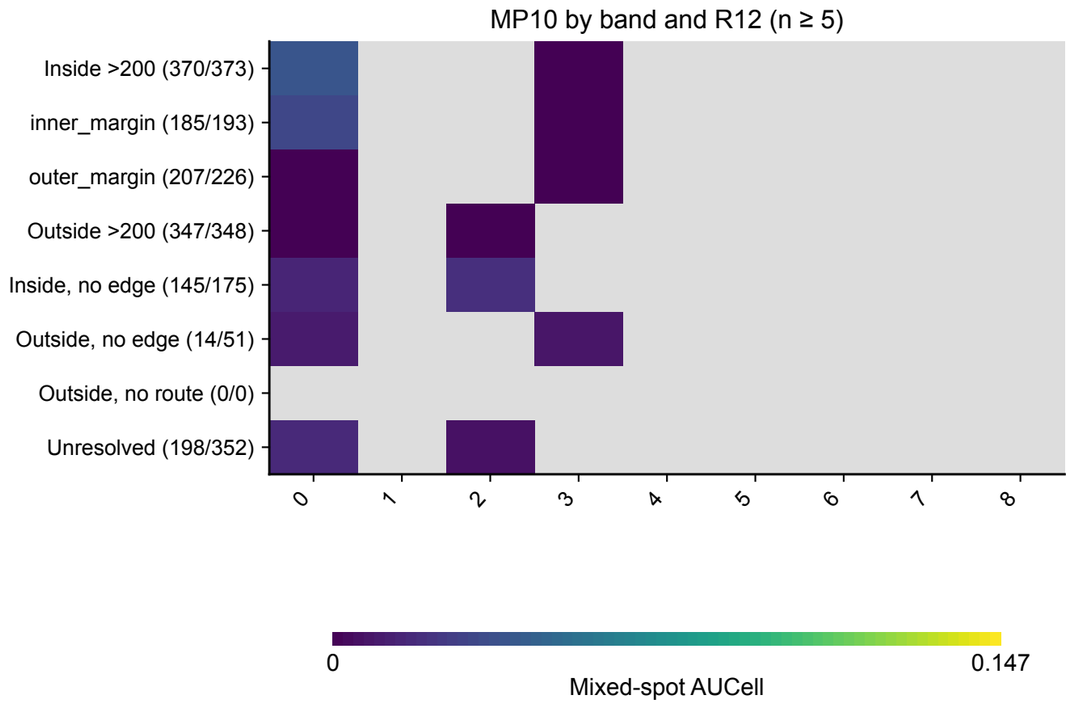
GSE319536_BC16 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable



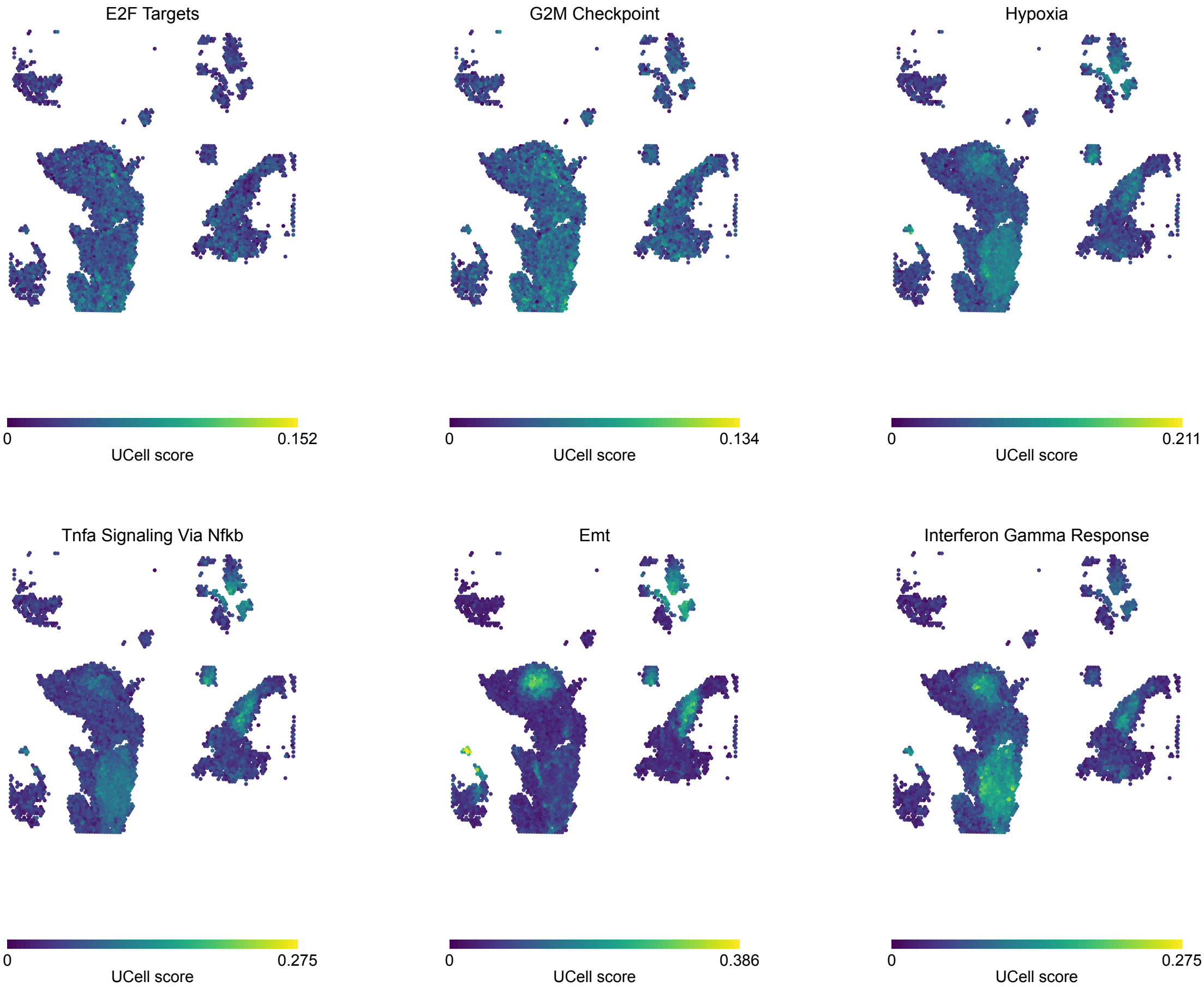






GSE319536_BC16 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections



GSE319536_BC16 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Il2 Stat5 Signaling

Tgf Beta Signaling

Cholesterol Homeostasis

Cholesterol Biosynthesis

Extracellular Matrix Organization

Co Inhibition By Ctla4

GSE319536_BC16 | Treg context across the full tissue

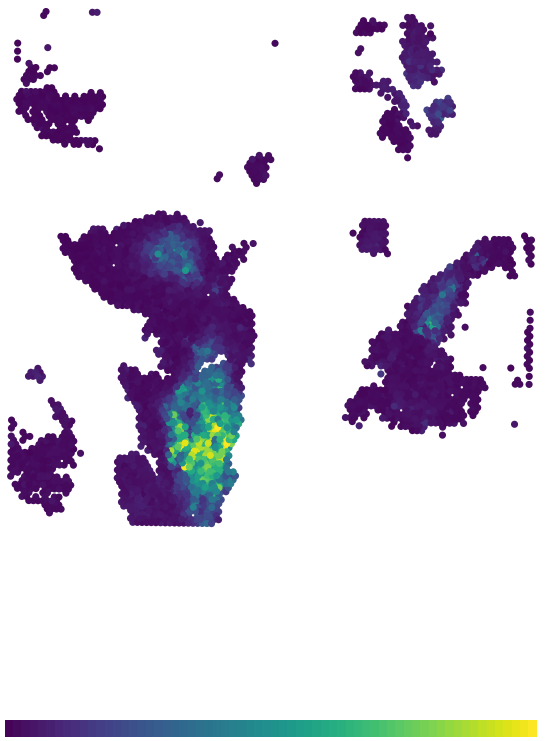
Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Treg composition

0 0.102

Fraction of all 30

Treg estimated abundance



0 2.85

Posterior mean estimate

Treg Resting Vs Activated Dn

0 0.0778

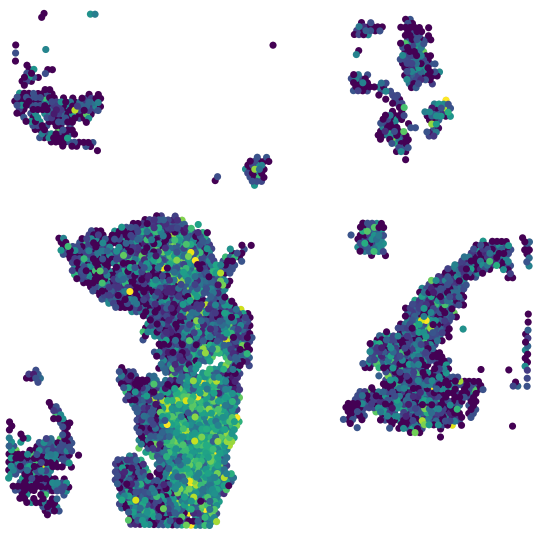
UCell score

Treg Resting Vs Activated Up

0 0.0609

UCell score

Co Inhibition By Ctla4



0 0.151

UCell score

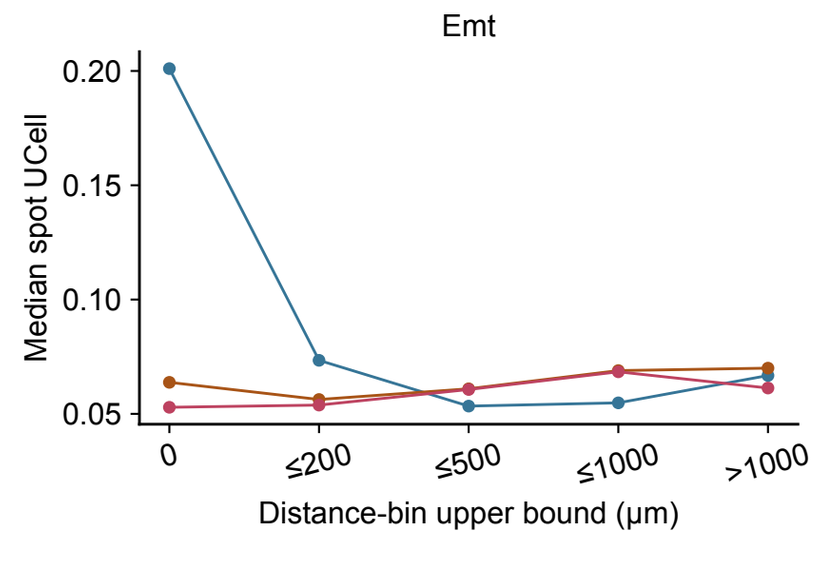
Il2 Stat5 Signaling

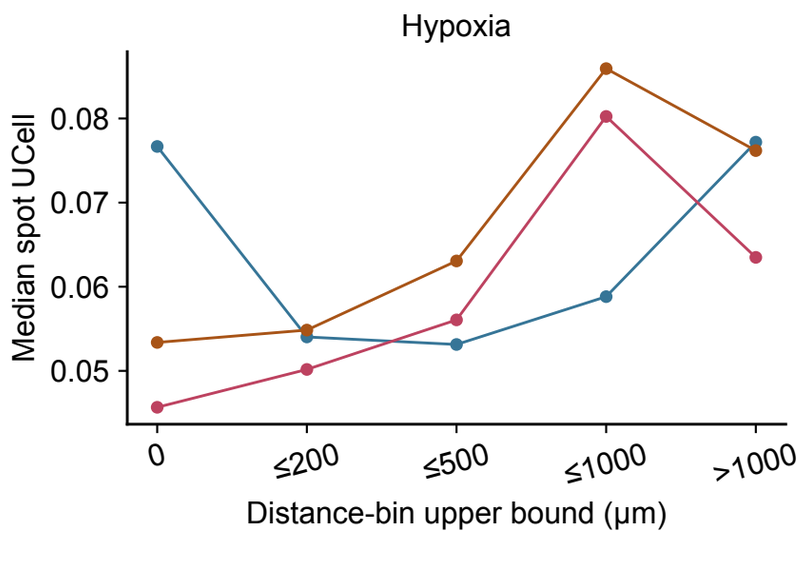
0 0.144

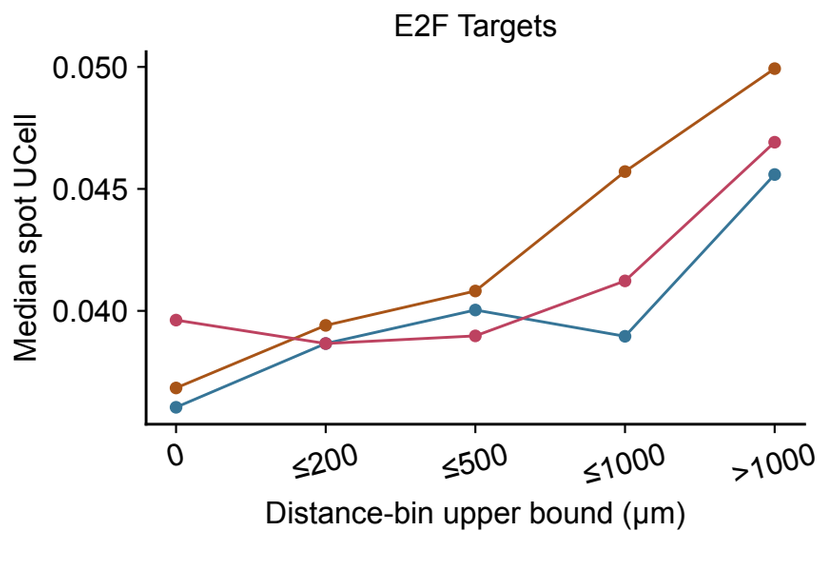
UCell score

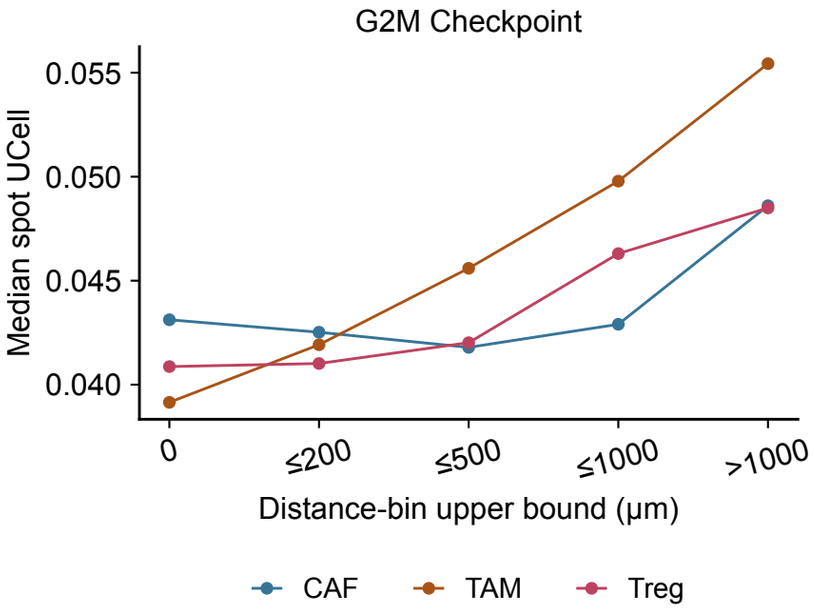
GSE319536_BC16 | Programs by target distance

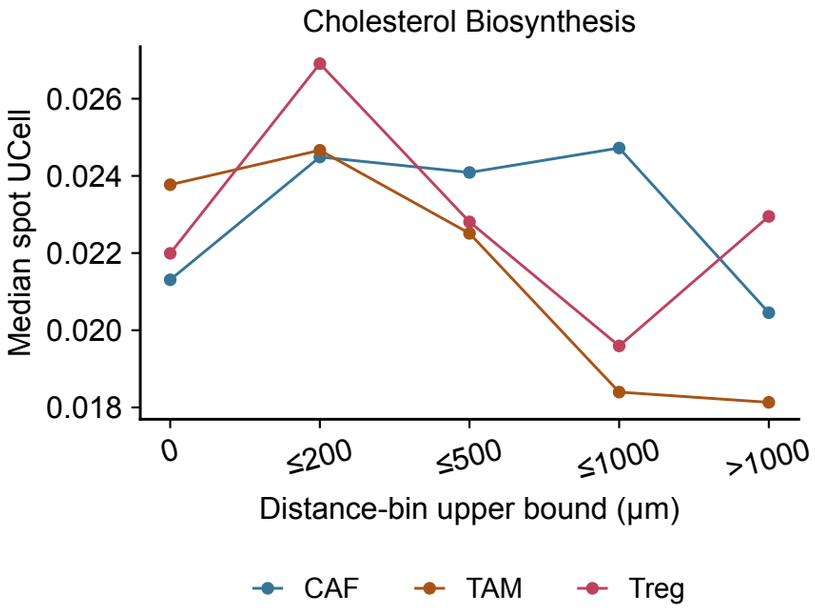
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

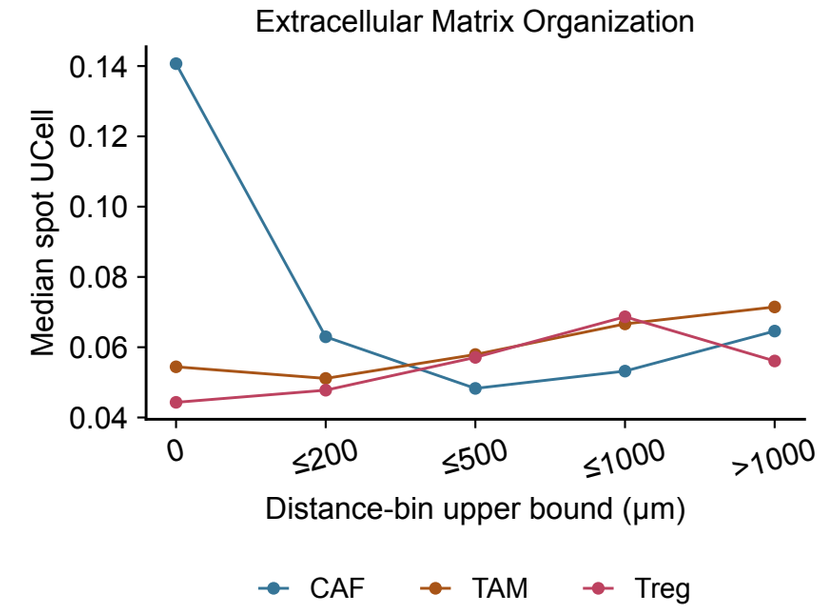






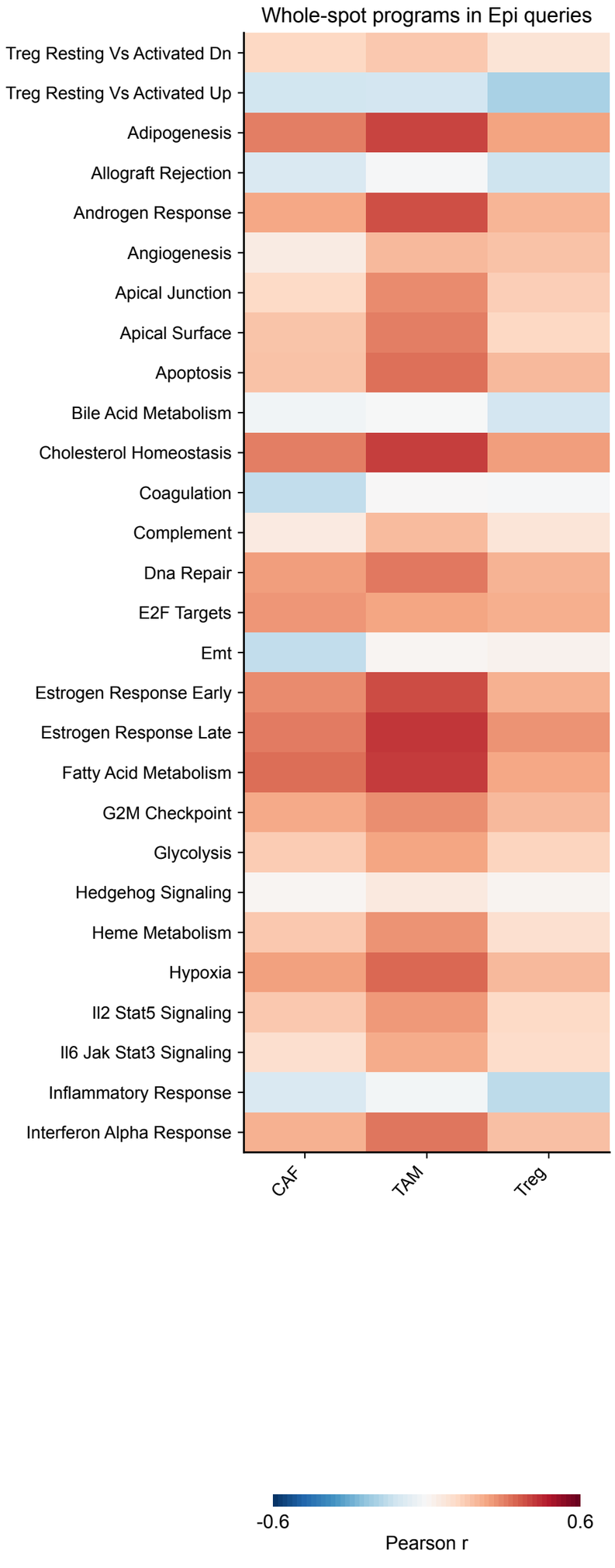


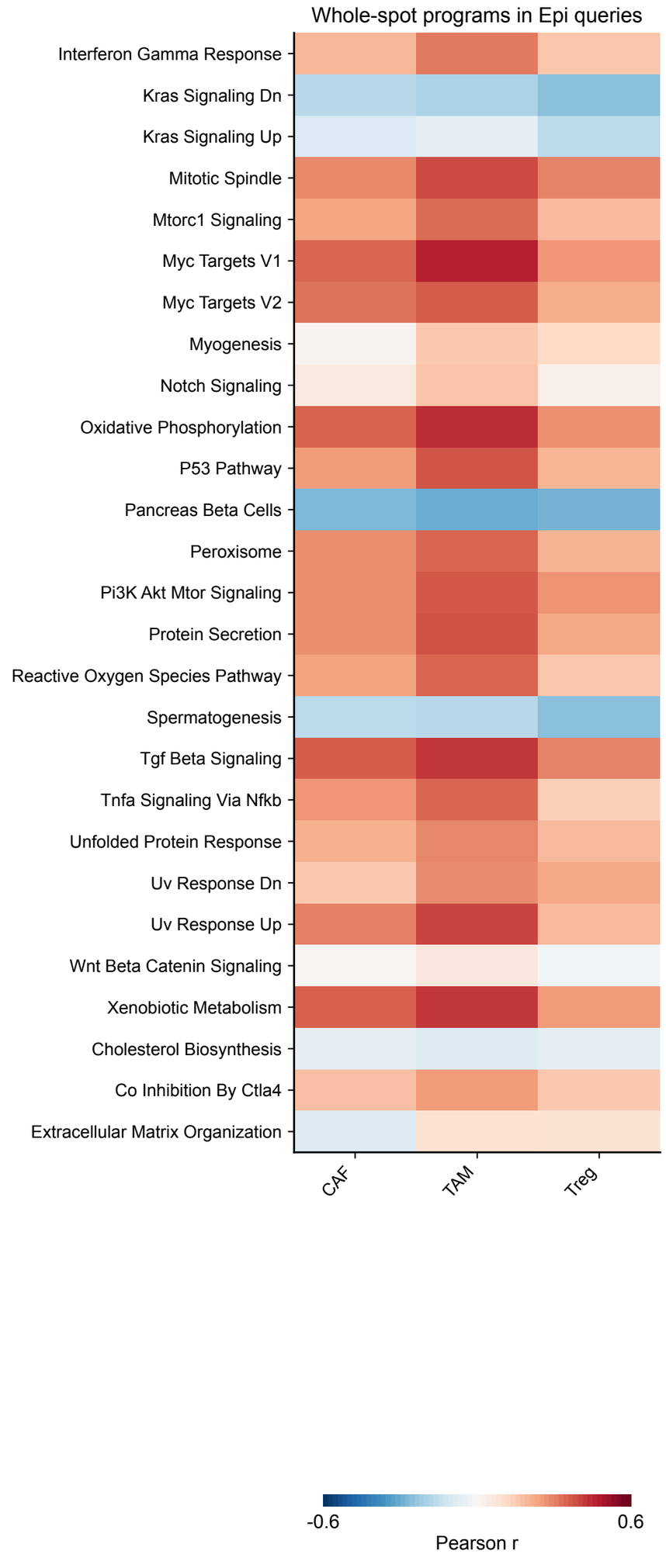




GSE319536_BC16 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

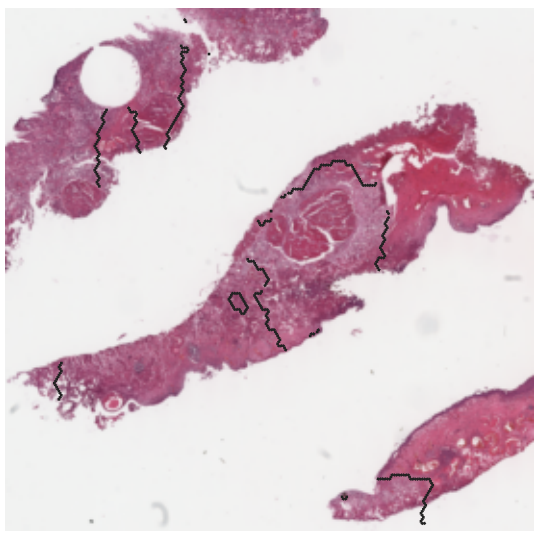




GSE319536_BC17 | Tissue context

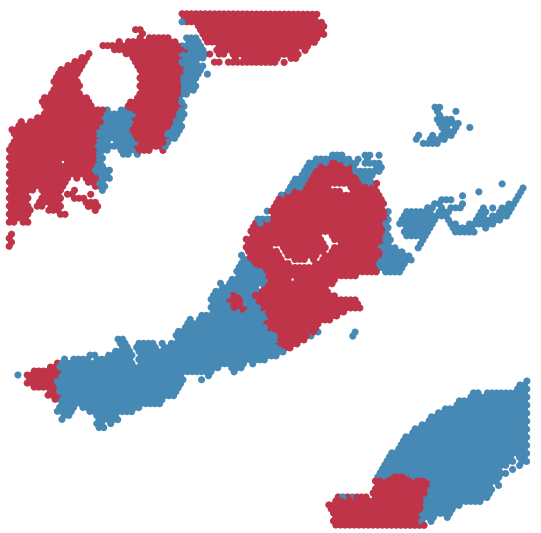
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



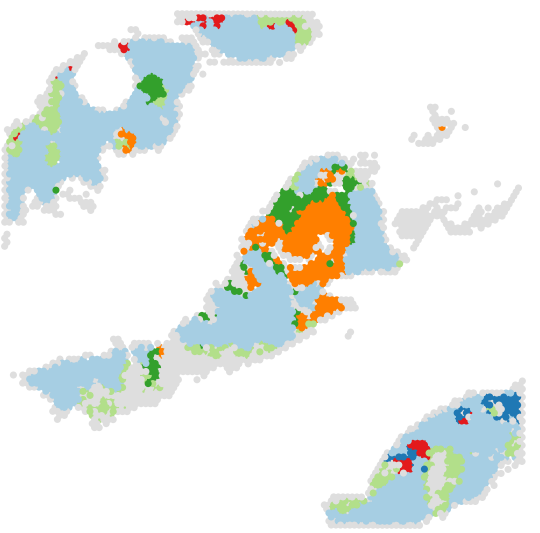
Black line: observed candidate outer edge

Final candidate location



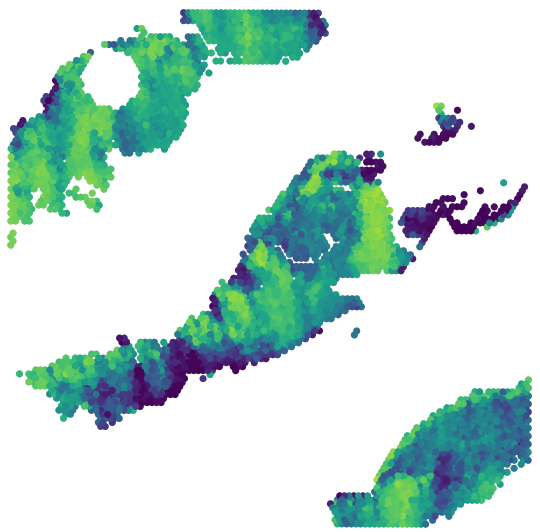
inside outside

Accepted R12 (original 0–8)



0 2 3 5 7 1

Epithelial support



0 1

Fraction of all 30

MP10 in Epi queries

0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)

0 0.14

Matched-model AUCell

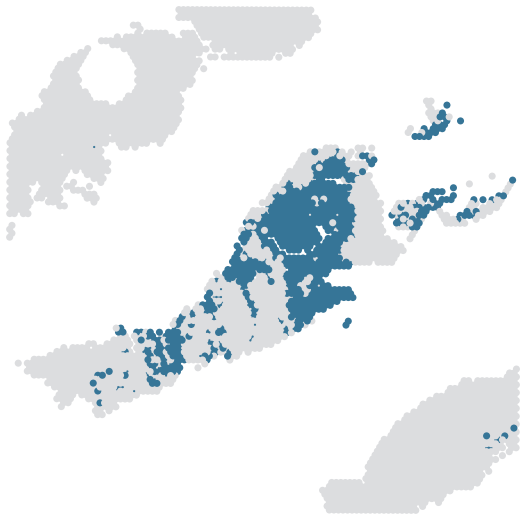
GSE319536_BC17 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

CAF composition

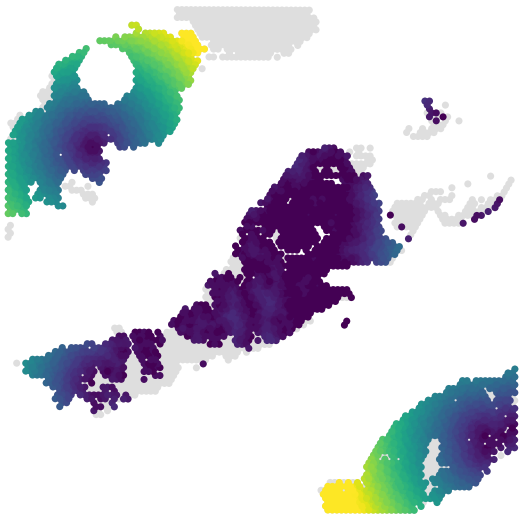
0 0.418
Fraction of all 30

CAF P80 targets



False True

CAF distance

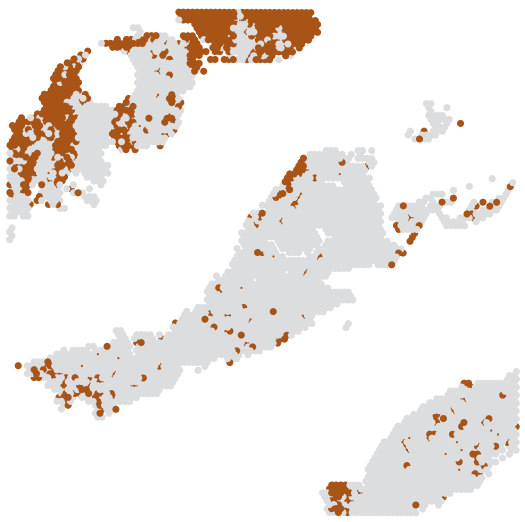


0 3.1e+03
Nearest spot center (μm)

TAM composition

0 0.291
Fraction of all 30

TAM P80 targets



False True

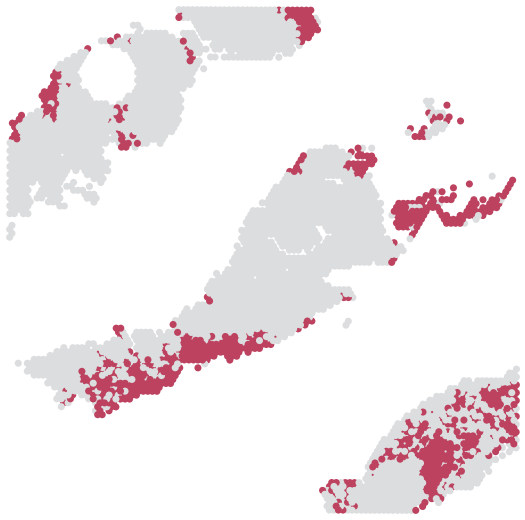
TAM distance

0 1.13e+03
Nearest spot center (μm)

Treg composition

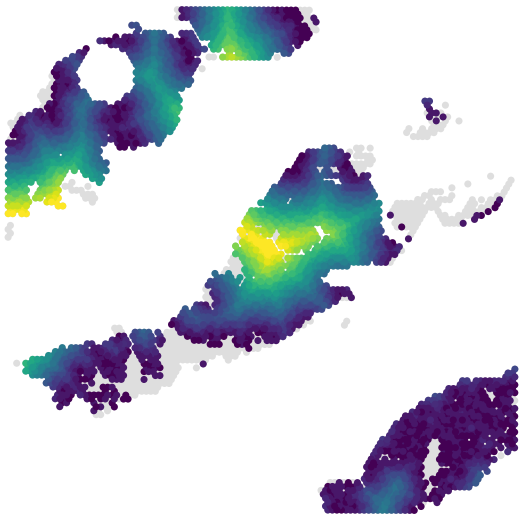
0 0.102
Fraction of all 30

Treg P80 targets



False True

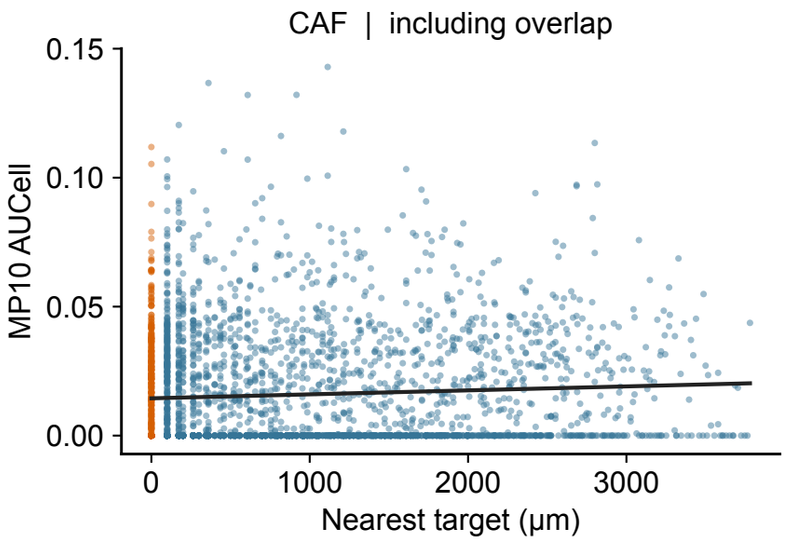
Treg distance



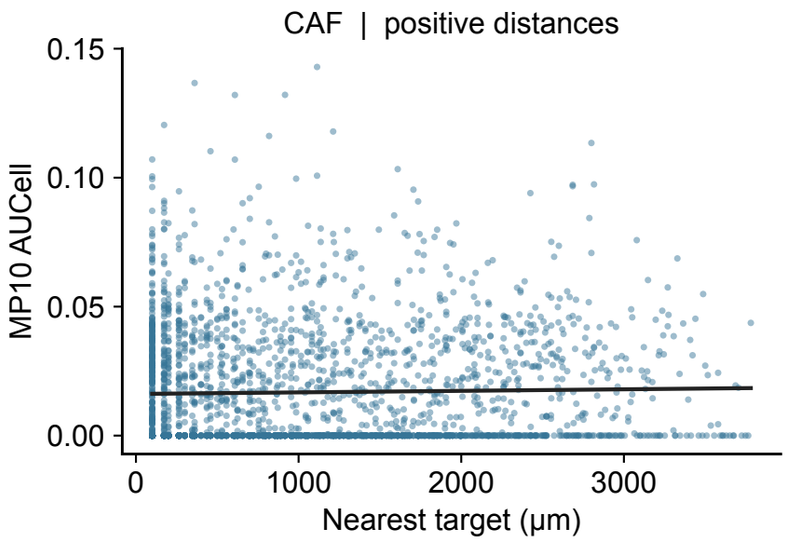
0 1.61e+03
Nearest spot center (μm)

GSE319536_BC17 | MP10 versus nearest target

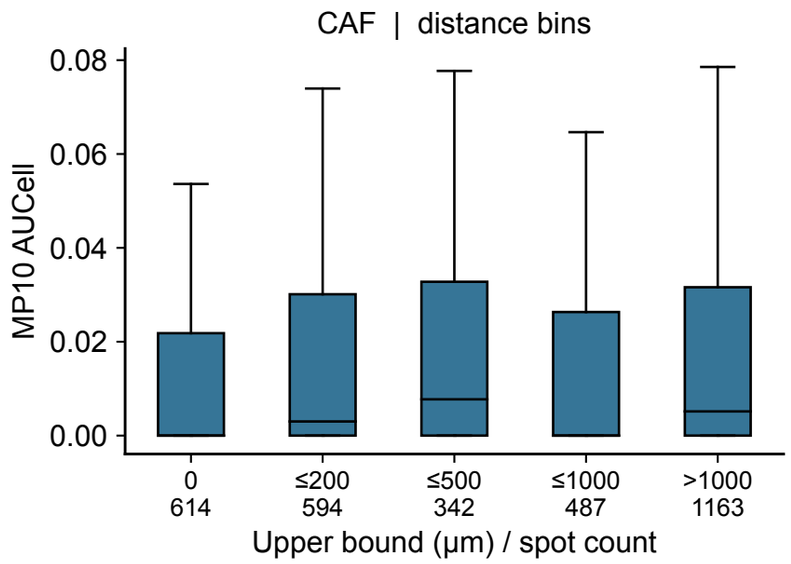
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



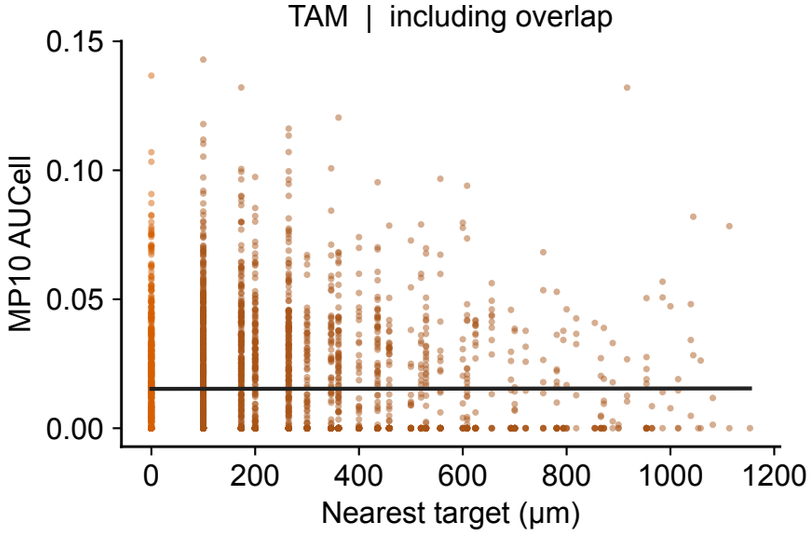
r +0.06 ρ +0.08
n 3200/3508 queries • 815 targets



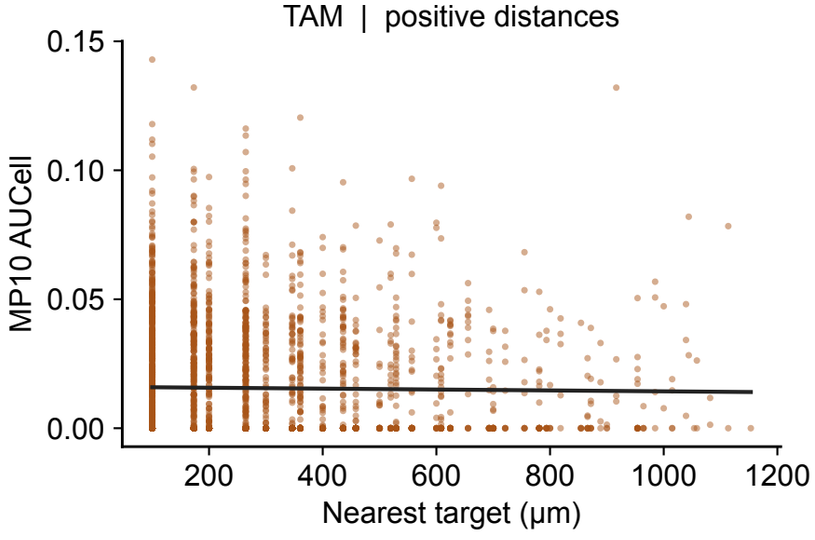
r +0.02 ρ +0.02
n 2586/3508 queries • 815 targets



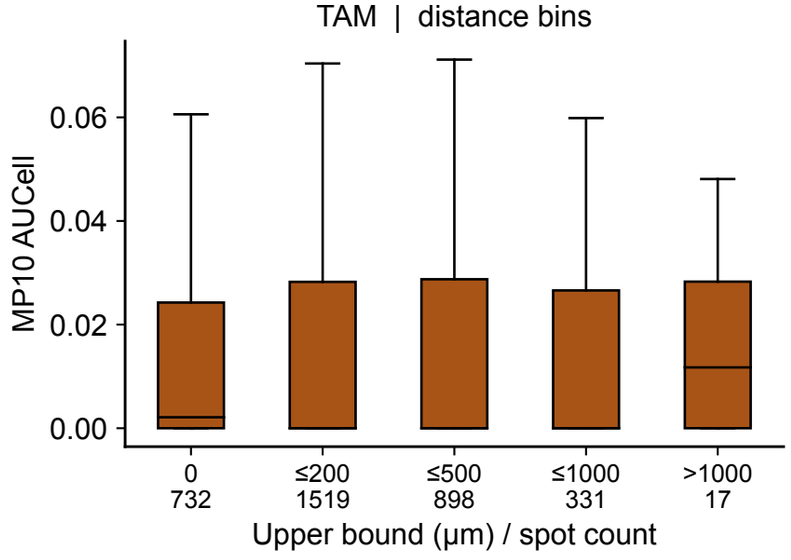
Orange = zero distance
Line = within-section linear fit



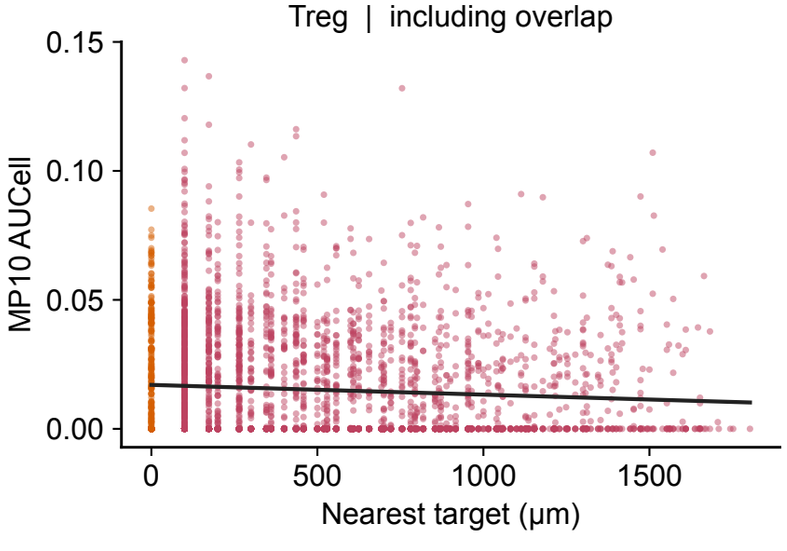
r +0.00 ρ +0.00
n 3497/3508 queries • 815 targets



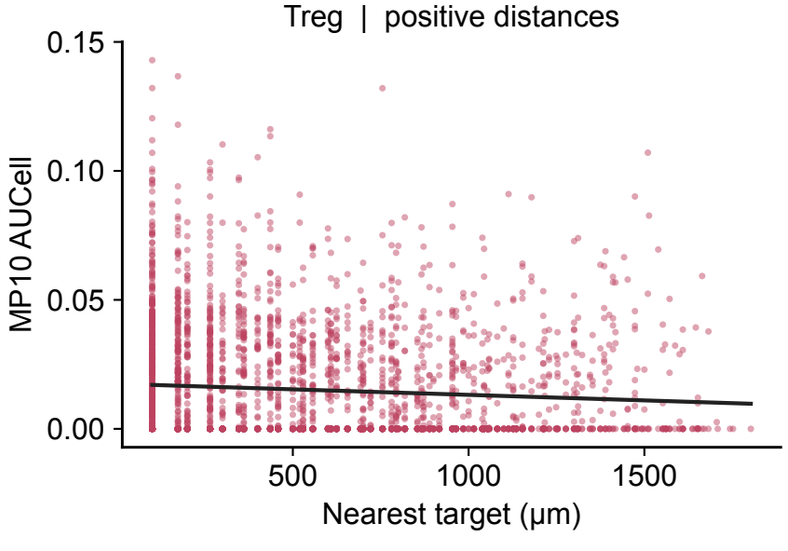
r -0.02 ρ -0.01
n 2765/3508 queries • 815 targets



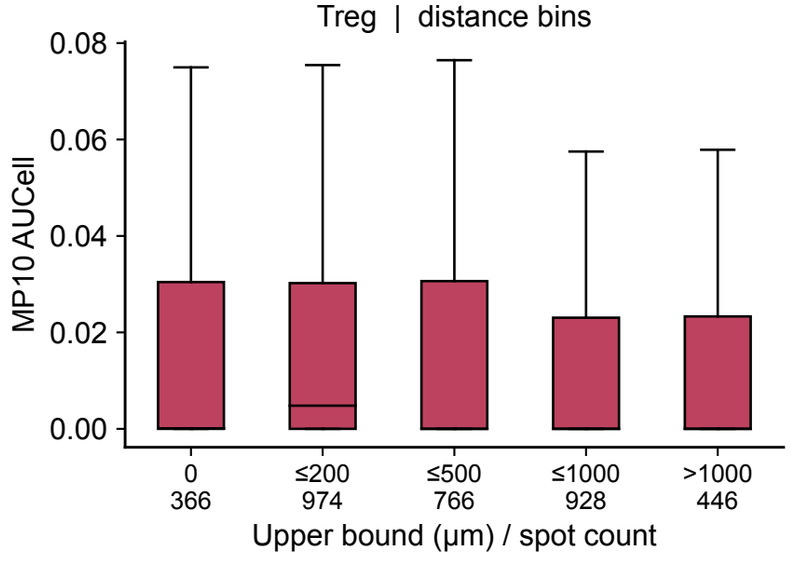
Orange = zero distance
Line = within-section linear fit



r -0.07 ρ -0.06
n 3480/3508 queries • 815 targets



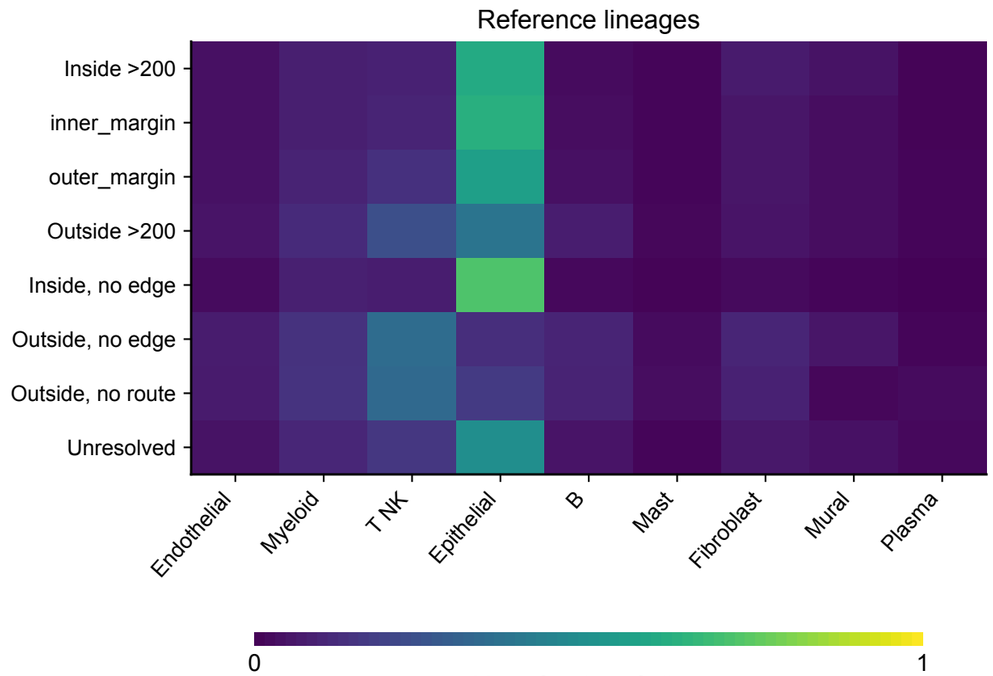
r -0.08 ρ -0.08
n 3114/3508 queries • 815 targets

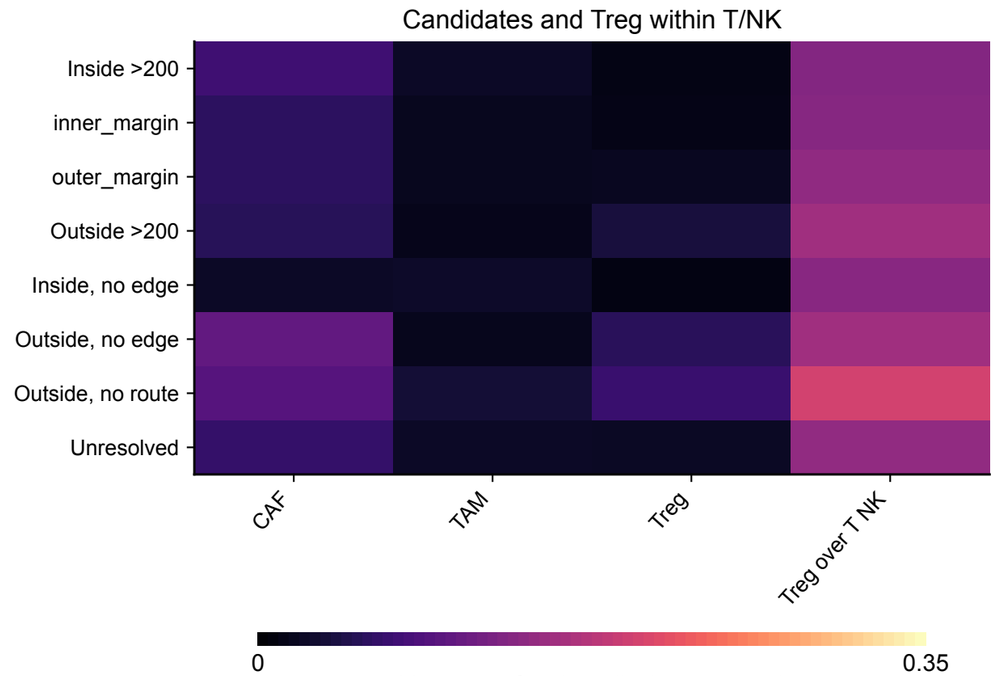


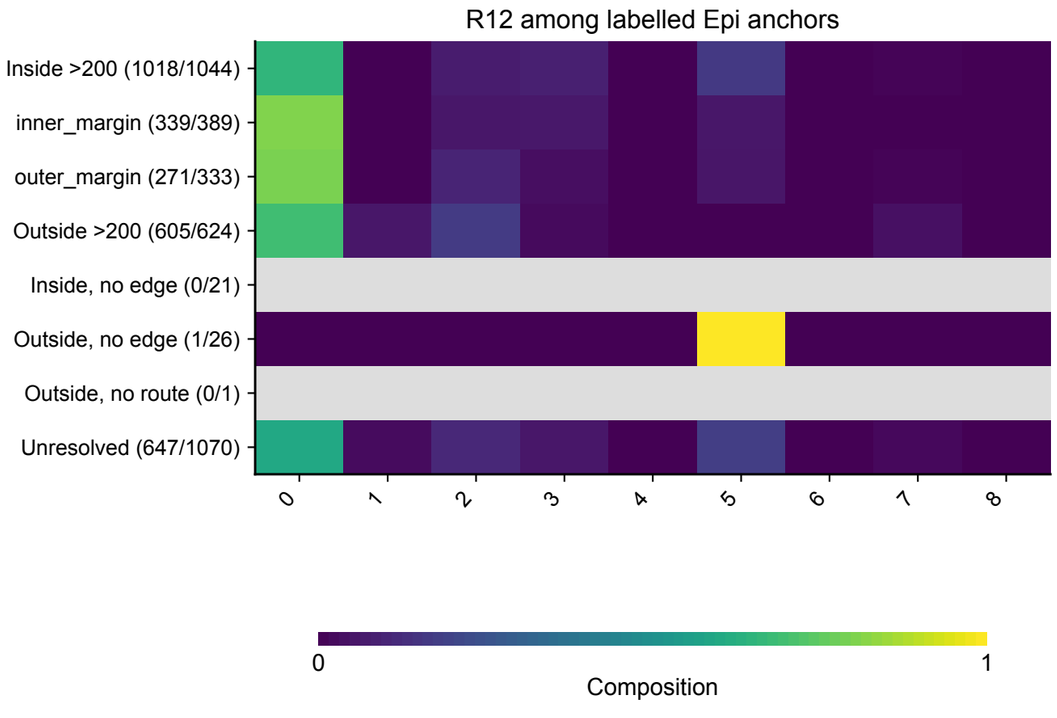
Orange = zero distance
Line = within-section linear fit

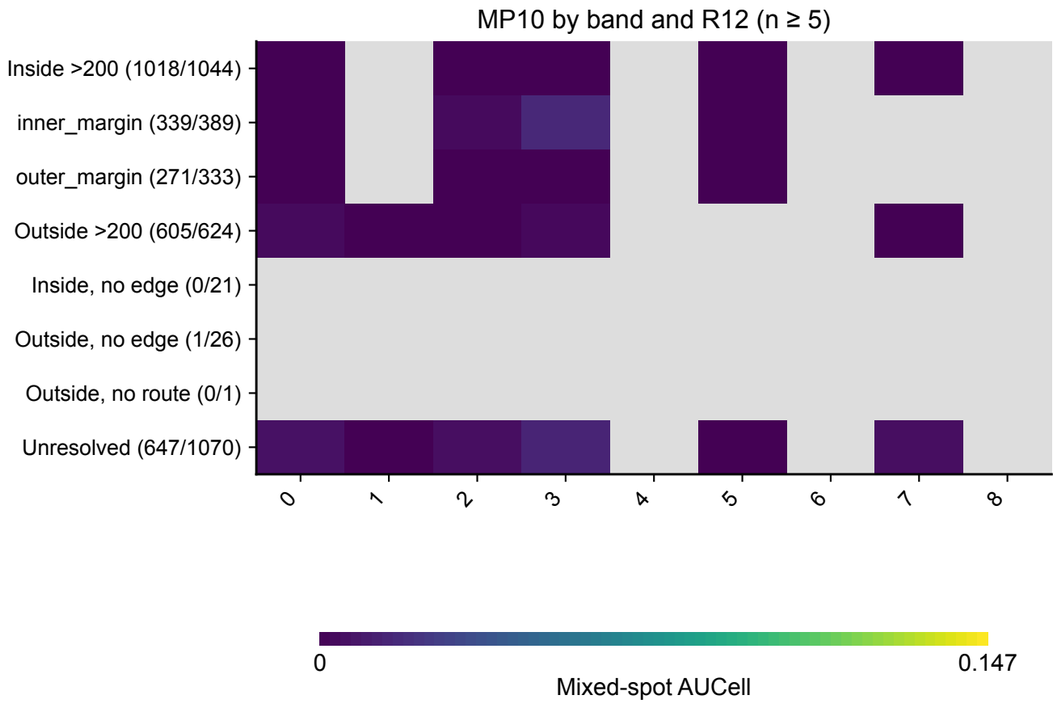
GSE319536_BC17 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable



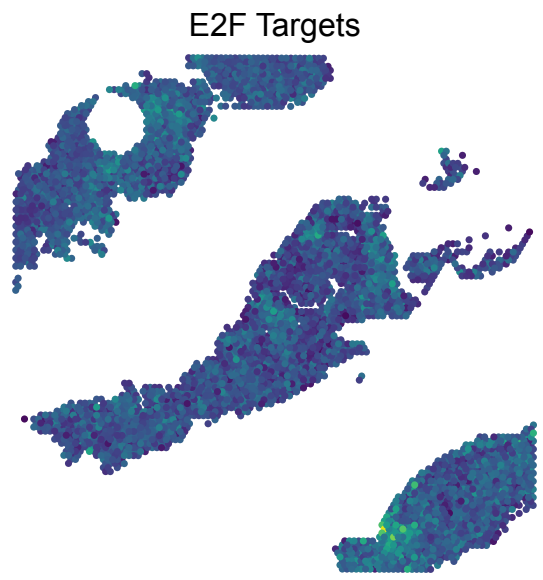


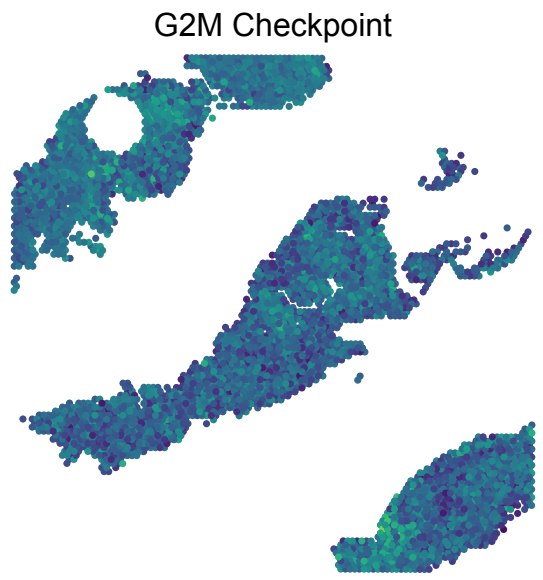


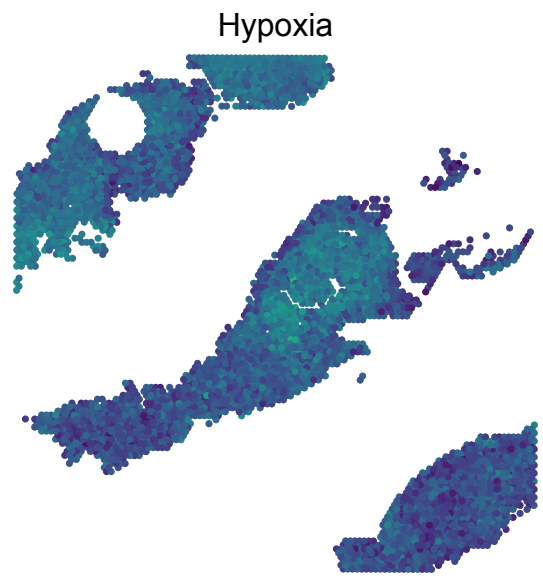


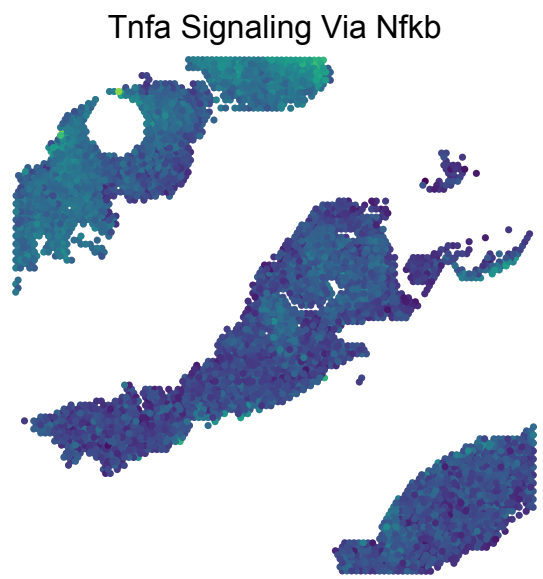
GSE319536_BC17 | Growth, stress and inflammation

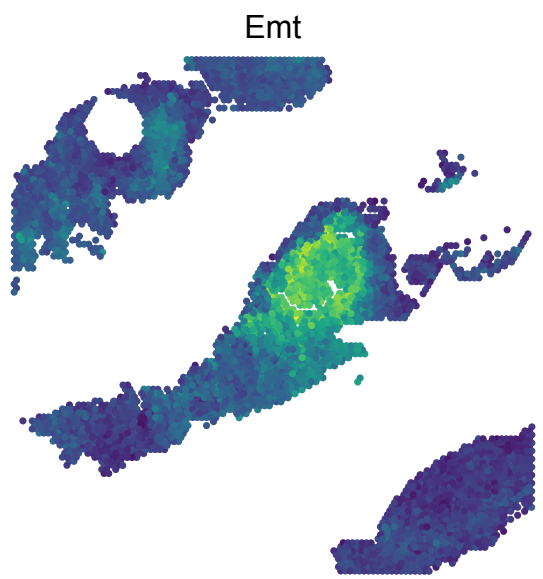
Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

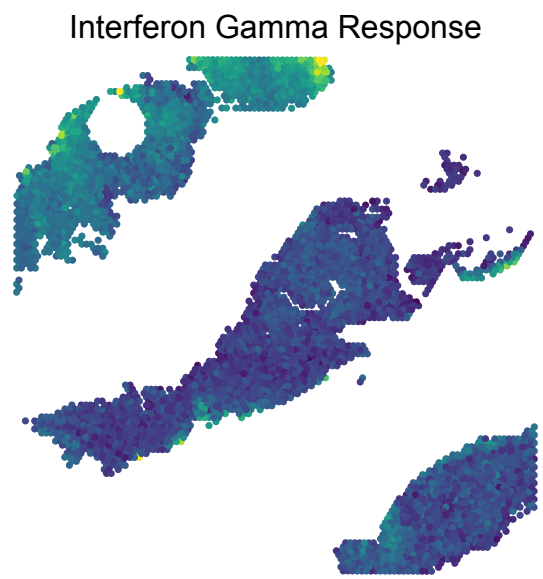












GSE319536_BC17 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Il2 Stat5 Signaling

0 0.144

UCell score

Tgf Beta Signaling

0 0.232

UCell score

Cholesterol Homeostasis

0 0.21

UCell score

Cholesterol Biosynthesis

0 0.166

UCell score

Extracellular Matrix Organization

0 0.247

UCell score

Co Inhibition By Ctla4

0 0.151

UCell score

GSE319536_BC17 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

Treg composition

0 0.102

Fraction of all 30

Treg estimated abundance

0 2.85

Posterior mean estimate

Treg Resting Vs Activated Dn

0 0.0778

UCell score

Treg Resting Vs Activated Up

0 0.0609

UCell score

Co Inhibition By Ctla4

0 0.151

UCell score

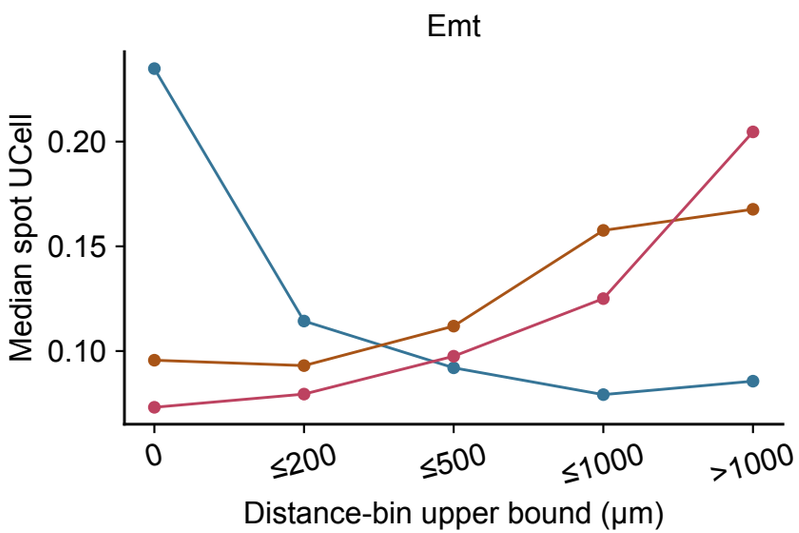
Il2 Stat5 Signaling

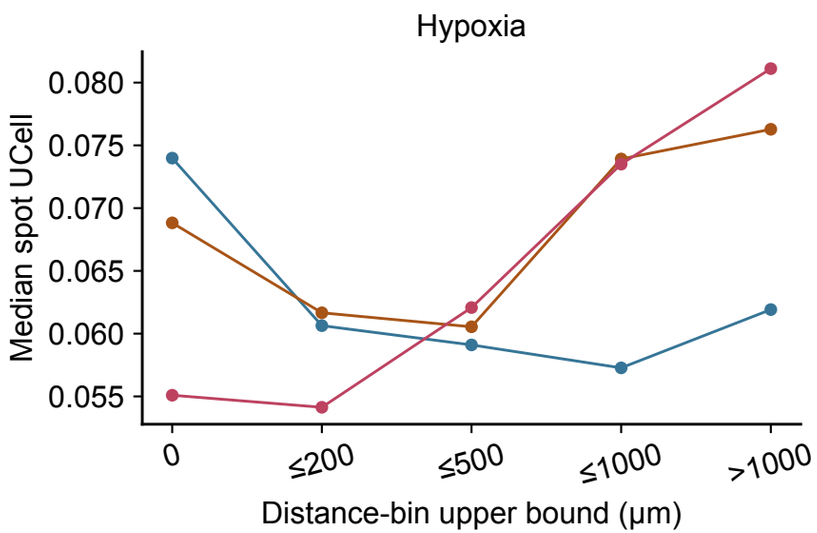
0 0.144

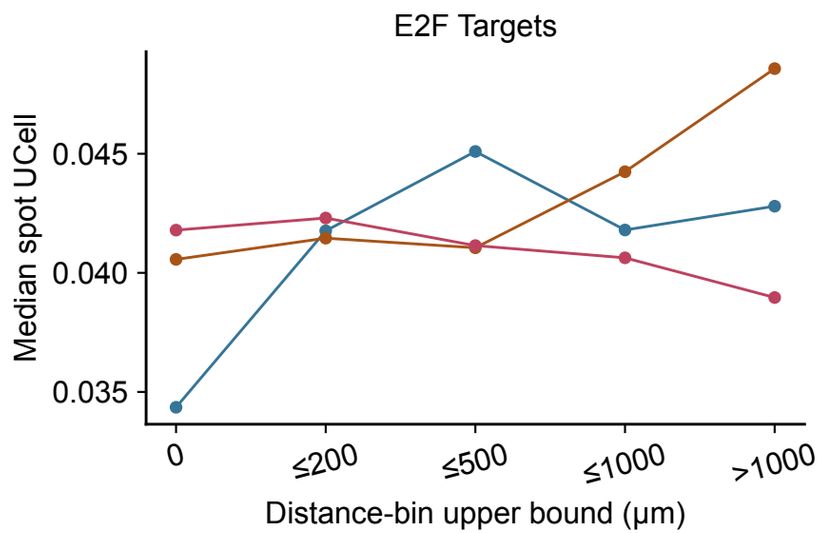
UCell score

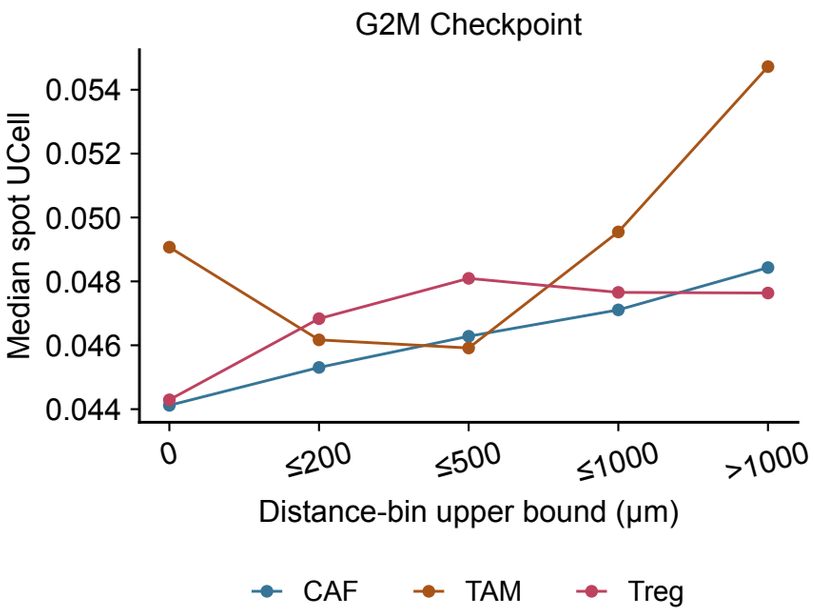
GSE319536_BC17 | Programs by target distance

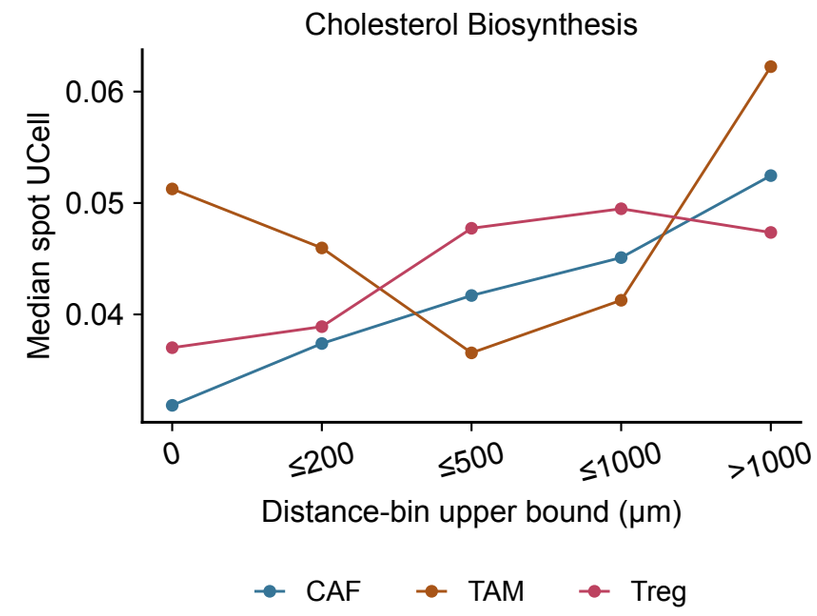
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

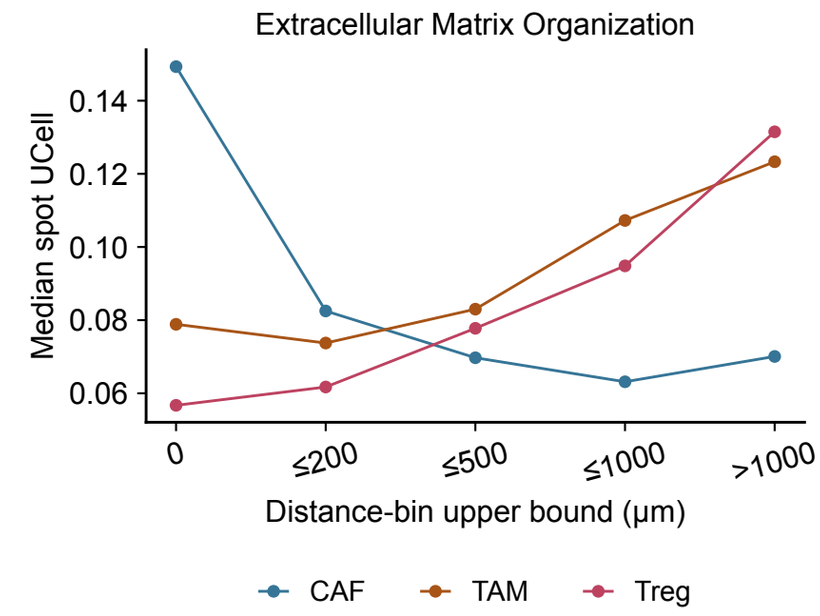






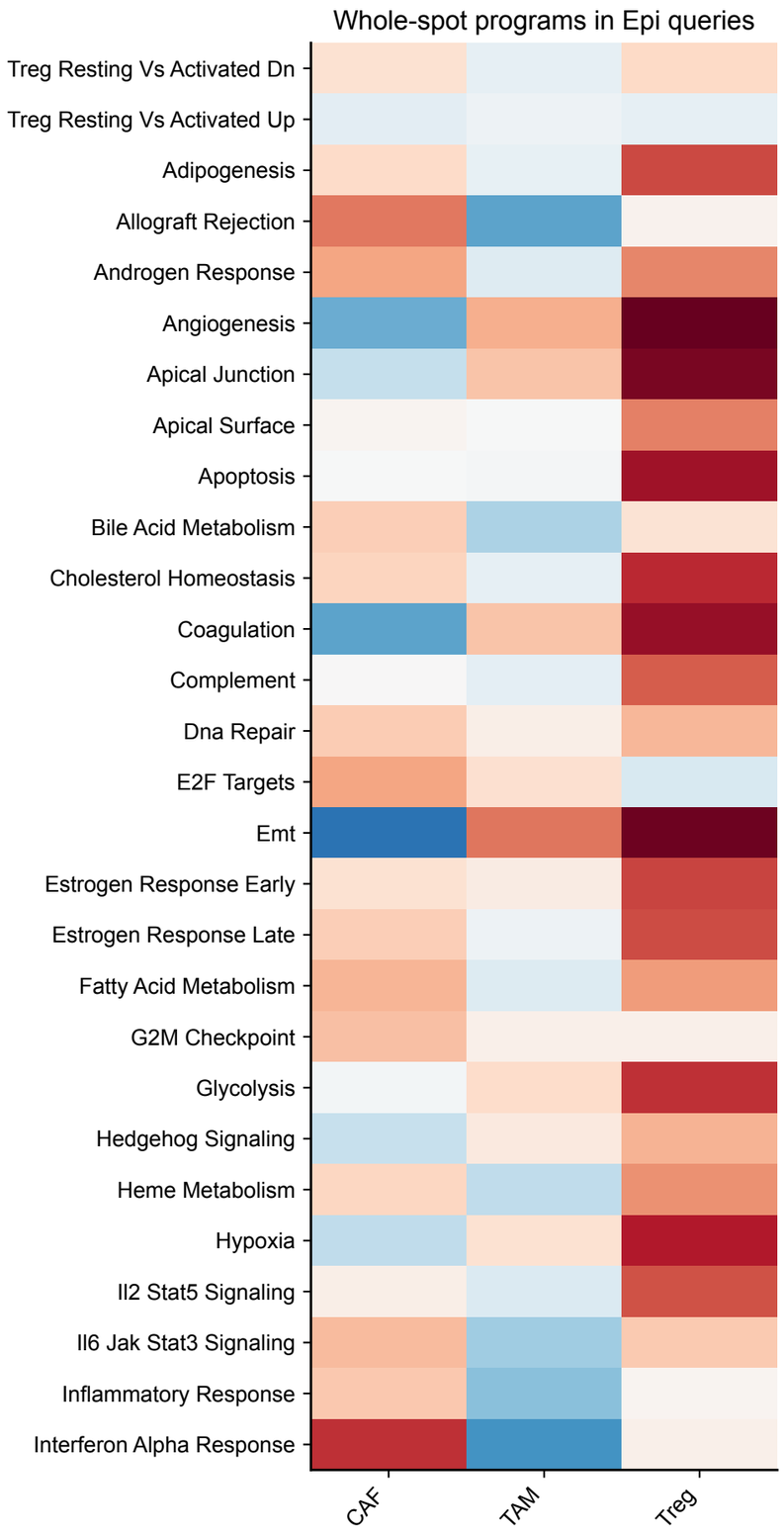


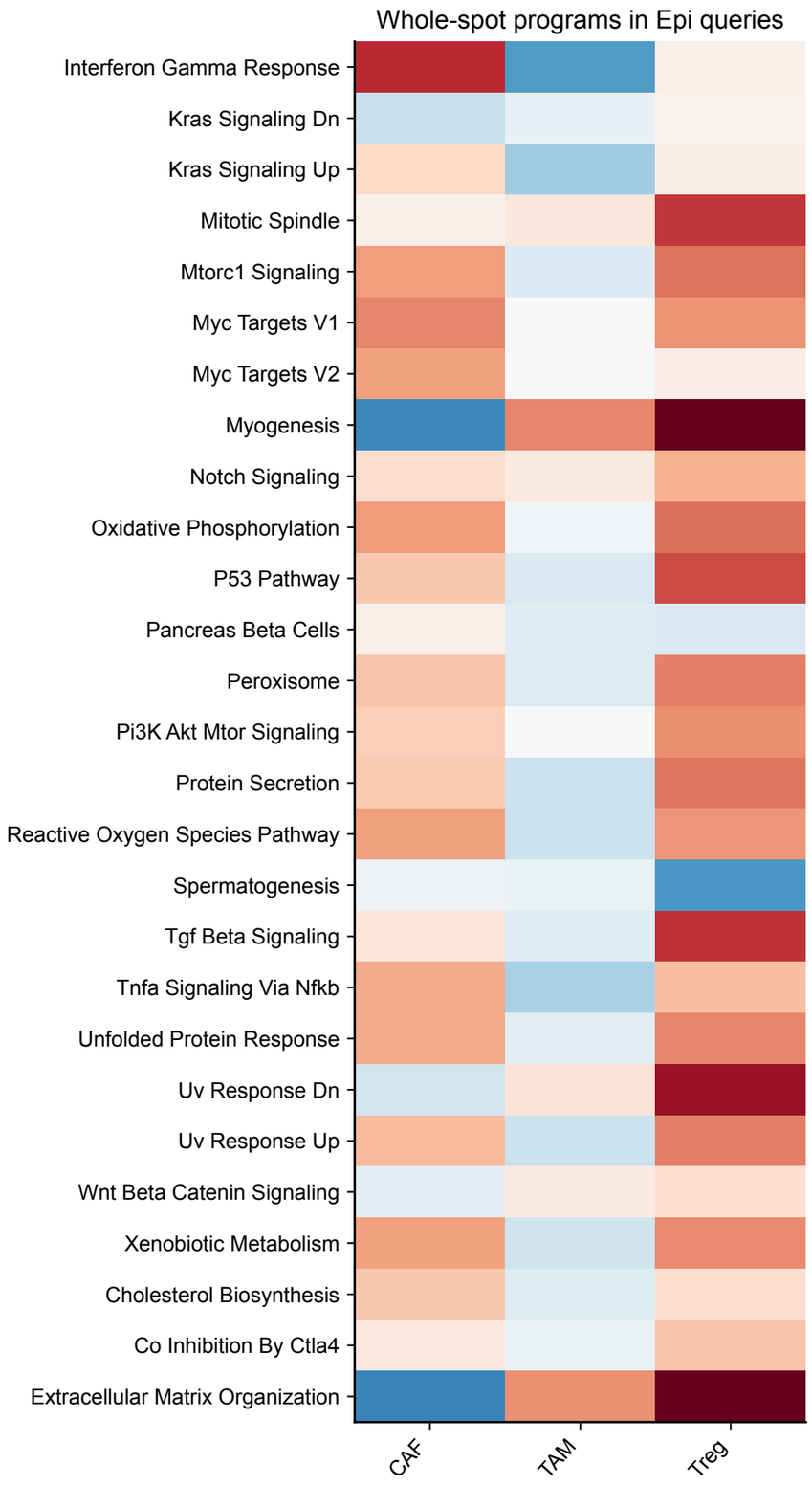




GSE319536_BC17 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

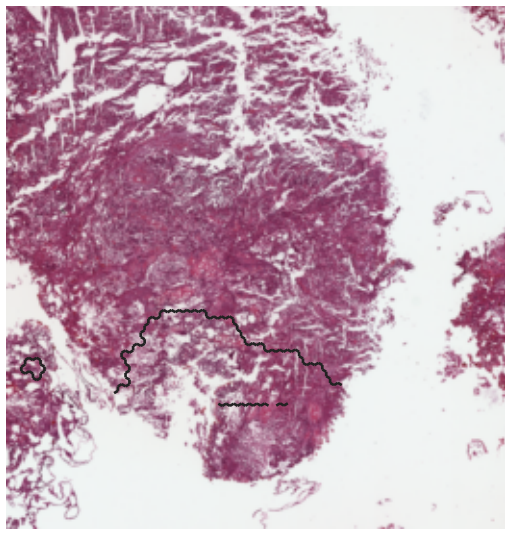




GSE319536_BC18 | Tissue context

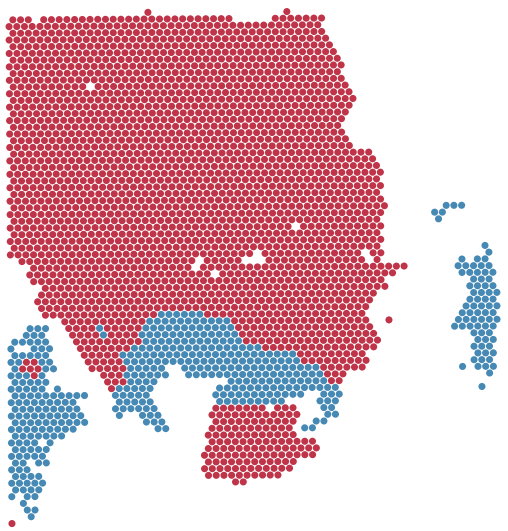
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



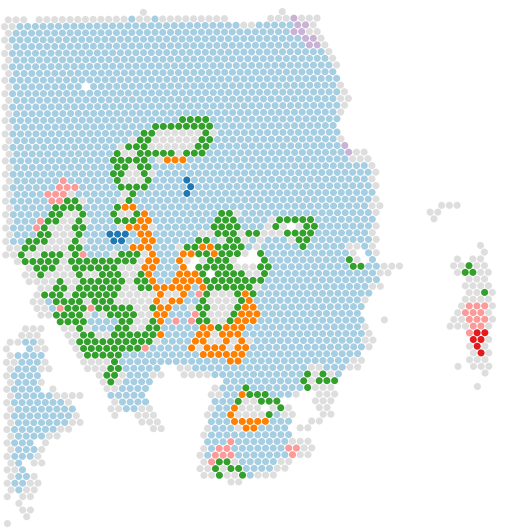
Black line: observed candidate outer edge

Final candidate location



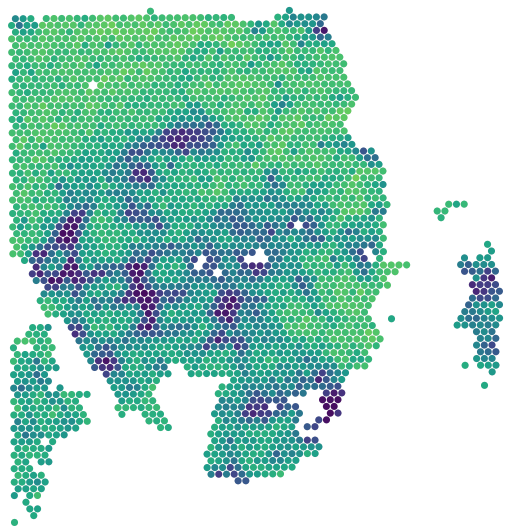
inside outside

Accepted R12 (original 0–8)



0 3 6 7 8
1 5

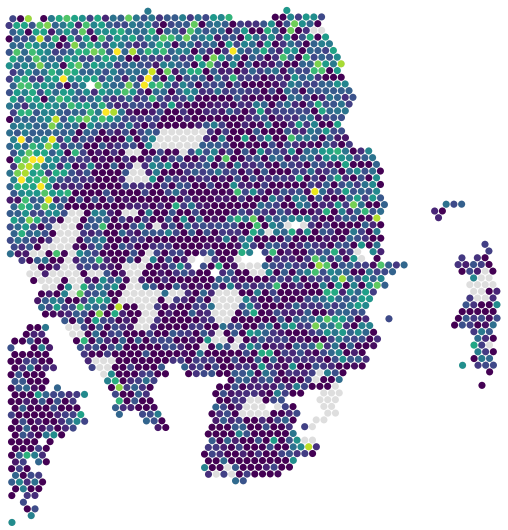
Epithelial support



0 1

Fraction of all 30

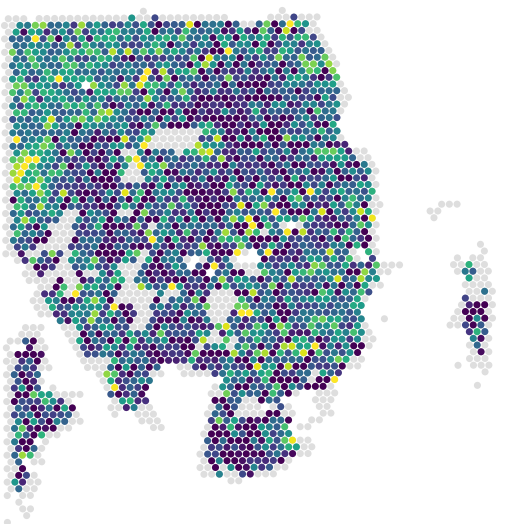
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



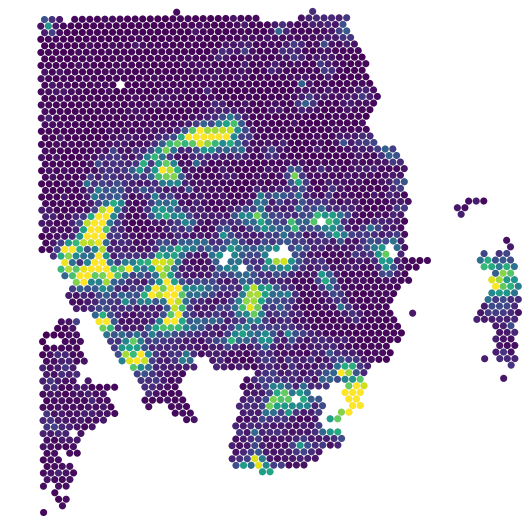
0 0.14

Matched-model AUCell

GSE319536_BC18 | Candidate neighborhoods

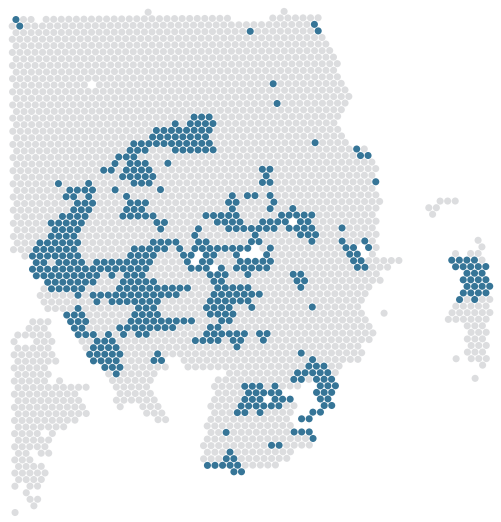
Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

CAF composition



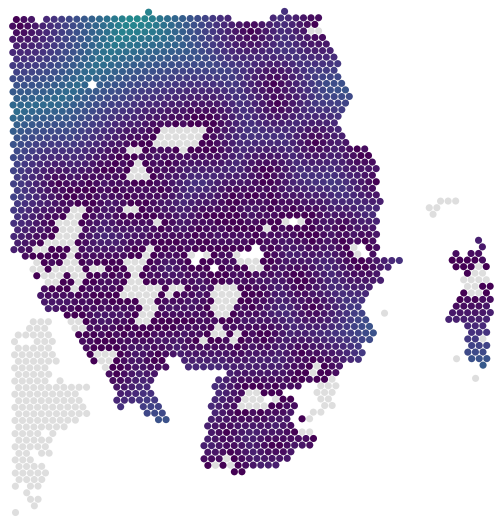
0 0.418
Fraction of all 30

CAF P80 targets



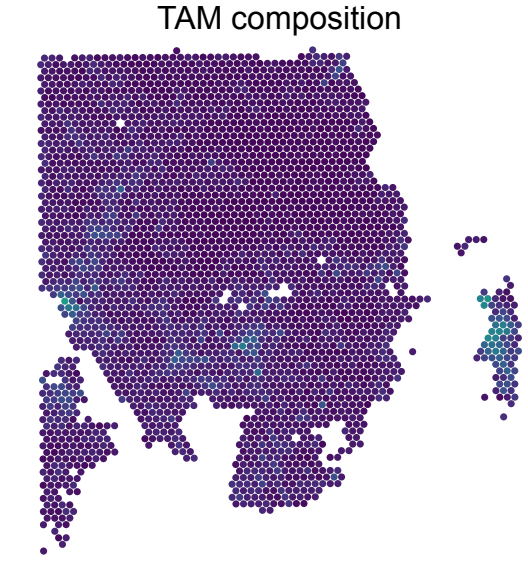
False True

CAF distance



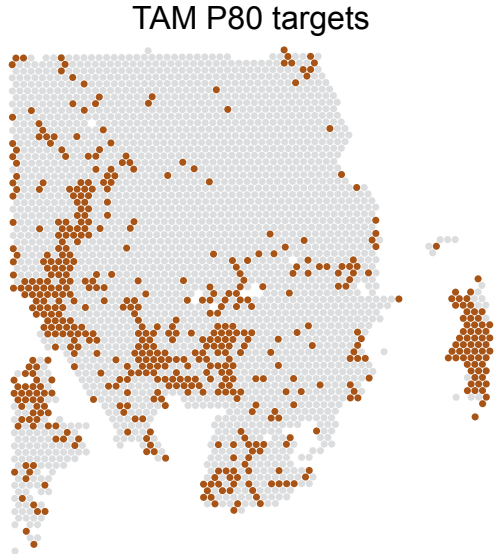
0 3.1e+03
Nearest spot center (μm)

TAM composition



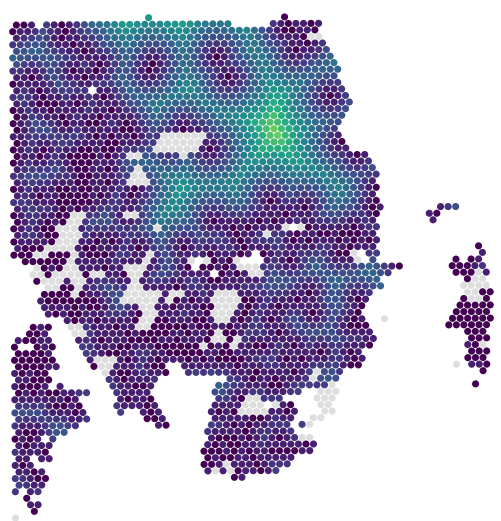
0 0.291
Fraction of all 30

TAM P80 targets



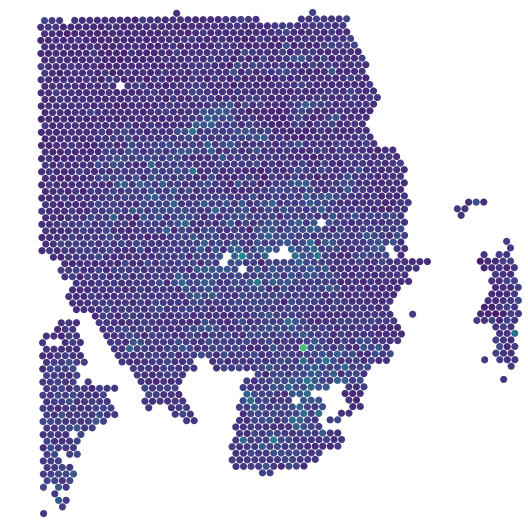
False True

TAM distance



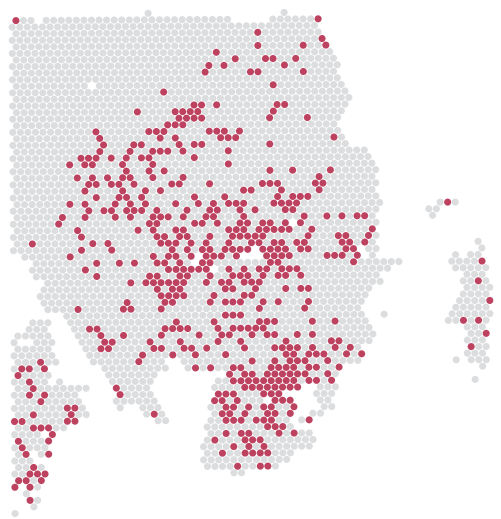
0 1.13e+03
Nearest spot center (μm)

Treg composition



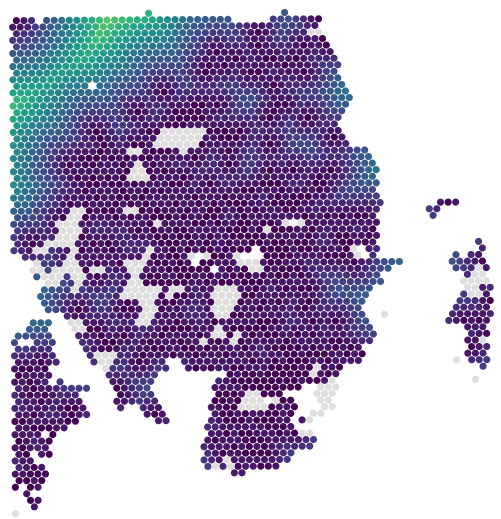
0 0.102
Fraction of all 30

Treg P80 targets



False True

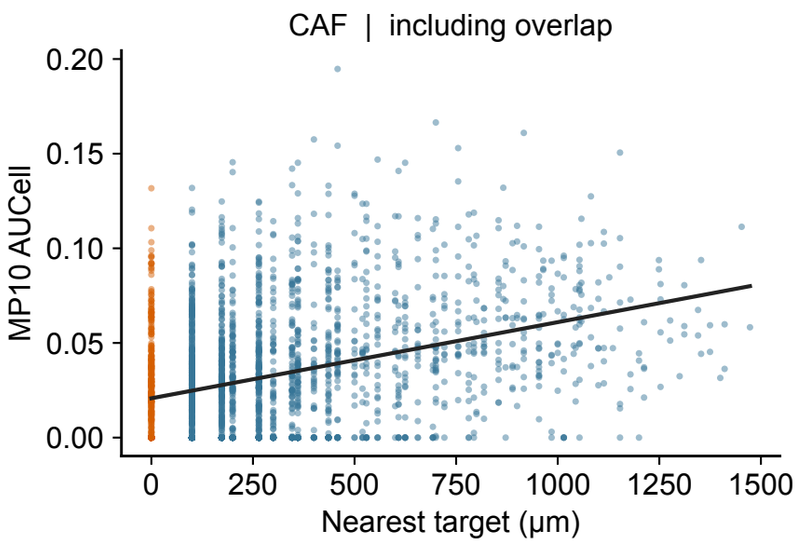
Treg distance



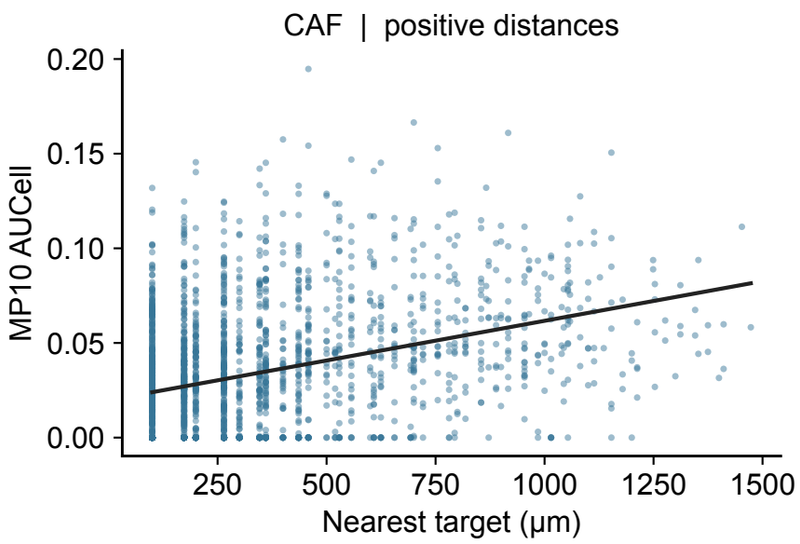
0 1.61e+03
Nearest spot center (μm)

GSE319536_BC18 | MP10 versus nearest target

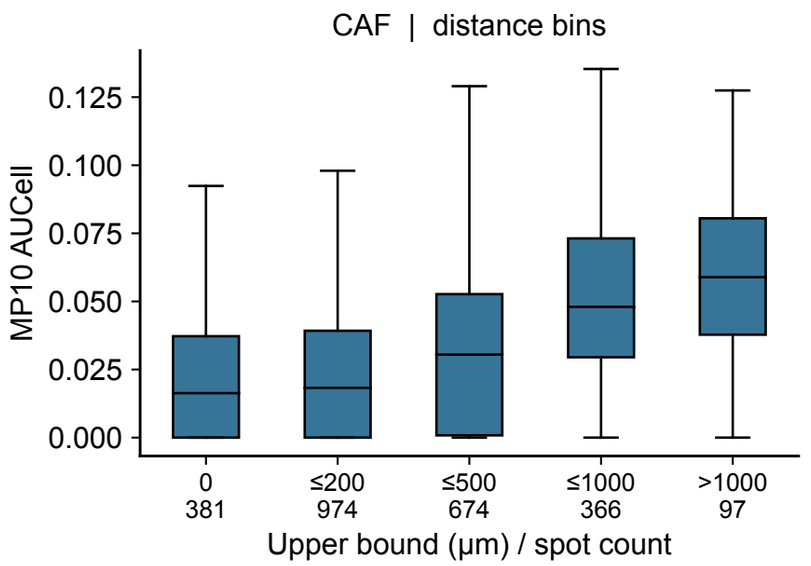
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



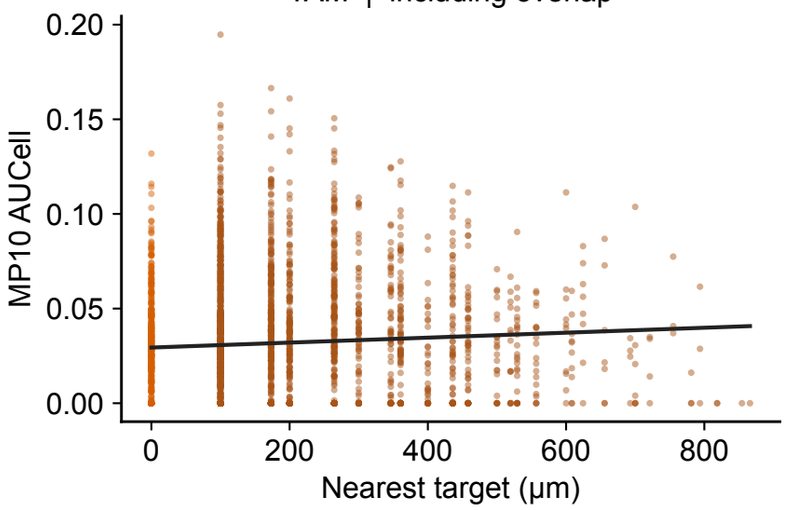
r +0.36 ρ +0.35
n 2492/2656 queries • 570 targets



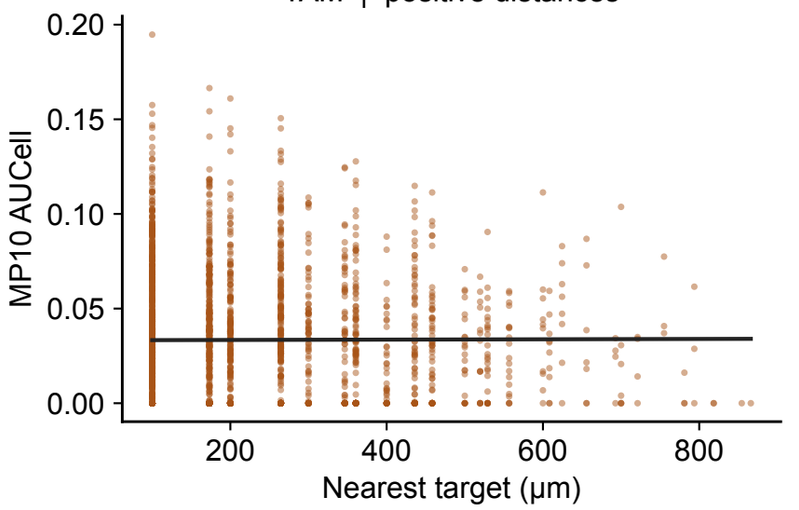
r +0.36 ρ +0.37
n 2111/2656 queries • 570 targets



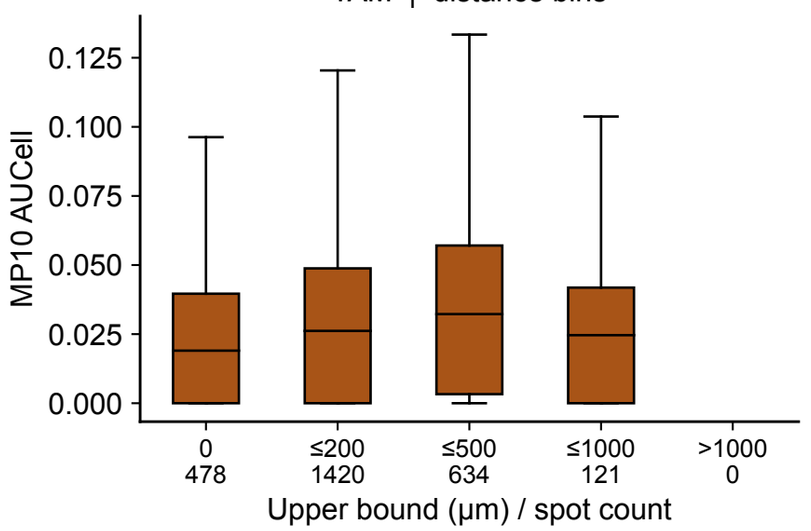
Orange = zero distance
Line = within-section linear fit



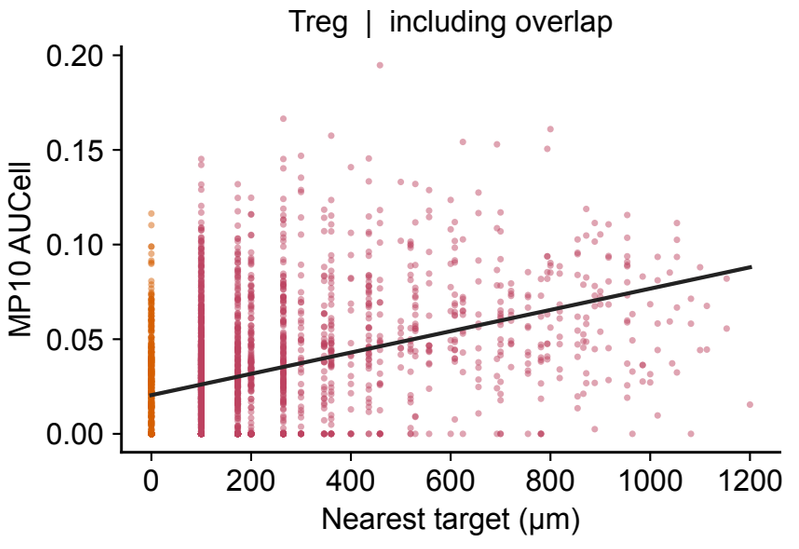
r +0.06 ρ +0.11
n 2653/2656 queries • 570 targets



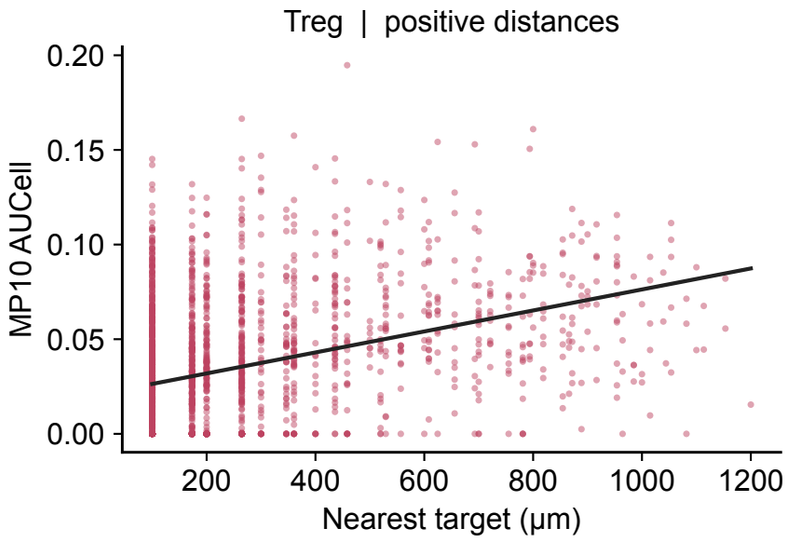
r +0.00 ρ +0.06
n 2175/2656 queries • 570 targets



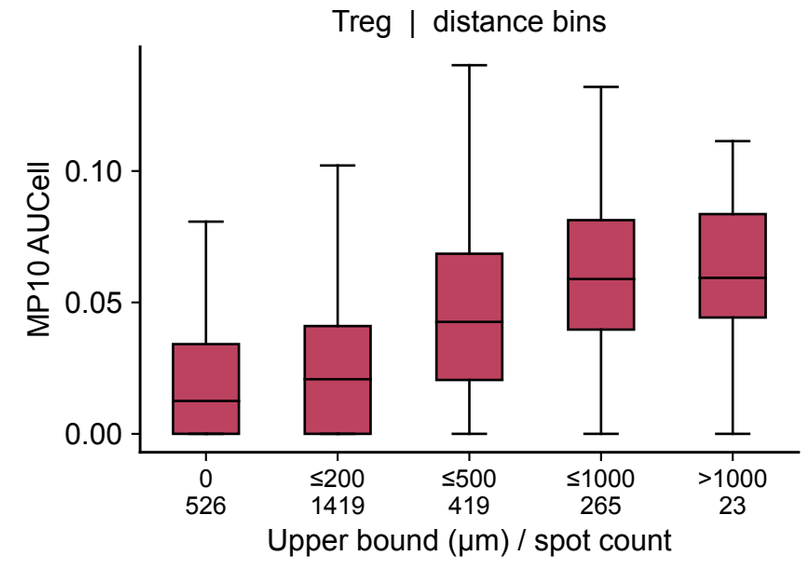
Orange = zero distance
Line = within-section linear fit



r +0.40 ρ +0.38
n 2652/2656 queries • 570 targets



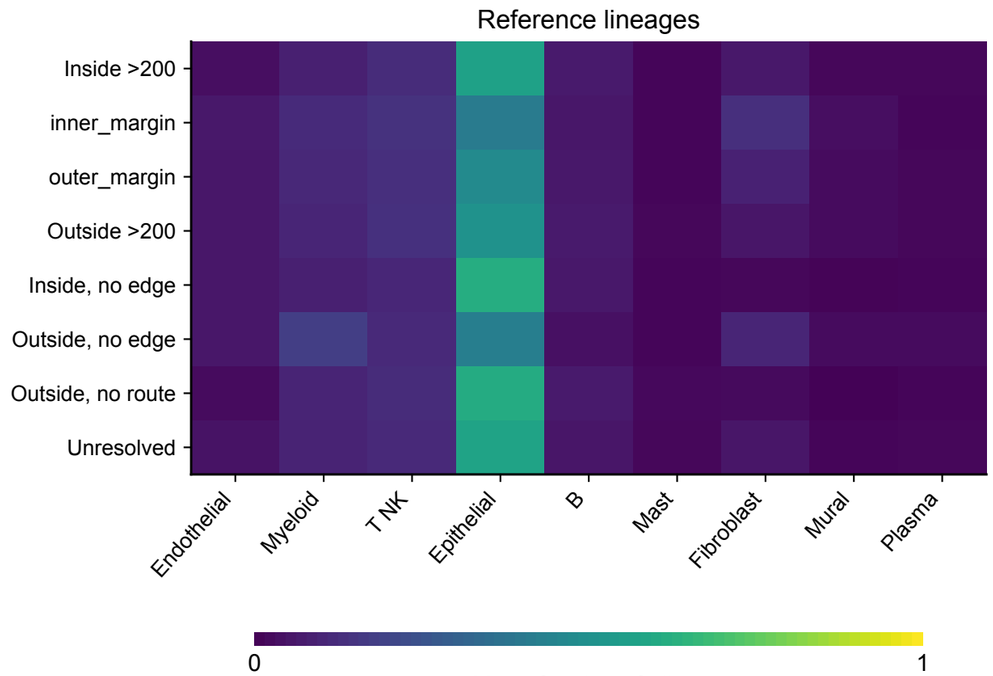
r +0.38 ρ +0.38
n 2126/2656 queries • 570 targets

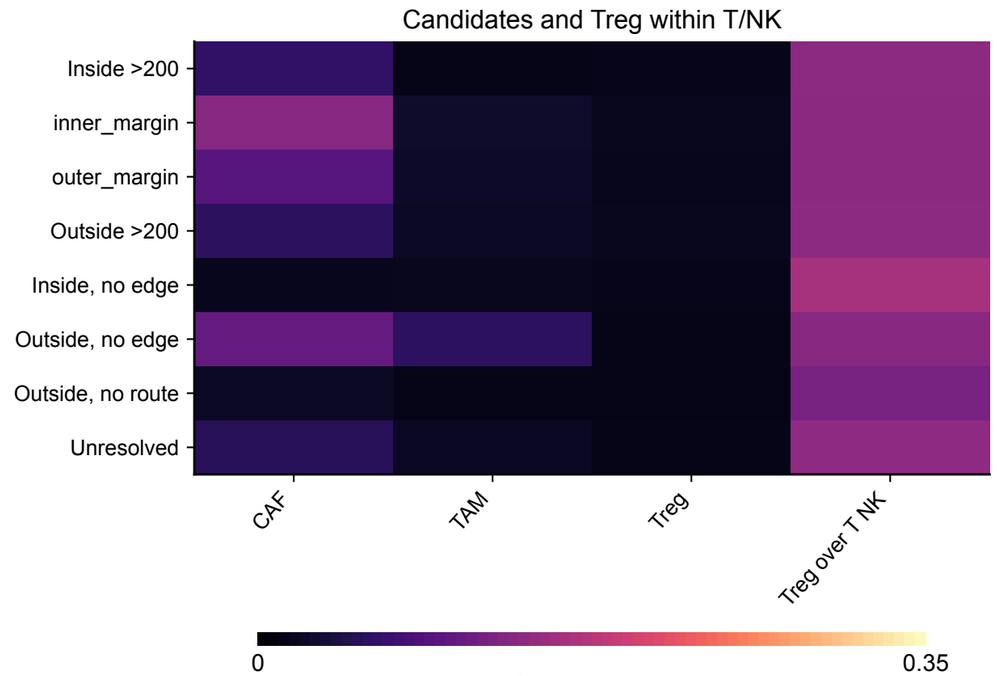


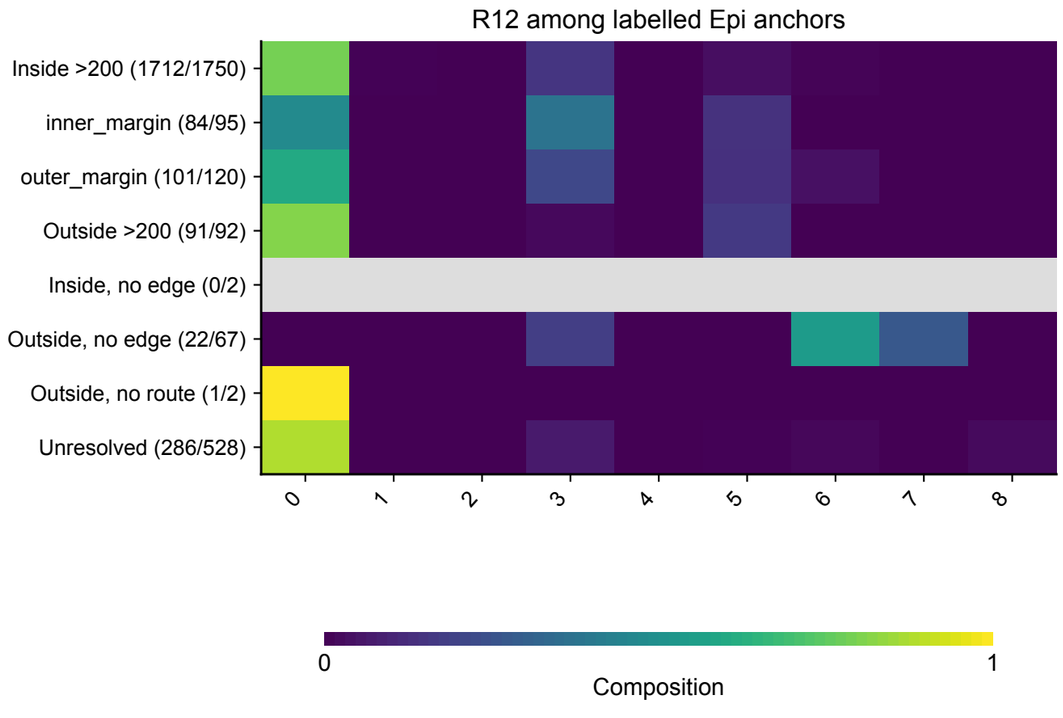
Orange = zero distance
Line = within-section linear fit

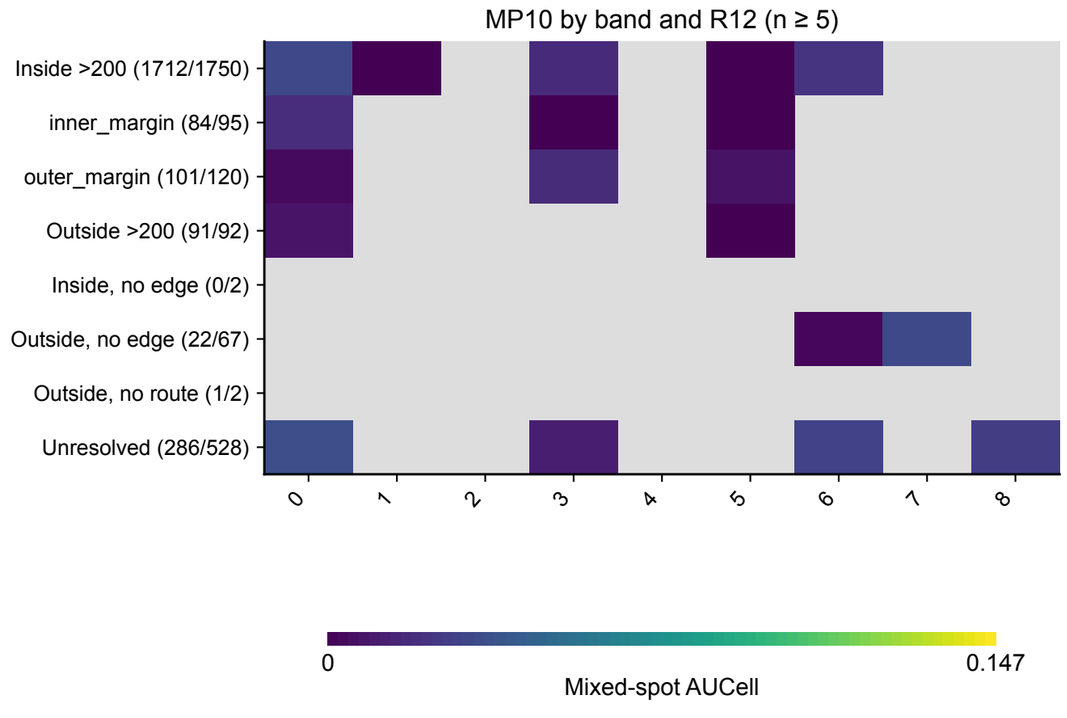
GSE319536_BC18 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





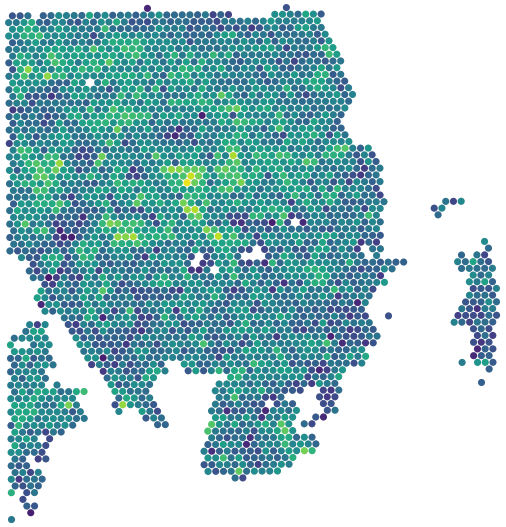




GSE319536_BC18 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

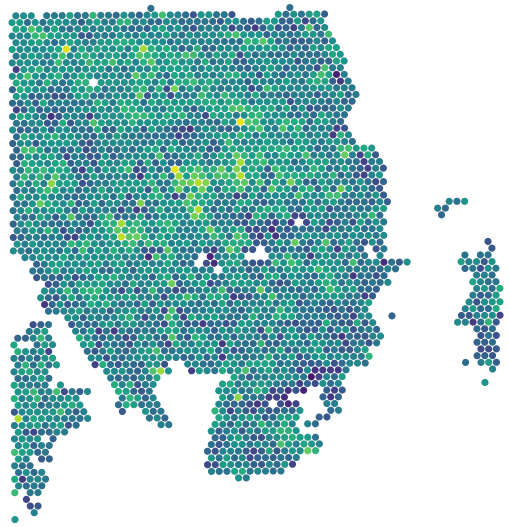
E2F Targets



0 0.152

UCell score

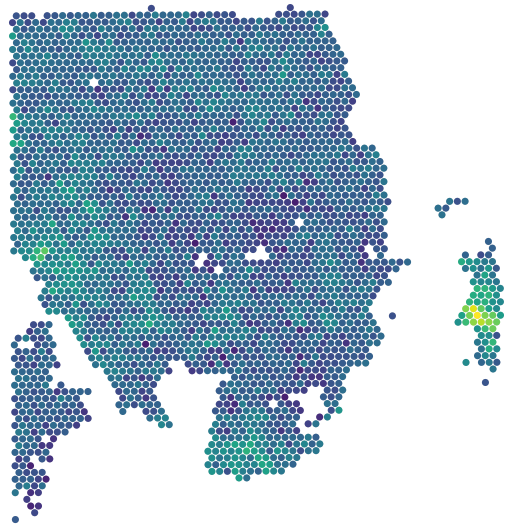
G2M Checkpoint



0 0.134

UCell score

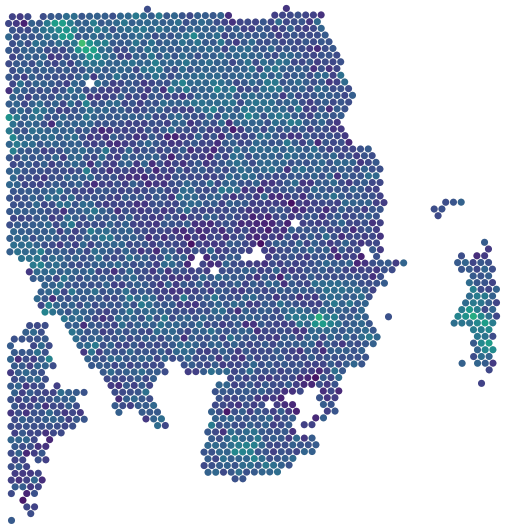
Hypoxia



0 0.211

UCell score

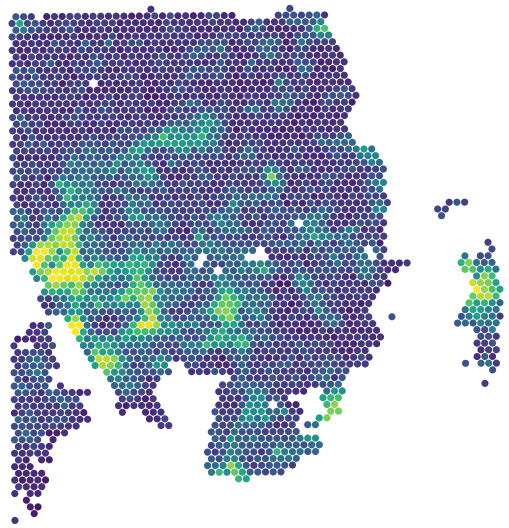
Tnfa Signaling Via Nfkb



0 0.275

UCell score

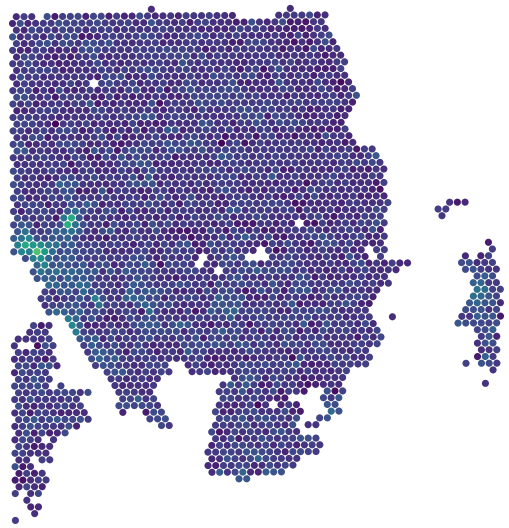
Emt



0 0.386

UCell score

Interferon Gamma Response



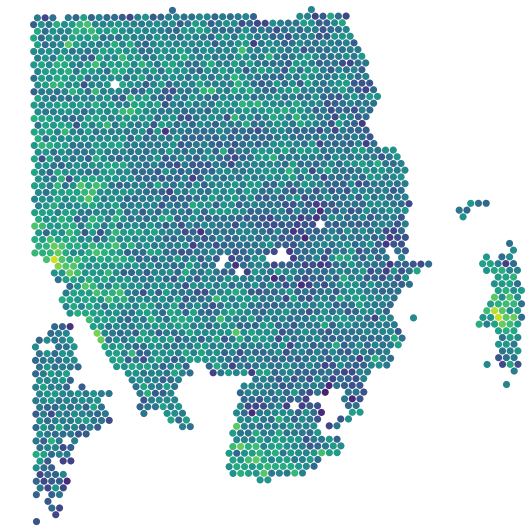
0 0.275

UCell score

GSE319536_BC18 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

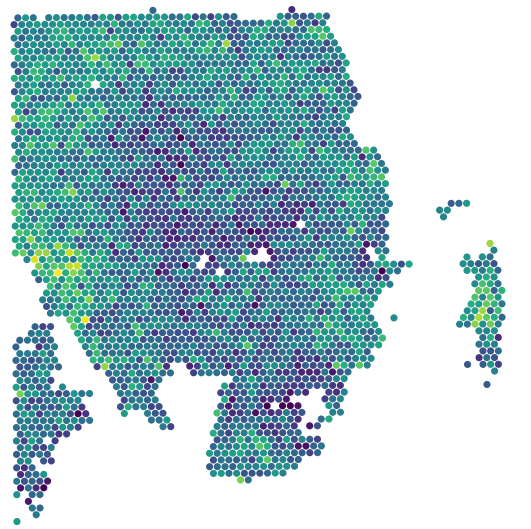
Il2 Stat5 Signaling



0 0.144

UCell score

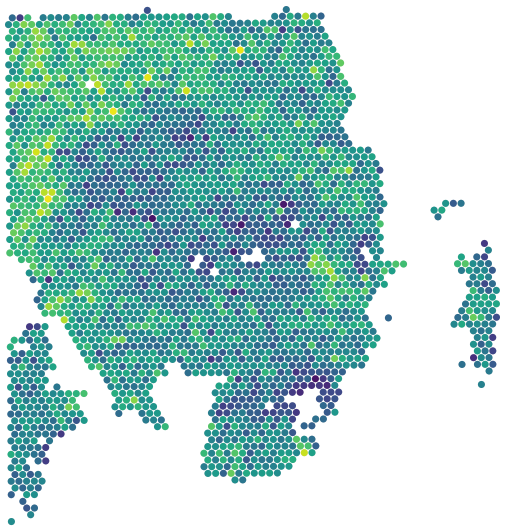
Tgf Beta Signaling



0 0.232

UCell score

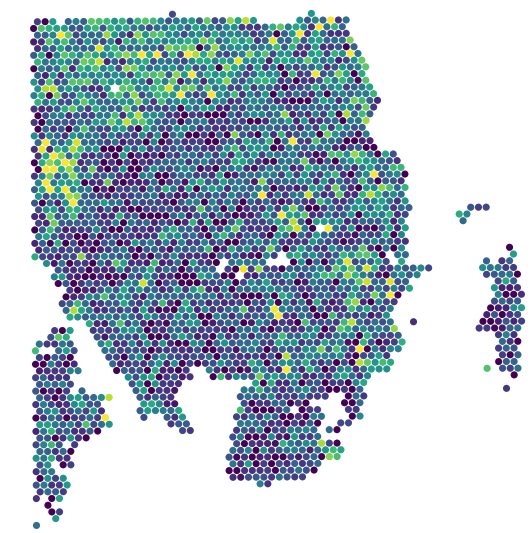
Cholesterol Homeostasis



0 0.21

UCell score

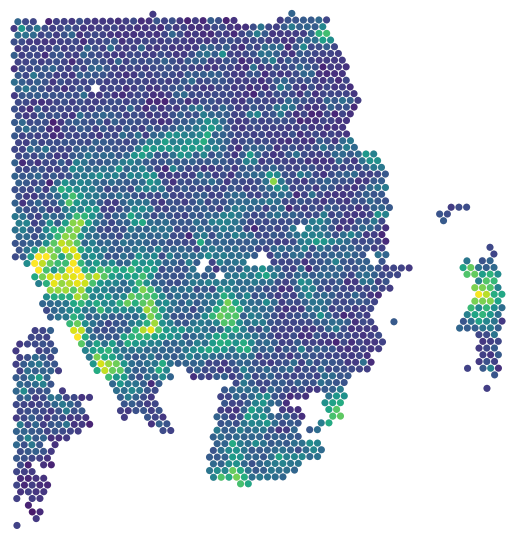
Cholesterol Biosynthesis



0 0.166

UCell score

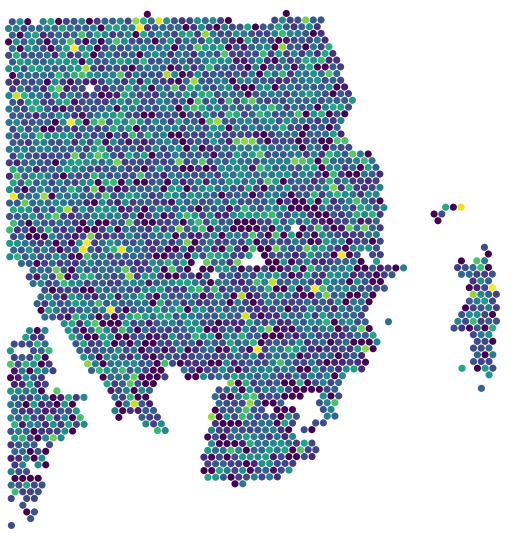
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



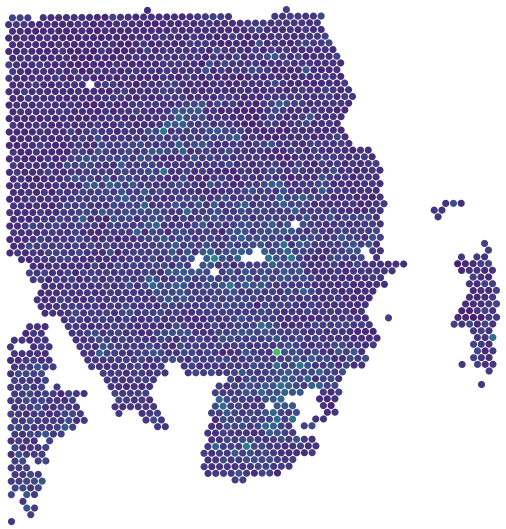
0 0.151

UCell score

GSE319536_BC18 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

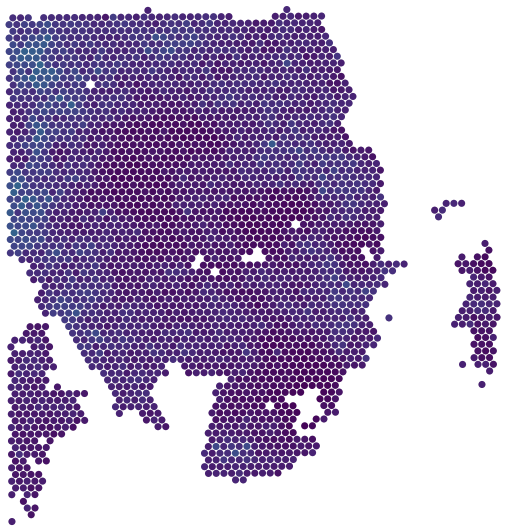
Treg composition



0 0.102

Fraction of all 30

Treg estimated abundance



0 2.85

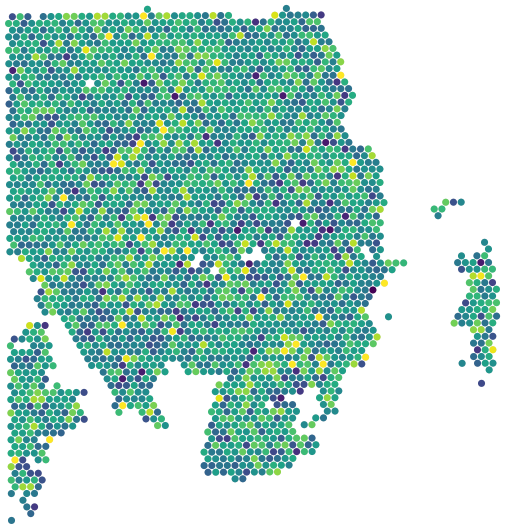
Posterior mean estimate

Treg Resting Vs Activated Dn

0 0.0778

UCell score

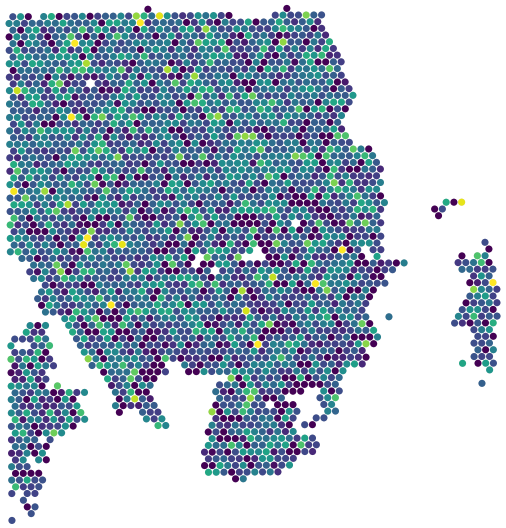
Treg Resting Vs Activated Up



0 0.0609

UCell score

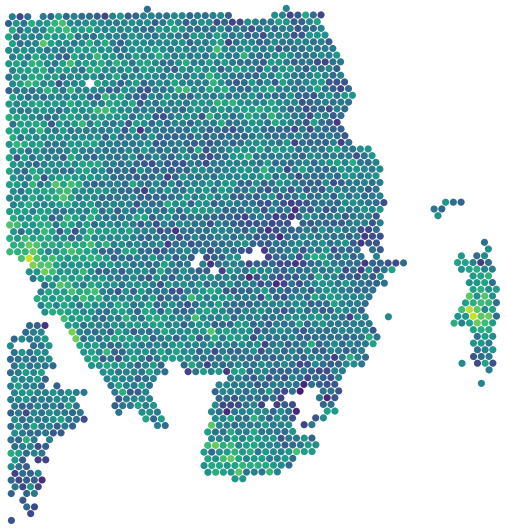
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

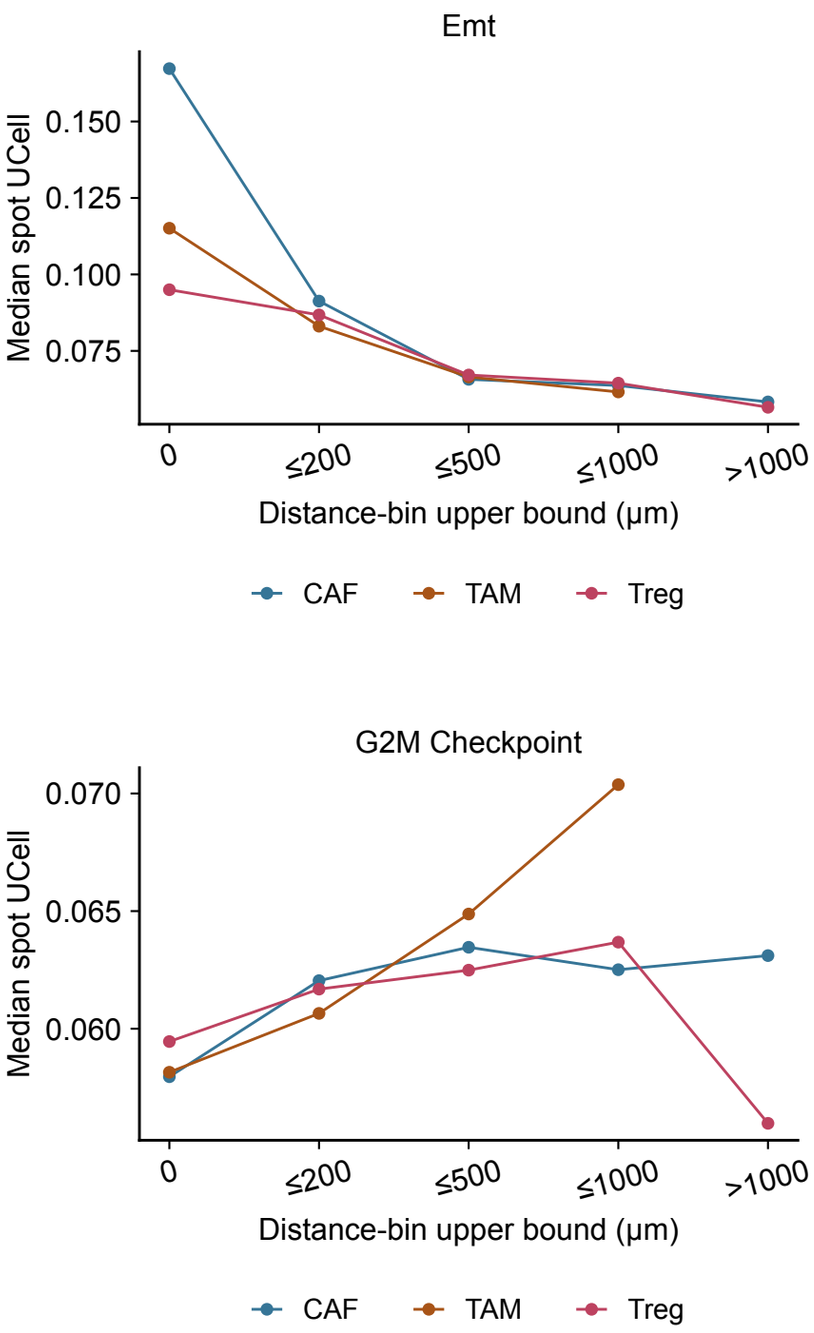


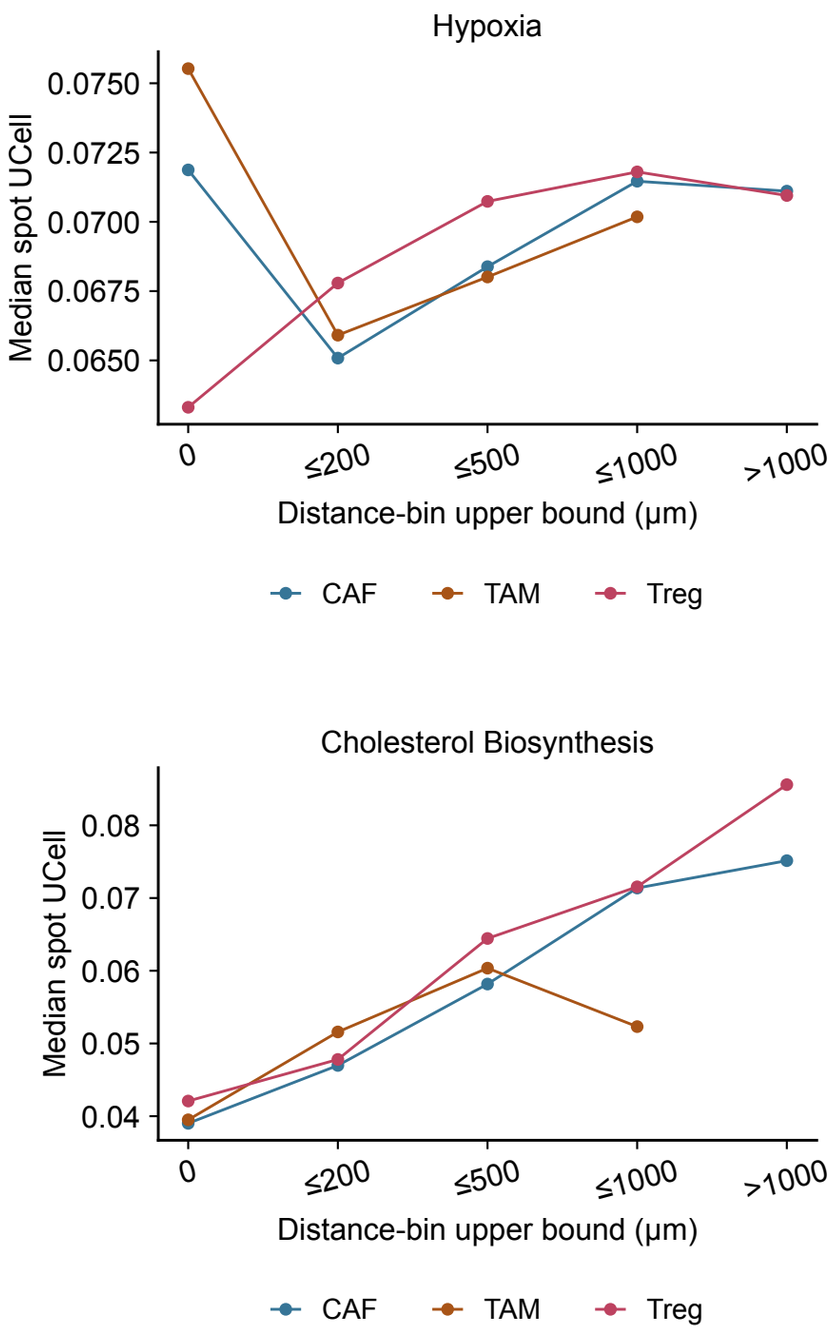
0 0.144

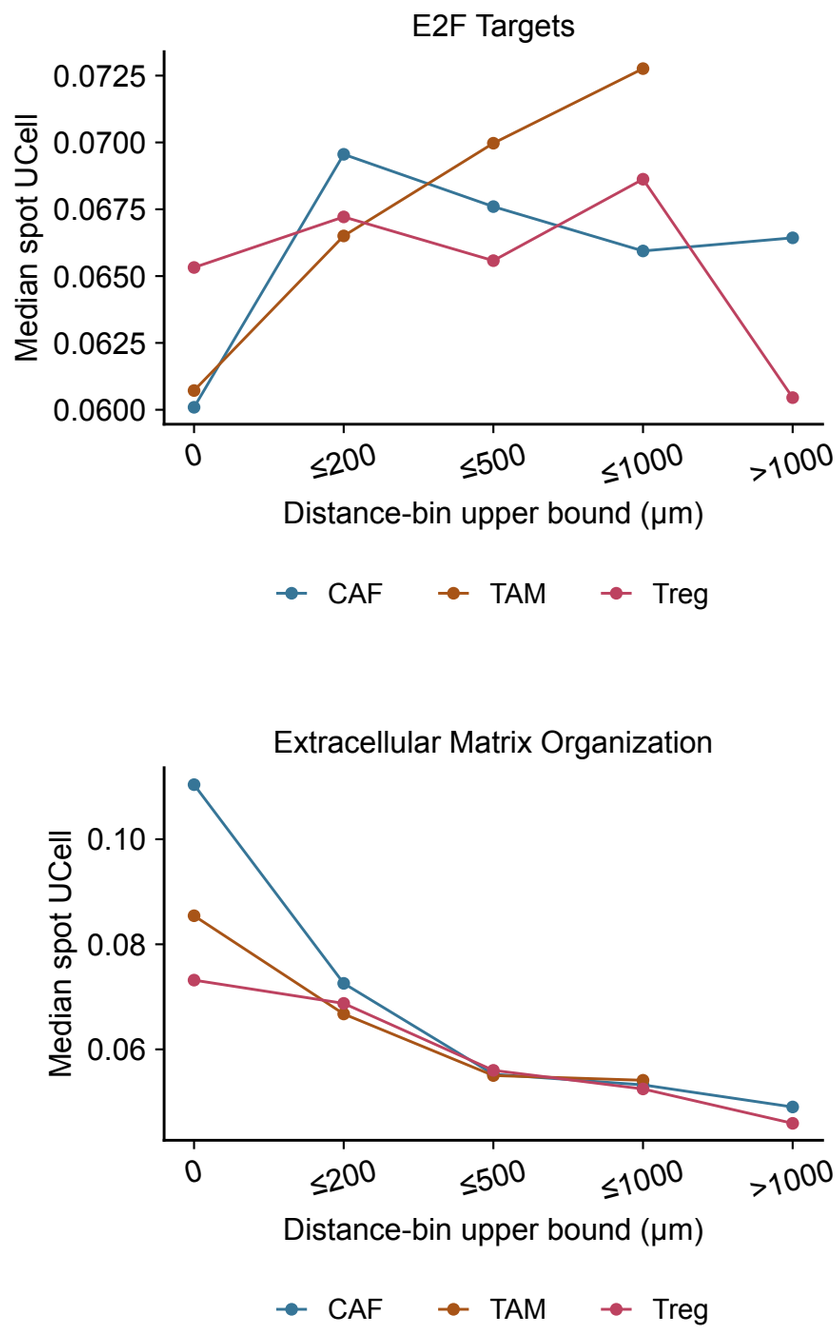
UCell score

GSE319536_BC18 | Programs by target distance

Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

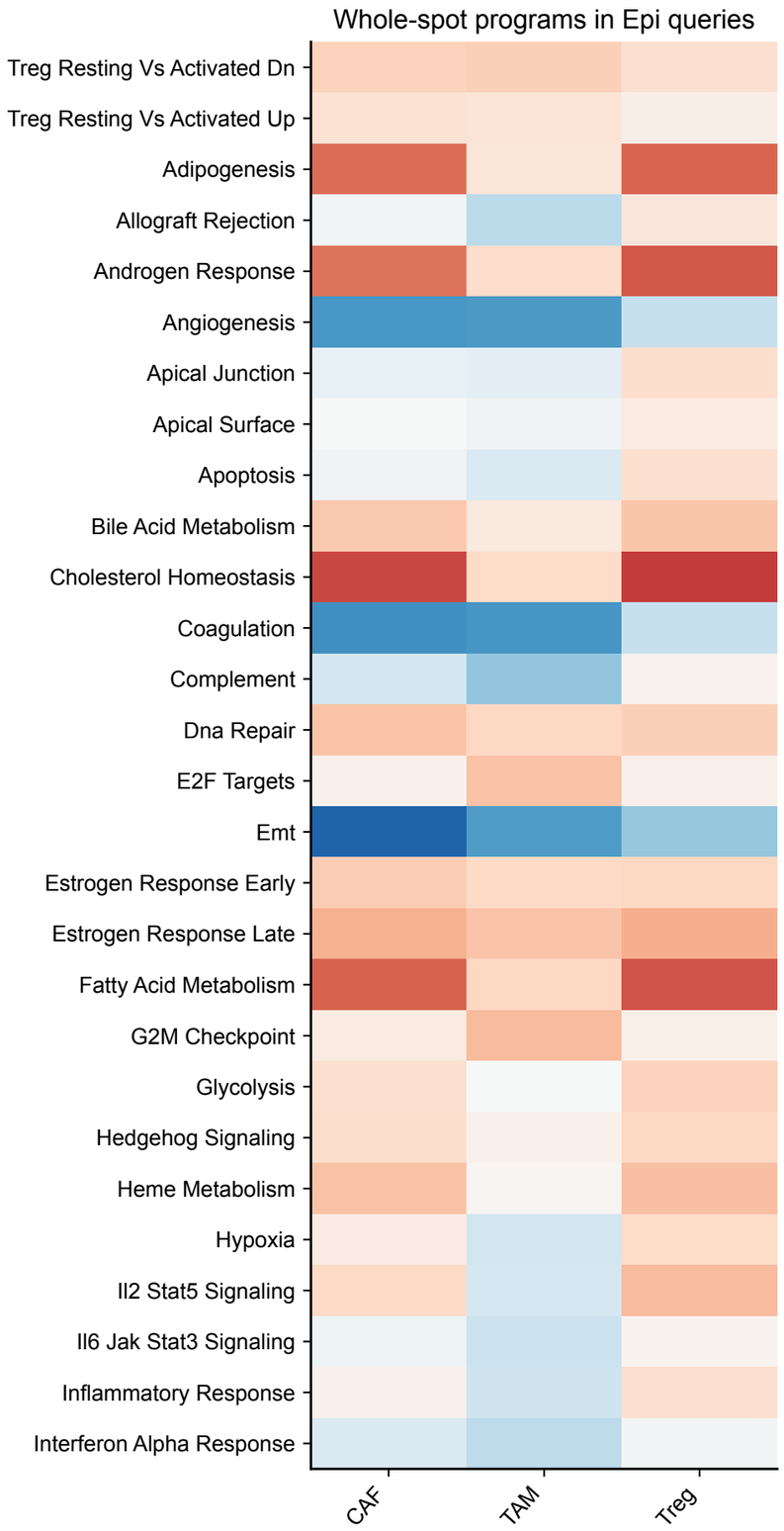


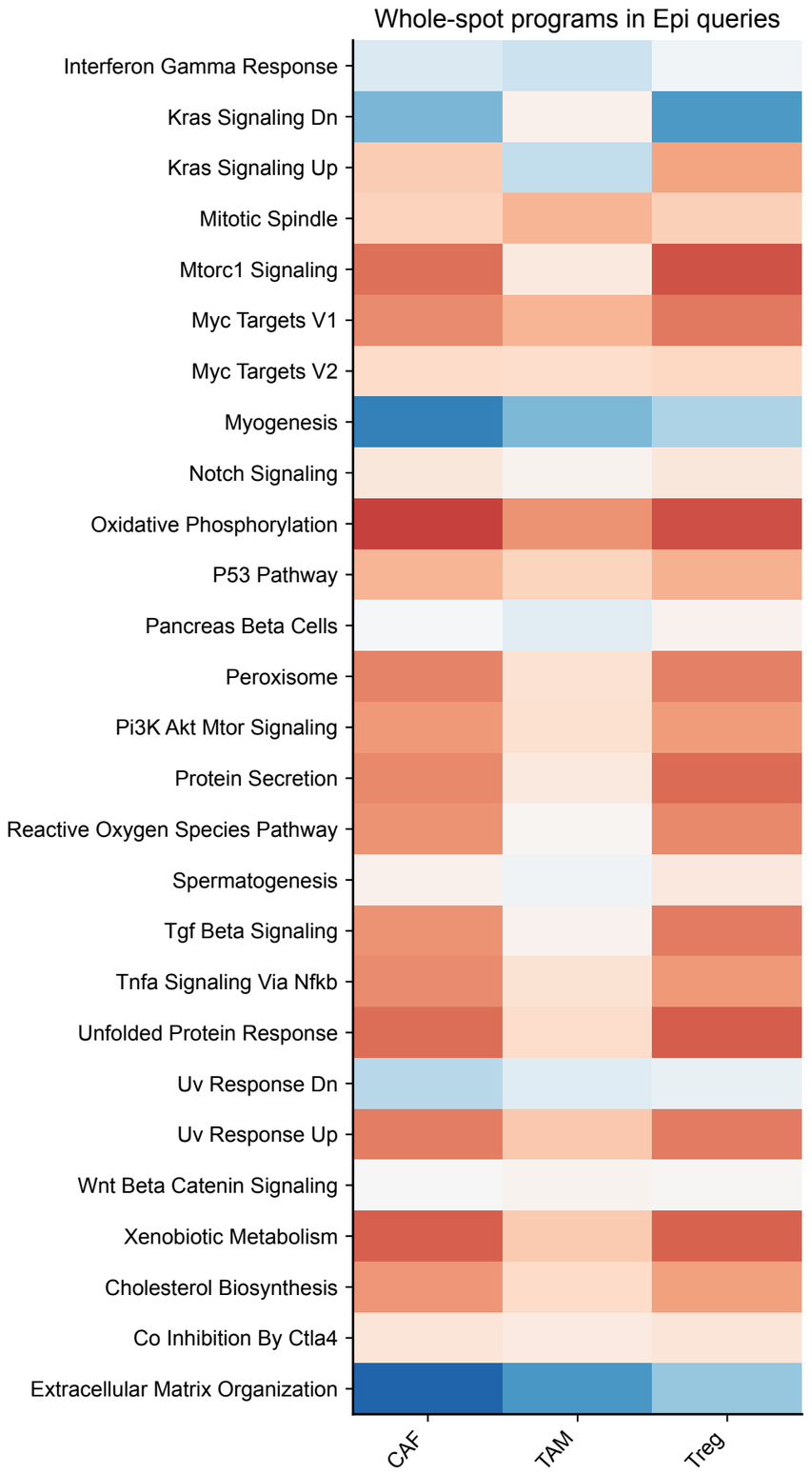




GSE319536_BC18 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

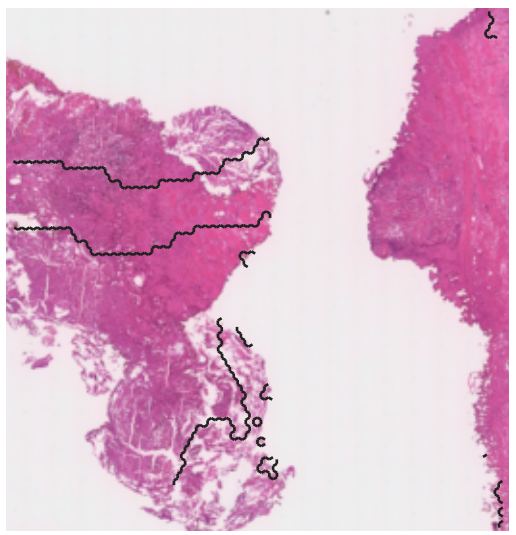




GSE319536_BC19 | Tissue context

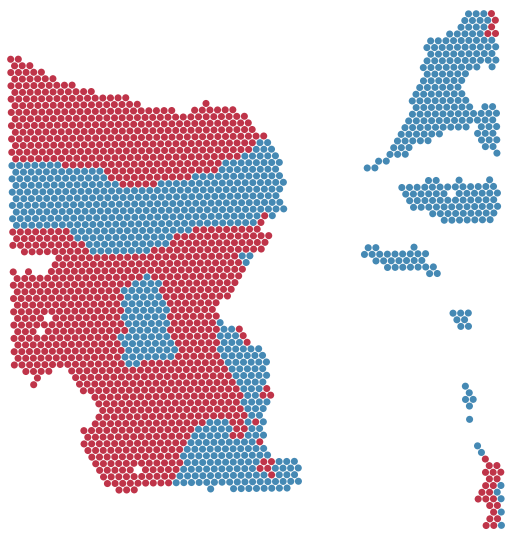
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



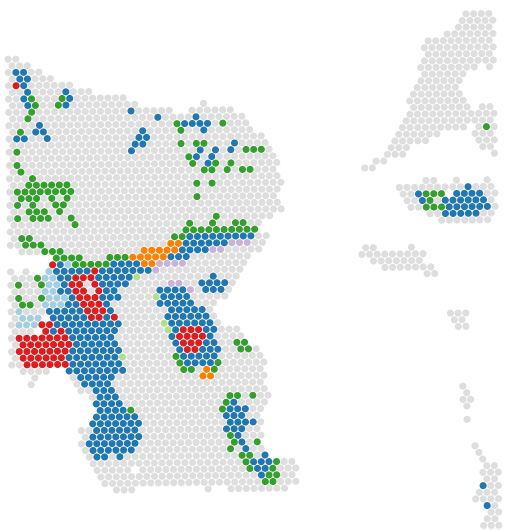
Black line: observed candidate outer edge

Final candidate location



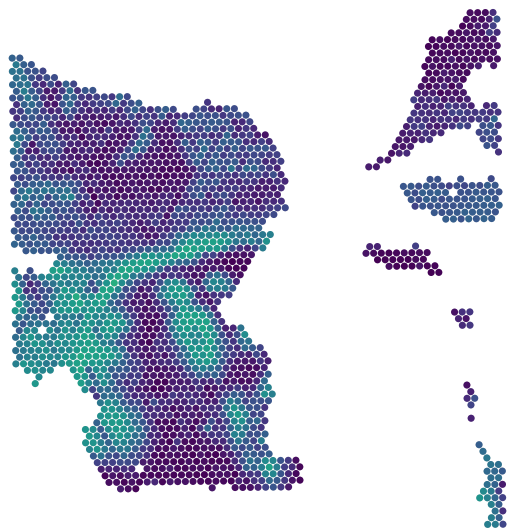
inside outside

Accepted R12 (original 0–8)



0 1 2 3 5 7 8

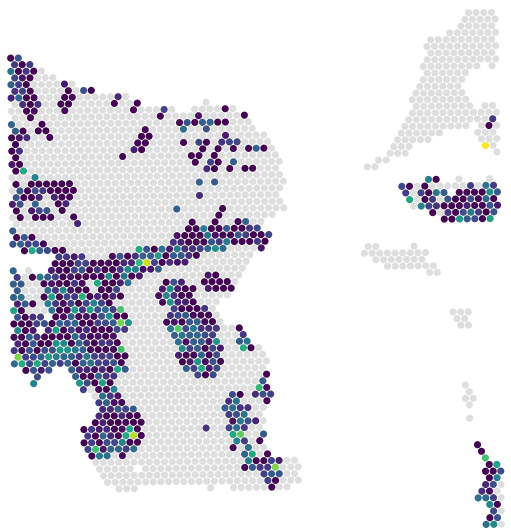
Epithelial support



0 1

Fraction of all 30

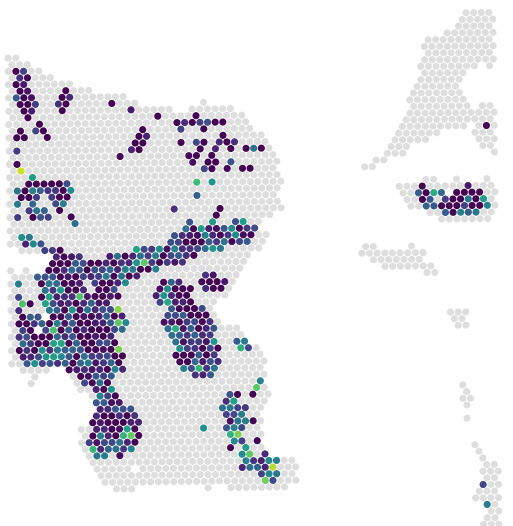
MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)



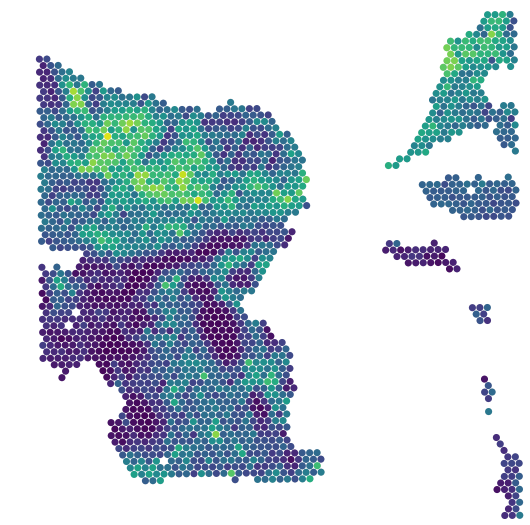
0 0.14

Matched-model AUCell

GSE319536_BC19 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

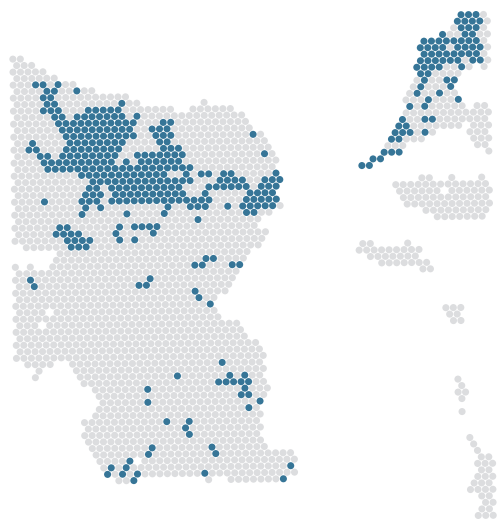
CAF composition



0 0.418

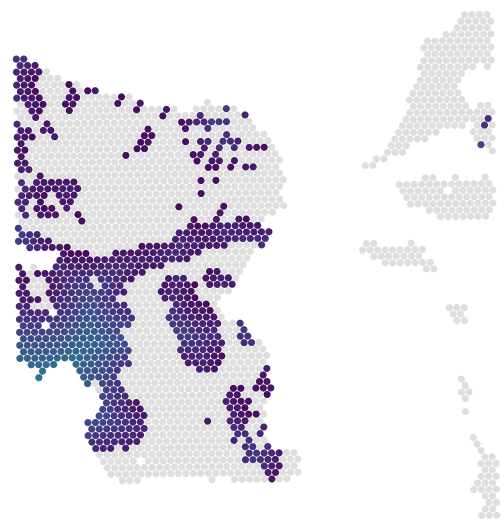
Fraction of all 30

CAF P80 targets



False True

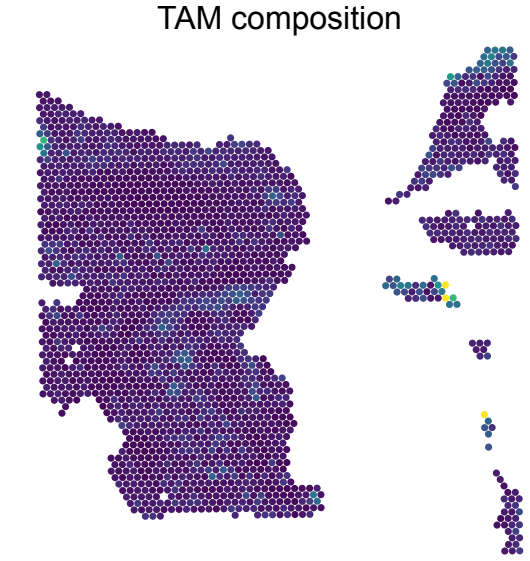
CAF distance



0 3.1e+03

Nearest spot center (µm)

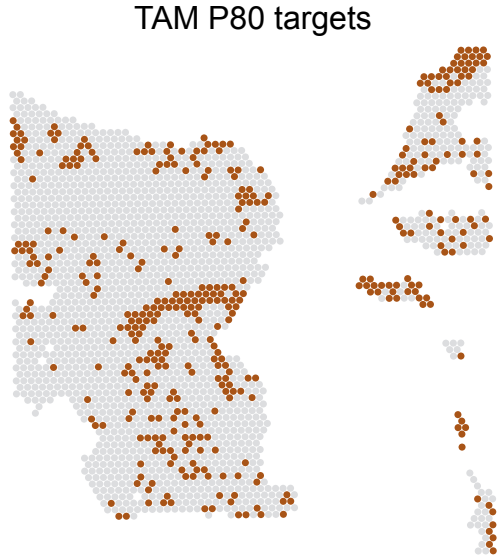
TAM composition



0 0.291

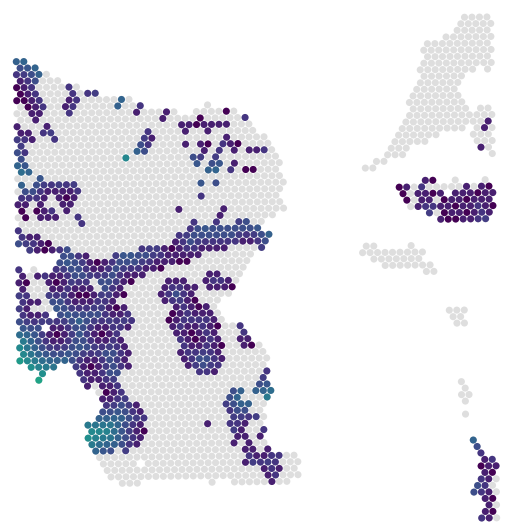
Fraction of all 30

TAM P80 targets



False True

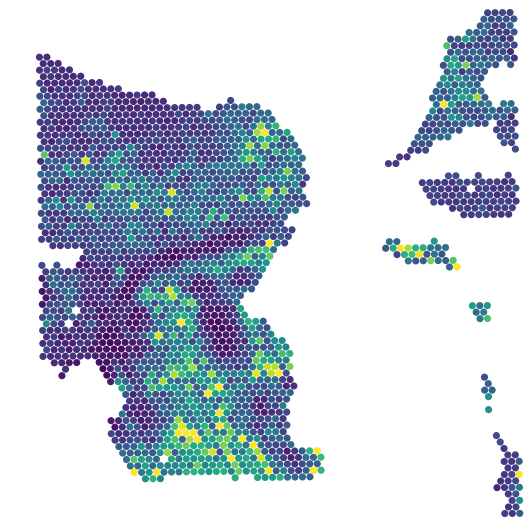
TAM distance



0 1.13e+03

Nearest spot center (µm)

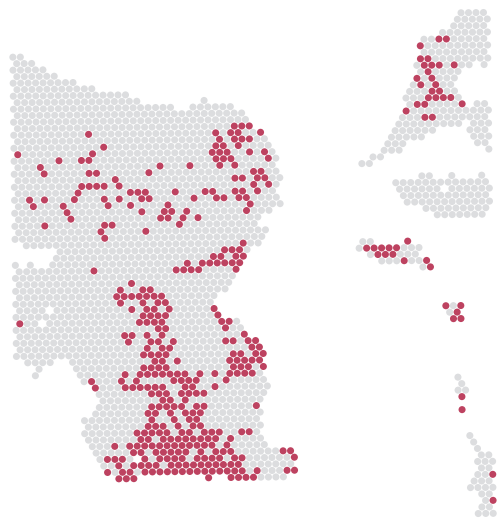
Treg composition



0 0.102

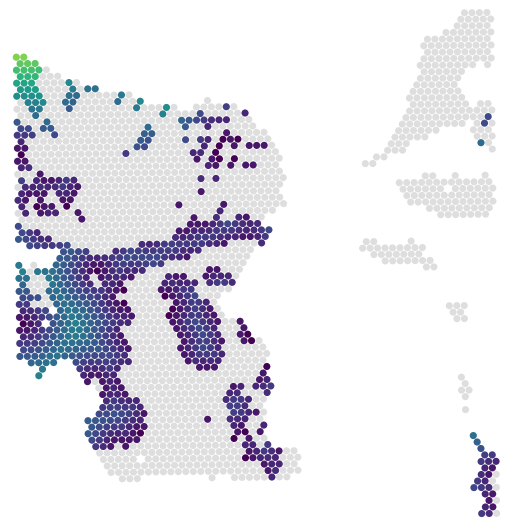
Fraction of all 30

Treg P80 targets



False True

Treg distance

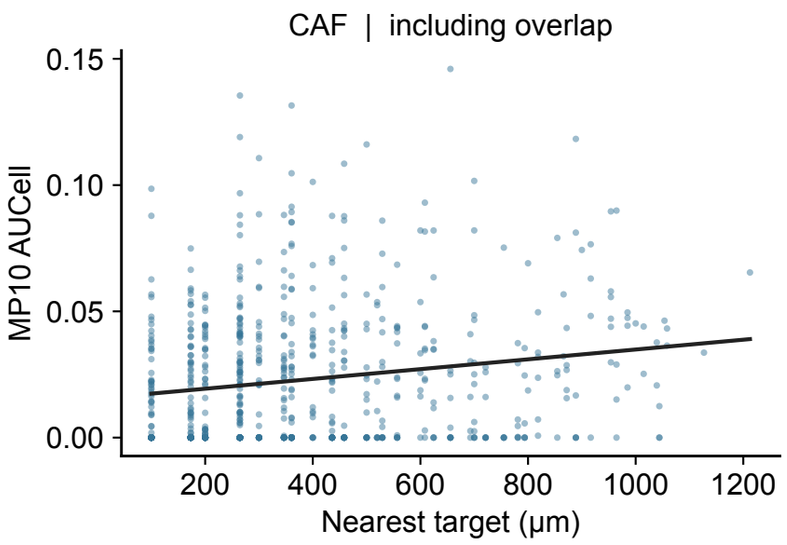


0 1.61e+03

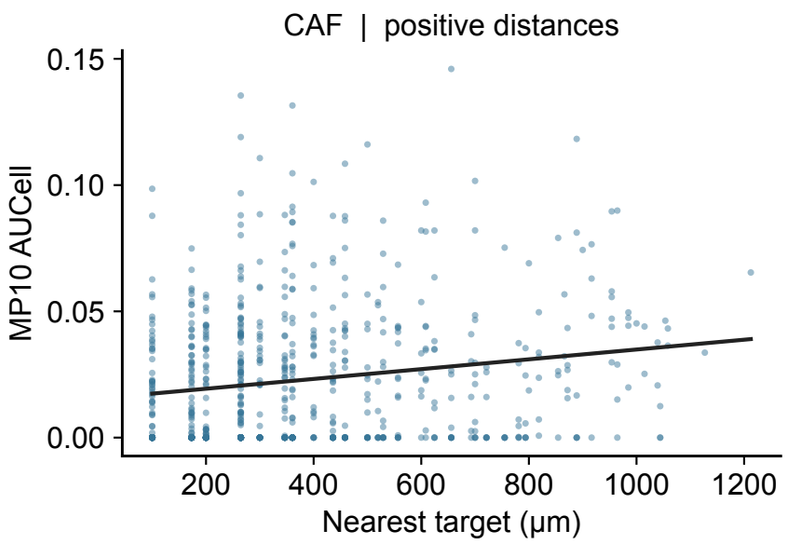
Nearest spot center (µm)

GSE319536_BC19 | MP10 versus nearest target

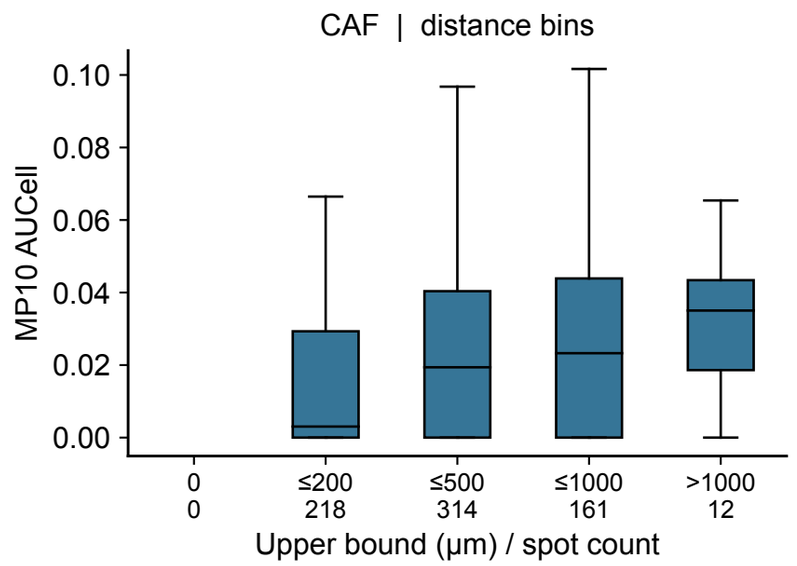
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



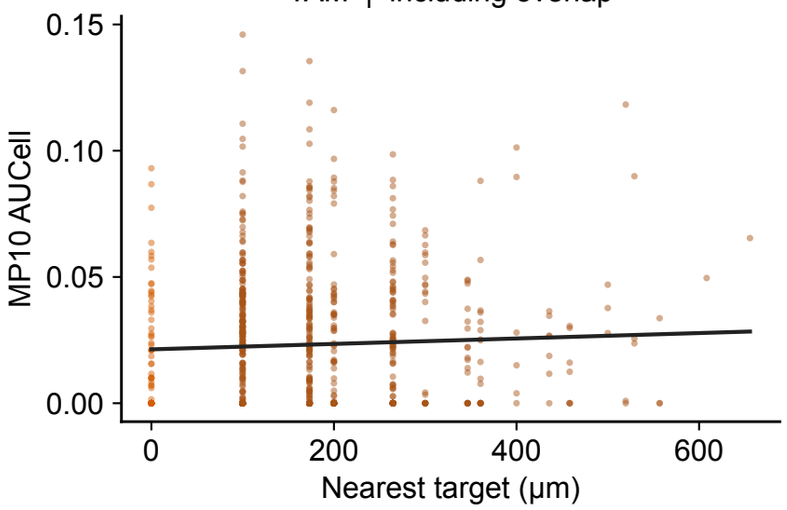
r +0.17 ρ +0.18
n 705/784 queries • 418 targets



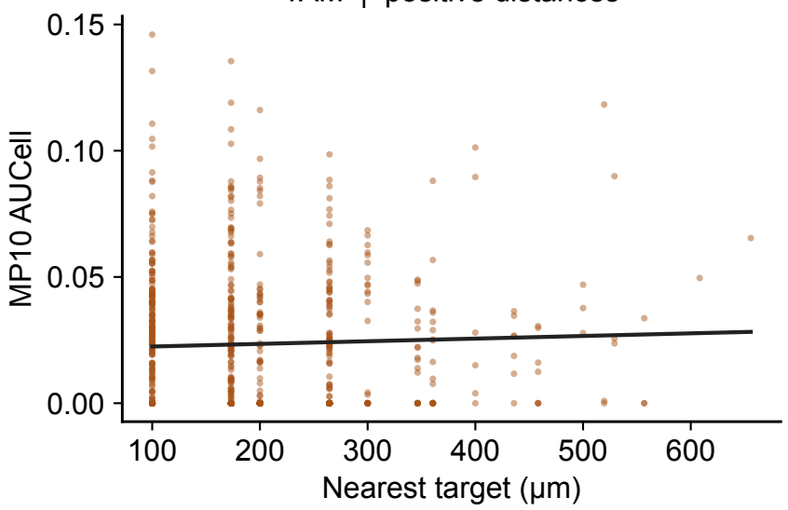
r +0.17 ρ +0.18
n 705/784 queries • 418 targets



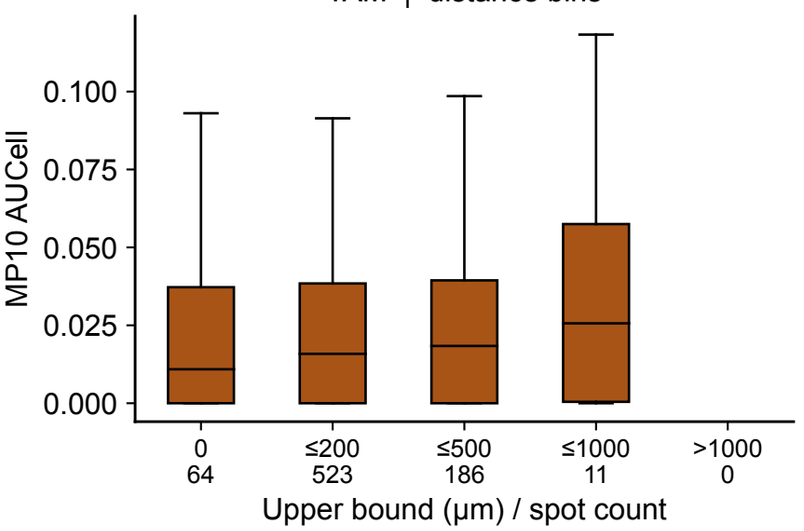
Orange = zero distance
Line = within-section linear fit



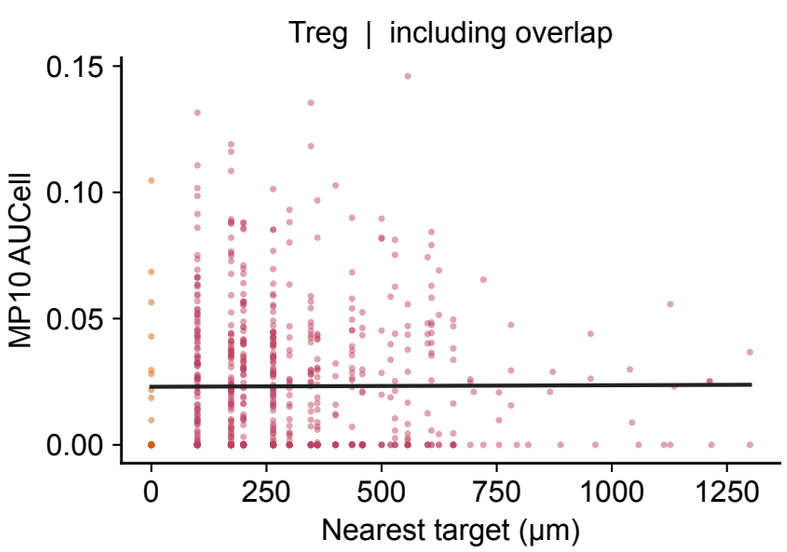
r +0.04 ρ +0.03
n 784/784 queries • 418 targets



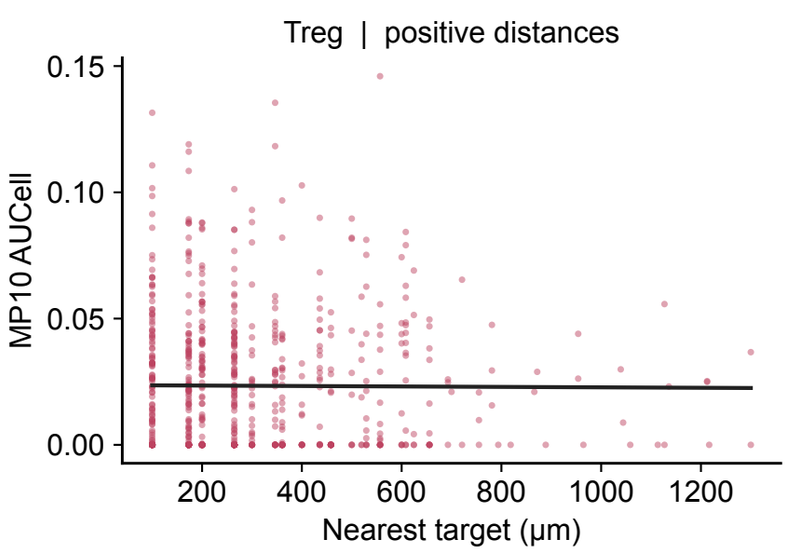
r +0.04 ρ +0.03
n 720/784 queries • 418 targets



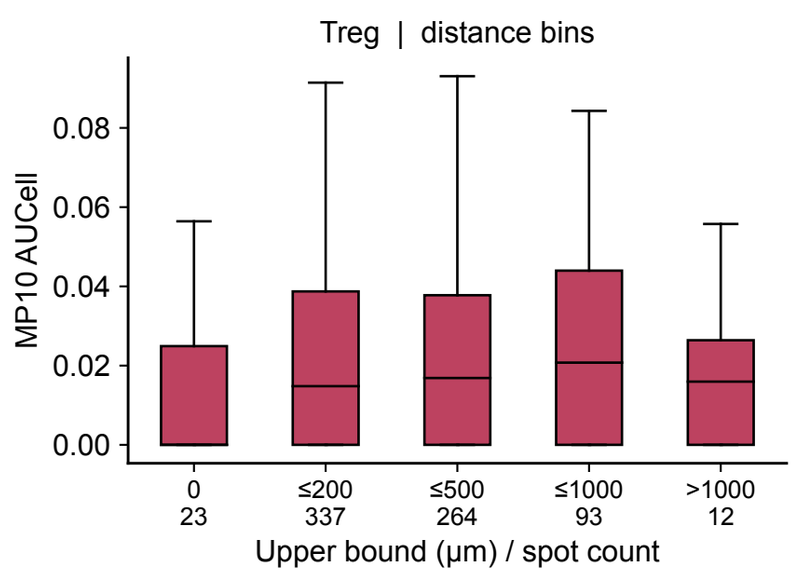
Orange = zero distance
Line = within-section linear fit



r +0.00 ρ +0.04
n 729/784 queries • 418 targets



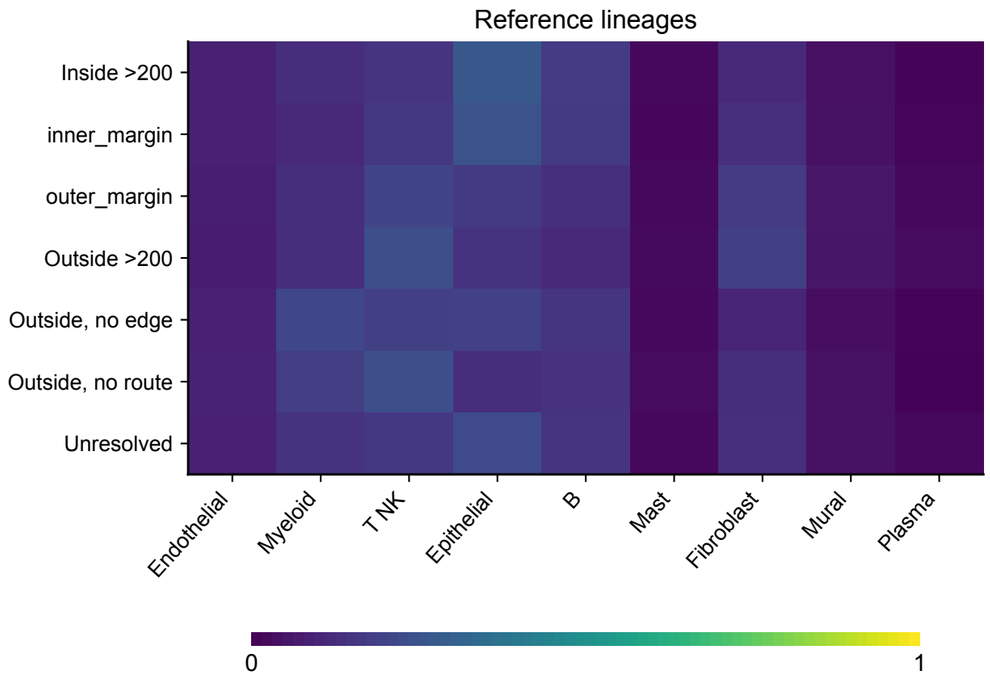
r -0.01 ρ +0.02
n 706/784 queries • 418 targets

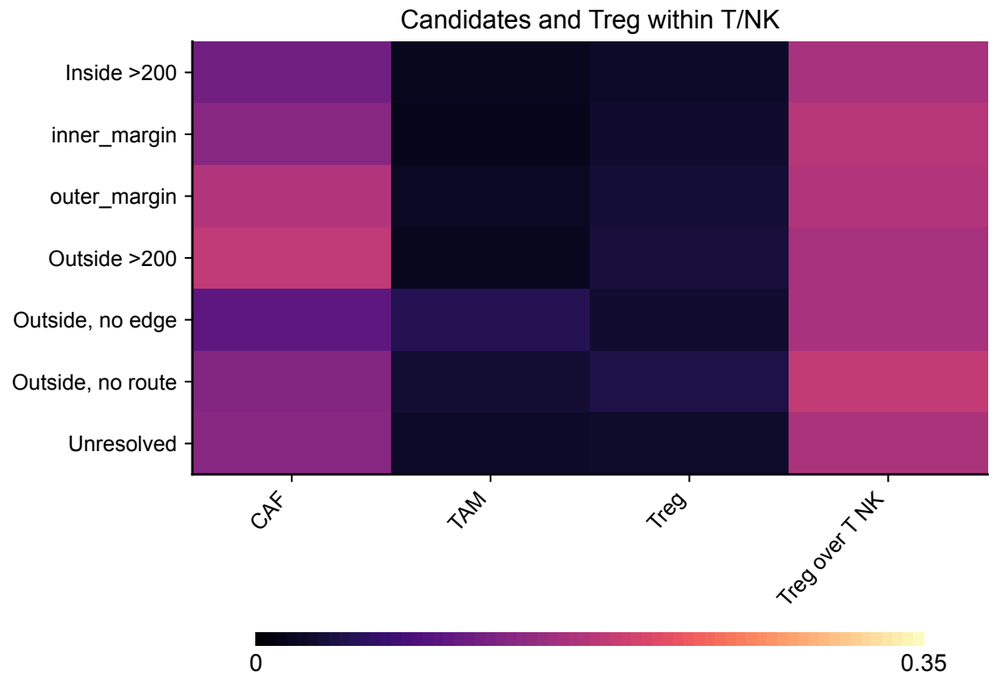


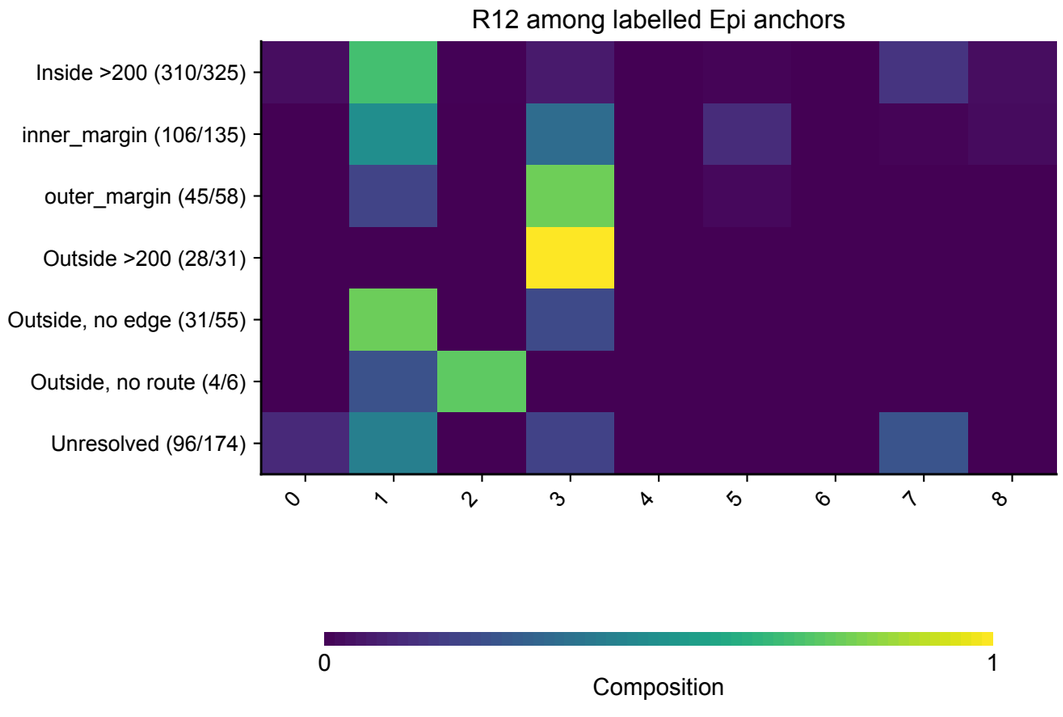
Orange = zero distance
Line = within-section linear fit

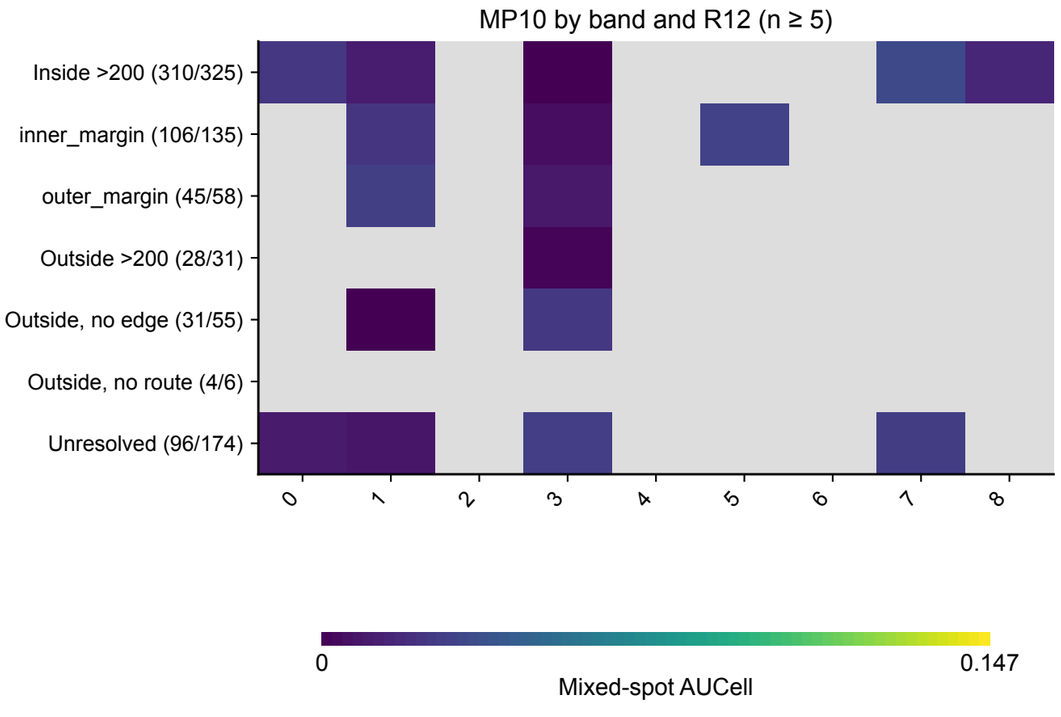
GSE319536_BC19 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





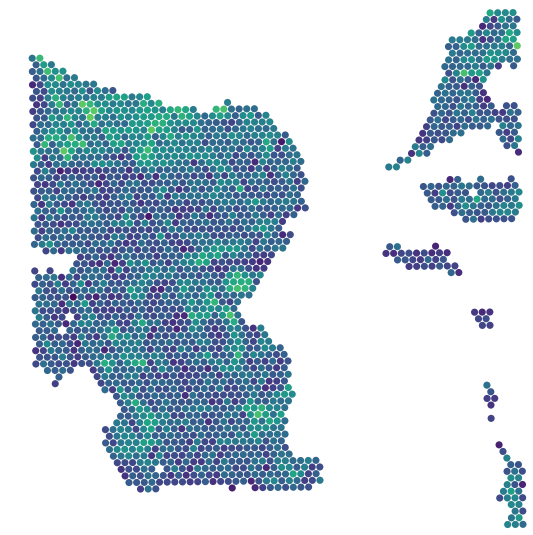




GSE319536_BC19 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

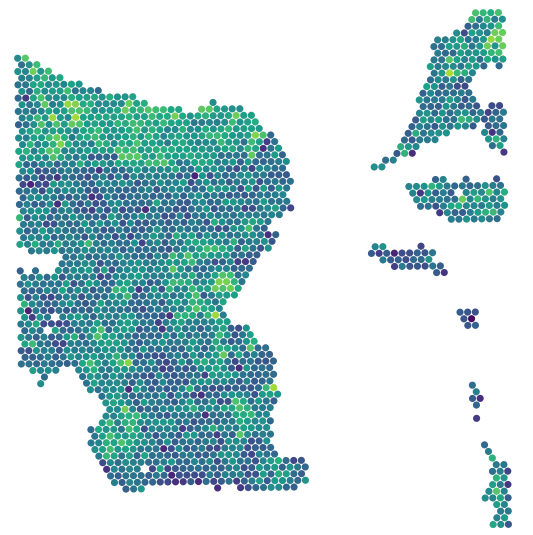
E2F Targets



0 0.152

UCell score

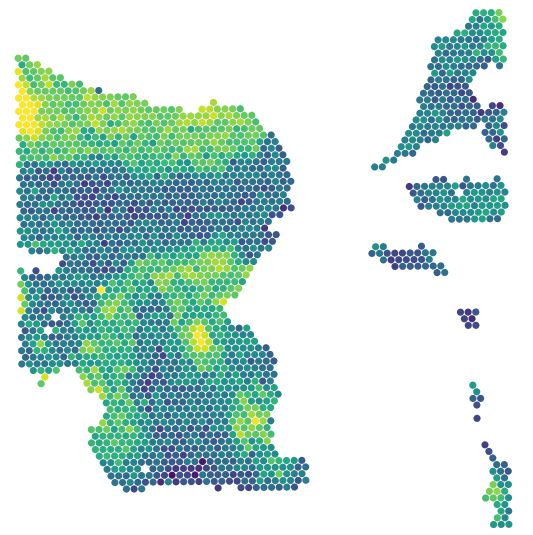
G2M Checkpoint



0 0.134

UCell score

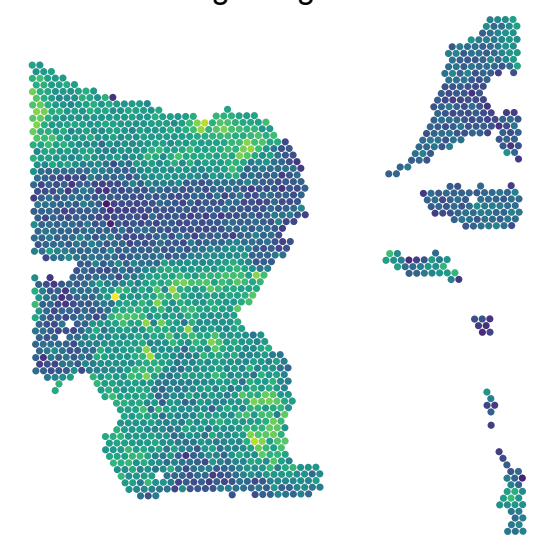
Hypoxia



0 0.211

UCell score

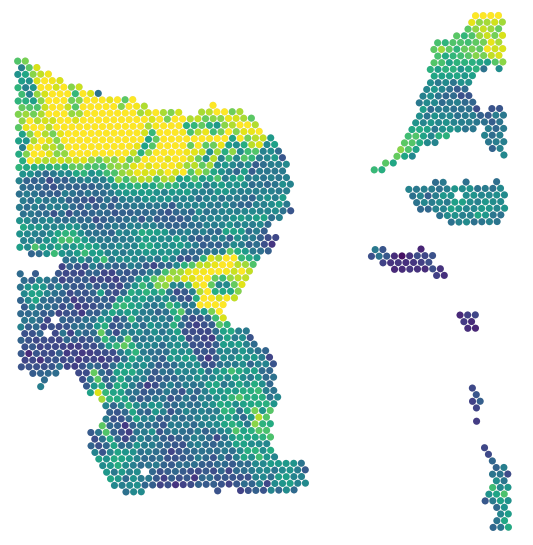
Tnfa Signaling Via Nfkb



0 0.275

UCell score

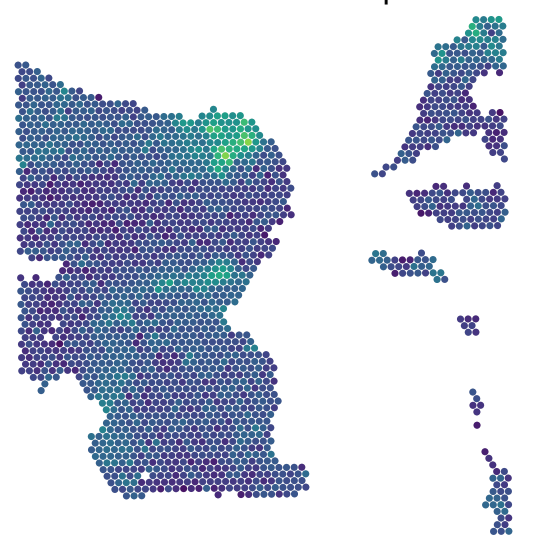
Emt



0 0.386

UCell score

Interferon Gamma Response



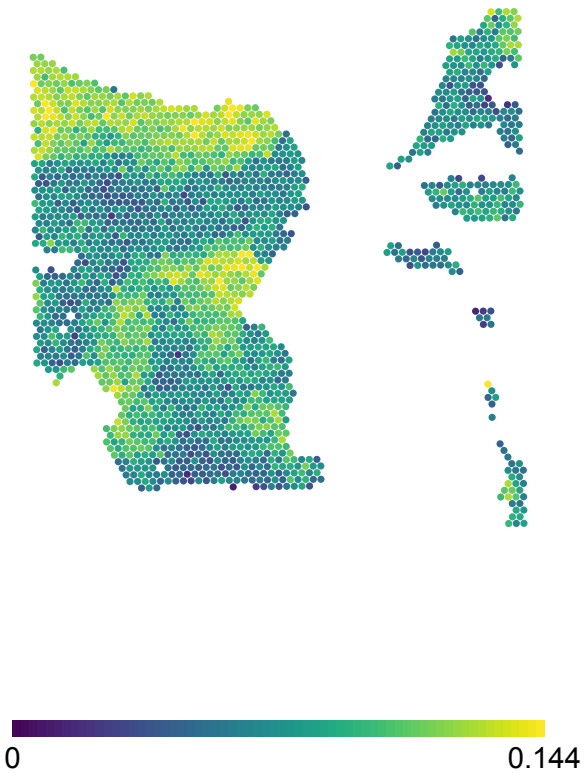
0 0.275

UCell score

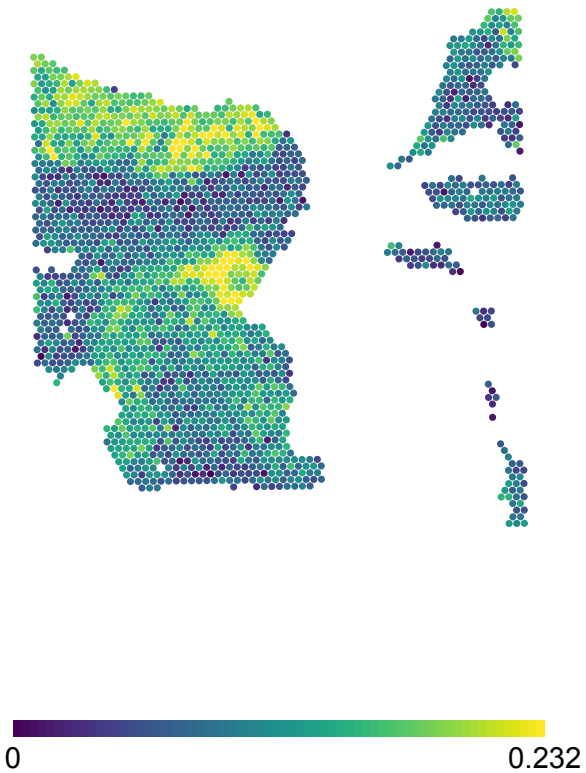
GSE319536_BC19 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

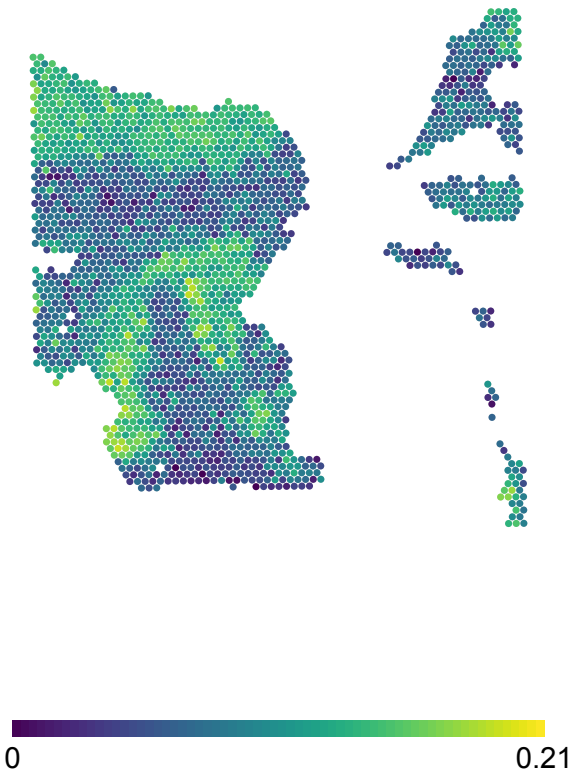
Il2 Stat5 Signaling



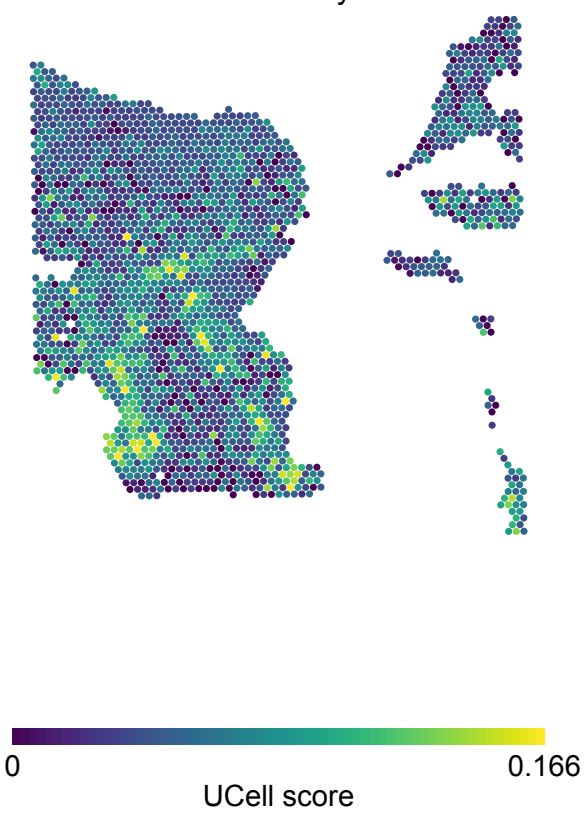
Tgf Beta Signaling



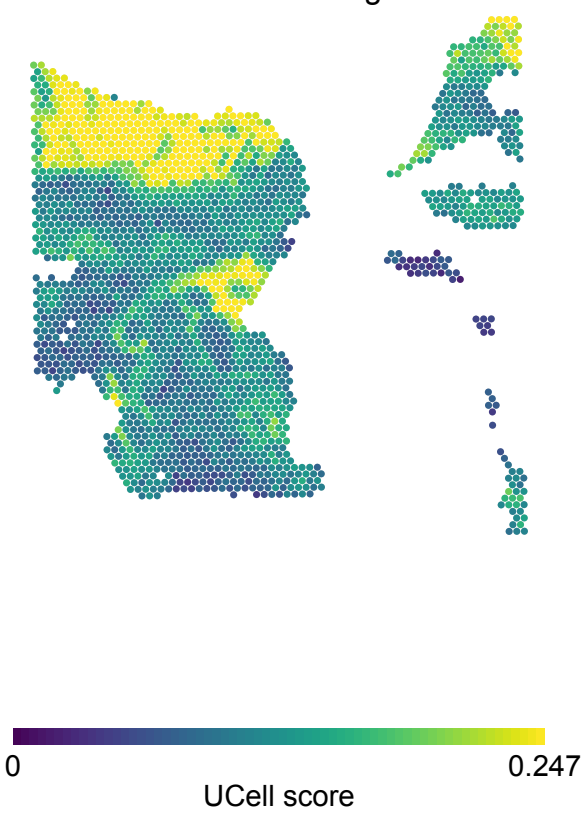
Cholesterol Homeostasis



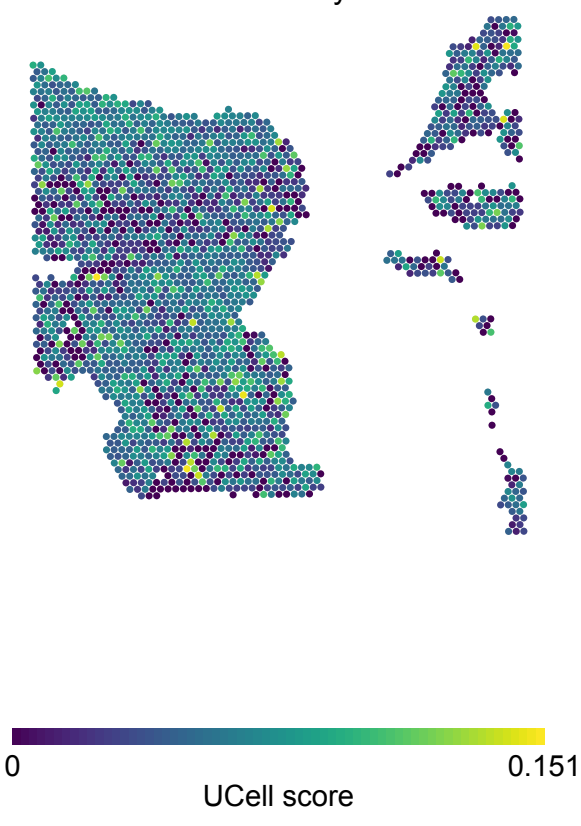
Cholesterol Biosynthesis



Extracellular Matrix Organization



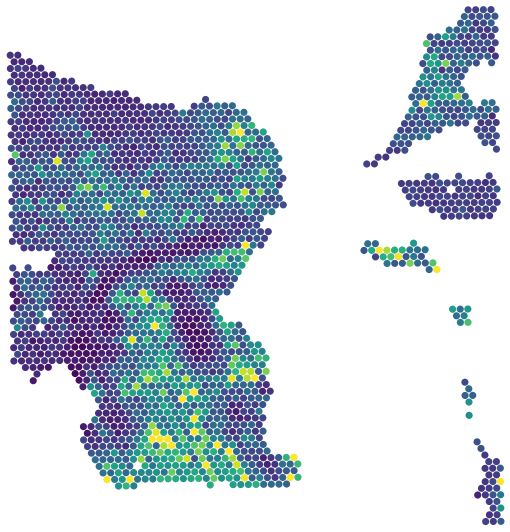
Co Inhibition By Ctla4



GSE319536_BC19 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

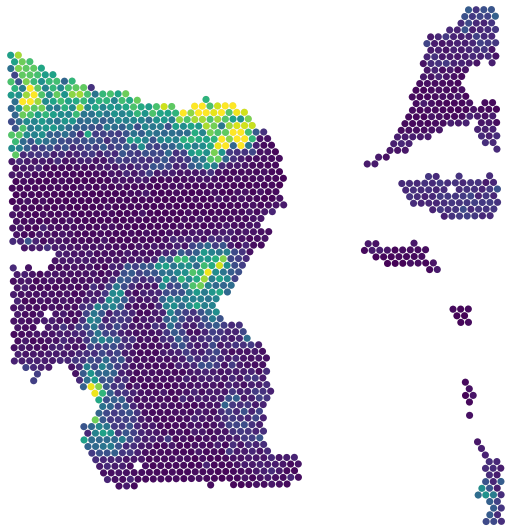
Treg composition



0 0.102

Fraction of all 30

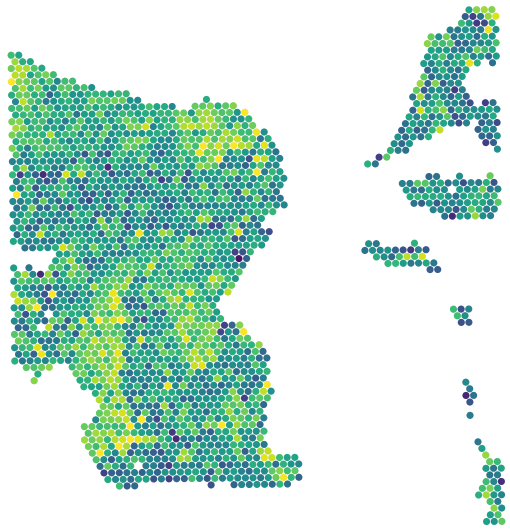
Treg estimated abundance



0 2.85

Posterior mean estimate

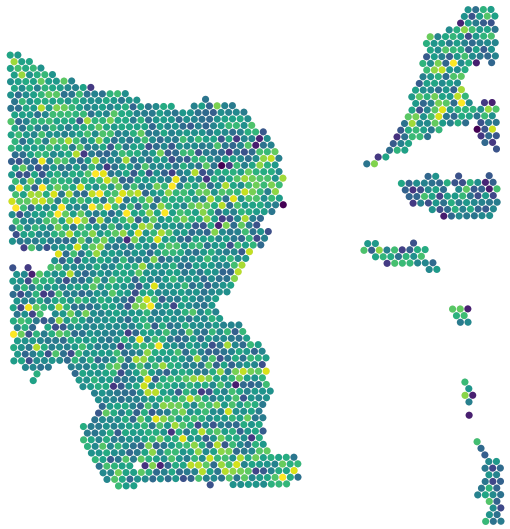
Treg Resting Vs Activated Dn



0 0.0778

UCell score

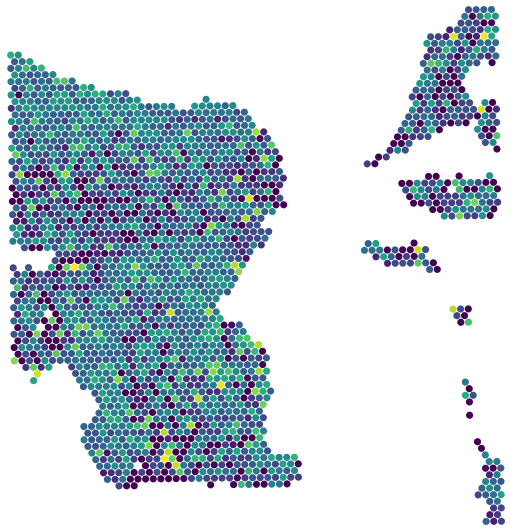
Treg Resting Vs Activated Up



0 0.0609

UCell score

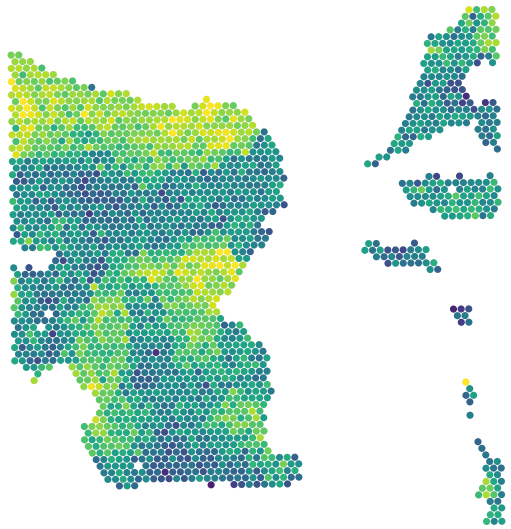
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

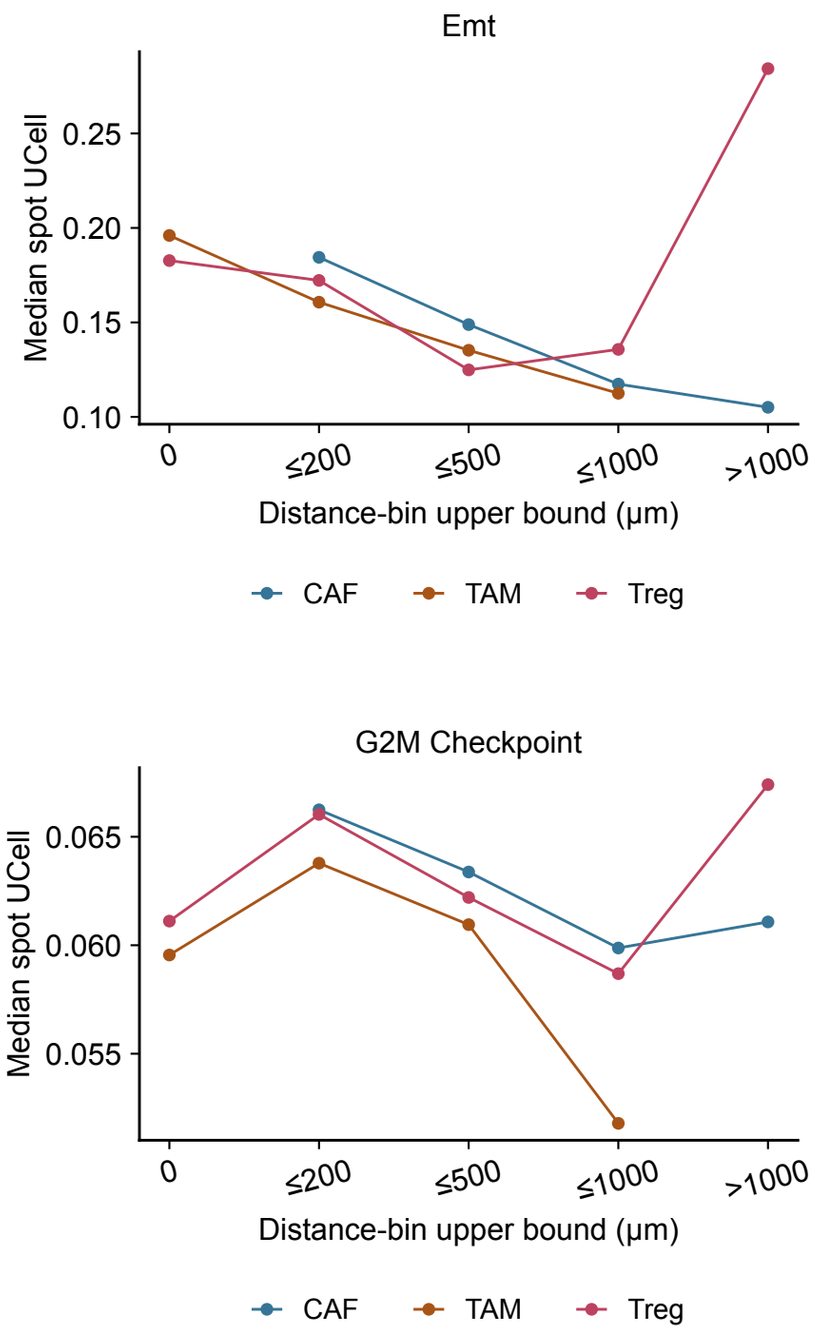


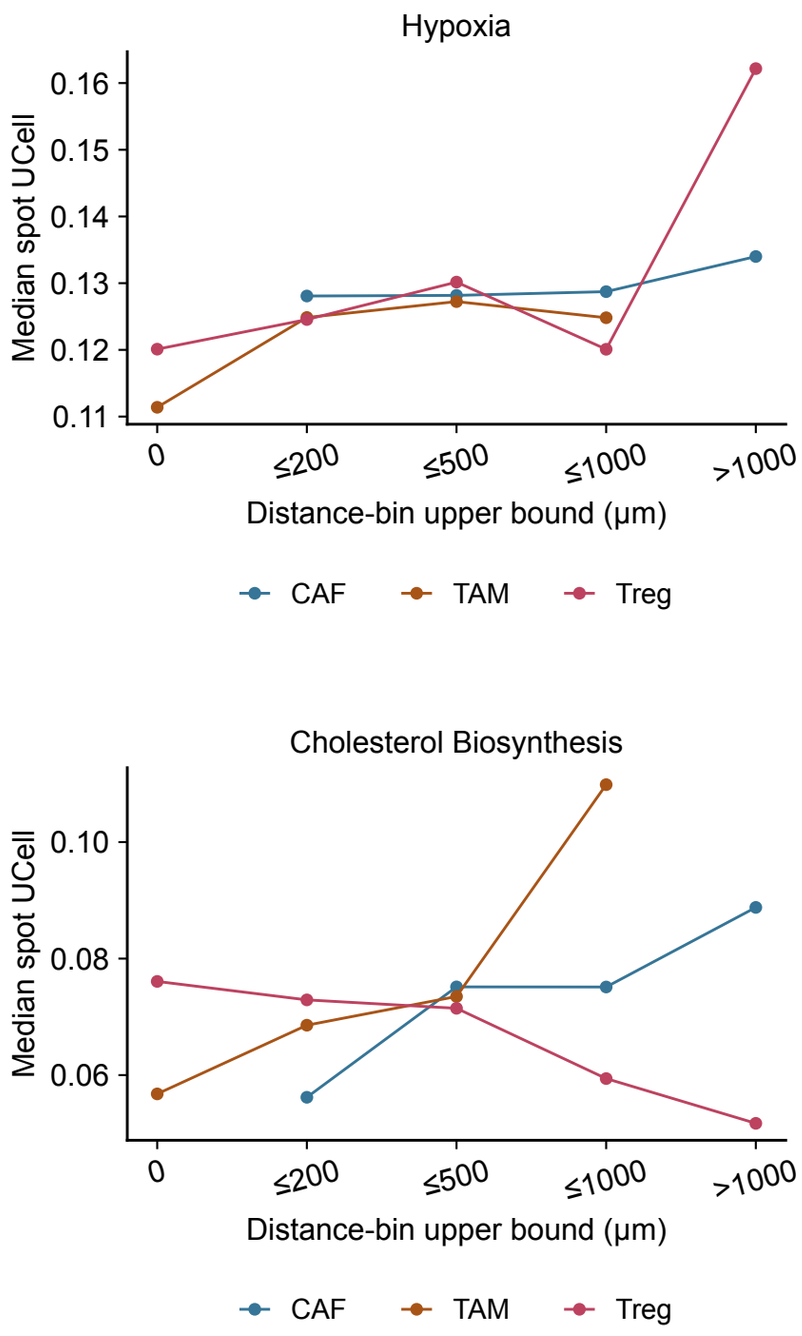
0 0.144

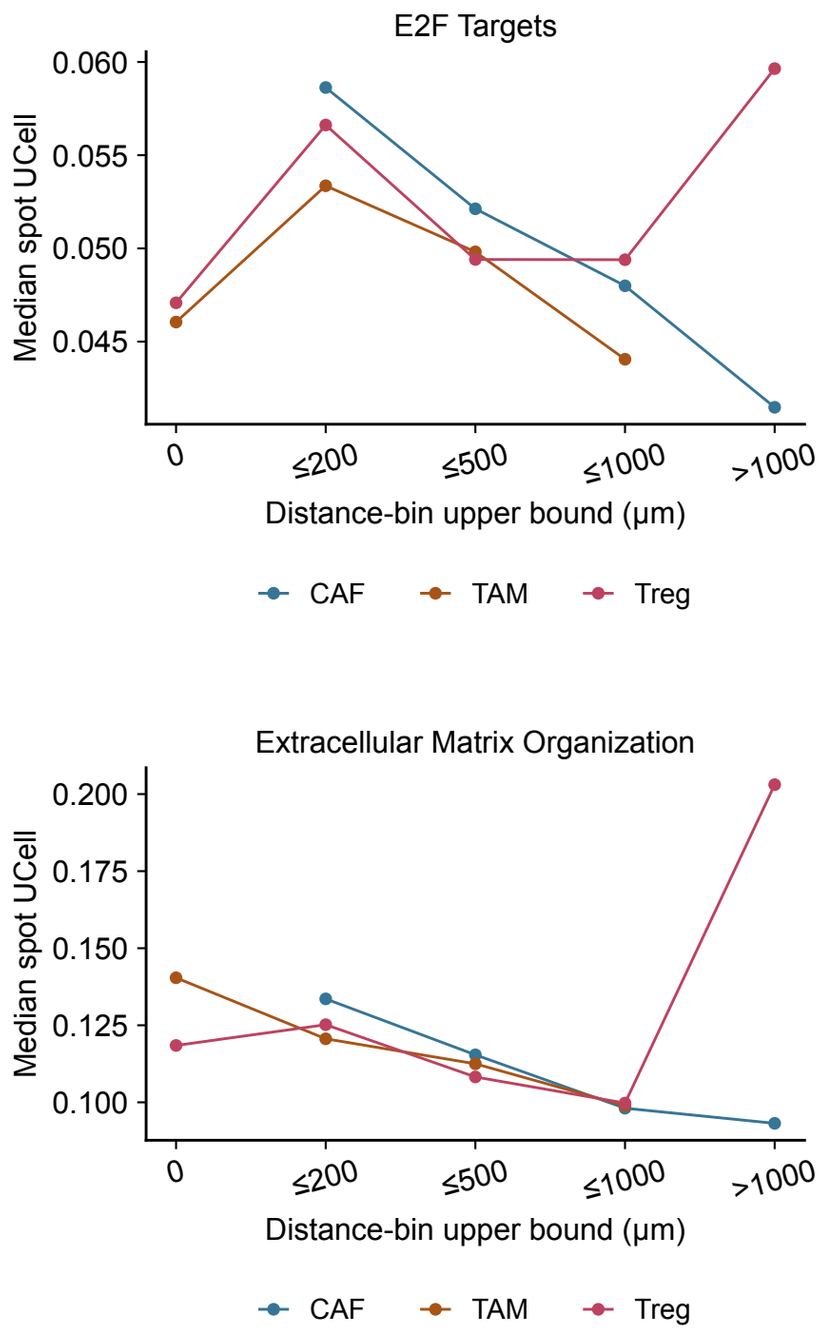
UCell score

GSE319536_BC19 | Programs by target distance

Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

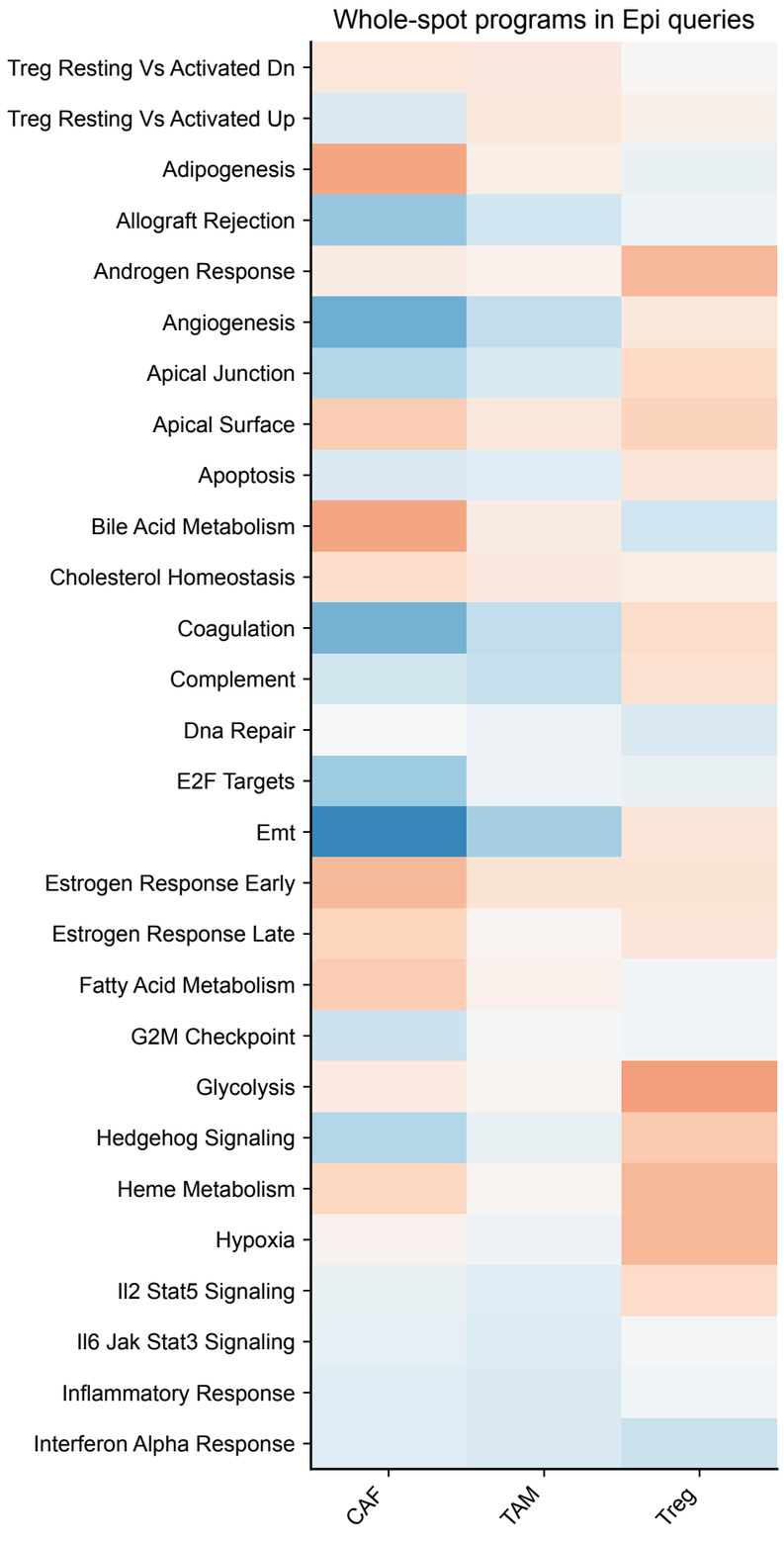


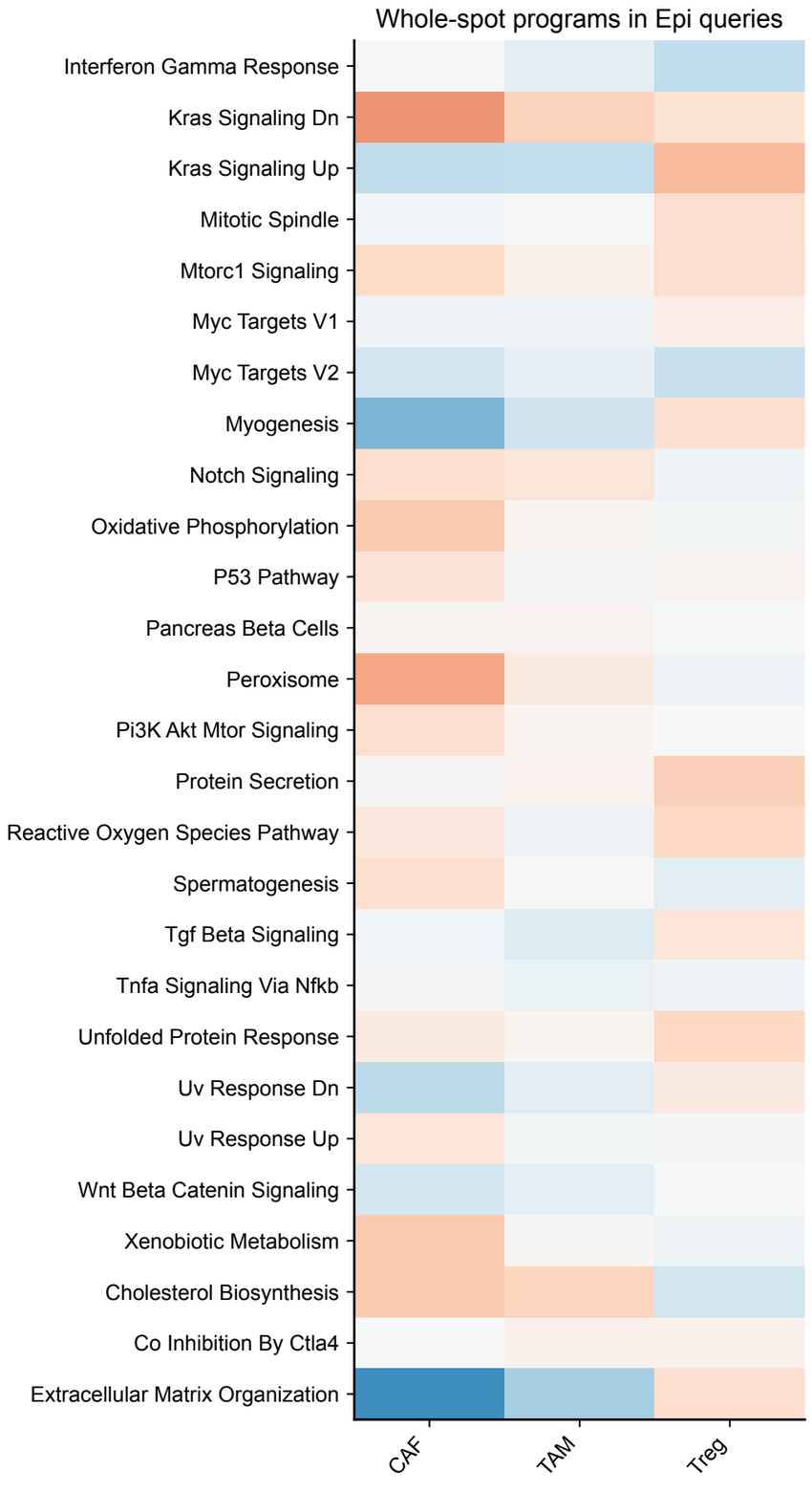




GSE319536_BC19 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6

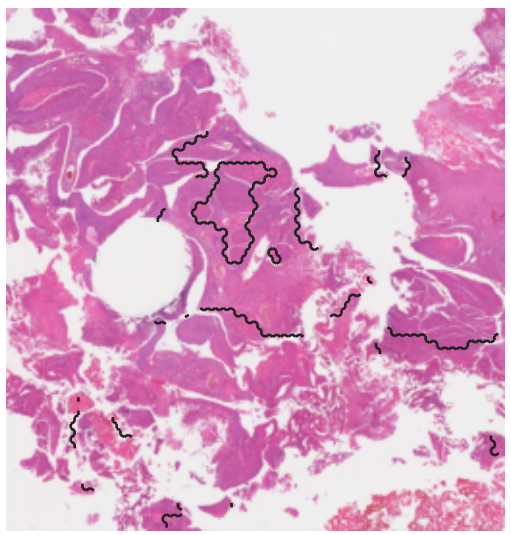




GSE319536_BC21 | Tissue context

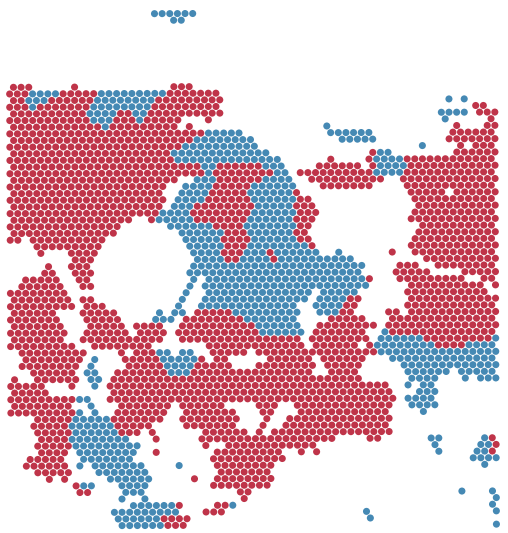
Aligned image pixels for display • frozen v3.4 regions • grey = outside displayed coverage

H&E and observed lesion edge



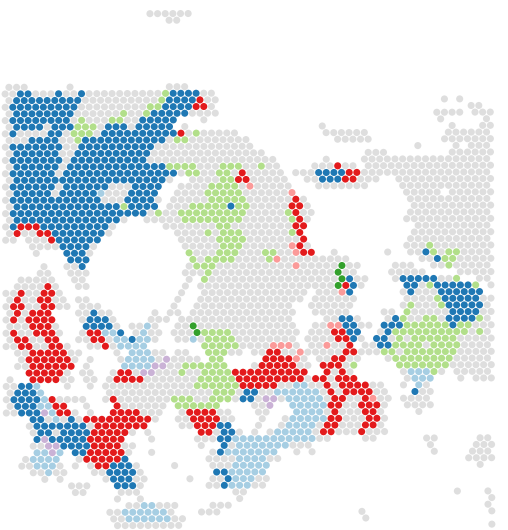
Black line: observed candidate outer edge

Final candidate location



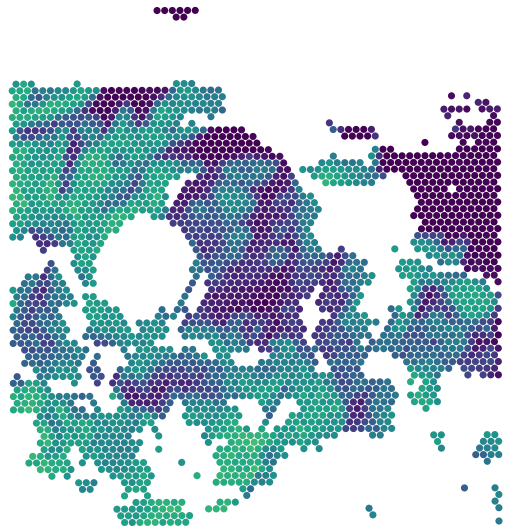
inside outside

Accepted R12 (original 0–8)



0 2 6 7 8
1 3

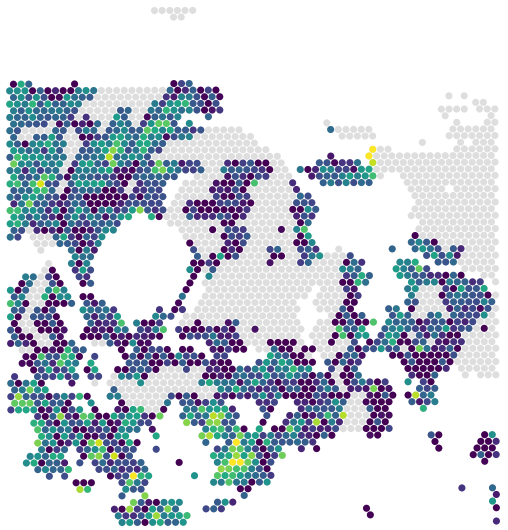
Epithelial support



0 1

Fraction of all 30

MP10 in Epi queries



0 0.147

Mixed-spot Top20 AUCell

Epi-component MP10 (valid only)

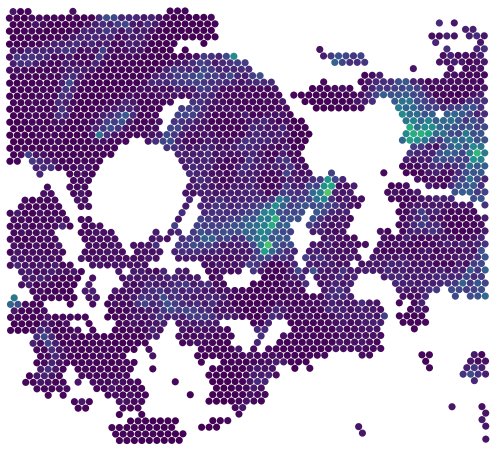
0 0.14

Matched-model AUCell

GSE319536_BC21 | Candidate neighborhoods

Columns: full-tissue composition, P80 enriched locations, nearest target distance in Epi queries

CAF composition

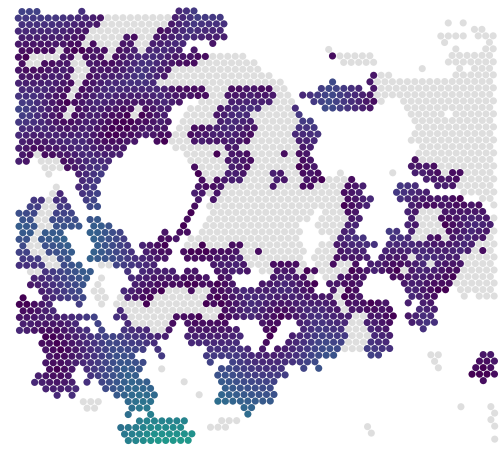


0 0.418
Fraction of all 30

CAF P80 targets

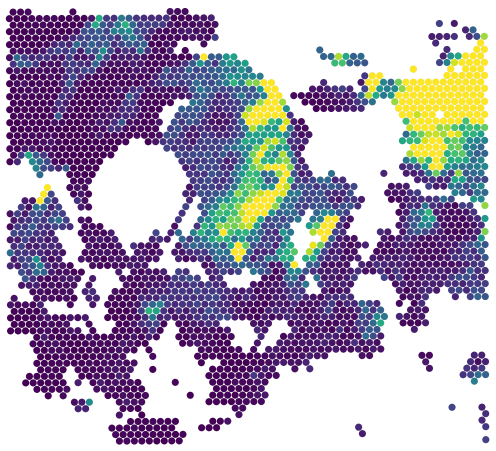
False True

CAF distance



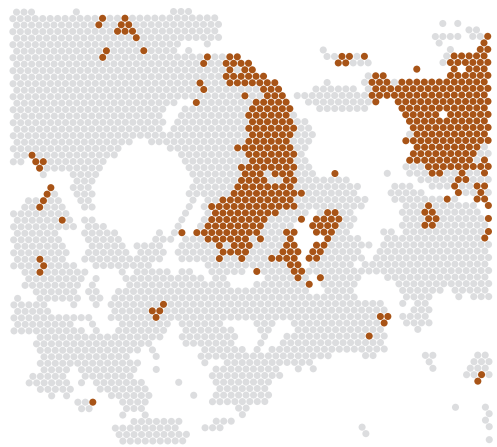
0 3.1e+03
Nearest spot center (μm)

TAM composition



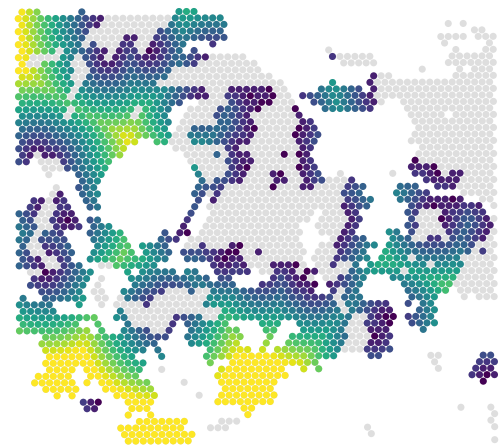
0 0.291
Fraction of all 30

TAM P80 targets



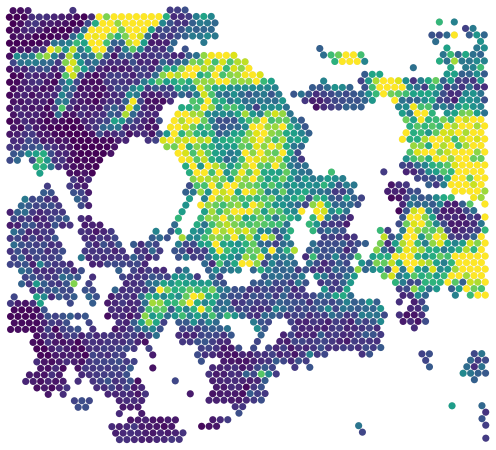
False True

TAM distance



0 1.13e+03
Nearest spot center (μm)

Treg composition

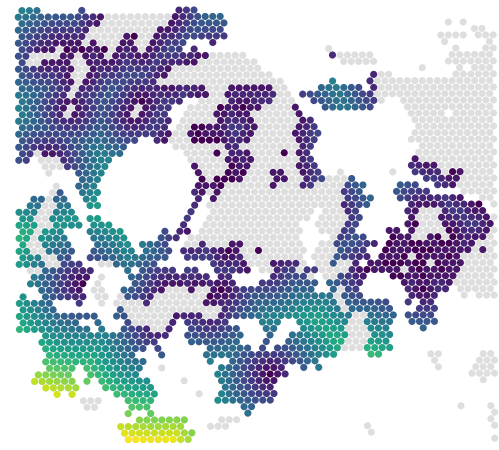


0 0.102
Fraction of all 30

Treg P80 targets

False True

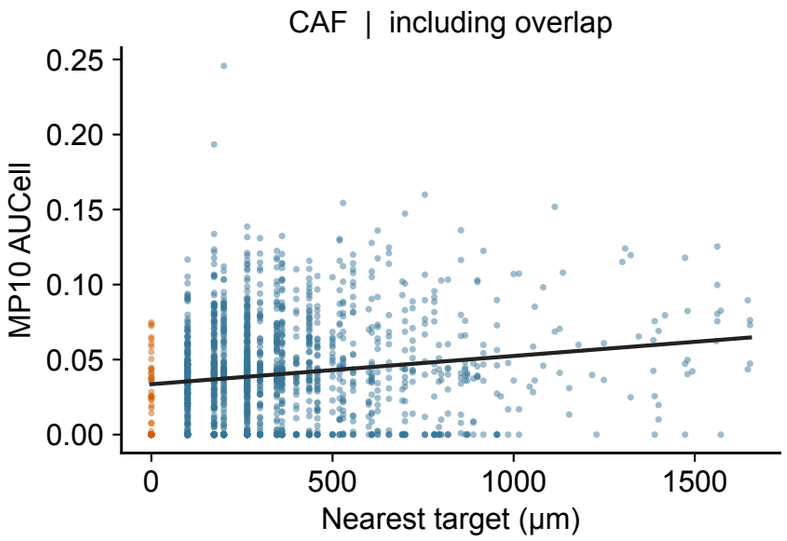
Treg distance



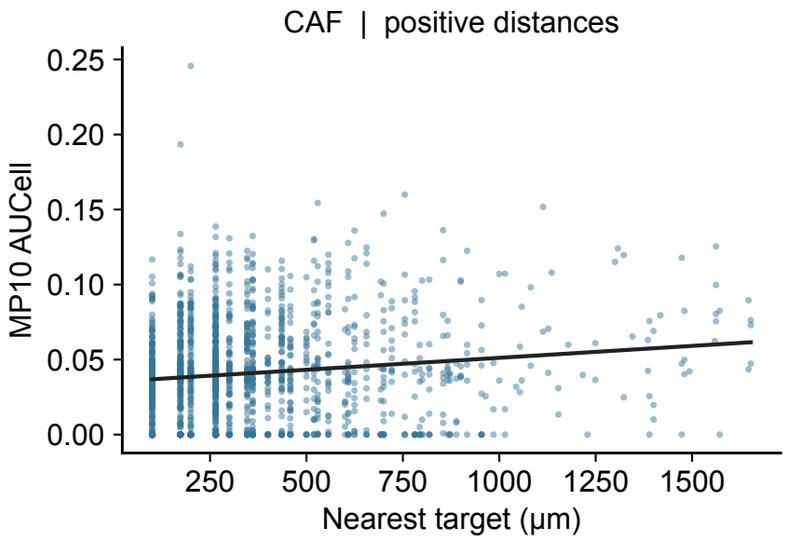
0 1.61e+03
Nearest spot center (μm)

GSE319536_BC21 | MP10 versus nearest target

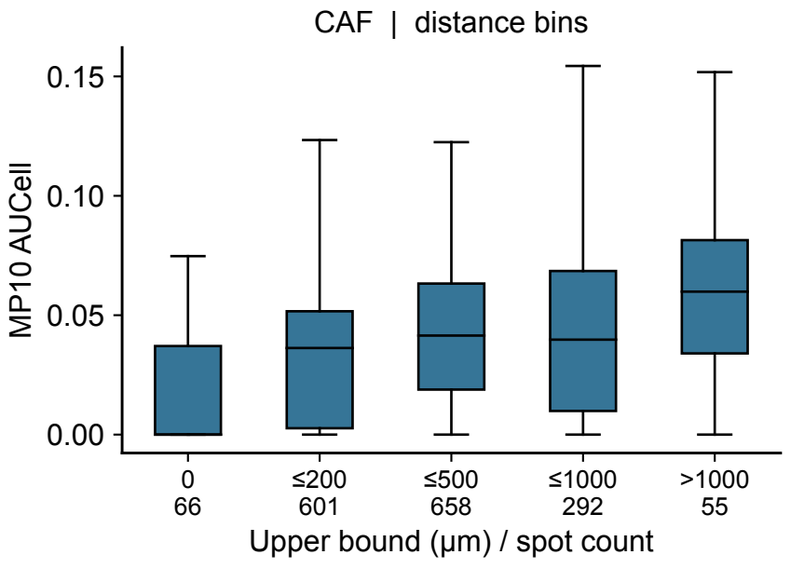
Mixed-spot Top20 AUCell in Epi queries • section P80, all tissue targets • same tissue component



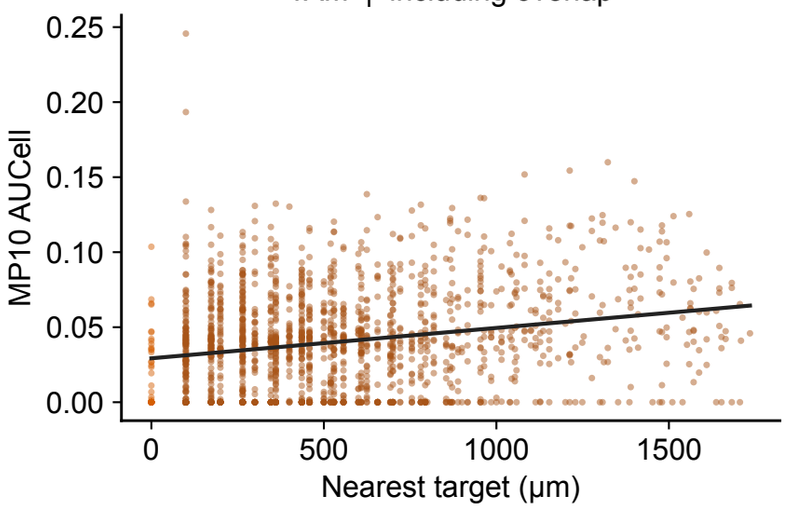
r +0.16 ρ +0.16
n 1672/1705 queries • 517 targets



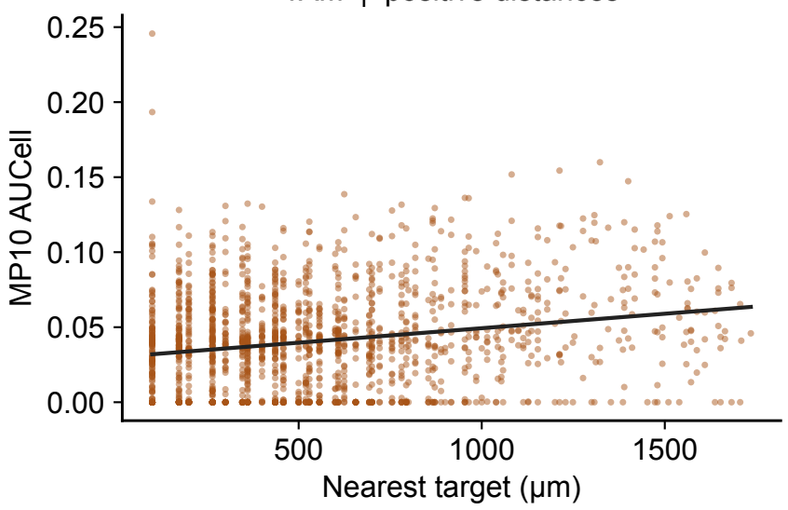
r +0.13 ρ +0.13
n 1606/1705 queries • 517 targets



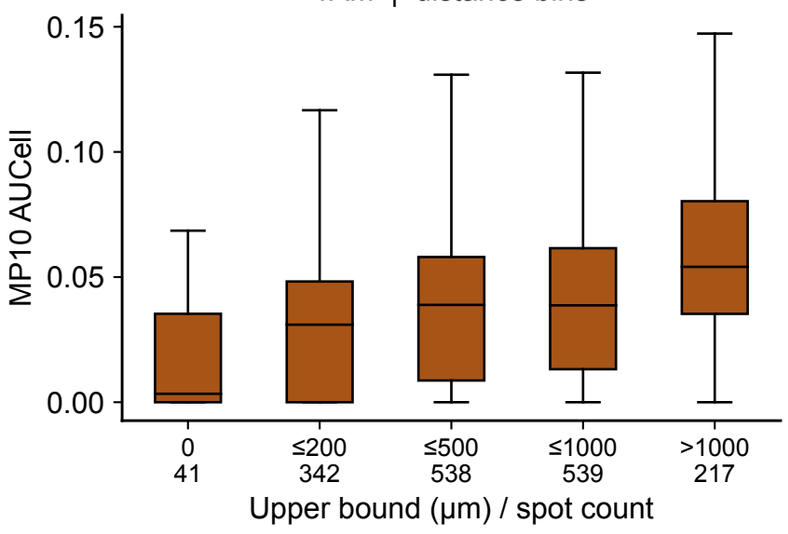
Orange = zero distance
Line = within-section linear fit



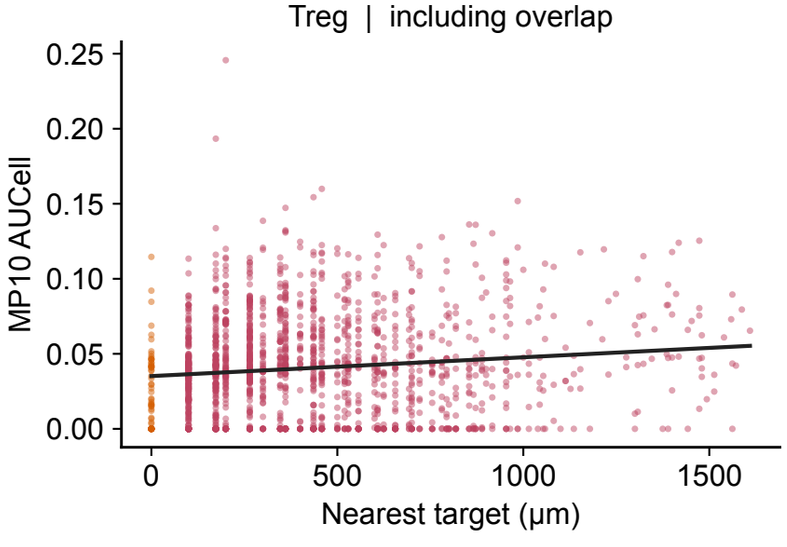
r +0.23 ρ +0.22
n 1677/1705 queries • 517 targets



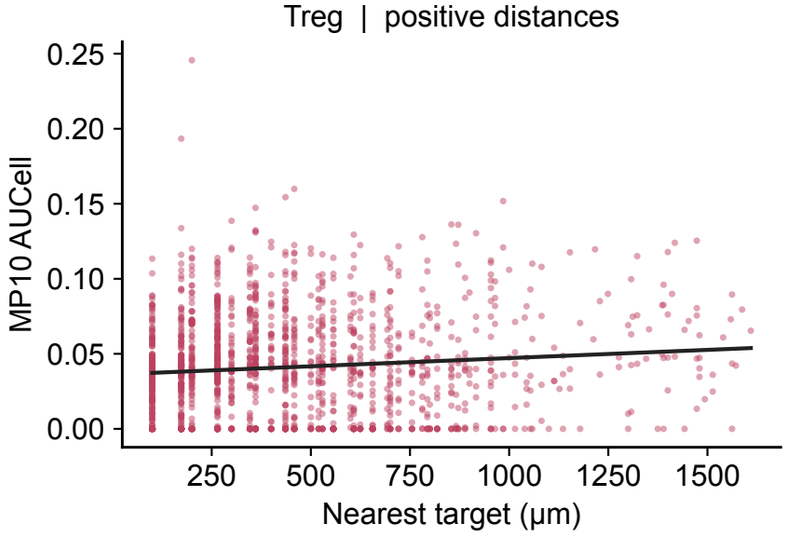
r +0.22 ρ +0.20
n 1636/1705 queries • 517 targets



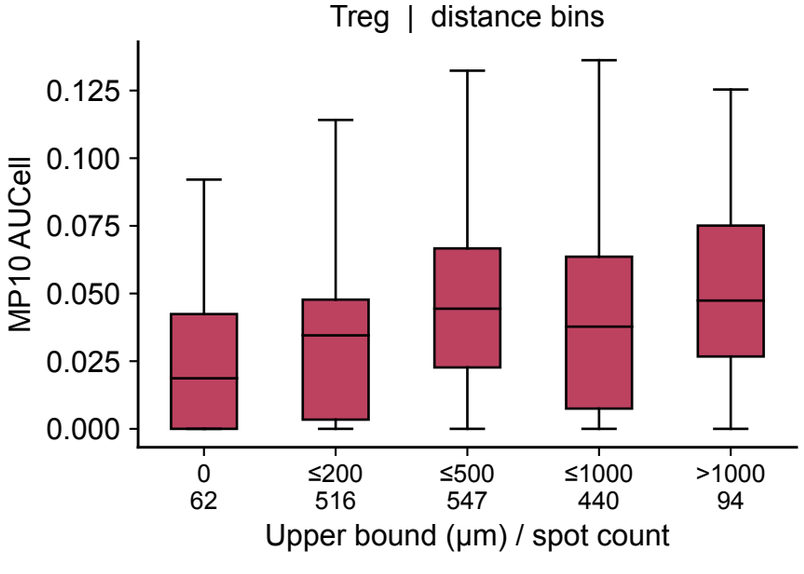
Orange = zero distance
Line = within-section linear fit



r +0.12 ρ +0.13
n 1659/1705 queries • 517 targets



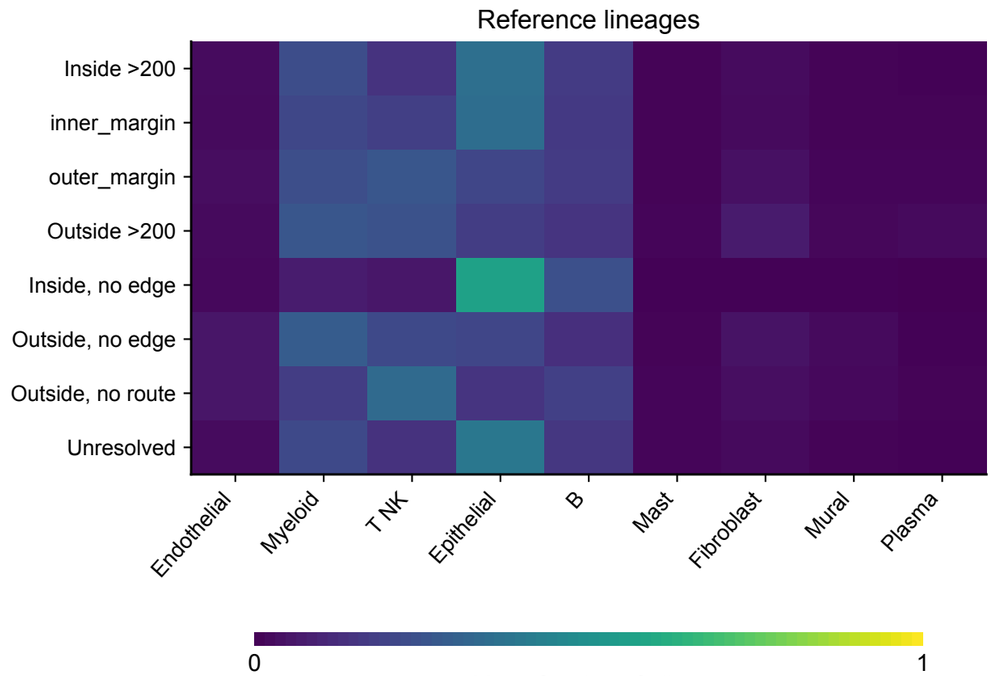
r +0.10 ρ +0.11
n 1597/1705 queries • 517 targets

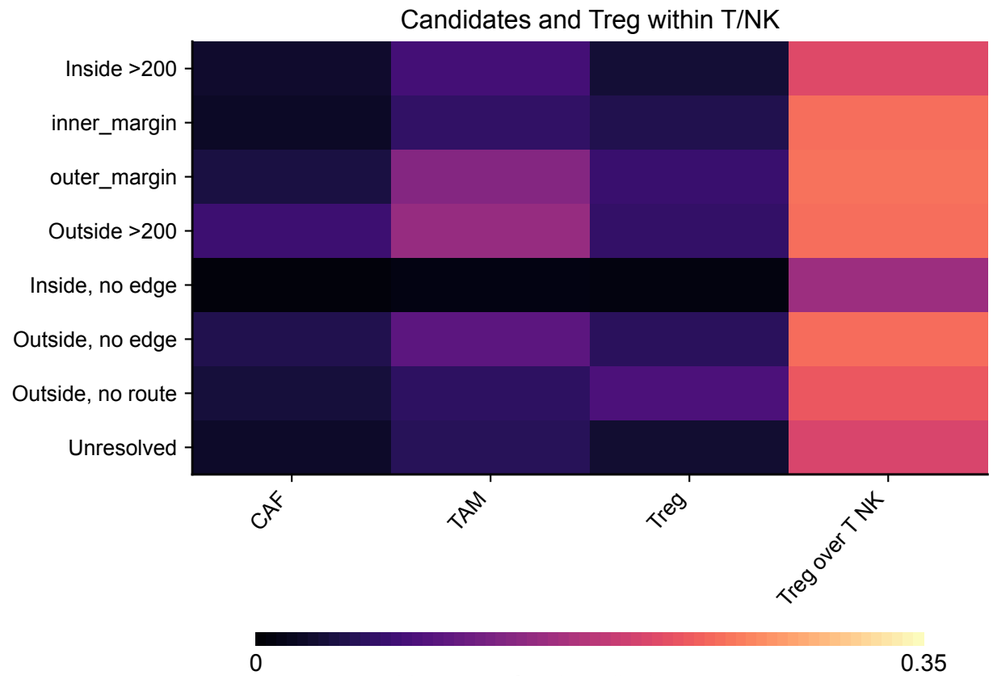


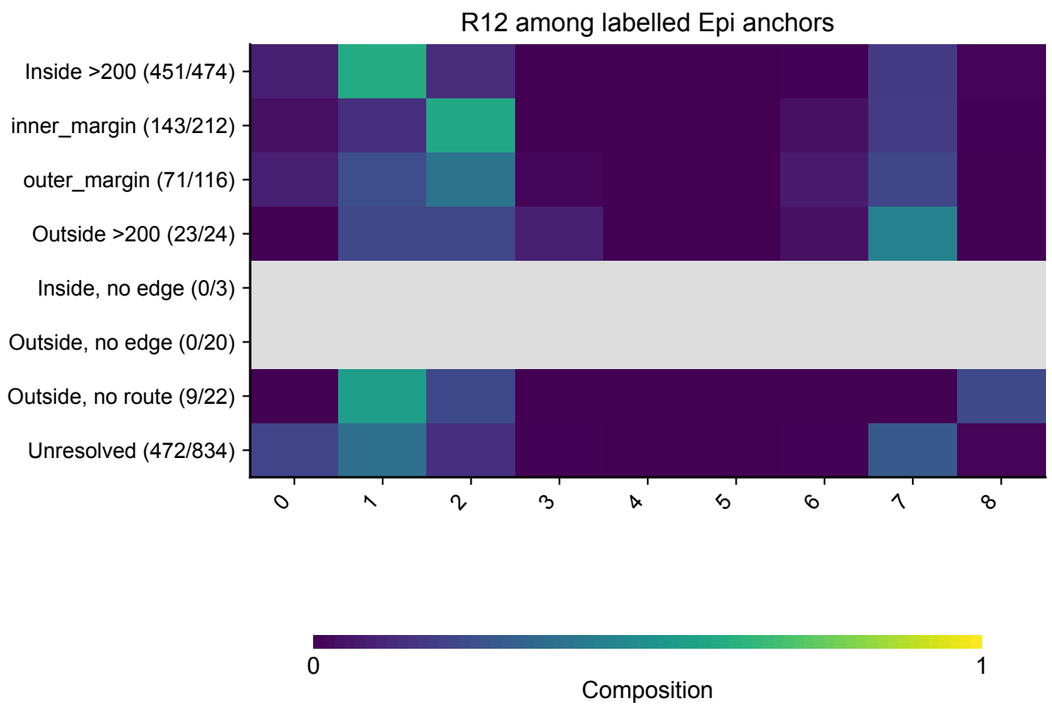
Orange = zero distance
Line = within-section linear fit

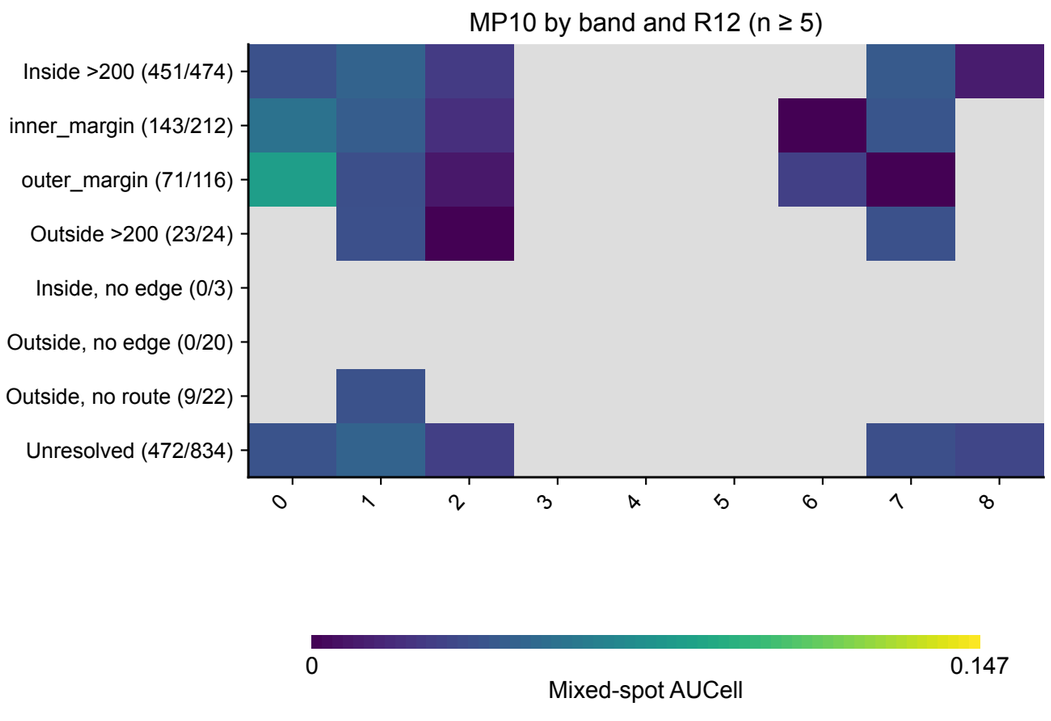
GSE319536_BC21 | Region composition

v3.4 200 µm bands • each mean gives equal weight to observed spots • grey = unavailable





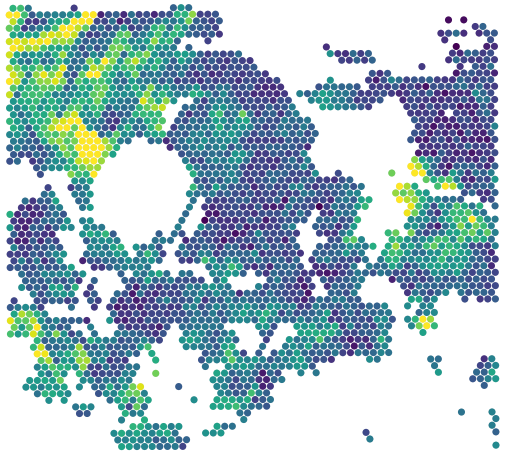




GSE319536_BC21 | Growth, stress and inflammation

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

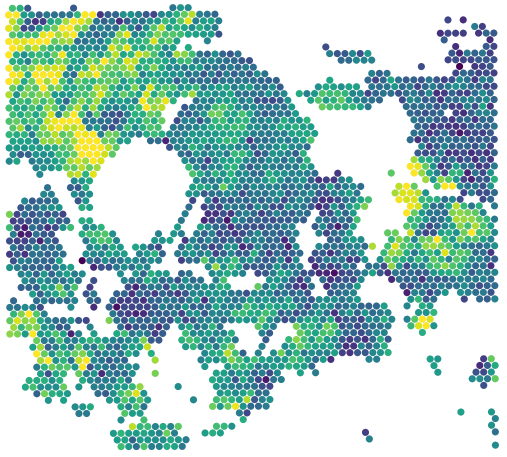
E2F Targets



0 0.152

UCell score

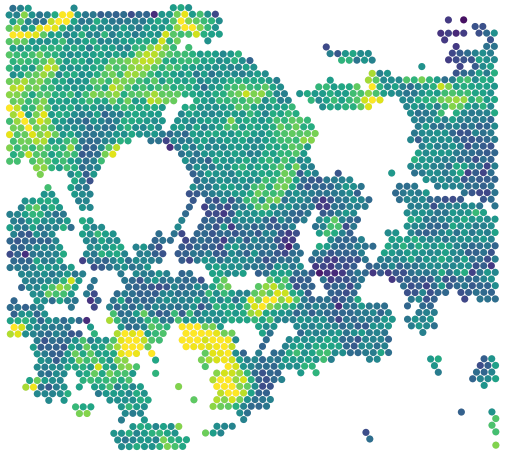
G2M Checkpoint



0 0.134

UCell score

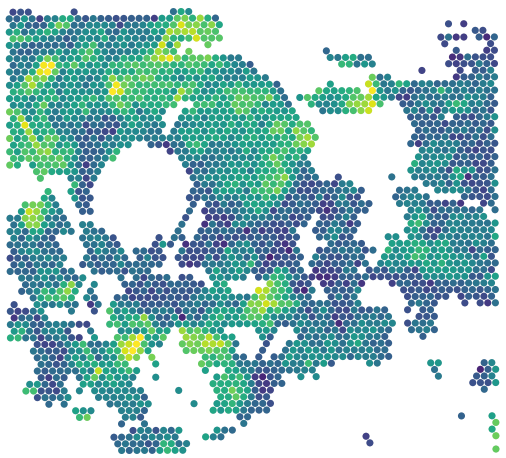
Hypoxia



0 0.211

UCell score

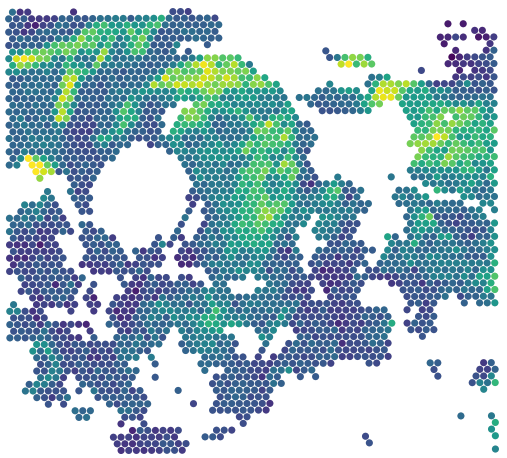
Tnfa Signaling Via Nfkb



0 0.275

UCell score

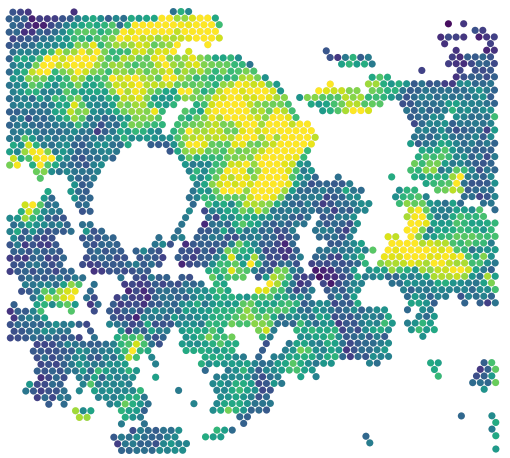
Emt



0 0.386

UCell score

Interferon Gamma Response



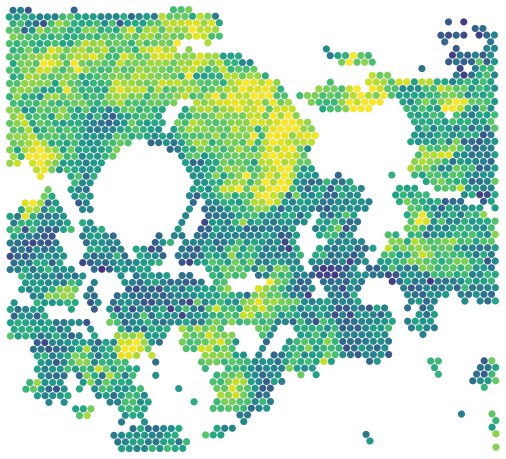
0 0.275

UCell score

GSE319536_BC21 | Immune, lipid and matrix programs

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

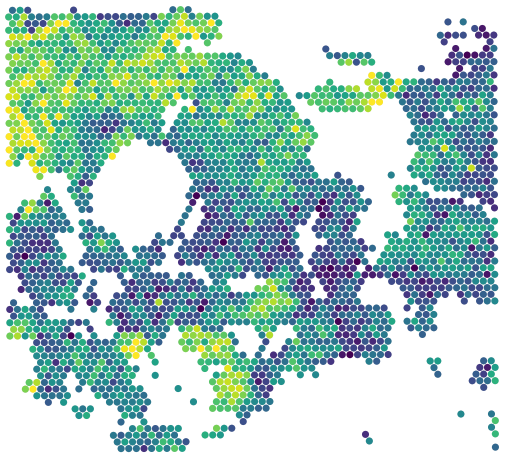
Il2 Stat5 Signaling



0 0.144

UCell score

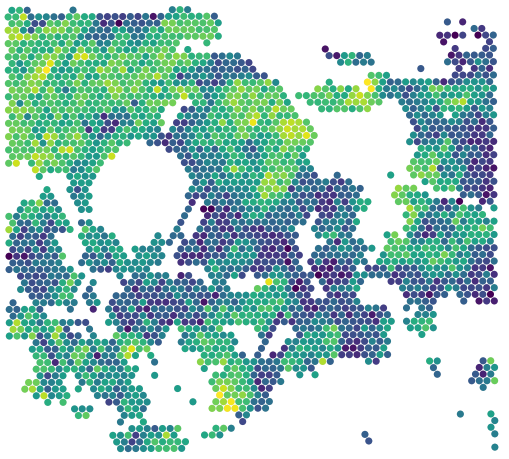
Tgf Beta Signaling



0 0.232

UCell score

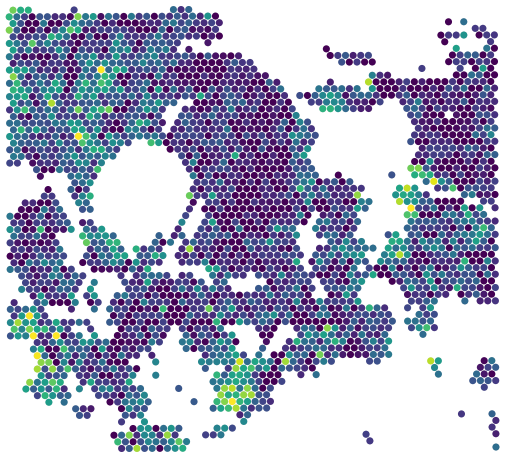
Cholesterol Homeostasis



0 0.21

UCell score

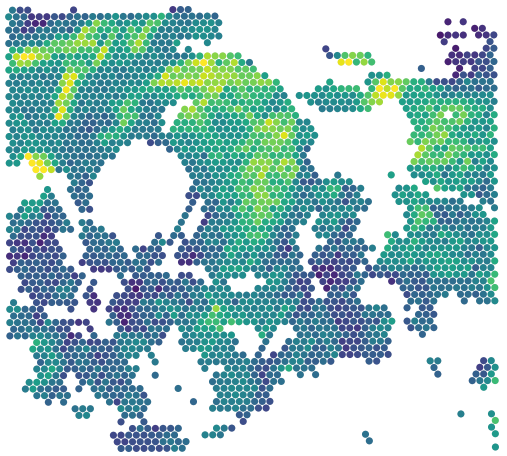
Cholesterol Biosynthesis



0 0.166

UCell score

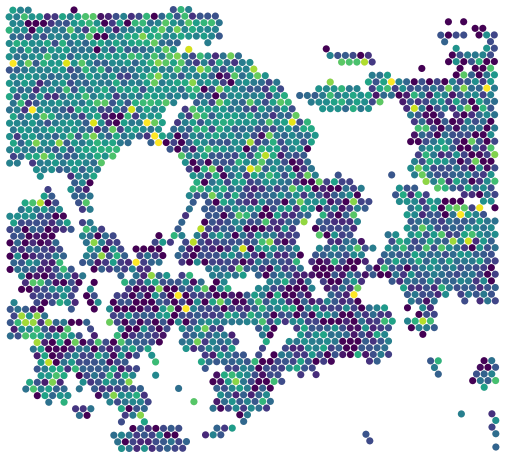
Extracellular Matrix Organization



0 0.247

UCell score

Co Inhibition By Ctla4



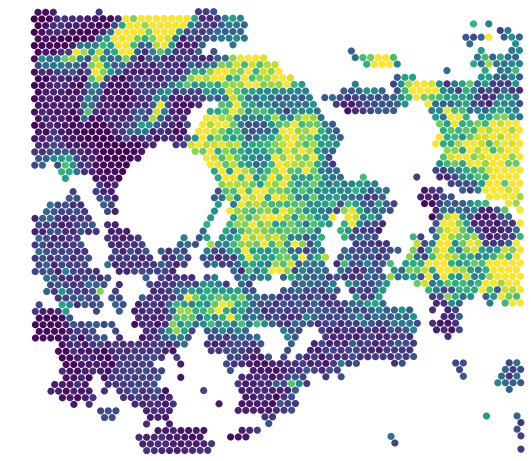
0 0.151

UCell score

GSE319536_BC21 | Treg context across the full tissue

Whole-spot RNA programs (UCell) • display scales fixed per feature across all sections

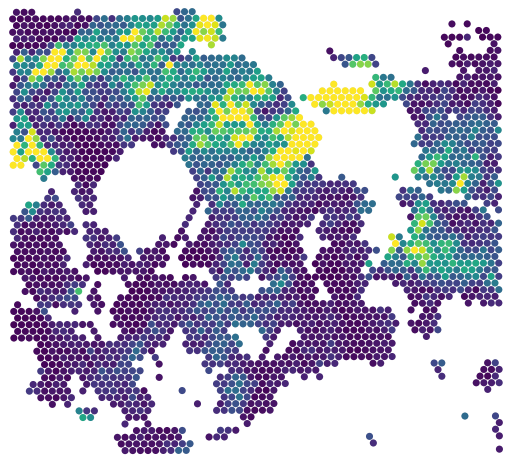
Treg composition



0 0.102

Fraction of all 30

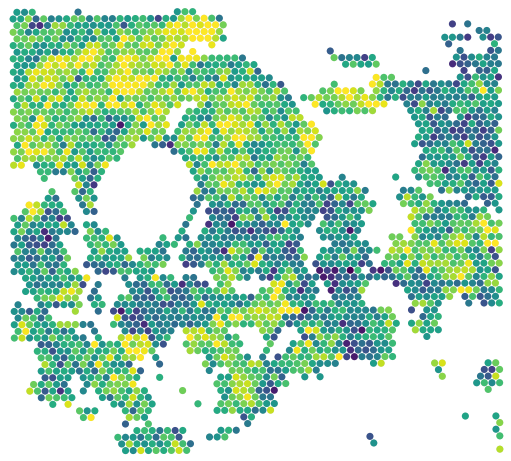
Treg estimated abundance



0 2.85

Posterior mean estimate

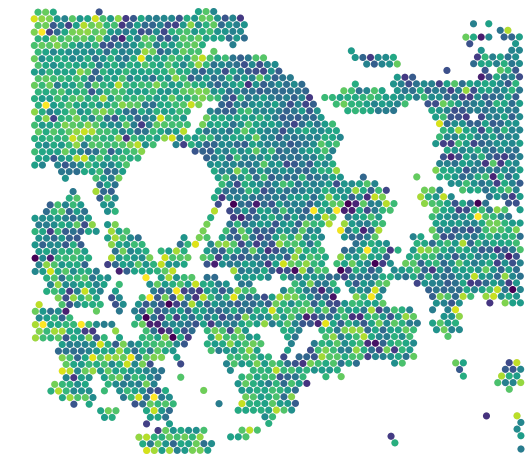
Treg Resting Vs Activated Dn



0 0.0778

UCell score

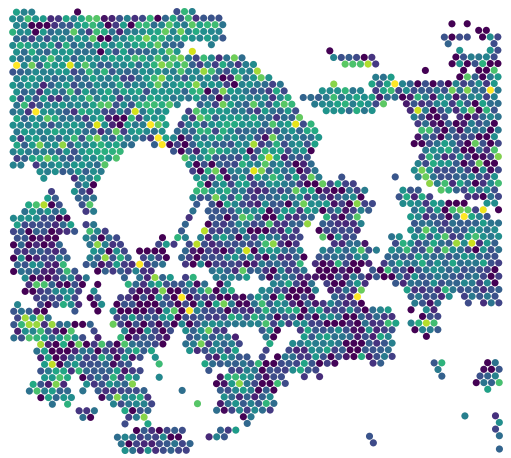
Treg Resting Vs Activated Up



0 0.0609

UCell score

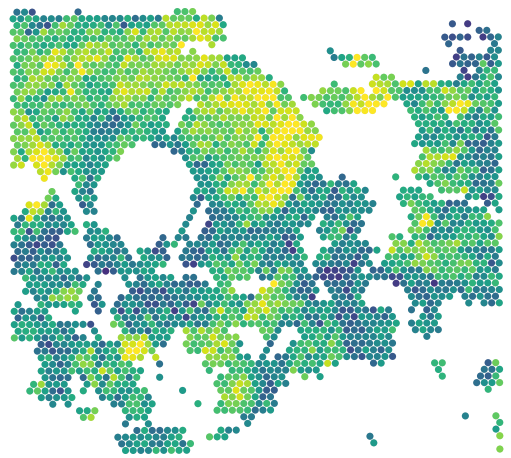
Co Inhibition By Ctla4



0 0.151

UCell score

Il2 Stat5 Signaling

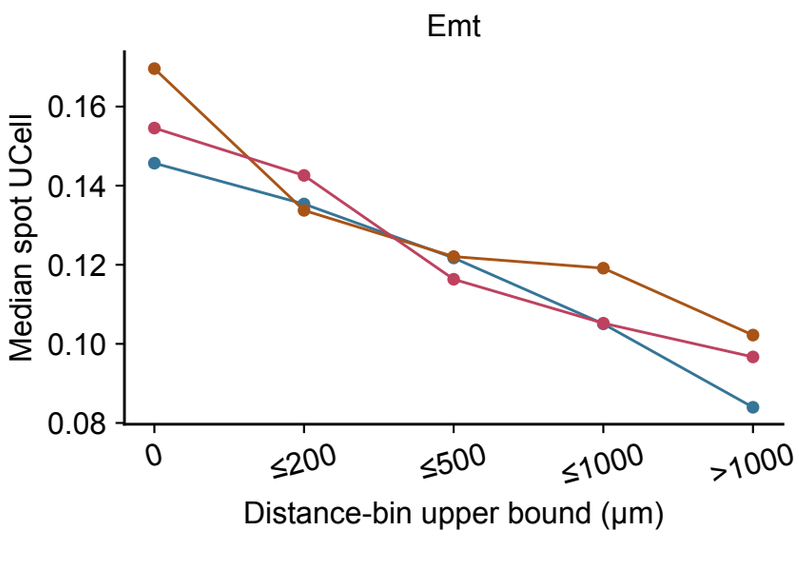


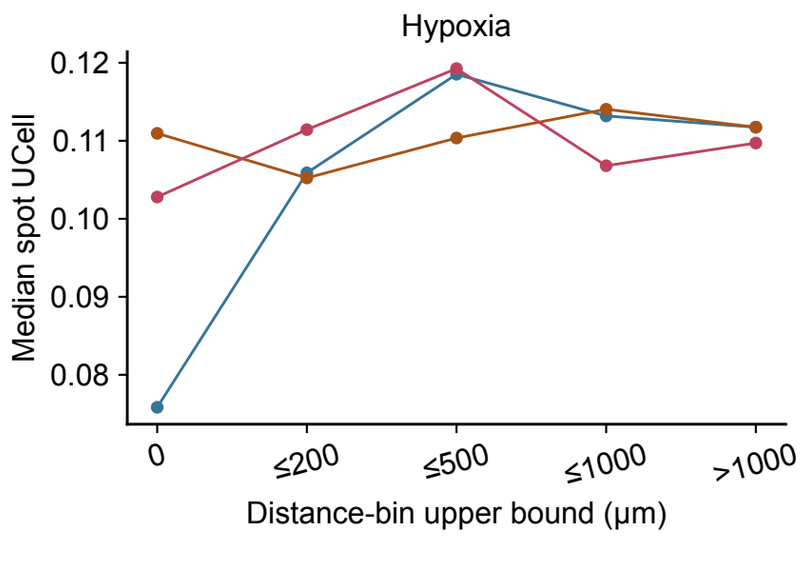
0 0.144

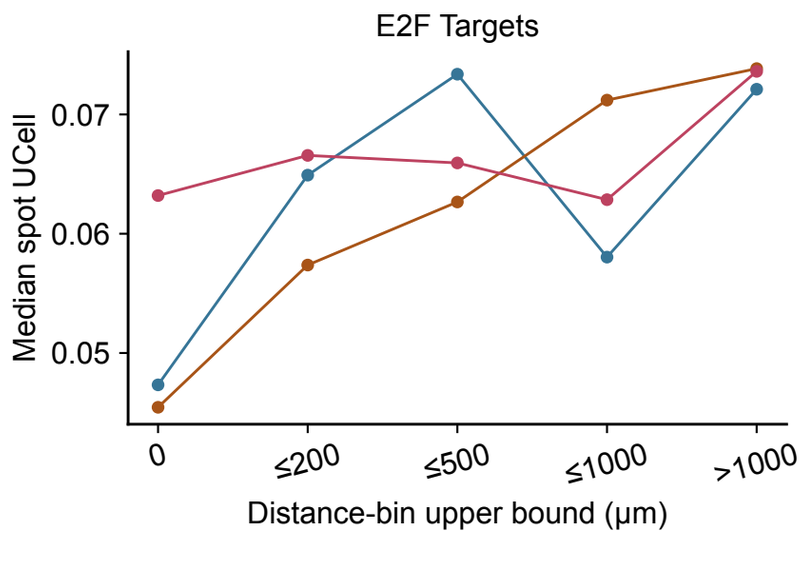
UCell score

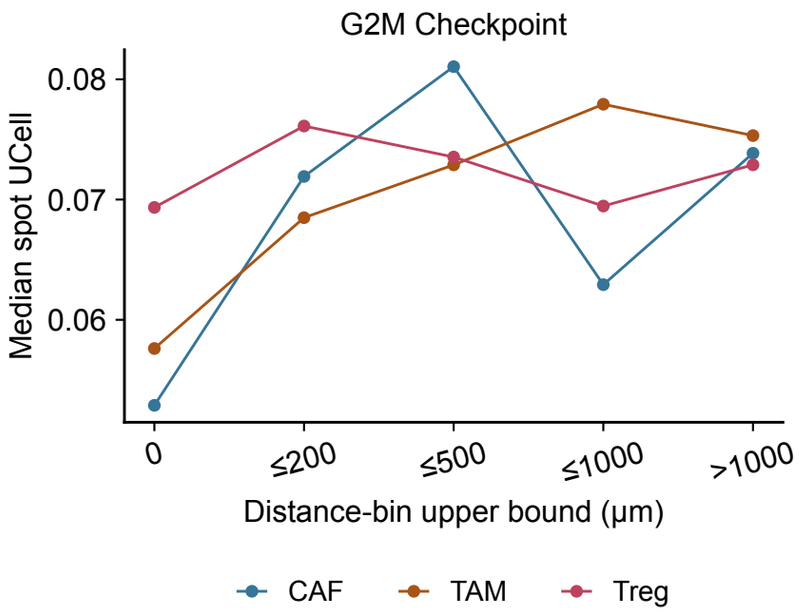
GSE319536_BC21 | Programs by target distance

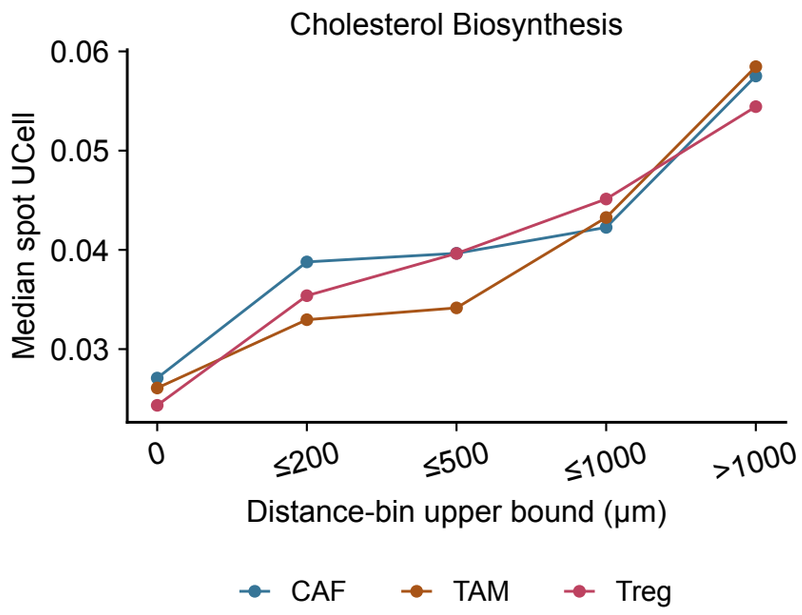
Epi queries • medians at observed distance bins • lines stop where a bin has fewer than 5 spots

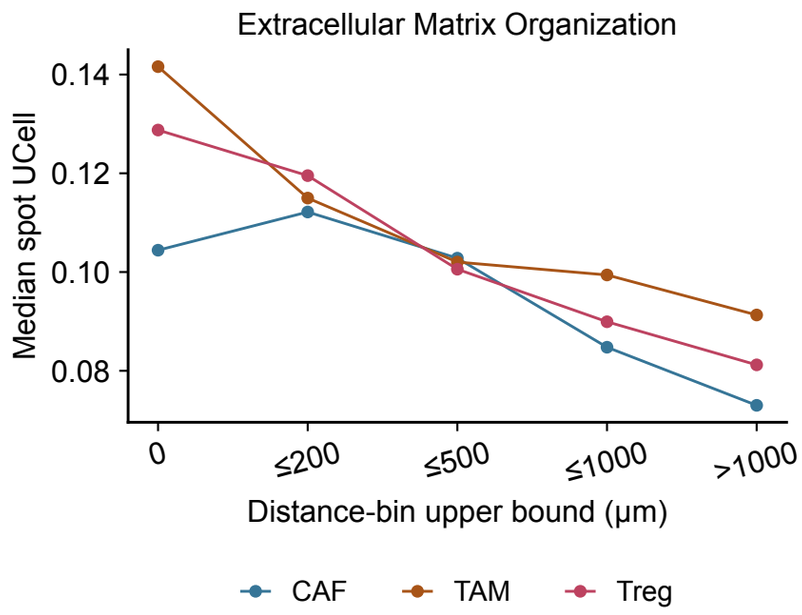












GSE319536_BC21 | Program–distance associations

All 55 UCell programs • one section • including zero distances • display cap ±0.6